

MEP Specifications

G+4P+28+RF

RESIDENTIAL DEVELOPMENT

ON PLOT 6117242, BUKADRA,

DUBAI – U.A.E.

PARKWAY BY PRESTIGE ONE

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Green Concept Engineering Consultants

جرين كونسپت للإستشارات الهندسية

T: + 971 4 2599 112 F: + 971 4 2599 113

E: info@greencec.ae Website: www.greencec.ae

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GENERAL REQUIREMENTS

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1.0 GENERAL

- 1.1 The Instructions to Tenderers, Form of Tender, Main Conditions of contract, Note on Pricing, the Preliminary Section and any Supplementary Conditions of Contract apply to and are part of this Specification and all subsections.
- 1.2 Sections of this Services Specification which relate to other Sections are not intended to supersede the specifications of these clauses, but to guide the Contractor in tendering and performance. In no event shall a quality less than that indicated here be acceptable.
- 1.3 The scope of works includes the supply, delivery to site, offloading, installation, testing, commissioning, demonstrating and staff training of the various complete systems outlined in this specification and on the drawings and documentation including liaison with the Main Contractor, Authorities, Master Plan Developer and their representatives and Consultants as necessary for the duration of the works and the defects liability period.
- 1.4 The contractor will be responsible for reaching an agreement on their Programme with the Main Contractor and coordinating every aspect of their design, submissions, approvals as well as installation, testing, commissioning and handover works with them.
- 1.5 Wherever the term authorities are indicated, it shall mean statutory authorities, the master plan developer or their representatives and services provider.

2.0 INTENT

- 2.1 Check issued drawings and accept responsibility for proper performance of all mechanical and plumbing systems in accordance with relevant standards and codes and design parameters listed hereinafter.
- 2.2 Provide all items, articles, materials, samples, operations, sundries, labour, supervision, guarantees, allowances for overhead and profit, etc., to achieve a fully functionable and acceptable system.
- 2.3 Consider the specifications as an integral part of the work together with the drawings. Consider any item or subject omitted from one, but mentioned or reasonably implied on the other as properly and sufficiently indicated and provide the same under the work of this section.
- 2.4 The Contractor is responsible for developing his own take off of materials and is to make this available as requested to the Consultant.

3.0 EXAMINATION OF SITE AND DRAWINGS

- 3.1 Visit the site of the proposed works and obtain all information as to existing conditions and limitations and all proposed works on adjacent sites and in adjacent areas which might affect the works on this site, whether by Private Individuals or by Government Authorities or others.
- 3.2 Examine the documents including the Specifications and Drawings of all other Sections before bidding and again before commencing any portion of the works.
- 3.3 Neither the Owner nor the Consultant will be responsible for any claim for extra work or expense resulting from the failure of the Contractor to be fully aware of Site Conditions, Drawing implications, etc.

4.0 PERMITS, FEES AND INSPECTIONS

- 4.1 Arrange for inspection of all work by the Authorities having jurisdiction over the works. This is to occur on an on-going basis throughout construction to avoid delays at the termination of the project. On completion of the work present to the Consultant, for the Owner, final unconditional approval certificates of the Inspecting Authorities and the master plan developer or their representatives.
- 4.2 Pay fees / charges as levied by statutory authorities for inspections, approvals, temporary services connections at no additional cost to contract unless noted otherwise. The Fee for the permanent connections shall be by the employer, however the contractor should make the payment to utilities authorities and will be reimbursed by the employer (client).
- 4.3 Before commencing any work on site, submit the required number of copies to the relevant Authorities (after coordination with present building Contractor) for checking and approval. Follow the progress of such drawings to permit the timely approval of them by the Authorities. Comply with any changes requested by the Authorities but notify the Consultant immediately of any such change for proper processing of these requirements. The Contractor, working on a regular basis with the Authorities is expected to know which changes are likely at the time he bids the job, and no extra cost will be allowed for such changes.

5.0 REVIEW OF TENDER DRAWINGS

- 5.1 The Tender / Issued for Construction drawings prepared by the Consultant primarily describe the design intent of the Works, the working principles of the systems, general arrangements, and principles of services co-ordination and the intended methods of installation.
- 5.2 The Tender drawings are accordingly part diagrammatic with runs of piping, ducts, cable, conduit and the like being shown to small scale and not necessarily indicating final exact installation positions.
- 5.3 The Tender drawings are intended to provide sufficient information, to facilitate the preparation of an estimate and tender to enable all other participants to appreciate the inter-relation of the works with other trades and disciplines.
- 5.4 When read in conjunction with the architectural and structural engineering drawings to enable the preparation of the final builder's work information, co-ordination, installation and shop drawings incorporating manufacturing drawings and final details of specified or selected plant and equipment.
- 5.5 The Tender drawings indicate equipment, pipework, ductwork, cable tray routes and trunking in sufficient detail to show how all the engineering services can generally be accommodated and coordinated with each other, the architecture and the structure. However, the Contractor shall be responsible for the detailed coordination of all engineering services.
- 5.6 The Contractor shall be responsible for ensuring that the Design Intent indicated on the Tender Drawings is in no way impaired due to additional pipe bends, deviations or changes in the service routes and plant layout.
- 5.7 Any dimensions are stated for elements that have a visual impact on the completed Works. The drawings will not necessarily show dimensions required to construct the Works nor will they accurately locate every item of equipment or plant ancillary which does not influence the visual impact of the completed works. The decision of the CA (Construction administration) as to what has a visual impact shall be final.

- 5.8 The Tender drawings are not intended to detail the building fabric or structure, nor are they based on survey drawings or installation drawings of any other party or Consultant.
- 5.9 Plant and equipment shown on the Tender drawings is based upon the manufacturer's catalogue information and is subject to verification by the Contractor.
- 5.10 Where sprinkler drawings are provided, the information shown on the tender drawings will indicate the design intent but this must be subject to verification by the approved installer who shall accept full responsibility for the design in accordance with the local civil defence (Authority Having Jurisdiction) and NFPA requirements.
- 5.11 Where it is necessary to survey an existing building or existing structure or services within that building to prepare coordinated installation and shop drawings the contractor is to allow for this within the contract price.
- 5.12 The Contractor shall interpret the design intent of the Tender drawings and reflect same in the co-ordination and installation drawings subject to the comments of the CA (Construction administration).
- 5.13 Where differences occur between the Tender and the co-ordination and installation drawings these differences shall be deemed to be included in the contract price for the Works and no additional cost shall be allowed due to any variation in layout, arrangement or detail unless the deviation is specifically covered by a CA (Construction administration) Instruction before the production of the drawings.
- 5.14 The Contractor shall obtain all information required to produce the co-ordination and installation drawings or requirements at no cost. Where it is necessary to survey an existing building or existing structure or services within that building to prepare coordinated installation and shop drawings the contractor shall undertake responsibility for obtaining that information. The cost of obtaining that information must be included in the contract price.
- 5.15 Any considered discrepancy or information which is incompatible or ambiguous between and within the tender documentation shall be drawn to the attention of the Engineer during the tender period, allowing sufficient time for a clarifying response and incorporation into the return tender or include for the most onerous option available. No allowance will be made for financial adjustment for discrepancies resolved following receipt of tenders.
- 5.16 In the event of the discrepancies appearing in the drawings, or between the drawings and Specification, or drawings and schedules, Tenderers shall refer the matter in writing to the CA (Construction administration) for interpretation.
- 5.17 The Contractor shall note that technical performance details, equipment schedules and/or equipment lists included in (or referred to by) this specification or referred may not reflect the final requirement and are thus subject to completion and/or revision or elaboration.
- 5.18 The Contractor shall not be entitled to any claim for extra monies, allowances or consideration (financial or other) in respect of any differences between the equipment performance and/or selection schedules or lists given in the project tender and construction documentation and the final selections procured by the contractor.
- 5.19 In the event of any contradiction between this specification (General Requirements) and any other specifications or documents referred to in this specification (General Requirements), the other specification shall take precedence; however, the Engineer shall be consulted for clarification.

- 5.20 If changes occur to Regulations, Statutory Instruments or Standards, the Drawings and Specification shall take precedence. The Contractor shall, however, refer any such discrepancies to the Engineer before assuming a particular interpretation, particularly in respect of Statutory Instruments, Regulations or Design Code.

6.0 INFORMATION TO BE SUBMITTED WITH TENDER

6.1 The following information shall be provided by the Contractor in his Tender:

- a. Details of any professional consultants employed to undertake required design services.
- b. Technical proposal for the system(s) which will be installed including method statement for the execution of the works.
- c. Schedules of proposed suppliers / manufacturers of products, equipment and plant.
- d. Initial technical submissions are to accompany the bid including full technical details of every item of equipment proposed for the project, with the appropriate figures and details highlighted in marker pen or similar in standard manufacturer's catalogues.
- e. After final checking of sizing after award of contract, resubmission will be made with any changes specifically noted.
- f. Method statements describing the organization and resources which the Contractor proposed and undertakes to provide to control the quality of the contract works. The statement must include the number and type of staff responsible for quality control, with details of their qualifications and duties.
- g. Details of any proposed Sub-contractors or Specialist Installers.
- h. The Contractor shall provide the complete working details of all interfaces with other services that he has included within his tender return. Failure to provide such details will result in the tender being void.
- i. Only wholly compliant tender submissions shall be acceptable. Any exclusions must be priced out or clearly identified to facilitate post tender analysis. The Contractor shall develop the concepts outlined in the tender information, sufficiently to submit a fixed price tender for the works.
- j. Any comments by the Tenderer on the efficiency of the design, the availability of plant, materials and labour and the time required for the completion of the work shall be made when the Tender is submitted.
- k. In the event that the Consultant requires further information on any of the proposed items the Contractor shall arrange visits to similar units which he has installed. In any event, all equipment proposed for the project shall be currently installed and working satisfactorily locally. The Contractor must be able to provide references of at least three satisfied Owners and Consultant with equipment of the particular size and model proposed.

7.0 CONTRACTOR'S WORKING DRAWINGS

- 7.1 Contractors working drawings or Shop Drawings shall be defined as a drawing showing the inter-relationship of two or more engineering services and their relation to the structure and building fabric and contain all the necessary information required by tradesmen on site to install the works.
- 7.2 The Contractor shall prepare and submit for approval of the Employers Agent all detailed working drawings of the scheme, plant and any other details necessary for the instruction of their workmen. Should the Contractor fail to comply with the requirements of this clause, they shall be responsible for any works of rectification including associated builders work; mechanical or electrical work etc., at their own cost if instructed by the Clients Agent.
- 7.3 The Contractor shall be responsible for dimensioning, setting out, detailed co-ordination of the services within the scope of his Contract, the design intent drawings provided at tender are not suitable for this purpose.
- 7.4 Coordination, installation and setting-out drawings dimensioned and to scale, shall be submitted for the Engineer's review to make clear the work intended and to show its relation to adjacent work and to the work of other trades. Three copies of such drawings shall be submitted for review, of which one will be retained by the Engineer.
- 7.5 The drawings must be comprehensive showing all details, dimensions, equipment, supports for services and sections through critical areas to properly coordinate all services and work of other trades. Unless accepted by the Engineer, all drawings must be to 1:50 scale.
- 7.6 Site services drawing shall be prepared to show all existing services, modification to existing services and all new services within the entire construction area. Drawings must show all connections of new or existing services to Municipal services and must be approved by the relevant authorities. Location of all existing services shall be carefully detailed and dimensioned on the drawings showing all sizes and invert elevations.
- 7.7 Drawings shall be prepared with due regard to the construction schedule and shall be submitted to the Engineer, allowing reasonable time for examination and review.
- 7.8 Work shall not proceed in areas involved until after final approval of all such drawings has been obtained. These drawings shall include all existing information.
- 7.9 Working Drawings shall indicate clearly all services and the materials and/or equipment actually being supplied, all details of construction, accurate dimensions, capacity, operating characteristics and performance.
- 7.10 Installation of any equipment shall not commence until after the Working Drawings have been reviewed and approved by the Engineer.
- 7.11 Working Drawings shall contain the following information:
- a. Project title.
 - b. Reference to the applicable drawing and specification article.
 - c. Contractor and supplier identification, addresses and telephone numbers.
 - d. Certification that the contractor has reviewed the submittal.
 - e. References to individual specification sections for additional information requirements.
 - f. Working drawing submittals shall be provided for each specific material, system, or equipment as identified herein.

- g. When requested, working drawings shall be supplemented by data explaining the theory of operation. The Engineer may also request that this information be added to the maintenance and operating manual.
- h. Where plant or equipment is to be installed within or adjacent to any building or structure, the Contractor shall take his own dimensions of the building or structure for the purpose of installing any plant and materials which are being supplied and fixed under this contract and shall be responsible for the accuracy of such dimensions. These dimensions shall be used in preference to any noted on the drawings.
- i. All dimensions shown on the Working Drawing shall be carefully checked and verified by the Contractor prior to commencing the installation. Each Working Drawing shall include a Certification from the Contractor that all related job conditions have been checked and that no conflict exists.

7.12 Contractor's Working Drawings shall be produced for all services described inclusive of, but not restricted to, the following:

- a. Layout of pipework and drainage sumps
- b. Layout of earthing electrodes and earth pits
- c. Builders-work drawings including details of plinths, bases, brackets, wall and floor openings, roof openings, including fan curbs and trimmer requirements.
- d. Details of access requirements throughout, suitable for testing, commissioning and maintenance.
- e. Schedule of electrical requirements listing manufacturer's specific data associated with all electrical plant and equipment being provided by the Contractor.
- f. Ductwork manufacturer's drawings.
- g. Air handling unit manufacturers dimensioned arrangement drawings.
- h. All packaged equipment certified drawings including fan coil units etc.
- i. Fan manufacturers dimensioned arrangement drawings.
- j. Specialist Automatic Controls Schematic and Wiring Diagrams
- k. Working drawings of pipework general arrangement along with Plant Room layouts for boiler house, ventilation plant rooms, etc. Pipework drawings to be 1:50 scale.
- l. Plant room details 1:20 scale, accompanied by cross-sections and elevations to a scale of at least 1:20. Drawings shall detail for each plant room the location of each unit, pipe routes and connections with valves and fittings, duct routes and connection with accessories, drain connections, electrical connections and controls. All drawings shall include explanatory notes and shall show the required sizes of pipes and ducts with and without insulation as applicable.
- m. The drawing should make allowances for installation working space and space to facilitate commissioning and maintenance.

- n. The drawing should be spatially coordinated i.e. there should be no physical clashes between the system components when installed.
- o. Critical dimensions, datum levels and invert levels should be provided together with dimensions in areas where tolerances are minimal.
- p. Spaces shown on the drawing should make allowance for the service at its widest point e.g. spaces between pipe and duct runs. Allowances for insulation, standard fitting dimensions and joint widths should be included on the drawing.
- q. Make allowance for those plant items specified by the designer and identified in the design specification.
- r. The drawing should indicate positions of main fixing points and supports where they have significance to the structural design.
- s. The drawing should demonstrate a feasible sequence of installation of the services.
- t. The drawing shall show the interconnection of electric components, panels etc. in accordance with the design and incorporating the details provided on manufacturer's drawings.
- u. The information shall indicate maximum electrical loading for each supply cable; cable termination facilities; and cable identification and all terminal numbers.
- v. The working drawings shall include all information provided by a manufacturer, fabricator or supplier for a particular project, and which is unique to that project and indicates a project specific representation of the product, components or plant items to be supplied. Examples include ductwork, prefabricated pipework, sprinkler systems, control and switchgear panels and associated internal wiring, prefabricated plant customized plant and equipment etc.
- w. Drawing provided by a manufacturer or supplier to indicate details of the product, components or plant items and which the manufacturer or supplier guarantees the supplied equipment will comply with.
- x. The working drawings shall include all controls diagrams, drawings and/or schematic details of all control components and instruments showing the layout with each item uniquely identified together with a description of the control's operation and details of the associated interlocking.
- y. The schematic and General Arrangement Drawings shall be prepared to describe the design of all engineering services, the working principles of the systems and the intended methods of installation together with sizes and capacities and rating of all components.
- z. Composite circuit and layout diagrams for electrical services shall detail not only all circuitry within main control panels but also connection diagrams for all external equipment such as starters, all forms of control devices for ventilation equipment, etc., together with all interconnecting wiring from the main point of supply onwards and all termination markings. All circuit and layout diagrams shall include explanatory notes and shall show the required sizes and type of all cables with the rating of such items on fuses, switches, etc.
- aa. Individual circuit and layout drawings from the various component manufacturers will not be accepted in lieu of composite diagrams.
- bb. Builder's work drawings shall show fully dimensioned, all foundations, bases plinths, sumps, holes and sleeves details required, and the overall size and weights of the plant concerned.

- cc. All drawings shall include explanatory notes and shall show the required sizes of pipes and ducts with and without insulation as applicable.
- dd. If abbreviations are employed for the designation of components, an integral schedule shall be provided on the drawings to explain the meanings of the abbreviations.
- ee. Individual equipment drawings from the various manufacturers will not be accepted in lieu of these composite plant room drawings.
- ff. Unless directed otherwise, drawings shall be prepared according to the Project's CAD / BIM Standard Manual, which shall be obtained at the time of tender for review and acceptance by the Contractor.
- gg. The Contractor shall allow for all necessary CAD / BIM resources for the production of drawings and make due allowance in the programme to revise the drawing as necessary to take account of the design development. For further information on the BIM requirements.

8.0 RECORD DRAWINGS

- 8.1 As the job progresses, mark on one set of prints to accurately indicate the status of installed work. Have the white prints available for inspection at the site at all times and present them for scrutiny at all job meetings. Transfer all information onto the set of final set of drawings. Obtain the Contractor's approval before forwarding these to the Consultant. Submit two sets of white prints to the Consultant for checking and approval.

Upon approval, submit one set of white prints and soft copy to the consultant for onward transmission to the client.

- 8.2 The completed Record Drawings and Schedules shall be signed by the Contractor as record drawings and shall be submitted to the Engineer at the same time as the Operating and Maintenance Manual is handed over for comment.
- 8.3 Record drawings of the final "as installed" layouts shall be issued in draft form to the CA (Construction administration) for examination 4 weeks prior to the Testing and Commissioning period to allow checking for accuracy.
- 8.4 Where portions of the work are to be concealed, draft copies of Record Drawings shall be supplied to the CA (Construction administration) before the work is concealed in order to facilitate checking and examination.
- 8.5 Record drawings shall be international A1 or A0 size printed on to best quality paper.
- 8.6 Record drawings shall also be reduced to international A3 size for inclusion in the O&M Manuals.
- 8.7 Record drawings shall be prepared in AutoCAD 2020 or later format to an internationally recognized and agreed project CAD layering convention. All services information shown shall be prominent and easily read when compared to architectural backgrounds.
- 8.8 A detailed list of CAD layers, external references or equivalent (if used) and list of files shall be provided with the electronic copies. If the drawings were produced from CAD drawings provided by the CA (Construction administration), the same layering system shall be used
- 8.9 Record drawing title blocks shall contain the following information:

The project name and site address.

Name, address and contact telephone/fax number for each member of the Contractor and the Consulting Engineer

Drawing title and building area that the drawing refers to.

Drawing number.

Drawing scales used.

Date of drawing issue.

The name of the person who produced the drawing.

The name of the person who checked and approved the drawing.

- 8.10 Each record drawing shall be endorsed with the words 'Record Drawing' in the bottom right hand corner adjacent to the title blocks.
- 8.11 Record Drawings shall show to the satisfaction of the Engineer the work as completed and shall record the following:
- a. The location, including level (if buried), of public service connections (if applicable) provided within the contract, whether installed by the Contractor or the appropriate Authority, together with the points of origin and termination, size and materials of pipes, line pressure, flow and other relevant information.
 - b. The layout, location and extent of all piped services showing pipe sizes, together with all valves for regulation, isolation and other purposes, drain cocks, test points, gauges, flow or pressure switches and other instruments.
 - c. Location, identity, size and details of all apparatus and controls equipment served by, or associated with, each of the various services.
 - d. The layout, location and extent of all chilled water pipe works, air ducts (if applicable), including those formed in builders work or otherwise outside the contract, showing dampers and other equipment, acoustic silencers, grilles, diffusers, other air terminals, balancing dampers, access panels, fire dampers, turning vanes, hand holes, test holes, gauges and instruments. Each duct and terminal should be marked with its size and air quantity flowing. Each terminal unit or grille shall have its duty clearly shown.
 - e. The location and identity of each room including spaces housing plant machinery or apparatus.
 - f. Detailed general arrangements of all machinery spaces, air handling plant rooms, tank rooms, electrical switch rooms and other plant or apparatus, including the location, identity, manufacturer, size and rating of each apparatus.
 - g. All necessary sections, elevations, isometrics and schematics of the plant spaces.
 - h. Control and wiring diagrams shall be provided incorporating details of each instrument and equipment item and written description of the sequence of operation of each

8.12 Floor layouts shall be to a scale of not less than 1:50.

8.13 Plant areas and equipment rooms, sections, shall be to a scale of not less than 1:20.

8.14 The drawings shall include symbols, valve charts, plans, sections, elevations and isometric interpretations to provide a comprehensive record of the installations.

- 8.15 Electrical drawings shall be in accordance with IEE Regulations 514-09 and with this Specification.

9.0 TEMPORARY SERVICE

- 9.1 Do not use any of the permanent service facilities during construction, unless specific written approval is obtained from the Consultant and the Owner or where specifically allowed elsewhere in the Contract Documents.
- 9.2 The use of permanent facilities for temporary services will not affect, in any way, the commencement day of the period in which the correction after completion is the Contractor's responsibility. Such period will commence only when the overall project is completed, and certificates of acceptance are issued.
- 9.3 In the event that mains power is not available or is not of sufficient capacity during construction, supply, install and operate diesel generator(s) of the correct capacity (which must be approved by the Consultant.)
- Any generator must have full safety features and must be maintained regularly to ensure site power at all times.
- 9.4 The Contractor is to ensure that permanent services are operating within the show flat to allow its use for the Client sales teams at the earliest opportunity, this must be identified and agreed with the main contractor.

10.0 COOPERATION

- 10.1 Confer with all trades installing equipment which may affect the work of this section and arrange equipment in proper relation with that equipment installed under all Sections of the Contract.
- 10.2 Furnish all items to be built in by others, in time, complete with all pertinent information, commensurate with the progress of the work.
- 10.3 Store materials neatly and out of the way and clean up all refuse caused by the work daily.

11.0 PROTECTION AND STORAGE

- 11.1 Protect the buildings and structures from damage due to carrying out of this work.
- 11.2 Protect all mechanical, plumbing & electrical works from damage. Keep all equipment dry and clean at all times.
- 11.3 Cover all openings in equipment and materials. Cover all temporary openings in ducts and pipes with polyethylene sheets or caps until final connection is made.

The quality of such cover must be determined with due regard to how long it may be until final connection.

- 11.4 Be responsible for and make good any damages caused directly or indirectly to any walls, floors, ceilings, woodwork, brickwork, finishes, services, roads, gardens, etc.

12.0 EXISTING WORK AND EQUIPMENT

Before the Contractor undertakes work in any area, he must prepare a list of deficiencies in that area which affect his works, or which could possibly be construed as being caused by himself if

not noted. In the event that such deficiency list is not prepared, then he shall be deemed responsible for such deficiencies. Any list shall be brought to the attention of the Consultant forthwith.

13.0 MANAGEMENT OF THE WORKS

- 13.1 The Subcontractor will maintain at site, as necessary for the performance of the Contract, qualified personnel and supporting staff, with proven experience in erecting, testing, and adjusting projects of comparable nature and complexity.

The Contractor team is to be headed by a suitably qualified Engineer who will act as the MEP Project Manager, he is to be assisted by Senior Engineers for Electrical, Mechanical, Public Health, Fire Alarm and Commissioning and covering each discipline as required and resident on the site full time supported by a suitable structure of supervision based on a standard hierarchy format.

- 13.2 Before commencing work the Contractor will submit details of the proposed Engineers and Supervisors, including copies of their Certificates. If in the Consultant's opinion the proposed Engineers or Supervisors are not adequately qualified or are otherwise unacceptable, the onus is on the Contractor to submit alternates until such approval is given.
- 13.3 Where the Contractor's or Subcontractor's staff is or becomes during the Contract deficient in the choice of adequate or acceptable personnel, the Contractor is to remedy the situation by appropriate measures.
- 13.4 Approval of the Contractor's or Subcontractor's Engineers or Staff shall in no way prevent the withdrawal of that approval at any time during the Contract should the Consultant so desire. In the event of such disapproval then the Contractor will be required to rectify the position as stated above within 14 days.
- 13.5 In the event of any negligent or severely detrimental behavior the Consultant has the right to order the removal from site of any Engineer, Supervisor, or worker on a "forthwith" basis.

The Contractor will remedy the situation on a similar forthwith basis and in any event will not be allowed to proceed with any area of work which as a result is left without proper supervision or technical backup until such time as the Consultant is again satisfied by the measures taken.

14.0 SCHEDULING

- 14.1 In the event that a completion date has been set before tendering it will be the obligation of the Contractor to provide a schedule adequate for all trades to meet the completion date.
- 14.2 If no completion date is specified it will be the obligation of the Contractor to provide to the Consultant a proposed schedule of work including mechanical trades, within 21 days of award of Contract. This schedule will be either approved by, or negotiated with, the Consultant.

15.0 STANDARDS AND AUTHORITIES

- 16.10 All equipment and installations shall be in accordance with local Codes and Regulations. In general, all equipment and installation shall also be in accordance with:

- AHRI : American Heating and Refrigeration Institute
- ASHRAE : American Society of Heating, Refrigeration and Air Conditioning Engineers
- ASTM : American Society of Testing Materials
- SMACNA : Sheet Metal and Air Conditioning Contractors National Association
- CIBSE : Chartered Institution of Building Services Engineers
- NFPA : National Fire Protection Association

- UL : Under writers Laboratory
- BS : British Standards
- EN : Euro Norms – European Standards
- IEE : Institution of Electrical Engineers
- IEC : International Electro-Technical Commission
- ISO : International Organization of Standardization
- DEWA : Dubai Electrical and Water Authority
- DM : Dubai Municipality
- DCD : Dubai Civil Defence
- UAE Fire Code (latest Edition)
- Uniform plumbing code of USA.
- National Plumbing Code
- Institute of Plumbing (UK)

- 16.11 Obtain approval of shop drawings from Authorities (Including the master plan developer or their representatives) having jurisdiction including but not limited to Drainage Department, Municipality, Ministry of Water and Electricity, Fire Brigade, Department of Health, and local telephone authority. All costs towards such approvals to be borne by the contractor.
- 16.12 All standards, referenced throughout all the sections of the specification, shall follow the latest editions.

17.0 WORKMANSHIP

- 17.1 Install equipment, ductwork, conduit and piping in a workmanlike manner to present a neat appearance and to function properly to the satisfaction of the Consultant. Install ducts and pipes parallel and perpendicular to the building planes. Install all piping and ductwork concealed in chases, behind furring, or above ceiling, except in unfinished areas. Install all exposed systems neatly and group to present a neat appearance.
- 17.2 Install all gauges, thermometers, etc., to permit easy observance.
- 17.3 Install all equipment and apparatus which requires maintenance, adjustment, or eventual replacement with due allowance for this.
- 17.4 Install control valves to guarantee proper sensing. Shield elements from direct radiation and avoid placing them behind obstructions.
- 17.5 Install all panels and boards, etc., to permit easy operation.
- 17.6 Include in the work all requirements of Manufacturers as shown on their drawings.
- 17.7 Replace all work unsatisfactory to the Consultant without extra cost and to the standard required by the Consultant. This applies to any item which is found to be defective in service during the DLP (Defects liability period) period.
- 17.8 Where height dimensions are given, the installation shall be reasonably expected to be within 1/8" (3mm) of the design dimensions. Where room centre or proportion dimensions are not shown these shall be reasonably expected to be located on site to within 1/2" (13mm) of actual position.
- 17.9 Items displaying a horizontal dimension or edge must be plumbed with a level and must be to the Consultant's satisfaction. Any apparent discrepancy between "level" electrical items and adjacent items MUST be reported immediately to the Consultant.

18.0 MATERIALS AND EQUIPMENT

- 18.1 All materials used on the project shall be brand new and of high quality. Obtain approval of all

manufacturers from the Consultant and the Owner.

- 18.2 "New" is defined as newly manufactured, "state of the arts," tested and proven item of equipment. Items which have been held in stock for any extended period of time by either the Manufacturer or the Supplier will be rejected.
- 18.3 All materials used on this project shall bear the third-party quality assurance stamp like British Standard Kitemark, AHRI, UL, FM, NFPA, WRAS, DVGW and/or other quality assurance authority as stated herein elsewhere.

19.0 CLEANING

- 19.1 Each day as the work proceeds and on completion, clean up, vacuum, wipe down, remove labels and markings from equipment and remove from the premises all rubbish, surplus material, equipment, machinery, tools, scaffolds, and other items used in the performance of the work. Clean out dirt and debris and leave the buildings clear and free from rubbish, waste, dust, dirt, debris and sand. Equipment is to be spotless and in 'Brand New' condition at handover complete with new or thoroughly cleaned filters and in a condition acceptable to the Consultant.
- 19.2 Where electrical items form part of the visible finish in the rooms, protect from over-painting, etc. and give all items a final cleaning before handing over of the project.

20.0 ACCESSIBILITY

- 20.1 Each item of equipment shall be located so as to be accessible for maintenance or repair without removing adjacent structures, equipment, piping, ducts, or other materials. For the large Air Handling units, the Contractor shall ensure that these can be built up on site from components which can be taken into the area. Any openings, framing or special lifting equipment which may be required shall be included by the General Contractor.
- 20.2 Cleanouts shall be located to permit rodding of all drain lines. These shall be located wherever possible external to occupied areas, and to minimize spillage problems during rodding.

21.0 BUILDERS WORK

- 21.1 Inform all other Divisions in time concerning required openings. Obtain the approval of the Consultant before doing any cutting.
- 21.2 Supporting members of any floor, wall or the building structure shall only be cut and in such a manner as approved by the Consultant. All reinstatement work must be done to the same standard as the original work.
- 21.3 Allowances shall be included in the Main Contract for the provision of Builder 's Work required for the MEPF Sub-Contract Works including:
- a. Formation of blockwork or concrete bases for engineering plant.
 - b. Casting-in conduits and pipe sleeves
 - c. Excavation and backfilling of trenches including the provision of sand/pea gravel both before and after the laying of services.
 - d. Installation of cable tiles marker tapes and cable markers, supplied by the Sub-Contractors. Provision of anchor thrust blocks. Formation of horizontal and vertical service ducts, covers and access panels as appropriate. Cutting/forming of holes and wall chases, etc., and making good.

- e. The Sub-Contractor shall be responsible for providing final Builder's Work details to the Main Contractor based on authorized / certified installation and manufacturer's drawings to facilitate the installation of the works.
- f. The Sub-Contractor shall provide information at such time to enable Builder 's Works provisions to be made according to the main construction program.
- g. The Sub-Contractor shall provide fully dimensioned drawings showing both size and position of Builder's Work making due reference to the final Structural Engineering and Architectural dimensioned detailed drawings.
- h. The sub-contractor shall undertake the redesign of the associated builder's work associated with any for approved alternative equipment or materials which subsequently vary the works in any way whatsoever.
- i. Provide final builders work details based on the installation, manufacturing and shop drawings to facilitate the installation of the works;
- j. (To include fully dimensioned drawings showing both size and position of builder's work making due reference to the structural engineering and architectural final dimensioned detailed drawings.);
- k. Detail all access requirements including access to false ceilings and ducts for maintenance. Provide fully dimensioned and annotated drawings;
- l. Undertake the redesign of the associated builder's work for approved alternative equipment or materials which subsequently varies the works in any way whatsoever;

22.0 INSERTS, SLEEVES, ESCUTCHEONS AND CURBS

- 22.1 Under this section supply necessary sleeves and other inserts to other trades.
- 22.2 Use only factory made, threaded or toggle type inserts as required for supports and anchors, properly sized for the load to be carried. Place inserts only in portions of the main structure and not in any finishing material.
- 22.3 Use factory made expansion shields where inserts cannot be placed, but only where approved by the Consultant and for light weights.
- 22.4 Do not use powder activated tools except with the written permission of the Consultant.
- 22.5 Supply and locate all inserts, holes, anchor bolts, and sleeves in good time when walls, floors, and roof are erected.
- 22.6 Pass insulation unbroken where pipe or duct is insulated. Size sleeves to provide adequate clearance all around and in accordance with specialist supplier's recommendations.
- 22.7 Use the following materials for pipe sleeves:
 - a) Through all interior walls above grade use uPVC Class C pipes, machine cut, flush with finished structure. For pipe crossing in fire wall use steel pipe (medium).
 - b) Through all exterior walls above grade use uPVC Class C pipes, machine cut, flush with finished structure inside and to suit flashing on outside. Puddle flange to be used as mentioned in clauses 24.2 & 24.3.

- c) Through all exterior walls below grade and all other waterproof walls use uPVC Class D sleeves, machine cut. Check flashing specification for further details.
 - d) Through all waterproof floors, janitor's closets, mechanical rooms, kitchens, roofs, use uPVC Class 150 sleeves, machine cut. Extend sleeves 100 mm above finished floor upwards and cut flush with underside of floor. Refer to flashing detail through waterproof floors.
 - e) Provide 100 mm high, 100 mm wide watertight concrete curbs with 20 mm chamfered edges around all pipes passing through waterproof floors except where furred in. Read Concrete specification before proceeding with curbs. All gaps to be filled up with waterproof type sealant.
 - f) Where uPVC pipes/ other pipework material passes through fire rated walls/partitions, provide fire rated sleeves.
- 22.8 Pack all sleeves between the insulated pipe and the sleeve or where un-insulated between the pipe and the sleeve with loose fibreglass insulation. Seal the annular space as follows:
- a) For all horizontal sleeves in exposed areas, use a seal equal or better fire rated than the wall to be sealed.
 - b) For all vertical sleeves through roofs, janitor's closets, equipment rooms, use permanently resilient silicone-based sealing compound, non- inflammable and waterproof (fire sealant). Ensure that the seal is compatible with floor and ceiling finishes. Check the room finishing schedules for further details and clarify if necessary, with the Consultant.
 - c) Cover exposed floor and wall pipe sleeves in finished areas with satin finish chrome or nickel-plated solid brass or with satin finished stainless steel escutcheons with non-ferrous set screws. Split cast plates of the screw locking type may not be used. Do not use stamped steel friction type split plates through roofs, provide curbs and sleeves as shown on drawings and to suit flashing requirements.
- 22.9 Use the following sleeving for ducts:
- For rectangular duct openings through walls and floors a removable hardwood/ GI box-out shall be provided of the required size. Soft wood or plywood will not be acceptable. Hardwood shall be further protected with anti-termite paint / solvent.
- Through fire walls, build fire dampers into wall, or make detailed fixing in accordance with DW 144 / NFPA 90A & UL 555/ SMACNA Standards.
- Through floors where ducts are not furred in or enclosed in a duct shaft, provide 100mm high and 100 mm wide watertight concrete curbs, with 25 mm chamfered edges all around. Extend sleeves where
- used flush to top of curb. Read Concrete Specification before proceeding. Through floors where duct is enclosed in a duct shaft or furred in, provide the watertight curbs at the extreme top and bottom only.
- 22.10 After ducts are installed, pack the opening and seal the packing as follows:
- a) Use fibreglass insulation for packing except through curbed concrete floors where asbestos braid must be used.
 - b) Seal the asbestos braid packing in openings through floors with permanently resilient

silicone base noninflammable waterproof compound. Press duct supports firmly down into caulking before bolting it down to curb.

- c) Through all vertical walls seal the fibreglass packing using permanently resilient silicone based waterproof sealant.

- 22.11 Brace duct sleeves and box-outs to retain their position and shape during the pouring of concrete and other work.
- 22.12 Provide bracing for each duct at every passage through structure to prevent sagging.
- 22.13 Cover exposed pipework, gullies, duct sleeves and openings in exposed areas. Use 100 mm galvanized steel escutcheons in the form of a duct collar or screw fixed pipework cap or similar approved protection. Over curbs extend the collar 30 mm down the side of the curb, similar to counter flashing. Fix collar in place with Cadmium plated screws.

23.0 ACCESS PANELS AND DOORS

- 23.1 Install all concealed mechanical equipment requiring adjustment or maintenance in locations easily accessible through access panels or doors. Install systems and components to result in a minimum number of access panels. Indicate access panels on as-built drawings.
- 23.2 Provide the respective Division of work with panels, doors or frames, complete with all pertinent information for installation.

Ensure that access doors are installed in a manner to match the building grids where applicable

- 23.3 Prepare detail drawings showing location and type of all access doors in coordination with other trades before proceeding with installation and hand these to the Contractor to obtain approval.
- 23.4 Size all access doors to provide adequate access and commensurate with the type of structure and Architectural finish. Should it be necessary for persons to enter, provide a minimum opening of 600 x 450mm. Coordinate with ID for access door size.
- 23.5 Ensure proper fire rating of access doors in fire separations
- 23.6 Lay-in type ceiling tiles, if properly marked may serve as access panels.
- 23.7 Provide panels in glazed tile walls of 2.6mm thick 304 alloy stainless steel, no. 4 finish, with recessed frame secured with stainless steel, countersunk, flush-headed screws. Coordinate with ID for panel size and finish.
- 23.8 Provide panels in plaster surfaces with dish shaped door and welded metal lath, ready to take plaster. Provide a plastic grommet for door key access.
- 23.9 Provide other access doors of welded 2.6mm thick steel, flush type with concealed hinges, lock and anchor straps, complete with factory prime coat. Submit details to Consultant for approval.

24.0 FLASHING

- 24.1 Flash all mechanical and electrical parts passing through or built into a roof, an outside wall, or a waterproof floor.
- 24.2 Provide uPVC or GI sleeves (depending on pipe material) with puddle flanges for services crossing waterproofed walls in basement.

- 24.3 Provide UPVC or stainless steel (depending on pipe material) puddle flanged connections where pipework penetrates walls of the water storage tank. The puddle flange shall have bitumen coating.
- 24.4 All flashing shall suit roof angle and shall extend a minimum of 400 mm on all sides. Leave the flashing as directed by the Contractor for him to build into the roofing system to render a completely watertight connection.
- 24.5 Provide counter flashing on all stacks, ducts, and pipes passing through roofs to fit over flashing or curb.
- 24.6 Flash pipes through waterproof floors as per approved detail.
- 24.7 Provide pipes and sleeves passing through outside walls with lead or copper flashings and as directed by the Consultant. All visual aspects of such sleeves to be approved by the Consultant.
- 24.8 Pay special attention to the waterproofing conditions of basements and walls and floors that may exist. Cooperate at all times with the waterproofing trade and do not cut or destroy any waterproofing seal without the consent of the waterproofing trade.

Provide piping sleeves passing through waterproof walls with asphalt roofing felt wrapped around so as to leave a 25 x 50mm recess on both sides of the wall. These recesses and the space between pipe and sleeve will be caulked by the General Contract, Moisture protection, or Waterproofing, Division.

25.0 GENERAL WELDING REQUIREMENTS

- 25.1 All welding shall be generally in accordance with ANSI B31. This will be modified where appropriate for other materials and may be relaxed or varied by order of the Consultant, provided that the Contractor has made a reasonably comprehensive request for an alternate.
- 25.2 Tack welds shall be performed by fully qualified welders and all tack welds shall be of a length equal to twice the pipe thickness and shall fully penetrate the pipe walls.
- 25.3 Where welding is carried out in the proximity of inflammable materials special precautions shall be taken to prevent risk of fire or other damage to the building fabric.
- 25.4 Where oxyacetylene cutting equipment or any welding plant is being used by an operative for any of the works defined in the contract documents, then fire extinguishers shall be supplied and carried as part of the equipment. The operators of cutting and welding equipment shall be trained in the use of the fire extinguishers which they carry, and all extinguishers shall be fully charged and ready for use. In all cases, extinguishers shall be positioned immediately adjacent to the position where cutting and welding is being carried out and shall be readily accessible for use in the event of an emergency.
Hot works permit shall be made in advance and approved by the safety manager.
- 25.5 All accommodation, benches, tools, welding plant, acetylene, oxygen or electricity, filler rods and electrodes, which are necessary for installations where welding is required shall be provided as part of this contract.
- 25.6 Welding shall not be done at a temperature of 5 degrees Centigrade or below unless the parent metal is preheated by torches or other approved means until it is warm to the hand (about 27°C) for about 150mm either side of the joint. No welding shall be done below minus 18 degrees C. After preheating, the heated portion including the welded joints shall be covered with muffs or suitable insulation materials to allow cooling free from draughts. Any open ends of the pipe or assembly shall be effectively sealed to prevent heat flow by convection.

- 25.7 All welded pipe assemblies shall be constructed so that individual welded joints do not affect each other. The distance between the centres of adjacent welds shall be not less than twice the bore diameter of the pipe.
- 25.8 No welded joints shall be left partially completed. Any joints tacked in position must be promptly finished within the working day. The Consultant will reject all work not done in accordance with this instruction.
- 25.9 Where work is rejected, pipes must be machine cut at least 150 mm either side of rejected welds and proper weld preparation must be used on the shortened sector. Where shorter fill-in sections are required because of such rejection and reworking, then new full-sized lengths must be supplied by the Contractor.
- 25.10 Where pipes with longitudinal seams are specified, pipe seams shall be arranged such that adjacent seams are opposed 45 deg. from each side of top dead centre and branches shall be made only with weldable fittings.
- 25.11 All filler metals which are coated shall be protected from excessive moisture changes. Filler materials or flexes which show any sign of deterioration shall not be used. If requested by the Consultant samples of filler rods to be used shall be submitted to him for approval before any work is done on site.

These may be submitted, at the Contractor's cost, to an independent testing laboratory for verification.

26.0 TESTS FOR WELDER QUALIFICATION

- 26.1 The purpose of the welder's qualification tests is to determine the ability of the welders to make sound and acceptable welds. Before any site welding on the contract is allowed, each proposed welder shall carry out the tests required in the presence of the Consultant, or the Employer's Insurance Inspector.
- 26.2 Any weld test specimens which have been suitably marked and approved shall be kept on site by a responsible person, so that they can be produced at any time, at the request of the Consultant.
- 26.3 All accommodation, benches, tools, welding plant, acetylene, oxygen, electricity, test pieces, filler rods, electrodes, facilities for cutting and grinding, polishing, bending and examining, which are necessary for welder's qualification tests shall be provided by the Contractor. In the absence of any items for inspecting the welds, the Consultant may submit the finished samples to an independent laboratory for testing at the Contractor's expense.
- 26.4 Under no circumstances shall a welder be employed on the contract, either on or off the site for welding operations other than those for which that welder is qualified.
- 26.5 Copies and records of all test reports shall be promptly given to and kept by the Consultant.
- 26.6 Even welders holding a certificate for welding are to be tested at this job site in full accordance with the specifications.
- 26.7 Tests for Qualification on Steel Pipes
- 26.7.1 The Contractor shall be responsible for the quality of welding and brazing and shall provide the testing of the welders employed. Certificates of such proficiency test, together with stamped samples provided for Engineer's review before welding work is commenced. The welder's proficiency standard to be equal to Lloyd's Standard Class 2, certified by an authorized testing laboratory paid for the contractor. The welders shall be certified for 5J/6J as minimum.

26.7.2 Qualification tests will be made for butt and branch welds in pipes. One test piece per welder will be required for the qualification of welding of pipes exceeding 100mm nominal diameter.

26.7.3 In addition, test pieces will be required for each position of a pipe. These positions will be horizontal and rotated while welding, horizontal and fixed while welding, vertical and fixed while welding for each range of nominal diameters specified.

27.0 TESTING OF WELDED PIPEWORK INSTALLATION

27.1 Testing of welded pipework installation shall be by either destructive or non-destructive test method.

27.2 The butt-welded test piece to be subjected to two root bend tests and two nick break tests of the welding.

The branch welded test piece will be subjected to two root bend tests and two fillet break tests of the welding.

The test piece weld to be radiographically and X-ray examined. Generally, 2 to 5% of the welded joints shall be tested. However, the Contractor shall allow for all the welds of the main risers pipe work and the pipework in ETS/ CHW pump room to be radiographically examined.

27.3 The Contractor shall assign an identification symbol to identify each welder and this will be used to stamp the work done by that welder.

27.4 The Contractor will compile a set of safety instructions concerning welding procedures for the welders and after review by the Engineer, these will be issued to the welding operatives and foremen.

27.5 Determination of piping sections to be tested will be designated by the Engineer. The timing of the radiographic examination of a weld or welds designated to be so examined will be confirmed in writing by the Engineer during the installation and commissioning.

Radiographic examination of welds will be carried out by a specialist employed by the Contractor subject to review by the Engineer. The specialist so employed will be responsible for the proper indexing, interpretation and reporting of the results of the radiographic tests.

27.6 Any weld, whether radiographically tested or not which are considered by the Engineer to be faulty, either in whole or in part, or which have failed under hydraulic or other tests so designated by the Engineer, will be removed and replaced by the Contractor.

27.7 The faulty part of a weld may be removed by flame gouging, chipping or grinding. Otherwise, the Engineer may instruct that the portion of the works containing the fault weld be cut out and replaced by the welding in of a new section of pipe or member.

27.8 Should any of the welder's work (as defined by the Engineer) be found to be faulty during radiographic tests, then that welder shall be removed from site and all other welds undertaken by that welder shall be checked by radiographic means and replaced as necessary at the expense of the contractor.

27.9 The extra radiography involved shall not be part of the inspection areas previously stipulated.

28.0 PAINTING, IDENTIFICATION AND VALVE TAGS

28.1 All painting of plant, equipment, storage vessels, and the like, and all surfaces to be painted in the areas where such items are installed, including generator area, condenser areas, etc., shall be carried out by an approved specialist painting contractor.

- 28.2 Painting shall be applied to all exposed pipework, ductwork and associated components, valves, fittings, etc., equipment, supports of any kind, insulation, plain mild steel, copper, or cast iron surfaces. Where movement is required between adjacent surfaces, request clarification from the Consultant.
- 28.3 The following items do not require site painting:
- (a) Insulation having any of the following finishes:
 - (1) Aluminium foil.
 - (2) Hot Dip Galvanized angles / channels
 - (3) Polyisobutylene sheeting / bitumen coating.
 - (4) Vinyl-glass/ resin-aluminium foil laminate.
 - (b) Equipment or plant or supports or frames delivered to site with any of the following finishes:
 - (1) Painted finish, other than primed only, provided that the finish is not damaged in any way.
 - (2) Stainless steel or plastic-coated steel.
 - (3) Stoved enamel.
- 28.4 All surfaces to be painted shall be prepared by thoroughly cleaning and removing all rust, grease, oil, dirt and surface corrosion, using wire brush, emery paper and/ or degreasing medium as required. The paint shall be applied in accordance with the maker's instructions and the type of paint to be used shall be in accordance with the following:
- Ferrous surface, one coat of Calcium Plumbate metal primer plus wash primers as necessary, followed by 2 undercoats and one finishing coat.
- Non-Ferrous surface, one coat of zinc chromate primer plus wash primers as necessary, followed by 2 undercoats, and one finishing coat.
- All paint shall be manufactured by approved manufacturer and type of paint shall be selected by the manufacturer based on the location i.e. Indoor / Outdoor.
- 28.5 For factory applied finishes, repainting or refinishing of any surfaces damaged during shipping, erection or construction shall be done using only factory supplied materials.
- 28.6 After insulation or final painting is completed, identify each piped and ducted service. Locate identification bands, stickers and arrows:
- (a) Behind each access door.
 - (b) At each change of direction on all joining pipes and ducts.
 - (c) At not more than 10 meters apart in straight runs of exposed pipes and ducts, but on both sides of sleeves.
 - (d) At not more than 10 meters apart in straight runs of pipes and ducts behind removable enclosures such as lay-in ceiling but on both sides of sleeves.

- (e) Above each floor or platform for pipes in shaft and vertical exposed pipes, preferably 1.5 meters above floor or platform level.
- 28.7 Identification Bands, stickers and arrows shall be made of self-adhesive plastic long lasting type. The stickers when used outdoor shall be weatherproof type suitable for the temperature in the country.
- 28.8 Identification of ductwork shall be carried out as per DW 144 standards coloured triangle with the type of service. Additionally, the AHU / FAN reference no shall be identified in the ducts.
- 28.9 All insulated or un-insulated pipework shall be identified as detailed above by means of PVC tape identification bands in accordance with British Code to BS 1710.

In addition, the name of the service and flow direction arrow shall be provided.

- 28.10 The access doors and fire dampers in duct shall be identified by a suitable sticker.
- 28.11 All identifying band colours and finishing colour of services shall be agreed with the Consultant prior to application.
- 28.12 All equipment shall have a nameplate secured to the item giving the following information: Equipment reference number (as indicated on the record drawings.)

Make / Model No.

System.

Room / Area served.

Duty / output information.

Power fed from:

The name plate shall be 100 x 100mm approximately, of white plastic 3 mm thick for indoor application and Aluminum or stainless steel plates for outdoor application, with the above information engraved in black lettering and the plate shall be secured by screws, bolts, clips, etc., as appropriate to the item concerned.

This plate is in addition to any name plate supplied by the manufacturer of the item giving detailed specification information for the equipment.

- 28.13 All main valves in plant room, roof, risers shall be numbered system wise, each with a unique number. The number shall be engraved in a tag 30mm diameter and suspended with a stainless steel chain from the valve handle.

The tag shall be of aluminium material for outdoors and traffolyte/plastic indoor.

The piping schematic diagrams shall have the valve numbers with valve chart and the same shall be enclosed as a part of operating instruction and shall be framed with glass and displayed in the relevant plant rooms.

29.0 FLUSHING AND CLEANING

- 29.1 Flush all fluid-carrying systems after completion with a stream of water or appropriate fluid at the highest obtainable pressure and velocity. Discharge the flushing fluid through all strainers and out through line sized valves with hose ends. Clean all strainers. Repeat flushing operation to the approval of the Consultant until the water is running clear, colorless and odorless and the conductivity is no more than 30ms/cm above the mains water.

- 29.2 If the water pressure normally available on site is, in the Consultant's opinion, inadequate to perform proper flushing, then the Contractor shall provide at his own expense, pumps, tanks, etc., to give pressure and volume flows to the Consultant's satisfaction.
- 29.3 For layer systems, the flushing and cleaning shall be carried out in sections of every three floors.
- 29.4 During flushing and cleaning all cooling coils, heat exchangers shall be bypassed in order to avoid debris / dirt going to coils. Permanent pumps shall not be used for flushing.
- 29.5 Domestic Water Sterilization: On completion of the potable cold water supply services system, including any parts of the existing service they shall be sterilized by the application of chlorine. The minimum requirements for chlorination shall be to flush the pipework thoroughly to remove dirt. Chlorine shall then be added to the storage tanks and water drawn from all draw-off points until chlorine is detected at each outlet point and to reach a concentration of 50ppm. The mains are then to be allowed to stand for a period of 24 hours, after which the outlets are to be tested for chlorine. If chlorine is not found to be present (at least 2 ppm), all the above shall be repeated until residual chlorine is detected. After this, the system is to be completely flushed with clean water and precautions taken to avoid subsequent contamination of the installation.

Final two samples to be taken to verify that chlorine concentration is not more than 0.4 ppm, and another sample for bacteriological analysis to confirm the sample is clear from bacteria. If any of these samples fails, then sterilization process shall be repeated again.

30.0 INSPECTION, TESTING AND ADJUSTING

- 30.1 All the works provided as part of this Contract shall be inspected and commissioned in accordance with all relevant Specifications, and Codes of Practice and to the entire satisfaction of the Consultant.
- 30.2 Carryout all tests specified. Carry out all tests required by Authorities having jurisdiction. Test equipment to the requirements of, and where necessary, in the presence of the Manufacturer.
- 30.3 Provide all equipment, labour, instruments, loading devices, incidentals, and pay for all fuel, power and sundries required to carry out the tests.
- 30.4 All installations shall be inspected and tested in sections as the work proceeds and on completion as composite systems and it shall be noted that the Consultant or any of the other relevant Authorities may require to inspect or test any equipment during manufacture at the Manufacturer's works. All necessary arrangements shall be made as part of this Contract.

This will generally not apply to specified items unless specifically noted in these documents, but may be necessary for alternate equipment, should this be considered at all.

- 30.5 All tests shall be arranged in cooperation with the Consultant and his Engineer and all other concerned parties and shall be subject to at least one-day notice in writing of the time, location and nature of the test to be performed. No test shall be considered valid unless the Consultant is present.
- 30.6 All necessary skilled and unskilled labour shall be provided for attendance during the tests (including pre- and post- test activities,) and the test media shall be provided and subsequently disposed of except where specifically stated otherwise.
- 30.7 The testing and adjusting is the contractual responsibility of the Contractor but actual performance of the tests is expected to be the sole responsibility of the Subcontractor.

- 30.8 Have all testing and balancing performed by only persons who are thoroughly versed in this type of testing and balancing and with proven ability. Submit names, complete with experience records, and references for approval of the Consultant.
- 30.9 Any defects occurring at any time during the test duration shall be made good and a complete re-test shall be carried out, all at no cost to the Contract.
- 30.10 Where failure occurs during a test, inspection or commissioning procedure which results in damage to the building fabric and/ or any services not provided as part of this Contract, or requires subsequent builder's work to be carried out, carry out all such repair work to the entire satisfaction of the Consultant at no cost to the Owner or the Consultant.
- 30.11 All test points shall be provided which are necessary to carry out the specified tests and commissioning procedures including facilities for measuring or monitoring temperature, pressure, pressure drop, volume flow, in-duct sound power or sound pressure, humidity, or other relevant conditions in both air side and water side systems. Such points shall be fitted with removable plugs, flanges, or other appropriate and approved devices.
- 30.12 Prepare test report forms for each test to be performed and submit these to the Consultant at least two weeks prior to the commencement of any tests.
- 30.13 Test only after the system installation has been completed and the system has been put into continuous operation. Perform the testing, adjusting, and balancing when outside conditions are commensurate with the design conditions for the given system. Add dummy loads to the system if outside and inside conditions are less severe than the specified points.
- 30.14 Prepare a complete list of instruments for each test containing for each instrument:

- Name of instrument Manufacturer.
- Scale and full scale accuracy.
- Date of last calibration test.
- Name of last calibrating company. All measuring instruments shall be recently calibrated with valid calibration certificates.

All instruments and consumables, such as recording paper, necessary for conducting the tests shall be provided, including but not limited to the following:

- Tube type velometer.
- Sloping tube manometer.
- Pitot tubes of various lengths.
- Mercury in glass thermometers.
- Weekly recording thermometers & digital thermometers.
- Weekly recording relative humidity meter.
- Anemometer for diffusers, with collector.
- Ammeter, voltmeter, wattmeter.
- Power factor meter.
- Insulation tester.
- Earth loop impedance tester.
- Tachometer.

- 30.15 Duplicate signed test certificates shall be provided after each test which will be countersigned by the Consultant who witnesses the test. The test certificate shall give the following particulars: -
- Apparatus or section under test.
 - Maker's number (if any).

- Nature, duration and conditions of test.
- Result of test.

No test shall be valid until the test certificate is provided.

- 30.16 Duplicate copies of test certificates carried out at Manufacturer's works shall be forwarded to the Consultant for approval prior to dispatch of the article to site.
- 30.17 No section of the works shall be insulated or in any other way concealed prior to testing and inspection and subsequent concealment where applicable shall only take place following written authority from the Consultant.
- 30.18 All necessary facilities, measuring and recording instruments including test pumps and gauges for inspection, testing and commissioning requirements shall be provided and shall be checked or calibrated as necessary before use.

The Consultant reserves the right to call for a demonstration of the accuracy of any instruments provided.

- 30.19 All representatives present during inspection, testing and commissioning shall be fully conversant with the system concerned and the method of system and instrument operation.

Manufacturer's or Specialist Subcontractors' representatives shall attend where specifically indicated elsewhere in the specification or where necessary to ensure full service and cooperation is available to the Consultant to enable the works to be tested and commissioned in accordance with the requirements of the specification.

- 30.20 All necessary precautions shall be taken to safeguard against damage during inspection, testing or commissioning. Any damage so caused shall be made good at no cost to the contract.
- 30.21 All tests shall last for the minimum time period stated or longer if necessary, to ensure all sections have been fully examined as required by the test.

One set of test instruments shall be made available during the warranty period to check the system whenever required.

31.0 TEMPORARY USAGE

- 31.1 The Consultant reserves the right to use any piece of electrical and mechanical equipment, device, or material installed under this section for such reasonable lengths of time as may be required to make a complete and thorough test of same before the final completion and acceptance of the work.

Such tests shall not be construed as acceptance of any part of the work, and it is agreed and understood that no claim for damage will be made for any injury or breakage to any part or parts of the above due to these tests, where caused by weakness, inaccuracy of parts, or by defective materials or workmanship.

32.0 INSPECTION OF THE WORK

- 32.1 The representatives of the Owner and the Consultant will make periodic visits to the site during construction to ascertain that the work is being executed in reasonable conformity with all plans and specifications but will not execute quality control at all times. Each Subcontractor must maintain the quality control as intended in the contract documents.
- 32.2 Correct all deficiencies immediately as noted during field inspections.

- 32.3 Request in writing that a final inspection of all services.

Do not issue this written request until:

All deficiencies noted during job inspections have been corrected.

All systems have been balanced and tested and are ready for operation. All balancing reports have been submitted and reviewed.

All instruction manuals have been submitted and reviewed.

The cleaning up is finished in all respects.

All spare parts and replacement parts specified have been provided and receipt of the same acknowledged.

33.0 OPERATING INSTRUCTIONS TO OWNER

- 33.1 Train the Owner's staff how to make minor adjustments, carry out necessary maintenance and how to operate each system.

For new equipment, provide the Consultant with three copies of complete operating and maintenance instructions for equipment at the time of delivery of the equipment.

Equipment will be rejected unless accompanied by instructions. Such documents must be received at least one month prior to the completion date of the relative section of the works.

Instructions shall be bound in a suitable loose-leaf booklet or binder, and shall include:

- a) Prints of the following drawings:
 - 1) List of all equipment installed.
 - 2) General layout.
 - 3) Wiring diagram of control panels.
 - 4) Non-dimensional layout, showing location of all electrical devices (Limit Switches etc.).
- b) Operation instructions, including start-up and shutdown procedure.
- c) Maintenance instructions, including preventative maintenance instructions for components of the equipment.
- d) Lubricating instruction and recommended cycle of lubrication.
- e) Complete parts list, showing manufacturer's name and catalogue number.
- f) Complete list of recommended spare parts for 2-year operation with price list for each item of equipment.
- g) Original catalogue
- h) System description, Equipment schedule
- i) Testing & commissioning reports
- j) As-Built drawings
- k) Warranty certificates
- l) Training records.

The contractor is to prepare a Master Index for discussion with the Consultant prior to finalizing the documents.

Portable tools and spare parts shall be correctly labelled and handed over to the Consultant.

The Contractor is to provide operating and maintenance instruction and training presentations and demonstration for the maintenance staff, these should be based on the content of the Operation and Maintenance manuals and in sufficient detail to talk the Contractor through the practicalities of the work on the site.

- 33.2 The section dealing with complete systems shall be subdivided into each service with a ready means of reference and detailed index. The function and manner of operation of each system shall be clearly described together with illustrations and line diagrams in schematic form showing the location and function of control valves, items of equipment and spaces or areas which are served by these items. The colour coding and identification systems employed shall be explained.

34.0 COMMISSIONING OF SERVICES

- 34.1 The Testing and Commissioning (T&C) of mechanical systems shall be carried out by either a specialized commissioning agency or by the same MEP contractor provided that a grade 6 T&C commissioning manager is available to oversee all T&C works are carried out in accordance with specifications. The grade 6 manager may be an in-house resource or an external specialist/party and is subject to consultant's approval.

The contractor shall be responsible to have a qualified T&C team with minimum grade 3 experience and have all the required calibrated tools, methods and instrumentations to complete the works in line with international codes and standards.

- 34.2 The Commissioning company and the commissioning manager shall be approved by the consultant and have regularly engaged in providing a testing balancing and commissioning service and who has been in continuous business for not less than 7 years employing fully trained staff having not less than 3 years dedicated experience.

A senior experienced Commissioning Engineer with minimum dedicated experience of 7 years shall be responsible for supervising and directing the activities for the testing and commissioning team.

- 34.3 All piping systems shall be flushed, chemically cleaned and then filled with treated water, or appropriate fluid, vented as necessary, and brought to operating conditions and the flows then regulated to the design values.
- 34.4 All water circuits shall be balanced by means of the regulating valves provided and flow rates shall be determined on a temperature and pressure drop basis.
Flow through pumps shall be measured by relating the pressure drop across the pump to manufacturers test curves. A copy of the test curve indicating the final operating point shall be forwarded to the Consultant.
- 34.5 All refrigerant systems shall be vacuumed, purged and charged according to A.S.H.R.A.E. standard methods. All safety controls and interlocks will be tested as part of the commissioning procedures.
- 34.6 Flow through coils shall be adjusted to give the design temperature difference. In general water systems shall be balanced in accordance with the guidelines specified in NEBB (National Environmental Balancing Bureau) Manual. Upon completion of balancing and testing operations for heating and cooling systems, temperature measurements shall be taken in all rooms and the

readings tabulated in schedule form together with hourly ambient temperature readings taken over the measuring period. Two copies of the schedule shall be forwarded to the Consultant.

- 34.7 All ventilating systems shall be commissioned in accordance with the procedures recommended in the AABC (Associated Air Balance Council) / CIBSE (Chartered Institution of Building Services Engineers) Manual. Duplicate schedules of commissioning results shall be forwarded to the Consultant and shall detail the recorded air volume and percentage deviation from design air volume, for each air input and extract terminal. Wet and dry bulb temperature measurements shall be taken in all rooms served by air supply systems and the results indicated on the schedule of commissioning results.

These shall be recorded together with external ambient wet and dry bulb temperatures measured at hourly intervals over the measuring period.

- 34.8 Sound pressure levels (dBA) shall be measured in all rooms containing supply or extract terminals, all plant rooms, all rooms immediately adjacent to plant rooms and all rooms located above or below plant rooms. The Consultant may at his discretion request a spectrum band analysis of sound pressure in any locations.

Sound pressure levels (NC) generated from drainage pipes in false ceiling of occupied areas (like Living/Dining rooms, bedrooms, open kitchen, etc.) except toilets and closed kitchens shall also be

tested in accordance with the specified noise levels of soundproof pipes / acoustically insulated uPVC pipes

- 34.7 All automatic controls shall be commissioned by the control's supplier.
- 34.8 Fault conditions shall be simulated, and all alarms and safety devices shall function correctly. Such proving tests shall be carried out in the presence of the Consultant and certificates shall be provided specifically detailing all check procedures which have been carried out.
- 34.9 Following proving tests all installations shall be left in working order ready for handover.
- 34.10 All electrical systems shall be tested and commissioned in accordance with IEE regulations and local authority codes.
- 34.11 On completion of all testing & commissioning, and prior to practical completion, undertake an Integrated Systems Test (IST).

35.0 SYSTEM ACCEPTANCE

- 35.1 The ultimate condition for system acceptance is that the Owner and Consultant have inspected the system and found it to be acceptable and indicated this in writing.

Issuance of the final payment certificate does not necessarily indicate system acceptance, neither does release of final payment holdback in whole or in part. The Consultant's acceptance may be contingent on any or all of the following if applicable:

- 35.1.1 Submit original copies of letters from manufacturers of all systems indicating that their technical representatives have inspected and tested the respective systems and are satisfied with the methods of installation, connections and operation.
- 35.1.2 Submit "as built" drawings and operation and maintenance manuals.
- 35.1.3 Train owner's maintenance staff.

35.1.4 Satisfactory completion of 12 months guarantee period.

36.0 CORRECTION AFTER COMPLETION

- 36.1 Remedy all work in accordance with the General Conditions of Contract for a period of one year from the date of acceptance of the completed work, or if so noted, for longer periods for parts of the work.
- 36.2 Attend immediately to any and all the defects occurring during the period defined above and repair in a manner to prevent recurrence. This contractor is responsible for all work required by other trades necessary to repair the works of this section, or necessary to repair damage caused by the failure of any part of this section.
- 36.3 Instruct all Suppliers and Manufacturers that guarantees on equipment will commence when the completed work is accepted and not from the date the equipment is put into operation. In the event that this condition is omitted by the supplier, or if subsequent cost to the Owner is involved, this Contractor shall be liable to the full extent of the guarantee as indicated in the specification.

37.0 GUARANTEES/ WARRANTY

- 37.1. The Contractor shall guarantee all material and workmanship for at least 12 months after the issue of final completion certificate and take over by the Engineer and Owner. Extended warranty for particular items/ equipment shall be as specified in related specification sections.
- 37.2. All warranties from equipment suppliers shall be vested in the Owner, regardless of whether the Contractor who supplied the equipment is still associated with the project or not. Detailed Warranty list for MEP equipment shall be provided by contractor for Engineer's approval.
- 37.3. Guarantees will be full guarantees and will include all overhead, profit, incidental charges and sundries.
- 37.4. Where damage is caused to any other item by any failure of the item guaranteed, then the guarantee shall also include the costs incurred in rectifying that damage.

38.0 MAINTENANCE

- 38.1. Maintenance is defined as the Contractual Liability to maintain the equipment in working condition till the end of Defects Liability Period (DLP).

39.0 APPENDIX- 1

O&M Manual Template	
Section	Content
General Project Information and Introductory Overview	Self-Explanatory
Contractual and Legal Information	Self-Explanatory
Emergency Information	Self-Explanatory
Health & Safety Information	Self-Explanatory
Scope of Works	System Description
	Description of Service and Design Intent
	Locations of Major Equipment
	Interface and Interconnections
Equipment Data	Manufacturers and Local Suppliers Data and Contact Details
	Technical Data and Equipment Schedule
	Energy Management
	List of As-Built Drawings
	Schematic Diagrams/Drawings
Operating Instructions	Installation and Commissioning
	Startup Instructions
	Normal Operation Instructions
	Emergency Shutdown
Maintenance Instructions	Maintenance Schedule
	Maintenance Procedures/Tasks
	Troubleshooting
Testing & Commissioning Reports and Data	Self-Explanatory
Warranties	Self-Explanatory
Spare Parts/Tools	Recommended Spare Parts/Tools List
	Handling and Storage of Spare Parts/Tools
Modification Information/Precautions	Self-Explanatory
Manufacturer's Literature	Self-Explanatory

END OF SECTION

MECHANICAL SECTION: 02

INSULATION

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1.0 INSULATION GENERAL

- 1.1 The whole of the insulation work shall be carried out by an approved specialist insulation Contractor. All allowances shall be included for arranging a specialist subcontract accordingly and for informing the specialist Sub-contractor of all conditions relating to the Contract and for coordinating his works with the remainder of the works.
- 1.2 All allowances shall be included for informing the specialist Sub-contractor of all details of the building structure, program arrangements, and other relevant details at the time of tender and for all necessary visits to site by the Sub-contractor or his workers.
- 1.3 All insulation shall be in accordance with British Standard Specifications (unless specified elsewhere) and Codes of Practice in all respects. Insulation meeting A.S.T.M. Standards will be considered if properly submitted.
- 1.4 In addition to complying with the relevant standards, all insulating material shall be free from Asbestos and with low VOC complying with DM Green buildings regulations. All insulation material shall be tested by Dubai Central Lab to confirm compliance with DM Green Buildings Regulations. Insulation system shall be approved by local Civil Defense.
- 1.5 Insulating materials shall be acceptable only if they are equal to or better than the grades or classes of fire resistance as follows:
- B.S. 476, Part 4, (Noncombustible grade.)
- B.S. 476 Part 5 Class P, (Not easily ignitable.)
- B.S. 476, Part 6, (Fire Propagation Index a maximum of 12.6.)
- B.S. 476, Part 7, Class 1, (Surface spread of flame.)
- B.S. 476 Part 9, (Production of emitted smoke shall not give more than 35% obscuration of the light beam.)
- All insulation finishes and coverings shall be classified as Class 1 (one) surface spread when tested in accordance with B.S. 476, Part 7.
- Alternatively, insulation systems shall be tested according to ASTM E84, by a testing agency acceptable to authorities having jurisdiction. They shall have a flame-spread index of 25 or less, and smoke-developed index of 50 or less for insulation installed indoors. For outdoor insulation, Flame-spread index of 75 or less, and smoke-developed index of 150 or less.
- Insulation materials shall have FM approval as per UAE Fire code requirements.
- 1.6 All adhesives, mastics, coatings, sealers and primers shall be classified as Class 1 surface spread when tested in accordance with B.S. 476, part 7, and shall have low VOC content in accordance with green buildings regulations.
- They shall not in any way attack the insulation or the surface to which the insulation is being applied and shall be suitable for the working temperatures.
- 1.7 Insulating materials shall have thermal conductivity values not more than those listed as follows:

MATERIAL	TYPE	THERMAL CONDUCTIVITY. W/M/K
Mineral wool	Sectional	0.04
Mineral wool	Slabs	0.04
Fibre Glass	All	0.034
-----MATERIAL		
TYPE	THERMAL CONDUCTIVITY. W/M/K	
Closed Cell (elastomeric foam)	All	0.038
Polyurethane	Sectional	0.025
Styrofoam	Rigid	0.026
Polyolefin	All	0.035

All conductivity figures are rated at an average temperature of 23°C.

- 1.8 The closed cell elastomeric foam insulation shall have following characteristics in addition to clause 1.5 above:

- a. Water vapour permeability - 0.09 ugm/Nh, BS EN ISO 844
(= 7000 to DIN EN ISO 12572)
- b. Density - 55 to 70 kg / m³ or 4lb / ft³ to 5lb / ft³
- c. Closed cell content - > 90%.
- d. Corrosion risk assessment - Din 1988-200 Certified.
- e. Noise reduction - Din 4109 up to 32 dB (A).
- f. Thermal conductivity of 0.037 w/m.k at 10°C and 0.038 w/m.k at 20°C to BS 874 PART 2 – 86 (DIN 52613, 52612).

The insulation thickness for pipes and ducts shall be verified by the specialist supplier/ manufacturer to confirm no condensation.

- 1.9 All material delivered to site shall be new and fully dried out and so maintained throughout the progress of the works.

All insulating materials shall be stored in storage sheds, and in accordance with the Manufacturer's recommendations.

- 1.10 In order to ensure that the insulation applied is in all respects in accordance with the specification, sections shall, as required by the Consultant, be cut from the finished insulation.

The Contractor is to allow in his price for the removal and replacement of two sections of each type of insulation.

If, however, defects are revealed, further sections shall be cut out for inspection, and all cutout sections shall be replaced at no cost to the Contract.

If further defects are revealed then the Consultant shall have the right, when in his opinion it is necessary, to issue instructions for any part or the whole of the insulation to be removed and replaced. The replacement with new insulation shall be to the satisfaction of the Consultant and the cutting out and replacing shall be at no cost to the Contract.

- 1.11 Particular attention shall be paid to the finished appearance of all thermal insulation which must present a neat and symmetrical appearance running true in line with pipe layouts, etc.
- 1.12 Any rough, irregular and badly finished surfaces shall be stripped down and re-insulated to the Consultant's satisfaction.
- 1.13 In certain cases the type of insulation specified will require a painted finish, in addition to identifying bands and any signs which are to be applied.
- 1.14 All systems are to have been tested and approved by the Consultant prior to installation of insulation.
- 1.15 Manufacturer shall have production process control in accordance with ISO 9002 or any other equivalent standard.
- 1.16 In case of heavy ducts and pipes wherein the supports are fixed directly on ducts / pipes the supports shall be also insulated with a thermal break / barrier.
- 1.17 Insulation for Chilled Water Pipes in Conditioned Areas (Interior): Insulation on both primary and secondary side of the ETS room shall be min. 50mm thickness or as per manufacturer recommendation, whichever is higher.
- 1.18** Insulation for Chilled Water pipes at Roof, Riser shafts and in Plant Room Areas (Exterior): Insulation on both primary side of the ETS room shall be min. 50mm thickness or as per manufacturer recommendation, whichever is higher.

2.0 CONCEALED COLD AIR DUCTS

- 2.1 All horizontal extract air ductwork within air-conditioned areas shall be bare GI without insulation.

The ductwork within non-air-conditioned areas shall be insulated with 32mm thick, 55-70kg/m³ density closed cell elastomeric foam insulation without any additional coating.
- 2.2 Insulate flexible connections and connections to diffusers with 25mm thick, 24kg/m³ density reinforced aluminium foil faced, flame resistant flexible fiberglass insulation. Overlap unto adjacent insulation and seal with adhesive duct tape to give good closure.
- 2.3 Extract duct from commercial kitchen hoods shall be insulated with 50mm thick, 48kg/m³ rockwool insulation with aluminium foil finish.
- 2.4 The thermal insulation performance shall be as specified in clause 1.0

3.0 EXPOSED COLD AIR DUCTS

- 3.1 For ductwork exposed in non-air-conditioned areas, insulate using 32mm closed cell elastomeric foam sheets without any additional coating, or chemically cross-linked closed cell self-adhesive polyolefin reinforced by alupet foil.
- 3.2 For ductwork exposed directly to sun light, insulate with 25mm thick, 55-70 kg/m³ closed cell elastomeric foam insulation sheets, UV resistant.

- 3.3 All ductwork in exposed areas (at roof, in car parking areas and mechanical rooms) shall be clad with aluminium plain sheet of 0.9mm thickness. The cladding in mechanical rooms shall be up to 2.5 m from finish floor level.
- 3.4 Where ducts penetrate the building shell, the duct shall be flashed and waterproofed before any insulation is applied.
- 3.5 The thermal insulation performance shall be as specified in clause 1.0

4.0 ACOUSTIC INSULATION

- 4.1 All low pressure ductwork shall be lined with acoustic insulation up to the first take off point, or 3 meters from the fan/ unit outlet.
- 4.2 Duct acoustic lining shall be 25 mm thick with high density acoustic liner.
- 4.3 Liner shall be tested to UL181 or approved equal for air velocity rating of 25m/s and air erosion rating of 63.5m/s.
- 4.4 Liner will be attached by a fire resistant adhesive such as Foster 81-99, or equivalent.
- 4.5 All abutting edges must be caulked.

5.0 APPARATUS CASINGS

- 5.1. Insulate apparatus casings which are not provided with insulation as follows:

Apply generally 50mm of rigid polyurethane insulation, 24kg/m³ density. On the underside of coil sections, apply 10mm thick polyurethane. All joints are to be sealed in cold adhesive compound.
- 5.2. Ensure that any access panels are insulated to the same standard and are openable.
- 5.3. For small units, cover the entire outside assembly with waterproof self-adhesive glass fibre tape. This must be at least 75mm minimum width and must overlap at least 50% on each application, to achieve watertight conditions.
- 5.4. For larger units, provide a weather resistant enclosure outside the insulation, subject to Consultant's approval.
- 5.5. Ensure that all equipment meets these requirements, regardless of manufacturer's standard practice, and add insulation as required to the above standard.
- 5.6. Where insulation, re-insulation or additional insulation is required to the casing of externally mounted air conditioning units, the insulation shall be covered with sheet Aluminium, of 0.7mm or thicker.
- 5.7. Insulation to drain pans is specified as part of the A.H.U. specification. Regardless of the Manufacturer's standard supply, the contractor shall reinsulate on site if necessary, so that the specification is complied with in full. This standard of insulation is also required for fan coils.

6.0 PIPE INSULATION

- 6.1 The following pipes (except those mentioned in clause 1.17 & 1.18 for ETS room and DC pipes) shall be insulated with closed cell elastomeric rubber foam insulation with following thickness:

- a) Chilled Water Pipes in Conditioned Areas:

Insulate with closed cell elastomeric foam sheet (without any coating) or chemically cross-linked closed cell self-adhesive polyolefin reinforced by alupet foil. Thickness of insulation shall be 19mm with 55-70 kg/m³ density for pipes up to 50mm dia.

Pipes 65mm to 150 mm dia., thickness of insulation shall be 25mm, and for pipes diameter above 150 mm insulation thickness shall be 32mm

- b) Chilled Water pipes at Roof, Car Park and in Plant Room Areas (including ETS room):

Insulate with closed cell elastomeric foam sheet without any additional coating. Thickness of insulation shall be 32mm with 55-70 kg/m³ density for pipes up to 50mm. Apply 0.9mm aluminium cladding plain sheet on top of insulation. Roof insulation shall be UV resistant.

Pipes 65mm diameter and above, thickness of insulation shall be 32mm.

Apply 0.9mm aluminium cladding plain sheet on top of insulation.

All insulated pipe work installed at roof, within plant rooms shall be cladded with 0.9mm thick plain aluminium sheets.

For Chilled water Riser pipes insulation follow clause 1.18

- c) Refrigerant Pipework:

Insulate with closed cell elastomeric foam sheet with insulation thickness of 19mm.

For exposed pipes at roof and in unconditioned areas insulate with 25mm thickness UV resistant insulation and apply 0.9mm aluminium cladding plain sheet.

- d) AC Condensate Drain Pipework:

AC condensate drain pipes embedded in block walls shall not be insulated

Insulate with closed cell elastomeric foam sheet with insulation thickness of 13mm.

For pipes at roof insulate with 25mm thickness UV resistant insulation and apply 0.9mm aluminium cladding plain sheet.

- e) Hot Water Pipework:

Hot water pipes embedded in block walls shall not be insulated.

Hot water pipe work (in PPR material) within bathrooms / Kitchens for local water heater installation shall not be insulated for the pipework within 7 m run from the water heater. However, for the hot water pipework more than 7m, it shall be insulated with 19mm thick closed cell elastomeric foam insulation sheet with no additional coating, or chemically cross-linked closed cell self-adhesive polyolefin reinforced by alupet foil.

All insulated pipe work installed at roof, within plant room shall be cladded with 0.9mm thick plain aluminium sheets.

f) Cold Water Pipework at roof and inside shafts

The cold water pipe work at roof exposed to sun light and the cooled water pipes inside shafts and within non air-conditioned areas shall be insulated with closed cell elastomeric foam sheet with insulation thickness of 19 mm and 55-70 kg/m³ density.

For pipes at roof provide 0.9mm aluminium cladding plain sheet on top of insulation.

g) Diesel Engine Exhaust Pipe:

The Diesel engine exhaust pipe and muffler shall be insulated with 100mm thick calcium silicate or rockwool insulation.

- 6.2. Pipe covering and insulation shall be manufactured by a recognized manufacturer and shall be neatly installed to the following specifications. The general intent is to provide a sealed insulation which will not permit sweating of the pipes and which will not retain moisture to the detriment of its insulating capability.
- 6.3. The insulation, in sizes indicated shall be applied over clean, dry surfaces. Adjoining sections of insulation should be butted firmly together with the longitudinal seam of the jacket located on the bottom half of the pipe.
- 6.4. Insulate and finish valves and fittings in the same manner and same thickness as piping in which such items are installed. Moulded, factory shaped sectional pipe covering, factory- or job-fabricated may be used subject to satisfactory visual checking by the Consultant.
- 6.5. Direct contact between pipe and hanger shall be avoided. Hangers shall pass outside of the sheet metal protection saddle, which shall cover a section of high-density insulation, of sufficient length to support the pipe without crushing the insulation. The vapour barrier shall be lapped over the saddle and securely cemented to it. Minimum thickness of metal saddle is 1.5 mm.
- 6.6. Rigid rubber inserts shall be provided at the pipe supports location with thickness to be equal to the pipe insulation thickness.
- 6.7. The insulation shall be installed in accordance with the manufacturer's printed instructions.
- 6.8. The thermal insulation performance shall be as specified in clause 1.0

7.0 LAGGING ADHESIVE

- 7.1. Adhesive shall be a flexible, fire resistive, fungicidal resistant compound suitable for vapour sealing insulated ducts and pipes.
- 7.2. Adhesive shall be suitable for indoor and outdoor use and in high humidity environments.
- 7.3. Water vapour permeance shall not exceed 0.05 perms at 0.030-inch dry film thickness when tested in compliance with ASTM F1249.
- 7.4. When tested for surface burning characteristics in compliance with ASTM E84, flame spread shall be 25 and smoke developed 50.
- 7.5. Adhesive shall be UL classified and shall meet or exceed the requirements of NFPA 90A and 90 B 25/50.

Adhesive shall have a VOC content complying with DM requirements.

8.0 ALUMINIUM CLADDING

- 8.1 Aluminium cladding shall be provided to ducts with neat 90 degree joints and finish. Material shall be with plain finish.
- 8.2 In pipework the longitudinal joints shall be cleated or held in position with stainless steel bands and clips. Elbows and other fittings shall be made with mitered section and riveted. All valves strainers, flexible connection etc. shall have easily removable boxes with stainless steel aluminium toggle clips.
- 8.3 In ductwork the joints shall be riveted or banded with stainless steel band and clips to avoid sagging. Boxes to be made around duct traverse joints.
- 8.4 All gaps shall be to a minimum and shall be sealed with silicon mastic suitable for outdoor.

9.0 SAMPLES

- 9.1 Submit samples of all types of insulation for approval. Submit pipe insulation samples 300mm long x 25mm inside x 125mm outside diameter, and flat samples 300mm x 300mm x 2mm, minimum size.
- 9.2 Submit with samples, the full technical brochures.
- 9.3 Where the specification calls for additional treatments, such as wrapping and waterproofing, submit also a completed sample of similar size.
- 9.4 All samples after approval are to be retained on site to be a reference for future work.

END OF SECTION

MECHANICAL SECTION: 03

CHILLED WATER PIPEWORK

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1.0 PIPEWORK MATERIAL GENERAL

- 1.1 All pipework shall be free from surface or general corrosion and without any signs of scaling, pitting or excess weathering. Any pipework so affected shall be replaced at no cost to the contract.

Each length of pipework shall have at least one coloured identification band or identifying mark, with name or symbol of manufacturer, standard material designation and schedule no, when delivered. All pipework shall be supplied in the manufacturer's straight random lengths, but not less than 6M long except where shorter specific or flanged lengths between fittings are actually required.

All plain mild steel pipework shall be supplied self-colored and externally varnished.

1.2 Chilled Water Service Pipework.

All pipework shall be black steel, ERW welded, medium weight to ASTM A53 SCH 40.

1.3 Chilled Water Fittings

A. For pipes diameter 65 mm and larger

1. Black steel to ANSI B16.9, ANSI B16.28 butt-welding ends.
2. Black steel with grooved ends suitable for mechanical jointing.

B. For pipes diameter 50 mm and smaller

1. Malleable iron to ANSI B16.3, BS 143 & BS 1256 Equivalent to DIN EN 10242 & ISO 49 with threads ends. threads to confirm to BS EN 10226 (ISO R 7 / 1)
2. Welding flanges for pipe 65 mm and larger: Steel to ANSI B16.5.
3. Unions for pipe 50 mm and smaller

Malleable iron ASTM A1 97, Class 150, with bronze to iron ground joint, ANSI B16.3.

Dielectric unions shall be used in case of joining dissimilar material,

4. Flange Accessories:

Gasket, canvas reinforced neoprene to BS 1737.

Bolts, square head machine with hexagonal nut, galvanized steel to BS 916.

5. Use "Y" pattern strainers, PN 16, with screens of type 316 stainless steel or Monel with approximately 0.79 mm perforations. Provide a valve blow-off connection in all caps 65 mm and larger. Maximum pressure drop with clean screen shall not exceed 6.89 kPa, with water velocity through inlet pipe of 1.5 m/s.

1.4 Condensate Drain Pipework.

All exposed condensate piping shall be galvanized medium weight, welded steel to ASTM A53 SCH 40, with screwed joints. Fittings will be screw type, forged.

All concealed condensate drain piping shall be uPVC Class D with solvent welded joints.

- 1.5 All pipework, pipework fittings, jointing materials and gaskets shall be stacked in storage sheds in accordance with the manufacturer's recommendations and as required elsewhere in this specification. The ends of all pipework shall be protected during transit and storage.
- 1.6 Metal pipework may be stacked in the open, provided that such stacks are, in the Consultant's opinion, adequately protected from weathering. The pipework shall be located in steel pipe racks and clear of the floor; the floor being formed from either creosoted timber sleepers or paving slabs. A watertight lightweight roof shall be provided to extend 450 mm beyond the edges of the stack and tarpaulin or heavy gauge Polythene sheeting shall be provided to cover all sides.
- 1.7 The pipes beyond stock rust shall not be used.
- 1.8 Pre-Insulated Chilled Water Pipework

All buried chilled water lines shall be pre insulated. The system shall be non-corrosive, structurally strong, completely waterproof, and entirely resistant to attacks by salts, water and all ground chemicals normally encountered in an underground environment. The system manufacturer shall have fabricated systems of the composition defined hereinafter for at least two years. All straight section, fittings, anchors, end seals and other accessories shall be factory prefabricated to job dimensions and designed to minimize the number of field welds. The system design shall be in strict conformance with ANSI B 31.1, Latest Edition. Pipe movement due to thermal expansion or contraction shall be accommodated with expansion loops or elbows.

All fittings shall be factory prefabricated and pre-insulated. Straight tangent lengths shall be added to all ends so that all field joints are at straight sections of pipe. Fitting covers shall be HDPE or fiberglass. HDPE covers shall be mitered, and extrusion welded, butt fusion welded, or molded. Fiberglass covers shall be chop sprayed. Gluing, taping or hot air welding shall not be allowed.

All insulation and jacketing materials for the field joint shall be furnished by the manufacturer.

The complete system shall be computer analyzed by the preinsulated piping system manufacturer to determine stress and anticipated thermal movement of the service pipe.

1.8.1 Service Pipe

MS Black steel pipes, ERW type to BS EN 10255 medium weight for sizes up to “6 diameter” & for sizes from 8” to 10” diameter shall be heavy weight. Alternatively, they shall conform to ERW, medium weight to ASTM A53 SCH40. All joints shall be butt welded for sizes 65mm and larger and socket welded for 50mm and below. Where possible, straight sections, shall be supplied in 12M random length.

1.8.2 Insulation

The pipe insulation shall be rigid polyurethane foam, cellular, CFC free, conforming to ASTM C 518, with a nominal density of 48 kg/m³. They shall have a closed cell content of 90-95%, K factor of 0.28W/m.K and compressive strength of 276 - 414 kPa. The insulation shall completely fill the annular air space between the pipe and the jacket. Support guides shall be factory installed to assure equal thickness of insulation around the pipe.

Insulation thickness shall be as follows:

Pipe size (mm Dia)	Up to 25	32-40	50- 200	Above 200
Insulation Thickness	25	40	50	80

2.0 PIPE INSTALLATION GENERAL

- 2.1 All pipework shall be arranged to set round piers and other obstructions and minor modifications shall be made as required by the Consultant to circumvent site difficulties.
- 2.2 All pipes and joints on both primary and secondary sides shall be protected by two coats of red oxide anti corrosion paint after proper preparation of pipes surface. Insulation shall be properly glued and prepared for cladding on the top of the pipes as specified elsewhere in this document.
- 2.3 Pipe shall be arranged to follow the contour of walls or beams or other building structure lines and all vertical pipework shall be plumb, without offsets and set as close as possible to any local projections consistent with maintaining adequate clearances for installation of wall plates or insulation.
- 2.4 Pipework shall be installed so as to give the following minimum clearances between the pipe (or pipe insulation where specified) and adjacent surfaces as follows: -

Walls	25 mm
Ceilings	50 mm
Finished floors; either: above top of skirting	50 mm
or (if greater)	150 mm
Adjacent pipes, both insulated	25 mm
Adjacent pipes in trench, both insulated	100 mm
Adjacent pipes, both uninsulated	150 mm
Adjacent pipes, one insulated	75 mm
Insulated pipes to adjacent conduit or trunking	100 mm
Adjacent electrical cables not in conduit or trunking	150 mm

Notwithstanding the above minimum clearances, sufficient space shall be allowed to facilitate easy application of insulating materials. Pipes shall not be enclosed in a common insulating covering.

- 2.5 Pipework shall be graded to ensure adequate draining and venting. Draining and venting facilities as detailed in the particular specification for individual services shall be fitted at all low and high points respectively and wherever else necessary to ensure that all sections and subsidiary sections can be drained and that no air locks can form.
- 2.6 The Consultant may at his discretion ask for the removal of installed pipework for examination. No extra payment will be made when such removal is called for.

If the pipework is found to have been installed in an unsatisfactory manner, then the complete installation shall be thoroughly inspected, and all unsatisfactory sections shall be removed and re-fixed in a proper manner.

- 2.7 During constructional work on all pipework services, care shall be taken to prevent any foreign matter entering the pipework. All open ends shall be capped with the appropriate pipework fittings. Wooden plugs and the like shall not be used. Valves fitted on the ends of pipework shall not be accepted as a means of preventing the ingress of foreign materials.

- 2.8 Failure to comply with these requirements shall mean that the Consultant shall have the right to instruct that pipework so left uncovered to be dismantled for such lengths as the Consultant requests, and the pipework blown through and/or cleaned at no cost to the contract.
- 2.9 All pipework fittings shall be installed in such a manner to ensure that air cannot be trapped, and that pipework can be drained.
- 2.10 Unequal tees shall be used in preference to equal tees with separate reducing fittings. Bushes shall not be used. Unequal tees shall be eccentric pattern on horizontal pipework, and concentric on vertical pipework. Eccentric reducer to be used for suction line of pump.
- 2.11 Segmented, or "cut-and-shut", or any other site manufactured bends or fittings shall not be installed in any section of the works.
- 2.12 The use of fire or cold pulled bends will not be permitted.
- 2.13 Where weldable steel pipework fittings to BS EN 10253 are to be used then branch bends, although not covered by this standard, will be accepted, but only at the specific location designated by the Consultant and subject to total approval by the Consultant.
- 2.14 Where fittings are connected to light or medium weight pipework they shall be 'medium' quality, and where connected to heavyweight pipework they shall be 'heavy' quality.
- 2.15 All fittings, valves, cocks etc., shall be manufactured from materials guaranteed proof against de-zincification unless specifically stated otherwise elsewhere.
- 2.16 Reducers on all vertical pipework shall be concentric. In all other positions eccentric reducers shall be fitted in a manner to maintain a level bottom and ensure that fluids are not collected at that point in the system. Where it is not possible to fit eccentric reducers the Consultant's approval shall be obtained before fitting concentric reducers. Reductions in all cases shall be made by the use of factory-made fittings.
- 2.17 Sufficient unions or flanges shall be provided to install and dismantle screwed or solvent jointed sections of pipework.

Unions on all steel pipework other than on steam installations, shall be of malleable iron construction with spherically ground bronze-to-bronze seats and shall have screwed ends to BS EN 10226.

Unions on uPVC systems shall be factory manufactured unions appropriate for the pipe.

- 2.18 All branches shall be made by easy sweep tees, twin elbows or sweep crosses. Bends shall be used wherever possible. All sweep fittings and all sweep bends shall be of the long radius pattern except where the use of these fittings would stand pipework too far from wall surfaces and make for unsightly appearance, in which case short sweep tees and elbows may be used provided that the Consultant's written approval is obtained beforehand. Thredolet, weldolet, or Elbowlet can be used for branches with diameters less than half the diameter of the main line.

3.0 PIPEWORK INSTALLATION, FLANGES

- 3.1 Flanges shall be provided on the pipework systems wherever necessary to connect to components, plant or equipment having flanged connections. In addition, flanges shall be provided where services are specified elsewhere as having flanged joints.
- 3.2 Flanges (including the associated nuts, bolts and washers) shall be to the minimum requirements of BS EN 1092 to suit the conditions within the pipework or to suit equipment, valves and other pipework components having flanged connection. Where these connections have flanges greater than the minimum requirements of BS EN 1092 then special mating flanges shall be provided to suit the

connection concerned. Where flanges to BS EN 1092 are not manufactured then flanges to BS 10 will be accepted.

- 3.3 Flanges shall be bolted up using hexagonal nuts and bolts manufactured from high tensile carbon steel in accordance with BS 916 using two flat steel washers, one each side of the jointed flanges.
- 3.4 Bolt threads shall not project more than 3.2 mm or less than 1.6 mm beyond the nuts when jointed up.
- 3.5 Blank flanges where required shall be manufactured from the same material and same thickness as the flanges to which they are mating.
- 3.6 Flanges shall be smooth machined across the full width of the flange and on the edge but may be rough turned on the back.
- 3.7 Flanges shall be mounted square with the axis of the pipe after all surface scale, oxides, grease, oil and dirt have been removed.
- 3.8 Care shall be taken to avoid distortion of the flange during welding.
- 3.9 Steel flanges shall be provided on mild steel pipework.
- 3.10 Cast iron flanges shall be provided on cast iron pipework.
- 3.11 Steel and gunmetal flanges shall be of the welding neck pattern, welded to the pipework in accordance with the general requirements for welding detailed elsewhere in the specification.
- 3.12 Flanges shall be finished or coated exactly as specified for the pipework on which they are to be fitted.
- 3.13 Gaskets for flange jointing shall be asbestos free, canvas reinforced neoprene full face gaskets to BS EN 1737, suitable for fluid to be handled and working temperatures and pressures.
- 3.16 Stub flange assemblies will only be permitted if specifically requested for a specific reason and if thought appropriate by the Consultant.
- 3.17 All items of plant and fan coil units etc., shall be capable of being disconnected from connecting pipework services by means of unions, union valves, flanges or flanged valves. Pipework shall be so arranged to allow the item to be removed.

4.0 PIPEWORK INSTALLATION - STEEL

Black steel sizes up to 50mm will be threaded and size 65 mm and above will be welded. Alternatively, grooved pipe mechanical joints may be used for all pipe sizes.

4.1 Threaded Joints.

Screw threads shall be clean and true.

In addition, all internal threads must be checked for quality and any with any indication of damage must be rejected and removed from site. All pipe shall be reamed or scraped to remove internal burrs after threading.

4.2 Welded Joints.

Welding shall generally be in accordance with the relevant Section of the General Specification. All steel elbows shall be of long radius pattern except where space conditions do not permit. The substitutions of short radius elbows must be approved by the Consultant.

4.3 Welding Piping:

4.3.1 Where so shown on drawings, specified or directed, welded joints, outlets and flanges shall be used. Welded joints may also be provided elsewhere, where approved by the Consultant, except on piping smaller than 80 mm, or at points where it may be explicitly specified or directed to leave flanged joints in order to facilitate future changes.

4.3.2 All welded joints (except pipe welded end-to-end) shall be made by use of forged one-piece welding flanges, caps, nozzles, elbows, branch outlets and tees as appropriate.

Cut samples shall be submitted for approval if directed. All such fittings etc., shall be of type which maintains full wall thickness at all points, ample radius and fillets, and proper bevels or shoulders at ends. "Weld-o-lets" may be used where standard fittings of required sizes are not available and elsewhere as approved. All job welding shall be done by the electric arc welding process in accordance with the following:

- 4.3.3 a) All joints 45° bevel type. Pipe shall be mill-beveled or machine-beveled under this section of the specifications.
- b) All scale and oxide removed with hammer, chisel or file and bevel left smooth and clean.
- c) Pipe lengths lined up straight with abutting pipe ends concentric vertical and eccentric for horizontal.

4.3.4 Both conductors from the welding machine shall be extended to locations at which welding work is being done. The leads from welding machine to locations of welding works shall be held together with tape or other approved means so as to prevent induced current in structural steel, in piping or other metals within the building.

The ground lead shall be connected to length of pipe with suitable clamp in such manner that welding current will not flow through joints in pipe, structural steel of building or steel pipe supports.

4.3.5 Weld metal is to be thoroughly fused with base metal at all sections. Welds shall be of sound metal, free from laps, slag inclusion or other defects.

4.3.6 All welders shall be approved by the Engineer for the service for which they are employed and on which they work.

4.3.7 Wherever welded piping connections to equipment, valves, or other units need maintenance, servicing, or require possible removal, the connection joint shall be flanged. Pressure rating of the pipe flanges shall match the pressure rating of the flanges on the equipment to which the piping connects.

4.3.8 All soldered joints shall be made with 95% tin and 5% antimony solder, having a melting point of not less than 238°C. All soldered joints for tubing larger than 50 mm in size shall be made with the simultaneous application of two or three blow torches.

4.4 Grooved Pipe Mechanical Joints

4.4.1 Grooved mechanical pipe couplings, fittings, valves and other grooved components may be used as an option to welding, threading or flanged methods. All grooved components shall be of one manufacturer and conform to local code approval and/or as listed by ANSI-B-31.9, ASME, UL/FM, IAPMO or BOCA. Grooved end product manufacturer to be ISO-9001 certified.

4.4.2 Mechanical couplings shall be rigid coupling for high pressure service. Couplings shall be cast of ductile iron conforming to ASTM A-536, Grade 65-45-12 or malleable iron conforming to ASTM A-47, Grade 32510. Flexible grooved couplings can be used in pump rooms and building expansion joints.

Mechanical coupling bolts shall be heat treated carbon steel track head conforming to physical properties of ASTM A-183, minimum tensile strength bolts shall be cadmium plated to ASTM B766-86.

Gaskets shall be Grade 'E' EPDM compound conforming to ASTM D-2000 designation 2CA615A25B 24F17Z. Temperature operating range - 30°F to +230°F (-34°C to + 110°C)

- 4.4.3 Fittings shall be full flow ductile or cast iron fittings, with grooves or shoulders designed to accept grooved end couplings.

Fittings shall be cast of ductile iron conforming to ASTM A-536, Grade 65-45-12, or malleable iron conforming to ASTM A-47, Grade 32510.

- 4.4.4 Standard fittings and coupling shall be provided with an alkyl enamel finish.

5.0 PIPEWORK INSTALLATION – uPVC – CONDENSATE PIPEWORK ONLY

uPVC piping system is to be installed in accordance with the manufacturer's recommendation, using appropriate pressure pipe and fittings.

Extreme care must be taken in the preparation of joints. All joints must be cut square. The burrs must be removed, and a 2 mm chamfer filed onto the outside at 45°.

The pipe is to be abraded using emery cloth to approximately the depth of the socket. In addition, the inside of the socket is to be roughened in a similar manner.

The roughened surfaces are to be cleaned using the proprietary cleaner from the system manufacturer.

The proprietary cement from the manufacturer of the system must be thoroughly stirred before each use. The cement is to be applied longitudinally to both the pipe and the fitting. Generally, two coats will be necessary, but care is to be taken to ensure that no excess solvent is left on the inside of smaller fittings.

Immediately after application of the cement the pipe must be pushed fully home against the stop in the fitting. The pressure must be maintained for about 30 seconds.

Wipe off any excess cement. Clean brushes in cleaner solvent and close lids on all solutions.

The system is to harden at least 24 hours before pressurizing.

6.0 PIPEWORK SUPPORTS

- 6.1 All pipework shall be adequately supported. All support installations shall be in accordance with relevant British Standard Specifications and Codes of Practice, except where modified or extended by the contract documents.
- 6.2 Supports shall allow free movement for expansion or contraction of pipework and shall be located to ensure that pipework branches or fittings are not fouled by the support during expansion or contraction of the pipework service.
- 6.3 Double banking of pipework from a single support position will be permitted, only for small pipes near fan coil units, provided the normal operating temperature of the fluids in the two pipes do not differ by more than 30°C., but only where space restrictions prohibit individual support. Triple banking will not be permitted.
- 6.4 Where double banking is necessary, the larger of the two pipes shall be uppermost, and where pipes are the same size but manufactured from different materials, then the pipe having the material with the

lowest coefficient of expansion shall be uppermost. Support intervals for double-banked pipework of different sizes shall relate to the smaller size.

- 6.5 Vertical rising pipes shall be supported at the base and the support shall withstand the total weight of the pipe and fluid contained.
- 6.6 Supports shall not be permitted which clamp the pipe so that it is in contact with building fabric or structure.
- 6.7 All supports shall be specifically designed for the outside diameter of the pipe concerned (including specified packings). Oversized brackets will be rejected.
- 6.8 Where non-ferrous pipework is to be supported using ferrous pipe clamps or rings, an approved plastic coating applied to the clamp or ring shall be used to prevent contact between the ferrous and non-ferrous surfaces.
- 6.9 All pipework shall be supported in accordance with the following table: -

Pipe Size (mm)	Horizontal Piping (Meters)			Vertical Piping (Meters)		
	Steel		uPVC	Steel		uPVC
15	2.1		0.7	2.4		1.3
20	2.1		0.7	3.0		1.5
25-32	2.4		0.9	3.0		1.8
40	2.7		1.0	3.6		1.8
50	3.0		1.0	3.6		2.0
65	3.3		1.0	4.5		2.0
80	3.7		1.3	4.5		2.5
100-125	4.0		1.5	5.0		3.0
150	5.2		2.0	6.0		3.5
200	5.5					
250	6.0					
300	7.0					
350	7.5					
400	8.0					
450	8.5					
500	9.0					

- a. Spacing does not apply where span calculations are made or where concentrated loads are placed between supports, such as flanges, valves, specialties, etc.

b. All spacing length are in meter.

- 6.10. Plastic pipework shall be supported using the pipework Manufacturer's standard support clip as detailed in the Schedule of Manufacturers.
- 6.11. Horizontal mild steel pipework shall be supported individually by clevis type hangers or roller hangers where indicated. Also split case clamp or U profile with channel support can be accepted.
- 6.12. Vertical steel pipework shall be provided with pipe alignment as recommended by supplier and with pipe anchors at locations as agreed with the Engineer. Contractor to carry out stress analysis to ensure proper installation and sizing of supports.
- 6.13. All chilled water pipe supports shall be galvanized.
- 6.14. All chilled water pipework services shall be supported around the insulation and not directly around the pipework. Care shall be taken to ensure the integrity of vapour sealing is not damaged in any way.
- 6.15. It shall be noted that in certain cases it will be necessary to adopt a combination of the support methods indicated, and when supports are required to be detailed to suit special site conditions or requirements, then these details shall be submitted to the Consultant for approval. Anti-vibration supports shall be fitted at all locations where pipework vibration is likely to be a problem. Anti-vibration mounts to be provided in plant room for the equipment and pipes.
- 6.16. Stress analysis to be provided for ETS Room, Pump Room, Risers, and for all pipes above 200 minimum diameter showing anchor points and pipe guides and the pipe support has to be designed based on the stress analysis calculation.

7.0 VALVES

- 7.1. Valves shall be provided and installed where required for the purpose of circulation control and isolation. Valves shall be of full bore size to suit the mains into which they are installed.
- 7.2. All castings shall be clean close-grained metal, free from rough projections. Screwed valves shall have heavy hexagon reinforcement threads, ample length threads and heavy shoulder to prevent over entry of pipes. Flanged valves shall have the flanges flat faced and of thickness conforming to the appropriate specifications for the respective services and drilled off-centre.
- 7.3. Valves shall have pressure rating as necessary for the service indicated on the drawing. Samples must be provided for inspection of the manufacturing methods.
- 7.4. Gate valves for sizes up to 50mm, shall be bronze, non rising stem, screwed bonnet, one-piece wedge. The stem shall have a lock shield to restrict unauthorized adjustment. The casting shall have large hexagon section at all threads to permit gripping to prevent line distortion during installation.
- 7.5. Gate valves sizes 65mm dia and above shall be cast iron body with bronze trim, inside screw, non rising stem, bolted bonnet, bronze trim.
- 7.5.1. ETS Room Butterfly Valves, 65 mm and Larger: Double Eccentric (DE) butterfly valves shall be manufactured in accordance with BS EN 593. Disc of the Valve shall be Double Eccentric Design
- 7.6. Globe valves size 50mm dia and below shall be bronze, rising stem, screwed bonnet and renewable dynamic disk.
- 7.7. Globe valves 65mm and up shall be bronze trimmed iron body valves, with outside screw and yoke, bolted bonnet, guide feature for disc seating. In ETS Room Non-Slam Check Valves (Solid Disc or

Ring Type): DN25-2200 shall be used on both primary and secondary side.

- 7.8 Check valves up to size 50mm shall be bronze, screwed ends, "Y" pattern with screwed access cap, regrindable seating, positive seating two-piece swivel disc. Check valve shall be swing type.
- 7.9 Check valves size 65mm and over shall be iron body with bronze trim, rotating disc with flexible hinge assembly, bolted access. Check valve shall be non-slam, swing type.
- 7.10 Balancing Valves up to size 50mm diameter shall be of bronze body/brass ball construction with glass and carbon filled TFE seat rings. Valves to have differential pressure readout ports across valve seat area. Read out ports to be fitted with internal EPT insert and check valve. Valve bodies to have 6mm NPT tapped drain/purge port. Valves to have memory stop feature to allow valve to be closed for service and then reopened to set point without disturbing balance position.

All valves to have calibrated nameplate to assure specific valve setting. Valves to be leak tight at full rated working pressure. All valves to be provided with moulded insulation to permit access for balance and read out.

- 7.11 Balancing valves size 65mm dia and above shall be of heavy duty cast iron flanged construction with flanged connections.

Valves size 65 and 80mm dia shall have a brass ball with glass and carbon filled TFE seat rings. Valves size 100mm dia and above shall be fitted with a bronze seat, replaceable bronze disc with EPDM seal insert, and stainless steel stem. Valves to have memory stop feature to allow valve to be closed for service and then reopened to set point without disturbing balance position. All valves to have calibrated nameplate to assure specific valve setting. Valves to be leak tight at full rated working pressure. All valves to be provided with moulded insulation to permit access for balance and read out.

- 7.12 Valves up to and including size 50mm shall be bronze body with screwed ends. Valves size 65mm and above shall be cast iron body bronze trim with flanged ends.
- 7.13 All valves shall be rated for a working pressure of 1.3 times the system pressure, the minimum rating being 16bar. High pressure valve shall be of cast steel construction.
- 7.14 In lieu of gate valves, butterfly valves can be used. Valves construction shall be ductile iron body, ductile iron disc, rubber encapsulated suited for the intended service. Valves shall be bubble tight, with memory stop.
The butterfly valve for pipe size of 200mm and above shall be fully lugged with gear box.
- 7.15 Certificate of Origin shall be provided for all valves.
- 7.16 In ETS Room chain-wheel operators, for valves 65 mm and larger, installed 1.8 m or higher above finished floor elevation. Extend chains to an elevation of 1.5 m above finished floor elevation.
- 7.17 The manufacturer shall warrant on supplied components for a period of 2 years from the date of handover for all the Valves.

8.0 Pressure Independent Control Valve (PICV)

Two-way pressure independent control valve (PICV) shall be installed on for each cooling coil as indicated on the drawings, particular specification and/or contract documents.

The valve is a combination of balancing valve and control valve and differential pressure control valve and shall be able to control the flow independent of the system pressure fluctuation.

The flow through the valve shall not change more than +/- 5% due to change in system pressure fluctuation across the valve in the selected operating range.

The valve should have test points for purpose of flow measurement based on Kv-methodology. If the test points are not available or flow measurement based on Kv-methodology is not possible separate orifice plate shall be provided and installed in series with each PICV.

Differential Pressure Control Element: shall be with membrane and not cartridge type.

Valve Specification:

Flow Characteristic: Equal Percentage and not linear. Equal percentage can be through the actuator with linear valve.

Stroke: $\geq 4\text{mm}$ and should not be lift limited on valves from 15 to 50 mm.

Body material:

15 to 50mm - Bronze or dezincification resistant copper alloy equivalent to CuZn33Pb2Si-S (CC751S) according to BS EN 1982.

65 to 150mm - Ductile Iron equivalent to EN-GJS-400-15 according to BS EN 1563.

End Connections:

15 to 50mm - Threaded Connection according to ISO 228.

65mm and above - Flange Connection according to BS EN-1092-2, type 21.

Diaphragm & Seals: EPDM or HNBR.

Valve Plug: Polyphenyl sulphide (PPS), dezincification resistant copper alloy or Stainless Steel. Spring: Stainless Steel.

Temperature Operating Range: -10 to 120°C .

Pressure class: 16bar or higher if required.

Maximum Differential Pressure Rating: 6 or 8bar as per actual maximum differential pressure in current application.

Valves shall be selected at minimum pressure drop of 600kPa.

Valves shall be selected to achieve the required design flow rate and within the working differential pressure range within the piping network.

In ETS Room all Control valves shall be having linear characteristics or equal percentage and will be decided according to the application and Engineer's decision.

Actuator Specifications:

The valve shall be fitted with a fully modulating actuator supplied by the valve manufacturer and calibrated at factory to ensure the compatibility.

Actuator shall accept 0-10VDC, 2-10VDC or 0-20mA, 4-20mA (with reversed option) control signal for accurate modulating control.

Actuator stroke shall be compatible with the stroke of the control valve for reliable and versatile operations.

Actuator shall have valve stroke adaptation and position indicator visible from all sides. Actuator shall have equal percentage characteristic.

Actuator for PICV size up to 32mm inside air conditioned area shall be rated IP42 and

in non-air conditioned area IP54.

Actuator for PICV size 40mm and above shall be rated IP54.

Actuator shall be suitable for ambient temperature ranging from 0 to 50°C.

Actuator shall be selected with adequate force/torque and shall include a 25% safety factor in addition to maximum force/torques within the working range of differential pressure required for the application. They shall be capable of closing valve against system pump shutoff head.

Power Supply: 24VAC/230VAC, 50 Hz.

All the actuator installed on air handling units or actuators installed on pipe diameter more than 50 mm shall be capable of providing the valve status (feedback) to the BMS system.

Identification:

Direction of flow shall be permanently marked on the valve body.

Identification aluminium tags shall be provided for all valves; tags shall be indelibly marked with part/model number, valve reference, production date and differential pressure range.

Warranty:

The manufacturer shall warrant on supplied components for a period of 2 years from the date of handover for all PICV's in the ETS room.

As a VE option, contractor can use FCU link valve package kit comprising of PICV, Isolating Valves, bypass line, strainer and flex connectors. Sample shall be submitted for engineer's approval.

9.0 STRAINERS

9.1 Furnish "Y" type strainers throughout the job unless specifically noted otherwise.

9.1.1 Strainers 50mm and smaller

Furnish screwed ends, screwed bronze cap and gasket in sizes 40mm and smaller and with a bolted cap on 50mm size. Screens on water service shall be made of Type 316 F stainless steel with screen perforation of 0.8mm. Furnish bronze strainers to BS EN 1982 in all piping upto 50mm diameter.

9.1.2 Strainers 65mm through 300mm.

Furnish flanged ends, bolted iron cap and gasket cast iron body to BS 1561 – GJL - 250. Screens on water service shall be Type 316 stainless steel with screen perforation of 0.6mm. Strainer shall be rated for PN16 (minimum).

9.2 The Y strainer shall have removable basket. Adequate spare baskets shall be provided for client use.

9.3 In ETS Room Suction Diffuser: heavy duty cast iron angle type body with stainless steel replaceable straightening vanes extending the full length of the orifice cylinder. Suction diffuser shall be provided with disposable fine mesh strainer and adjustable support foot to carry weight of suction piping, suitable for a working pressure 2070kPa rating. Suction diffuser to be provided with davit arm for sizes more than 300mm for easy removal of strainer and the I-bolt material to be stainless steel.

10.0 GAUGES AND THERMOMETERS

10.1 Wet service thermometers shall be straight shank mercury fill insertion type. They shall have a stem

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length of 80mm and a scale length of 225mm, and the capability of adjusting the angle of the scale to the stem. Provide copper or brass separable wells for each thermometer. Pressure gauge shall be with stabilizer damping movement.

- 10.2 Air service thermometers shall be straight shank mercury fill insertion type. They shall have a stem length of 150mm and a scale length of 225mm, and the capability of adjusting the angle of the scale to the stem.
- 10.3 Pressure gauges shall have 100mm diameter cast Aluminium case, black figures, forged brass sockets with phosphor bronze bushed rotary type movement and Bourdon tube. Gauges shall be wet type and complete with impulse dampening insert and T-handle gauge cock.
- 10.4 Where the line size is 100mm or less the gauge is to be installed into a "Tee" branch, with reducers. For sizes above 125mm a "threadolet" weld-on connection may be used, subject to proper procedures and requirements of the welding specifications.
- 10.5 Range of gauges shall be selected such that the operating point is almost midway of the selected range.
- 10.6 All pressure gauges must be provided with syphons.

11.0 AIR VENTS

- 11.1 Provide air vents at all high points in supply and return piping.
- 11.2 Automatic air vents shall have gunmetal or brass bodies, nonferrous or stainless steel floats and guides and non corrodible valves and seats. Each automatic air vent shall be controlled by a lock shield valve. Air release pipes shall be run to discharge at the nearest visible point - to be agreed with the Engineer.
- 11.3 In ETS Room Use automatic air vent with combined air intake and air release function (triple function) for the header pipes. High quantity air release during water filling, low quantity air release during operation under pressure and to vent high quantity of air during draining the pipeline.

12.0 OTHER ACCESSORIES

12.1 FLOW SWITCHES

- 12.1.1 Flow switches shall be provided at locations as agreed with the Engineer.
- 12.1.2 Flow switches shall be paddle type with bronze construction for all parts in contact with water.

Electrical termination box shall be weather protected to IP 65 and suitable for installation in ambient case temperatures up to 85 deg. C.

12.2 EXPANSION COMPENSATORS

- 12.2.1 Expansion compensators for 75mm diameter pipe and larger shall be pack less bellow type with equalizing rings, stainless steel bellow, limit stops, internal telescoping sleeves and carbon steel bevelled welding ends.
- 12.2.2 Expansion compensators for 65mm diameter and smaller shall be pack less bellows type with stainless steel bellow, anti-torque device, limit stops, guides and threaded pipe ends.

12.3 FLEXIBLE PIPE CONNECTORS

The flexible connector shall be made of multi-layer in nylon tire cord fabric reinforcement with EPDM cover and liner. Straight connectors shall have minimum one sphere and shall be according to manufacturer's recommendations. Connectors 50mm and smaller may have threaded ends. Larger sizes shall have baked enamel ductile iron floating flanges. There shall be a moulded in ductile iron reinforcing ring.

Standard duty flexible connector shall be rated for 1723kPa at 76.6°C, and 1130kPa at 121 °C.

Where flexible connectors are connected to unanchored piping or isolated equipment, provide control cables and rods when pressure exceeds the maximum recommended for this application by the manufacturer.

Flexible hoses shall be stainless steel braid and carbon steel fittings.

12.4 UNIONS

12.4.1 All unions in steel pipes shall be minimum 1200kPa malleable iron, screwed, with brass to iron ground joints.

12.4.2 All unions in copper pipe shall be cast bronze, wrought copper or wrought bronze, with threaded or solder-joint tube ends.

12.5 TEST PLUGS

12.5.1 Provide test plugs of solid brass or stainless steel at locations shown on Drawings and as required for the efficient and proper testing and commissioning of the system. Plugs shall be capable of receiving either a temperature probe or pressure probe. Fittings shall have a valve core of neoprene suitable for temperatures to 100°C, and shall be rated for zero leakage for highest working pressures of the respective systems.

12.5.2 Provide two (2) test kits each consisting of a minimum 65mm dial face pressure gauge, with pressure ranges expected within the system; 25mm dial thermometer, -4 to 52°C.; and a standard gauge adapter all contained within a carrying case. The pressure gauge and thermometers shall be capable of easy recalibration.

13.0 FLOW MEASUREMENTS

13.1 Provide in the main incoming supply line a feature to enable flow measurements to be made. Provide all equipment and, if indirect measurements are used, provide all calibration equipment, charts etc.

13.2 Provide Binder Sockets suitable for the application of flow sensors at all locations shown or necessary for all balancing purposes.

13.3 Provide water meter to monitor the quantity of water being added to the system by the make-up unit.

14.0 EXPANSION PROVISIONS AND ANCHOR POINTS

14.1 All sections of pipework installations shall be installed in such a manner as to allow expansion and contraction for the pipework, without causing undue stress in any part of the installation.

14.2 The stress in the pipework shall be kept below the yield point. Care shall be taken to prevent branch connections becoming anchor points.

14.3 Wherever possible expansion and contraction shall be absorbed by natural offsets and changes in

direction of pipe runs. Provide anchors, pipe guides and expansion loops where shown or required, to the Consultant's approval. Do not use screwed fittings on expansion loops.

- 14.4 Wherever it is not possible to accommodate expansion and contraction of the installations as described above, expansion devices shall be supplied, erected and connected into the pipework installations, and the position of these devices shall be to the approval of the Consultant. 'Articulated' expansion devices shall be fitted on steel pipework systems.
- 14.5 The exact location and working details including anchor loads of all expansion devices, guides, anchors, and all associated equipment shall be submitted to the Consultant for approval prior to commencement of the installation, carrying the manufacturer's confirmation that these are in accordance with his requirements.
- 14.6 Connections to items of plant and equipment shall be made so that no stress is placed on the equipment or its' connections. All expansion devices shall be carefully erected in full accordance with the manufacturer's recommendation and instructions and be approved by the manufacturer for the particular application concerned.
- 14.7 No system of expansion control shall be accepted where the closure of movement exceeds the amount recommended by the manufacturer when operating from the cold to the upper limit working temperature. The minimum temperature difference for calculation purpose shall be 40°C for interior pipes, and 50°C for exposed or external runs.
- 14.8 All expansion devices shall be cold drawn by a distance equal to half the total expansion and all contraction devices shall be cold compressed by a distance equal to half the total contraction. Cold draw shall not be applied until the anchor installations have been completed and approved by the Consultant.
- 14.9 All anchor points shall be treated as main anchor points, the practice of utilizing a less substantial anchor for "intermediate positions" shall not be permitted.
- 14.10 Where the installation is required to be tested in sections, extra anchor points shall be installed where necessary, for the protection of the expansion devices.
- 14.11 Pipework between anchors shall wherever possible be straight, but where this is not possible, the bracing of all guides shall be adequate to overcome the resultant turning moment produced by the offset forces, but the Consultant shall be informed before the work commences.
- 14.12 Each axial expansion device shall be guided as close as possible on both sides of the joint, but in no case more than 1 1/2 diameters away. A further set of guides shall be installed at a distance of not more than 15 diameters away from the axial expansion device unless the manufacturer imposes more stringent requirements.
- 14.13 The positioning of anchors and guides immediately adjacent to angular expansion devices shall be in accordance with the manufacturer's recommendation for the application concerned and shall be submitted to the Consultant for approval before installation commences. The remainder of the pipework shall be set in guides at a maximum of twice the maximum interval for support spacing's specified elsewhere for the pipe size to which the expansion device is fitted.

15.0 WATER SYSTEM TREATMENT (CLEANING)

- 15.1 The chilled water system shall be prepared as detailed elsewhere herein. The system will then be further treated.
- 15.2 Prior to testing and commissioning, the consultant shall be provided with water conditioning programme, to control water quality. Cleaning shall be supervised, and programme of conditioning administered by the conditioning company.

- 15.3 For Pre-commissioning cleaning, the system shall be dosed with the prescribed amount of the non-acid cleaning agent and a surfactant as advised by the chemical company.
- 15.4 Cleaner shall be run into the system and retained for a period of 72 hrs. or more as advised by the chemical company. At the end of this period the system shall be drained and flushed with clean water.
- 15.5 All strainers and trapping points shall be inspected, and any debris removed.
- 15.6 System shall be refilled with clean water, re-circulated or run through for a further four hours, and again drained and flushed. The flushing shall continue till the effluent is clear, colourless, odorless, free from suspended solids and such that the iron level is not more than 1 ppm and the conductivity is no more than 30 $\mu\text{S}/\text{cm}$ greater than the makeup water. Permanent pumps shall not be used for flushing.
- 15.7 Upon completion of flushing all strainers shall be inspected and any debris removed.
- 15.8 The system shall be refilled with clean water and a specified amount of inhibitor added.
- 15.9 The Chemical Company representative shall conduct an analysis of the system's water after the cleaning operation and shall submit a report to the Consultant.

Note: The Mechanical Contractor shall advise on the holding capacity of the systems to be cleaned to the Chemical Company, in order that the amount of cleaner required may be calculated.

16.0 START UP CHEMICALS

- 16.1 The supply of all chemicals for system startup shall be the scope of this contract. Chemicals sufficient for contractual maintenance period (where specified) shall be arranged by the Contractor.

17.0 BTU METERS

- 17.1 The energy meter shall be made up of an Ultrasonic flow meter, two temperature sensors, energy calculator, and serial communication ports. Each Meter shall be equipped with Modbus module that transmits data through Modbus network back to the central Modbus master, on which all meters' data is stored in central work station which can be used to print the meter readings and prepare custom made bills for consumption.
- 17.2 The BTU energy meter shall have the following specifications:
- A. Ultrasonic meter type, with calibration certificates required for each meter.
 - B. Battery powered calculator.
 - C. Modbus transmission signal to/from the meter.
 - D. A proofed/guaranteed 10+ year's battery life.
 - E. Relevant for horizontal and vertical installations.
 - F. Stainless steel temperature sensors.
 - G. Stainless steel sensors pockets.
 - H. Battery and temperature sensors shall be replaceable without the need to re-calibrate or replace the entire meter.
 - I. Battery life shall be 10 years.
 - J. 16 bar Pressure rate for water side components.
 - K. Integrated Data memory and loggers
 - L. LCD display.
 - M. Set/reset function
 - N. High accuracy.
 - O. ModBus master shall be able to automatically identify the meters by searching the network.
 - P. ModBus master shall be fast, standardized, up to date communication interfaces, built-in

- memory & ability to read and configure remotely.
- Q. ModBus Master shall be able to connect more devices or to build up larger Modbus networks.
- R. PC workstation with laser printer and billing software with license.

17.3 Installation of BTU Meters:

The installation of chilled water meters involves the following activities:

To install Ultrasonic BTUs as shown on the drawings and provide the necessary provision for the meter installation which consists of:

- Reducers to match the Meter sizes
- Isolation Valves
- Necessary plumbing accessories, if any required

Meters supplier shall supervise installation and provide necessary guidance to installation contractor engineers/technicians to enable them to install the meters as per installation manuals.

- 17.4 On primary side of ETS Room Electromagnetic Type Stainless steel tube/stainless steel conical raised electrodes, electromagnetic induction type meter, producing AC signal proportional to liquid flow. Magnetic flow to current converter shall be of the solid-state feedback type. The integrator can be remotely mounted and should have communication over the communication bus capability, and complete with LCD display. Accuracy and repeatability shall be as per BS EN 1434. Assembly shall be complete with integrator mounting accessories and three meters' connection cable. Hard wiring for all flow meters is to be done. The accuracy of the flow transmitter shall be +/-0.5% of calibrated span. The repeatability of flow transmitter shall be 0.25% of calibrated span. It shall detect zero flow. All circuits for input, output and power supply shall be galvanically isolated from each other.

18.0 VIBRATION ISOLATION

See Ductwork Specification.

19.0 PRESSURE TESTS

See relevant sections of the General requirements.

20.0 WELDING TESTS

See relevant sections of the General requirements.

21.0 SUBMITTALS

- 21.1 Technical submissions are to include the schematic for components actually to be installed showing flow rates and accompanied by pressure drop calculation for the system.

- 21.2 Technical brochures and information on all components: -

Pipework,
Pipe fittings.
Pipe supports.
Valves.
Strainers.
Gauges and accessories.
Air vents.
Flow measuring instruments.
Expansion joints.

Anchor points.
Test forms.
Insulation.
Water treatment

21.3 In the event that the Consultant requires further information on any item, samples must be provided as necessary.

22.0 **AIR SEPARATOR** (Where Applicable)

Air separator shall be constructed of steel designed and fabricated per ASME Section VIII Division I with a maximum working pressure rating of 250 PSIG at 375°F units up to 65 mm connection sizes shall have threaded connections and units 75mm and above connection sizes shall have flanged connections. Units shall have separate top connections for system expansion tank and air venting, and bottom connection for blow down cleaning. Air separator shall be furnished with a removal stainless steel strainer with a free area not less than five times the cross-sectional area of the connection's pipes.

END OF SECTION

MECHANICAL SECTION: 04

PLUMBING WORKS

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1.0 SERVICES DESCRIPTION**1.1 System Design**

The plumbing system shall be designed in accordance with the recommendations of the Chartered Institution of Building Services Engineers and to the requirements of the Local Authorities. The system shall be installed as detailed in the drawings.

2.0 PIPEWORK INSTALLATION: GENERAL

2.1 Pipe materials shall be as specified elsewhere herein. The Contractor shall install unions or pairs of flanges in the pipework system at the following locations: -

- (a) To enable the disconnection of all items of plant, equipment, pumps and valves for maintenance purposes.
- (b) To enable sections of pipework to be subsequently dismantled and fixed without disturbing other pipework, plant, equipment and building fabric.
- (c) Where indicated on the drawings.

2.2 Pipework shall be run neatly, parallel to walls or grid lines on plan and shall, where necessary, set around columns and projections. All vertical pipe shall be plumb. Direction changes in pipe groups shall be set out from a common centre point.

2.3 All screwed and flanged joints shall be easily accessible.

2.4 Where pipes are fitted around columns or piers, the direction change shall be made using short sets and shall be supported from the floor on the centre line of the column. (floor support only applicable in the mechanical /parking area)

2.5 It shall be the responsibility of the Contractor to ensure that the pipework systems are at least 150 mm from any electrical conduit, cable or wiring.

2.6 Pipes entering or leaving trenches or ducts, shall be square to them. All pipework shall be fixed in such a manner as to be readily accessible for tools, welding operations or removable without dismantling adjacent pipework.

2.7 Where pipework passes through walls, etc., the Contractor shall ensure that these pipes are not subsequently bedded-in by making good by other trades.

2.8 Pipework passing through brick or concrete shall be enclosed in a PVC pipe sleeve of such diameter that the pipe maybe removed without cutting or deforming the pipework. In particular, where plastic pipes pass through a fire rated wall, fire arresting sleeves shall be used with a packing of intumescent strip between the pipe and sleeve.

2.9 Pipework passing through internal or external dry sheet construction walls shall pass through formed holes of such size that the pipework may be removed without cutting or deforming the pipe.

2.10 All holes through such walls shall be closed after the erection and testing of the pipework by means of removable split closing plates on both sides of the wall. All closing plates shall be manufactured from galvanized sheet steel of not less than 16g (Gauge) thickness in the case of brick or concrete walls. All closing plates shall be painted or finished in accordance with the requirements of this specification.

3.0 STORAGE AND PROTECTION

- 3.1 Generally, storage shall comply with manufacturer's recommendations.
- 3.2 The Contractor shall ensure that all pipework and fittings delivered to the site for inclusion in the Works are stored in an approved manner to avoid deterioration due to accidental damage and atmospheric conditions.
- 3.3 The plastic pipework components should, at all times, be stored in a location offering protection from direct sunlight, i.e. from ultraviolet light and the associated heating effect, it being remembered that materials close to the ground will be heated by reflection in sandy locations.

- a) The storage site should be flat, level and free from sharp stones, flints etc.
- b) Pipes should not be stacked to heights exceeding the following:

Pipe size inch/mm	Max. Stacking Height
Up to 3 in or 90 mm	10 x pipe size
4 in-6 in or 110-160 mm	6 x pipe size
8 in or 200 mm or greater	4 x pipe size

N.B: - Smaller pipes should only be nested inside larger pipes which are adjacent to the ground.

- c) Plastic pipes should be protected from direct sunlight and heat of the sun by providing shaded areas.
- 3.4 Any part of the pipework installation which, in the opinion of the Engineer, is sub-standard, unduly distorted, marked by tools or in a deteriorated condition shall be removed and replaced at the expense of the Contractor.
- 3.5 Before final erection, all pipework, shall be free of scale, rust, Gas or other forms of corrosion. Such defects shall be removed, and the pipework made good in a manner approved by the Engineer. Any stoppage or restriction occurring after the pipework systems have been put into operation which is attributable to neglect by the Contractor shall be rectified by the Contractor at his own expense.
- 3.6 The open ends of all pipes prepared for welding, shall be protected by plastic plugs. Plugs of paper, cardboard, cotton waste, hessian or similar material will not be permitted.
- 3.7 Open ends of all pipes intended for threading shall be provided with either a protective cap or screwed socket to protect the threaded ends of pipe in addition to the plastic plug.
- 3.8 Plastic Pipes shall be protected from direct sunlight. Where pipes are exposed to sunlight, it should be painted by a Titanium Dioxide based industrial quality white paint or as recommended by the pipe manufacturer for UV resistant paint. Pipes shall be lightly abraded, cleaned, wiped with white spirit and painted with white under coat and followed by white gloss paint.
Chemical cleaner and solvent cement shall be kept in a ventilated and air-conditioned store.

4.0 GRADIENT AND DRAINAGE OF PIPELINES

- 4.1 All pipework shall be erected in such a manner that air may be vented and liquids drained from the whole or isolated sections of the system. Except where specified elsewhere herein, 32mm diameter key-operated cocks with hose unions shall be fitted at all low points of water systems to ensure complete drainage.

- 4.2 Pipelines shall be graded to fall in the direction of flow unless indicated otherwise on the drawings or elsewhere herein.
- 4.3 At low points in water pipework systems, all vessels and on the dead side of each pair of isolating valves in the distribution mains, drain points are to be provided in accessible positions. These shall be fitted with drain cocks of a type specified elsewhere herein.

5.0 PIPEWORK SUPPORTS

- 5.1 All pipework shall be adequately supported. All support installations shall be in accordance with relevant British Standard Specifications and Codes of Practice, except where modified or extended by the contract documents.
- 5.2 Supports shall allow free movement for expansion or contraction of pipework and shall be located to ensure that pipework branches or fittings are not fouled by the support during expansion or contraction of the pipework service.
- 5.3 Double banking of pipework from a single support position will be permitted, provided the normal operating temperature of the fluids in the two pipes do not differ by more than 30 degrees C., but only where space restrictions prohibit individual support. Triple banking will not be permitted.
- 5.4 Where double banking is necessary, the larger of the two pipes shall be uppermost, and where pipes are the same size but manufactured from different materials, then the pipe having the material with the lowest coefficient of expansion shall be uppermost. Support intervals for double-banked pipework of different sizes shall relate to the smaller size.
- 5.5 Vertical rising pipes shall be supported at the base and the support shall withstand the total weight of the pipe and fluid contained.
- 5.6 Supports shall not be permitted which clamp the pipe so that it is in contact with building fabric or structure.
- 5.7 All supports shall be specifically designed for the outside diameter of the pipe concerned (including specified packings). Oversized brackets will be rejected.
- 5.8 Where non-ferrous pipework is to be supported using ferrous pipe clamps or rings, an approved plastic coating applied to the clamp or ring shall be used to prevent contact between the ferrous and non-ferrous surfaces. Where cast iron rollers are to be used in contact with copper pipework, these shall be copper-plated.
- 5.9 Pipework shall be supported individually by hangers consisting of rubber lined malleable split rings with malleable iron sockets or steel clevis type hangers.
- 5.10 Pipework shall be supported as recommended by the manufacturer as required in the specification and in particular on both sides of all changes in direction. Pipework shall be supported such that no load is imposed upon plant machinery. Pipework shall be supported at intervals not exceeding the following:
- 5.11 , Steel or heavy gauge copper pipework.
- | | | |
|--------|---|---------|
| (15mm) | - | (1.8 m) |
| (20mm) | - | (2.5 m) |
| (25mm) | - | (2.75m) |
| (32mm) | - | (2.75m) |
| (40mm) | - | (3.0 m) |
| (50mm) | - | (3.0 m) |
| (65mm) | - | (3.6 m) |

- (80mm) - (3.6 m)
 (100mm) - (4.5 m)

- 150mm and above at intervals not exceeding the recommendations of BS 3974 Part 1.
 5.12 light gauge copper Plastic water pipework .

- | | | | | | |
|--------------|---|---------|------------|---|----------|
| (15mm) | - | (1.2 m) | (15 to 50) | - | (1.05 m) |
| (20mm) | - | (1.5 m) | (65) | - | (1.02 m) |
| (25mm) | - | (1.8 m) | (80) | - | (1.35 m) |
| (32mm) | - | (2.2 m) | (100) | - | (1.75 m) |
| (40mm) | - | (2.5 m) | (150) | - | (2.00 m) |
| (50mm) | - | (2.7 m) | (200) | - | (2.04 m) |
| (80mm) | - | (2.7 m) | | | |
| (100 & over) | - | (3.0 m) | | | |

All pipe supports shall be manufactured in accordance with BS. 3974 Part 1.

6.0 PIPE EXPANSION

- 6.1 Pipework shall be supported in such a manner as to permit free movement due to expansion and contraction. Pipework supports shall be arranged as near as possible to joints and changes in direction and each support shall take its share of the load. The spacing of the supports shall not exceed the intervals given elsewhere herein. Where there are two or more pipes the support spacing may be based on the centres required by the smallest bore pipework.
- 6.2 Vertical rising pipework shall be supported at the base or as indicated to carry the total weight of the riser. Branches from risers shall not be used as a means of support for the riser.
- 6.3 Provision for movement due to expansion and contraction shall be made either by loops, special expansion joints or by changes in direction of the pipework. Supports at such points shall be arranged to ensure that all movement is taken up by the loop, joint or change in direction of the pipework. Contractor shall submit calculations for pipes thermal expansion for engineer's approval.

7.0 AIR VENTING

- 7.1 Automatic air eliminators shall be provided at all high points in water pipework system. They shall be installed at the highest points of the sections they are intended to vent.

Insect screen shall be provided on all open vents and overflows.

- 7.2 Air vents on water systems shall be of float type, air eliminator of approved manufacture, having bottom inlet and top outlet with screwed connections. They shall have a stainless steel float valve and valve seat, incorporate a stainless steel check valve and shall be so designed as to allow the internal parts to be removed for maintenance or inspection without disturbing pipework. Air vents shall be suitable for the system pressures and temperatures. An air release pipe shall be connected to the air outlet and taken to the nearest suitable discharge point.

8.0 WATER HAMMER ARRESTOR (Where Applicable)

- 8.1 Each cold / hot water end circuit shall have automatic water hammer arrestor. arrestor shall have NPT solid hex brass adapter for easy installation. It can be installed with a standard pipe tee vertically, horizontally or at any angle. The water hammer arrestor shall be Factory air charged and permanently capped and solder sealed. PDI Listed Air pre-load is 60psi (414 kPa)
- 8.2 Contractor is responsible for the sizing and location of Water Hammer arrestors required for the project.

9.0 PUDDLE FLANGES

- 9.1 Where pipework passes through the external walls of the buildings or trenches below ground level, the Contractor shall supply and cast or build puddle flanges into the structure. Link seal type puddle flange shall be used for below ground external pipe work passing from outside to inside the building.
- 9.2 Puddle flanges are to be manufactured from the same material as the pipework of which they form a part.
- 9.3 Each puddle flange shall comprise a length of pipe, flanged or screwed at each according to diameter with an undrilled slip-on flange welded on the outside at a point where it will be located mid-way in the thickness of the wall. The puddle flange is to be painted externally with two coats of bituminous paint before being built into the structure.

10.0 PIPEWORK MATERIAL

- 10.1 Main incoming cold water pipe shall be in uPVC Class E to BS 3505 (BS EN 1452).
- 10.2 All above ground horizontal cold water distribution pipe(after PRV), hot water piping and fittings except those embedded in builder's work shall be PPR pipes to DIN – 8077, SDR 7.4 and fittings to DIN - 16962, PN 20. Refer clause 12.0 for specification.
- 10.3 All the vertical drops of cold and hot water pipes embedded in builder's work shall be in PEX pipes to DIN 16892 & 16893 with DVGW / WRAS approval. PEX pipes shall run within 25 mm corrugated PVC conduit of red or blue color depending on pipe service (hot or cold water). Refer clause 11.0 for specifications.
- 10.4 The cold water pipe riser material (water transfer pipe and gravity distribution pipe up to floor branch PRV) shall be as per below table. Refer other clauses for detailed specification of each material.

Item Description	Remarks	Pipe Material
Cold water transfer pipe riser	Up to 9 bar pressure	PPR to DIN 8077, SDR 6 and fittings to DIN 16962 PN25
Cold water transfer pipe riser	For 9 – 12 bars pressure	PPR-CT PN25
Cold water transfer pipe riser	Above 12 bars pressure	Hard Copper to BS EN 1057 table X water tube drawn temper, with copper solder joint fittings to BS EN 1254.
Cold water Gravity Pipe riser	Up to 100 mm diameter	PPR to DIN 8077 / SDR 6 and fittings to DIN 16962 PN25
Cold water Gravity Pipe riser – Entire riser (up to floor branch PRV)	Above 100 mm diameter	HDPE - SDR9
Heating water pipework between boilers and indirect water heaters / pool plate heat exchangers	All pipe sizes	Black steel, ERW welded, medium weight to ASTM A53 SCH 40.

- 10.5 The HDPE pipes and fittings shall be High density polyethylene (PE100) with a pressure rating of SDR9 (20 bar) at 20 °C.
The material shall conform to ISO 161-1, ISO 3607, ISO 4427, ISO 8770, ISO 8772, prEN 12201, prEN 13224, DIN 8075, DIN 19535, DIN 19537, WIS 4.32.13.
The pipe shall have WRAS / DVGW approval.
The pipe shall have a density of 0.95 g/cm³ and shall be UV stabilized.
The Ends shall be Plain for electro-fusion welding with a protected wire without medium contact.

- 10.6 All valves, fittings and accessories in contact with potable water shall be certified safe for drinking water by international certifying committee like WRAS, DVGW, NFS or equivalent.
- 10.7 Heating Water Service Pipework.

All pipework for the Heating water between boilers and the indirect water heaters and plate heat exchangers shall be black steel, ERW welded, medium weight to ASTM A53 SCH 40.

10.7.1 Heating Water Fittings

A. For pipes diameter 65 mm and larger

1. Black steel to ANSI B16.9, ANSI B16.28 butt-welding ends.
2. Black steel with grooved ends suitable for mechanical jointing.

B. For pipes diameter 50 mm and smaller

1. Malleable iron to ANSI B16.3, BS 143 & BS 1256 Equivalent to DIN EN 10242 & ISO 49 with threads ends. threads to confirm to BS EN 10226 (ISO R 7 / 1)
2. Welding flanges for pipe 65 mm and larger: Steel to ANSI B16.5.
3. Unions for pipe 50 mm and smaller

Malleable iron ASTM A1 97, Class 150, with bronze to iron ground joint, ANSI B16.3.

Dielectric unions shall be used in case of joining dissimilar material,

4. Flange Accessories:

Gasket, canvas reinforced neoprene to BS 1737.

Bolts, square head machine with hexagonal nut, galvanized steel to BS 916.

5. Use "Y" pattern strainers, PN 16, with screens of type 316 stainless steel or Monel with approximately 0.79 mm perforations. Provide a valve blow-off connection in all caps 65 mm and larger. Maximum pressure drop with clean screen shall not exceed 6.89 kPa, with water velocity through inlet pipe of 1.5 m/s.

10.7.2 Heating Water Pipes installation

For the pipe installation refer the specifications in section 03 clause 2.0 to 4.0

11.0 PEX PIPES

11.1 General

Cold and hot water distribution pipework embedded within builder's work inside walls of bathrooms, kitchen shall be composed of:

- Crosslinked polyethylene pipe according to DIN 16892 and 16893 pulled in a corrugated conduit of ordinary polyethylene.
- Fittings and valves of bronze construction/ DZR

11.2 Pipes

11.2.1 The inner pipe shall be a cross link polyethylene pipe withstanding up to 75 °C at a max. pressure of

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10 bar without deformation or damage. Short time temperature loading up to 95°C at 10 bar.

11.2.2 The pipe shall be resistant to all natural constituents of drinking water to DIN 2000, disinfectants and cleaning agents to DVWG – W291 and DIN 2000, anti-corrosion agents to DIN 1988 T4 (E) and common building materials, such as mortar, cement and plaster.

11.2.3 The pipe manufacturer shall offer minimum of 50 years guarantee for the pipes and 10 years guarantee for the complete system.

11.2.4 All pipes shall be stamped on equal intervals showing clearly the name of the manufacturers along with the pressure and temperature rating of these pipes.

11.2.5 All pipes shall be laid so that the 16mm pex water pipe can be replaced, if necessary.

11.3 Fittings & Accessories

11.3.1 Pipes shall be connected to the different fittings via a wall box of reinforced plastic fitted with DZR brass elbow suitable for 15mm or 20mm threaded connection. The box should be suitably designed in order to enable the replacement of existing pipes in the event of their damage.

11.3.2 All valves, box elbows, tees, bends shall be of DZR brass material. Connection shall be of a cone grip unions type allowing full flow capacity, minimum pressure loss, easily detachable with torque clearly defined for a fool proof installation. Contractor shall use proper tools for assembly as recommended by manufacturer, i.e. assembly pliers, Ratchet torque wrench, cutters, etc.

11.3.3 All fittings shall be WRAS or DVGW approved.

11.3.4 Manufacturer approved union to be used connect PEX from Main horizontal pipe at ceiling level.

12.0 POLYPROPYLENE (PPR) PIPES & FITTINGS

PP-R (POLYPROPYLENE-RANDOM POLYMER)

12.1 PPR Pipes shall be SDR 7.4 P.P.R. Green color,

12.2 SDR 7.4 P.P.R. pipes shall withstand a maximum constant working pressure 20.4 bars (296 psi) at 20°C with a lifetime expectancy of 50 years.

12.3 Reserved.

12.4 PPR Pipes shall be manufactured in accordance with

- WRAS (Water Regulations Advisory Scheme), U.K.
- NSF 61 & 14 (National Sanitation Foundation), U.S.A.
- DVGW (German Association for Gas and Water) – Germany
- ASTM F 2389
- DIN ISO 15874
- DIN-8077 – 8078
- BS 6920
- DIN 16962

12.5 PPR fittings shall be manufactured in accordance with DIN-16962 and shall be manufactured by ISO Certified Company.

Only Injection molded fittings will be accepted.

- 12.6 The polymer used for the manufacturing of the products must be approved by Worldwide Known Quality Body Certifier such as, DVGW, SKZ, ISO and NSF Standard 61 (Cold & Hot 180 F/ 82 C).
- 12.7 The pipes & fittings used for conveying potable water or likely to come into contact with potable water shall not constitute a toxic hazard, shall not support microbial growth and shall not give rise to unpleasant taste, or odor, cloudiness or discoloration of water.
- 12.8 PPR shall be virgin (no mix batching).
- 12.9 The polymer used for the manufacturing of the products must contain metal deactivators to reduce the iron ions in the water to prevent corrosions.
- 12.10 PPR pipes. Below table are for indicative only with minimum outer diameter of the PPR pipe.

PPR PIPE SIZE - SDR 6		
PPR Pipe size shown in the drawing (Inner diameter) "mm"	PPR pipe size (outside diameter) "mm"	Minimum wall thickness "mm"
15	25	4.2
20	32	5.4
25	40	6.7
32	50	8.3
40	63	10.5
50	75	12.5
63	90	15
75	110	18.3
80	125	20.6
110	160	26.6

PPR PIPE SIZE - SDR 7.4		
PPR Pipe size shown in the drawing (Inner diameter) "mm"	PPR pipe size (outside diameter) "mm"	Minimum wall thickness "mm"
15	25	3.5
20	32	4.4
25	40	5.5
32	50	6.9
40	63	8.6
50	75	10.3
63	90	12.3
75	110	15.1
80	125	17.1

110	160	21.9
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Criteria shown above are indicative. PN-20 or suitable higher rating shall be adopted depending on building height, system operating and test pressures at the derated temperature condition.

12.11 FITTINGS

12.11.1 All PPR fittings shall be from the same manufacturer.

12.11.2 Fitting shall be one of the following; poly fusion welding or electro-fusion welding and comply with the International Standard requirements.

12.11.3 Fittings for PPR shall be approved to DIN 16962 or equivalent.

12.11.4 Fitting metal bodies should be made of brass material by using dezincification resistance brass.

13.0 MULTI-LAYER POLYPROPYLENE (PPR-CT) PIPES & FITTINGS

PP-RCT (POLYPROPYLENE RANDOM CRYSTALLINE STRUCTURE TEMPERATURE)

13.1 PP-RCT Pipes and fittings shall be manufactured in accordance with

- ASTM F 2389
- DIN ISO 15874
- DIN-8077 – 8078
- DVS 2207-2208
- DIN EN 10226

13.2 Beta Nucleated ready-made colored PP R Crystallinity Temperature compound or equal can be used.

13.3 Service life of the materials shall meet the testing requirements as set in EN ISO 15874.

13.4 All metal inserts shall be from a corrosion resistant, lead free DZR brass material. DZR brass shall be third party tested and test reports shall be provided from international certified laboratories.

13.5 Pipes shall be manufactured in accordance with DIN 8077/8078 and DIN EN ISO 15874.

13.6 The pipe inner diameter and all pressure data and safety factors shall be declared along the DIN EN standards regarding the mentioned material classes. The PP-RCT pipes pressure rating shall be PN25.

13.7 The pipes shall have DVGW and WRAS/ NSF certification.

13.8 The pipes installed exposed to sun shall be UV-protected.

13.9 Fittings shall be manufactured in accordance with DIN EN ISO 15874. All fittings shall be injection molded.

13.10 PP-RCT pipes and fittings shall be joint with socket fusion or electro-fusion welding.

14.0 VALVES

- 14.1 Valves shall be provided to isolate each item of plant and each main branch.
- 14.2 The Contractor shall make all reasonable attempts to ensure the manufacture of valves is consistent throughout the whole of the installation in order to minimize stock holding of spares.
- 14.3 Valves and cocks of any tank or vessel not within a secure building shall be lockable.
- 14.4 Any valves and cocks exposed to public view, shall be of the easy clean pattern and have a polished finish.
- 14.5 Valves in plant room, tank rooms etc., shall be of the hand wheel pattern or lever type
- 14.6 All valves shall be located so as to provide easy access for operation and servicing. Control valves shall normally be installed in horizontal lines with stems vertical.
- 14.7 Each such regulating valve shall be fitted with a regulation indicator calibrated in tenths of one revolution of the hand wheel.
- 14.8 Globe valves shall be used on service pipelines where regulation is required and shall be supplied and fitted in positions indicated on the Contract Drawings. Valves shall be of the same nominal bore as the pipeline in which they are indicated
- 14.9 Globe and oblique valves used for circuit regulation shall have characterized plug discs.
- 14.10 Hand wheels / lever shall be designed for operation at a temperature suitable for safe handling. They shall be of the open wheel type manufactured from malleable iron or die cast aluminium alloy. Handles shall have 'open/shut' direction indication. Bronze valves shall be Die cast Aluminum handwheel and for Cast Iron valves, Cast Iron / Malleable iron Hand wheel.
- 14.11 Provide cold water supply to water heaters with check valves.
- 14.12 Provide angle valves for incoming water supply to all fittings.
- 14.13 Angle valves shall be marked with red and blue tags for hot/cold water respectively.
- 14.14 Globe valves up to 50mm diameter shall have threaded ends with bronze/DZR body and trim renewable dynamic disk, screwed bonnet suitable for re-packing under pressure. Globe valves 65mm and larger shall be cast iron body, bronze Trim, Flanged ends outside screw, renewable seat and disc.
- 14.15 Gate valves up to 50mm diameter shall be bronze/DZR body with threaded ends bronze wedge disc, high quality packing and gland, non-rising stem with screwed bonnet. Aluminum Die cast hand wheel.
- Gate valves 65mm diameter and larger shall be butterfly valve type with flanged cast iron body, non-rising stem, bronze trim wedge disc type complete with hand wheels, bolted bonnet and stuffing box.
- Butterfly valves for pipe sizes 150 mm and above shall be fully lugged type.
- Fully lugged Ductile Iron body, Stainless steel disc and stem, EPDM Seat.
- 14.16 Check valves up to 50mm diameter shall be threaded bronze pattern swing type with renewable disc and screw-in cap.
- Check Valves 65mm diameter and above shall be flanged cast iron swing type with renewable seat and disc components.

- 14.17 Pressure reducing valves (PRV) shall be pilot controlled, hydraulically operated, diaphragm type with a low by-pass capability. The low-flow by-pass capability shall be achieved by using a balanced direct acting PRV as an integral part of the main valve. At very low flows when the main valve is almost completely closed, to prevent the possibility of cavitation the direct acting valve shall by-pass the main valve and maintain flow.

PRVs up to 50 mm shall be direct acting PRVs and shall be of bronze construction including the trim and above 50 mm shall be cast iron. The pressure reducing valves shall be suitable for maximum working pressure that exist within the system and downstream pressure should be site adjustable between 2 and 4 bar. Pressure gauges shall be fitted at inlet and outlet of PRV

- 14.18 Float Valves shall be bronze equilibrium float valves, 80mm diameter and above shall be flanged end, flat faced and drilled to suit BS EN 1092 PN16. Float valves up to 50mm diameter shall be screwed end BS EN ISO 228 parallel and shall be provided complete with back nut.
- 14.19 Bib Taps: Bib-cocks shall be in accordance with BS 1010. They shall be lock shield type provided with hose union nosepiece and hand wheel operated.
- 14.21 Valves shall always be installed in accessible locations to permit easy operation and maintenance.
- 14.22 Special care must be exercised in the installation of screwed valves to avoid straining their bodies and preventing the gate or seat from closing tight. The wrench shall always be applied to the side being attached to the valve. When attaching pipe to a valve already in place, a second wrench shall be used to hold the valve while the pipe is being tightened. A pipe shall not be screwed so far into a valve as to damage the seat.
- 14.23 Unless otherwise indicated, valves on systems with working pressure up to 12bar shall be rated for 16 Bar. Valves on systems with working pressure higher than 12bar shall be rated for 20bar. Valves rated for 20bar shall be cast steel construction or approved equal.
- 14.24 All cast iron valves shall be coated with epoxy internally and externally to average DFT 300 microns for protection against corrosion of body components.

Heavy valves shall be supported separately in such a way that no weight is exerted to piping components.

15.0 STRAINERS

- 15.1 The Contractor shall install strainers in the pipeline before the following items of equipment: -
- (a) Solenoid valves.
 - (b) Automatically controlled valves.
 - (c) Pumps.
- 15.2 Strainers shall be suitable for withstanding the maximum operating temperature and pressure of the pipeline in which they are installed.
- 15.3 Install strainers in pipeline before all equipment as recommended by equipment manufacturers.
- 15.4 Strainers

Strainers size up to 50mm diameter shall be bronze body with threaded ends and open mesh stainless steel screen.

The strainers size 65mm diameter and larger shall be of the straight Y- type. The strainers shall be complete with cast iron / steel body and cap steel screen rods and standard perforated stainless steel screens.

16.0 JOINTING OF PIPES

- 16.1 uPVC pipe jointing sequence shall be carried out as per the written recommendations of the manufacturer and it must be stated that any omission or variation from the jointing procedure will result in a marked reduction in the efficiency of the joint. It is essential in the jointing process that only cleaner and solvent cement recommended by the manufacturer is used to make joints.
- 16.2 When making a joint in high ambient temperature conditions the following should be observed: -
- (a) Store the cement in a cool shady place.
 - (b) When possible, protect the pipe which is about to be installed from direct sunlight.
- 16.3 Copper Pipe Work shall be joined by brazed joints. Joints to sanitary fittings, valves etc. Dielectric joint with neoprene or similar insulator shall be fitted at junctions between copper and galvanized mild steel. Direct joints between copper and steel will not be permitted.
- 16.4 Ferrous pipes shall be screwed or welded. All welds shall be in accordance with the relevant Piping Design Code given elsewhere herein. Methods, other than those covered by the standards, maybe used provided that their suitability is demonstrated and that their use is the subject of agreement with the Engineer. Welder qualification with welding trade certificate to be submitted for engineer approval.

17.0 NOT USED**18.0 NOT USED****19.0 ELECTRIC WATER HEATER**

Provide and install in the positions indicated in the drawings electric water heaters.

The heaters shall be of welded steel construction with corrosion resistant inner lining having guaranteed minimum life of 5 years. Contractor & Manufacturer to provide 2 years warranty for the complete system.

An outer casing of enameled steel shall be provided, with a minimum of 25mm of cork, glass fiber or foamed plastic insulation between water chamber and outer casing. The water chambers shall be tested to 8 bar and the water inlet shall be fitted with a safety unit comprising stopcock, non-return drain cock and relief valve. The relief valve discharge shall be taken to drain or to atmosphere. Storage water heater shall be fitted with a dial type thermometer and adjustable thermostat.

The heating elements shall be suitable for 230 volts' single phase 50 Hz supply.
Magnesium anode shall be provided.

The required capacities are shown on the drawings.

20.0 PACKAGED BOOSTER PUMP SETS

- 20.1 The booster pump set shall be a factory manufactured works tested.
- 20.2 The pump set shall incorporate centrifugal pumps with one as duty pump and the other as standby. Each pump shall have a duty as scheduled.

- 20.3 Unless otherwise stated the pumps will be of horizontal multistage, inline construction with direct-coupled motors and formal WRAS/DVGW approval for use on potable water systems (including valve and accessories and pressure gauge).
- 20.4 The pumps will have stainless steel bodies with stainless steel shaft and impellers and fitted with tungsten carbide faced mechanical seals.
- 20.5 The pump motors are to be of squirrel-cage, aluminium construction with totally enclosed fan cooling. The motors shall have class F insulation and IP55 protection, with performance in line with the requirements of IEC34-1. EFF2 motors are to be fitted as standard with the option of high efficiency EFF1 motors over 0.75kW.
- 20.6 The control panel shall be complete with star-delta starters, door interlocked isolator, indicator lights, fuses, main isolator, duty select switch, test pushbutton, adjustable 0-6minute timer for each duty pump, anti-condensation heater a cable numbering, all in steel enclosure IP 65.
- The control panel shall have LED screen to reflect all parameters: Cut off & cut in pressure, water flow, motor frequency (Hz), current, etc.
- 20.7 The diaphragm pressure vessel shall be constructed of a steel shell with polypropylene lining on internal wetted areas and a butyl rubber diaphragm.
- 20.8 All interconnecting pipe work shall be copper to BS EN 1057 Table X/ with a gate valve on the suction and discharge of each pump and a silent check valve on the discharge branches of each pump.
- 20.9 A pressure switch for each of the pumps and a system pressure gauge should be mounted on a common Aluminium Pressure Manifold to eliminate using small bore fittings, to ensure easy replacement of pressure switch or gauge, and to ensure that the pump set arrangement is compact and has only one common pressure signal source.
- 20.10 Demand shall be met in the first instance from the pressure vessel. When the vessel pressure has fallen to a pre-set minimum (the "cut-in" pressure) the first duty pump shall start. In the event of the failure of the duty pump the standby pump shall automatically take its place.
- As demand reduces again and the pressure builds up to a pre-determined cut-out pressure, the pump will stop. An adjustable timer for each pump shall be fitted in the control panel to ensure frequent switching cannot occur.
- 20.11 Local assembly for the pumps shall be acceptable subject to the following:
- The assembly shall be carried out by the authorized local agent.
 - All QA/QC procedures, performance or any applicable tests to be considered during assembly process same as adopted in the main factory and accordingly all the related certificates to be provided.
- 20.12 After final assembly of the complete packaged unit, the automatic controls shall be fully tested in the workshop to simulate site operating conditions and piping tested to the operating conditions. The unit shall be cleaned down and painted one undercoat and one coat of protection paint.

21.0 WATER TRANSFER PUMPSET

The unit shall be a packaged assembled and tested unit complete with one duty/ one assist/ one (unless otherwise indicated in the equipment schedule) stand-by pump, interconnecting pipework in copper/bronze valves, safety devices and prewired control panel.

Pump set shall in general comply with the requirements of clause 20.0.
Control of pumps shall be as follows:

- Pumps switching shall be controlled by float switches installed in the roof storage tank.
- A low level float switch installed in the underground storage tank shall prevent dry running of pumps.

22.0 PACKAGED WATER PUMPSET – VARIABLE SPEED

22.1 General

- 22.1.1 The pumping unit shall be a variable speed system. The pumping system shall include, as a minimum, the programmable pump logic controller, adjustable frequency drive, remote sensor / transmitters, motors and pumps.

The pump manufacturer shall be responsible for the successful operation of the system supplied by him.

- 22.1.2 The manufacturer of the packaged pumping system shall provide a minimum of one day's training at the job site for the owner's personnel the operation and maintenance of the packaged pumping system.

- 22.1.3 The variable speed pumping system shall be as manufactured by the pump manufacturer. A pump manufacturer is defined as a company who is regularly engaged in the design and manufacturer of the pumps specified, herein. The pump manufacturer shall have been in the business of manufacturing complete pumping systems for a minimum of ten (10) years and shall catalog variable speed pumping systems.

Local assembly for the pumps shall be acceptable subject to the following:

- The assembly shall be carried out by the authorized local agent.
- All QA/QC procedures, performance or any applicable tests to be considered during assembly process same as adopted in the main factory and accordingly all the related certificates to be provided.

- 22.1.4 The system consists of variable speed pump (s) sized to meet system duty condition as indicated on the pump schedule, adjustable frequency drive(s), bypasses, a pump logic controller, and sensors / transmitters all as described herein.

- 22.1.5 At the time of submittal for approval, the pump manufacturer shall provide a complete system profile analysis which shall include as a minimum: the pumps performance (variable speed pump curves) and the operating characteristics in the system (system curve).

- 22.1.6 This system profile analysis shall include pump motor and adjustable frequency drive efficiencies, staging points, horsepower, and kilowatt / hour draw.

- 22.1.7 Check valve shall be used at pump discharge with Gate valve.

- 22.1.8 Pumps shall be stainless steel vertical multi-stage type as specified in clause 20.0 above. Pump shall be rated for a minimum of 175 PSIG working pressure and or match with system working pressure. Casings shall have gauge ports, vent and drain ports at top and bottom of casing.

- 22.1.9 Headers shall be copper – type L.

- 22.1.10 Motors shall be TEFC, Class 'F' insulation, 2900 RPM suitable for 415V/3PH/50HZ supply.

22.2 Pump Logic Controller

- 22.2.1 The pump logic controller shall be provided, and the operator interface flush mounted in the door of the packaged pumping system control cubicle. All components within the pump logic controller panel shall be U.L. listed or recognized components. The pump logic controller shall meet FCC Part 15, for Class A computing devices.

The pump manufacturer shall supply and be responsible for the operation of a programmable solid stage variable speed pump logic controller. The controller shall function to a proven program that safeguards against damaging hydraulic conditions including:

- a) Motor overload
- b) NPSHR above NPSHA
- c) Hunting

- 22.2.2 The adjustable frequency drives shall be controlled by a direct digital pump logic controller with discrete input signal selection. Use of a multiplexer for multiple sensor inputs is not acceptable. All individual sensor / transmitters shall be individually wired to the pump logic controller for continuous scan and comparison function. The pump logic controller shall be microcomputer based and hold its software in EPROM. Online field modified data entries, such as set point, shall be stored in RAM memory.

EPROM memory storage shall prevent accidental loss of data due to voltage surge or spike. RAM memory shall be backed up by EPROM. In the event of a complete power outage, all factory preset values remain stored and shall be able to be recalled by the operator.

- 22.2.3 The pump logic controller shall be capable of receiving up to two process variable analog input signals. It will then select the analog signal that has deviated the greatest amount from its set point. This selected signal will be used as the command feedback input for a closed loop hydraulic stabilization function to minimize hunting. Each input signal may be programmed for a different set point and set points shall be able to be changed in the field by keypad entry through password security.

- 22.2.4 The pump logic controller shall be capable of displaying an additional analog input corresponding to flow. The analog input signal shall be the feedback and the set point shall be the command. The closed loop hydraulic stabilization function shall have the capability of performing the traditional proportional, integral and derivative modes. The proportional function shall have range and be infinitely programmable with that range.

Rates shall be in percent gain. The integral function shall also be infinitely programmable and compatible with the adjustable frequency drive (s) AFD(s) and the hydraulic system. It shall be entered as repeats / minute. Derivative shall be entered in units of seconds and be fully programmable within the range allowing infinite entries with that range.

- 22.2.5 The scan and compare rate that selects the command set point and process variable signal shall be continuous and automatically set for optimum operation. Each sensor shall be scanned every 20 milliseconds.

- 22.2.6 The pump controller shall have offline and online diagnostic software. Offline diagnostics shall consist of CPU, EPROM, EEPROM and RAM memory test. It shall have self-calibration of all analog inputs with diagnostics, display test, program test, and all I/O tested and status of above displayed. The controller shall include a watchdog circuit to provide system protection / operation in the event of a controller failure. In the event of a controller failure, the system shall be able to be operated manually across the line operation.

Online diagnostics include offline diagnostics and operational diagnostics including but not limited to individual pump fault (s), AFD fault (s), overloads and sensor / transmitter fault(s).

- 22.2.7 Further fault information may be accessed by interrogating the pump logic controller through the front mounted keypad. In case of failure, the controller shall fail in a predetermined safe operating mode. Reset after failure shall not result in a no flow condition unless all equipment fails.
- 22.2.8 The pump logic controller and sensor / transmitters shall be self-powered. The pump logic controller shall be powered by 230 / 1Ph / 50 Hz. The incoming power and I/O circuitry shall reject electromagnetic (EMI) and radio frequency (RFI) interference. All digital outputs shall be electrically isolated.
- 22.2.9 All external transmitters furnished by the pumpset manufacturer shall be powered by the pump logic controller through a quality integral 24-volt power supply. Overvoltage and short circuit protection shall be on board.
- 22.2.10 The pump logic controller shall operate the pump (s) in a predetermined manner as indicated in the sequence of operation section.
Failure of any pump, motor or drive in the system shall cause the controller to not only display the fault conditions, but alternate or transfer the operation to another operating unit as indicated in the sequence of operation section. The pump logic controller shall be capable of utilizing individual differential pressure switches to determine pump failure.
- 22.2.11 Pump, AFD and overload trip faults shall not clear automatically. They must be manually reset after correction of the fault. All above faults shall be displayed before starting. First faults shall always be displayed. Faults shall not trigger erroneous fault messages.
- 22.2.12 The pump logic controller shall be self-prompting. It shall display all messages in plain English. The messages shall include, but not be limited to, how to obtain default values, when to turn it on, and if it is in the active remote control mode.
- 22.2.13 The operator interface shall have the following:
- (a) Multi fault memory and recall.
 - (b) On screen help functions.
 - (c) LEDs (pilot lights) and switches.
 - (d) Light and switch shall also be displayed in plain English messages scrolled continuously.
- 22.2.14 The operator interface shall be capable of remotely being started / stopped.
- 22.2.15 The operator interface and process controller units shall be modular. They shall be able to be interchanged with locally available parts without further programming required.
- 22.2.16 The readout shall be bright LCD super twist backlit capable of displaying any set point value, I input or output signal (s), and alarm function in engineering units. There shall be forty 6mm characters in a two-line display.
- 22.2.17 The pump logic controller shall be capable of operating in automatic, manual or offline diagnostic modes.
- 22.2.18 Three levels of password and software security shall be provided for protection of field programs and use accessible features.
- 22.2.19 The pump logic controller shall provide hardware communications to the building management system.

22.3 Adjustable Frequency Drive

22.3.1 Drive shall be of a microprocessor based PWM design, employing a diode bridge rectifier to convert 3 phase AC to a fixed DC voltage. No input transformer may be used for either power supply or DC bus power within the drive. A constant displacement power factor of 0.95 shall be at all speeds and rated loads.

22.3.2 The DC bus shall consist of a choke and a capacitor bank to eliminate line disturbances. A switching power supply operating off the DC bus will provide excellent under voltage and ride through capability.

Control and inverter logic shall be isolated from the power section.

22.3.4 Bipolar power transistors shall be used in the inverter section converting a fixed DC voltage to a 3 phase adjustable frequency output with a minimum 97% efficiency at rated load.

22.3.5 For maintenance purpose, the drive shall be capable of starting, stopping and running with stable operation with the motor completely disconnected (no load).

22.3.6 Drive must protect itself against all phase to phase and phase to ground faults, removal of load, single phasing, power outages, and prevent starting with wrong phase rotation.

22.3.7 Protection is defined as normal shut down with no component damage.

22.3.8 The drive shall employ adjustable torque limit control which will override the speed command and decrease the frequency while maintaining the correct volt/hertz ratio whenever the load level surpasses the drive design level.

Torque limit shall be settable from 50 to 115% of full drive rating. When torque limit engages, the output frequency is steadily reduced until the load reduces to the design capacity. At that point the speed will stabilize. If the load reduces further the drive will reaccelerate to the preset speed.

Electronic motor overload provides electronic equivalent to thermal motor overload and shall be adjustable from 60 to 100% of the drive rating.

The drive shall be capable of being started by a 2 wire control circuit, a 230 Volt AC signal or application of input power.

Adjustments shall include:

Current Limit	Min / Max Speed	Motor Boost
Accel Time	Decel Time	V/Hz Ratio

Speed Reference signals shall be:

Potentiometer input	10,000 ohms
Voltage input	0 to 10 VDC
Current input	0 to 20 or 4 to 20 m Amp.

Analog output shall be a non-isolated 0 to 5 VDC, selectable for frequency or amperage.

The drive shall provide as standard 3 relays to indicate drive run, drive fault, and current limit condition.

Full RS 232 communications shall be provided as optional equipment.

The variable frequency drive including all factory installed option shall have ETL or UL agency approvals.

VFD specifications

For detailed VFD specifications, refer electrical specification volume.

Input voltage	:	230V/1PH/50Hz 415V/3PH/50HZ
Input Range	:	+/- 10%
Speed Range	:	20:1 Maximum or as specified.
Ambient Temperature	:	0 – 40°C
Humidity	:	0 – 95 % Non-Condensing
Altitude	:	to 3300 feet above sea level
Efficiency	:	96% or greater at rated load
Power Factor	:	0.95 regardless of speed or load.
Adjustments	:	Min. / Max. Speed, accel / decel independently adjustable to min. 120 sec., current limit, 12 T.
Drive Protection	:	Over temperature, under / over voltage, phase to phase short circuit, phase to ground short circuit, overcurrent, current limit, decel limit.
Status Indication	:	Drive enabled, power fault indication for under / over voltage, over current, motor trip, accel / decel limit.

Sequence of Operation

With the pumpset in the “Automatic” mode and the duty pump selector on pump No. 1, pump No. 2 assist the pump No. 3 standby. Duty pump No. 1 shall operate through the AFD which is being controlled by a signal generated by the pump logic controller. Pump No. 2 will start if duty is not met and pressure cannot be maintained. Pump No. 2 shall generally remain as assist mode. Pump No. 3 remain standby mode.

In the event of a system differential pressure failure, due to duty pump or overload fault, the pump logic controller will automatically initiate a timed sequence of operation to start the backup pump in the variable speed mode. A solid red indicator light shall signal this condition and a message shall be displayed on the operator interface of the pump logic controller indicating the failure.

In the event of a system differential pressure failure, due to an AFD fault, the pump logic controller shall automatically initiate a timed sequence of operation to start the selected pump across the line. A solid red indicator light shall signal this condition and message shall be displayed on the operator interface of the pump logic controller indicating the failure.

A subsequent overload trip or duty pump failure shall automatically start a timed sequence of operation to start the standby pump across the line.

When in the “bypass” mode and the pump selector on “Pump No.1”, Pump No. 1 shall operate across the line at constant speed with motor overload and ground fault protection. A solid red indicator light shall signal this condition. The standby pump shall be locked out.

22.3.9 In the event of an overload fault while in the “bypass” mode, the backup pump shall be locked out.

In the event of the failure of a zone sensor / transmitter, its process variable signal shall be removed from the scan / compare program. Alternative zone sensor / transmitters, if available, shall remain in the scan / compare program for control.

22.3.10 The zone number corresponding to the failed sensor / transmitter shall be displayed on the operator interface of the pump logic controller.

22.3.11 In the event of failure to receive all zone process variable signals, the AFD shall maintain 100% speed, reset shall be automatic upon correction of the zone failure.

22.3.12 Pump, AFD, overload fault shall be continuously scrolled through the display on the operator interface of pump logic controller until the fault has been corrected and the pump logic controller has been manually reset.

22.3.13 The Packaged booster pump set shall have 2 years' warranty.

23.0 PACKAGED WATER FILTRATION & SOFTENING UNIT (Where Applicable)

23.1 Pre Chlorination

The raw water supplied by the Municipality shall be chlorinated by means of Sodium hypochlorite solution. The Sodium Hypochlorite Dosing set shall comprise of dosing tank (PVC), Electronic metering pump (PP).

DATA SHEET FOR THE CHLORINATION DOSING SYSTEM

Description	Data	Chlorination Details
Type of Chemical		Sodium Hypochlorite solution
<u>Dosing tank details</u>		
Number		1
Type		Vertical, Circular
Material of tank		Polyethylene
Capacity		As per the design requirements
Number		As per the design requirements
Type		Electronic Metering
Material of pump		PVC
Capacity		0 – 20Lph

Chlorination with PH system for main water tanks located at basement floor shall be provided as per DM requirement in order to control Chlorine level not more than 0.5 PPM and PH from 7.4 - 8.5

23.2 Auto Multimedia Filtration Unit

Type of Treatment for Removal of Suspended impurities

23.2.1 Vessel:

The filter vessels shall be fabricated from high grade steel and internally lined with epoxy paint. The External surface of the vessels shall be protected with a primer undercoat and Epoxy paint.

The vessel shall be designed for 100PSI working pressure and tested to 150PSI. The vessel shall be supplied internally coated with Zinc Epoxy and externally painted.

23.2.2 Valves:

Automatic Solenoid driven Hydro / Pneumatic Diaphragm / Butterfly valve to control the operating steps. The valves shall be made of Carbon Steel. The water treatment equipment supplier as per the requirement shall arrange necessary compressed airline.

23.2.3 Frontal Piping

The frontal piping of the filter unit shall be uPVC surface protected with U.V. resistant paint.

23.2.4 Media

The Multimedia Sand Filter bed shall consist of a top layer of a uniformly graded granular form of anthracite. The intermediate layer shall consist of three grades of Silica sand. The under bed shall consist of a layer of gravel, which is a granular material of extra ordinary density available in graded sizes for fine filtration.

Charge of carefully graded filtering Sand and Anthracite Media complete with supporting gravel underbed.

23.2.5 Pressure Gauge

Inlet / Outlet gauge shall be fitted to give indication of pressure drop through filter.

23.2.6 Controls

The filters shall be provided with Specialized PLC Controllers with timer mechanism to enable to program the backwash cycle of the filter at any time of the day. The PLC Controller shall complete with Soft Keyboard and LCD display and shall give signals to the Solenoid driven hydro / Pneumatic Butterfly valves.

23.3 DUPLEX SOFTENER

23.3.1 The unit shall consist of softener vessels for continuous supply of the softened water. The plant shall be capable of producing zero temporary hardness at the rate as scheduled.

23.3.2 The Duplex softener should consist of two tanks in Polyester Fibre Glass with suitable coating internal and external of the construction. Pressure rating of the tanks is 10bars.

23.3.3 Control valve mounted on the vessel shall be constructed from brass for strength, durability and corrosion resistance to ensure longer life. The controller shall be equipped with volume based Electronic controller in which regeneration is initiated after specified volume of water has passed.

23.3.4 Setting conditions for volume based regeneration shall be programmed depending on raw water hardness.

23.3.5 Provision for Manual initiation of regeneration shall be provided.

23.3.6 The interconnecting pipe system shall be specially designed for the units and hydraulically balanced to allow uniform load on each resin tank. These shall be dimensioned to ensure minimum pressure loss in the pipe system.

23.3.7 The unit shall come with a brine tank of PVC / Fiberglass material.

24.0 G.R.P. STORAGE TANKS

24.1 This section covers for the supply, delivery, installation, testing and commissioning and leaving in

complete working order water storage tanks and cisterns, as indicated on the Contract Drawings. The GRP tank panels shall be insulated (tank shall be insulated only, if tank is expose to sunlight) in all sides with 40 mm thick rigid PUF insulation.

- 24.2 The tanks shall be glass reinforced plastic construction.
- 24.3 Each tank shall be provided with a heavy duty cover formed from standard panels and complete with manholes and vent.
- 24.4 The termination of vent pipes into cisterns and tanks shall permit easy removal of the covers.
- 24.5 Tanks shall be of the type and capacity shown in the schedules.
- 24.6 Jointing materials shall be suitable for the liquid to be stored. In the case of drinking water, jointing materials shall be of a non-toxic nature.
- 24.7 All facilities for pipe connections shall be made at the manufacturer's works. Pipe connections for up to 50mm nominal bore pipework shall be BSP screwed sockets. Pipe connections for pipework 65mm nominal bore and above shall be studded pads suitable to receive counter flanges to BS 10.
- 24.8 Where shown tanks shall be divided into equal compartments using standard panels and components. Divisions shall be stayed to enable one compartment of the tank to be emptied whilst the other compartment is in service.
- 24.9 All tank covers shall be provided with purpose made manholes of a minimum diameter of 458mm with hinged or bolted lids. Where tanks are provided with internal ladders separate manholes shall be provided to the ladder and to the ball float valve or float switch. Tanks with no internal ladder shall be provided with one manhole positioned to allow access both to the tank interior and to the ball float valve.
- 24.10 Tank cover(s) shall also be provided with cowls to adequately ventilate the interior.
- 24.11 Tanks 2000mm deep and over shall be provided with internal GRP and external galvanized steel access ladders suitably protected against corrosion and bolted to the tank walls. In elevated positions external tank ladders shall be fitted with safety hoops.
- 24.12 Each tank shall be fitted with a delay action float valve. Ball float valves shall be of the sizes indicated and suited to the pressure available.
- 24.13 Overflow pipes shall be twice the bore of the ball valve fitted 32mm size, whichever is the greater. Overflow pipes shall be run to discharge outside the building via floor gully to the nearest manhole.
- 24.14 Vent cowls and overflows shall be fitted with a suitable gauze screen to prevent insects and vermin from entering the tank. Overflow pipe shall be minimum 100 mm dia.
- 24.15 The tanks shall be sectional pattern, glass reinforced plastic manufactured by the hot press moulded process, constructed with plain diaphragm panels 1m square.
- 24.16 Tanks to have heavy duty covers with hinged ventilated cover. Jointing of panels to be with flexible sealing strip and stainless steel bolts. External support of tanks in excess of 1.84m deep shall be formed from rolled I section steel members to BS EN 10056-1, hot dipped galvanized to BS EN ISO 1461. The design of the corsetry is to comply with BS 449, Part 1.

- 24.17 All tanks shall be assembled on the supporting bases in accordance with the manufacturer's instructions, using torque wrenches to prevent damage to the sections due to over-tightening of nuts and bolts.
- 24.18 Detailed construction drawings of all tanks shall be submitted to the Engineer for approval prior to manufacture.
- 24.19 The Contractor shall provide and install a layer of bituminized felt for insertion between the tank and its bearers.
- 24.20 All metallic parts in continuous contact with water shall be of stainless steel 316.

Commissioning and Testing of Cisterns and Tanks

- 24.21 All tanks operating at atmospheric pressure shall be tested under "tank full" conditions for tightness and structural soundness.
- 24.22 All cold water tanks shall be subject to a standing leak test for twenty-four hours, during which time all joints shall be carefully examined. Any defects shall be rectified immediately and the test repeated.
- 24.23 WRAS Product Approval Certificate must be submitted. The certificate must clearly mention the complete tank with panels, gaskets and accessories.

G.R.P. Tanks (Particular Specifications)

- 24.23 The Contractor shall supply and assemble at site sectional water tank of hot pressed GRP insulated panels (Glass Reinforced Plastic) provided with lockable manhole cover air vent with overflow and all piping connections. The panels shall comply with BS EN 13280 and shall have convex shape to assist a complete draining of water of the following characteristics: -
- (c) Specific gravity of 1.8
 - (b) Tensile strength of not less than 850 kg/cm²
 - (c) Flexural strength of 1800 kg/cm² to ISO 178.
 - (d) Impact strength of 80 kg/cm² to ISO 179
 - (e) Thermal expansion of 2.1 x 10⁻⁵°C to ASTM D696.
 - (f) Thermal conductivity of 0.6W/m² C to BS EN 13280.
 - (g) Water absorption less than 0.2% to ISO 62.
 - (h) Glass content more than minimum 30%
 - (i) No light transmission
 - (j) Gasket of synthetic rubber
 - (k) Bolts, washers and nuts of stainless steel 316 / A4.
 - (l) Frame angles and frame plates of hot dipped galvanized steel.
 - (m) External ladder of hot dipped galvanized steel.
 - (n) Internal ladder of GRP or S.S. according to Manufacturer's standards.

- (o) Tie rods of stainless steel 316
- (p) G.R.P. panel thick not less than 8mm
- (q) Resin cover of 2mm
- (r) Glass tube level indicator for all compartments.
- (s) Insulation 40mm low density polyurethane foam (CFC free) with 2mm GRP skin cover for all 4 sides, top and bottom panels. The insulated tanks shall be provided where exposed to sun, the tanks within plant rooms shall not be insulated.

25.0 DIGITAL WATER FLOW METERS

The digital water meter shall be a battery –powered electromagnetic flow meter with a very high accuracy even at very low flow rates without any moving parts and enclosed within IP 67 enclosure.

The meter shall be equipped with an internal data logger which can read out via IRDA or RS232 interface with Modbus and via radio frequency or GSM/GPRS.

All the data shall be interfaced with BMS.

The main feature of the meter as per the followings:

- 20 years' battery life span
- Accuracy better than $\pm 0.5\%$
- Wide measuring range
- LCD display
- IP67 protection
- M-bus interface
- Powder coated cast aluminum casing
- Temperature from 0°C up to 80°C
- PN16

26.0 NOT USED

27.0 HYDROSTATIC TESTING OF WATER PIPING SYSTEMS

- 27.1 All water piping systems shall be hydrostatically tested for at least 1.5 the working pressure and not less than 10bars, for ensuring complete tightness under the test pressure and for the duration of time as specified under the respective system concerned.
- 27.2 Systems may be tested as a whole or in sections to facilitate the progress of the work.
- 27.3 No part of any piping system shall be tested to pressure less than the specified test pressure measured at the highest point of the system.
- 27.4 Care shall be taken not to subject any equipment, apparatus or device to a pressure exceeding its prescribed test pressure as obtained from its name plate data or from manufacturer's published data. Pressure tests shall be applied before connecting piping to equipment. Relief valves instruments, automatic air vents and all devices that might be damaged by the test pressure shall be removed, disconnected or blanked off.
- 27.5 No pressure shall be applied against the closed gate of gate valves. All valves shall be in the open

position but not completely back seated during testing. End valves shall be capped.

- 27.6 In testing flanged piping, temporary blank flanges shall be installed and firmly anchored to accommodate all developed end thrust.
- 27.7 All piping that can be damaged by end thrust developing from hydrostatic testing shall be properly anchored during testing especially at changes of direction.
- 27.8 The piping system to be tested shall be closed by plugging and blanking all openings in the system in an approved manner and filled slowly with water making sure to vent all entrapped air. Plugs shall be released temporarily to ensure that water has reached all parts of the system.
- Pressure shall be applied to the system by means of a hand pump drawing from a water container. The pump discharge shall be connected to the system through a globe valve, check valve and recently calibrated pressure gauge of suitable range to have the test pressure read in the middle of the range.
- 27.9 After the test pressure is reached, the pump shall be blocked off by closing the globe valve and the variations of pressure in the system monitored on the pressure gauge.
- 27.10 The test pressure shall be maintained on the system for 24 hours during which time the system shall have no noticeable drop in pressure.
- 27.11 While the system is under pressure, a careful inspection shall be made of all pipes and joints and if any leaks in joints or evidence of defective pipe or fitting is disclosed the defective work shall be corrected immediately by replacing defective parts with new joints and materials. No makeshift repairs or application of any repair compound will be permitted.
- 27.12 After the correction is made the pressure test shall be repeated until a completely tight system is ensured.
- 27.13 The test pressure shall be released slowly so as not to produce shocks and sudden contractions that might damage the piping.
- 27.14 All piping shall be tested to at least 1.5 times the system working pressure.

28.0 TESTING OF EQUIPMENT

- 28.1 After installation, all equipment shall be tested for performance and proper functioning and for demonstrating to the full satisfaction of the Engineer that the equipment meets the design requirements as shown on the Drawings and as specified.
- 28.2 Should any equipment prove defective, unsatisfactory or does not comply with the Specification, the Contractor shall correct, repair or replace such equipment or his own account as directed by the Engineer. After the correction is made, subsequent tests shall be performed until the equipment is proven to fulfil the requirements to the entire satisfaction of the Engineer.

29.0 REJECTION OF EQUIPMENT

Equipment shall be rejected if:

- (a) It has a capacity less than required.
- (b) Its electric motor is overloaded.
- (c) It causes objectionable noise or vibration.

30.0 TESTING AND ADJUSTMENT OF CONTROLS

All controls and instruments shall be operated simultaneously with the equipment or system that they control and tested for proper, accurate and dependable functioning and operation.

All controls and instruments shall be calibrated and adjusted to the satisfaction of the Engineer.

31.0 GOVERNMENT AND LOCAL AUTHORITIES

Perform at appropriate times during the currency of site work, all tests required by Government and Local Authorities who may from time to time have jurisdiction over the work and obtain the necessary certificates of approval. Local regulations shall be strictly followed.

32.0 DOMESTIC WATER PIPING STERILIZATION AFTER COMMISSIONING

On completion of the potable cold water supply services system, including any parts of the existing service they shall be sterilized by the application of chlorine. The minimum requirements for chlorination shall be to flush the pipework thoroughly to remove dirt. Chlorine shall then be added to the storage tanks and water drawn from all draw-off points until chlorine is detected at each outlet point with a concentration of 50 ppm. The mains are then to be allowed to stand for a period of 24 hours, after which the outlets are to be tested for chlorine. If chlorine is not found to be present or detected with a concentration of less than 0.4 ppm, all the above shall be repeated until residual chlorine is detected. After this, the system is to be completely flushed with clean water and precautions taken to avoid subsequent contamination of the installation.

32.1 Follow the following procedure for sterilization of water piping and storage water tanks:

- a. Prior to starting work, verify that system is complete, flushed and clean.
- b. Ensure pH of water to be treated is between 7.4 and 7.6 by adding alkali (caustic soda or soda ash) or acid (hydrochloric acid).
- c. Add disinfectant, free chlorine in liquid, powder or tablet form to obtain 50 ppm residual.
- d. Fill the system piping with chlorinated water and bleed water from outlets to ensure distribution and test for disinfectant residual at minimum 15 percent of outlets.
- e. Maintain disinfectant in system for 24 hours.
- f. During the disinfections period install temporary warning signs in both Arabic and English at conspicuous places throughout the Project against using the chlorinated water.
- g. If final disinfectant residual tests less than 0.4 ppm, at any outlet, repeat treatment.
- h. Flush disinfectant from system until residual equal to that of incoming water or 1.0 mg/L.
- i. Take samples no sooner than 24 hours after flushing, from remote outlet of each riser and from water entry and analyze in accordance with AWWA C651. Submit result of analysis to Engineer for approval.

32.2 The Chlorination procedure during commissioning stage shall consider the following:

- a. The final water samples should be tested for both microbiological and Legionella species reports.
- b. Repeat the procedure if biological examination shows contamination.
- c. Swimming pool and water feature legionella prevention shall be considered by the pool contractor.
- d. During operation the following shall be considered.

- Legionella control and management manual should be prepared by the specialist company.
- Quarterly monitoring and inspection of water piping system & water tanks. Chlorination to be performed every 6 months or as recommended by the specialist supplier.
- Take samples of water tank and conduct bacteriological test.
- If Legionella is identified, the whole water distribution system shall be cleaned and disinfected as per HSE standards.
- Record for sample cleaning disinfection shall be kept always with contractor/ FM Company.

END OF SECTION

MECHANICAL SECTION: 05

SOIL, WASTE, VENT & STORM WATER SYSTEMS

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1.0 SYSTEM DESCRIPTION**1.1 System Design**

This shall be a gravity drainage system designed in accordance with the recommendations of the Chartered Institution of Building Services Engineers and to the requirements of the Dubai Municipality / Local Drainage Department.

1.2 System Description**Sewer Drainage Network**

1.2.1 The system shall be gravity type with DOUBLE stack together with vent installation. All work shall be installed and tested in accordance with BS EN 12056-2 code or equivalent EN standard for sanitary works.

1.2.2 Surface water drainage network

Rainwater from the roof shall be piped down via separate pipe work to connect as shown on drawing.

1.2.3 Balcony drain from the balconies shall be piped down and connected to the waste pipes through running traps. Contractor shall ensure that, U-Trap is getting wastewater / condensate water to wet the trap and provide water seal year around.

1.2.4 Condensate drain from Air-conditioning equipment shall be connected to nearest floor drain or to waste stack through U-trap. Contractor to coordinate with A/C layouts for exact locations. Condensate drainpipe shall be insulated as specified in section 02.

INTERNAL DRAINAGE**2.0 INSTALLATION**

2.1 Vent pipes shall extend through roof and terminate 2m as per DM above it with vent cowl. Vent pipes passage through the roof shall be made watertight by proper flashing. Vent pipes shall be at least 5m away from any air intake to the building.

2.2 All changes of direction shall be gradual and not abrupt, 45° fittings shall be used wherever possible, and 90° fittings shall be of the long sweep type. All unnecessary turns and off sets shall be carefully avoided and run as directly as possible from the sanitary fixtures to the vertical stacks.

2.3 Concealed pipes shall be installed in such a manner as to permit easy accessibility for maintenance.

2.4 All pipes shall be fixed in neatly arranged lines, and adequately pitched horizontal lines to allow the system to be properly vented and drained. Air pockets, traps and sags shall be carefully avoided.

2.5 Supports, clamps and hangers shall be made of galvanized steel with rubber internal rubber seal, fixed with drilled plugs. Cutting and pinning of fixings will not be permitted.

2.6 Run building drains at following slope: For pipe diameter up to 100 mm dia at 1.66 % (1:60), pipe diameter 150 mm at 1.1 % (1:90), and 200 mm dia pipes at 0.8 % (1:120) downward in the direction of flow. Pitch branch connections to stacks from fixtures at 1.66 % (1:60) where possible.

2.7 Provide all the required appurtenances to make the drainage system complete in compliance with code requirements including traps, pipe fittings, hangers, and the like.

2.8 Wherever possible, vent stack offsets shall be made with 45° fittings.

- 2.9 Take special care in setting roof drains to ensure that they are set at an elevation which will preclude formation of puddles.
- 2.10 Install connections to roof drains in conjunction with the roofing specified under civil works, so that the building is adequately protected during construction from damage by storm water.
- 2.11 No short radius bends to be used. Use short "Tee-Wye" fittings in vertical piping only.
- 2.12 Any piping passing through roofs shall be so arranged to be a minimum of 300mm (1 ft.) from walls or other obstructions so as to permit proper flashings which are provided by another trade.
- 2.13 Where drainage pipework crosses fire rated partitions, walls and floors, provide proprietary fire rated intumescent sleeves with a fire rating equal to or greater than the fire rating of the respective wall or floor.
- 2.14 Horizontally long running soil & waste shall be vented from the top end, weather shown or not on layouts.

3.0 CLEANOUTS ON SOIL WASTE VENT AND STORM WATER INSTALLATION

- 3.1 Cleanouts shall be installed at each change of direction of drainage pipes, greater than 45°, inside the building, and where indicated on the drawings. Cleanouts shall be not more than 10m apart in horizontal lines. A cleanout shall be provided at or near the foot of each vertical waste or soil stack.
- 3.2 Cleanouts on concealed piping shall be extended through and terminate flush with finished wall or floor. Pits or chases may be left in the wall or floor, provided they are of sufficient size to permit removal of the cleanout plug and proper cleaning of the system.
- 3.3 Where it is necessary to conceal a cleanout plug, a heavy-duty covering plate shall be provided, which will permit ready access to the plug.
- 3.4 Cleanout plugs shall be of heavy duty with seal and lock. Final finish shall be to the approval of the Architect.
- 3.5 Cleanouts shall be so installed that there is a clearance of not less than 45cm for the purpose of rodding and cleaning.
- 3.6 Provide cleanouts at foot of all stacks, changes of directions, at the ends of branch runs, in straight runs as required and where indicated. Terminate as specified under "Cleanouts".

4.0 MATERIAL

- 4.1 All above ground Drainage pipes and fittings (soil, waste, vent pipes) including shaft risers shall be sound proof polypropylene drainage pipes and fittings as specified in clause 4.8 below.

The soil/waste/vent pipes in Basement car parking floors shall be either Polypropylene pipes or uPVC to BS EN 1329-1 (BS 4514) for pipe sizes 80mm dia and above and MuPVC (cPVC) / to BS EN 1455 & BS EN 1566-1 or ABS material for pipe sizes 50mm dia and smaller. Push fit joints shall be used for the gravity drainage above 100 mm dia, and solvent welded joints for the pressure pipes for sump pump discharge pipes and gravity drainage pipes below 100 mm dia.

Rainwater pipes shall be HDPE SDR26 to BS EN 1519-1, joined with butt-fusion or electro-fusion sockets methods.

- 4.2 Drainage pipes and fittings (soil, waste, rainwater) running within false ceiling void of occupied areas (like living/dining rooms and bedrooms) shall be provided with acoustic lagging insulation of 25 mm thick 90-100 Kg/m³ foil faced Rockwool/fiberglass insulation with 2mm thick barrier material of barium

sulphate loaded limp vinyl sheet or equal approved. Refer Drainage details drawing for method of installation of the acoustic lagging insulation.

- 4.3 Drainage pipes for commercial Kitchens shall be HDPE pipes withstanding high temperatures to BS EN 1519.
- 4.4 The AC condensate drainpipes within false ceiling void shall be in uPVC class D with solvent cement joints.
- 4.5 Drainage pipes buried under ground up to manhole shall be uPVC to BS EN 1401-1 (BS EN 13598).
- 4.6 Sewage pipe between manholes shall be uPVC to BS EN 1401-1 (BS EN 13598) protected with concrete encasement.

Discharge pressure pipes from sump pump/sewage lift pump set shall be uPVC class E to BS 3505 (BS EN 1452).

- 4.7 All pipes under building, pipe running externally with cover less than 700 mm, and those subject to traffic especially under roads shall be protected with reinforced concrete.
- 4.8 Soundproof piping system shall be made from mineral supplemented polypropylene. The pipes shall consist of three layers (polypropylene-mineral additives-polypropylene) with a sound insulation layer, density of the mineral filler shall not be less than 1.9gr/cm³ for soil and waste, or one thick layer pipe made of mineral-reinforced polypropylene.
Soundproof drainage pipe system shall comply with following standards:

- DIN 4109 soundproof, absorption standards.
- DIN 4102, B2. Self-extinguishing flameless.
- DIN 19560 / DIN EN 1451. Hot water resistance, 90°C (Long term), 100°C (short term).

4.8.1 Physical characteristics

Mineral Filler Density - 1.9 g/cm³ DIN EN ISO 1183

Elongation at break - 50%

Tensile strength - 20N / mm²

E-modulus - 3800 N / mm² for 1 Layer and 2300 N/ mm² for 3 Layer

Linear expansion - 0.09 mm / m. K

- 4.8.2 The soundproof pipes and fittings shall be self-extinguishing flameless for the fire behavior and complies with DIN 4102, B2.

The soundproof drainage pipes shall have the following wall thickness. However, as per the manufacturer's standards for special pipe material if the wall thickness is lesser and still can achieve the required noise level of 20 dBA as per the standard EN 14366 the same shall be acceptable.

Sr. No.	Outside dia (mm)	Wall thickness (mm) 1 Layer	Wall thickness (mm) 3 Layer
1.	58	4.0	2.1
2.	78	4.5	2.5
3.	110	5.3	3.4
4.	160	5.3	4.9

5.	200	6.2	6.2
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- 4.8.3 The floor trap shall also be of soundproof material to DIN 4109.
- 4.8.4 The elbows of the soil pipe coming out of WC above habitable spaces shall be provided with acoustic lagging material for at least 1 m from each side in order to ensure achieving the specified resultant noise level of NC 30 in the occupied rooms.
- 4.9 Drainage pressure pipework from sump/sewage lifting pumps shall be in uPVC pressure pipe class E (15 bars) to BS 3505
- 4.10 The rainwater down pipe terminated at ground floor level for free discharge shall have stainless steel flapper at pipe outlet, if there is a stormwater network available, then the rainwater pipe can connect directly to this.

5.0 FLOOR / ROOF & BALCONY DRAINS / GULLIES / GREASE SEPARATOR

- 5.1 Floor drains unless otherwise indicated shall have uPVC traps. The floor trap shall have minimum water seal of 7.5 cm for outlets greater than 50 mm as per BS EN 12056-2. The open area of strainer shall be at least two thirds of the cross-sectional area of the drain line to which it connects. Floor drains shall have heavy duty stainless steel SS 316 cover (subject to Engineer Approval) with a removable strainer. Floor drain shall have built in rodding eye.

All floor drains / cleanouts must be coordinated with floor tiling layout. Funnel floor drains, where specified and for condensate drain in plant rooms shall include a Nickel bronze funnel secured to the floor grating.

Dry floor drains (within water meter rooms, shafts) shall include a rubber seal with anti-microbial compound to be installed within the extension of the uPVC floor trap below the cover to prevent the odor transfer in case of no water seal.

Car parking drain outlet shall be 300x300 mm heavy duty ductile iron grating and body with 100 mm pipe outlet. The car parking drain shall have suspended sediment bucket.

All ramps at parking floors shall have 300 mm heavy duty ductile iron gratings at bottom of ramp. The grating shall have sediment bucket with strainer at the outlet.

- 5.2 Balcony drain shall be with uPVC body with stainless steel SS 316 cover to ID approval.
- 5.3 Clean outs

Install cleanouts in aboveground piping and building drain piping according to the following, unless otherwise indicated:

1. Size same as drainage piping up to DN100. Use DN100 for larger drainage piping unless larger cleanout is indicated.
2. Locate at each change in direction of piping greater than 45°.
3. Locate at minimum intervals of 15 m.
4. Locate at base of each vertical soil and waste stack.
5. For cleanouts located in concealed piping, install cleanout wall access covers, of types indicated, with frame and cover flush with finished wall.
6. For floor cleanouts for piping below floors, install cleanout deck plates with top flush with finished floor.

Clean out material:

For internal wet areas (Toilets and Kitchens, etc.): Stainless steel SS 316 with square or round adjustable cover. All cleanouts must be coordinated with tiling layout.

For car parking areas and plant rooms: Heavy-duty cast iron cover.

Clean out at high level shall be uPVC screw cap and washer.

5.4 Rainwater outlets

- a) Rainwater outlets shall be cast iron body and dome with special coating material to suit the outdoor application, the outlet shall have flashing clamp device with integral gravel stop. Rainwater outlets shall be selected to suit the type of roof where installed and shall be provided with a no hub outlet for mechanical jointing to rainwater pipe. Rainwater outlets shall be provided with perforated extension assemblies, where required to match the roof construction.
- b) Scupper type rainwater outlets shall be cast iron body with special coating material with reversible back or bottom outlet, oblique grate with combination frame and membrane flashing clamp.
- c) External catch basins shall be a concrete pit of 60x60x60 cm (LxWxD) with 60x60 Heavy duty ductile iron (HDDI) grating and GRP perforated basket within the pit. The outlet pipe size from the external catch basin shall be as shown on drawings.
- d) The internal catch basin shall be similar to above but with 30x30 cm grating size and concrete pit.

- 5.5 Flower bed drain outlet shall be with combination membrane flashing clamp / gravel guard, low silhouette dome and stainless steel (SS 304) mesh over dome with bottom connection to uPVC drain pipe. No floor trap is required.

5.6 AUTOMATIC GREASE SEPARATOR

- 5.6.1 The Automatic Grease Separator shall be manufactured according to EN 1825-1, body to be made of 100% corrosion free polyethylene construction, with integrated sludge trap according to specify norms.
- 5.6.2 The Separator shall be oval / rectangular in shape.
- 5.6.3 Automatic switching between shredding, mixing and disposing shall be via actuator with program controlled shredder - mix - disposal system.
- 5.6.4 Macerating pump (where applicable) shall be able to handle the kitchen sludge/effluent efficiently, including grease and sludge, bones, meats, shells, plastics, etc without the need of any kind of pre filtration of sludge via strainer baskets.
- 5.6.5 Efficient suction of sludge shall be provided for a proper cleaning frequency in the form of wedge shaped bottom of the separators, wherever possible.
- 5.6.6 The macerating pump capacity of nominal 3kW, 37m³ per hour at 1 bar or higher capacities depending on the project requirement with respect to disposal pipe runs.
- 5.6.7 Closure valves to be provided for easy removal of macerating pumps for any maintenance purposes.
- 5.6.8 Top mounted water spray nozzle for efficient cleaning of the insides of the separator body. Program controlled solenoid valves of 1" for hot (if available) and cold water connections for the rinsing stage.
- 5.6.9 Air gap shall be provided in the filling line.
- 5.6.10 Disposal flange of PN10 pressure rating to be provided for coupling with disposal truck during the disposal stage.

- 5.6.11 Sonic control level sensing system with ultrasonic sensor technology to measure grease level thickness in real time and emit an alarm for notification of disposal.
- 5.6.12 Control panel should be able to be connected to the BMS system via volt free contacts.
- 5.6.13 Sampling chamber to be provided for easy access to the effluent water for the authorities for sample water test.
- 5.6.14 Remote control for the access of the function of disposal routine to be provided if the grease separator and the terminal point of the disposal flange is far away from each other.
- 5.6.15 Remote control and the disposal coupler should be enclosed and protected in a SS box.

5.7 AUTOMATIC GREASE SEPARATOR

- 5.6.1 The Automatic Grease Separator shall be manufactured according to EN 1825-1, body to be made of 100% corrosion free polyethylene construction, with integrated sludge trap according to specify norms.
- 5.6.16 The Separator shall be oval / rectangular in shape.
- 5.6.17 Automatic switching between shredding, mixing and disposing shall be via actuator with program controlled shredder - mix - disposal system.
- 5.6.18 Macerating pump (where applicable) shall be able to handle the kitchen sludge/effluent efficiently, including grease and sludge, bones, meats, shells, plastics, etc without the need of any kind of pre filtration of sludge via strainer baskets.
- 5.6.19 Efficient suction of sludge shall be provided for a proper cleaning frequency in the form of wedge shaped bottom of the separators, wherever possible.
- 5.6.20 The macerating pump capacity of nominal 3kW, 37m³ per hour at 1 bar or higher capacities depending on the project requirement with respect to disposal pipe runs.
- 5.6.21 Closure valves to be provided for easy removal of macerating pumps for any maintenance purposes.
- 5.6.22 Top mounted water spray nozzle for efficient cleaning of the insides of the separator body. Program controlled solenoid valves of 1" for hot (if available) and cold water connections for the rinsing stage.
- 5.6.23 Air gap shall be provided in the filling line.
- 5.6.24 Disposal flange of PN10 pressure rating to be provided for coupling with disposal truck during the disposal stage.
- 5.6.25 Sonic control level sensing system with ultrasonic sensor technology to measure grease level thickness in real time and emit an alarm for notification of disposal.
- 5.6.26 Control panel should be able to be connected to the BMS system via volt free contacts.
- 5.6.27 Sampling chamber to be provided for easy access to the effluent water for the authorities for sample water test.
- 5.6.28 Remote control for the access of the function of disposal routine to be provided if the grease separator and the terminal point of the disposal flange is far away from each other.
- 5.6.29 Remote control and the disposal coupler should be enclosed and protected in a SS box.

6.0 SEWAGE LIFT PUMPSET (Where Applicable)

- 6.1 Electric submersible sewage pumps shall be suitable for wet sump or dry pit installation and for intermittent or continuous operation.
- 6.2 Pump shall be suitable for a maximum depth of submersion of 10m with the outlet directed vertically upwards.
- 6.3 Motor shall be of the synchronous squirrel cage type, IP68 rating, hermetically enclosed within the stator housing. Stator shall be wound with double lacquered wire and subsequently triple dipped to provide insulation in accordance with IEC 85 temperature Class F (155°C). Stator winding over temperature shall be prevented by bi-metallic thermal switches or thermistors embedded in each of the stator phase windings. Motor shall be designed for a maximum of 10 starts/hour. Motor casing shall be with integral ribs for maximum heat dissipation.
- 6.4 Cable entry glands shall be suitably designed to effectively seal the junction box from the surrounding liquid at depths up to 10m and to mechanically restrain the cable to avoid distortion of the gland rubbers and to prevent kinking of the cable adjacent to the cable entry.
- 6.5 Motor shaft shall be of stainless steel and mounted on robust pre-greased rolling element bearings. The shaft seal assembly shall comprise of two independent mechanical face seals mounted in tandem with an interposing oil chamber to cool the seals and to act as an additional leakage barrier. Seal combination shall be high wear resistant tungsten carbide/tungsten carbide.
- 6.6 Motor and pump casing constructed of cast iron.
- 6.7 Pump casing shall be smooth profiled volute geometry for clog free operation fitted with replaceable inlet wear rings.
- 6.8 Impeller shall be dynamically balanced single/multi-channel type depending on the pump selection and duty specified.
- 6.9 The shredding system shall consist of spiral bottom plate with stationary cutter ring with cutting edges and a shredding rotor located below the impeller for optimum blockage free running.
Pumps shall be of non-clog design,
- 6.10 A mixing valve mounted directly on the pump casing shall be automatically activated to provide efficient mixing of the sump content prior to discharge.
- 6.11 Pumpset shall be supplied with a control panel complete with breakers, fuses, starters, duty/assist/stand-by pump selector switch, pump on/off/fault indicator lights, auto change over facility etc. Panel enclosure shall be to IP 65.
- 6.12 Pump operation shall be automatic controlled through float switches installed in the sewage collection pit.
- 6.13 Pumps shall be supplied with all accessories including foot bends, guide rails, stainless steel lifting chains, float switches, etc.

7.0 SUMP PUMPS

- 7.1 Sump pumps shall be vertical with fully floodable submersible motor pumps in close coupled design, single stage with level control complying with UL 778 or equal. Motor shall be jacket cooled single phase, IP 67 with integrated temperature switch, cable and shock proof plug.

- 7.2 Motor casing shall be cast iron steel absolutely tight and corrosion proof. Motor bearings shall be maintenance free, sealed for life.
- 7.3 Pump casing shall be cast iron, impact and abrasion resistant. Impeller shall be cast iron. Pump shaft shall be chrome steel.
- 7.4 Pump shall be supplied with control panel complete with IP 55 enclosure, integrated fuse elements, hand-off-automatic switch, motor contactor, duty pump selector switch, run / fail lamps for individual pumps etc. Pumps shall be controlled by a unit mounted float control.
- 7.5 Provide a horizontal swing non-return valve in the discharge from each pump. The valve shall be equipped with a lever weight for quick closure.
- 7.6 All guide bars, guide bar supports, fixings, lifting chains drop rods for fixing float switches, and pipe supports etc. within the sump shall be stainless steel.
- 7.7 The sump pump control panel shall have an audible alarm for the overflow level at sump pit.

EXTERNAL DRAINAGE

8.0 GENERAL REQUIREMENTS

- 8.1 Pipe connections to manholes, collection tanks and percolating pits shall be made in a completely watertight and approved manner.
- 8.2 Pipes shall be kept clean until final acceptance of the work. Exposed ends of all incomplete lines shall be closed with wooden plugs and adequately secured at all times when pipe laying is not actually in progress.
- 8.3 Pipes shall be installed on a good foundation and adequate means taken to prevent settlement. Pipes laid in trenches shall be provided with a solid uniform bearing throughout their entire length.
- 8.4 Pipes shall not be buried at less than 60 cm below finished grade for protection against mechanical damage. Pipes shall not be run closer than 1m to building bearing walls and footings for protection against building settlement.
- 8.5 All pipes shall be laid to a uniform slope. Slopes of sewer drain shall be limited to 3% maximum. The free vertical drop of a sewer pipe into a manhole shall be limited to 45cm between the invert level of the pipe opening and the bottom of the manhole. Where conditions necessitate that the drop would exceed 45cm at the maximum slope of 3% a drop manhole shall be used.
- 8.6 Trenches shall be kept free of water by pumping, use of well points, underdrains or other approved means during pipe laying operations so that all pipe joints are made in the dry.
- 8.7 Precautions shall be taken to protect incomplete work from floating due to storms or from any other cause. All pipelines or structures not stable against uplift during construction shall be well braced or otherwise protected.
- 8.8 All completed underground lines shall be subject to the inspection and approval of the Engineer. All pipes shall be true to line and grade. The full circle of the pipe shall be visible at the manholes.

9.0 INSTALLATION

- 9.1 The drains shall be laid truly straight in line and to an even gradient.
- 9.2 Excavations shall be made true and even to falls. The bottoms being trimmed to correct level and well rammed. Remove mud, rock projections, boulders and hard spots and replace with approved fill material well consolidated.

- 9.3 Minimum width of trench shall be 300mm greater than external diameter of pipe.
- 9.4 Any trenches excavated in error to a greater depth than required shall be in-filled back to the required level with concrete mix, at the Contractor's expense.
- 9.5 Before laying, all pipes and components shall be checked for defects and joint spaces cleared of dirt.
- 9.6 Socketed pipes to be laid with sockets uphill.
- 9.7 Flexible joints shall be made in strict accordance with manufacturer's instructions.
- 9.8 Where lubrication of the joint is required the pipe manufacturer's recommended lubricant shall only be used.
- 9.9 Spigot and socket cement joints where required shall be made with ring of tarred gasket well caulked in socket and remaining socket space filled with 1:3 cement mortar, finished in 45° fillet.
- 9.10 Lay and compact bed of granular material to provide 100mm thickness over the full width of the trench. Scoop out locally at pipe sockets where socketed pipes are used. Adjust pipes to line and level and ensure that pipe barrels rest uniformly on the bedding.
- 9.11 Add granular side fill uniformly up each side of pipes compacting by hand.
- 9.12 Any trench sheeting should be lifted before the fill is compacted.
- 9.13 The granular material shall be compacted in 100mm layers by hand up to 100mm minimum distance above top of pipes.
- 9.14 Granular material shall be locally available crushed rock graded in relation to pipe size as follows: -
- | | | |
|-----------------------------|---|---------------------------|
| 10mm nominal size aggregate | - | 100mm dia. pipes |
| 10- or 14-mm ditto. | - | 150mm dia. pipes |
| 10, 14 or 20mm. ditto. | - | 200mm dia. pipes & above. |
- 9.15 During bad weather, or in wet fine-grained soils such as clays, silts or sands, it is important to prevent the trench bottom being churned up by mean working in the trench. In such cases a blanket of granular material 75mm thick laid over the trench bottom immediately after excavation, or alternatively a sealing layer of weak concrete 50mm thick, is required.
- 9.16 Provide concrete bed Grade 'A' (CP 301 Table 1 Specified Works Cube Strength at 21 N/mm² at 28 days) to drainpipes and concrete infill (Concrete Mix 1:15) where drain track is below level of bottom of foundation base and within 900mm horizontal distance from foundation base. The concrete infill to be carried up to the level of the bottom of the foundation base.
- 9.17 Drain tracks beyond 900mm of adjacent foundations and where the bottom of the trench is lower than a depth beneath the foundation equal to the horizontal distance between the nearside of the trench and the foundation less 150mm, shall be infilled with concrete up to that depth.
- 9.18 Concrete bed and surround shall incorporate movement joints at each drainpipe joint.
- 9.19 Each joint shall form a plane surface in the concrete above and below the drain and vertical to the drain and centre line, and shall separate the lengths of concrete with 13mm thick resilient fibre board or expanded polystyrene pre-cut to pipe diameter and concrete cross-section height and width and left in site.

- 9.20 Drains below roadways, car parks, and any area subject to vehicular traffic where less than 900mm of cover shall be bedded and surrounded with concrete Grade 'A' (CP 301 Table Specified Works Cube Strength at 21 N/mm² at 28days) 150mm minimum thickness all round with provision for movement joints.
- 9.21 Granular fill shall be laid in 100mm layers and had compacted to a level 300mm minimum above top of pipe followed by main backfill material placed and compacted in 300mm layers any trench sheeting being withdrawn as the work proceeds. Heavy mechanical compactors shall not be used until there is at least 300mm cover over the pipes.
- 9.22 Construction vehicles shall not be allowed to cross drain trenches until the final surface is placed except where timber sleepers or steel plates are positioned to bridge the trench.

10.0 MANHOLES

- 10.1 All manholes shall be constructed to the requirements of the Drainage Department of Dubai Municipality.
- 10.2 Covers are to be in site concrete filled, with surface finish to match surrounding floor. All covers to be locking and sealed with a clear opening of 600mm x 600mm.
- 10.3 All covers shall be of ductile iron /cast iron and shall be painted with solvent free epoxy paint.
- 10.4 All manholes and gully traps shall be vented as per drainage department requirements. Gully trap covers shall have clear opening of 300 x 300 mm. All manhole covers and frames shall comply with BS EN 124. All covers shall be black bitumen coated ductile iron with fiber tray.

10.5 Internal Manholes:

All internal manhole covers (inside building) shall be bolt down airtight double seal recessed type, they shall be in situ concrete filled with surface finish to match surrounding floor. Covers shall be medium duty, locking and sealed.

10.6 External Manholes:

Covers in paved areas shall be medium duty, double seal and recessed type. They shall be in situ concrete/pavement filled with surface finish to match surrounding floor.

Covers subject to vehicular traffic shall be heavy duty, single seal, non-rock with square push fit seal plate completes with lifting key-holes.

GRP sealing plate of 6mm shall be provided for all sewer manholes.

- 10.8 Manholes to be protected by timber box of same internal dimensions to avoid collection of waste of building materials in manholes (i.e. concrete & sand) during construction period.
- 10.9 Covers and frames shall be fixed in such a way as to match adjacent surface levels with tolerances not exceeding + or - 3 mm. Before the maintenance period expires, check all manhole covers and readjust them if they are in excess of above-mentioned tolerances.

11.0 GULLY GRATES

Where indicated on drawings provide kerb gully unit (side inlet) of ductile iron hinged grate, road retaining bar with clear opening size 532x405mm. Unit to be black bitumen coated and shall be designed to deter ingress and help prevent blockage. A GRP grid shall be provided as standard.

Storm water / car park gratings shall be heavy duty ductile iron (side inlet) kerb gully, non-rock clear openings size as indicated on drawings, grey - epoxy coated and shall comply with BS EN 124.

12.0 TESTING OF DRAINAGE AND VENT PIPE SYSTEMS

- 12.1 Before the sanitary fixtures are installed, drainage pipe systems shall be subjected to a water pressure test to ensure and prove their tightness and to a flow test to ensure their freedom from obstructions.
- 12.2 The water pressure test shall be applied to the system in its entirety or in sections. All openings in the piping shall be tightly closed with special cast iron pipe plugs or other suitable means and the system filled with water to the point of overflow from the highest opening. The plugs shall be temporarily opened to make sure that all air has been vented and that water has reached all parts of the system.
- 12.3 No section shall be tested to less than a 3m head of water. In testing successive sections at least, the upper 3m of the next preceding section shall be tested so that no joint or pipe, except the uppermost 3m of the whole system, shall have been subjected to a test of less than a 3m head of water.
- 12.4 The water shall be kept in the system or in the portion under test for at least 4 hours before inspection starts. While the system is under pressure, a careful inspection shall be made of all pipes and joints and if any leaks in joints or evidence of defective pipe or fitting is disclosed the defective work shall be corrected immediately by replacing defective parts with new joints and materials. No makeshift repairs or application of any repair compound will be permitted.
- 12.5 After the correction is made the pressure test shall be repeated until the system is proved tight.
- 12.6 Underground drainage pipes shall be tested by plugging the end of the pipe and filling with water to a minimum head of 3m. The test pressure shall be maintained for 24 hours. Pipes and joints shall be inspected and approved before backfilling the trench.
- 12.7 All drainage systems shall be tested for proper flow to ensure their freedom from any obstruction. The Contractor shall disassemble, clear, repair and re-assemble obstructed piping at his own expense. After re-assembly the piping shall again be subjected to the pressure test.

END OF SECTION

MECHANICAL SECTION: 06

FIRE FIGHTING SYSTEM

1.0 FIRE FIGHTING SYSTEM**1.1 Scope of Work**

This specification covers the supply, delivery and complete installation on site and in full working order of the equipment specified in this document and/or shown on the drawing

The contract works shall include fire protection pipework and equipment as specified herein and as shown on the drawings. The complete system shall be installed by a Specialist Contractor approved by Local Civil Defence. All material used for firefighting works shall be approved by the Local Civil Defence with valid approval certificates.

Fire protection contractor shall carry out hydraulic calculation to confirm the pipe sizes and pump head.

Full particulars, performance curves and illustrations of the equipment, complying as closely as possible with this specification, shall be provided with the tender, but any deviations from the specification must be fully and clearly indicated in the tender submission schedules.

1.1.1 The Contractor shall provide and install the following:

- The complete fire pumping station, including the firewater pumps (pumps, diesel engines drive, pump controllers, skid & pump), the jockey pump (pumps, electric drive, electric drive controllers), the piping & hanging, the valves, check-valves, relief valves, fittings, the instrumentation, the storage fire water tanks and feeding controllers (valves & controllers) and fuel tank for diesel fire pump.
- The firewater mains (sprinkler, standpipe & hydrant), including the piping, the fittings, the hanging, the block-valves & the hydrants.
- The wet sprinkler systems, including the control valves, the piping, the hanging, the heads, the instrumentation.
- The hose reels (standpipes, hose connections, external hose streams).
- The portable extinguishers.
- Clean agent fire suppression system for the rooms mentioned in the drawings.
- The marking of all the safety equipment.

1.1.2 The Contractor shall be responsible for all the following aspects of the provided systems:

- The checking of the tender drawings and any calculations supplied with the drawings. The Contractor shall check each drawing and satisfy himself that the design is correct for the equipment offered and will meet all applicable rules and regulations
- The provision of any additional information required by the Contractor's installation engineers who are particular to the equipment being supplied e.g. fabrication drawings, equipment data sheets, commissioning instructions, etc.
- The provision of drawings and equipment data related to the systems as required by the Engineer.
- The supply of all required equipment and other fixtures associated with the systems.
- Shop testing of equipment and provision of inspection facilities.

- The provision and fittings of suitable wall boxes, transits and plates where pipe work or cable passes through walls, etc...as well as fire rating reinstatement materials.
- The supply of as built drawings and as built calculations.
- The provision of scaffolding, ramps and any other equipment required to enable the systems to be installed, tested and commissioned correctly.
- A demonstration to key personnel in the use and testing of the systems and equipment supplied.
- Operating and maintenance manuals.

1.2 Codes and Standards

All the systems shall be designed and installed in accordance with the requirement of International Codes and Standards and Local Authority having Jurisdiction. They shall be particularly in accordance with the following NFPA / BS EN / VdS standards:

- ❖ NFPA 10, Standard for Portable Fire Extinguishers
- ❖ NFPA 11, Standard for the Low-, Medium- and High-expansion foam
- ❖ NFPA 11A, Standard for Low-, Medium- and High-expansion foam system
- ❖ NFPA 13, Standard for the Installation of Sprinkler Systems
- ❖ NFPA 14, Standard for the Installation of Standpipe and Hose Systems
- ❖ NFPA 17, Standard for dry chemical extinguishing systems
- ❖ NFPA 17A, Standard for wet chemical extinguishing systems
- ❖ NFPA 20, Standard for the Installation of Stationary Pumps for Fire Protection
- ❖ NFPA 22, Standard for Water Tanks for Private Fire Protection
- ❖ NFPA 24, Standard for the Installation of Private Fire Service Mains and their Appurtenance
- ❖ NFPA-25, Standard for Inspection, Testing, and Maintenance of Water Based Fire Protection Systems
- ❖ NFPA-70, National Electric Code
- ❖ NFPA 72, National Fire Alarm and Signalling Code
- ❖ NFPA 75, Standard for the Fire Protection of Information Technology Equipment
- ❖ NFPA 92A, Standard for Smoke-Control Systems Utilizing Barriers and Pressure Differences
- ❖ NFPA-101, Life Safety Code
- ❖ NFPA-110, Standard for Emergency and Standby Power systems
- ❖ NFPA-170, Standard for Fire Safety and Emergency Symbols
- ❖ NFPA 204, Standard for Smoke and Heat Venting
- ❖ NFPA-220, Standard for Types of Building Construction
- ❖ NFPA-221, Standard for High Challenge Fire Walls, Fire Walls and Fire Barrier Walls
- ❖ NFPA-2001, Standard on Clean Agent Fire Extinguishing systems
- ❖ NFPA 5000, Building Construction & Safety Code
- ❖ OSHA, Code of federal regulations, Title 29, Part 1926
- ❖ BS EN 3 for portable fire extinguishers
- ❖ BS EN 1866 for mobile fire extinguishers
- ❖ BS EN 671-1 for fixed fire fighting systems

- ❖ BS EN 54 Standard for fire detection and fire alarm system.

1.3 Pipes and Fittings

1.3.1 Pipe works

- 1.3.1.1 All above ground pipes shall be of factory galvanized steel to ASTM A53, Sch. 40, Welded steel pipes (ERW) and fittings.

1.3.2 Threaded Fittings

Fittings for 2" diameter and below shall be 300psi (malleable iron), galvanized & threaded. They shall be UL listed and FM approved.

1.3.3 Grooved Fittings

Fittings 2-1/2" diameter and above shall be galvanized mechanical grooved type. These fittings shall be UL listed and FM approved with minimum working pressure of 300psi.

1.3.4 Pipe Hangers

All hangers shall be made of steel. Hangers shall be Galvanized, and UL/FM approved. The spacing of the hanger supports shall comply with NFPA requirement.

- 1.3.4.1 Pipes supports shall be designed to maintain their integrity in case of fire exposure.
- 1.3.4.2 All Above ground piping / equipment's hangers, supports and sway braces shall be fabricated in accordance with ANSI B31.1. The Hangers, rod supports and supporting devices and accessories shall be capable of supporting at least twice the live loads of the product being supported. Fire pipe riser /support shall be approved by DCD.
- 1.3.4.3 Hangers shall be adjustable clevis type, of yoke and lower U strap design with cross bolt and manufactured from galvanised steel. Cross bolt shall have a double locking nut. Vertical adjustable supporting rod shall have a load nut below the yoke and a locking nut above the yoke.

1.4 Fire Protection Service Valves

All Fire protection service valves (Gate valve, Check valves & Butterfly valves) shall be from one source.

Minimum Pressure rating for Standard-Pressure Piping: 200psi (1200kPa)

Minimum Pressure rating for high-Pressure Piping: 300psi (2070kPa)

All valves and related components shall be rated for 200psi 300psi, depending on location of valves and actual pressure conditions at site.

1.4.1 Gate Valve

Gate / Isolation valve shall be O S & Y type or non-rinsing type as indicated in the drawing.

Sizes up to 2" and below shall be threaded type. The valve body shall be cast bronze and wheel shall be cast iron. The valve shall be UL listed and rated for 200 PSI working pressure.

Sizes 2-1/2" and above shall be cast Iron O S & Y / non-rising gate valves as and where indicated in the drawing. The valve body shall be cast iron. The valve shall be UL listed & FM approved and rated for 200 PSI minimum working pressure.

Use non-UL listed valves where applicable (where pressure exceeds UL listing range).

Isolating valves shall be UL 262 Listed and shall be fitted to the installation in order to shut the valve, the spindle must turn clockwise. The hand wheels of all stop valves shall be clearly marked to indicate which direction the wheel is to be turned to close the valve. An indication shall also be provided which shows whether the valve is open or shut.

Each main isolation valves shall be installed with a tamper / supervisory switch and interconnected to the building fire alarm system or secured open by a padlock or riveted strap.

1.4.2 Check Valve

Check valves shall be swinging type or split clapper type with flanged / grooved ends as required by the drawing. The valve body shall be cast iron. The valve shall be UL listed & FM approved and rated for 200 PSI minimum working pressure.

Use non-UL listed valves where applicable (where pressure exceeds UL listing range).

1.4.3 Butterfly Valve

Butterfly valves shall be wafer type / grooved type with tamper switch. The valve body shall be made of ductile Iron EPDM encapsulated / Nylon-11 coated. The valve shall be UL listed & FM approved and rated for 200 PSI minimum working pressure.

1.4.4 Pressure Reducing Valve

Use class 150 / 250 / 300 pressure reducing valve, as per the drawing / requirements, where required. The body material shall be cast iron / cast steel as applicable. The valve shall be externally site adjustable. The disc seat / main disc piston shall be stainless steel.

1.4.5 Air Release Valve

Air release valve shall be made of copper alloy to BS EN 1982. The valve shall be 1" size and tested to 350 psi.

1.5 Fire Protection System Specialty Equipments

1.5.1 Fire Hose Cabinets:

Fire hose cabinets shall be provided where shown on the drawings in accordance with this Specification.

1.5.1.1 Recessed Type Fire Hose Cabinet

Provide recessed type fire hose cabinet where shown in the drawing. The recessed fire hose cabinet shall be a masonry enclosure to accommodate the fire equipment with stainless steel door.

Door and architrave shall be made of stainless steel with finish as indicated in the drawing and to the approval of the engineer/ ID.

1.5.1.2 Surface Type Fire Hose Cabinet:

Provide surface type cabinet where shown in the drawing. The cabinet shall be in mild steel. Door, architrave and back box shall be of electro-galvanized mild steel painted with red electrostatic powder coating where mild steel cabinet is indicated in the drawing. They shall be manufactured by ISO 9001 Company duly certified by LPCB / BVQI and approved by Local Civil Defence. The Fire cabinet shall house the fire equipment as shown on details drawings and specified below.

1.5.2 Automatic Hose Reels:

Provide and install as indicated on the drawings automatic swinging hose reels. The reels shall comply with BS EN 671 and comprise a reinforced rubber hose, 30m long x 25 mm. The outer end of a hose shall be fitted with a “jet / spray / shut-off” control nozzle. The inner end of the hose shall be connected through the reel to the water supply.

The hose reel drum shall be made of 1.2 pre-treated electro-galvanized steel sheets, red painted with electrostatic powder coating. Hose reels shall have additional rib reinforcement to give extra strength & to avoid twisting and swing through 180 degrees. The hose shall have two synthetic rubber layers and textile reinforcement. The working pressure of the hose shall be 12 bar minimum and bursting pressure shall be 50 Bar. The hose nozzle shall be brass chrome plated/ with (JET/SPRAY/SHUT OFF) type.

The Fire hose reel shall be approved by the Civil Defence Department.

Each hose reel shall deliver not less than 24 LPM when the two remotest reels are in use simultaneously, providing a jet of approximately 6 m in length. Pressure reducing devices shall be provided whenever necessary to limit the pressure to 4.5 bar.

Downstream of all pressure reducing valves inside the fire hose cabinet shall be fitted with approved UL listed pressure gauge with stopcock.

Provide the following where automatic hose reel is indicated in the drawing: The reels shall incorporate a valve that automatically turns on the water supply when more than 3 m of hose is unreeled and shuts off again when the hose is re-wound onto the reel. A manual shut-off valve, lock shield type shall also be installed on the service pipe to each hose reel.

An inspection plate shall be supplied with each reel indicating the method of use in English and Arabic.

Brief operating instructions should be clearly displayed on or by the side each hose reel e.g., for automatic hose reel

“TO OPERATE”

- 1) RUN OUT HOSE
- 2) TURN ON CLOCK AT NOZZLE.

1.5.3 Landing Valve

Landing valves shall be installed inside the building, for the use of fire brigade, where shown on the drawings. The valve shall be of brass / Gunmetal construction with approved finish. Landing valves shall be oblique / straight pattern, brass, 65mm size, bore fitted with 65 mm instantaneous female coupling conforming to BS 336 and have a brass blank cap secured by a suitable length of chain. The hand wheel shall be provided in each landing valve with clear marking of OPEN and CLOSED directions (opening shall be anti-clockwise). The whole valve fittings shall be sound construction and shall have a minimum working pressure of 200 PSI. Provide landing valve with integral PRV to limit the pressure to the requirements of the Local Civil Defence, where shown in the drawing. The Landing valve shall have UL & FM approval or BSI KITE Mark approval.

1.5.4 Fire Hose

Fire hose UL listed, synthetic of 65 mm diameter and 30 m long equipped with male coupling for the connection with the landing valve shall be provided where indicated in the drawing. The hoses shall have following construction features: EPDM rubber lining, Jacket of polyester wrap and filtered

thread. Fire hose shall have minimum working pressure of 15 Bar and bursting pressure of minimum 45 Bar.

1.5.5 Fire Brigade Inlet

The inlet breeching piece shall be of horizontal pattern and any of the following types, as indicated in the drawing. It shall be FOUR WAY with 150 mm flanged outlet and 4 nos. of 65mm instantaneous male inlets, complete with blank caps and chains.

The breeching inlet shall have a non-return (check) valve to protect the inlet, couplings complying with BS336 and 25 mm drain valve to drain the raiser. The breeching inlet shall be BSI KITE/LPCB approved or UL Listed/FM approved.

All parts including valve body, chain and caps shall be brass / gunmetal if it is UL Listed and valves body shall be made of ductile iron body with brass inlet valves polished chrome plated if it is BSI kitemark/LPCB approved.

The breeching inlet shall be enclosed in an approved breeching inlet cabinet (Cabinet shall be of mild steel construction) with wired glass glazed front panel. They shall be constructed in such a way the operation shall be possible from inside without the aid of a key, after the glass has been broken. It should be large enough for hose to be connected to inlets even if the door cannot be opened and the only means of access is by breaking the glass.

The breeching piece shall be conspicuously indicated by the words: "FIRE BRIGADE – WET MAIN INLET" or "WET RISER INLET" in block letters on the inner face of the glass, both on English and Arabic.

1.5.6 PORTABLE FIRE EXTINGUISHERS

Provide and install fire extinguishers as indicated on the drawings.

1.5.6.1 Carbon Dioxide Fire Extinguisher

Carbon dioxide fire extinguishers shall be manufactured in accordance with BS EN 3 (portable) and shall be operated by means of a lever / squeeze handle operated valve provided with a safety pin which shall avoid inadvertent operation and the valve shall be capable for controlling partial discharge. The cylinder shall be made of high grade seamless carbon steel pipe and it must be coated externally by electrostatic polyester powder quating and oven baking to resist corrosion. The cylinders shall be free from leakage during hydrostatically pressure tested to 250bar. The temperature shall range from -20°C to 60°C.

The suction tube shall be Aluminium. All extinguishers shall carry instructions for operation and maintenance in English and Arabic.

1.5.6.2 ABC Dry Powder Fire Extinguisher:

Dry powder fire extinguishers shall be manufactured in accordance with BS EN 3 / NFPA 10 for portable extinguishers and shall be operated by means of a lever/squeeze handle operated valve provided with a safety pin which shall avoid inadvertent operation and the valve shall be capable for controlling partial discharge. The cylinder shall have minimum working pressure of 14bar. The temperature shall range from -20°C to 60°C. It must be coated externally by electrostatic polyester powder and oven baked for corrosion resistance. The cylinder shall be made special cold rolled steel sheet with Argon/CO₂ automatic welding process and the suction tube shall be made of PVC material. All extinguishers shall carry instructions for operation and maintenance in English and Arabic.

1.5.7 Fire Blankets

Provide fire blankets in Kitchens where indicated in the drawings. Fire blanket shall be manufactured from 430 gm/m² glass fiber material which shall withstand temperature up to 550°C. The size shall be 4 feet x 4 feet and it shall have Local Authority approval.

1.6 LOCATION PLAN

The MEP sub-Contractor shall provide and fix in an approved position a plan of the risk upon which shall be clearly indicated the location of the main stop valves.

1.7 VALVE LABELS AND CHARTS

The MEP sub-contractor shall label each valve. The number shall relate to a framed glass covered valve chart to be supplied and installed in each sprinkler chamber / fire pump house, as required by the engineer in charge. The charts shall indicate the function of the valves and shall be to the Civil Defence and engineers' approval. No two valves on the system shall bear the same number.

1.8 CLEANING AND TESTING

Tubes and all items of equipment shall be delivered, stored and maintained in closed storage with their open ends effectively plugged, capped or sealed. All fittings, valves and sundry items shall be stored in clean bins or bagged and stowed in suitable racks. All such stored items shall be maintained under weatherproofed cover to be supplied by the MEP sub-contractor until they are ready for incorporation in the works. Particular care shall be taken to ensure that electrical equipment and components are kept clean and dry.

Before installations are handed over or subjected to the inspection and tests, the entire installation shall be thoroughly cleaned, both internally and externally.

All fire protection installations shall be flushed out with clean water. During the flushing out, provision shall be made to exclude any items of plant, which could be damaged by the cleaning operation. The entire operation shall be carried out to the satisfaction of the Engineer.

The fire protection system shall be tested in accordance with the NFPA codes and the relevant standards, with particular reference the requirements set forth in this section:

All testing shall be performed under the work of this section. All services required for testing shall be a responsibility of the work of this section. The MEP sub-Contractor shall notify the Engineer of all tests, 48 hours prior to testing.

The fire protection piping System shall be tested hydrostatically for not less than 2 hours and at one time of the system pressure without any leaks or as directed by the local fire department.

If the systems are tested in sections, the connection to the previously tested section shall be included.

All tests that may be required by the local fire authorities shall be performed under the work of this section, in the presence of their representative, if so required.

All leaks shall be corrected, and the system re-tested until no leaks are found, at no cost to the client.

Functional tests of all water flow alarms are a responsibility of the works of this Section.

When the various systems are completed, operation tests shall be run on all equipment to demonstrate proper operating conditions. These tests shall be run under the observation of the inspector and the client's operating engineer. The client's operators shall be instructed during this period. Operation tests shall be performed under the work of this section.

Should any piece of equipment or apparatus of any material or work fail in any of these tests, it shall be immediately removed and replaced by a perfect material, and the portion of the work replaces shall again be tested under the work of this Section.

Cost to repair any damages to the building construction occasioned by pipe leaks or defective materials shall be become under the works of this section, at no cost to the client. All corrective work shall commence immediately after damage occurs.

2.0 AUTOMATIC SPRINKLER PROTECTION SYSTEM

2.1 GENERAL

The design of the sprinkler system indicated on the drawing shall be installed by the contractor strictly in accordance with NFPA – 13 and the requirements of the local Civil Defence. Spacing of sprinklers shall suit the hazard of the occupancy as shown in the drawing.

The system shall be complete with automatic fire pumping station and valves, water motor alarm and gong, pressurized water mains, main distribution pipes, distribution pipes, range pipes, hangers and supports, sprinkler heads and main drain system.

All isolating valves related to water tank, pump room and main distribution line shall be OS and Y type and shall be monitored by fire alarm system for open / close position.

2.2 MAIN CONTROL VALVES

The system shall be provided with a set of installation control valves, main stop valve, alarm valve and test valve.

The main stop valve should be strapped and double padlocked in the open position.

The alarm valve should be closed in the static position with the pressure below and above the valve, at normal level and no water flow. In the event of a pressure drop above the valves due to activation of a sprinkler head, water is admitted to the branch pipe connected to the water motor alarm and the gong.

2.3 SPRINKLER HEADS

Pressure Rating for Residential Sprinklers: 175 psig (1200kPa) maximum. Where there is exceed in system pressure than the rated above rated pressure, an approved UL listed pressure restricting device shall be provided.

The glass bulb sprinkler heads shall be quick response type. The type of sprinkler head shall be as shown on drawings. Semi recessed sprinkler head shall be used in general for all false ceiling areas, rated for 68°C (154°F).

Concealed type where shown on drawings shall be with face plate, that to match with ceiling finish.

Pendant type where shown on drawings shall be rated for 68°C (154°F).

Upright type sprinkler head shall be provided for mechanical plant rooms and car parking areas (where false ceiling is not provided) and rated for 79°C (174°F). Also, upright sprinkler heads shall be provided for ceiling voids where electric water heater is provided (as per UAE Fire Code), rated for 68°C (154°F).

For the rooms where no false ceiling, side wall type (extended coverage type as required) rated for 68°C (154°F). These shall be located such that there is no interference with the discharge pattern and to be maintained at 75-150mm below ceilings.

For Sauna & Steam rooms, high temperature sprinkler heads (135 C) shall be provided.

For other types of sprinkler heads refer description on the sprinkler layout design drawings.

The sprinklers and lines shall not be spaced too close together, in order to prevent an operating sprinkler from wetting and thereby delaying operation of adjacent sprinklers.

In locations where sprinkler heads are liable to be operated or damaged by accidental blows, they should be protected by stout metal guards, care being taken to ensure that the normal operation of the sprinkler head in the event of fire is not thereby impeded.

The contractor shall supply adequate quantity of spare sprinkler heads (10nos of each type used in the project) with necessary tools of the correct temperature rating to the employer for use during an emergency. These spare sprinklers boxes shall be stored in pump rooms.

Sprinkler heads shall be installed strictly as per the manufacturer instructions and in compliance with NFPA standards and UAE Fire Code latest edition.

2.4 VALVES

2.4.1 Ball valves:

50mm and smaller: Bronze, 2-piece, threaded or sweat ends, standard port, blowout proof stem, chrome plated ball, and glass reinforced seats, UL approved at with minimum rating PN20 or higher as per site requirements.

2.4.2 Gate valves:

50mm and smaller: Outside screw and yoke gate valves, 1,400kPa (14bar), bronze body, bronze mounted, screwed bonnet, rising stem, solid wedge, with normally open supervisory/tamper switch with double wire leads.

65mm and larger: Outside screw and yoke gate valves, 1,400kPa (14bar), cast iron body, bronze mounted, bolted bonnet, rising stem, solid wedge, with normally open supervisory/tamper switch with double wire leads and rising stem.

EPDM resilient seat, EPDM seals, nickel plated ductile iron disc. Valve assembly to be bubble tight to 14bar with no downstream flange/pipe attached. Use cap screws for removal of downstream piping while using the valve for system shutoff.

2.4.3 Supervisory/tamper switches:

For O S & Y valve or butterfly valve installations, UL/FM listed/approved, to monitor position of valve, tamper resistant cover screws, single or double SPDT switch contacts, corrosion resistant, for indoor or outdoor use, NEMA 4 & 6P enclosures.

2.4.4 Spring loaded check valves:

50 and smaller: Bronze body, threaded ends, bronze trim, stainless steel spring, stainless steel centre guide pin, 1,400kPa (14bar) teflon seat unless only bronze available.

65 and larger: Cast or ductile iron body, wafer or globe type, bronze trim, bronze or EPDM seat, stainless steel spring, stainless steel stem if stem is required, 16bar pressure rating.

2.4.5 Pressure-Restricting Valves:

The valve shall be rated for 2760kPa minimum, brass construction. Include, female NPS inlet and outlet; adjustable setting feature; and straight or 90° angle pattern design. All fittings from fire pumps

to the pressure restricting valve shall be selected to have higher pressure rating than the pump shut off pressure, i.e. 140% of total system pressure.

2.4.6 Drain valves:

20mm min. two or three-piece bronze body ball valve; threaded ends, chrome plated bronze ball; glass filled teflon seat; teflon packing and threaded packing nut; blowout-proof stem; with minimum rating PN20 or higher as per site requirements with hose thread outlet and cap.

2.4.7 Alarm check valves:

Cast or ductile iron body, flanged or grooved ends, 1,600 kPa (16 bar) minimum working pressure, bronze grooved seat with O-ring seal, single hitch pin and latch design. Provide trim for bypass, drain, electric sprinkler alarm switch, pressure gages, precision retarding chamber, and drip cup assembly piped to floor or hub drain, fill line attachment with strainer.

The alarm check valve shall be UL 753 listed conforming to NFPA-13. Retarding device of approved design to obviate false alarms due to mains pressure variations where necessary or where indicated. Alarm valves shall be complete with Water Motor Alarm Gong, all required auxiliary valves and drains and with pressure switch for wiring into the building electric alarm system.

Factory assembled water motor shall consist of body, impeller, gasket and cover. Body shall be cast from highly corrosion resistant aluminium magnesium alloy.

2.5 FLOW SWITCHES

A vane type water flow switch with metal enclosure with adjustable pneumatic retard and electrical characteristics to be compatible with alarm system. Flow switches shall be UL/FM approved.

2.6 PRESSURE SWITCHES

Pressure actuated switch with field adjustable settings, metal enclosure and electrical characteristics compatible with alarm system. Pressure switches shall be UL/FM approved.

2.7 LOCAL ALARM BELL

Weatherproof electric alarm bell with red painted metal housing, mounting base and gong; solenoid operator; weatherproofing O-ring seal and electrical characteristics compatible with alarm system. Alarm bells shall be UL/FM approved.

2.8 PRESSURE GAUGES

Stainless steel of not less than 9 cm in diameter, double strength glass window, black lettering on a white background, phosphor bronze bourdon tube with bronze bushings, recalibration from the front of the dial, 99% accuracy over the middle half of the scale, 99.5% accuracy over the remainder of the scale. Include bronze 3-way globe valve with plugged outlet for Fire Inspector's test gauge. Pressure gauges shall be UL/FM approved.

2.9 FIRE PUMP TEST METER

Fire Pump Test Meter, UL/FM approved, incorporating a calibrated venturi and attached GPM meter, to be installed on the discharge side of the fire pump, to accurately measure pump performance. Test Meter shall be supplied with grooved ends for installation with grooved end couplings.

3.0 FOAM WATER SPRINKLER SYSTEM

The MEP sub-contractor shall include in his tender for the complete installation, testing and commissioning of the foam water sprinkler system as detailed on the tender drawings and in this specification, to the standards and approval of Local Civil Defence. The MEP sub-contractor shall include for all labour, materials, tools and tackle, complete with all supports and fixings back to approve building structural members. The MEP sub-contractor shall be responsible for all dimensions and location of other services, and the correct setting out of the installation.

3.1 GENERAL

The generator room, fuel tank room and all other areas indicated in the drawing shall be protected with a foam water sprinkler system to match the requirements set forth in this specification, the codes mentioned below and to the requirements of Local Civil Defence authority.

The MEP sub-contractor shall employ a specialist contractor approved by Local Civil Defence to design, supply, install, test and commission the foam water sprinkler system.

3.2 DESIGN CODES

NFPA 16
NFPA 13
UL 162.

3.3 SUBMITTALS

Product Data: For each type of product indicated. Include the following:

- 1 Piping, valves, fittings, and hangers.
- 2 Bladder tanks and proportioning devices.
- 3 Foam concentrate.
- 4 Discharge devices. Include flow characteristics.
- 5 Monitoring and alarm devices. Include electrical data and supervision method.
- 6 Strainers.
- 7 Test connections.
- 8 Auxiliary details and information such as working plans prepared according to NFPA 16 and approved by authorities having jurisdiction. Include hydraulic calculations complying with NFPA 13.
- 9 Schematic wiring diagrams for all components, clearly indicating all required field electrical connections.
- 10 Design Calculations: For amount of foam concentrate required for each hazard area.
- 11 All information necessary for the complete supply and installation of the proposed foam water sprinkler system.
- 12 Hydraulic calculations for piping to ensure adequate pressure for system operation.

3.4 SYSTEM DESCRIPTION AND OPERATION

Engineered, fixed, deluge, automatically actuated, low-expansion, AFFF fire-extinguishing system for flammable-liquid fires. System includes diaphragm proportioning tanks and devices as described in NFPA 16. Section Includes:

1. Concentrate piping and piping specialties.
2. Bladder tanks and proportioning devices.
3. Foam concentrate.
4. Discharge devices.
5. Monitoring and alarm devices.

AFFF: Aqueous film-forming foam.

AR-AFFF: Alcohol-resistant aqueous film-forming foam.

Standard Working Pressure of Piping-System Component: Listed for at least 175 psig (1200 kPa).

Unless authorities having jurisdiction have stricter requirements, minimum design parameters are as follows:

1. Solution: 3 percent foam-water solution.
2. Sprinkler Spacing: Maximum coverage area of 100 sq. ft. (9.5 m²) per sprinkler, and maximum 12-foot (3.7-m) spacing.
3. Design Density: Minimum 0.16 gpm/sq. ft. (0.108 L/s per m²).
4. Foam Supply: Minimum 10-minute discharge time.
5. Water Supply: Minimum 60 minutes.
6. Remote Area: Open-sprinkler systems shall discharge over the entire system area.
7. Sprinkler Temperature Rating: Maximum 250 to 300°F (121 to 149°C) at a roof or ceiling, and 135 to 170°F (57 to 77°C) for intermediate sprinklers.

Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

Discharge Devices: Not less than 20 percent of amount of each type installed.

Foam Concentrate: Not less than 200 percent of amount installed.

Specialties shall comply with NFPA 16, be compatible with the foam concentrate, and be designed to be drained and cleaned.

Foam-Concentrate Storage Tanks: Buna-N, bladder-type proportioning tank complying with UL 162 and ASME Boiler and Pressure Vessel Code: Section VIII; designed for use with foam-concentrate pumps and for specific type of foam concentrate used. Include bladder, internal piping, fill and drain, glass sight gage, piping, and valves. Contain concentrate in the bladder.

Foam-Concentrate Pumps (Where applicable): Listed for use in foam-water systems according to NFPA 20. Include supply side pressure relief valve and drain cock or valve.

Proportioning Controllers: Venturi type complying with UL 162 and of capacity to match design at minimum and maximum flow.

Concentrate Control Valves: Water-operated ball or deluge valve designed to open with flow through the proportioning controller.

Concentrate Strainers: Bronze body and stainless-steel mesh strainer with minimum 0.125-inch (3.2-mm) perforations to remove solids that would block system components.

Pressure Gages: Comply with UL 393; with 3-1/2-inch (90-mm) minimum-diameter dial, 0- to 300-psig (0- to 2070-kPa) dial range, and caption "WATER" or "CONCENTRATE" on dial face.

R-AFFF liquid concentrate, complying with NFPA 11 and UL 162, for making foam-water fire extinguishing foam solution.

Discharge devices shall be UL listed or FM Approved.

Sprinklers: Open, air-aspirating type; UL listed or FM Approved and listed for use with type of foam concentrate used.

Spray Nozzles: For foam water; include foam generator and distributing deflector to distribute foam or water.

Valve Supervisory Switches: Single pole, double throw, with normally closed contacts complying with UL 753. Switch shall signal fire-alarm panel or releasing panel when valve is in other than fully open position.

Pressure Switches: Single pole, double throw, UL listed or FM Approved and complying with UL 753. Switch shall signal an alarm condition at the fire-alarm panel or releasing panel when switch is in other than fully open position.

Flow Switches: Single pole, double throw, UL listed or FM Approved and complying with UL 753. Switch shall signal an alarm condition at the fire-alarm panel or releasing panel when switch is in other than fully open position.

3.5 INSTALLATION

Install tanks level and plumb, according to layout drawings, original design, and referenced standards. Maintain manufacturer's recommended clearances. Arrange units so controls and devices needing service are accessible. Install seismic restraints for tanks only if mandated by authority.

Install flanges, flange adapters, or couplings for grooved-end piping on piping, valves, apparatus, and equipment having NPS 2-1/2 (DN 65) and larger end connections.

Ream ends of pipes and tubes and remove burrs.

Remove scale, slag, dirt, and debris from inside and outside of pipes, tubes, and fittings before assembly.

Dissimilar-Material Piping Joints: Make joints using adapters compatible with materials of both piping systems and with foam concentrate.

3.6 CHARGING THE SYSTEM

Fill proportioning tanks with foam concentrate after field quality-control testing is complete and satisfactory results have been achieved.

3.7 TESTS UPON COMPLETION

Engage the services of a qualified professional engineer to inspect installed fire-extinguishing systems, prepare installation report, and certify that installation complies with the Contract Documents, calculations, and requirements of authorities having jurisdiction.

4.0 PRE-ACTION SPRINKLER SYSTEM

The MEP sub-contractor shall include in his tender for the complete installation, testing and commissioning of the pre action sprinkler system, detailed on the tender drawings and in this specification, to the standards and approval of Local Civil Defence. The MEP sub-contractor shall include for all labour, materials, tools and tackle, complete with all supports and fixings back to approve building structural members. The MEP sub-contractor shall be responsible for all dimensions and location of other services, and the correct setting out of the installation.

4.1 GENERAL

The lift machine rooms and or all other areas indicated in the drawing shall be protected with a pre action sprinkler system to match the requirements set forth in this specification, the codes mentioned below and to the requirements of Local Civil Defence authority.

The MEP sub-contractor shall employ a specialist manufacturer to design, supply, install, and test and commission the pre action sprinkler system.

4.2 SUBMITTALS

After award of contract and prior to fabrication, Fire Protection System Contractor shall submit to the Engineer complete design submittals for approval. Design submittals package shall include the following:

Product Data: Manufacturer's catalogue data sheets containing details of technical information, system and component operating instructions for the following;

Sprinklers heads, escutcheon and guards, including sprinkler flow characteristic, temperature rating, finish and mounting orientation.

Listed alarm / monitoring switches that include size, pressure rating, electrical rating, etc.

Schematic wiring diagrams for all components, clearly indicating all required field electrical connections.

All information necessary for the complete supply and installation of the proposed pre action sprinkler system.

4.3 SYSTEM DESCRIPTION AND OPERATION

A double interlocked pre-action system shall be provided with the method of release being an electric solenoid valve and a pneumatic actuator. The electric solenoid valve will open upon activation of the electrical supplemental detection system. The pneumatic actuator shall open upon loss of supervisory air due to activation of a sprinkler. Both devices must activate before the deluge valve will open. Opening of the deluge valve shall not be dependent on release device order of activation. The riser shall be a listed and approved assembly provided with all necessary appurtenances to complete the system. A rubber seated check valve shall be installed downstream of the deluge valve and prior to the supervisory air connection.

System shall be installed in accordance with drawing and specifications, however, no requirements of NFPA 13 shall be violated. Material and labor necessary to fully comply with drawings and specifications and with rules, regulations, ordinances shall be furnished by the Contractor.

Scope of work:

1. Hydraulic design of sprinkler system
2. Shop drawings
3. Riser Assembly
4. Air Compressor
5. Release Control Panel
6. Smoke/Heat Detectors

4.4 INSTALLATION & CHARGING

- A. Supply and install a pre-action sprinkler system assembly coordinated with the releasing panel. The unit shall be UL listed and FM approved. All system components shall be tested and listed for their use together.
- B. The system shall be a double interlocked pre-action with electric-pneumatic release containing all hydraulic, and pneumatic components required for the control of a pre-action system. The system shall be fully interfaced with the releasing panel
 - 1. Deluge Valve complete with Schedule 40 galvanized steel Trim rated at 250 psi.
 - 2. Release trim with solenoid valves, pneumatic actuator and each supervisory device required.
- C. Air Supply: Air compressor and supervisory trim shall be provided and adjusted for the selected configuration. The air supply must be regulated and of the proper size in order to be able to restore normal system air pressure within 30 minutes.
- D. System Drain: The drain collector of the system shall be connected to an Open Drain. The drain piping shall not be restricted or reduced and shall be of the same diameter as the drain collector.

4.5 TESTS UPON COMPLETION

Engage the services of a qualified professional engineer to inspect installed fire-extinguishing systems, prepare installation report, and certify that installation complies with the Contract Documents, calculations, and requirements of authorities having jurisdiction.

5.0 FIRE PROTECTION PUMP SET**5.1 SCOPE OF WORK**

The MEP sub-contractor shall include in his tender, for the complete design, supply, installation, testing and commissioning of fire pump set. The fire pump set shall serve the fire protection systems indicated in the drawing, required as per the specification and acceptable to the Local Civil Defence.

5.2 GENERAL

The fire pump shall be a fully automatic unit consisting of electric motor driven pump to act as duty pump, diesel engine driven pump to act as standby pump and electric motor driven jockey pump for maintaining the pressure in the system. All the equipment shall be suitable in all respects for the local climatic conditions.

- 1. Fire pumps shall be designed in accordance with UL 448.
- 2. All the accessories of the pump set shall be as per NFPA 20.
- 3. Fire pumps shall be UL / FM approved.
- 4. ESL certificate for the Fire pump shall be provided by the Fire fighting specialist contractor as per Civil Defence requirements.

5.3 DEFINITIONS**5.3.1 Fire Pump Unit:**

UL / FM approved pump set consisting of an electric and diesel driven fire pumps, driver, controller and accessories

5.3.2 Fire Pump Design:

Centrifugal pump characterized by a housing that is split parallel to the shaft or vertical end suction.

Vertical turbine pump shall be provided if the water tank is below the pump room.

5.3.3 Jockey Pump

An electric motor driven pump used to maintain water pressure in fire protection systems, vertical multistage pump.

5.3.4 Fire Pump Controller:

A group of devices that serve to govern, in some predetermined manner, the starting and stopping of the fire pump driver as well as monitoring and signalling the status and condition of the fire pump unit.

5.4 SYSTEM PERFORMANCE REQUIREMENTS

Provide fire pump units, jockey pump complete with controller associated piping as shown on tender drawings. All system components shall be rated for a pressure of minimum and shall exceed the system working pressure / shut off pressure.

5.5 QUALITY ASSURANCE**5.5.1 Manufacturer's Qualifications**

Manufacturing firms regularly supplying fire pump units and whose products are listed by product name and manufacturer in UL's "Fire Protection Equipment Directory", FM's "Fire protection Approval Guide", approved by Civil Defence Authorities and comply with other requirements indicated. Jockey pump shall not be UL listed

5.5.2 Code Compliance

All the equipment and installation shall comply with NFPA 20 "standard for the installation of stationary pumps for fire protection".

All electrical equipment and installation methods shall comply with NFPA 70 "national electrical code", article 695.

5.5.3 Manufacturer's Tests:

Furnish test certificates for the pumps provided. This should match to the condition for which the system is designed for.

5.5.4 Submittals

Provide the following in the post order submittals.

1. Product for main fire pump, standby fire pump, jockey pump, their controllers and accessories.
2. List of special tools and spares.
3. Shop drawing indicating dimensions, weights, required clearances and methods of assembly of components.
4. Wiring diagrams of control panels clearly indicating field connections.
5. Characteristic curve of all pumps.
6. Hydraulic calculation to verify proper selection of pump head.
7. Material approval certificates from local Civil Defence.

5.5.5 Delivery, Storage and Handling:

Store fire pump units, jockey pump and its controller in clean dry place to protect them from weather, dirt, fumes, water, construction debris and physical damage.

Comply with manufacturer's rigging and installation instruction for unloading and moving them to final location.

5.6 MAIN FIRE PUMPS & CONTROLLER

5.6.1 Fire Pumps:

Provide horizontal split case or vertical discharge end –suction pumps complete with drivers to the capacity as indicated on drawing.

Fire pumps shall be capable of furnishing not less 150% of rated capacity at not less 65% of rated head. Shut off head shall not exceed 140% of rated head.

Entire Fire pump unit shall be painted to red colour before shipping.

Pump's shaft shall be stainless steel with bronze sleeve.

Wear rings shall be replaceable type, bronze.

Provide name plate with all relevant information.

5.6.2 Motor Drivers:

Provide NEMA MG1, TEFC squirrel cage induction motor, with service factor 1.15 for main fire pump.

Electric motor of the main fire pump shall be UL listed.

The locked rotor current shall not exceed the values specified in NFPA 20.

Terminal and wiring of the motor shall be compatible with controller of the pump.

5.6.3 Diesel Engine Driver:

Provide an approved heat exchanger cooled diesel engine as driver for standby fire pump.

Diesel engine shall be provided with following accessories.

1. Facility for manual operation.
2. Cooling system with all accessories.
3. Dual batteries.
4. Fuel system. Size of the fuel tank shall be as NFPA 20 requirements.

5. Silencer.

The exhaust pipe shall be flexibly mounted, insulated and further protected with hammer finish aluminium cladding.

5.6.4 Pump Accessories:

The following accessories shall be included with each fire pump unit.

- A. Automatic air release valve.
- B. Casing relief valve for electric motor driven pump
- C. Suction and discharge pressure gauges for all pumps
- D. Eccentric tapered suction reducer
- E. Concentric discharge increaser.
- F. Flow meter.
- G. Main relief valve
- H. Waste cone closed type.

5.6.5 Electrical Motor Controller:

The fire pump's controller shall be completely assembled and tested at the factory, listed (UL listed / FM approved) for electric motor driven fire pumps, primary resistance type, with auto transfer switch and have the following features:

1. Isolating means and circuit breaker.
2. "Power on" Pilot lamp.
3. "Pump running" Pilot lamp.
4. "Power failure" Pilot lamp.
5. "Phase reversal" Pilot lamp.
6. "Pump motor temperature alarm" Pilot lamp.
7. Volt free contacts to fire alarm system for Pump running, Power failure and Phase reversal.
8. Suitable for manual and automatic operation.
9. Push buttons for starting and stopping the fire pump.
10. Push button for silencing alarm.
11. Minimum run timer to prevent short cycling.
12. Emergency run mechanical control.
13. Pressure switch.
14. Audible alarm.
15. Voltmeter and ammeter on the front side of the panel.

Controller shall be suitable for wall mounting or freestanding type

Enclosure of the controller shall meet the requirements of NEMA 12, drip tight terminals of the controller shall be sized to suit the incoming power cables.

5.6.6 Diesel Engine Controller:

The fire pump's controller shall be completely assembled and tested at the factory, listed (UL listed / FM approved) for diesel engine driven fire pumps, primary resistance type, with auto transfer switch and have the following features:

1. "Automatic" pilot lamp.
2. "Fail to start" pilot lamp.
3. "Battery failure" pilot lamp.
4. Suitable for manual and automatic operation.
5. Push buttons for starting and stopping the fire pump.

6. Push button for silencing alarm.
7. Pressure switch.
8. Audible alarm.
9. Voltmeter and ammeter on the front side of the panel.

Controller shall be suitable for wall mounting or freestanding type.

Enclosure of the controller shall meet the requirements of NEMA 12, drip tight terminals of the controller shall be sized to suit the incoming power cables.

5.7 JOCKEY PUMP & CONTROLLER

5.7.1 Pump:

Jockey pump shall be supplied by approved fire pump manufacturer and shall be vertical multistage, factory assembled and tested for capacity as indicated on drawings. Construction of jockey pump shall be as given below.

1. Impeller: Stainless steel.
2. Shaft Stainless steel
3. Casing: Cast iron or stainless steel.
4. Connections: flanged or threaded

5.7.2 Controller:

Provide across-the-line type controller UL listed, factory assembled and tested suitable for combined manual and automatic operation.

Enclosure shall be rated for NEMA 12 drip tight.

Controller shall have following features

- a) "Power on" pilot light.
- b) Disconnect switch.
- c) Hand-Off-Auto selector switch.
- d) "Pump running" pilot light
- e) Pressure switch.
- f) Volt free contacts for fire alarm system for pump running.

5.8 EXAMINATION

Examine areas, equipment foundation and conditions.

Examine the elevation of the piping connections from the fire water tank.

Proceed with installation only after unsatisfactory conditions have been corrected.

5.9 INSTALLATION

Install pumps on foundations indicated and arrange to provide access for periodic maintenance.

Follow manufacturer's recommendations for installation.

Suction and discharge pipe sizes shall meet or exceed Table 5.25 of NFPA20 - 2003. (comply to UAE FLS code Table 9.3.a.: Pipe Sizes Connecting to Centrifugal Fire Pump)

Wiring from the controller to the motor of main fire pump and jockey pump shall be in steel conduit or protected from damage and fire.

Piping to pressure switch (sensing line) shall be stainless steel 316

5.10 ALIGNMENT

Align the pump and driven shafts after complete unit has been levelled on foundation and grout has set and foundation bolts have been tightened

After the alignment is correct, fill the base plate completely with grout.

Make pipe connections and check the alignment.

5.11 FIELD QUALITY CONTROL

Provide services of factory authorized service representative to supervise field assembly of components, installation of fire pump units and jockey pump units, including piping and electrical connections and field acceptance tests.

Perform field acceptance test for all fire pump units and jockey pump in accordance with the requirements of Local Civil Defence.

5.12 COMMISSIONING

Provide services for manufacturer's representative for start-up and to demonstrate and train owner's representative

Train owner's maintenance personnel on procedures and schedules related to start up, shutdown, troubleshooting, servicing and preventive maintenance

Lubricate all bearings.

Start the pump and check the operation the driver of each pump (ensure the fire water tank is filled with water before starting the pump).

5.13 SPARES

The installing fire protection specialist contractor shall supply the owner with quantities of spare parts LIST

5.14 AS BUILT DRAWING / OPERATIONS & MAINTENANCE MANUAL

The contractor shall submit the following during testing and commissioning of fire pump set.

- 1) All maintenance manuals / catalogues of installed equipment's.
- 2) As built drawing of fire pump set in hard and soft copy.

5.15 WARRANTY

All fire fighting and sprinkler system including fire pump set components furnished, and installed under this contract, shall be guaranteed against defects in design, materials and workmanship for the full warranty period which is standard with the manufacturer, but in no case less than one (1) year from the date of system acceptance, except for that equipment having a specified warranty period.

6.0 CLEAN AGENT GAS EXTINGUISHING SYSTEM (FM200):

6.1 DESCRIPTION OF WORK

The MEP sub-contractor shall include in his tender for the complete installation, testing and commissioning of the clean agent gas extinguishing system (FM 200), detailed on the tender drawings and in this specification, to the standards and approval of Local Civil Defence. The MEP sub-contractor shall include for all labour, materials, tools and tackle, complete with all supports and fixings back to approve building structural members. The MEP sub-contractor shall be responsible for all dimensions and location of other services, and the correct setting out of the installation.

6.2 GENERAL

The computer room (server room), telecommunications room, electrical room, BMS Room, central control room and all other areas indicated in the drawing shall be protected with a clean agent fire extinguishing system to match the requirements set forth in this specification, the codes mentioned below and to the requirements of Local Civil Defence authority.

The MEP sub-contractor shall employ a specialist manufacturer to design, supply, install, and test and commission the clean agent fire extinguishing system.

6.3 DESIGN CODES

The clean agent fire extinguishing system shall comply with the latest edition of the following standards:

NFPA 2001	-	Clean Agent Fire Fighting Systems
NFPA 70	-	National Electric Code
NFPA 72	-	National Fire Alarm Code
NFPA 75	-	Standard for the Fire Protection of Information Technology Equipment

6.4 SUBMITTALS

- A. Manufacturer's product technical data sheets, installation & start-up instructions and ratings for all the equipment's pertaining to clean agent firefighting system.
- B. Shop drawing for each area clearly indicating the agent storage tanks, location of all plant and equipment, pipe runs with sizes and lengths, control panels with terminal wiring ports, detectors, conduit runs, manual pull stations, abort stations, audible and visual alarm, etc.
- C. Auxiliary details and information such as hydraulic flow calculations from a UL listed computer program supporting the pipe size and routing shown in the drawing, door holders, special sealing requirements, operation and control documentation, damper operation, alarm devices, shutdown functions, remote signaling, agent discharge times etc., required for successful operation of cleaning agent system.
- D. Schematic wiring diagrams for all components, clearly indicating all required field electrical connections.
- E. All information necessary for the complete supply and installation of the proposed clean agent firefighting system.

6.5 PRODUCT DELIVERY, STORAGE AND HANDLING

- A. All materials used for the installation must be new, free from all defects and imperfections unless otherwise indicated. Handle all materials and components carefully to prevent damage, breaking, denting and scouring.
- B. Store the components in clean dry place. Protect from weather, dirt, fumes, water, construction debris and physical damage.

6.6 SYSTEM DESCRIPTION AND OPERATION

The system shall provide a clean agent gas concentration of 7% by volume, minimum.

The cylinder shall be actuated only by the burst of rupture disc, which is maintenance free, actuation through solenoid valve is not acceptable.

Fire sensing element in each area protected by clean agent system optical smoke detectors installed for a maximum coverage of 25m². These detectors shall be installed in all three voids namely room, sub-floor / false floor and false ceiling of the protected area. Detection shall be wired to provide a double knock signal using cross-zoned detection method of operation.

The automatic operation of each protected area shall be as follows:

- A. Actuation of one zone (detector) shall perform the following functions
 - 1) Illuminate a warning lamp on the control panel.
 - 2) Actuates alarm bell.
 - 3) Energize visual indicator.
- B. Actuation of a second detector, within the system, shall perform the following functions
 - 1) Illuminate a pre – discharge lamp on the control panel.
 - 2) Energize strobe horn.
 - 3) Starts time delay sequence.
 - 4) Enable the system abort sequence.
 - 5) Shut down the HVAC system and / or close dampers.
 - 6) Close all doors to the protected area.
- C. After completion of the time delay sequence the agent shall be discharged and the following shall occur:
 - 1) Illuminate a pre – discharge lamp on the control panel.
 - 2) Energize a system fired lamp on the control panel.
 - 3) Energize visual indicator outside the hazard in which the discharge occurred.
 - 4) Energize the system fired audible device.
- D. The system shall be capable of being actuated by manual discharge devices located at the exit of each area. Operation of a manual device shall duplicate the operation of the second detector described in the above except the time delay and abort functions (both shall be by – passed). The manual discharge station shall be of electrical actuation type and shall be supervised at the extinguishing control panel.

6.7 PRODUCTS**6.7.1 Electrical Components & Accessories:**

- A. **EXTINGUISHING CONTROL PANEL:** It shall be single hazard and double knock type. Each hazard shall have its own extinguishing panel. The panel shall include the following over and above the requirements for their proper functioning
- 1) 3 audible circuits.
 - 2) Diagnostic LED's for easy trouble shooting.
 - 3) Solid state field programmable time delay 0 to 60 second.
 - 4) Protective covers over high voltage terminals.
 - 5) Connections to power a remote annunciation.
 - 6) Cross – zone principle of agent discharge.
- B. **MANUAL RELEASE SWITCH:** This is a dual action device for manual discharge of the agent bypassing the time delay and abort functions. They shall be provided near the entrance of each protected hazards. These devices are intended to be used by the human occupants to control the discharge of the extinguishing agent manually.
- C. **ABORT STATION:** An abort station shall be located next to each manual release switch and shall be monitored by the control panel. Upon activation of the abort switch a signal shall be generated at the main control panel, producing a visual indication for their actuation. This shall functionally prevent the discharge of clean agent for a preset time delay period.
- D. **ALARM BELL:** Fire alarm bell shall be provided that shall operate to give warning to occupants in case that automatic fire detectors sense the present of fire (first stage detection).
- E. **HORN / STROBE:** Audible horn & strobe light shall be provided to give information to the personnel / occupants when the extinguishing agent is about to discharge (by horn) and when it has been fully discharged (by strobe).
- F. **SMOKE DETECTORS:** Provide and install optical type smoke detectors.
- G. **CONTROL & MONITORING CABLE:** Cables shall be fire rated (FP – 200 / approved equal). The cable shall be laid through conduits of sufficient size.

6.7.2 Mechanical Components & Accessories:

- A. **PIPES AND FITTINGS:** The pipes shall be galvanized steel as per ASTM A53, Gr. B, approved, ductile iron material, threaded / grooved type and rated for minimum 300PSI working pressure.
- B. **GAS CYLINDERS:** Gas cylinder shall be manufactured from high strength alloy steel, conforming to NFPA 2001 and tested to a pressure 700PSI. They shall consist of rupture disc assembly to get operated by a gas cartridge actuator.
- C. **DISCHARGE NOZZLES:** Provide and install agent discharge nozzles with 360° / 180° discharge pattern. The specialist subcontractor shall determine the number, location and type of the nozzles as per the recommendation of manufacturer.
- D. **GAS CARTRIDGE ACTUATOR (GCA):** Proved gas cylinders shall utilize a fast-acting rupture disc assembly. The GCA shall consist of stainless-steel body containing 2 sets of bridge wires and a charge of smokeless powder. The actuation shall occur when the powder

gets ignited (when the electrical current is sent through the bridge wires) and generates the necessary pressure to open the rupture disc.

6.8 EXECUTION

System installation in general shall be conducted by experienced technicians and supervised by engineer who has successfully installed a fire extinguishing system of the same type and designed as specified herein. The services of the qualified manufacturer's representatives, experienced in the installation and operation of the system being provided shall be furnished to supervise the preliminary testing, including final testing, system adjustment and instructions.

6.9 TEST UPON COMPLETION

Upon completion of the installation, the system shall be subject for functional and operational test (no gas discharge) including the in place of each field devices. When all the required corrections have been rectified, the system shall be re – tested and the representative of the owner shall be notified of the readiness for final inspection. At this time, any and all required test shall be repeated, and any defects corrected until the system is found to be acceptable.

7.0 FIRE STOP SYSTEM

7.1 SUMMARY

Applications of fire stop systems include:

- Penetrations for the passage of duct, cable, cable tray, conduit, piping, electrical bus ways and raceways through fire-rated vertical barriers (walls and partitions), horizontal barriers (floor / ceiling assemblies), and vertical service shaft walls and partitions.
- Openings and penetrations in fire-rated partition or walls containing fire doors.
- Multiple electrical boxes located within the same stud cavity.
- All materials shall conform to relevant North American or European Standards and shall be with valid approval certificate from Local Civil Defence.

7.2 DEFINITIONS

Fire stopping: Material or combination of materials used to retain integrity of fire-rated construction by maintaining an effective barrier against the spread of flame, smoke, water and hot gases through penetrations in fire rated wall and floor assemblies.

7.3 SYSTEM DESCRIPTION

Performance Requirements: Provide fire stop systems that have been manufactured and installed to maintain performance criteria stated by manufacturer without defects, damage or failure.

7.4 SUBMITTALS

- General: Submit listed submittals in accordance with Conditions of the Contract and Submittal Procedures Section.
- Product Data: Submit product data including manufacturer's product sheet, for specific products.
- Shop Drawings: Submit shop drawings showing layout, profiles and product components.

- Quality Assurance Submittals: Submit the following:

Certificates: Product certificates stating that materials comply with specified performance characteristics and physical requirements.

Manufacturer's Instructions: Manufacturer's installation instructions.

7.5 QUALITY ASSURANCE

- Source Limitations: Obtain fire stop through-penetration and joint systems from a single manufacturer.
- A manufacturer's direct representative (not distributor or agent) to be on-site during initial installation of fire stop systems to train appropriate contractor personnel in proper selection and installation procedures. This will be done per manufacturer's written recommendations published in their literature and drawing details.
- Proposed fire stop materials and methods shall conform to applicable governing codes having local jurisdiction.
- Fire stop Systems shall not reestablish the structural integrity of load bearing partitions/assemblies or support live loads and traffic. Installer shall consult the structural engineer prior to penetrating any load bearing assembly.

7.6 DELIVERY, STORAGE, AND HANDLING

Deliver materials in manufacturer's original, unopened, undamaged containers with identification labels.

Store materials protected from weather and damage and at temperature conditions in compliance with manufacturer's requirements.

Before handling, read product data sheets and material safety data sheets. Do not use damaged or expired materials.

7.7 PRODUCTS

7.7.1 Penetrations of fire-resistant assemblies

"Penetrations of pipes, tubes, conduits, wires, cables, ducts and vents through fire resistant walls and shaft enclosures, and floor, ceiling, and roof elements of fire-resistant assemblies shall be protected in accordance with the following:

7.7.2 Non-combustible penetrations

"Non-insulated metal ducts and non-combustible pipes shall be protected by an approved through-penetration protection system. The annular gap around the penetrates shall be packed with mineral wool / rock wool and the ends shall be filled and sealed with Fire stop silicone sealant to an appropriate depth.

7.7.3 Combustible penetrations

Metal ducts with combustible jackets or insulation shall be protected by an approved through penetration protection system. The annular gap around the duct shall be packed with mineral wool / rock wool and the ends shall be filled and sealed with Fire stop intumescent sealant to an appropriate

depth.

7.7.4 Single cable or bunched cables

The annular gap around cable or bunches of cables shall be packed with rock wool / mineral wool and the ends shall be filled and sealed with intumescent sealant to an appropriate depth.

7.7.5 Cable trays crossings:

Permanent system Fire stop intumescent sealant shall be applied around and between the cables. The opening shall be sealed with towable fire resistant cement-based mortar.

Temporary system where future changes are expected ready-to-use, intumescent flexible block firmly seated shall be built up inside the opening sealing it completely. Fire stop intumescent sealant shall be applied around and between the cables.

Non-metal pipe crossings less than 50mm diameter: The annular gap around the penetrates shall be packed with mineral wool / rock wool and the ends shall be filled and sealed with Fire stop intumescent sealant.

Non-metal pipe crossings of 50mm diameter and more: The annular gap around the penetrates shall be packed with mineral wool / rock wool and the ends shall be filled and sealed with Fire stop intumescent sealant to an appropriate depth.

The pipes shall be provided with a pre-assembled graphite based intumescent jacket contained in a metal housing, fixed around the pipe with metal fixings.

7.8 EXECUTION

7.8.1 Manufacturer's Instructions

Comply with manufacturer's product data including product technical bulletins, product catalog installation instructions and product carton instructions.

7.8.2 Examination

Verification of Conditions: Examine areas and conditions under which work is to be performed and identify conditions detrimental to proper or timely completion.

7.8.3 Preparation

Prepare surface to receive fire stop system products in accordance with manufacturer's instructions for surface preparation.

Verify penetrations are properly sized and in suitable condition for application of materials.

Comply with manufacturer's recommendations for temperature and humidity conditions before, during and after installation of fire stopping.

7.8.4 Installation

Comply with manufacturer's instructions for installation of through-penetration or construction joint fill materials.

Seal all holes or voids made by penetrations to ensure an air and water-resistant seal.

Consult with mechanical engineer, project manager, and damper manufacturer prior to installation of fire stop systems that might hamper the performance of fire dampers as it pertains to duct work.

7.8.5 Field Quality Control

Examine sealed penetration areas to ensure proper installation before concealing or enclosing areas.

Keep areas of work accessible until inspection by applicable code authorities.

7.8.6 Protection

Protect materials from damage on surfaces subjected to traffic.

END OF SECTION

MECHANICAL SECTION: 07

AIR CONDITIONING AND VENTILATION WORKS

1.0 DESIGN PARAMETERS**1.1 Location:**

- i. Dubai, United Arab Emirate
- ii. Longitude: 55.2° E
- iii. Latitude: 25.2° N

1.2 Outdoor Condition of the Building:

	Dry Bulb Temperature (DB)	Wet Bulb Temperature (WB)	Relative Humidity (RH)	Daily Range
Summer	46°C (115°F)	29°C (85°F)	30%	13.8°C (56.8 °F)
Winter	13°C (55.4°F)	10.5°C (50.9°F)	65%	

1.3 Indoor Condition of the Building:

1. Indoor design temperature = 22-24°C
2. Heating in winter season shall not be considered.
3. Relative humidity control shall not be considered. However, it shall be within the range of 50% +/- 5%

1.4 Margin of Safety:

- As per DM Green Building Regulations the safety factor applied for cooling load calculations must not be greater than:
 - Sensible Heat 10%
 - Latent Heat 5%

1.5 Building U-Values:

As per DM Green Building Regulations the following U-Values shall be considered in the AC calculations. The final parameters shall be according to Authority regulations and project design requirements.

- External wall U-Value: 0.57 W/m² K
- External roof U-Value: 0.30 W/m² K
- Glazing U- Value: 1.9 W/m² K (considering the glass area is less between 40 - 60% of total external wall area).
- Glazing Shading Coefficient: 0.32
- Glazing Shading Coefficient for Ground Floor Retail: 0.76

1.6 Ventilation, Occupancy and Electrical Lighting & Power Requirements:

- The Occupancy level, lighting density and equipment power heat gain in each space shall be as per ASHRAE Standard 62.1-Latest Edition, and for the Lighting Power Density as per ASHRAE Standard 90.1- Latest Edition where applicable.

1.7 Pressure Philosophy:

- Apartments shall be maintained under positive pressure.
- Common corridors shall be maintained under positive pressure.
- Retails shall be maintained under positive pressure.
- Toilets, Kitchen and garbage rooms shall be maintained under negative pressure.
- Lobbies, lift lobby shall be maintained under positive pressure.

1.8 Air Filtration:

- Air filtration shall be based on ASHRAE 52.2.
- Indoor fan coil units shall have 25 mm thick washable filters at 30% efficiency (MERV 6)
- Fresh Air handling units shall have two banks of filters: panel filter at 30% (MERV 6) and bag filter at 80% efficiency (MERV 11)

1.9 Ductwork / Pipe Sizing Criteria:

- AHU / FCU Coil Face Area → Air Velocity not to exceed 2.5 m/s (8.2 fps).
- Air Duct Sizes- Supply air → Pressure drop not to exceed 0.5 Pa/m (0.06 inch /100 ft).
- Air Duct Sizes- Extract air → Pressure drop not to exceed 0.6 Pa/m (0.08 inch /100 ft).
- Air Outlet Sizes → Air Velocity through net Free Area not to exceed 2.0 m/s (6.6 fps)
- Chilled water pipe sizing for pipes up 50mm ϕ → Water Velocity not to exceed 1.35 m/s (4.4 fps) Pressure drop not to exceed 300-600 Pa/m (6 ft / 100 ft).
- For Pipes above 50mm ϕ and up to 200 mm ϕ → Pressure drop not to exceed 400 Pa/m (4ft / 100ft; velocity not to exceed 2.4 m/s (8fps)).
- For Pipes above 200 mm ϕ → Velocity not to exceed 2.5 m/s (8.2 fps).
- Coil Connection Pipes to be sized for full flow rate of respective coils

1.9.1 The following maximum air Velocity in ductwork to be considered

Duct Selection	Duct Max Velocity (m/s)
Main Branch	7.5
Sub Branch	5.0
Terminal Branch	3.5
Terminal Unit – FCU	≤ 2.5
Kitchen Exhaust	10
Risers	10

1.10 Noise Levels

- Bedroom NC 30-35
- Main Lobby, Corridors & Living Room NC 40
- Office NC 40
- Retail NC 35-40
- Gym (Health Club) Area NC 35-40
- Plant rooms / Car Parking NC 50-55

1.11 Chilled water-cooling Coil

- For the building provided with EMPOWER District Cooling network, the following chilled water temperatures shall be considered for the HVAC system:

DC Primary Side		Building Secondary Side	
Primary Return Temp.	Primary Supply Temp.	Secondary Return Temp.	Secondary Supply Temp.
13.5°C (56°F)	4.5°C (40°F)	14.5 °C (58°F)	5.5°C (42°F)

Chilled water circuits shall be with variable flow system.

Primary side chilled water control is solely within Empower scope and responsibility.

The chilled water temperatures to be verified according to the District Cooling Provider guidelines.

- Maximum velocity of water through pipework : 2.13 m/s.

Pipes to be sized for full flow rate of all cooling coils.

Maximum coil face velocity : 2.5 m/s.

1.13 Pressurization Philosophy

Fresh air requirements to each zone shall be generally determined by the level of dirty extract to that same zone, or to a value as required for the levels of occupancy in each zone.

Areas requiring no dirty extract shall be pressurized delta positive or maintained neutral as recommended by ASHRAE.

Areas requiring dirty extract, shall be pressurized delta negative.

The whole building is positively pressurized to avoid any hot air infiltration.

1.14 Air Intake System

Fresh air intake to each air handling unit shall incorporate a sand trap louver.

Air intake louvers shall, in general, be located as far away as possible from extract louvers and drainage vent stacks to avoid short circuiting of dirty air.

2.0 PLAT HEAT EXCHANGER (At Tertiary Circuits)

- 2.1 Furnish and install where indicated on drawings, plate-and-frame heat exchangers of the duties as scheduled. All Plate heat exchangers shall be certified according to AHRI Liquid to Liquid Heat Exchanger Certification Program and listed on the AHRI directory site (www.ahridirectory.org).
- 2.2 The units shall be rated for maximum system temperature and working pressure and tested at a minimum of 1.5 times maximum system working pressure at full differential pressures. Unit must receive certified factory inspection prior to shipment. A copy of the inspection report shall be furnished to the contractor.
- 2.3 The sizing of the heat exchangers is to allow for the specified duties along with a spare extension capacity of 10 % for possible future loads.

- 2.4 Plate-Frame Exchanger to be constructed of the following materials:
 - 2.4.1 Plate Material: Plates to be AISI 316 stainless-steel with 2B finishes with minimum thickness of 0.5 mm and the gaskets shall be of nitrile rubber.
 - 2.4.2 The structural frame consisting of both stationary and movable covers, upper and lower carrying bars, outrigger and compression bolting shall be fabricated of carbon steel. The frame assembly's upper carry and lower guide bars shall be designed to allow for minimum of 10% plate expansion as per manufacturer's recommendation.
 - 2.4.3 The upper T-section hanger and the lower guide bar shall be of stainless steel and provide a positive steering and locking mechanism for plate-to-plate alignment.
 - 2.4.4 All steel components of the exchangers are to be adequately protected as per the manufacturer's recommendation with acrylic polyurethane or epoxy paint to protect it from the elements.
 - 2.4.5 Gaskets to be furnished so as to conform to the respective maximum system working pressures, temperatures and water treatment of each water system.
 - 2.4.6 Nozzles to be rated for 150 percent of maximum system operating pressure and to be of the flanged type. Nozzle velocities shall not exceed 3 m/s at the maximum rate of heat transfer as scheduled on the Drawings.
- 2.5 The following items shall be installed on the units:
 - 2.5.1 Two lifting lugs located on top.
 - 2.5.2 Roller assembly on the movable cover for ease of opening.
 - 2.5.3 Ball bearing assemblies for ease of tightening and untightening the unit.
 - 2.5.4 All carbon steel fabricated components shall be painted with Epoxy paint or Acrylic polyurethane finish on all exterior surfaces.
 - 2.5.5 The heat exchangers furnished shall have sufficient surface area to perform the specified thermal duty plus additional surface area for fouling as per AHRI certification requirement.
 - 2.5.6 Plate-Frame Exchanger to be equipped with fully thermally insulated, stainless steel shroud enclosure panels supplied by the unit manufacturer. The enclosures are to be manufactured in sections to allow easy dismantling for inspection and maintenance and to include locking catches and handles.

Manufacturer to supply pre-fabricated insulated cover and drip tray. Removable and reusable panel type insulation cover for the plate heat exchanger to enable expedient removal and replacement of insulation to facilitate removal and cleaning of plates. To facilitate assembly and disassembly the insulation pieces shall be held together with stainless steel "suitcase" type latches.
- 2.6 Provide a 5-year warranty for the plate heat exchangers including servicing and parts
- 2.7 Install heat exchangers in accordance to the manufacturers written instructions. Furnish and install plate and frame heat exchangers designed with manufacturer's standards and guaranteed to perform to the capacities and pressure drops at the fluid temperatures and flow rates as scheduled. The maximum allowable water pressure drops for the cold and hot sides shall be less than 40 kPa.
 - 2.7.1 All exchangers are to be installed on a base which should be a minimum of 150mm in height and the base should extend beyond the frame by no less than 300mm on all sides.

- 2.7.2 Manufacturers recommended clearances for servicing, maintenance and installation of piping shall be adhered to at all times.

3.0 AIR HANDLING UNITS

- 3.1 Manufacturers of air handling units shall have the following certifications as a proof of quality assurance:

- 3.1.1 ISO 9001:2015 certification for product quality.

- 3.1.2 Manufacturers of air handling units shall have active Eurovent certification (proof of certification must be submitted from the respective website on the date of submission) as per BS EN 1886 as per below minimum standards:

- | | | |
|----|---|-----------|
| a) | Casing strength to | Class D1 |
| b) | Casing air leakage minus 400Pa to | Class L1 |
| c) | Casing air leakage plus 700Pa to | Class L1 |
| d) | Thermal transmittance | Class T2 |
| e) | Thermal bridge factor | Class TB2 |
| f) | Filter frame leakage | Class F9 |
| g) | Acoustic Insulation as mentioned below: | |

HZ	125	250	500	1000	2000	4000	8000
dB	9.2	10.5	12.4	13.5	20.4	34.7	42

- 3.1.3 Insulation shall be CFC free Polyurethane type
- 3.1.4 Airtight seal material shall meet NFPA 90A requirements for flame spread and smoke generation
- 3.1.5 Casing sheet metal shall withstand 500 hours salt spray exceeding ASTM B117
- 3.1.6 Air filters performance shall be according to ASHRAE Standard 52.2 or EUROVENT Standards.
- 3.1.7 Heating Elements (where applicable) shall be manufactured in accordance with NEC and UL 1996 - Electric Duct Heaters.
- 3.1.8 Fan performance shall be tested and certified to DIN and AMCA Standard
- 3.1.9 Fan motor shall be of IE2 energy grade and conform to IEC Standard.
- 3.1.10 Energy Recovery Wheels shall be AHRI or Eurovent certified.
- 3.1.11 A service light must be provided in all accessible sections of the AHU.

3.2 CASING CONSTRUCTION

- 3.2.1 The unit construction shall consist of a frame and panels. The profile shall be with Aluminium Pentapost, 3 mm thick with thermal break profile.
- 3.2.2 The frame shall hold a 50 mm double-skin casing wall with panels, doors, inspection hatches and removable center posts.
- 3.2.3 Unit shall be constructed of a complete frame with easily removable panels. Removal of any panel shall not affect the structural integrity of the unit, thereby facilitating maintenance work.

- 3.2.4 Unit shall have thermal break profiles to minimize the conduction path from the inside of the casing to the outside. The construction should reduce machine noise, absorbing vibration and avoiding air leakage.

3.3 PROFILE FRAME AND PANEL CONSTRUCTION

- 3.3.1 Heavy-duty profile frame made from Extruded Anodised Aluminium, fully welded and hermetically sealed at the ends. Profile shall be more robust and can be repeatedly disassembled. Profile shall be coated in according with ASTM A 653/A 653 M-01
- 3.3.2 Profiles made from Aluminium shall be anodized and suitable for corrosive environment.
- 3.3.3 The standard casing wall construction shall consists of 0.8mm internal and external skin with 50 mm injected polyurethane, fire retardant according to DIN 4102 class B2, 40kg/m³ density, 0.19 W/m.°C K value
- 3.3.4 The doors and the panels shall be formed by bending the panel external sheet metal over the internal sheet metal so that closed panel structure resistant to water and steam will be obtained. Bended external sheet metal structure of the panel (preventing the formation of sharp edges) also reduces the risk of any possible injuries during mounting and maintenance.
- 3.3.5 Panel shall be tested for strength at ± 1000 Pascal with deflection < 1 mm/m and without permanent deflection at ± 2500 Pascal, confirmed to BS EN 1886 Class D1.
- 3.3.6 Panels shall be manufactured from steel sheets treated with weather and scratch-resistant coating on both sides, frames and roof are painted with highly durable heat hardened polyester based powder paint with 50 microns film thickness.
- 3.3.7 The paint shall conform to ASTM D-2201 and ASTM D6386 for gloss, adhesion, impact resistance, withstand up to 1000 hours salt spray resistance test, exceeding 500 hours ASTM B117.
- 3.3.8 Thermal Transmittance, U-value tested at 0.521W/m².K confirmed to Eurovent BS EN 1886 Class T2.
- 3.3.9 Thermal Bridging Factor kb 0.62 conformed to Eurovent BS EN 1886 Class TB2.
- 3.3.10 Fasteners shall be zinc coated comply with minimum 1000 hours salt spray resistance test ASTM B117.
- 3.3.11 Panels and profile frames shall be assembled with a closed cell self-adhesive polyethylene seal installed in between them to prevent air leakage, reduce vibration noise and minimized thermal losses.
- 3.3.12 Closed cell self-adhesive poly-ethylene seal shall avoid external water from penetrating inside the unit and shall block heat infiltration and avoid external contamination into the conditioned space area inside the unit.
- 3.3.13 The construction of panels to the profile frames shall provide minimum air leakage Class L1. Casing leakage tested and conformed to Eurovent Standard BS EN 1886.
- 3.3.14 A base shall be provided under the unit to avoid contact of the bottom surface of the unit with the others surfaces during transportation and installation, and to ensure required body strength. The height of the base shall be 100 or 150 mm.
- 3.3.15 Roof curb profiles, where the unit will be carried shall be of closed welded section and thus minimize the risk of its deformation during transportation.

- 3.3.16 Roof curb shall be from heavy gauge (2mm) Aluminium Anodized sheets, longitudinally welded with the profile, connected with 3 mm crossbar to form strong and rigid structure and shall be supplied with multiples holes to suit optimum hoisting stability.
- 3.3.17 In case of unit installation at roof with direct sun exposure, unit shall be provided with whether proof canopy which prevents water penetration on top surfaces of the AHU when installed outdoor. Silicone shall be applied to the profile joints on the outer surface of the unit to increase the insulation against water. Construction shall be of heavy gauge (G16) galvanized steel grade G90 and powder painted RAL7035 with 50 microns thick.

3.4 DOORS AND ACCESS PANELS

- 3.4.1 Panels shall be hinge type with handle and lock that can only be opened with a special key. Access door panel shall be equipped with metal hinges with plastic cover. It shall be with EPDM rubber sealed at door perimeter which provides an effective airtight and protection from heat gain.
- 3.4.2 Access door type shall be at the Fan Section, filter section and the Mixing Box Section. View port shall have double skin.
- 3.4.3 Unit shall have hinged type safety access panels accessible internally and externally.
- 3.4.4 Removable access panels shall be fixed to the profile frames with bolts which provide additional access to ease mobility for major service work on large and heavy component such as coil, fan and motor.

3.5 FAN SECTION AND DRIVE ARRANGEMENT:

- 3.5.1 Fans shall be double inlet double width, forward or backward curved centrifugal type fans.
- 3.5.2 Fans shall have sound power level measurement according to AMCA 300 & 301. Sound power level measured values shall be converted to other operating speeds with a calculation procedure described in the BS 848 Part 2 standard. Outlet side sound power levels shall be according to ISO 5136, DIN 45635 Part 9, BS 848 Part 2 Chapter 6 or AMCA 330-97.
- 3.5.3 Fan shall not exceed a maximum outlet velocity of 12 m/sec.
- 3.5.4 Fan wheel shall be statically and dynamically balance within fan operating speed in compliance to DIN ISO 21940 with grade G4. Shaft diameters are selected to achieve a safety factor for critical speed ≥ 1.25 higher than the maximum operating speed.
- 3.5.5 Blades of the fans shall be from galvanized steel or steel treated with polyester powder coat.
- 3.5.6 Fan scroll from hot dip galvanized steel JIS G3302 Grade SGCC with no electrical spot welding as the scroll back is joined to the side plates with the Pittsburgh lock forming system which prevents any oxidation starting from the welding spots.
- 3.5.7 Bearings shall be for heavy-duty service with nominal life expectancy of 40,000 L10 operating hours (according to DIN ISO 281 Section 1). Bearing shall be life lubricated type
- 3.5.8 Motor design and performance shall be IE2 class according to IEC Standards. Fan motor shall be able to operate at maximum 50% of rated speed. At 50% of rated speed, the fan motor shall deliver 80% of rated torque and 40% of rated power.
- 3.5.9 Standard fan's motor shall be a Totally Enclosed Fan Cooled type (TEFC).

- 3.5.10 Motor's frame shall be of cast iron. Motor protection shall be with IP 55, motor shall be protected against dust and protected against jets of water from all direction.
- 3.5.11 Insulation shall be Class F (T max 155 °C) and Temperature Rise Class B (ΔT 80K). Motor shall able to operate at 50°C operating temperature.
- 3.5.12 Motor shall have a thermal reserve to spare for overheating when induced by voltage fluctuation at $\pm 10\%$, compliance to IEC 60034-1.
- 3.5.13 Thermal Overload Protection "Normally Closed Thermostat" (PTO) shall be a standard accessory. PTO shall provide general surveillance for non-transient overload. Bi-metallic strip with normally closed contact is indirectly heated when line current passes through the strip and shall shut down or restores supply circuits as necessary. PTO connection shall be mounted to control circuit and suitable for manual or automatic reset.
- 3.5.14 Positive Temperature Coefficient thermistor (PTC) shall provide general surveillance for transient overload.
- 3.5.15 Fan and motor assemble on a common base, FMB, fabricated from heavy gauge galvanized steel.
- 3.5.16 Motor rail shall be a slide rail type for flexible belt tensioning. Fan and motor assembly are totally isolated from AHU casing structure by isolators and flexible canvas. Rubber isolators shall be capable to achieve standard vibration isolation efficiency up to E85.
- 3.5.17 Spring isolator with 250-micron powder painted and with 50mm free deflection shall be capable to achieve critical vibration isolation efficiency E95 up to E99.9. Fan shall be isolated to discharge panel with fire retardant fan flexible connection.
- 3.5.18 Fan drive package shall consist of fan and motor pulleys and assembled with Wedge belts. Variable pitch pulley shall be available at motor drive up to 15HP (11kW).
- 3.5.19 Standard belt Service Factor designed at minimum SF1.1 and flexible to reach up to SF1.5.
- 3.5.20 The static resistance of the various systems has been calculated by the Engineer and the Sub-Contractor shall be required to check and recalculate the resistance in the system taking into consideration the resistance through the equipment that is finally approved and satisfy himself that the resistance figures for the fan is adequate. If necessary, the total fan resistance shall be adjusted and agreed with the Engineer. No claim shall be allowed by the Engineer if the Sub-Contractor fails to fully acquaint himself with the capabilities of the various systems. The fan shall have provision for connecting variable frequency device.
- 3.5.21 All motors shall be rated for use with variable frequency drive (VFD) shall be Inverter duty" or "drive duty" motors, compatible with the drive to which it is connected. Use of motor with a VFD shall not adversely affect the operation, useful life, or warranty of the motor.
- 3.6 COIL SECTION (CHW)
 - 3.6.1 Coils shall be fully enclosed within the coil section.
 - 3.6.2 Removable panel shall be assembled at both left and right side of coil section.
 - 3.6.3 Coil panels shall be bolted externally; air tight sealed, easy removal and quick reassemble to facilitate access for service and maintenance.
 - 3.6.4 Coils shall be of the cartridge type, mounted on a condensate drain pan with a slide in-out coil track so that the complete assembly is wholly removable for easy maintenance.

- 3.6.5 Coils shall be fabricated and tested in accordance with ASHRAE/ANSI 15 Safety Standard for Refrigeration Systems (latest edition).
- 3.6.6 Coil assembly shall sit within coil section face area and seal to the casing wall to avoid air by-pass, and shall be plate by-pass of removable type for easy access to header section.
- 3.6.7 Coils shall be housed in factory fabricated frame independent of unit casing and shall be individually removable. Multiple coils shall be mounted on independent support structure. Directly stacking coils using coil casing as structural support for other creep is not acceptable.
- 3.6.8 Drain pan shall be made of Galvanised steel with Epoxy coating. Fabricate pans and drain connections to comply with ASHRAE Standard 62.1 Latest Edition.
- 3.6.9 Drain pan shall be mounted in slanted angle at one side for complete and positive drainage; drain connection shall be at one side.
- 3.6.10 Eliminator shall be provided for coil face velocity more than 2.5m/sec.
- 3.6.11 Mineral reinforced PVC eliminator shall be provided which are highly efficient and can trap moisture molecules at much higher velocity.
- 3.6.12 Fins shall be a collar type, assembled with 12.7 mm OD copper tubes and mechanically bonded to copper tube for greater heat transfer coefficient.
- 3.6.13 Coil construction materials shall be of Seamless copper tube of 12.7 OD with 0.4mm wall thickness and aluminium fin 0.12mm thick.
- 3.6.14 Coil frames shall be made from galvanized steel.
- 3.6.15 Headers shall be Copper and connection shall be with threaded end Copper stub.
- 3.6.16 Header's connections shall be sealed internally and externally with rubber seal to avoid air leaking through gap at coil panels.
- 3.6.17 Coils shall be "counter" flow and equipped with vent and drain.
- 3.6.18 The coil shall be tested at 300-psig submerged under water, designed to operate up to 230 psig (16 Bar) at 200°F (93°C).
- 3.6.19 The manufacturer shall guarantee that each coil can meet the specified duty.
- 3.6.20 Coils shall be pitched and installed as required in the manufacturer's installation instructions.
- 3.6.21 Anti-corrosion coating to be applied on the cooling coil for FAHU's only.
- 3.7 AIR FILTERS
- 3.7.1 The pre-filter section shall be 50mm thick, MERV 8.
- 3.7.2 Pre-filter frame shall be made from galvanized steel.
- 3.7.3 Pre-filter shall be in flat form and should be cleanable and washable.
- 3.7.4 The bag filters shall be at 95 % efficiency (MERV 14). They shall comprise of a corrosion resistant frame housing the filter media. The media cartridge shall be pre-formed and shall consist of synthetic

- media and shall be supported with galvanized steel frame. Filter shall be capable to operate under high humidity conditions.
- 3.7.5 Soft bag filter length shall not be less than 500mm.
- 3.8 FILTER MOUNTING FRAME:
- 3.8.1 Mounting frame shall be made from galvanized steel and filter can be accessed on to the mounting frame from “front” (front or rear access or existing frame mounting) or “slide” (side mounting) type.
- 3.8.2 Frame mounting frame or Slide rails should be sealed for a minimal air leak ISO EPM1, $\geq 80\%$ as per ISO 16890-2016.
- 3.9 HEAT EXCHANGER THERMAL HEAT WHEEL
- 3.9.1 Highly efficient heat recovery wheels shall be incorporated in FAHU having efficiency not less than 70%. Both Enthalpy and Sensible wheels shall be provided as shown on drawings.
- 3.9.2 The Enthalpy Heat recovery wheel (HRW) shall be able to recover sensible and latent energy.
- 3.9.3 The wheel shall have purge mechanism to provide definite separation of airflow minimizing the carryover of bacteria, dust and pollutants from the exhaust air to the supply air.
- 3.9.4 HRW shall have cross contamination protection devices with non-contact synthetic brush seal and fixed and should have adjustable purge sector.
- 3.9.5 Wheel speed shall be less than 25 RPM. HRW shall be factory fitted.
- 3.9.6 Wheel housing shall be made of galvanized steel, cartridge type, and slide-in and slide-out installation into heat wheel section.
- 3.9.7 Face flatness of the wheel shall be maximized in order to minimize wear on inner seal surfaces and to minimize cross leakage.
- 3.9.8 Rotor shall be constructed of alternating layers of flat and corrugated media. Wheel layers should be uniform in construction forming uniform aperture sizes for air flow.
- 3.9.9 Wheel construction shall be fluted or formed honeycomb geometry so as to eliminate internal wheel bypass.
- 3.9.10 Wheel layers that can be separated or spread apart by air flow are unacceptable due to the possibility of channelling and performance degradation.
- 3.9.11 The minimum acceptable performance shall be as specified in the drawings/submittal.
- 3.9.12 The desiccant material shall be a molecular sieve, and specifically a 3A or smaller molecular sieve to minimize cross contamination.
- 3.9.13 The wheel frames shall consist of evenly spaced steel spokes, galvanized steel outer band and rigid center hub. The wheel construction should allow for post fabrication wheel alignment.
- 3.9.14 The wheel seals shall be full contact nylon brush seals or equivalent. Seals should be easily adjustable.
- 3.9.15 Cassettes shall be fabricated of heavy duty reinforced galvanized steel or welded structural box tubing. Cassettes shall have a built-in adjustable purge section minimizing cross contamination of supply air. Bearings shall be inboard, zero maintenance, permanently sealed roller bearings, or

alternatively, external flanged or pillow block bearings. Drive systems shall consist of fractional horsepower A.C. drive motors with multilink drive belts.

3.9.16 The wheel shall be AHRI certified and must bear the AHRI certification stamp.

3.10 DX COOLING COIL AND CONDENSING UNIT (Where Applicable)

Direct expansion coils of capacities as indicated on equipment schedules shall have aluminium fins with belled collars and bonded to 13mm OD copper tubes by mechanical expansion. Coils shall be provided with brass distributors with solder type connections and shall have a minimum of two distributors. Coils shall have full face active with intertwined circuit. Suction and discharge connections shall be at the same end. After testing, coils shall be dehydrated and charged with dry air. Coils in contact with outside air (either mixed or full) shall be provided with factory applied baked heresite coating and anti-fungal paint.

The condenser coils shall be constructed with internally finned seamless copper tubes having a "W" configuration and arranged in a staggered row pattern and mechanically expanded into lanced and rippled aluminium fins with full fin collars for higher efficiencies. The space between fins is given by a collar that will increase the surface area in connection with the tubes, protecting them from ambient corrosion.

The coils will have an integral subcooler circuit that provides sufficient sub cooling to effectively eliminate the possibility of liquid flashing and increase the unit's efficiency of 5-7% without increasing in power absorption.

The condenser coil shall be leak-tested and submitted to a pressure test with dry air.

The condensing units shall be air-cooled type incorporating heat exchanger coils manufactured from copper tubes and aluminium fins, factory treated to reduce the effect of atmospheric corrosion. The unit casing shall be manufactured from polyester powder coated baked enamel finish sheet steel in order to have a high corrosion resistance and to protect against salt laden environment close to where the units may be installed. The air outlet shall have plastic coated wire guards.

The compressor shall be scroll type with fixed and orbiting scrolls. The low pressure refrigerant shall be fed directly into the scroll and the discharge refrigerant shall cool the motor windings and place the compressor shell under discharge pressure.

The refrigerant used in the units shall be R410a / R407c or R134a. Use of R-22 refrigerant shall not be allowed.

The fans used in conjunction with the condenser coils, shall be helical type with aerofoil blades for higher efficiencies and lower noise. Each fan shall be protected by a fan guard. The air discharge shall be vertical and each fan must be coupled to the electrical motor, supplied as standard to IP54 and capable to work to ambient temperatures of -40 C to +55 C. They shall have individual overload protection via a disconnect switch.

The unit must have refrigerant circuits completely independent with one compressor per circuit. Each circuit shall include an: thermostatic expansion device (electronic device shall be available as option), compressor discharge shut-off valve, liquid line shut-off valve with charging connection, replaceable core filter-drier, sight glass with moisture indicator and insulated suction line. Suction line shut-off valve should be available as option.

Field power connection, control interlock terminals, and unit control system should be centrally located in an electric panel (IP 54). Power and starting controls should be separate from safety and operating controls in different compartments of the same panel. Starting will be star/delta type.

Power and starting controls should include fuses and contactors for each compressor winding and fan motors. Operating and safety controls should include energy saving control; emergency stop switch; overload protection for compressor motor; high and low pressure cut-out switch.

3.11 ELECTRIC HEATING COILS (Where Applicable)

Electric heating coils (where applicable) of capacities as indicated on schedules shall be heavy construction of 304L stainless steel with absolute safety finned tubes. Maximum specific power shall be 4 watts/Cm².

Heater battery shall be supplied with manual reset over heat thermostat, prewired to terminal strip.

3.12 IONIZATION AIR QUALITY SYSTEM

Ionization air quality system shall be provided after the cooling coil of all AHUs & FAHUs as specified in clause 20.0 in this section.

4.0 FAN COIL UNITS

4.1 Description: Factory-packaged and -tested units rated according to ANSI/AHRI 440, and ASHRAE 33. Manufacturer shall be quality system certified to ISO 9002.

4.2 Coil Section Insulation: 12mm thick, closed-cell foam complying with ASTM C1071 and attached with adhesive complying with ASTM C916.

- a. Fire-Hazard Classification: Insulation and adhesive shall have a combined maximum flame-spread index of 25 and smoke-developed index of 50 when tested according to ASTM E84.
- b. Airstream Surfaces: Surfaces in contact with the airstream shall comply with requirements in ASHRAE 62.1-Latest Edition.

4.3 Main and Auxiliary Drain Pans: Galvanized steel with Epoxy coating. Fabricate pans and drain connections to comply with ASHRAE 62.1- Latest Edition.

4.4 Chassis: Galvanized steel where exposed to moisture.

4.5 Cabinet: 1mm steel with exposed corners and edges rounded, easily removed panels, glass fiber or closed cell foam insulation and integral air inlet and outlet plenums.

- a. Supply-Air Plenum: Sheet metal plenum finished and insulated to match the chassis with mill-finish, aluminum, double-deflection grille.
- b. Return-Air Plenum: Sheet metal plenum finished to match the chassis.
- c. Mixing Plenum: Sheet metal plenum finished and insulated to match the chassis with outdoor- and return-air, formed-steel dampers.
- d. Exposed-to-sun installations shall be with AHU-like construction.
- e. Externally installed units in parking or under shade shall be double skin construction.

4.6 Finish:

- a. Concealed ducted type: Galvanized steel.
- b. Exposed Cabinet type: Factory baked enamel colour as selected on visible surfaces of enclosure or cabinet.

4.7 Filter: Easily removed 25mm permanent washable type, located to filter air before coil, Filter class MERV 6.

4.8 Water Coil: Evenly spaced aluminium fins mechanically bonded to copper tubes designed for system pressure as per design drawings and building pressure requirements. Provide drain pan under cooling coil, easily removable for cleaning, with drain connection. Pressure test at factory to 25 bar for half an hour. Provide test certificates prior to delivery to site.

4.9 Fan and Motor Board: Removable.

1. Fan: Forward curved, double width, centrifugal; directly connected to motor. Galvanized steel fan scrolls.
2. Motor: Permanently lubricated multispeed; resiliently mounted on motor board. Comply with requirements in Clause 13 – Electric Motors.
3. Wiring Termination: Connect motor to chassis wiring with plug connection.

4.10 Basic Unit Controls:

- a. Control voltage transformer.
- b. Wall-mounting thermostat with the following features:
 1. Fan-speed switch. (Low / Med / High)
 2. On/Off switch.
 3. Mode selector switch. (Cool / Fan / Auto)
 4. Room and Set point temperature indication.
 5. Communication capability with a centralized system for public area FCU.

5.0 CHILLED WATER PUMPS

- 5.1 The Contractor shall calculate the head and flow requirement for all pumps, based on his actual layouts as approved shop drawings and shall submit calculations for Engineer's approval.
- 5.2 Pumps shall be end suction type unless otherwise indicated. The efficiency of the pumps shall be maximum applicable as per project specification and shall not be less than 75%. Primary side CHW Pumps shall be Double-suction Horizontal Split case pumps.

The chilled water pump should be capable to run at 14 Hz without any cavitation in the pump impeller or overheat of electrical motor winding.

Hydraulic calculations must be submitted along with the pump material submittal and no approval shall be granted by any party prior to finalize and approve the pump head calculations.

Motor KW selection and calculation must be attached in the material submittal.

Pumps performance curves at different frequencies highlighting the system curve with operating points at 100% , 75% , 50% and 25% must be attached in the material submittal

5.3 Casing

Pumps shall have the casing divided either on the horizontal or vertical centre line. The casing halves shall be accurately machined, bolted and doweled together. A non-asbestos type gasket material shall be furnished between the casing halves. The casing material shall be close grained cast iron with a minimum tensile strength of 2300 kg/cm². Removal of the upper casing half shall permit removal of the complete rotating assembly without disturbing piping connections.

Casings shall be designed for scheduled working pressure and shall be hydrostatically tested. Suction and discharge flanges shall be drilled to ANSI Standards and be machined flat face. Pumps shall be fitted with bronze renewable case wear rings indexed with a dowel pin for fixed positioning.

5.4 Impeller

The bronze impeller shall be an enclosed type double suction design, hydraulically and dynamically balanced. The impeller is to be securely mounted on the stainless steel shaft, and attached with a stainless steel impeller key. The impeller shall be locked in position by threaded sleeves.

5.5 Bearings

The stainless steel pump shaft shall be adequately supported by the pump bearings to limit the shaft deflection to 0.05mm.

Bearings shall be the ball type, grease lubricated and locked to the shaft with positive locks of ample size to withstand any axial thrust loads.

5.6 Shaft Seal

The pump manufacturer shall recommend the proper mechanical seal based on the pressure, temperature and liquid outlined on the equipment schedule.

5.7 Shaft Sleeves

Bronze shaft sleeves shall be firmly attached to the pump shaft through threading and locking means. Shaft sleeve design shall prevent corrosion and wear to the shaft.

5.8 Base, Coupling and Guard

- a. The pumps shall be mounted on a cast iron base and directly connected through a heavy duty flexible coupling to a horizontal motor. The base shall be provided with a drip rim. The pump manufacturer shall provide coupling guard which shall be mounted between the pump and motor and attached firmly to the base.
- b. Differential Pressure Indicating Transmitter across suction and discharge of the pump and strainer shall be installed only on primary side of ETS Heat Exchangers.

5.9 Motors

The motor shall be sized to operate continuously without exceeding the horsepower rating regardless of the flow and head throughout the entire range of operation. Motors shall be TEFC, 1450 RPM and have IE3 efficiency. Windings shall have Class 'F' insulation and shall have thermistors in windings to protect against excessive temperature rise. Motors shall be from USA / Europe. Motors shall be selected for the pumps to operate at any point of the pump curve.

5.10 Variable Speed Pumps.

The secondary and tertiary chilled water pumps units shall be variable speed systems. The pumping system shall include as a minimum, the programmable controller, adjustable frequency drives, remote sensors, motors and pumps.

5.11 The pump manufacturer shall take “Unit Responsibility” for the successful operation of the system supplied by him.**5.12 The pump manufacturer shall supply the complete variable speed system and shall have a minimum of ten years’ experience with variable speed pumping systems.**

5.13 At the time of submittal for approval, the pump manufacturer shall provide a complete system profile analysis which shall include as a minimum the pumps performance (variable speed pump curves) and the operating characteristics in the system (system curve).
This system profile analysis shall include pump motor and adjustable frequency drive efficiencies, load profile, staging points, horsepower, and kilowatt / hour draw.

5.14 Variable Speed Pump Controller

5.14.1 The pump manufacturer shall supply and be responsible for the operation of a programmable solid state variable speed pump controller. The controller shall prevent hunting.

5.14.2 An internal P.I.D or proportional, integral, derivative controller shall be furnished.

5.14.3 The controller shall control pump speed in response to one remote set point sensor analog input. The set point shall be resettable by the operator through keypad entry.

5.14.4 The variable speed pump controller shall be capable of performing diagnostics to verify sensor input on the readout device.

5.14.5 The readout device shall incorporate alpha/numeric ½ inch display in bright LED or fluorescent characters. Operator interphase shall be through a sealed membrane type keypad. Operator access shall be protected with key lock security to allow only authorized program changes.

5.14.6 The controller components shall be mounted in a NEMA 1 enclosure.

5.15 Adjustable Frequency Drives

5.15.1 The adjustable frequency AC motor drive (AFD) shall be of the pulse width modulated (PWM) inverter type. The AFD shall be designed to convert the 50Hz input power to adjustable frequency output power.

The variable frequency power and logic unit shall be completely solid state. The unit shall transform 400 volt 3 phase 50Hz. input power into frequency and voltage controlled 3 phase output suitable to provide positive speed control to standard induction motors. The speed control shall be step less throughout the range under variable torque load on a continuous basis. The adjustable frequency control shall have a diode bridge / capacity input designed to provide high, constant power factor of 0.98 regardless of load or speed and eliminates SCR line noise. Gate turnoff (GTO) device shall be employed for high reliability in the output power switching circuit.

The AFD shall be capable of operating any standard squirrel cage induction motor with load rating within the capacity of the AFD. At any time, it shall be possible to substitute a standard motor in the field without requiring modification of the AFD.

Equipment will be designed and manufactured in accordance with applicable current NEMA and IEEE recommendations and be designed for installation per NEC. Equipment will have UL and / or CSA approval.

The AFD shall be a dedicated variable torque design for specific use with centrifugal loads.

5.15.2 Self Protection and Reliability Features.

- a. Current Limit to 110% of drive rating.
- b. Instantaneous Overcurrent Trip.
- c. Under voltage Trip.

- d. Over temperature Trip.
- e. Short circuit protection – phase to phase to ground faults phase rotation insensitive.
- f. Momentary Power Loss.
- g. Transient Protection – against all normal transients and surges in incoming power line.
- h. Orderly shutdown – in the event of any of the above conditions, the drive shall be designed to shut down safely without component failure.

5.15.3 Standard Features.

- a. Enclosure – oil tight enclosure.
- b. Drive logic shall be microprocessor based.
- c. Stand-alone operation – to facilitate start – up and troubleshooting procedures.
- d. Fused input protection.
- e. Input power – 400V + 10% 50 Hz, 3 phase.
- f. Output power – 0.415V, 2-50 Hz, 3 phase.
- g. Ambient temperature
Operating 0°C to 40°C (32°F to 104°F)
Storage -20°C to 60°C (-4°F to 140°F)
- h. Altitude – 3,300 ft. without derating.
- i. Isolated signal inputs.
- j. Frequency stability

Output frequency will be held to +0.1 % of maximum frequency regardless of load, +10% input voltage change or temperature changes within the ambient specification.

Indicating lights:

Power on	Current limit
Zero speed	Under voltage
Enabled	Overvoltage
Over temperature	Overcurrent

- k. Speed reference signal – 0-5 VDC or 4-20 MA.
- l. Start / stop Control – Controlled decel stop
- m. Customer contacts – SPDT contact shall be provided as required.
- n. 115-volt AC power supply.
- o. Minimum and maximum speed control.
- p. Adjustable Accel / Decel – independently adjustable 10-100 second.

q. Hand - off - Auto switch (H – O – A)

r. Auto restart – after power outage.

5.15.4 Provide One (1) year manufacturer's unconditional warranty for CHW pumps including spare parts and manpower.

5.15.5 All pumps delivered to site must have a factory performance test report confirming the all parameters. Only for Locally assembled pumps such performance test shall be witnessed by the engineer.

6.0 DYNAMIC PRESSURIZATION UNIT

6.1 Supply and install pump controlled dynamic pressurization unit with integrated automatic water make-up and atmospheric de-aeration system for heating & cooling water systems.

6.2 The unit shall be two pumps and two motorised spill valve units for maintaining system pressure. The two stainless steel multi-stage pumps shall act as duty/standby/assist with automatic run-time change over sequences. Both pumps shall be provided with individual adjustable soft start /stop function.

6.3 There shall be two electrically actuated fail-safe spill ball valves with duty/assist-standby function. The self-optimizing feature of the spill valves, in operational modes shall open/close in terms of percentage and should continuously adapt to the current system pressure and shall not be affected by dirt volume in the inlet strainers. The use of solenoid valves and/or mechanical valves for the unit spill function shall not be used avoid the pressure peaks.

6.4 A safety relief valve shall be included and set at 5 bar to act as a pressure safe guard for the basic and secondary vessels which must have a design pressure maximum of 6 bar. This safety relief valve is exclusively provided for the above-indicated purpose and not to be considered as a safety relief for the connected HVAC system which must be equipped with proper safety relief valves which are adapted to the maximum allowed system pressure.

6.5 The unit shall be mounted on a base permitting an easy assembly on site and shall be provided with inlet strainers/pump check valve and lockshield ball valves for individual isolation and system connection lines.

6.6 The device shall have EC marking and suitable for use in surroundings where a low sound level is required.

6.7 The system pressure is required to be monitored by a touchscreen microprocessor controller. The control unit shall read the system pressure via a pressure transducer and shall maintain the pressure within the close tolerance limit of +/- 0.2 bar.

6.8 The controller shall feature a TFT colour display which is integrated into a sturdy plastic cabinet (IP 54 protection class) and as standard directly mounted on the hydraulic unit. Communication facilities of the controller shall consist of:

- resistive colour touchscreen,
- two RS485 interface provisions for data exchange and BMS connection,
- Volt-free output for common fault signalling and low water level
- Two electrically isolated analog outputs for system pressure and water content of the basic vessel
- Input terminal for the evaluation of a contact water meter pulse.

6.9 The controller screen shall have the graphical display of system pressure, the content of the expansion vessel and the operating & error messages. These messages can be transferred to BMS system using the soft communication such as ModBus, BacNet or voltfree contacts.

- 6.10 The controller shall feature a parameter and error memory. This way the latest 20 error messages can be chronologically displayed with an indication of date and time of occurrence. In case of error message occurs, the root cause & remedy shall be displayed on the controller screen.
- 6.11 The degassing of a partial flow of the installation water shall be processed according to an optimized time schedule with selected table degassing programmes in the controller. After a period of continuous degassing (adjustable and depending on the installation volume) at the start up, the controller shall switch automatically to an interval degassing mode.
- 6.12 Basic/Secondary Vessel: Supply and install bladder type expansion vessel - unpressurized built and tested according to EN 13831 and EU pressure equipment directive 2014/108/EC.
- 6.13 The vessel shall be manufactured of steel with grey external, anti-corrosion paint protection. The high quality of butyl bladder shall be used to prevent the vessel contents from permeation of oxygen. The vessel shall be equipped with an electrode connected to a relay to indicate any rupture/damage to the bladder.
- 6.14 Vessel shall be supplied 2 x flexible connection set to suit the hydraulic module, automatic air vent with non- return valve to avoid the ingress of air (deaeration), level transducer (tank volume) and swan neck (anti-vacuum). Vessels shall stand vertically with steel profile or tubular feet.
- 6.15 Buffer vessel shall be provided to compensate for minor pressure changes and prevent frequent switching of the pump and the spill valve.
- 6.16 Supply and install combined backflow preventor with contacts water meter assembly with a valve for the direct connection of make-up units for heating and cooling systems with the water mains.
- 6.17 The unit shall consist of - two isolation ball valves, - type; BA backflow preventer (RPZ valve) according to DIN 1988/T4 (FA) respectively EN 1/17 (BA), provided with integrated dirt strainer. - water meter with volt free pulse output, - wall bracket supplied for horizontal mounting.
- 6.18 The total added water volume shall be indicated on the display and an error message shall be activated in the event of overshooting the maximum makeup volume (added up). This shall also be the case when the established input make-up quantity exceeds against an input volume in one filling cycle, or if the make-up system because of malfunction lets water through without being activated by the control.
- 6.19 The size of the dynamic pressurisation unit shall be calculated based on a system volume, static head or system working pressure and permissible system temperature.

7.0 DIRT SEPARATOR

- 7.1 Furnish and install In- Line type dirt separator suitable for a maximum pressure of 10 bar and if necessary 16 or 25 bar shall be installed according to the operating pressure and 110°C operating temperature and sized for the full capacity /flow.
- 7.2 The entering velocities not to exceed 1.2 m/s at a specified flow rate. Units specifically designed for high-velocity systems may have an entering velocity of up to 3.0 meters per second and shall be used in case of pipe size or space constraint.
- 7.3 The separator must confirm to design as per Pressure Equipment Directive 2014/68/EU article 4 paragraph 3
- 7.4 The vessel shell diameter to be of a larger cross-section than the inlet/outlet pipe diameter to reduce the flow speed in the separator. Inlet and outlet to be full size.

- 7.5 The core element is a tube mesh construction made of copper wire and the flow to be guided through an area with a greater cross-section than the connection dimensions in order to reduce the flow. The ensuing turbulence caused by the tube mesh causes heavy materials to move in an undetermined direction. Depending on the volume flow, density, and volume, parts of these sludge particles are supported in their natural breakaway motion and guided to the bottom section of the casing. The capacity to capture large amounts of sludge results in longer intervals before cleaning becomes necessary. The magnetic rod shall be inserted in the copper wire tube mesh to separate the iron particles. The Dirt separator shall be able to remove effectively dirt particles to 5 microns.
- 7.6 The units shall be designed to prevent clogging and to maintain a constant pressure drop across the unit irrespective of the amount of dirt trapped.
- 7.7 Demountable flange unit shall be provided in which the lower section can be opened, and the internal mechanism removed for cleaning and inspection.

8.0 CLOSE CONTROL UNITS (Where Applicable)

8.1 DESCRIPTION

A Description: Packaged, single chilled water coil or dual coil (chilled water and direct expansion with air cooled condensing unit) as shown on drawings, factory assembled, pre-wired and pre-piped unit, consisting of double skin cabinet, fans, filters, sound attenuators (on supply and return air), humidifier, electric re-heater, and controls. Complete factory charged with refrigerant R134a.

Unit configuration shall be as per schedule on drawing.

B. The capacity of the close control unit shall be sized based on 20 C at 50 % RH room conditions.

8.2 CABINET AND FRAME

A Structural Frame: 10 gage (3.3 mm) welded steel suitably braced for rigidity, capable of supporting compressors and other mechanical equipment and fittings with welded tubular steel floor stand with adjustable legs and vibration isolation pads.

B Doors and Access Panels: 18 gage (1.2 mm) galvanized steel with poly-urethane gaskets, hinges to allow removal of panels, and concealed fastening devices.

C Insulation: Thermally and acoustically line cabinet interior with 1 inch (25 mm) thick 1½ pound density acoustic duct liner.

D Finish of Exterior Surfaces: Baked-on textured vinyl enamel.

8.3 EVAPORATOR FANS AND MOTORS

A. Fans: Double inlet, forward curved centrifugal fans, statically and dynamically balanced, on steel shaft with self-aligning grease lubricated ball bearings, and V-belt drive.

B. Motor: TEFC, class 'F' insulation.

C. V-Belt Drive: Cast iron or steel sheaves, dynamically balanced, keyed, variable and adjustable pitch motor sheave, minimum of two matched belts, drive rated minimum 2.0 times nameplate rating of motor.

8.4 COMPRESSORS

- A. Type: Semi-hermetic with suction gas cooled motors, vibration isolators, thermal overloads, oil sight glass, manual reset high pressure switch, pump down low pressure switch, suction line strainer, reversible oil pumps, 1450 rpm.
- B. Compressors: Individually serviceable without dismantling other components or removing unit from service.
- C. Refrigeration Circuits: Two, each with hot gas mufflers, thermal expansion valve with external equalizer, liquid line solenoid valve, liquid line filter-drier, refrigerant sight glass with moisture indicator, service shut-off valves and charging valves and accumulator sized for liquid seal under light load.

8.5 EVAPORATOR COILS

- A. Alternate row circuits, direct expansion cooling coils of seamless copper tubes expanded into aluminium fins in A-frame configuration.
- B Mount coil assembly in stainless steel drain pan.

8.6 REMOTE CONDENSERS

- A. Air Cooled: Air cooled refrigerant condenser consisting of corrosion resistant cabinet, copper tube aluminum fin coils arranged for two circuits, multiple direct drive propeller fans with permanently lubricated ball bearings single phase motors with internal overload protection; refer to Section 15065. Refer to Section 15675. Provide capacity control by cycling fans.

8.7 CHILLED WATER COILS

- A. Seamless copper tubes expanded into aluminum fins with two way PICV Control valve.

8.8 FILTERS

- A. Media: Pleated, lofted, non-woven, reinforced cotton fabric; supported and bonded to welded wire grid; enclosed in cardboard frame; 2 inch (50 mm) nominal thickness.
- B. Rating, ASHRAE Std 52.1:
 - 1. Dust spot efficiency: 25-30 percent.
 - 2. Weight arrestance: 90-92 percent.
 - 3. Initial resistance at 500 FPM (2.54 m/sec) face velocity: 0.30inch WG (75 Pa).
 - 4. Recommended final resistance: 1.0-inch WG (250 Pa).

8.9 REHEAT/HEATING COILS

- A Heating Coils: Enclosed fin electrical elements arranged for minimum of two stages.
- B Circuit Protection: Primary and secondary thermal cut outs, differential air pressure switch, and manual reset overload protection, branch circuit over current protection.

8.10 HUMIDIFIER

- A. Infrared Type: High intensity quartz lamps mounted above stainless-steel evaporator pan, serviceable without disconnecting water, drain, or electrical connections; pre-piped and utilizing

condensate water from cooling coils with stainless steel or brass float valve mechanism; located in bypass air stream; with flush cycle timer and solenoid drain valve.

B. Evaporative Pan Type: Stainless steel pan and cover, serviceable without disconnecting water, drain, or electrical connections; pre-piped with stainless steel or brass float valve mechanism; electric heating coil and low water cut-off switch; with flush cycle timer and solenoid drain valve.

C. Electrode Steam Type:

1. Type: Self-contained, replaceable cylinder, microprocessor controlled electrode steam generating unit.
2. Cylinders: Disposable, polypropylene plastic with field adjustable steel electrodes.
3. Plumbing Components and Valve Bodies: Plastic, linked by flexible rubber hosing, with water fill with air gap and solenoid valve incorporating built-in strainer, pressure reducing and flow regulating orifice, and drain with integral air gap on drain.
4. Cabinet: Steel, 16 gage (1.5 mm) with enamel finish, with hinged and lockable access door.
5. Control: Fully modulating control to provide gradual 0 to 100 percent capacity with field adjustable maximum capacity; high water probe.
6. Drain Cycle: Field adjustable drain duration and drain interval.
7. Steam Distributor: Stainless steel steam dispersion tube.

8.11 ELECTRICAL PANEL

- A. Control Cabinet: NEMA 250; Type 2 enclosure, UL listed, with piano hinged door, grounding lug, combination magnetic starters with overload relays, circuit breakers and cover interlock, and fusible control circuit transformer.
- B. Disconnect Switch: Non-automatic molded case circuit breaker with handle accessible with panel closed and capable of preventing access until switched to "off" position.

8.12 ELECTRONIC CONTROL SYSTEM

- A. Solid state with start button, stop button, temporary loss of power indicator, manual reset circuit breakers, temperature control humidity control, and monitor panel.
- B. Monitor Panel: Back lighted with no visible indicator lights until operating function is activated; indicators include cooling, humidification, loss of air flow, change filters, high temperature, low temperature, high humidity, low humidity, high head pressure (each compressor), and low suction pressure (each compressor).
- C. Temperature and Humidity Control Modules: Solid state plug-in with adjustable set point, "push-to-test" calibration check button, and built-in visual indicators to indicate mode of operation.
- D. Location: Through hinged door in front of unit; isolated from conditioned air stream to allow service while system is operating.

8.13 MICROPROCESSOR CONTROL SYSTEM

- A. Logic Circuitry: Microprocessor shall continuously monitor operation of process cooling system; continuously digitally display room temperature and room relative humidity; sound alarm on system malfunction and simultaneously display problem. When more than one malfunction occurs,

display fault in sequence with room temperature, remember alarm even when malfunction cleared, and continue to display fault until reset.

B. Malfunctions: Power Loss, Loss of Air Flow, Clogged Air Filter, High Room Temperature, Low Room Temperature, High Humidity, Low Humidity, Smoke/Fire, Compressor No. 1 - Overload, Compressor No. 1 - Low Pressure, Compressor No. 1 - High Pressure, Compressor No. 2 - Overload, Compressor No. 2 - Low Pressure, Compressor No. 2 - High Pressure, Water-Under-Floor, and Supply Fan Overload.

C. Light Emitting Diodes Display: Control Power On, System On, Humidification, De-humidification taking place, Compressor No. 1 operating, Compressor No. 2 operating, Heat or Reheat operating, Economy Cooling.

D. Push Buttons: Provide to STOP process cooling system, START process cooling system, SILENCE audible alarm, push-to-test LED indicators, and display room relative humidity.

E. Remote Signalling: Provide termination for remote signalling of system status and alarms.

8.14 CONDENSATE PUMP

A. Factory mounted condensate pump complete with sump, motor and automatic control.

8.15 WATER SENSOR

A. Solid state water sensor mounted under the unit to sense the presence of water.

B. Connection to microprocessor panel to activate an audible alarm upon sensing water.

C. Message display "Water detected in under floor area" upon activation.

9.0 NOT USED

10.0 DESSICANT DEHUMIDIFIER For Indoor Pool (Where Applicable)

10.1 GENERAL

The **Dehumidifier** is the key component in maintaining the critical requirement of moisture and humidity in the process area.

The dehumidifier shall be of the industrial type and manufactured to the highest **consistency** and **reliability** and **ecology** standards and the dehumidifier shall, therefore, be manufactured in an ISO 9001 and ISO 14001 certified manufacturing facility employing CNC sheet metal manufacturing.

The dehumidifier shall be fully factory assembled, and shall be fully functionally tested at the manufacturer's facility prior to shipment.

The dehumidifier manufacturer shall have sufficient factory trained service personnel in major metro towns to provide a quick and efficient service back up, whenever required.

The **dehumidifier** shall be of the rotary type, employing the principle of chemical adsorption to remove moisture from the air on a continuous basis by use of desiccant synthesized on fluted type media in rotor form.

The dehumidifier, designed for industrial applications, shall conform strictly to the specifications given hereunder.

10.2 ELIGIBILITY CRITERIA

The dehumidifier manufacturer shall have an experience of manufacturing and applying industrial type rotary desiccant dehumidifiers for at least 20 years.

The manufacturer should also have in house rotor testing facility to test the performance of the rotor at the given inlet conditions.

The bidder should have adequate manpower/required strength for carrying out the design engineering, procurement, manufacturing, assembling, testing and commissioning of the system.

The bidder has to submit their previous experience and reference approval copies for similar unit configurations.

10.3 SCOPE OF WORK

The contractor shall furnish, install and place in operation, acceptable to the engineer, all dehumidifier units with flexible connectors, filters, volume dampers, control panel, and all necessary controls for proper operation of the system, as shown on contract drawings and as specified herein.

The dehumidifier unit shall be arranged to provide continuous dehumidification with constant outlet temperature and humidity for a given inlet condition. The unit shall be a two compartment type using “**Ecodyr**” fluted (honeycomb) desiccant media with process and regenerative sections.

10.4 CONSTRUCTION

The **dehumidifier** shall be capable of sustained operation of either process or reactivation fan without energized react heat for prolonged periods of operation without damage to the humidity transfer media or bed contactor.

The **dehumidifier** shall consist of a high efficiency “**Ecodyr or equivalent**” desiccant rotor/bed mounted on a horizontal fixed shaft arrangement. It shall have positive sealing of process and reactivation air streams to allow for independent airflow with no balancing required. The dehumidifier shall be fully automatic, factory assembled package unit, complete with reactivation heaters, filters, motors, fans, desiccant bed drive assembly, access panels, volume dampers, weatherproof or dust-tight automatic electrical panel, desiccant rotor/bed, and all component auxiliaries as recommended by the manufacturer for safe, unattended, automatic operation.

The **dehumidifier** shall not require desiccant charging, pneumatics, or field erection of fans unless specifically requested by the engineer.

The unit should be constructed with a structurally integral welded frame work completed with double insulated panel work to ensure 100% vapor tightness and thermal resistant structure capable of having a ‘G’ Value ≥ 5 .

The dehumidifier framework/casing shall be made from tubular stainless steel hollow section, duly welded, and shall be of industrial design and robust body of industrial quality, for maximum durability, life, and strength, with advanced component welding and joining techniques.

Within the industrial quality welded SS framework, shall be provided double skin insulated panels. The external skin/panel shall be pre coated sheet and at least 0.80mm thick. The inside skin/panel shall be GI and at least 0.80mm thick. The insulation, between the panels, shall be glass fiber, or rockwool, with a minimum density of 48kg/cu.mt.

The dehumidifier framework/casing together the sandwich panels, with advanced component welding and jointing technique, ensure a vapour tight (zero vapour leakage) construction, to achieve a leakage standard complying with SMACNA class 6, which allows a maximum leakage of 16 cfm/100 sq.ft of the casing surface at a pressure of 4” (100 mm) water gauge.

The entire equipment shall be Single Skid mounted and should not require any further foundation at site.

The dehumidifier unit casing shall be compact and space saving in design, and, for ease of service and inspection, shall have test points for air pressure and air flow measurement. An observation window shall be provided to permit visual inspection of the bed while the unit is in operation.

The “*Ecodry*” desiccant honeycomb rotor media shall be adsorbent, non-toxic, non-flammable, fully water washable, having metal silicate desiccant synthesized in-situ, on an inorganic fiber substrate, with net organics less than 2%. The **desiccant rotor** shall have a perimeter flange, and shall have “teeth” located around the perimeter, to ensure a positive and slip free rotation with a chain drive mechanism. The desiccant rotor shall rotate on integral long life bearings supported by a sample fixed shaft design which allows a simple slide out of the bed along with the centre shaft, for ease of maintenance and inspection. The desiccant rotor shall have long life with adequate air filtration.

Special high quality silicon bulb seals, dividing the process and reactivation airstreams, shall be designed to positively seal the two air streams for inter-compartmental air and moisture leakage, at upto 6 to 8” W.G. differential pressure. The seals shall be made from high temperature silicon rubber with PTFE composition coated face for low friction, durability, and long life. For maximizing the seal life, the perimeter seals should not contact the face of the fluted media, but seal the smooth perimeter flange of the rotor, thus ensuring that the seals will have a long life and will not be cut, torn, or otherwise damaged. Seals should not be fastened, screwed, or rivetted, but should be of the slip on type so that the seals are simple and easy to replace with no tools required.

The reactivation heat source, built into the dehumidifier casing/unit, shall be electric and installed with filters, as detailed in the plans.

Both process and reactivation fan and motor assemblies shall be factory mounted as an integral part of the dehumidifier. The process and reactivation fan blowers shall be designed for a minimum external static pressure of 1.2 in. and 1.0 in. W.G. respectively.

Plug fan shall be provided for **process air**.

TEFC (totally enclosed fan cooled) type motors shall be provided for process and reactivation air.

The **control panel** shall be housed integrally with-in the unit casing and shall have factory wired fan motor starters, and auxiliary controls. Starting and stopping of the dehumidifier unit shall be sequenced by a humidistat.

10.5 DESICCANT ROTOR

The desiccant honeycomb rotor media shall be adsorbent, non-toxic, non-flammable, fully water washable and PH Neutral. The substrate of the rotor shall not be made from asbestos or any synthetic material, and shall not have any toxic desiccants impregnated like lithium chloride, etc.

The desiccant media shall have in-situ synthesized metal silicate desiccant on an inert inorganic fiber substrate.

The active desiccant material shall be at least 80% of the media weight, so as to ensure high performance and minimal heat carry over.

The net organics in the honeycomb media shall not exceed 2%.

The desiccant rotor shall have integral long life bearings supported by a simple fixed shaft design to allow a simple slide out of the rotor/bed.

The desiccant media shall have a perimeter flange which should encircle the entire perimeter so as to allow greater durability and to roll the rotor on the ground, without damage. The perimeter flange should be smooth and consistent to serve as a perimeter seal surface, thus ensuring long life for the perimeter seal, without being cut, torn or otherwise damaged.

The desiccant media shall not fracture due to repeated temperature and moisture cycling and all the materials of construction shall be non toxic.

The surface of the media shall have a special edge hardening so as to ensure a smooth surface and long life of both the media and the seal contacting it.

The desiccant media shall not use any organic burn-off process, as this shall weaken the media structure.

10.6 FILTERS

All air entering the dehumidifier must be filtered. Filters shall have minimum rating of EU4 and of a size and capacity as recommended for air shown.

10.7 VOLUME DAMPERS

All volume dampers on the dehumidifier shall be of multiblade, opposed blade type with locking quadrants. Dampers shall be factory installed at the outlets of both process and reactivation air fans. The damper shaft exposed at the quadrant shall be notched to indicate the position of damper blades.

10.8 PACKAGEABILITY

The dehumidifier unit shall be modular in design so as to be able to connect with factory standardized post cool modular boxes. These modular boxes shall have the same construction method as the unit casing for material of construction, water vapour leakage, industrial use, and paint work.

Unit must be provided with Inbuilt Pre-mixing box for Treated fresh air and Room return air. Pre mixing box also must be same construction method as the unit casing for material of construction, water vapour leakage, and paint work.

10.9 POST COOLING COIL SECTION

Unit should include Post Cooling Coil

Cooling coils should be made of copper tubes / Aluminum fins

Coil performance shall be rated in accordance with ARI Standard 410-2001.

Cooling coils shall be cartridge type mounted for side access.

All cooling coils shall be leak tested with 350 PSIG air pressure under water

Post-Cooling coil shall be designed for maximum face velocity of 600 and the material of construction for tubes is 1/2" OD Copper with thickness of 0.41mm, casing material shall be 16 gauge galvanized iron, Coil header with Mild steel adapter and fins are in 36 gauge Aluminium with 10-12 FPI.

Coil tubes shall be copper mechanically expanded into Aluminium Corrugated Fins. . No soldering or tinning shall be used in the bonding process. Coils shall be designed to maximize the use of the unit cross section area.

Coil casing frame shall be manufactured of galvanized Sheet with drain holes in the bottom channels to ensure condensate drainage stacked coils shall have intermediate drain channels to allow top coil condensate drainage

DRAIN PAN

Units with cooling coils shall have a dual pitch sloping drain pan with a bottom connection to eliminate stagnation of condensate. The drain pans shall be corrosion – resistant, insulated to prevent condensation and installed right below the cooling coil.

10.10 ELECTRICALS & SAFETY DEVICES

The control panel shall be factory pre-wired for the system operation and shall be supplied with fuse/MCB's contactors, starters, control transformer, overload protection, and all the necessary components to ensure a continuous automatic operation.

The unit shall be provided with safety interlock option, so that the dehumidifier cannot run if access panels or the control panel door are open/ removed.

In accordance with IEC regulation, one independent thermostats shall be provided for automatic heater cut off.

Smart-2.0 control system having 7" colour touch panel for easy interface and having following features:-

1. Latest PLC controller to supervise the operation, fault management along with display messaging of system parameters and faults.
 2. Continuous data log of system parameters and on line display of operating parameters. (P in, P out, R in, Rout temperatures, RH / Dew point for P in and Pout, relative pressure (process & react). This can be done easily by putting USB pen drive to the system. The file generated can be opened in MS-excel for further analysis.
 3. RS485 communication ports
 4. Ethernet port
 5. Graphical representation of data on touch panel
 6. BACnet/Modbus communication protocol
 7. Remote start / stop facility to switch on/off the dehumidifier through BMS
 8. Programmable security code to provide unauthorized mishandling
 9. Intelligent power circuit switching to minimize voltage drops in unit supply system
 10. Real time clock / calendar
 11. Battery backup of settings
- Option for increasing additional sensors and Analog outputs

11.0 VARIABLE OR CONSTANT VOLUME TERMINAL UNITS (Where Applicable)

11.1 Terminal Box.

- 11.1.1 Furnish and install single duct, variable or constant air volume terminals of the sizes and capacities shown in the plans.
- 11.1.2 Terminals should be certified under the ARI Standard 880-98 Certification Program and carry the ARI Seal.
- 11.1.3 The terminal casing shall be minimum 22 gauge galvanized steel, internally lined with 1" thick 1-1/2" lbs/ cu.ft density insulation which complies with UL 181 and NFPA 90A. Unit shall incorporate thermally isolated primary inlet (where applicable) and be capable of accepting 50° F primary air at the inlet without casing condensation at ambient air conditions of 80% and 60% relative humidity. The discharge connection shall be slip-and-drive construction for attachment to metal ductwork. The casing shall be constructed to hold leakage to the 4% at 3 in. wg. from maximum cfm rated for the unit. A 3'-0" long sound attenuator shall be provided per Specification. A control transformer shall be provided and factory mounted on the unit.
- 11.1.4 The damper shall be 20 gauge steel with shaft rotating in Delrin or bronze oilite self-lubricating bearings. Nylon bearings are not acceptable. Shaft shall be clearly marked on the end to indicate damper position. The damper shall incorporate a mechanical stop to prevent overstroking, and a synthetic seal to limit close-off leakage.
- 11.1.5 Actuators shall be capable of supplying at least 35 in./lb. of torque to the damper shaft, and shall be mounted externally for service access. Terminals with internal actuator mounting or linkage connection must include gasketed access panel, removable without disturbing ductwork. Casing with access panel shall be constructed to hold leakage to the 4% at 3 in. wg. from maximum cfm rated for the unit.
- 11.1.6 At an inlet velocity of 2000 fpm, the differential static pressure required to operate terminal size shall not exceed .10" wg. for the basic terminal or .15" wg. for the terminal with integral attenuator.

11.2 CONTROLS

- 11.2.1 The terminals shall be equipped with pressure independent electronic controls (Type II) which can be reset to modulate airflow between minimum and the maximum cfm as scheduled.
- 11.2.2 The terminals shall incorporate multi-point, centre-averaging velocity sensors. A minimum of four measuring ports must be parallel to the take-off point from the sensor. Sensors with measuring ports in series are not acceptable. The sensor must provide a minimum differential pressure signal of .03" wg.

at inlet velocities of 500 fpm. The sensor must provide control signal accuracy of +/-5%, with the same size inlet duct at any inlet condition.

- 11.2.3 Control devices shall be factory set for the scheduled minimum and maximum flow rates. Flow measuring taps and flow curves shall be supplied with each terminal for field balancing air flow. Each terminal shall be equipped with labelling showing unit location, size, minimum and maximum cfm set points, damper fail position, and thermostat action.

12.0 CHEMICAL TREATMENT FOR CHILLED WATER CIRCUITS.

12.1 CLEANING & PRE-TREATMENT CHEMICALS

a) Pre-treatment Chemical

This product shall be a liquid blend of Sequestrant and Dispersant designed for the gradual on or offline cleaning of non-potable water systems to remove grease, debris and mil scale. The chemical shall be dosed during pre-commissioning of the system, normally during start up.

b) High Level Passivation

The Corrosion & Scale Inhibitor shall be dosed at double the normal levels for two to three days to insure the rapid development of corrosion inhibitor films.

12.2 OPERATING PARAMETERS

The cooling tower parameters shall be maintained and operated under the following conditions or as per Cooling Tower Manufacturer's requirement, whichever is stringent:

PARAMETER	OPERATION RANGE
PH	8.2 – 8.8
Total Dissolved Solids (mg/l)	1500 Max
Total Suspended Solids (mg/l)	10 -50
Total Alkalinity (mg/l)	200 -300
Zinc (mg/l)	1 - 3
Phosphonate (mg/l)	– 2
Inorganic Phosphates (mg/l)	6 -8
Total Hardness (mg/l)	200 -300
Total Iron (mg/l)	<3
Chlorides (mg/l)	< 400
Oil & Grease (mg/l)	- 10
Corrosion Rate	– 5 mils per year
Total Viable Count	x 10 ⁵
Sulphate Reducing Bacteria	100 / 100 ml
Scaling	– 30 mg / cm ² / year
Mg X SiO ₂ (mg/l)	< 35,000
Silica (mg/l)	100 -125

12.3 GUARANTEES:

- | | | | | |
|--------------------|---|----------------|---|------------|
| a) Corrosion rates | - | For Mild Steel | : | 3 - 5 MPY. |
| | - | Copper | : | 0.5 MPY. |
| | - | SS | : | < 0.1MPY. |

- b) Scaling : 15-30mg/cm²/year.
- c) Total Viable Count : 5 x 10⁵
- d) Sulphate Reducing Bacteria's : 100/100 ml.

12.4 FLUSHING AND CLEANING

Flush all fluid-carrying systems after completion with a stream of water or appropriate fluid at the highest obtainable pressure and velocity for a period of two hours. Discharge the flushing fluid through all strainers and out through line sized valves with hose ends. Clean all strainers, repeat flushing operation to the approval of the Consultant until no foreign matter collects in the strainers. Drain all tanks and clean tanks.

If the water pressure normally available on site is, in the Consultants opinion, inadequate to perform proper flushing, then The Contractor shall provide at his own expense, pumps, tanks, etc. to give pressure and volume flows to the Consultant's satisfaction.

Ensure that all valves including control valves are fully open during the flushing and cleaning. Before starting up any fans, inspect and dean the inside of all air handling systems including fans, ducts, coils, terminal units, to ensure that they are free from dust and debris. Carry out additional cleaning operations of systems as specified in other sections of this Section.

12.5 CHILLED WATER SYSTEM TREATMENT (CHEMICAL CLEANING)

The chilled water system shall be prepared as detailed elsewhere herein the system will then be further treated.

Prior to testing and commissioning, the consultant shall be provided with water conditioning program, to control water quality. Cleaning shall be supervised, and program of conditioning administered by the conditioning company.

For Pre-commission cleaning, the system shall be dosed with the prescribed amount of the non-acid cleaning agent and a surfactant as advised by the chemical company.

Cleaner shall be run into the system and retained for a period of 72 hrs. or more as advised by the chemical company, at the end of this period the system shall be drained and flushed with clean water.

All strainers and trapping points shall be inspected, and any debris removed.

System shall be refilled with clean water, re-circulated or run through for a further four hours, and again drained and flushed. The flushing shall continue till the effluent is clear, colourless, odourless, free from suspended solids and such that the iron level is not more than 10 ppm and the conductivity is no more than 30 µs/cm greater than the makeup water.

Upon completion of flushing all strainers shall he inspected, and any debris removed.

The system shall be refilled with clean water and a specified amount of inhibitor added.

The Chemical Company representative shall conduct an analysis of the systems water after the cleaning operation and shall submit a roped to the Consultant.

Note: The Mechanical Contractor shall advise on the holding capacity of the systems to be cleaned to the Chemical Company, in order that the amount of cleaner required may be calculated. The

mechanical Contractor shall provide automatic closing system complete with chemical tank, pump and control panel.

13.0 ELECTRIC MOTORS

13.1 Unless otherwise specified in the equipment specifications, all motors supplied and installed as a part of mechanical works shall conform to the following requirements:

13.2 Motors shall be of high efficiency (IE2 as per IEC 60034-30-1), and shall be constructed and type tested in compliance with relevant sections of BS EN 60034-6 and BS EN IEC 60034-3. These shall be of totally enclosed fan cooled (TEFC) type to IP55 either with foot-mounting or flange-mounting suitable for its application. Unless otherwise specified, motors shall be designed for continuous running duty type S1 for operation during 3 years without maintenance. They shall be of such size as to drive adequately their associated equipment under the range of operating conditions, without overloading.

Windings shall be insulated with Class F insulation material to BS EN 60085, and temperature rise of windings shall be class B (I.E. maximum of 80°C. as measured by resistance method).

13.3 Electric motors up to and including 0.37 Kw shall be suitable for operation at 240 volts single phase. Motors rated above 0.37 Kw shall be of the 3 phase squirrel cage induction type with drip proof housing. Motors located directly in air streams shall be totally enclosed, fan cooled. All motors and equipment shall be tropicalized.

13.4 All motors shall be fitted with a power factor correction capacitors rated at 85% of the motor magnetizing KVA. Providing a power factor of not less than 0.93. This connection shall be in accordance with the B.S. Code of Practice. The capacitors shall be directly connected to the motor terminals. This connection shall be made by the equipment manufacturer. The capacitors shall be complete with discharge resistors.

13.5 All the motors shall be provided with a substantial earthing terminal tapped into the frame or clamping device located on the cable end box side of the motor. When air inlets are protected by a screen, such a screen shall be of corrosion resistant material.

13.6 Terminal connections shall be constructed in such a way that direct contact between screws, bolts or nuts and the conductor is avoided.

13.7 The terminals shall be alphabetically or numerically marked in such a way that the cyclic phase sequence in time of the supply corresponds to the sequence of terminal markings, the direction of rotation is clockwise when facing the driving end.

13.8 The terminal box shall be of sturdy construction with ample space for connecting cables. An earthing bolt of adequate size shall be provided inside the terminal box for connecting to the cable earthing. This will be in addition to the earthing bolt provided in the motor enclosure. The terminal box shall be adequately weather proofed with a suitable non sticking reusable non-hygroscopic gasket material. These terminal boxes also shall be provided with suitable cable sealing end boxes or cable glands.

13.9 The rating plate or plates shall be made of a corrosion resistant metallic material, fixed to a non-removable part of the frame, and shall be marked with the appropriate items in accordance with IEC 60034-1, whereby the values given shall be those actually measured.

13.10 In addition to these items the following information shall be indicated:

- a) Type of enclosure in accordance with IEC 60034-5 (IP classification).
- b) Locked rotor torque on % of rated torque

Locked rotor current in % of rated torque

- c) Allowable running-up time in seconds (ART) bearing type, size, fit, re-greasing period and type of grease.
- d) Net weight.
- e) Purchase order number.

13.11 All motor shafts shall be provided with a key way unless the motor is of the flange mounted type in which case they shall be complete with adjustable rails.

13.12 The frame shall be painted in accordance with manufacturer's standards. However, the painting shall at least consist of one coat of primer and two coats of oil-resistant finishing paint.

14.0 MOTOR STARTERS, CONTROLS SWITCHGEAR

All starters and other control components shall be supplied and installed under this section but the contractor shall ensure that all components supplied shall comply with the requirements of the relevant sections of the electric specifications.

15.0 PANEL BOARDS AND MOTOR CONTROL CENTER

Construction of all panel board, motor control centers, control panels etc. shall comply with the relevant sections of the electrical specifications.

15.1 Air handling unit / fan starter panels:

Each air handling unit shall be supplied with a starter cubicle including the following:

- a) A door interlocked isolator handle.
- b) A three position HAND/OFF/AUTO control switch.
- c) A "motor run" lamp (green).
- d) A "motor trip" lamp (red).
- e) Motor speed selector switch
- f) Motor starter assembly.
- g) Control fuse.
- h) Filter pressure drop (filter clog indication)
- j) Inter-connection with fire alarm system and extract fans.
- k) Power fuses with carriers and bases.
- l) Moisture elimination heating coil.
- m) Heating element contactors and controls.
- n) Heating element 'energized' lamp (Green)

- o) Heating element 'tripped' lamp (Red)

Note:

- Control switch "Off" - Motor shall not run.
- Control switch "Test" - Motor shall run except for local lock-off stop buttons.
- Control switch "Auto" - Motor shall start and stop under dictation from remote control circuit.

Starter panels, where installed outdoor, shall be of the weatherproof type in IP 65 enclosure.

15.2 Fan Coil Unit Control Enclosure

Control components of each fan coil unit like contactors, transformers, controllers, etc. shall be housed within an enclosure located besides the unit within the ceiling void.

15.3 Motor Control Centre

Motor control centre shall be of the cubicle type, floor mounted and shall contain the following:

- Incoming as shown on the schematic diagram.
- Busbars for each phase, neutral and earth.
- Outgoing as shown on the schematic diagram.
- Starter as indicated on the schematic diagram.
- Pilot light for each phase.
- Pilot light for each motor in the plant room.
- Push button (start-stop) for each motor and plant room.
- One ammeter with selector switch for each motor in plant room.
- One ammeter for each phase for the incomer power supply.
- Voltmeter with selector switch.
- All required interlock wiring.
- Motor duty selector switch.
- Phase monitor relay for each motor.
- Low voltage protection device.
- Bank of capacitors for power factor improvement as required by DEWA.

Where motor control centres are installed outdoor, the panel shall be of the weatherproof type (IP 65).

Motor control centre construction and components shall comply with the requirements as set out in the relevant sections of the electrical specifications.

16.0 FANS CENTRIFUGAL

- 16.1 Furnish, install and connect up complete, all supply, return, exhaust, transfer and exhaust fans. All fans shall be of the non-overloading centrifugal type.
- 16.2 Fans shall have direction of rotation, discharge direction indicated on the casing. Fan arrangement shall be selected to suit the space conditions.
- 16.3 Fan housing shall be rigidly built and braced. Where fan scroll is 480 mm or more in width, provide access door with frame and gasket. All access doors shall be so fabricated that the inner surface is flush with the inside of the scroll. Raised frame doors of the pan type shall be provided on all fans where insulation is required. The doors shall be secured to the frame by hand-grip latches and shall be provided with lift handles. Bolted doors are not acceptable,
- 16.4 Fans shall not produce excessive noise as compared to units of like size and power when used in conjunction with the specified vibration isolation. Fan manufacturers shall furnish for approval for each fan, certified sound power ratings with an octave band analysis and also the volume- horsepower-pressure characteristic curves from shut-off to free delivery. Wheels shall have ample strength and shall be statically and dynamically balanced to avoid vibration and shall have blades designed to ensure quiet efficient operation.
- 16.5 Fans, unless otherwise indicated, shall be belted to respective motors by "V" belt drives. Sheaves shall be cast steel. Drives requiring 2 grooves or more shall have variable pitch motor sheaves and companion type fan sheaves. Belts shall have a rated capacity of not less than 150% of the motor horsepower. All "V" belt drives for integral horsepower motors shall have not less than two belts and shall have sufficient belt capacity to drive fans with one belt broken. Sheave ratio shall be selected so that top fan speed is not less than 105% and not more than 110% of rated speed. Drives for fractional horsepower motors may have one belt rated at 200% of motor horsepower. If to balance an air system, a fan other than the one provided should be required, necessary changes shall be made in the "V" belt drive by the Contractor at his expense. Supply and install belt guards of perforated metal, or as approved, for all sheaves and belts. Belt guards shall be grommetted at fan and motor shafts for ease in taking tachometer readings.
- 16.6 Each fan motor shall be sized to drive its respective fan when fan is operating at a speed (due to pulley adjustment) of 10% in excess of that required to meet fan performance, and when the fan requires the maximum power at this speed.
- 16.7 All fans shall carry the AMCA, or similar approved Certification Association performance and construction seal.
- 16.8 All fans shall be factory coated with one coat of primer and one coat of machine enamel. The interior of all fans and all wheels shall be painted as per manufacturer's standard.
- 16.9 All fans discharging directly up through the roof shall be equipped with a 38 mm casing drain which shall be piped to the nearest floor drain.
- 16.10 All fans scheduled to be arranged as belted vent sets shall have overhung wheels and motors mounted on a common base with the bearings. All such belted vent sets shall be furnished with weatherproof motor housings where located outdoors.
- 16.11 All double width, double inlet fans shall be provided with galvanized steel inlet guard screens.

17.0 FANS, AXIAL

- 17.1 Axial flow fans shall be capable of giving the design flow when tested to BS 848.
- 17.2 Each fan shall be supplied with a suitably rated contactor / starter / isolator of approved pattern.
- 17.3 Fan casings shall be constructed of mild steel plates with angle stiffeners, with the casing hot dip

galvanized after manufacture. The inlets and outlets of the axial flow fans shall be flanged for connection to the system. A drain plug shall be fitted and the casing designed to permit removal of the impeller.

- 17.4 An access door is to be provided on the casings of all fans. Casings shall cover both impeller and motor so that fans can be removed without disturbing adjacent ductwork or other components of the system. A flameproof external terminal box shall be fitted on the casing.
- 17.5 All lubrication points are to be extended to the outside of the casing and in a position that will permit access in relation to the adjacent plant, services or building structure.
- 17.6 Impellers shall be die cast in Aluminium alloy and X-rayed during manufacture. The impellers shall be capable of running continuously at 20% in excess of the rated speed. The impeller shall be keyed and locked onto the shaft, which shall be statically and dynamically balanced and tested at over speed before dispatch from the Manufacturer's works.
- 17.7 Flexible connections shall be supplied and installed at both inlet and outlet of each fan.
- 17.8 The fan bearings shall be of the sleeve type wherever possible. The bearings are to be truly aligned and rigidly mounted on to the casing.
- 17.9 Graphs of the performance curve of each fan are to be forwarded to the Consultant together with a test report on the sound level at three diameters in a free field condition prior to the order being confirmed.
- 17.10 Blade angle shall be adjustable over at least a 30-degree range, with markings at the root to indicate the blade angle.
- 17.11 Axial flow fans are to be driven by electric motors of commercially silent pattern carrying a Makers guarantee in this respect. The motors shall be totally enclosed, fan cooled.
- 17.12 All motors are to be positioned to permit effective ventilation of the motor and all component parts of the fan and motor are to be suitable to withstand the temperature conditions expected in the fan.
- 17.13 Electrical connections to the motors shall be in a totally enclosed terminal box secured to the exterior of the casing. Wiring within the axial flow fans shall be suitable for the conditions within the casings.
- 17.14 A suitable steel support is to be provided for each fan, and the frame is to be fabricated from rolled steel channel with adequate cross members for bolting the fan into position. The frame shall be of welded construction with anti-vibration mountings.
- 17.15 The anti-vibration mountings are to be rubber in shear pattern and of a type that are bolted to both the fan and the support.

18.0 PARTICULAR FAN

18.1 SMOKE EXTRACT FANS

- 18.1.1 Smoke extract fans shall comply with the requirements BS EN 12101-3. They shall be approved and listed by the AHJ.
- 18.1.2 Fans and ancillaries shall be constructed to withstand the rated temperatures for the required duration as indicated and shall be suitably accredited to BS EN 12101-3. Motors and drives shall be similarly constructed or protected and ventilated to ensure that they are not subjected to fire conditions.
- 18.1.3 Units shall be suitable for normal operation in ambient conditions up to 50°C and high temperature conditions up to 400°C for 2 hours.

- 18.1.4 Impeller with High efficiency air foil blades, variable pitch angle in still position, in cast aluminium. Balancing according to UNI ISO 21940-11 norm.
- 18.1.5 Axial flow fans shall be of the long casing pattern with a heavy gauge, steel sheet casing flanged and drilled at each end. Casing shall be epoxy coated and fully airtight.
- 18.1.6 Motors shall be pre-wired with high temperature rated cable to an external terminal box through high temperature flexible conduits. All components shall be rated for the operating conditions.
- 18.1.7 Fans shall incorporate inlet and outlet flanges with pre-drilled bolt holes, and an external terminal box to IP55. The motor shall be foot mounted and protected to IP55 (shall be either 4, 6 or 8 poles), class H insulated and shall have sealed for life bearings.
- 18.1.8 Axial fan motors shall have non-overloading characteristics within the normal working range.
- 18.1.9 Fans shall be suitable for either vertical or horizontal mounting. They shall be mounted on purpose made supports incorporating anti-vibration mountings and mounting feet.
- 18.1.10 Air-tight inspection doors shall be provided giving access to fan impeller and motor and other components requiring regular servicing or maintenance.
- 18.1.11 Rotors shall be statically and dynamically balanced.
- 18.1.12 Motors in the air stream shall be totally enclosed air over or totally enclosed fan cooled squirrel cage induction
- 18.1.13 Control Panel for the Smoke Extract Fans shall be interfaced with the Fire Alarm Panel.
- 18.2 SINGLE FAN EXTRACT UNITS
 - 18.2.1 Unit casing shall be manufactured in epoxy coated aluminium alloy and shall house fan assemblies, each comprising double inlet forward curved centrifugal impellers running in an individual scroll. Impellers may be either direct driven or belt driven depending upon the fan duty. Motors shall be manufactured to BS EN IEC 60034-3, TEFV type with sealed for life bearings. Motor insulation shall be Class 'F'.
 - 18.2.2 Fans shall be fitted with air flow sensors and shall discharge into a common plenum chamber through a linked shutter system. Motors and flow sensors shall be prewired to fitted isolator accessible from outside.
 - 18.2.3 Units shall be supplied for either roof mounting or plant room use as indicated on the drawings. Units shall be supplied with auto changeover panels with duty / stand-by selector switch, run / fail indication lamps, cyclic relay for duty sharing. Starter / Auto changeover panels shall be incorporated within electrical power supply panels as indicated on electrical schematic drawings.
- 18.3 IN-LINE FAN EXTRACT UNITS
 - 18.3.1 Unit casing shall be manufactured in epoxy coated aluminium alloy and shall house twin fan assemblies, each comprising double inlet forward curved centrifugal impellers running in an individual scroll. Impellers may be either direct driven or belt driven depending upon the fan duty. Motors shall be manufactured to BS EN IEC 60034-3, TEFV type with sealed for life bearings. Motor insulation shall be Class 'F'.

18.4 STAIRWELL PRESSURIZATION FANS

- 18.4.1 Blowers shall be efficient, forward curved axial fan designed in strong, die formed steel housings.
- 18.4.2 Housing shall be of epoxy coated, reinforced, galvanized steel.
- 18.4.3 Fan motor shall be totally enclosed squirrel cage. Motor carcass shall be constructed of aluminium alloy of cast iron dependent upon the temperature of operation. The grade of motor insulation shall be suitably selected to meet the requirements of high temperature smoke venting. Motors shall have as safety margin of an average of twice the expected survival times. Motors shall be either single speed with a pressure relief damper, along with monitoring pressure sensor (low rise building, up to 10-storey) and VFD without pressure relief, including pressure sensors, for high rise building (more than 10-storey).
- 18.4.4 Motor bearings and greases shall be selected to provide long life at normal ambient temperature and still survive the emergency condition during this lifetime.
- 18.4.5 Motors shall be totally enclosed with class 'F' insulation. Fire resistance requirements shall follow the requirement of AHJ.

18.5 JET FANS

- 18.5.1 Car park jet fans shall consist of (in the direction of air flow):
- Bell mouth inlet with low loss guard
 - 50mm thick double wall silencers on the fan inlet and outlet.
 - Aerofoil Axial Flow Jet fan and motor.
 - Mounting bracket to support the fan from the ceiling
 - Discharge deflectors (If needed).
- 18.5.2 In general construction of fans shall be as per specification for standard axial fans.
- 18.5.3 Fan motors must be sized to have the power to allow the fan to handle air at normal ambient temperature to allow periodic testing at ambient temperature. Fan motors that have been de-rated to the power required to handle the hot fire smoke are not acceptable as the fans must be capable of handling relatively cool smoke during the early stages of the fire without overloading the motor.
- 18.5.4 Fans shall be rated for once only operation of 400°C for 2hour.
- 18.5.5 Car park fans shall be provided by manufacturer approved by the local civil defence authority.
- 18.5.6 Fire rating of fans should be independently tested in accordance with BS EN 12101-3. Necessary documentation as proof must be submitted to the Engineer.
- 18.5.7 Fans shall be driven by 2 motors.
- 18.5.8 The Jet fans shall have 2 speed. The low speed shall be for normal ventilation at low CO concentration and the high speed shall be for ventilation at high CO concentration and in case of fire.
- 18.5.9 The CO concentration shall be measured via CO sensors (Carbon Monoxide Detectors) to be located in the car parking area as shown on drawings and according to manufacturer's recommendations.
- 18.5.10 The CO sensors shall be connected to a central control panel which will control the operation of Jet fans.
- 18.5.11 The specifications of the Carbon Monoxide Detectors shall be as follows:

- Measuring Gas: Carbon Monoxide, Enhanced CO + radicals Detection.
- Sensor element (cell):- Electrochemical Diffusion, Electrolyte type Gel based (no leak, no membrane rupture, no orientation problem & no false alarm)
- Measuring Range: – 0-300 ppm (Standard)
- Connection: – Two and Three wire sensor
- Response time T90:- ≤ 50 Sec
- Warm-up time (First Time):- ≤ 3 Min.
- Protection Class Available in : IP44/54/65/66 (Flameproof and weatherproof)
- Stability & resolution: – ± 3 ppm
- Sensor life expectancy:- ≥ 5 years
- Calibration Interval: 1 Year.
- Sensor Coverage Area: – 400 m² to 500m²
- Humidity non-condensing : 15-90% RH
- Working temperature :-10°C to 50°C
- Power supply:- 18 – 28 VDC (Overload protected)
- Analog output signal :- Standard: 4-20 mA Linear Output, also available in 2-10VDC
- Flammability: – UL 94: V0
- Warranty: 12 Month only on manufacturing defects
- Guidelines Approval : EMC Directives 2004/108/EC, CE.

18.5.12 The Jet Fans shall be reversible type where required to change air flow direction based on the prevailing wind direction.

19.0 ECOLOGY UNIT FOR KITCHEN EXTRACT

19.1 General

Supply and install all Ecology Unit with capacities as shown in the schedule of equipment and on drawings. The space available for the unit shall be confirmed with the civil and architectural proposals and sizes of units shall be selected to fit into the spaces available.

19.2 Quality Assurance

The equipment manufacturer shall strictly adhere to following standards and specification:

- a. ISO 9001-2008 certificate required.
- b. AMCA certificate required.
- c. UL/EUROVENT certificate for filters required.
- d. European make motor to be used with CE conformity certificate.

19.3 Unit Construction

The unit casings shall be of double skinned panels. Casing shall be assembled with self-supporting modular panel elements with an integrated base frame made of zincated steel and sections along upper sides of the units.

Each Assembly shall be of compact construction. Framework is made of extruded anodized anticorodal aluminum profile, supported by high efficiency thermal break and reinforced nylon corner joint. All sections shall comprise a heavy duty framework formed from hollow rectangular aluminum sections of preformed Penta-post.

Double skin construction and sandwiched with superior polyurethane insulation between the panel interior and exterior skins. The inner panel shall be of 0.6 mm thick stainless steel construction and outer panel thickness shall be 0.6 mm with pre-coated galvanized steel. The outside of the outer skin shall be finished with a 50-micron thick coat of polyester based powder for additional protection

The base frame of the units shall be made from sendzimir galvanized sheet metal for size with largest dimensions up to 2500 mm, and hot dip galvanized U-profile for larger units. Service doors shall be provided with special gasket and locking device. Access doors 400mm and larger shall be provided with hinges and handles for external and internal opening

Access doors with width smaller than 400mm shall be removable type fitted with quick release flap turn lock with spring pressure arrangement. Sealant between panels shall be an anti-fungicide sealant material.

Door and access panels shall be hinge type with handle and lock - access door panel equipped with hard nylon hinges, nylon handle and metal lock. EPDM rubber sealed at door perimeter. This access door type shall be at the Fan Section.

Vibration Isolation: The Unit shall have internal vibration isolation system by mounting fan, motor and drive assembly on spring isolators. The fan discharge shall be connected to the air handling unit casing through canvas connection to prevent vibration transfer. In addition to the above the entire unit shall be mounted on additional vibration isolators

19.4 Filter Section

A. First stage of filtration:

Three stage filtration systems, which includes

Stage: 1 Plated filter

1) MERV Rating: MERV7

Stage: 2 Bag filter

2) MERV Rating: MERV15

Stage: 3 Electrostatic Precipitator

Air is drawn by the motor/blower through a filter bank which traps large dust particles. The remaining particles, some as small as 0.1 microns, pass into a strong electric field (ionizing section) where the particulate receives an electrical charge. The charged particles then pass into a collector plate section made up of a series of equally spaced parallel plates. Each alternate plate is charged with the same polarity as the particles, which repel, while the interlaying plates are grounded, which attract and collect.

Stage: 4 Activated carbon filter

The filters are housed in a rigid sheet metal epoxy coated casing which may be used on their own or assembled into a multi-bank arrangement in the factory or on-site. The honeycomb structured filter cells are loaded with high quality activated carbon obtained from coconut shells used in pure granular form. Their inherent strength prevents void and gap formation while the high weight of carbon in each cell maximizes filter life. The improved environmental conditions which result from control of odours and noxious vapours range from reduced fatigue, improved efficiency and fewer IAQ complaints to higher employee productivity.

19.5 Filter cells shall be of standard sizes and shall be obtained from reputed a European manufacturer. The filters shall be sealed against the filter frame using a permanently elastic gasket to a standard compatible with the filter efficiency.

19.6 All primary and secondary filters shall be installed in galvanized sliding rails.

19.7 Pressure drop tapings shall be fitted to allow manometer or filter monitor to be connected. Filter materials shall be flame-retardant, incombustible, and non-odorous and offer no sustenance to vermin. Each filter section has to be provided with manometer or other pressure measurement device/component.

19.8 Fan and Fan Motor

Fans shall be double inlet, double width, forward curved centrifugal type for up to 700Pa total static pressures and backward curved non overloading type for above 700Pa total static pressure, with galvanized steel casings and fan scroll. Fans shall be tested in accordance with AMCA 300 and 301. Backward curved impeller should be coated with 60-micron epoxy painting of high quality.

The fan shall be connected to the outlet opening by means of an airtight flexible connection. Fans shall not exceed a maximum outlet velocity of 12.0m/sec.

The degree of protection shall be IP55 with mounting method B3 and Class F insulation

Fan bearing shall be re-lubricated type with extended lube lines to terminate outside the fan section. The bearing shall be selected for life of 200,000 Hours operation. Fan drive shall be rated at 150% of the maximum motor power of the units and shall be fitted with adjustable belt tension arrangement.

The Fan section shall be supplied with double plastic glass inspection window sealed with rubber from the inside and outside panel skin.

Safety micro switch in fan section shall be provided in accordance with CEN Standard.

Light, switch with internal wiring and Isolation Switch outside the Fan Section in accordance with CEN Standard.

The fan motor shall be suitable for operating 400V, 3Ph, 50Hz electrical power supply. The fan motor shall be wired to the safety isolation switch or connection box. Motor shall be TEFC with minimum performance rating of IE1 class according to IEC standards. Motor shall have thermal reserve for overheating in case of voltage fluctuation in compliance with IEC 60034-1 standard.

The contractor shall select power input and speed of the fan subsequent to ascertaining system static pressure in accordance with pressure drop calculations to the approval of the Engineer.

The fan & motor shall be mounted on a common, torsionally rigid, aluminium base frame with motor mounted on slide rails for adjustment of belt tension

The fan and motor assembly shall be isolated from the casing by means of spring anti-vibration mountings selected to suit the speed of the fan and designed for 90% isolation

All motors rated for use with variable frequency drive (VFD) shall be compatible with the drive to which it is connected. Use of motor with a VFD shall not adversely affect the operation, useful life, or warranty of the motor.

20.0 NOT USED

21.0 AIR CURTAINS

The Air Curtain shall be suitable for doors dimension as shown on design drawings. The unit shall be made of Slim design, simple mounting, high durability and powerful airflow. The unit shall be complete with a polycarbonate cabinet of colour acceptable to Architect for the exposed units. However, for the concealed type the unit shall be mounted within false ceiling void and shall have supply air linear grille/slot diffuser at the false ceiling level. The unit shall be provided with adjustable damper to regulate the airflow.

The unit motor shall be class F insulation, totally enclosed, rated for continuous duty with lifetime lubricated sealed ball bearings and protected with thermal overload switch. The unit motor shall have two-speed control for high and low speed control.

The unit shall act as a closed door and prevent intrusion of external hot, humid or impure air. The unit shall comprise of centrifugal silent fans and cleanable air filters.

The unit dimension shall be selected to suit the door width and height.

22.0 DX SPLIT UNITS

22.1 General

The split system A/c units shall be supplied and installed as per schedule shown in the air conditioning drawings. The outdoor units shall be installed outdoor (at roof / in car parking) and the refrigerant piping shall be taken through shafts.

The equipment shall be capable of handling minimum suction lift of 98 feet (29.9m).

The capacities shall be as mentioned on drawings. The split units shall be able to work continuously at 52°C without mechanical failure.

All A/C equipment shall be with power factor not less than 0.9.

Units shall have only ozone friendly HFC (Hydrofluorocarbons) refrigerant R-407C, R-410A or approved equal. The system must meet and exceed the minimum energy efficiency requirement of ASHRAE 90.1 2019 standard. All split units must have efficiency (EER Value) more than 11.0 when rated at AHRI conditions.

22.2 Indoor & Outdoor Fan Coil Units:

The indoor units' type shall be as shown on drawings. Evaporator coils shall be constructed with aluminium fins mechanically bonded to refrigeration quality copper tubes. The unit shall be insulated with rubber insulation that is 10 mm thickness & 32 kg/m³ density. The units shall be equipped with Aluminium media cleanable type permanent filters with 1-inch aluminium frame.

Main and Auxiliary Drain Pans: Galvanized steel and powder painted. Fabricate pans and drain connections to comply with ASHRAE 62.1-Latest Edition.

22.3 Air Cooled Condensing Units:

The units shall be pre charged, prewired and installed on concrete bases in the roof of the building as shown on the drawings. Each unit shall consist of a single enclosed completely weather proof heavy duty steel casing coated with hardwearing epoxy resin paint.

Compressors shall be fully hermetic, equipped with crank case heaters, internal overloads, spring vibration isolators and muffler. Access to controls and compressors shall be via full width removable door fitted with quick release fasteners. The condenser coils shall be manufactured from refrigeration quality copper tubes with mechanically bonded aluminium fins. The condenser fans shall be large vertical, or side discharge slow speed propeller types vertically mounted into the unit, protected by a corrosion resistant fan guard. Motors shall be totally enclosed type immolated to class F. All compressors shall have five years guarantee. Units shall have only ozone friendly HFC (Hydrofluorocarbons) refrigerant R-407C, R-410A or approved equal. The system must meet and exceed the minimum energy efficiency requirement of ASHRAE 90.1 2019 standard. All DX units must have efficiency (EER Value) more than 11.0 when rated at AHRI conditions.

22.4 Room Thermostat:

Thermostat shall be of digital type for the indoor units. The following minimum features shall be available in selected brand /model.

1. 24-hour time control of system stop and start.
2. Liquid-crystal display indicating temperature, set-point temperature, time setting, operating mode, and fan speed.
3. Fan-speed selection, including auto setting.

22.5 REFRIGERANT PIPEWORK

The interconnecting refrigerant piping shall be manufactured from seamless quality phosphorous dioxide soft/medium drawn copper to BS EN 12449, ASTM, DIN 12449/8905, fully tested by the eddy current method, dehydrated and capped. The copper refrigerant pipes shall be nitrogenized Type L ACR 700 psi rated for 410 A application.

The copper tubing shall be of a suitable thickness to be capable of withstanding pressure test as detailed in the current revision of BS EN 378.

The copper tubing shall be suitably selected with a wall thickness capable of withstanding the system operating pressures on Refrigerant R410A, all as described within the outdoor unit's installation manual.

Pipework shall be supported and not secured at regular intervals not exceeding 1.5m. Preferably to create a method of adequate support and containment, piping should be laid on suitably supported galvanized cable tray, insulated and held in position with plastic coated steel banding. Due care and attention shall be taken at all times to allow pipework to freely expand and contract naturally.

Leak Test

Test high- and low-pressure side piping of each system at not less than the lower of the design pressure or the setting of pressure relief device protecting high and low side of system.

- a. System shall maintain test pressure at the manifold gage throughout duration of test.
- b. Test joints and fittings by brushing a small amount of soap and glycerin solution over joint.
- c. Fill system with nitrogen to raise a test pressure of 1035kPa
- d. Remake leaking joints using new materials and retest until satisfactory results are achieved.

System Charging

Charge system using the following procedures:

- a) Install core in filter-dryer after leak test but before evacuation.
- b) Evacuate entire refrigerant system with a vacuum pump to a vacuum of 67Pa. If

vacuum holds for 12 hours, system is ready for charging.

- c) Break vacuum with refrigerant gas, allowing pressure to build up to 14kPa.
- d) Charge system with a new filter-dryer core in charging line. Provide full-operating charge as per the manufacturers recommended weight of refrigerant.

23.0 SUBMISSIONS

- 23.1 Initial technical submissions are to accompany the bid including full technical details of every item of equipment proposed for the project, with the appropriate figures and details highlighted in marker pen or similar in standard manufacturer's catalogues.
- 23.2 After final checking of sizing after award of contract, resubmission will be made with any changes specifically noted.
- 23.3 Technical submissions for the following are particularly required to be submitted.
 - Plate Heat Exchangers
 - Chillers
 - Cooling Towers
 - Pumps
 - Pressurization Unit
 - Air handling units.
 - Fan coil units.
 - Motor Control Centre.
 - Fans.
 - Ecology Units.
- 23.4 In the event that the Consultant requires further information on any of the proposed items the Contractor shall arrange visits to similar units which he has installed. In any event, all equipment proposed for the project shall be currently installed and working satisfactorily locally. The Contractor must be able to provide references of at least three satisfied Owners and Consultant with equipment of the particular size and model proposed.

END OF SECTION

MECHANICAL SECTION: 08

DUCTWORK AND AIRSIDE EQUIPMENT

1.0 DUCTWORK MATERIAL

- 1.1 Make all ductwork (except what is mentioned in clause 1.2 below) of galvanized sheet steel conforming to BS EN 10143 or ASTM A653/653M, G90 (Z275). The galvanizing shall be carefully done, and the sheets shall be of such quality that they may be bent flat on themselves with no fracture to the coating or the base metal.

All duct sizes shall be verified by contractor against actual selected machine flow and to be resized, if necessary, to limit total system flow resistance and noise. Total system resistance and fan power shall be limited in accordance with ASHRAE 90.1-Latest Edition.

All duct work with in three meter of fan coil units' outlets shall be acoustically lined.
For GI ductwork insulation refer to clause 02.

- 1.2 The following ductwork shall be of pre-insulated phenolic foam ductwork as specified in clause 1.9 below:
- Fan coil units' ductwork.
 - Fresh air supply branch ductwork to floor levels after VCD.

Ducts shall be with valid approval certificates from Local Civil Defence.
All other ductwork (extract air branch to floor levels after VCD, risers, at roof and mechanical floors) shall be in galvanized sheet steel.

- 1.3 Flexible ductwork shall be UL Listed, lightweight aluminium laminated duct suitable for low and medium pressure systems. The aluminium laminated construction shall encapsulate a high tensile steel wire helix between two layers of 0.9-micron thick aluminium plus 12-micron thick polyester. Flexible ducts shall be with factory applied thermal insulation and vapour barrier. Insulation shall be 25mm thick, 16 kg/m³ density.
- 1.4 All supports, stiffeners, flanges etc. used in ductwork construction and installation shall be hot dip galvanized.
- 1.5 Commercial Kitchen extract ductwork (Where applicable) shall conform to NFPA 96 and shall be air and watertight welded construction manufactured from black steel sheets with anti-corrosion paint. Provide access panels 508x508mm for personnel access and if this size is not possible, access panels for thorough cleaning of ducts shall be provided every 3.7 m for cleaning of ducts on horizontal runs and every two floors on vertical risers. Kitchen extract ductwork shall be insulated as specified in Section 2. Ducts shall be with valid approval certificates from Local Civil Defence.
- 1.6 Smoke extract Fire rated ductwork shall be galvanized steel without additional coating. The construction shall conform to BS 476 Part 24 for stability and integrity up to 2 hours. Ducts shall be with valid approval certificates from Local Civil Defence.
- 1.7 Commercial Laundry extract ductwork (where applicable) shall be air and water tight construction manufactured from aluminium sheets in accordance with BS 1470 and insulated as specified in Section 2.
- 1.8 Ductwork for AHUs serving indoor pools and for commercial laundry & dishwashers extract ductwork shall be in Aluminum sheets.
- 1.9 Ductwork, with equipment attached to it, with an external static pressure exceeding five hundred (500)Pascal (Pa) and all ductwork exposed to external ambient conditions or within unconditioned spaces must be pressure tested prior to occupancy in accordance with methods stated in DW 143 & SMACNA latest standards. Test pressure shall be mutually agreed with engineer and or must be higher than the system working pressure. The test shall be conducted prior to applying any filed joint thermal insulations.

1.9 PHENOLIC PREINSULATED DUCTWORK:

- 1.9.1 The pre-insulated ductwork system shall be installed for the indoor duct installation except the duct risers within shafts and outdoor ductwork which shall remain in GI material.

Pre-insulated ductwork shall be made of aluminium & closed-cell phenolic panels that are UL 181 listed.

- 1.9.2 Non-metal Ductwork fabricated from closed-cell Phenolic panels, UL Listed Class 1 Air Duct, to Standard for Safety UL 181 (Factory-Made Air Ducts & Air Connectors).
- 1.9.3 The ductwork shall have a maximum flame spread index of 25 without evidence of continued progressive combustion and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Ducts shall be with valid approval certificate from Local Civil Defence.
- 1.9.4 All insulation materials shall be manufactured with a blowing agent that has zero Ozone Depletion Potential (ODP) and Low Global Warming Potential (GWP<5).
- 1.9.5 All insulation materials shall be CFC / HCFC Free.
- 1.9.6 The vapor barrier shall be granted by the reinforced aluminum foils covering both faces of the hard foam.
- 1.9.7 Non-metal Ductwork and plenums shall be fabricated according to manufacturer's construction method, using adequate materials and accessories. All the insulation panels shall have to be printed with the name of the manufacturer and the production date.
- 1.9.8 The non-metal ductwork shall be thermoset phenolic, physiologically and chemically inert and insoluble, vermin proof, fungus proof, anti-bacterial and non-metabolisable.
- 1.9.9 The panels shall have the following characteristics:

	Air-conditioned Areas	
Minimum Thickness of Panels	20	mm
Minimum Thickness of Aluminium	80	Microns
Density of the foam	55 - 60	Kg/m ³
Thermal Conductivity	0.021	W/m.K

Both sides of the aluminium foils shall be lacquered with a 2 g/m² weatherproof and ultraviolet ray's protection polyester lacquer.

- 1.9.10 The Phenolic duct construction shall be according to SMACNA standards or equivalent. Ductwork and plenums shall be fabricated & joined in accordance with manufacturer's instructions.

The joints between the ducts to be installed in air-conditioned areas and plant rooms shall be made, using aluminium flange with aluminium slide-in bayonet.

The joints between the ducts will be completed with the installation of adhesive gasket to reduce air leakage.

Ductwork shall be installed, using supports, as described in SMACNA or DW144 & according to manufacturer's requirements. Maximum distance between supports shall not exceed:

- a) 4000mm for ducts with section not exceeding 1200 x 1000mm
- b) 2000mm for ducts with section exceeding 1200 x 1000mm

1.9.11 The Phenolic ductwork shall have the following properties:

- Thermal Conductivity: 0.021 W/m.K
- Operating temperature: -26°C to 85°C
- Minimum compressive strength at 10% compression (BS EN 826): 200 kPa
- Nominal density of insulation: 55-60 kg/m³
- Minimum closed cell content of insulation: >90%

Design Pressure: Positive of 1000 Pa and Negative of 750 Pa

Fire retardant characteristics shall be as follows: Class 0 according to BS 476 Part 6 & 7.

2.0 DUCTWORK SPECIFICATION

2.1 Ductwork fabrication and hanging methods shall be as described in the "Equipment" Volume of the American Society of Heating, Refrigerating and Air Conditioning Engineers Guide and Data Book, latest edition, and/or current edition of applicable manuals published by the Sheet Metal and Air Conditioning Contractors National Association Inc., where methods described in these volumes are not at variance with the requirements of any authority having jurisdiction, or do not conflict with methods described hereafter. Ductwork shall not be shop fabricated until work has been job measured and interference situations coordinated.

2.2 Any Contractor who is more familiar with British Standards may refer to B.S. CP. 352 and DW 142. Any discrepancy will be clarified by the Consultant on request.
In any event, the standard of work required is the best available, and the specifications are to be so interpreted.

3.0 DUCTWORK GENERAL

3.1. All dimensions shall be checked on site before ductwork manufacture is commenced.

3.2. The whole of the ductwork installation shall be carried out by an approved specialist in ductwork manufacture and installation. No ductwork shall touch the building structure or building finishes direct but shall be isolated with insulating spacer.

3.3. The fabrication shall be carried out in a neat and workmanlike manner with all ductwork true in size and cross-section, braced and stiffened as specified and with all internal and external surfaces free from projections and sharp edges.

3.4 At each main branch in ductwork and at each fan discharge and suction, provide sufficient number of Pitot tube test holes for balancing systems. Also provide test holes for traverse fan discharge and at all equipments. Test holes shall be located within easy reach of catwalks or ladder. Each test hole shall have 20 mm clear opening, provided with a metal ring plate with a threaded hole in the boss, and matching screwed head plug. Where these plugs are installed in insulated ductwork, provide an extension collar against which the insulation can be finished.

- 3.5. Reinforced holes shall be provided where thermometers, manometers, thermostats, gauges, damper rods etc., occur in ductwork. Extended collars shall be provided for the reinforced holes where these occur on insulated ductwork. Where copper tubing passes through ductwork, or casing, provide a rubber grommet to prevent damage to copper tubing.
- 3.6. Ductwork shall be rigidly suspended or supported from building structure. Expansion type concrete inserts shall be placed so that the fastener is in shear rather than tension. Powder actuated fasteners placed by an explosive charge will not be accepted. Angle type trapeze hangers with rod supports spaced at 2 meters maximum shall be used. "C" type beam clamps will not be accepted. Provide necessary steel angle iron required for bracing of ductwork or equipment, and for supporting ductwork from building structure.
- A layer of felt strip 12.5 mm compressed thickness shall be provided between any support member which is designed to clamp or grip the duct (e.g. circular duct band clip) or on which the duct is to rest. All supports shall be hot dip galvanized.
- 3.7. Increase in duct size shall be gradual. Where width or largest dimension of a duct is over 450 mm, duct shall be stiffened by bending in a break across corners in both directions. Ducts shall be self-supporting and complete in themselves. Single thickness partition between ducts will not be accepted. Visible internal portions of duct outlets to grilles and registers shall be painted in dull black.
- 3.8. All necessary allowances and provisions shall be made in the installation of the ducts for the structural framing of the building and when changes or offsets are necessary, the required cross-sectional areas shall be maintained. All of these changes however, shall be approved, and installed as directed at the time.
- 3.9. During installation the open ends of ducts shall be protected with blank, flanged sheet metal baffles, securely attached to prevent debris and dirt from entering.
- 3.10. Where ducts are shown connecting to masonry openings and / or along the edges of all plenums at floors, walls, etc., provide a continuous 30 x 30 x 4 mm galvanized angle iron which shall be bolted to the structure and made airtight to same by applying caulking compound on the angles before they are drawn down tight. The sheet metal at these locations shall be bolted to the angle iron framing.
- 3.11. All air ducts, casings, plenums etc., shall be constructed of lock forming quality prime galvanized steel sheets, which are free from blisters, slivers, pits, imperfectly coated spots etc. No second quality sheet metal allowed.
- Where damage (or rusting) has occurred on galvanized duct work the affected section shall be made good by painting with two coats of zinc-rich paint and approved finishing paint, or where the damage in the Consultant's opinion cannot be made good, then a new section of ductwork shall be provided at no cost to the contract.
- 3.12. Duct shall be constructed using double or Pittsburgh lock corner seams. All seams shall be hammered down and made airtight by applying sealant before hammering down. For transverse joints, refer to current ASHRAE guide for low pressure ductwork.
- 3.13. Support the vertical ducts installed in the various shafts at each floor level with galvanized supporting irons of approved size. Install these angles across the width of the shaft, with their ends attached to angle irons securely anchored into the masonry walls of the shaft or attached to the framing of the floor openings. The ducts shall be bolted to these supporting angle irons.
- 3.14. Ensure that all openings required through floors, walls, partitions etc., for the duct system are provided in the exact location required.
- 3.15. Each piece of ductwork shall be wiped inside and out before installation and all open ends shall be capped and sealed to prevent entrance of dirt during construction. Ensure that ductwork systems are

clean and free from dirt, dust, grime, debris etc., before initial operation of fans. Fans shall not be operated until the filters are installed and approval from the Consultant has been obtained.

- 3.16 The bottom joint and 150 mm of vertical joint on outside air intake ducts and mixing chamber ducts shall be soldered and made watertight. Provide drain connection and run copper drain pipe to nearest floor drain.
- 3.17 All fixing devices including nuts, bolts, and washers etc., used in the construction or support of galvanized ductwork shall be Sherardized, or cadmium-plated.

4.0 LOW PRESSURE DUCTWORK

- 4.1 This applies to ductwork with mean velocities less than 2,000 fpm (10 m/sec) and static pressures of 2 inches water gauge, (500 Pascal's) or less.
- 4.2 Rectangular low pressure ductwork shall be fabricated from prime quality, re-squared, tight coat galvanized steel sheets of the following gauges.

Size of Duct (Milli metres)	Gauge of Sheet Metal.
300 mm less in width/depth	No.26 US Std. Gauge (0.6 mm)
300 to 750 mm width/depth	No.24 US Std. Gauge (0.7 mm)
750 to 1350 mm width/depth	No.22 US Std. Gauge (0.85 mm)
1350 to 2100 mm width/depth	No.20 US Std. Gauge (1.00 mm)
2100 mm and over	No.18 US Std. Gauge (1.3 mm)

- 4.3 Reinforcing and joints shall be in accordance with ASHRAE Guide and Data Book, Equipment Volume, 1988 Chapter 1.

Particular note is to be made of table 5, giving rigidity classification, and, showing joint constructions. Up to 400 mm duct size is to have a "Standing S" joint as column 3 of Table 5A. Size 400 to 1200 mm is to have a "Standing S-angle reinforced" as column 6 of Table 5A.

Size above 1200 mm is to have a "Standing S-angle reinforced" as column 6 of table 5A but, with intermediate reinforcing angle as table 6.

- 4.4 Low pressure suction and discharge plenum chambers shall be fabricated from 1.3 mm (18 gauge) galvanized steel with galvanized angle iron framework and bracing.
- 4.5 In square elbows and in elbows where radius is less than 1-1/2 x width of duct, sheet metal deflector vanes shall be installed the full height of the duct, being securely riveted in place. All vanes shall be double thickness vanes of two gauges heavier than the duct in which they are installed, and shall be factory made, not site fabricated. Vanes shall be tack welded to vane rail. For vane lengths over 1000 mm, tack weld vanes to 10 mm tie-rod at mid-span.
- 4.6 Circular low-pressure ductwork (where applicable) shall be fabricated from prime quality as per DW 142 or SMACNA Standards.

5.0 FLEXIBLE DUCT WORK - LOW PRESSURE

- 5.1 Flexible ductwork shall be manufactured with a two-layer aluminized polyester inner core, surrounded by 25 mm thickness of 24 kg/m³ density fibreglass, all wrapped in a reinforced multilayer aluminium

polyester outer jacket.

Ductwork shall meet the standards of NFPA 90A and be U.L. listed or to meet BS 476 and BS 413. Maximum flame spread rating - 12 (Class 1). Maximum smoke developed rating - 50.

5.2 Duct clamps shall be with standard GI jubilee clamp or cable ties.

5.3 Flexible duct installation shall be in accordance with manufacture's instruction. Joints between factory insulated flexible ducts and field insulated ductwork shall be sealed and taped under this section.

6.0 FLEXIBLE CONNECTION

6.1 Flexible connectors at inlet and discharge to air handling equipment (AHU / FCU) shall be pre-assembled 24 gauge (0.7 mm) galvanized steel to ASTM A660/A660M with minimum of 75 mm wide (exposed) fabric. For higher pressure applications 25 mm of width of fabric shall be used for each 25 mm of static pressure.

6.2 Flexible connectors attached to acoustically treated ductwork shall be insulated with 25 mm fibreglass insulation. At equipment collar caulk between collar and flange on housing with 10 mm thick permanently flexible sealant.

Care shall be taken to ensure that ducts on both sides of the connection are independently supported and that no "bridging" occurs.

6.3 All flexible connection will be by an approved manufacturer, with ratings of fire spread, strength etc., listed by a recognized Testing Authority. All flexible connectors shall be UL certified.

6.4 The fabric shall be rated for use up to 200 degrees F (93 degrees C) working temperature with a tensile strength of not less than 100 psi (690 Kilopascals).

6.5 The material shall also be impervious to moisture, dimensionally stable, and shall not rot.

7.0 ACCESS PANELS

7.1. Duct access panels shall be minimum 450 x 300 mm. unless restricted by duct dimensions. All access panels shall be constructed from galvanized steel sheets, 25 mm apart with necessary reinforcing inside for rigidity, with space filled with glass fibre insulation. Panels shall be made airtight with a continuous neoprene rubber gasket.

7.2. Openings in ductwork shall be provided with continuous galvanized reinforcing bars, which on insulated ductwork, shall be extended to the face of the insulation. 'Small' panels shall be provided with at least two self tightening camlocks. All panels shall have drawer type handle.

7.3. Apparatus access doors shall be minimum 600 x 1500 mm with angle or channel frame. Provide two 75 mm strap hinges with brass pins; and two handles minimum, operable from inside and outside. All access panels shall be constructed from double thickness 18 gauge (1.3 mm) galvanized steel, with 25 mm insulation between sides with necessary reinforcing for rigidity. Doors shall be made airtight with "tadpole" gaskets.

7.4. Provide access panels where shown, required and directed and in the following locations:

- Bottom of all duct risers.
- Next to outside air intakes and outlets.
- At each fire damper.

- Into apparatus casings to facilitate maintenance and cleaning of all components.

8.0 BALANCING DAMPERS

- 8.1 Fabricate in accordance with SMACNA construction standards, and as indicated.
- 8.2 Fabricate single blade dampers for duct sizes to 9.5 x 30 inch (240 x 760 mm). Fabricate multi-blade damper of opposed blade pattern with maximum blade sizes 12 x 72 inch (300 x 1825 mm).
- Frame shall be manufactured from min. gauge 22 Galvanized steel with formed channel for flanged connections. Blades shall be of Galvanized steel min gauge 20 aerofoil double skin galvanized sheet with rubber gasket on tips to minimize leakages.
- 8.3 Assemble centre and edge crimped blades in prime coated or galvanized channel frame with suitable hardware. Except in round ductwork 12 inches and smaller, provide end bearings. On multiple blade dampers, provide oil-impregnated nylon or sintered bronze bearings.
- 8.4 Provide locking, indicating quadrant regulators on single and multi-blade dampers. Where rod lengths exceed 30 inches (750mm) provide regulator at both ends. On insulated ducts mount quadrant regulators on stand-off mounting brackets, bases, or adaptors.
- 8.5 Provide actuators, motors, all accessories required where motorized dampers are specified. Provide volume control dampers as shown on the Drawings and as required by Contract Administrator and Commissioning specialist, whether they are 'in-house' to the Contractor or not in the various branches of the ductwork for use in balancing the system. Dampers are to be installed separate and independent from the registers hereinafter specified to be set behind supply air grilles.
- 8.6 Provide multi-blade dampers in ducts above 600 mm in width or 600 mm in height. On all dampers located above inaccessible plaster or drywall ceilings provide remote gear operators. Provide volume dampers of the quadrant type, of heavy construction, pivoted to turn easily and provided with approved operating and locking devices mounted on outside of the duct in an accessible place.

9.0 BACKDRAFT DAMPERS

- 9.1 Backdraft dampers shall be low leakage with parallel blades and neoprene edge seals.
- 9.2 Dampers frames shall be constructed from galvanized sheet steel with Aluminium blades. Blade stub shafts shall be brass with PVC bearings. Sealing strips on blades shall be polyester foam.
- 9.3 Pressure relief dampers shall be multi-parallel blade with weighted arm closing assist. The frame shall be anodized Aluminium channel sections with formed Aluminium blades. Maximum blade length shall be 100 mm, and polyester foam seating strips shall be incorporated on blade edges. Bearings shall be in PVC with non-corrodible shafts.

10.0 SPLITTER DAMPERS

- 10.1 In each low-pressure system take off where opposed blade duct dampers are not specified, splitter dampers shall be provided. The only justification for not providing splitters is in a system which can be demonstrated to be sized by a static regain program and which is run-in high-pressure fittings.
- 10.2 Splitters shall consist of hollow blades in a vane rail assembly, made from galvanized steel by a recognized manufacturer.
- 10.3 A lockable quadrant adjustment lever shall be located outside the insulation and marked clearly to show vane position.

11.0 FIRE DAMPERS

- 11.1 Provide, where required by ordinances or codes, fire dampers made to British standard BS 476-24 or NFPA 90A, and complete with angle iron frame, 72°C fusible link, pivot rods, and spring catches. Enlarge duct sections around fire dampers, to allow 100% unrestricted duct area while in open position. Provide approved type access doors, with airtight gaskets, for inspection and servicing of fire dampers. Fire dampers shall hold integrity and stability for 1½ or 2 hours rated depending on wall fire rating and shall be labelled by manufacturer.
- 11.2 Provide fire dampers in all ducts as required by the latest UAE FLS Code, in the following locations and in accordance with Fire & Life Safety drawings for fire walls location:
- 11.3 Where fire dampers are located remote from fire partition, duct between fire damper and partition shall be encased in double metal lathe and plaster or other fireproofing acceptable to Authorities having jurisdiction.
- 11.4 Fire damper construction shall be of 1.2mm galvanized steel casing, 0.8mm galvanized steel blades, fitted with catch plates, stainless steel closure springs and stainless-steel seals. Blades shall be out of the air stream. Fusible links shall be rated for a fusing temperature of 72°C unless otherwise specified.
- 11.5 Damper installation frame shall be supplied by the fire damper manufacturer.
- 11.6 Dampers shall be UL listed or approved by an equal certification authority, and valid approval certificate from Local Civil Defence.
Dampers shall be tested to BS 476 part 20, 1987 by an independent testing laboratory.

12.0 SILENCERS

- 12.1 Silencers shall be installed when required to achieve the specified noise levels.
- 12.2 At shop drawing stage the contractor will undertake calculation for every unit and every area and determine the insertion loss required to meet the stated noise criterion. Supply and return duct noise must be considered as well as duct breakout noise.
- 12.3 Materials of construction shall be galvanized sheet metal and mineral fibre acoustic fill which is inorganic, inert, moisture and vermin resistant. Silencers shall be so constructed as to prevent erosion and pregnability of the acoustic fill.
- 12.4 Silencers shall have available certified test data concerning insertion loss. This must be available when required, in advance of ordering units. Where a silencer is made under license, or part assembled locally, then tests must also be made after assembly and witnessed by the engineer.
Locally manufactured silencers shall be with an international certifying affiliated company which shall perform periodic testing & inspections on manufacturer's products.
- 12.5 The static pressure loss of any required silencer must be considered in air handling equipment original selection.

13.0 VIBRATION ISOLATION (FOR ALL SECTIONS) & SEISMIC SUPPORTS

- 13.1 All plant and equipment shall be isolated from the building structure in such a manner that noise and vibration is not transmitted through the structure.

Anti-vibration equipment will be manufactured by a specialist company acceptable to the Consultant, and with standard ratings information available.

Anti-vibration equipment shall comply with Seismic code for zone 2B only if mandated by authority, where applicable according to specialist recommendations.

The following requirements are minimum requirements and must be confirmed by the specialist manufacturer's representative as being adequate for the particular equipment proposed by the Contractor.

Seismic Supporting system shall be considered for the following heavy equipment at Mechanical floors and roof:

Water Tanks, Water pumps, Fire pumps, Chilled water pumps, Heat exchangers, FAHUs and AHUs, Fans, Electrical panels.

Also, Seismic supports shall be considered for the bus bar risers and steel pipe risers in shafts.

For the detail of Seismic support system Refer the AC details drawings.

13.2 The Specifications of Seismic supports for the above-mentioned equipment and risers shall be as follows:

1. Two layers of 3/4" (19mm) thick neoprene pad consisting of 2" (50mm) square waffle modules separated horizontally by a 16 (1.5mm) gauge galvanized shim. Load distribution plates shall be used as required.
2. Bridge-bearing neoprene mountings shall have a minimum static deflection of 0.2" (5mm) and all directional seismic capability. The mount shall consist of a ductile iron casting containing two separated and opposing molded neoprene elements. The elements shall prevent the central threaded sleeve and attachment bolt from contacting the casting during normal operation. The shock absorbing neoprene materials shall be compounded to bridge-bearing specifications.
3. Sheet metal panels shall be bolted to the walls or supporting structure by assemblies consisting of a neoprene bushing cushioned between 2 steel sleeves. The outer sleeve prevents the sheet metal from cutting into the neoprene. Enlarge panel holes as required. Neoprene elements pass over the bushing to cushion the back panel horizontally. A steel disc covers the inside neoprene element and the inner steel sleeve is elongated to act as a stop so tightening the anchor bolts does not interfere with panel isolation in 3 planes. Bushing assemblies can be applied to the ends of steel cross members where applicable. All neoprene shall be bridge bearing quality.
4. A one piece molded bridge bearing neoprene washer/bushing. The bushing shall surround the anchor bolt and have a flat washer face to avoid metal to metal contact.
5. Spring isolators shall be free standing and laterally stable without any housing and complete with a molded neoprene cup or 1/4" (6mm) neoprene acoustical friction pad between the baseplate and the support. All mountings shall have leveling bolts that must be rigidly bolted to the equipment. Spring diameters shall be no less than 0.8 of the compressed height of the spring at rated load. Springs shall have a minimum additional travel to solid equal to 50% of the rated deflection. Submittals shall include spring diameters, deflection, compressed spring height and solid spring height.
6. Restrained spring mountings shall have an SLF mounting within a rigid housing that includes vertical limit stops to prevent spring extension when weight is removed. The housing shall serve as blocking during erection. Installed and operating heights are equal. A minimum clearance of 1/2" (12mm) shall be maintained around restraining bolts and between the housing and the spring so as not to interfere with the spring action. Restraining Bolts shall have a neoprene bushing between the bolt and the housing. Limit stops shall be out of contact during normal operation. Since housings will be bolted or welded in position there must be an internal isolation pad. Housing shall be designed to resist all seismic forces.
7. Spring mountings built into a ductile iron or steel housing to provide all directional seismic snubbing. The snubber shall be adjustable vertically and allow a maximum of 1/4" (6mm) travel in all directions

before contacting the resilient snubbing collars.

8. Air Springs shall be manufactured with upper and lower steel sections connected by a replaceable flexible nylon reinforced neoprene element. Air spring configuration shall be multiple bellows to achieve a maximum natural frequency of 3 Hz. Air Springs shall be designed for a burst pressure that is a minimum of three times the published maximum operating pressure. All air spring systems shall be connected to either the building control air or a supplementary air supply and equipped with three leveling valves to maintain leveling within plus or minus 1/8" (3mm). Submittals shall include natural frequency, load and damping tests performed by an independent lab or acoustician.
9. Restrained air spring mountings shall have an MT air spring as described in Specification 8, within a rigid housing that includes vertical limit stops to prevent air spring extension when weight is removed. The housing shall serve as blocking during erection. A steel spacer shall be removed after adjustment. Installed and operating heights are equal. A minimum clearance of 1/2" (12mm) shall be maintained around restraining bolts and between the housing and the air spring so as not to interfere with the air spring action. Limit stops shall be out of contact during normal operation. Housing shall be designed to resist all seismic forces.
10. Hangers shall consist of rigid steel frames containing minimum 1 1/4" (32mm) thick neoprene elements at the top and a steel spring with general characteristics as in specification 5 seated in a steel washer reinforced neoprene cup on the bottom. The neoprene element and the cup shall have neoprene bushings projecting through the steel box. To maintain stability the boxes shall not be articulated as clevis hangers nor the neoprene element stacked on top of the spring. Spring diameters and hanger box lower hole sizes shall be large enough to permit the hanger rod to swing through a 30E arc from side to side before contacting the rod bushing and short circuiting the spring. Submittals shall include a hanger drawing showing the 30E capability.
11. Hangers shall be as described in 10, but they shall be supplied with a combination rubber and steel rebound washer as the seismic up stop for suspended piping, ductwork, equipment and electrical cable trays. Rubber thickness shall be a minimum of 1/4" (6mm). Submittals shall include a drawing of the hanger showing the installation of the rebound washer.
12. Hangers shall be as described in 10, but they shall be pre compressed and locked at the rated deflection by means of a resilient seismic up stop to keep the piping or equipment at a fixed elevation during installation. The hangers shall be designed with a release mechanism to free the spring after the installation is complete and the hanger is subjected to its full load. Deflection shall be clearly indicated by means of a scale. Submittals shall include a drawing of the hanger showing the 30E capability.
13. Seismic Cable Restraints shall consist of galvanized steel aircraft cables sized to resist seismic loads with a minimum safety factor of two and arranged to provide all-directional restraint. Cables must be pre stretched to achieve a certified minimum modulus of elasticity. Cable end connections shall be steel assemblies that swivel to final installation angle and utilize two clamping bolts to provide proper cable engagement. Cables must not be allowed to bend across sharp edges.
14. Seismic solid braces shall consist of steel angles or channels to resist seismic loads with a minimum safety factor of 2 and arranged to provide all directional restraint. Seismic solid brace end connectors shall be steel assemblies that swivel to the final installation angle and utilize two through bolts to provide proper attachment.
15. Steel angles, sized to prevent buckling, shall be clamped to pipe or equipment rods utilizing a minimum of three ductile iron clamps at each restraint location when required. Welding of support rods is not acceptable.
16. All-directional seismic snubbers shall consist of interlocking steel members restrained by a one-piece molded neoprene bushing of bridge bearing neoprene. Bushing shall be replaceable and a minimum of 1/4" (6mm) thick. Rated loadings shall not exceed 1000 psi (.7kg/mm²). A minimum air gap of 1/8"

(3mm) shall be incorporated in the snubber design in all directions before contact is made between the rigid and resilient surfaces. Snubber end caps shall be removable to allow inspection of internal clearances. Neoprene bushings shall be rotated to insure no short circuits exist before systems are activated.

17. All directional seismic snubbers shall consist of interlocking steel members restrained by shock absorbent rubber materials compounded to bridge bearing specifications. Elastomeric materials shall be replaceable and a minimum of 3/4" (19mm) thick. Rated loadings shall not exceed 1000 psi (.7kg/mm²). Snubbers shall be manufactured with an air gap between hard and resilient material of not less than 1/8" (3mm) nor more than 1/4" (6mm). Snubbers shall be installed with factory set clearances. The capacity of the seismic snubber at 3/8" (9mm) deflection shall be equal or greater than the load assigned to the mounting grouping controlled by the snubber multiplied by the applicable ΔG force. Submittals shall include the load deflection curves up to 1/2" (12mm) deflection in the x, y and z planes.
18. Stud wedge anchors shall be manufactured from full diameter wire, not from undersized wire that is rolled up to create the thread. The stud anchor shall also have a safety shoulder which fully supports the wedge ring under load.

Female wedge anchors are preferred in floor locations so isolators or equipment can be slid into place after the anchors are installed. Anchors shall be manufactured from full diameter wire, and shall have a safety shoulder to fully support the wedge ring under load.

19. Vibration isolation manufacturer shall furnish integral structural steel bases. Rectangular bases are preferred for all equipment. Centrifugal refrigeration machines and pump bases may be T or L shaped where space is a problem. Pump bases for split case pump shall include supports for suction and discharge elbows. All perimeter members shall be steel beams with a minimum depth equal to 1/10 of the longest dimension of the base. Base depth need not exceed 14" (350mm) provided that the deflection and misalignment is kept within acceptable limits as determined by the manufacturer. Height saving brackets shall be employed in all mounting locations to provide a base clearance of 1" (25mm).
20. Vibration isolation manufacturer shall furnish rectangular steel concrete pouring forms for floating and inertia foundations. Bases for split case pumps shall be large enough to provide for suction and discharge elbows. Bases shall be a minimum of 1/12 of the longest dimension of the base but not less than 6" (150mm). The base depth need not exceed 12" (300mm) unless specifically recommended by the base manufacturer for mass or rigidity. Forms shall include minimum concrete reinforcing consisting of 1/2" (12mm) bars welded in place on 6" (150mm) centers running both ways in a layer 1 2" (38mm) above the bottom. Forms shall be furnished with steel templates to hold the anchor bolts sleeves and anchors while concrete is being poured. Height saving brackets shall be employed in all mounting locations to maintain a 1" (25mm) clearance below the base. Wooden formed bases leaving a concrete rather than a steel finish are not acceptable.
21. The horizontal thrust restraint shall consist of a spring element in series with a neoprene molded cup as described in specification 5 with the same deflection as specified for the mountings or hangers. The spring element shall be designed so it can be preset for thrust at the factory and adjusted in the field to allow for a maximum of 1/4" (6mm) movement at start and stop. The assembly shall be furnished with 1 rod and angle brackets for attachment to both the equipment and the ductwork or the equipment and the structure. Horizontal restraints shall be attached at the centerline of thrust and symmetrical on either side of the unit.
22. Housekeeping pad anchors shall consist of a ductile iron casting that is tapered and hexagonal, smaller at its base than at its top. The upper portion shall have holes for rebar to pass through. The anchor shall be continuously threaded from top to bottom for the attachment of soleplates. Housekeeping pad anchors shall be attached to the structural slab using a stud wedge anchor.

23. Pipe Riser isolation: Risers shall be suspended from specification 10A hangers or supported by specification 5 mountings, anchored with specification 25 anchors, and guided with specification 26 sliding guides. Steel springs shall be a minimum of 0.75" (19mm) except in those expansion locations where additional deflection is required to limit load changes to ∇ 25% of the initial load. Submittals must include riser diagrams and calculations showing anticipated expansion and contraction at each support point, initial and final loads on the building structure, spring deflection changes and seismic loads. Submittal data shall include certification that the riser system has been examined for excessive stresses and that none will exist in the proposed design.

ELECTRICAL SERVICES:

1. All electrical bus ducts, cable trays and ladder trays shall be restrained with specification type 12, seismic cable restraints or specification 13 seismic solid brace restraints.
2. Transverse restraints shall occur at 30' (9m) intervals or both ends if the electrical run is less than the specified interval. Transverse restraints shall be installed at each electrical services turn and at each end of the electric run.
3. Longitudinal restraints shall occur at 60' (18m) intervals with at least one restraint per electric run. Transverse restraints for one electric section may also act as a longitudinal restraint for a duct for an electric section connected perpendicular to it if the restraints are installed within 4' (1.2m) of the intersection of the electric run and if the restraints are sized for the larger electric run.
4. All floor mounted transformers, motor starters, switchgears and substations must have a resilient media between the equipment mounting hole and the anchor bolt. Anchor bolts shall be designed in accordance with section 1.06 seismic forces. Neoprene bushings shall be specification type 4 and anchor bolts shall be specification type 18 or 19.
5. Wall mounted panels, transformers and motor starters shall be mounted with specification type 3 bushings. Floor mounted panels shall be mounted on specification type 4 bushings. Anchor bolts shall be specification type 18 or 19.
6. All generators shall be mounted on a Specification type 21 concrete inertia base, with Specification type 5 spring isolators and Specification type 17 seismic snubbers.
7. Connection to the structure must be made with a non-friction connection

13.3 Piping.

All piping connected to isolated equipment shall be supported with isolation hangers for at least the first three points of support on all sides of the equipment.

The first 3 supports shall have a static deflection of 2 x the equipment deflection, except that the deflection of this point shall not exceed 50 mm (with pipe work full).

The fourth and fifth supports shall have deflections of 25 mm each.

For suspended piping, springs will be box type spring hangers with snubber, level adjustments, and colour coding.

For low level piping mounted on roller systems, the isolators will be open type springs with equal stiffness in horizontal and vertical planes.

- 13.4 In addition, Flexible connection shall be included at inlet and outlet to all items of equipment.

Flexible connections will be of full line size and will be flanged with gaskets and mating flanges.

Connectors shall be twin sphere manufactured with multiple plys of nylon tire cord fabric and neoprene in hydraulic rubber presses and vulcanized.

- 13.5 Ductwork.

(See clause 6.0 for flexible connection.)

14.0 DUCT DRAINS AND DRIP PANS

- 14.1 Provide drip pans under cooling coils mounted in ducts in built up plenums.

Drip pan shall be constructed of not less than 18 gauge steel (1.2 mm) hot dip galvanized after fabrication, with all joints welded or soldered. Sides of pans shall be not less than 100 mm upstream and 300 mm downstream of coil.

Pan width shall extend 50 mm outside coil headers and/or return bends to collect any condensation released from these surfaces

Pan shall be equipped with one or more 25 mm drain which must be run to drainage system as indicated by the Consultant.

- 14.2 Drain pans for fan coil units shall be factory manufactured and supplied with the fan coil unit. Drain pan must be factory insulated on the sides in addition to on the bottom, with at least 20 mm rigid insulation.

- 14.3 Drain lines will have traps and falls as specified in the plumbing section.

15.0 DIFFUSERS AND REGISTERS

- 15.1 Diffusers, registers and grilles shall be arranged for flush mounting in lay-in type ceilings and over lap mounting in plaster, mineral tile and similar ceilings, with concealed fixings unless otherwise directed.

- 15.2 Grilles, register and diffuser locations shall be adjusted to suit reflected ceiling drawings, or Consultant's site instructions. All grilles, registers, diffusers, louvres shall be from one manufacturer.

- 15.3 Provide plaster frame for grilles, and diffusers installed in plaster ceilings.

- 15.4 All diffusers, grilles and registers shall be supplied completely factory powder coated. Finish colour shall be to the approval of the Architect / ID. The interior of all grilles and diffusers is to be factory painted matt black.

- 15.5 All supply grilles and diffusers shall have black colored opposed blade balancing dampers. All will have foam rubber sealing band around the edge to seal to the structure. All pivots will be round section, not of formed sheet, and not relying on a spring steel locking wire.

- 15.6 Unless otherwise specified basic grilles and diffuser materials shall be Aluminium extruded sections. Sections in the airstream shall be carefully selected to minimize turbulence.

- 15.7 All grilles and diffusers supplied on this project shall be tested and rated in accordance with ASHRAE Standard 70, ADC 1062: GRD-84 and ISO 3741.

- 15.8 12.5mm centers pressed into a notched steel retaining bar. The core can be either welded into the outer frame, or, where the grille is used in a sill application, held in the outer frame by spring clips fixed to the core retaining bar.

The outer frame shall be 35mm deep and shall have a visible flange 25.4 mm wide. Mitered end caps shall be welded to give a near invisible joint.

The grill shall be complete with an opposed blade damper painted matt black and shall be fixed with universal mounting brackets. Both the damper and the fixing brackets shall be accessible through the face of the grille.

Continuous grilles shall be provided with positive alignment strips, which fit into special keyways extruded into the frame of the grille to ensure clean unbroken lines.

15.9 Ceiling Diffusers shall be multi-cone giving 4 way horizontal discharge.

The three centre cones of the diffuser shall be manufactured from pressed aluminium, with the remaining cones and the outer frame fabricated from extruded aluminium welded at the corners to give near invisible joints.

One-, two-, and three-way pattern cores shall be used as indicated on schedules. All cores shall be interchangeable.

The core shall be removable without the use of special tools, but for safety, shall be fixed to the outer frame by a small length of chain.

The diffuser shall be complete with an opposed blade damper painted matt black.

**15.10 Wall registers shall be double deflection fabricated from aluminium, the front vanes being horizontal, the rear vanes vertical. This grille shall be complete with an opposed blade damper painted matt black and adjustable from the face of the diffuser.
Both sets of vanes shall be fully adjustable without the use of special tools.**

15.11 Eggcrate return or extract grille shall be provided with a steel lattice core of 12.7mm x 12.7mm openings, giving a free area of 90%.

The core shall be fixed into an extruded aluminium frame, with welded corners and a 25mm face flange.

The grille is complete with an opposed blade damper painted matt black and adjustable through the face of the diffuser.

15.12 Vision proof door transfer grilles (Where applicable) shall consist of a steel core with inverted-vee type blades, and an extruded aluminium frame with matching rear flange.

The frame shall be adjustable from 28mm to 60mm to suit the door width.

Transfer grilles for 'LIGHT TIGHT' applications such as dark rooms shall be with two vision proof cores back to back.

Transfer grilles on fire rated doors shall include a block of intumescent material as a fire stop.

15.13 Linear Slot diffusers shall provide an unobtrusive continuous air diffusion with a pleasing aesthetic appearance. Hairline butt joints shall ensure clean unbroken linear runs for active and dummy sections.

The diffusers shall be complete with pattern control blades, fully adjustable from face of diffuser through 180° and shall be fitted with end caps at each end.

The diffusers shall be with hit and miss sheets or OBD to adjust air flow.

The diffuser members shall be constructed from high quality aluminium extrusions to BS EN 573 / BS EN 755 / BS EN 12020 while the pattern control blades shall be of black rigid PVC.

- 15.14 Exhaust valves shall be manufactured from highly quality sheet steel spinings protected by a stove enameled or powder coated paint finish. Flanges shall be fitted with sealing gaskets.
- 15.15 Provide square uniform height plenum for ducted return and exhaust application of scheduled neck size.

16.0 EXTRACT AIR LOUVRES

- 16.1 Louvres shall be extruded Aluminium frame with Aluminium blades of not less than 2 mm thickness and shall be firmly fixed so as not to vibrate. Unsupported blade width shall not exceed 1800 mm.
- 16.2 Behind each louvre shall be an insect mesh screen 76 x 6 mm made from 2 mm diameter wire. The screen will be clamped by a 20 mm frame and will be firmly fixed to the outer edges of the louvre. The screen and frame shall be hot dip galvanized after fabrication.
- 16.3 The connection to the louvre shall be flexible and shall ensure no duct load is transmitted to the louvre.
- 16.4 Louvres shall be provided with powder coated finish to the approval of the Architect.

17.0 SAND TRAP LOUVRE FOR AIR INTAKE

- 17.1 Sand trap louvres shall have a double deflection inlet passage to separate sand from incoming air by means of centrifugal forces.
- 17.2 Separation efficiency particle size 350-700 shall not be less than 90% at a face velocity of 1m/s and not less than 70% at a face velocity of 2m/s .The selections shall be for a face velocity not exceeding 1.5m/s during normal operation.
- 17.3 Sand trap louvre shall be of aluminium construction, self cleaning and maintenance free. The base of the louvre shall have self-emptying sand holes.
- 17.4 Pressure drop at 2 m/s average face velocity shall not exceed 120 Pascal's.
- 17.5 Insect mesh shall be included.
- 17.6 Sand louvres shall be provided with powder coated finish to the approval of the Architect.
- 17.7 Sizing of sand louvers shall consider minimum 85% efficiency.

18.0 FILTERS

- 18.1 Pre filters
Filter Media shall be of the nonwoven cotton fabric type. The filter media shall have an average efficiency of 25-30% on ASHRAE 52.2. It shall have an average arrestance of 90-92% in accordance with that test standard.

Media support Grid shall be a welded wire grid with an effective open area of not less than 96%. The welded wire grid shall be bonded to the filter media to eliminate the possibility of media oscillation and media pull away.

The media support grid shall be formed in such a manner that it affects a radial pleat design, allowing total use of filter media.

Enclosing frame shall be constructed of a rigid, heavy duty, high wet strength beverage board, with diagonal support members bonded to the air entering and air exiting side of each pleat, to ensure pleat stability. The inside periphery of the enclosing frame shall be bonded to the filter pack, thus, eliminating the possibility of air bypass.

18.2 Bag Filters

Bag filters shall be high performance extended area disposable type filters. Each filter shall consist of high density glass microfibre media with a chemically bonded backer, individual pockets and a corrosion resistant galvanized steel enclosing frame. The filter unit shall be completely factory assembled and shall be classified by underwriter's laboratories.

Pockets shall consists of glass micro fibers chemically bonded to a reinforced backing. They shall be sewn with a tapered stitch which forms a 'V' or wedge shaped pocket resulting in uniform velocity in the passages of the air entering and air exiting sides of the filter. The pockets shall be equipped with a minimum of 40 support points per square foot of the filter media.

All stitching points shall be completely sealed with foam seal or equivalent. The pockets shall be chemically adhered around the periphery of the galvanized steel retainers. Retainers shall have rolled edges to reduce possible cuts to media, or lacerations to installers.

Enclosing frame shall be constructed of a 'J' return channel of galvanized steel.

19.0 MOTORIZED SHUT OFF DAMPERS

19.1 The Contractor shall furnish and install, control dampers as required for the proper functioning of the system.

19.2 All control dampers shall be parallel blade.

19.3 Casing shall be in galvanized sheet steel to BS EN 10143 Z2 G275 M.

19.4 Blades shall be of natural anodized extruded aluminium to BS EN 573 / BS EN 755 / BS EN 12020.

19.5 Blade tip seals silicon rubber, temperature resistant up to 300°C.

19.6 Side seals shall be stainless steel to BS 1449 Grade 302 – 525.

19.7 External blade linkage shall be stainless steel to BS EN 10277 304-S 15.

19.8 Blade spindles shall be in stainless steel to BS EN 10277 304 S11 supported in plain brass bearings.

19.9 Shut off dampers shall have a leakage rate not exceeding 200 m³ / hr / m² at 1000 Pa.

19.10 Motorized dampers used in smoke extract ducts shall be low leak type confirming to UL 555S. The actuator shall be fixed in the factory and certified for proper operation. Dampers shall be with valid approval certificate from Local Civil Defence.

20.0 MOTORIZED FIRE & SMOKE DAMPER (MFSD)

20.1 Motorized Combination Fire & Smoke Damper shall be used as a Motorized Smoke Damper for Smoke Applications.

20.2 The Dampers along with the Actuator shall be UL listed for the intended service. Listing shall comply to UL 555S & UL 555.

- 20.3 The Motorized Fire & Smoke Dampers shall have 3 Hours Fire Rating and Class I, II, III Leakage.
- 20.4 Frame shall be made of 16G (1.5 mm) thick Galvanized Iron Sheet.
- 20.5 Blades shall be made of 16G (1.5 mm) thick Galvanized Iron Sheet, Single Skin for parallel operation.
- 20.5.1 MFSD shall have a Thermal Resetting Device suitable for 72°C (162°F) as standard, which shall send Open/Close signal to Damper as per requirement.
- 20.6 Mounting sleeve shall be of length 400 mm.
- 20.7 Motorized Combination Fire & Smoke Dampers product shall be approved by Local Civil Defence.
- 20.8 Control Panel shall be interfaced with the Fire Alarm Panel.

21.0 SUBMISSIONS

21.1 Technical Submissions

For any minor revisions in architectural layouts, contractor shall be responsible to carry out part redesign for such area and submit necessary calculations for Engineer's approval before proceeding with the shop drawings.

- 21.2 The Contractor will prepare full shop drawings, including sections, of distribution systems and equipment.

21.3 Hardware Submissions.

The Contractor will submit catalogue information for all distribution equipment including, but not limited to: -

Duct work - Low pressure.

Flexible duct work - Low pressure

Flexible connections.

Access panels.

Balancing dampers.

Motorized dampers.

Backdraft dampers.

Splitter dampers.

Turning vanes.

Fire dampers.

Silencers.

Vibration isolators.

Diffusers, grilles and registers.

Louvres, Sand trap louvres.

Filters.

- 20.4 The Contractor will submit samples for any or all of the above as requested by the Consultant after receipt of the catalogues.

- 20.5 Samples will definitely be required of these items which are exposed, such as diffusers, grilles, louvres. The samples must be of size, specification and finish to be relevant to the project.
- 20.6 Where a Country of Origin is given, this refers to the Head Office in the case of International Corporations. However, for each product not manufactured in that Country, separate approval must be obtained from the Consultant.

END OF SECTION

MECHANICAL SECTION: 09

LIQUIFIED PETROLEUM GAS INSTALLATION (LPG)

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1.0 GENERAL**1.1 Reference**

Conform to General Requirements for Mechanical Services of Division Fifteen.

1.2 Description of Work

- 1.2.1 Furnished and install, test and commission the LPG installation as indicated on drawings within the buildings. Contractor is responsible for connection to the LPG site system provided by others. The following technical specifications apply to the specialist LPG supplier except where details for connecting to this distribution system within or immediately external to the building by Contractor apply. Gas detection within building by contractor in accordance with Civil Defense and Specialized Gas Company requirements.

Installation, testing and commissioning of LPG system shall be carried out by a specialist Contractor having Civil Defense approval. All items, design proposals, drawings shall be submitted to the Engineer for approval, and have his approval prior to supply, manufacturing or installation as described in the technical specifications.

The LPG detailed shop drawings and LPG installation shall be carried out by a specialist LPG contractor approved by Local Civil Defence Authority.

- 1.2.2 This section specifies the requirements for the LPG components.
- 1.2.3 Provide electrical work related to this section and complying with requirements of electrical specifications.
- 1.2.4 LPG design shall be coordinated with Specialist Gas Company.

1.3 Quality Assurance

- 1.3.1 Manufacturer's Qualifications: Firms regularly engaged in the field of LPG Gas Piping work and distribution components with characteristics, sizes and capacities required, whose products have been in satisfactory use in similar service for not less than 5 years.
- 1.3.2 Professional Engineer Certification : Approval/ compliance certification shall be submitted from the following agencies to certify that the professional engineer is legally qualified to practice in the jurisdiction where project is located and who is experienced in providing engineering services of the kind indicated:
- i. Local Civil Defense Authority
 - ii. NFPA 54
 - iii. NFPA 58
 - iv. British Standards Code of Practice LPG ITA.
 - v. I.G.E.

1.4 Submittals

- 1.4.1 Product Data: Submit manufacture's product specifications, installation and startup instructions.
- 1.4.2 Shop Drawings: Submit shop drawings indicating tank locations, weight loadings, required clearances and LPG pipe work distribution complete with valves, fittings and controls.
- 1.4.3 Submit the following elements of the LPG Distribution and Storage System: Pipe and fittings, valves, dielectric unions, strainers, meters, instruments, storage tanks, Leak detection and alarm system.

- 1.4.4 Steel Pipes: Statement from manufacturer on his letterhead that the pipe furnished meets the international standards specified and contained in this section. Subcontractor shall submit the Mill certification papers, also known as material test reports, for the pipe furnished for this project, in English. Code numbers on these papers to match the code numbers stenciled on the pipe. Chemical analysis indicated on the mill certification papers to meet or exceed the requirements of the referenced International Standard specification.
- 1.4.5 Certificates to be provided for welding procedure with qualification of welders including list of welder's names and symbols.
- 1.4.6 Wiring Diagrams: Submit manufacturer's electrical requirements for power supply wiring. Submit manufacturer's ladder-type wiring diagrams for interlock and control wiring. Clearly differentiate between portions of wiring that are factory – installed and portions to be field – installed.
- 1.4.7 Maintenance Data: Submit maintenance data and parts lists for each component control and accessory, including “trouble – shooting” maintenance manual.
- 1.4.8 List of spare parts for two years operation and maintenance of complete system.
- 1.4.9 Contractor to carry out planned preventive and breakdown maintenance of LPG system for one year from date of taking over.

1.5 Product Delivery, Storage and Handling

- 1.5.1 Handle LPG System components carefully to prevent damage, breaking, denting and scoring. Do not install damaged components, replace with new.
- 1.5.2 Store LPG system components in clean dry place. Protect from weather, dirt, fumes, water, construction debris, and physical damage.
- 1.5.3 The peak ambient design temperature to be considered is 50 deg. C.

2.0 PRODUCTS

All LPG products used in the project shall have from Local Authorities approval.
The Contractor shall be responsible for obtaining all drawing and installation approvals from the Local Authorities

2.1 LPG Storage Tank (Where Applicable)

The LPG Tank and accessories will be supplied by the Specialist Gas Company.

2.2 Piping, Fittings and Valves

2.2.1 General

This section includes basic design provisions and material specifications for pipe, tubing, pipe and tube fittings, valves used to connect container appurtenances with the balance of the L P Gas system.

2.2.2 Pipe

- a. Below ground Distribution pipe work shall be seamless steel to ASTM A53, A 106 with Factory applied polythene sleeve or HDPE/MDPE SDR11.
- b. Above ground distribution pipe work shall be Seamless Carbon steel Pipe to ASTM A53 Gr.B or copper to BS 2871 Table X or Y

- c. Final connections to equipment shall be soft copper tubes to BS 2871 Table X or Reinforced Flexible Hose steel over braided.
- d. The Pipe selection should match the natural gas service in future.

2.2.3 Pipe and Tubing Fittings

- a. Pipe joints in steel, brass or copper pipe shall be welded or brazed.
- b. The requirements of the Local Authorities shall be accommodated, including pipe-in-pipe installations within the building. Extremely ventilated shafts shall be incorporated to comply with civil defense requirements.
- c. Fittings used at pressure higher than container pressure, such as on the discharge of liquid transfer pumps, shall be suitable for a working pressure of at least 350 psig (2.4 Mpa gauge).
- d. Fittings used with liquid L.P-Gas, or with vapor LP-Gas at operating pressures over 125 Psig (0.9 Mpa gauge) shall be suitable for a working pressure of at least 250 Psig (1.7 Mpa gauge).
- e. Fittings for use with vapor LP-Gas at pressure not exceeding 125 psig (0.9 Mpa gauge) shall be suitable for a working pressure of 125 psig (0.9 Mpa gauge).
- f. Soldering or brazing filler material shall have a melting point exceeding 1,000 deg. F (538 deg. C).

2.2.4 Valves, Other than Container Valves

- a. Pressure containing metal parts of valves (except appliance valves), including manual positive shut off valves, excess flow check valves (either manually or automatically operated), used in piping systems shall be of steel, ductile (nodular) iron, malleable iron or Brass. Ductile iron shall meet the requirements of ANSI/ASTM A 395 or equivalent and malleable iron the requirements of ANSI/ASTM A 47 or equivalent. All materials used including valve seat discs, packing, seals and diaphragms shall be resistant to the action of L P Gas under service conditions.
- b. Valves shall be suitable for the appropriate working pressure, as follows:
 - i. Valves used at pressures higher than container pressure, such as on the discharge of liquid transfer pumps shall be suitable for a working pressure of at least 350 psig (2.4 Mpa gauge). [400 psig (2.8 Mpa gauge) WOG valves comply with this provision. Valves should be UL listed.
 - ii. Valves to be used with liquid L P Gas, or with vapor L P-Gas at pressures in excess of 125 psig (0.9 Mpa gauge), but not to exceed 250 psig (1.7 Mpa gauge), shall be suitable for a working pressure of at least 250 psig (1.7 Mpa gauge).
 - iii. Valves (except appliance valves) to be used with vapor L P Gas at pressures not to exceed 125 psig (0.9 Mpa gauge) shall be suitable for a working pressure of at least 125 psig (0.9 Mpa gauge).
- c. Manual shut off valves, excess flow check valves and backflow check valves used in piping systems shall comply with the provisions for container valves.
- d. Emergency shut off valves shall be approved and incorporate all of the following means of closing.
 - i. Automatic shut off through thermal (fire) actuation. When fusible elements are used they shall have a melting point exceeding 250 deg.F (121 deg.C)

- ii. Manual shut off from a remote location.
- iii. Manual shut off at the installed location.
- e. Solenoid valves shall be intrinsically safe construction suitable
- f. Valves selection should match the Natural Gas conversion in future.

2.3.5 Gas Meters (Where Applicable)

- a. Installation and application of gas meters shall be in accordance with the relevant clauses in NFPA 54 and IGE/GM/8 in compliance with ESMA requirements and must have design pressure same as relevant pipe work. All Gas meter must have valid calibration seal for acceptable test standards.
- b. Gas Meters shall be selected for the maximum expected pressure and permissible pressure drop.
- c. Vapor meters of the tin or brass case type of soldered constructions shall not be used at pressure in excess of psi (7 KPa).
- d. Vapor meters of the die cast or iron case type shall be permitted to be used at any pressure equal to or less than the working pressure for which they are designed and marked.
- e. Gas meters shall be located in ventilated spaces readily accessible for examination, reading, replacement or necessary maintenance.
- f. Gas meters shall not be placed where they will be subjected to damage, such as adjacent to a driveway, under a fire escape, in public passages, halls or where they will be subjected to excessive corrosion or vibration.
- g. Gas Meters shall be located at least 1 m from sources of ignition.

2.3 Gas Leak Detection System

A gas leak detection system shall be installed to monitor for gas leaks at different locations of the gas network.

2.3.1 Control Panel

- a. The central control panel shall be located in the control room located on the site and shall easily be accessible in the event of an emergency. (Geca or Equivalent.)
- b. The central panel shall have a modular design incorporating input alarm/display module for each gas detector. Each channel shall have indication for low alarm at 15% LEL, high alarm at 30 % LEL and fault at 0-100% LEL indicator.
- c. Main LPG supply shall be shut off at 30% LEL indication.
- d. The concentration of the gas shall be displayed on a common indicator with selector switches to display the concentration of the desired channel or zone indicator.
- e. Below the gas control panel is the system control panel which shall have the following indications:
 - AC healthy
 - DC healthy
 - Sprinkler on
 - Gas pressure status

- Gas Solenoid Valve operated
- Common fire signals
- Gas leak high
- Gas Percentage at Tank

The panel shall have the following switches:

- Sprinkler manual operate
- Gas manual shut off
- Test lamp
- System reset

2.3.2 Gas Detectors

The gas detectors shall use a pellistor sensor to detect flammable gases and vapors. The sensor shall be housed in a stainless steel, flame proof housing, fitted onto an explosion proof junction box.

2.3.3 Sound Flasher

A sound flasher unit shall be installed in the vicinity of the panel located in a visible area.

2.3.4 Cables

MICC or Fire Proof cables of at least 1.5 sq. mm area shall be used to interconnect between each detector and the control panel.

2.3.5 Logic of operation

When low-level gas leak is detected in any zone, local internal alarm and indication shall show on the control panel. When high level gas leak is detected in any zone, the following sequence shall occur:

- a. The solenoid valve in the gas distribution channel shall be closed automatically.
- b. External alarm shall be operated.

Connection to the fire alarm system:

In the event of a fire alarm signal, the following sequence shall occur.

In the event of an emergency, it shall be possible to bypass the automatic system and operate the system manually, this shall include:

- a. Manual shut off/on of the gas through the emergency shut-off push button.
- b. Manual operation of the sprinkler system On/Off
- c. Manual reset.

3.0 EXECUTION

3.1 General

- 3.1.1 All pipe work, pipe supports and brackets shall be painted with 2 undercoats and 2 finishing coats of yellow color paint.
- 3.1.2 The corridor Gas Pipes should be by Pipe in Pipe work. The sleeve Pipe shall be Steel sch-10.
- 3.1.3 Gas detectors should be provide for Tank Yard , PRDP , Pipe in Pipe (Each 18 mtr or each 3 Joints.) , Air well and Kitchens

- 3.1.4 LPG piping shall be painted with a specific color different than other pipes. Orange or yellow colors are recommended. Valves shall also be painted. Direction arrows to indicate the flow of gas in the system shall be painted.
- 3.1.5 The system in parts and as a whole shall be tested to ensure that it will operate safely, without leaks and to the satisfaction of the Civil Defense Authority.
- 3.1.6 Commissioning of the system shall be tested to ensure that all the equipment and the whole system are in good working order.
- 3.1.7 The LPG contractor shall accommodate any acceptance of the system.
- 3.1.8 The gas network shall be leak tested using gases such as nitrogen or argon. Oxygen or flammable gases will not be used. The leak test shall include all the different pressure stages of the system including:
- a. The filling line operating pressure up to 20 bar.
 - b. The Tank and high pressure stage operating pressure 16.5 bar
 - c. The first stage regulator operating pressure 1.5 bar.
 - d. The second stage pressure, operating pressure 1.5 bar.
- 3.1.9 Test pressure shall be at least 1.5 times the operating pressure. The leak test duration shall be 1 hour. The specified test pressure shall be monitored for a minimum of one hour during which the soap water method shall be used to leak test every joint. If no leak is detected and the pressured gauge indicates no drop in the test pressure then the system will be proven to be leak free.
- 3.1.10 If a leak is found, the leaking joint shall be repaired or the leaking equipment shall be replaced. The testing procedure shall be repeated to prove the complete system as leak free.
- 3.1.11 A test report shall be submitted to the Company for approval.

3.2 Testing and Commissioning

- 3.2.1 Before installations are handed over or subjected to inspection and testing, the entire installation shall be thoroughly cleaned, both internally and externally.

3.2.2 Purging

Vessels and their ancillary equipment shall be purged until the oxygen level is reduced below that which will support combustion before commissioning. When purging into or out of service is required this shall be carried out in accordance with LPGA Code of practice No. 17.

Prior to Purging of LPG system, LPG reticulation shall be flushed with Nitrogen gas.

- 3.2.3 LPG shall not be introduced into a pipe work system unless it has passed a leak test for the entire system to be commissioned, and signed records are available.
- 3.2.4 Purging of pipe work systems shall only be carried out by or under the supervision of competent persons. Pipe work should be purged with an inert gas e.g. nitrogen, so that the oxygen level is reduced to less than 9% by volume, before introducing LPG. Purge gas should be introduced in a controlled manner using an adjustable pressure regulator.

- 3.2.5 When commissioning liquid systems, sudden chilling of the system from rapid vaporization shall be avoided by purging with LPG vapor up to vapor pressure before introducing liquid, or by pre-pressurization with the inert gas.
- 3.2.6 During the introduction of LPG the gas shall be released at a safe location and must not be allowed to accumulate in any confined space.
- 3.2.7 The introduction of LPG may be under taken in sections if necessary, provided suitable purge gas release points have been provided.
- 3.2.8 LPG storage tank and its piping shall be painted in yellow color as per DCD requirements.
- 3.2.9 Contractor to provide fencing with lockable gate surrounding the LPG storage tank as per DCD requirements.

Note:

The Pipes, Valves, Gas Meters and Hoses selection should match with Natural Gas codes and Standards. The above said Items should work in Natural Gas in future.

END OF SECTION

MECHANICAL SECTION: 10

BUILDING MANAGEMENT SYSTEM

PART 1 – GENERAL**1.1 WORK INCLUDED**

- 1.1.1 Compliance with sections 01, General Requirements and documents referred to therein.
- 1.1.2 Conform to General Requirements for Electrical services
- 1.1.3 Conform to General Requirements for mechanical services.

1.2 SCOPE OF WORKS

The BMS shall consist of the following elements:

- 1.2.1 Microcomputer based remote control panels interfacing directly with sensors, actuators and environmental delivery systems (i.e., HVAC equipment, lighting systems, etc.).

The system shall have an inbuilt spare capacity of minimum twenty percent at all levels of control and network to allow for future expansion.

Two-wire communication networks to allow data exchange between remote panels and the central Building Management computer.

One personal computer (PC) based central and associated operator station(s) and software functioning as the primary operator interface for the BMCS.

Electronic and electric controls for all items indicated on drawings and described hereinafter including sensors, control valves and thyristers.

Submittals, data entry and electrical installation, programming, interfacing with other systems, start up, test and validation, instruction of Employer's representative on maintenance and operation, as built documentation, and system warranty.

1.1 SYSTEM DESCRIPTION

- A. A modern integrated Building Management System (BMS) solution will be provided for automatic control, global monitoring and energy conservation by optimizing all the available mechanical and electrical systems. The Integrated Building Management System (IBMS) system shall use a Client Server architecture based around a modular PC network, utilizing industry standard operating systems, networks and protocols.
- B. BMS core is the field controllers which is connected to the all other field devices and equipment. These controllers are microprocessor-based type Direct Digital Control (DDC) shall be responsible of energy management, equipment monitoring and control. DDC's shall be connected all together through a standard very high-quality communication network to the Operator Interface Workstation.
- C. Data and programs should be stored centrally on a file server. User-specific information such as passwords, access rights etc. shall be administered centrally. The file server also shall facilitate and accelerate the handling of software updates and the modification of project data. The consistent, central archiving of alarms, historical data, log-data etc. shall provide a comprehensive over all view of the system.
- D. The system should be capable of supporting the following functions in future if required by the end user:

1. BMS shall support extended connectivity to the cloud, where Smart Phones/Mobile Applications are allowed to monitor and manage the building remotely and to receive alarm and event notifications.
 2. Also, the BMS shall have the capability to allow occupants with smart phones to use Mobile Applications for building access, air conditioning control, path finding, and meeting rooms booking.
 3. BMS should allow Integration with a diverse range of devices, enterprise systems, Internet and Intranet sources allowing the building management of key facility control and information.
 4. The maintenance & service provided after hand-over shall make use of smart analytic algorithms in the cloud to provide predictive service scheduling. The smart algorithms shall send alarms and alerts to service team to attend to problems that are building up and predicted to occur in near future.
- E. BMS should use industry standard hardware with Windows 2016 Server, SQL 2016, IE11, Windows 10, HTML5, Cyber Security, BACnet 1.12* & LNS IzoT, Arch improvements, LonMark®, OPC® and Modbus®. or latest.

1.3 PERFORMANCE OBJECTIVES

The Contractor must note that the Specification describes in general terms the control philosophy for various systems and the means of achieving the same. These are the minimum requirements only. In view of the rapid changes in the field of controls and related technology, the Contractor shall include for the latest technology and systems available from the specified manufacturer at the time of tender. The Contractor shall allow the costs related to loading the latest version of software available at the time of handing over of the installation.

The Building Management System (BMS/BMCS) shall be comprised of Network Area Controller or Controllers (NAC) within the facility. The NAC shall connect to the Employer's local or wide area network, depending on configuration. Access to the system, either locally in each building, or remotely from a central site or sites, shall be accomplished through standard Web browsers, via the Internet and/or local area network. Each NAC shall communicate to LonMark/LonTalk (IDC) and/or BACnet (IBC) controllers.

1.4 DESIGN PARAMETERS

The proposed system and components shall be compatible with a wide variety of control devices using BACnet/IP (preferred), LON, Ethernet TCP/IP, ModBus, KNX and OPC standard for open system communications and other current international communication standards. The BMS shall be capable of being extended with controllers on the BACnet protocol and the LON bus. The BMS shall allow integrating future BACnet controllers on the process level.

Using appropriate hardware, the system shall be Listed by Underwriters Laboratories Inc (ULI) or approved equal for use in energy management (category PAZX), critical process (category QVAX), security (category APOU), and as the primary control and monitoring device for smoke control (category UUKL) and fire alarm systems (category UOJZ).

1.5 OPEN, INTEROPERABLE, INTEGRATED ARCHITECTURES

- 1.5.1 The intent of this specification is to provide a peer-to-peer networked, stand-alone, distributed control system with the capability to integrate both the ANSI/ASHRAE Standard 135-2016 BACnet and LonWorks technology communication protocols in one open, interoperable system.
- 1.5.2 The supplied computer software shall employ object-oriented technology (OOT) for representation of all data and control devices within the system. In addition, adherence to industry standards including ANSI / ASHRAE™ Standard 135-2016, BACnet, Lontalk, and LonMark to assure interoperability between all system components is required. For each LonWorks device that does not have LonMark

certification, the device supplier must provide an XIF file for the device. For each BACnet device, the device supplier must provide a PICS document showing the installed device's compliance level. Minimum compliance is Level 3; with the ability to support data read and write functionality. Physical connection of BACnet devices shall be via BACnet Ethernet or BACnet MSTP.

- 1.5.3 All components and controllers supplied under this contract shall be true "peer-to-peer" communicating devices. Components or controllers requiring "polling" by a host to pass data shall not be acceptable.
- 1.5.4 The supplied system must incorporate the ability to access all data using standard Web browsers without requiring proprietary operator interface and configuration programs. An Open DataBase Connectivity (ODBC) or Structured Query Language (SQL) compliant server database is required for all system database parameter storage. This data shall reside on a supplier-installed server for all database access. Systems requiring proprietary database and user interface programs shall not be acceptable.
- 1.5.5 A hierarchical topology is required to assure reasonable system response times and to manage the flow and sharing of data without unduly burdening the customer's internal Intranet network. Systems employing a "flat" single tiered architecture shall not be acceptable:
- a) Maximum acceptable response time from any alarm occurrence (at the point of origin) to the point of annunciation shall not exceed 5 seconds for network connected user interfaces.
 - b) Maximum acceptable response time from any alarm occurrence (at the point of origin) to the point of annunciation shall not exceed 60 seconds for remote or dial-up connected user interfaces.
- 1.5.6 The field controller(s) shall be provided for manipulation of data from the field controllers for onward communication to the Management Level. It shall also perform management functions and event-related communication.
- 1.5.7 The Control and Interlock system for the process control shall comprise microprocessor based controllers together with the necessary input and output modules. All the field devices shall be connected to the respective input and output modules.

Use of dumb input and output modules shall not be permitted.

1.6 QUALITY ASSURANCE

- 1.6.1 The system shall be installed by competent mechanics, regularly employed by the BMCS manufacturer with full responsibility for proper operation of the BMCS including debugging and proper calibration of each component in the entire system. Supplier shall have an in-place support facility within 10 miles of the site with technical staff, spare parts inventory and all necessary test and diagnostic equipment.

1.6.2 Codes and Approvals:

The complete BMS/BMCS installation shall be in strict accordance to the national and local electrical codes and the electrical section of these specifications. All devices designed for or used in line voltage applications shall be UL Listed or approved by an equal regulatory authority.

All microprocessor based remote DDC devices shall be UL916 Listed or approved by an equal regulatory authority.

European EMC directive 89/336/EEC/EMC including all its amendments. Tested standard BS EN 50561 & BS EN 55032 – Emission and BS EN IEC 61000-6-1 – Immunity.

European Low Voltage Directive for Safety 73/23 EEC including all amendments, Tested Standards BS EN 60950 and BS EN 60730-1.

European General Product Safety Directive 92/59/EEC./ Tested standards – BS EN 60730-1.

All microprocessor based remote F&S devices shall be UL864 Listed or approved by an equal regulatory authority.

All BMCS central equipment shall be UL 864 Listed or approved by an equal regulatory authority.

All BMCS central and remote equipment used in security applications shall be UL 1076 Listed or approved by an equal regulatory authority.

All BMCS central and remote equipment used in access control applications shall be UL 294 Listed or approved by an equal regulatory authority.

All electrical environmental control and monitoring devices shall be UL 429 and/or UL 873 Listed or approved by an equal regulatory authority.

1.6.3 All electronic equipment shall conform to the requirements of FCC regulation Part 15, Section 15 governing radio frequency electromagnetic interference and be so labelled.

1.6.4 System shall be designed to provide satisfactory operation without damage at 110% and 85% of rated voltage and at + 3 hertz variation in line frequency.

1.6.5 System shall be designed to provide static, transient, and short circuit protection on all inputs and outputs. Communication lines shall be protected against incorrect wiring, static transients and induced magnetic interference. Bus connected devices shall be a.c. coupled or equivalent so that any single device failure will not disrupt or halt bus communication.

1.6.6 All real time clocks missing shall be battery backed.

1.6.7 Plant Control Processors shall comply with ISO 9000.

1.7 ACCEPTABLE BID PROCEDURES

1.7.1 Technical Proposal: It is the intent of this specification to define a state-of-the-art distributed computerized software Integrated Building Management and Control System which is user friendly, has known reliability, is extremely responsive, and which is to be designed, installed, implemented, and supported by a local office of the manufacturer by people skilled in providing functional and efficient solutions to building system needs.

To allow evaluation of vendors and systems, a detailed technical proposal shall be provided and formatted strictly in accordance to this outline. Vendor's standard literature not complying to this format and content requirement will not be considered or evaluated. Proposal requirements shall be as follow:

Proposed system complete with location and block diagram including central computer type and memory, peripherals, communication interfaces, all LAN cards, all active hubs and repeaters, network layout, distributed peer bus connected control/monitor panels with location, listing of equipment directly connected to a DDC controller, and secondary network drivers and connected systems.

Provide copies of required UL listing cards.

1.7.2 Compliance Checklist - Provide a specification paragraph-by-paragraph listing of strict word-for-word GCE 596: (G+4P+28+RF) Residential Development on Plot No. 6117242 in Bukadra, Dubai – U.A.E.

PARKWAY BY PRESTIGE ONE

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compliance or non-compliance. For each item of conditional or non-compliance, spell out the vendor's substitute response to the feature excepted.

Example:

Paragraph	Comply	Conditional Comply	Non-Comply	Justification for Non-Comply
2.08 a	Full Compliance	X		
2.08 b	Cull Compliance	X		
2.08 c	Covers are polished	X		

1.7.3 Computer based central controller features.

Day-to-day interactive operator interface description with sample

- a) English/Graphic displays, penetration scheme for display/command and logical configuring, and examples of alarms and data in various modes (Disabled, fixed, not responding, etc.)
- b) Short cut penetration schemes for direct graphic, point, and command access
- c) Color graphic system description with sample system displays, color graphic penetration and command schemes, graphic creation means, library of symbols, and curve plot
- d) Interactive data editing scheme for modification of system data base and parameters including operators, peripheral assignments, system configuration, text, time schedules, point monitoring limits, event-initiated control, and control application program parameters.
- e) Electronic mail/messaging description and samples
- f) Data access control and security scheme
- g) Report descriptions including selection scheme and sample reports for setting up and displaying extended trend file reports
- h) Sample sequence of operation and specified flow chart display
- j) Sample custom report
- k) Sample text alarm message
- l) Sample of system and point descriptor text
- m) Technical description of all hardware components
- n) Sample alarm report
- o) Sample alarm acknowledgement report
- p) Sample text terminal menu display

1.7.4 Distributed control panel point architecture, memory, real time clock.

Functional description

Scan/Report speed and alarm detection speed
Control and survival capabilities
Life safety interfacing

1.7.5 Communications network architecture, points, and panels

- a) Protocol description, host independent, bus-cut reaction
- b) Bus limitations, distances between active links and hubs, etc.
- c) Number of devices on bus without repeater.
- d) Error checking, collision prevention, recovery
- e) Controller internal architecture, communication cards, board-to-board communications, secondary bus drivers.
- f) Point sharing/update scheme
- g) Meantime end-to-end responsiveness for a large fully operational system for analog alarm report, digital alarm report, panel failure report, host command execution, event command (within one panel and panel-to panel)

1.7.6 Available services for full system maintenance, software update and modifications, hardware spare parts, adds and changes, system training and training classes

1.7.7 Contractor experience/capability, size

- a) Organization with key brief resumes
- b) 15 years history of local support of management, engineering, installation, and maintenance locations and business
- c) Local experience on similar jobs with references
- d) Local facility description

1.7.8 Vendor's standard specification data literature.

1.8 SUBMITTALS

1.8.1 Product Data: Submit manufacturer's technical product data for each control device furnished showing dimensions, weights, capacities, performance characteristics, and electrical characteristics, finishes of materials, installation instruction and startup instruction.

1.8.2 Shop Drawings: Submit shop drawings for each electrical control system, containing the following information:

Schematic flow diagram of system showing fans, pumps, coils, dampers, valves, and control devices.

Label each control device with setting or adjustable range of control.

Indicate all required electrical wiring. Clearly differentiate between portions of wiring that are factory installed and portions to be field installed.

Provide details of faces of control panels, including controls, instruments, and labeling.

Include written description of sequence of operation.

Include a complete list of components for each proposed panel.

1.8.3 Submit catalogues for all controls and accessories.

1.8.4 Maintenance Data: Submit maintenance instructions and spare parts lists, include this data, product data, and shop drawings in maintenance manuals.

1.9 DELIVERY, STORAGE AND HANDLING

Provide factory shipping cartons for each piece of equipment, and control device. Maintain cartons through shipping, storage and handling as required to prevent equipment damage, and to eliminate dirt and moisture from equipment. Store equipment and materials inside and protected from weather.

1.10 MANUALS

1.10.1 The following manuals shall be provided:

An Operators Manual with coloured graphic explanations of operator functions. Although operator functions are to be self-instructional and menu-interactive from the keyboard terminal, this manual shall be for off-line study and refresher use.

Computerized printouts of all PCP (Plant Control Processor) data file construction including all point processing assignments, physical terminal relationships, flowcharts of graphic program. etc.

A user manual for the integrated spreadsheet package.

A manual including revised As Built documents of all shopping and materials as required under the paragraph "SUBMITTALS" on this specification.

Five sets of Operators Manuals and As Built Manuals shall be provided to the Employer.

1.11 TRAINING

1.11.1 All training shall be by the BMCS manufacturer and shall utilize specified manuals, as-build documentation, and the on-line help utility. The following training shall be repeated quarterly during the warranty period.

1.11.2 Operator training shall include two initial six-hour sessions encompassing:

Sequence of Operation review.

Sign on-Sign off.

Selection of all displays and reports.

Commanding of points, keyboard and mouse mode.

Modifying English text.

Use of all dialogue boxes and menus.

Modifying alarm limits and start-stop times.

System initialization.

Download and initialization of remote panels.

Purge and/or dump of historical data.

Use of Portable Operators Terminals.

Troubleshooting of sensors (determining bad sensors).

Password modification.

Creation and modification of cardholder database.

1.11.3 Supervisor training shall include an additional two hour session encompassing:

- Password assignment/modification.
- Operator assignment/modification.
- Operator authority assignment/modification.
- Point disable/enable.
- Terminal and data segregation/modification.
- Use of portable operator terminal.
- Use of spreadsheet package with system data.
- Creation and modification of site-specific user-defined cardholder fields.

1.11.4 Programmer training shall include two additional three hour sessions encompassing:

- Software review of Sequence of Operation and flowcharts.
- Use of diagnostics.
- System maintenance procedures.
- Review of initialization.
- Upload/download and off-line archiving of PCP (Plant control Processor) and PC
- System software.
- Graphic creation.

Programmer training shall be for two Employer personnel and shall be scheduled by the Employer with two week notice anytime during the warranty period.

1.12 WARRANTY

1.12.1 All components, system software, parts and assemblies supplied by the BMCS manufacturer shall be guaranteed against defects in materials and workmanship for one year from acceptance date.

1.12.2 Labor to troubleshoot, repair, reprogram, or replace system components shall be furnished by the BMCS contractor at no charge to the Employer during the warranty period.

1.12.3 All corrective software modifications made during warranty service periods shall be updated on all user documentation and on user and manufacturer archived software disks.

PART 2 - PRODUCTS

2.1 PROGRAMMABLE CONTROLLERS

2.1.1 BUILDING CONTROLLER

A. General Requirements

1. BACnet Conformance

- a. Building Controller shall be approved by the BTL as meeting the BACnet Building Controller requirements.
- b. Please refer to section 22.2, BACnet Functional Groups, in the BACnet standard, for a complete list of the services that must be directly supported to provide each of the functional groups listed above. All proprietary services, if used in the system, shall be thoroughly documented and provided as part of the submittal data. All necessary tools shall be supplied for working with proprietary information.

2. Building controller shall be of scalable design such that the number of trunks and protocols may be selected to fit the specific requirements of a given project.
3. The controller shall be capable of panel-mounted on DIN rail and/or mounting screws.
4. The controller shall be capable of providing global control strategies for the system based on information from any objects in the system, regardless if the object is directly monitored by the building controller module or by another controller.
5. The controller shall be capable of running up to six (6) independent control strategies simultaneously. The modification of one control strategy does not interrupt the function or runtime others.
6. The software program implementing the DDC strategies shall be completely flexible and user-definable. All software tools necessary for programming shall be provided as part of project software. Any systems utilizing factory pre-programmed global strategies that cannot be modified by field personnel on-site, using a wide area network (WAN) or downloaded through remote communications are not acceptable. Changing global strategies using firmware changes is also unacceptable.
7. Programming shall be object-oriented using control function blocks and support DDC functions. All flowcharts shall be generated and automatically downloaded to controller. Programming tool shall be supplied and be resident on workstation. The same tool shall be used for all controllers.
8. The programming tool shall provide means to graphically view inputs and outputs to each program block in real-time as program is executing. This function may be performed using the operator's workstation or field computer.
9. Controller shall have 6,000 Analog Values and 6,000 Binary Values.
10. Controller IP configuration can be done via a direct USB connect with an operator's workstation or field computer.
11. Controller shall have at a minimum a Quad Core 996Ghz processor to ensure fast processing speeds.
12. Global control algorithms and automated control functions shall execute using a 64-bit processor.
13. Controller shall have a minimum of 1 GB of DDR3 SDRAM on a 533Mhz bus to ensure high speed data recording, large data storage capacity and reliability.
14. Controller shall support two (2) on-board EIA-485 ports capable of supporting various EIA-485 protocols including, but not limited to BACnet MS/TP and Modbus.
 - a. Ports are capable of supporting various EIA-485 protocols including, but not limited to BACnet MS/TP and Modbus.
15. Controller shall support two (2) ports—each of gigabit speed—Ethernet (10/100/1000) ports.
 - a. Ports are capable of supporting various Ethernet protocols including, but not limited to BACnet IP, FOX, and Modbus.
16. All ports shall be capable of having protocol(s) assigned to utilize the port's physical connection.
17. The controller shall have at a minimum four (4) onboard inputs, two (2) universal inputs and two (2) binary inputs.

18. Schedules

- a. Building controller modules shall provide normal seven-day scheduling, holiday scheduling and event scheduling.
- b. Each building controller shall support a minimum of 380 BACnet Schedule Objects and 380 BACnet Calendar Objects.

19. Logging Capabilities

- a. Each building controller shall log as minimum 2,000 objects at 15-minute intervals. Any object in the system (real or calculated) may be logged. Sample time interval shall be adjustable at the operator's workstation.
- b. Logs may be viewed both on-site or off-site using WAN or remote communication.
- c. Building controller shall periodically upload trended data to networked operator's workstation for long-term archiving if desired.
- d. Archived data stored in database format shall be available for use in third-party spreadsheet or database programs.

20. Alarm Generation

- a. Alarms may be generated within the system for any object change of value or state (either real or calculated). This includes things such as analog object value changes, binary object state changes, and various controller communication failures.
- b. Each alarm may be dialed out as noted elsewhere.
- c. Alarm log shall be provided for alarm viewing. Log may be viewed on-site at the operator's terminal or off-site using remote communications.
- d. Controller must be able to handle up to 2,000 alarm setups stored as BACnet event enrollment objects, with system destination and actions individually configurable.

21. Demand Limiting

- a. Demand limiting of energy shall be a built-in, user-configurable function. Each controller module shall support shedding of up to 1,200 loads using a minimum of two types of shed programs.
- b. Load shedding programs in building controller modules shall operate as defined in section 2.1.J of this specification.

22. Tenant Activity Logging

- a. Tenant Activity logging shall be supported by a building controller module. Each independent module shall support a minimum of 380 zones.
- b. Tenant Activity logging shall function as defined in section 2.1.K of this specification.

B. BACnet MS/TP

1. BACnet MS/TP LAN must be software-configurable from 9.6 to 115.4Kbps
 - a. Each BACnet MS/TP LAN shall support 64 BACnet devices at a minimum.
 - b. All proprietary object types, if used in the system, shall be thoroughly documented and provided as part of the submittal data. All necessary tools shall be supplied for working

with proprietary information.

C. BACnet IP

1. The building controller shall comply with Annex J of the BACnet specification for IP connections. This device shall use Ethernet to connect to the IP internetwork, while using the same Ethernet LAN for non-IP communications to other BACnet devices on the local area network (LAN).
2. Must support interoperability on WANs and campus area networks (CANs), and function as a BACnet Broadcast Management Device (BBMD).
3. Each controller shall support at a minimum 128 BBMD entries.
4. BBMD management architecture shall support 3,000 subnets at a minimum.
5. Shall support BACnet Network Address Translation.
6. All proprietary object types, if used in the system, shall be thoroughly documented and provided as part of the submittal data. All necessary tools shall be supplied for working with proprietary information.

D. Expansion Ports

1. Controller shall support two (2) expansion ports.
 - a. Combining the two on-board EIA-458 ports with fully loaded expansion ports, the controller shall support six (6) EIA-485 trunks simultaneously.
2. Expansion cards that mate to the expansion ports shall include:
 - a. Dual port EIA-485 card.
 - b. LON network card.

E. Power Supply

1. Input for power shall accept between 17 and 30VAC, 47 and 63Hz.
2. Optional rechargeable battery for shutdown of controller including storage of all data in flash memory.
3. On-board capacitor will ensure continuous operation of real-time clocks for minimum of 14 days.

F. Controller shall be in compliance with the following:

1. UL 916 for open energy management
2. FCC Class B
3. ROHS
4. IEC 60703
5. C-Tick Listed

G. Controller shall operate in the following environmental conditions:

1. -4 to 149 °F (-20 to 65 °C) without optional battery, or 32 to 122 °F (0 to 50 °C) with optional battery.

2. 0 to 95% relative humidity (RH), non-condensing.

2.1.2 CENTRAL PLANT AND AIR HANDLER APPLICATION CONTROLLERS

- A. Provide one or more native BACnet application controllers for each air handler and provide native BACnet application controllers as needed for central plant control that adequately cover all objects listed in object list. All controllers shall interface to building controller through either MS/TP LAN using BACnet protocol, or Ethernet LAN using BACnet over Ethernet or BACnet TCP/IP. No gateways shall be used. Controllers shall include input, output and self-contained logic program as needed for complete control of units. Controllers shall be fully programmable using graphical programming blocks. Programming tool shall be resident on operator workstation and be the same tool as used for the building controller. No auxiliary or non-BACnet controllers shall be used.
- B. BACnet Conformance
 1. Application controllers shall be approved by the BTL as meeting the BACnet Advanced Application Controller requirements.
 2. Please refer to section 22.2, BACnet Functional Groups, in the BACnet standard, for a complete list of the services that must be directly supported to provide each of the functional groups listed above. All proprietary services, if used in the system, shall be thoroughly documented and provided as part of the submittal data. All necessary tools shall be supplied for working with proprietary information.
 3. Standard BACnet object types supported shall include, as a minimum, Analog Input, Analog Output, Analog Value, Binary Input, Binary Output, Binary Value, Multi-state Values, Device, File, and Program object types. All proprietary object types, if used in the system, shall be thoroughly documented and provided as part of the submittal data. All necessary tools shall be supplied for working with proprietary information.
- C. Application controllers shall include universal inputs with 12-bit resolution that accept 3K and 10K thermistors, 0–10VDC, Platinum 1000 ohm RTD, 0–5VDC, 4–20mA and dry contact signals. Any input on a controller may be either analog or digital with a minimum of three inputs that accept pulses. Controller shall also include support and modifiable programming for interface to intelligent room sensor with digital display. Controller shall include binary and analog outputs on board. Analog outputs with 12-bit resolution shall support either 0–10VDC or 0–20mA. Binary outputs shall have LED indication of status. Software shall include scaling features for analog outputs. Application controller shall include 20VDC voltage supply for use as power supply to external sensors.
 1. All outputs must have onboard Hand-Off-Auto (HOA) switches and a status indicator light. HOA switch position shall be monitored. Each analog output shall include a potentiometer for manually adjusting the output when the HOA switch is in the Hand position.
 2. The position of each and every HOA switch shall be available system wide as a BACnet object property.
- D. All program sequences shall be stored on board application controller in EEPROM. No batteries shall be needed to retain logic program. All program sequences shall be executed by controller up to 20 times per second (minimum of 10 times per second) and capable of multiple PID loops for control of multiple devices. All calculations shall be completed using floating-point math and system shall support display of all information in floating-point nomenclature at operator's terminal.
 1. The following control blocks shall be supported:
 - a. Natural Log

- b. Exponential
 - c. Log base 10
 - d. X to the power of Y
 - e. Nth square root of X
 - f. 5th Order Polynomial Equations
 - g. Astronomical Clock (sunrise/sunset calculation)
 - h. Time based schedules
- E. Programming of application controller shall be completely modifiable in the field over installed BACnet LANs or remotely using modem interface. Operator shall program logic sequences by graphically moving function blocks on screen and tying blocks together on screen. Application controller shall be programmed using programming tools as described in operator's terminal section.
- F. Application controller shall include support for intelligent room sensor (see Section 2.10.B.) Display on intelligent room sensor shall be programmable at application controller and include an operating mode and a field service mode. All button functions and display data shall be programmable to show specific controller data in each mode, based on which button is pressed on the sensor. See sequence of operation for specific display requirements at intelligent room sensor.
- G. Schedules
- 1. The controller shall support a minimum of 3 BACnet Schedule Objects and have a real time clock on board with battery backup to maintain time through a power loss.
- H. Logging Capabilities
- 1. Controller shall support a minimum of 50 trendlogs. Any object in the controller (real or calculated) may be logged. Sample time interval shall be adjustable at the operator's workstation.
 - 2. Controller shall periodically upload trended data to system server for long-term archiving if desired. Archived data stored in (MS Jet Database or SQL) database form and shall be available for use in third-party spreadsheet or database programs.
- I. Alarm Generation
- 1. Alarms may be generated within the controller for any object change of value or state (either real or calculated). This includes things such as analog object value changes, and binary object state changes.
 - 2. Alarm log shall be provided for alarm viewing. Log may be viewed on-site at the operator's terminal or off-site using remote communications.
 - 3. Controller must be able to handle up to 25 alarm setups stored as BACnet event enrollment objects, with system destination and actions individually configurable.
- J. The controller processor shall be a 32-bit processor.
- K. The packaging of the controller shall provide operable doors to cover the terminals once installation is complete. The housing of the controller shall provide for DIN rail mounting and also fully enclose circuit board.

2.2 UNITARY CONTROLLERS

- A. Provide one native BACnet application controller for each piece of unitary mechanical equipment that adequately covers all objects listed in object list for unit. All controllers shall interface to building controller through MS/TP LAN using BACnet protocol. No gateways shall be used. Controllers shall include input, output and self-contained logic program as needed for complete control of unit.
- B. BACnet Conformance
1. Application controllers shall, as a minimum, support MS/TP BACnet LAN types. They shall communicate directly using this BACnet LAN at 9.6, 19.2, 38.4 and 76.8 Kbps, as a native BACnet device. Application controllers shall be approved by the BTL as meeting the BACnet Application Specific Controller requirements and support all BACnet services necessary to provide the following BACnet functional groups:
 - a. Files Functional Group
 - b. Reinitialize Functional Group
 - c. Device Communications Functional Group
 2. Please refer to Section 22.2, BACnet Functional Groups in the BACnet standard, for a complete list of the services that must be directly supported to provide each of the functional groups listed above. All proprietary services, if used in the system, shall be thoroughly documented and provided as part of the submittal data. All necessary tools shall be supplied for working with proprietary information.
 3. Standard BACnet object types supported shall include, as a minimum, Analog Input, Analog Output, Analog Value, Binary Input, Binary Output, Binary Value, Device, File, and Program Object Types. All proprietary object types, if used in the system, shall be thoroughly documented and provided as part of the submittal data. All necessary tools shall be supplied for working with proprietary information.
- C. Application controllers shall include universal inputs with 10-bit resolution that can accept 3K and 10K thermistors, 0–5VDC, 4–20mA, dry contact signals and a minimum of 3 pulse inputs. Any input on controller may be either analog or digital. Controller shall also include support and modifiable programming for interface to intelligent room sensor. Controller shall include binary outputs on board with analog outputs as needed.
- D. All program sequences shall be stored on board controller in EEPROM. No batteries shall be needed to retain logic program. All program sequences shall be executed by controller 10 times per second and shall be capable of multiple PID loops for control of multiple devices. Programming of application controller shall be completely modifiable in the field over installed BACnet LANs or remotely through modem interface. Operator shall program logic sequences by graphically moving function blocks on screen and tying blocks together on screen. Application controller shall be programmed using same programming tools as building controller and as described in operator workstation section. All programming tools shall be provided and installed as part of system.
- E. Application controller shall include support for intelligent room sensor (see Section 2.10.B.) Display on room sensor shall be programmable at controller and include an operating mode and a field service mode. All button functions and display data shall be programmable to show specific controller data in each mode based on which button is pressed on the sensor.
- F. Intelligent Room Sensor with Touchscreen for FCU and VAV Controllers

Hardware

- a. Room sensor shall include:
Backlit touchscreen LCD digital display
Temperature sensor
Humidity sensor
Programmable Status Light indicator
CO2 sensor or BACnet MS/TP communication up to 115.2kbps
- b. Temperature sensor shall be a Uni-Curve Type II thermistor with an accuracy of +/- 0.36 °F (0.3 °C) at calibration point over the range of 32 to 158 °F or better.
- c. Humidity sensor shall have an accuracy of +/-3% from 10 to 90% relative humidity (RH) or better, non-condensing.
- d. The intelligent room sensor's Status Light indicator shall have a minimum of four (4) colors (blue, red, amber and green) that will cast a glow onto the wall below the sensor to be used as visual indicator to the occupants of the condition of the system. The color and on/off state of the Status Light indicator shall be fully programmable.
- e. CO2 sensor shall have an accuracy of +/- 30 ppm over the range of 0–5000 ppm or better.
- f. CO2 sensor shall utilize Automatic Baseline Correction to maintain sensor calibration without the need for manual calibration.
- g. The user shall interact with the smart sensor using a touchscreen, with no buttons allowed.
- h. The intelligent room sensor shall have provisions for a tamper proof installation requiring tools to be removed from the wall.
- i. The touchscreen shall have a surface hardness of Mohs 7 or greater to prevent being easily scratched.
- j. Controller shall function as room control unit, and allow occupant to raise and lower setpoint, and activate terminal unit for override use—all within limits as programmed by building operator.

2. Display Content

- a. The intelligent room sensor shall simultaneously display room setpoint, room temperature, and outside temperature at each controller.
- b. The intelligent room sensor shall have the ability to add or remove from the display time-of-day, room humidity, and indoor air temperature to customize the view for the customer.
- c. The intelligent room sensor must have the capability to show temperatures in degrees Fahrenheit or degrees Celsius.
- d. A communication loss or improper communications wiring shall be displayed on the LCD screen to aid in trouble shooting.
- e. Information about the version of firmware shall be displayable on the LCD screen.

- f. A cleaning mode will be provided to allow for the touchscreen to be cleaned without inadvertently making changes to system parameters.
- g. The intelligent room sensor shall have the ability to display the status of a lighting zone and control the on/off state of the zone from the touchscreen using a tenant-accessible display page.
- h. The intelligent room sensor shall have the ability to display the status of a window zone (e.g., blinds) and control the on/off state of the zone from the touchscreen using a tenant-accessible display page.
- i. After Hours Override shall:
 - Override time may be set and viewed in 30-minute increments.
 - Override time countdown shall be automatic but may be reset to zero by occupant from the sensor.
 - Time remaining shall be displayed.
 - Display shall show the word “OFF” in unoccupied mode unless a function button is pressed.

3. Other Modes

- a. The intelligent room sensor shall also allow service technician access to hidden functions for advanced system configuration. This functionality shall be accessed-protected with a configurable PIN number.
- b. Field Service Mode shall allow access to common parameters as dictated by the application’s sequence of operations. The parameters shall be viewed and set from the intelligent room sensor with no computer or other field service tool needed.
- c. If the intelligent room sensor is connected to VAV controller, Balance Mode shall allow a VAV box to be balanced and all air flow parameters viewed. The balancing parameters shall be viewed and set from the intelligent room sensor with no computer or other field service tool needed.

4. Intelligent Room Sensor shall be in compliance of the following:

- a. UL Standard for Safety 916
- b. FCC Part 15.107 & 109, Class B, CFR47-15
- c. EMC Directive 89/336/EEC (European CE Mark)

2.2.1 NOT USED

2.2.2 TOUCH SCREEN COMMUNICATING THERMOSTAT

A. BACnet Conformance

- 1. Touch screen communicating thermostats shall be approved by the BTL as meeting the BACnet Application Specific Controller requirements.
- 2. Touch screen communicating Thermostats shall, at a minimum, support MS/TP BACnet LAN types. They shall communicate directly through this BACnet LAN at 9.6, 19.2, 38.4 and 76.8

Kbps, as a native BACnet device.

3. Standard BACnet object types supported shall include, as a minimum, Analog Input, Analog Output, Analog Value, Binary Input, Binary Output, Binary Value, Device, File, and Program Object Types.
4. All proprietary object types, if used in the system, shall be thoroughly documented and provided as part of the submittal data. All necessary tools shall be supplied for working with proprietary information.

B. Touch screen Communicating Thermostat hardware shall:

1. Include a 32 Bit processor
2. Include a backlit touch screen for the user interface, buttons are not allowed.
3. Include Three (3) universal inputs with 12-bit resolution that can accept 3K and 10K Type II thermistors, 0-10VDC, 0-5 VDC, 4-20mA, and dry contact signals. Inputs on controller may be either analog or digital.
4. Include built-in temperature sensor.
5. Include built-in humidity sensor.
6. Include Six (6) relay outputs on board.
7. Include Two (2) analog outputs with 12-bit resolution. Each auto-detecting for 0-10 V or 4-20 mA control signals.
8. Meet the requirements of Listed Underwriters Laboratory for Open Energy Management Equipment (PAZX) under the UL Standard for Safety 916.
9. Meet the requirements of EMC Directive (European CE Mark) EN 60950.
10. Meet the requirements for FCC Part 15, Class B.
11. Be powered by 24VAC power.

2.3 USER INTERFACES - GENERAL

2.3.1 Overview

The design of the BMS shall be capable of providing a range of user interfaces including:

Main Server(s) and Clients

Panel PCs for plant room applications

Hand-Held and Pocket PCs

All communications between the user and the system shall be in clear language, without reference to special code or codes.

It shall be a fundamental requirement that the user interfaces are easy to use and intuitive throughout. The use of window techniques such as "drag and drop" is preferred.

2.3.2 Password Access

BMS user interfaces shall be accessible after logging on by entering a password.

All user interfaces shall use the same access regime such that each user will retain their access criteria and access levels across all user interfaces.

It shall be possible to grant or deny access to each of the various functions for individual users.

2.3.3 Help facility

A complete 'help' facility shall provide information about the system and its use. This shall be made available to the user.

The information provided shall be in simple clear language.

2.4 BMS TERMINAL & CLIENT TERMINALS

2.4.1 General System Architecture

The BMS shall be based on open system architecture and shall support industry standard protocols like Lon Works, BACnet, Modbus, OPC, EIB, ODBC, XML, SOAP and Advance DDE. The system shall be designed on a distributed computing principle, assuring that no single point of failure exists in the system. Furthering this concept, the system shall comprise of the following discrete and segregated layers:

Management Layer – Comprising of supervisory database servers and Human Machine Interfaces connected on a BACnet/IP network

Supervisory Layer – Comprising of BACnet gateways, and is applicable only when the Automation Layer comprises of controllers with Lonworks protocols

Automation Layer – Comprising of BACnet/LonMark controllers managing / monitoring various electromechanical system including field level components

The data communication between the Automation/Supervisory and Management layers shall be through Ethernet Network using TCP/IP protocol.

A BACnet / IP or BACnet / Ethernet router connects the Automation / Supervisory Layer to the Management Layer; whilst a BACnet/IP or BACnet / Ethernet gateway performs the translation of Lon protocol based Automation Layer to the BACnet based Supervisory Layer.

2.4.2 The Management Layer

The Management Layer shall comprise of a distributed network of Database Servers, Operator Stations, Print Servers, and BACnet Servers etc. The system shall use a Client Server architecture in which the Database Servers shall have the requisite Clients to access data from the Automation/Supervisory Layer Controllers. Furthermore, the Database Servers shall act as data providers for Operator Workstations and other data consumers like Central Monitoring System, ERP systems, Computerized Maintenance Management Systems, Financial Systems etc. The Database Servers shall have the capability to provide data to these systems via BACnet Server, OPC Server, Advance DDE Server, XML, SOAP, or ODBC.

The management layer shall be able to deal with the high data traffic with complex data structures between management stations and between the management station and the process level.

The management layer equipment and services shall be able to support current and future building

management standard in order to allow backward and forward compatibility.

The network technology must be based on BACnet/IP and Ethernet, and be compatible with the latest LAN, WAN technology. The Database Server Computers and Operator Workstation hardware shall be capable of interfacing to an IEEE 802.3 Standard Local Area Network (LAN).

The Ethernet network infrastructure including cabling, active and passive components shall be provided. The BACS vendor shall co-approve the patch panels and cables prior to order by the MEP contractor (as per structured cabling specifications). All active and passive components after the patch panels required to interface to the BACS system shall be provided by the BACS vendor in his scope.

The following sections define the functional requirements of each of the components on the Management Layer:

1.2 PERFORMANCE REQUIREMENTS

- A. Building Management System (BMS) shall use an open Engineer and fully support a multi-vendor environment. To accomplish this effectively, the BMS shall support open communication protocol standards and integrate a wide variety of third-party devices and applications. The system shall be designed for use on the Internet, or intranets using off the shelf, industry standard technology compatible with other owner provided system.
- B. System shall be modular in nature and shall permit expansion of both capacity and functionality through the addition of sensors, actuators, controllers and operator devices, while re-using existing controls equipment.
- C. System design shall eliminate dependence upon any single device for alarm reporting and control execution.

BMS shall be hierarchical in structure; each level shall be capable of functioning autonomously and shall be operated independently.

- D. Server shall meet at minimum the following requirements:
 - 1. Intel® Xeon® Processor, latest available family's version at the time of ordering shall be provided, 16 Gigabytes of installed DDR4 RAM, spare slots capable of 24GB total RAM.
 - 2. Three 1 Terabyte hard disks with 8ms. or faster access time for a hardware based "Fault Tolerant Redundant Array of Independent Disks" (RAID) to a minimum of RAID 5.
 - 3. Dual Gigabit Network Interface Card Server Adapter.
 - 4. Latest Windows operating system Server 2016 or later.
 - 5. Database shall be SQL Server 2016 or later

Software

The BMS system server shall be based around the Microsoft Windows 64 bit multitasking environment, Windows 2016 Server. The BMS system shall be a true 64 bit application to take advantage of Microsoft Windows 2016 enabling technologies. Any 16-bit system running on the Microsoft platform (such as those originally based on MSDOS and Microsoft Windows 3.x) shall not be acceptable.

Standard services supported by the server computer operating system will include the following:

Multi-tasking Multi-user support

TCP / IP Network Support

Graphic Display Building Editor

Application software.

Hot Backup Server

This facility shall enable the system server to operate in high availability architecture with no single point of failure. To achieve this, the facility must provide specific features.

The system must be capable of running a pair of similarly configured computers in a hot backup configuration where at any point in time, one is the acting primary and the other the acting as the Hot Backup. An online database duplication mechanism must be supported.

Scanning I/O on two separate systems and processing independently is not acceptable.

The database duplication must be performed on a per-transaction basis for two reasons:

- a. To ensure that the duplicated Backup database is consistent at all time with the primary database.
- b. To avoid unnecessary loading of field devices caused by duplicate polling.

It must be possible to remove one of the redundant systems for maintenance without interrupting operation, and upon its reinstatement, re-synchronize the database, again without interruption to system operation. A method of manually initiating a fail over must be provided to assist with such maintenance operations.

Failure of either system must be announced audibly and visually via the alarming subsystem.

To accommodate recoverable faults, the failed system must be able to reboot automatically after non-fatal errors and assume the role of acting as hot backup automatically.

The above specification is a minimum requirement. Vendor to propose the exact hardware based on his system requirements.

2.4.2.2 OPERATOR'S WORKSTATION

- A. General structure of workstation interaction shall be a standard client/server relationship. Server shall be used to archive data and store system database. Clients shall access server for all archived data. Each client shall include flexibility to access graphics from server or local drive. Server shall support a minimum of 50 simultaneous clients.
- B. BACnet Conformance
 1. Operator workstation shall be approved by the BTL as meeting the BACnet Advanced Work Station (AWS) requirements.
 2. Please refer to Section 22.2, BACnet Functional Groups, in the BACnet standard, for a complete list of the services that must be directly supported to provide each of the functional groups listed above. All proprietary services, if used in the system, shall be thoroughly documented and provided as part of the submittal data. All necessary tools shall be supplied for working with proprietary information.
 3. Standard BACnet object types accessed by the AWS shall include as a minimum: Analog Value, Analog Input, Analog Output, Binary Value, Binary Input, Binary Output, Calendar, Device, Event Enrollment, File, Notification Class, Program, and Schedule object types. All

proprietary object types, if used in the system, shall be thoroughly documented and provided as part of the submittal data. All necessary tools shall be supplied for working with proprietary information.

4. The AWS shall comply with Annex J of the BACnet specification for IP connections. Must support remote connection to server using a thick client application. This device shall use Ethernet to connect to the IP internetwork, while using the same Ethernet LAN for non-IP communications to other BACnet devices on the LAN. Must support interoperability on wide area networks (WANs) and campus area networks (CANs). AWS shall support Foreign Device Registration to allow temporary workstation connection to IP network.

C. Data Displays

1. Data displays shall render all data associated with project as called out on drawings and/or object type list supplied. Graphic files shall be created using digital, full color photographs of system installation, AutoCAD or Visio drawing files of field installation drawings, and wiring diagrams from as-built drawings.
2. Data displays shall render all data using iconic graphic representations of all mechanical equipment. System shall be capable of displaying graphic file, text, and dynamic object data together on each display and shall include animation. Information shall be labeled with descriptors and shall be shown with the appropriate engineering units. All information on any display shall be dynamically updated without any action by the user.
3. Data display frame shall allow user to change all field-resident AWS functions associated with the project, such as setpoints, weekly schedules, exception schedules, etc., from any screen, no matter if that screen shows all text or a complete graphic display. This shall be done without any reference to object addresses or other numeric/mnemonic indications.
4. Analog objects shall be displayed with operator modifiable units. Analog input objects may also be displayed as individual graphic items on the display screen as an overlay to the system graphic.
5. All displays and programming shall be generated and customized by the local use energy management and control system (EMCS) supplier and installer. Systems requiring factory development of graphics or programming of DDC logic are specifically prohibited.
6. AWS shall be supplied with a library of standard graphics, which may be used unaltered or modified by the operator. AWS shall include a library of equipment graphic components to assemble custom graphics. Systems that do not allow customization or creation of new graphic objects by the operator (or with third-party software) shall not be allowed.
7. Data display frame shall include customizable and persistent tree navigation for building, equipment and system diagnostic centric display organization.
8. Each display may be protected from viewing unless operator credentials have the appropriate access level. An access level may be assigned to each display and system object. The menu label shall not appear on the graphic if the operator does not have the appropriate security level.
9. Data displays shall have the ability to link to content outside of the EMCS system. Such content shall include but is not limited to launching external files in their native applications (for example, a Microsoft Word document) and launching a web browser resolving to a specified web address.
10. The AWS shall have the ability to support 20 concurrent web clients.
11. Data displays shall support:

- a. Graphic items with custom geometry that offer both color gradient shading and variable opacity in scale to system variables and range setpoints.
- b. Clear and custom geometry navigation buttons to provide intuitive navigation.
- c. Graphic files in JPG, PNG, and GIF file types.
- d. Viewing of 1,024 system data points in a single screen.

D. Password Protection

1. Provide security system that prevents unauthorized use unless operator is logged on. Access shall be limited to operator's assigned functions when user is logged on. This includes displays as outlined above.
2. AWS shall provide security for a minimum of 200 users. Each user shall have an individual User ID, Username, and Password. Entries are alphanumeric characters only and are case sensitive (except for User ID). User ID shall be 0–8 characters, Username shall be 0–29 characters, and Password shall be 4–8 characters long.
3. Each user shall be allowed individual assignment of only those control functions, menu items, and user-specific system start display, as well as restricted access to discrete BACnet devices to which that user requires access.
4. All passwords, usernames, and access assignments shall be adjustable online at the operator's terminal.
5. Users shall also have a set access level, which defines access to displays and individual objects the user may control. System shall include 10 separate and distinct access levels for assignment to users.
6. The system shall include an Auto Logout feature that shall automatically logout user when there has been no keyboard or mouse activity for a set period of time. Time period shall be adjustable by system administrator. Auto Logout may be enabled and disabled by system administrator. Operator terminal shall display message on screen that user is logged out after Auto Logout occurs.
7. The system shall permit the assignment of an effective date range, as well as an effective time of day, that the User IDs are permitted to authenticate.

E. Operator Activity Log

1. An Operator Activity Log that tracks all operator changes and activities shall be included with AWS. System shall track what is changed in the system, who performed this change, date and time of system activity, and value of the change before and after operator activity. Operator shall be able to display all activity, sort the changes by user and also by operation. Operator shall be able to print the Operator Activity Log display.
2. Log shall be gathered and archived to a hard drive on AWS as needed. Operator shall be able to export data for display and sorting in a spreadsheet.

F. Scheduling

1. AWS and web client shall show all information in easy-to-read daily format including calendar of this month and next. All schedules shall show actual ON/OFF times for day based on scheduling priority. Priority for scheduling shall be events, holidays and daily, with events being the highest.
2. Holiday and special event schedules shall display data in calendar format. Operator shall be

able to schedule holidays and special events directly from these calendars.

3. Operator shall be able to change all information for a given weekly or exception schedule if logged on with the appropriate access privileges.
4. AWS shall include a Schedule Wizard for set up of schedules. Wizard shall walk user through all steps necessary for schedule generation. Wizard shall have its own pull-down selection for startup or may be started by right-clicking on value displayed on graphic and then selecting Schedule.
5. Scheduling shall include optimum start based on outside air temperature, current heating/cooling set points, indoor temperature and history of previous starts. Each and every individual zone shall have optimum start time independently calculated based on all parameters listed. User shall input schedules to set time that occupied set point is to be attained. Optimum start feature shall calculate the startup time needed to match zone temperature to set point. User shall be able to set a limit for the maximum startup time allowed.
6. Schedule list shall show all schedules currently defined. This list shall include all standard, holiday and event schedules. In addition, user shall be able to select a list that shows all scheduled points and zones.
7. Display of all three schedules must show all ON times for standard, holiday and event schedules in different colors on a given day. In addition, OFF times for each must also be shown in additional colors. User shall be able to select from standard calendar what days are to be scheduled and same display shall show all points and zones affected. User shall be able to set time for one day and select all days of the week that shall be affected as a recurrence of that same schedule for that given day.
8. Any displayed data that is changeable by the operator may be selected using the right mouse button and the schedule shall then be selectable on the screen. Selection of the schedule using this method shall allow the viewing of the assigned schedule allow the point to be scheduled.
9. Schedule editor shall support drag-n-drop events and holidays onto the schedule calendar.
10. Schedule editor shall support drag-n-drop events default to a two-hour period, which can then be adjusted by the user.
11. Schedule editor shall support drag-n-drop holidays default for OFF all day and can be edited for multiple-day holidays.
12. Schedule editor shall support the view of affected zones when adding or editing timed events of a schedule.

G. Alarm Indication and Handling.

1. AWS shall provide visual, printed, and email means of alarm indication. Printout of alarms shall be sent to the assigned terminal and port. Alarm notification can be filtered based on the User ID's authorization level.
2. Web client shall display a persistent alarm state for the system regardless of the data view including points in alarm but not acknowledged, and points that have gone into alarm and returned to normal without being acknowledged.
3. Alarm History shall provide log of alarm messages. Alarm log shall be archived to the hard disk of the AWS. Each entry shall include a description of the event-initiating object generating the alarm. Description shall be an alarm message of at least 256 characters in length. Entry shall include time and date of alarm occurrence, time and date of object state

return to normal, time and date of alarm acknowledgment, and identification of operator acknowledging alarm.

4. Alarm messages shall be in user-definable text (English or other specified language) and shall be delivered either to the operator's terminal, client or through remote communication using email (Authenticated SMTP supported).
5. AWS shall include an Alarm Wizard for set up of alarms. Wizard shall walk user through all steps necessary for alarm generation. Wizard shall have its own pull-down selection for startup or may be started by right-clicking on value displayed on graphic and then selecting alarm setup.
6. AWS shall support color-coded indication of current alarms as follows:
 - a. Red indicator shows number of active alarms that have not been acknowledged.
 - b. Yellow indicator shows number of alarms that are still active but have been acknowledged.
 - c. Blue indicator shows number of alarms that have returned to normal but have not been acknowledged.
 - d. Color-coded indicators, when selected by the user, navigate to a pre-filtered view of alarm history.
 - e. Alarm history can be filtered by color-coded indicator states.
7. Alarm annunciation includes navigation link to a user-selected display or URL.
8. User can silence audible annunciation for the current session.
9. User can disable auto-refresh of alarm annunciation for current session.
10. Any displayed data that is changeable by the operator may be selected using the right mouse button and the alarm shall then be selectable on the screen. Selection of the alarm using this method shall allow the viewing of the alarm history or allow the creation of a new alarm.

H. Trend log Information

1. AWS shall periodically gather historically recorded data stored in the building controllers and store the information in the system database. Stored records shall be appended with new sample data, allowing records to be accumulated. Systems that write over stored records shall not be allowed unless limited file size is specified. System database shall be capable of storing up to 50 million records before needing to archive data. Samples may be viewed at the web client. Operator shall be able to view all trended records, both stored and archived. All trend log records shall be displayed in standard engineering units.
2. AWS shall be capable of trending on an interval determined by a polling rate, or change-of-value.
3. AWS shall be able to change trend log setup information. This includes the information to be logged as well as the interval at which it is to be logged. All operations shall be password protected. Viewing may be accessed directly from any and all graphics on which a trended object is displayed.
4. AWS shall include a Trend log Wizard for setup of logs. Wizard shall walk user through all necessary steps. Wizard shall have its own pull-down selection for startup or may be started by right-clicking on value displayed on graphic, and then selecting Trend logs from the displayed menu.

5. AWS shall be capable of using Microsoft SQL as the system database.
6. Any displayed data that is changeable by the operator may be selected using the right mouse button and the trend log shall then be selectable on the screen. Selection of the trend log using this method shall allow the viewing of the trend log view.
7. Trend log viewer shall provide:
 - a. Software that is capable of graphing the trend-logged object data shall be included.
 - b. Access and ability to create, edit and view are restricted to users by user account credentials
 - c. Specific and repeatable URL defines the trend log(s) that comprise the view.
 - d. Call out of trend log value at intersection of trend line and mouse-over vertical axis.
 - e. Trend log and companion logs can be configured to display on one of two independent vertical scales.
 - f. Click zoom for control of data set viewed along either graph axis.
 - g. User-specifiable start and end dates as well as a fast scroll features that supports click zoom of macro scale view of the data for quickly finding data set based on visual signature.
 - h. User export of the viewed data set to MS Excel.
 - i. Web browser-based help.
 - j. Optional min/max ranges (Upper Control Limits, Lower Control Limits) for each value.
- I. Energy Log Information
 1. AWS shall be capable of periodically gathering energy log data stored in the field equipment and archive the information. Archive files shall be appended with new data, allowing data to be accumulated. Systems that write over archived data shall not be allowed unless limited file size is specified. Display all energy log information in standard engineering units.
 2. All data shall be stored in database file format for direct use by third-party programs. Operation of system shall stay completely online during all graphing operations.
 3. AWS operator shall be able to change the energy log setup information as well. This includes the meters to be logged, meter pulse value, and the type of energy units to be logged. All meters monitored by the system may be logged. System shall support using flow and temperature sensors for BTU monitoring.
 4. AWS shall display archived data in tabular format form for both consumption and peak values. Data shall be shown in hourly, daily, weekly, monthly and yearly formats. In each format, the user shall be able to select a specific period of data to view.
- J. Demand Limiting
 1. AWS shall include demand limiting program that includes two types of load shedding. One type of load shedding shall shed/restore equipment in binary fashion based on energy usage when compared to shed and restore settings. The other type of shedding shall adjust operator-selected control setpoints in an analog fashion based on energy usage when compared to shed and restore settings. Shedding may be implemented independently on each and every zone or piece of equipment connected to system.
 2. Binary shedding shall include minimum of five (5) priority levels of equipment shedding. All loads in a given priority level shall be shed before any loads in a higher priority level are shed.

Load shedding within a given priority level shall include two methods. In one, the loads shall be shed/restored in a “first off-first on” mode, and in the other the loads are just shed/restored in a “first off-last on” (linear) fashion.

3. Analog shed program shall generate a ramp that is independently used by each individual zone or individual control algorithm to raise the appropriate cooling setting and lower appropriate heating setting to reduce energy usage.
4. AWS shall be able to display the status of each and every load shed program. Status of each load assigned to an individual shed program shall be displayed along with English description of each load.

K. Tenant Activity

1. AWS shall include program that monitors after-hours overrides by tenants, logs that data, and generates a bill based on usage and rate charged for each tenant space. Tenant Activity program shall be able to assign multiple zones, from a list of every zone connected to system, to a particular tenant. Every zone is monitored for after-hours override usage and that data logged in AWS. Operator may then generate a bill based on the usage for each tenant and the rate charged for any overtime use.
2. Configuration shall include entry of the following information for use in logging and billing:
 - a. Tenant’s contact name and address
 - b. One or multiple tenant zones that make up a total tenant space, including a separate billing rate for each separate zone
 - c. Minimum and maximum values an event duration and event limit
 - d. Property management information
 - e. Overall billing rate
 - f. Seasonal adjustments or surcharge to billing rate
 - g. Billing notification type including, but not limited to printer, file and email
 - h. Billing form template
3. Logging shall include recording the following information for each and every tenant event:
 - a. Zone description
 - b. Time the event begins
 - c. Total override time
 - d. Limits shall be applied to override time
4. A tenant bill shall be generated for a specific period using all the entered configuration data and the logged data. User with appropriate security level shall be able to view and override billing information. User shall be able to select a billing period to view and be able to delete events from billing and edit a selected tenant activity event’s override time.

L. Reports

1. AWS shall be capable of periodically producing reports of trend logs, alarm history, tenant activities, device summary, energy logs, and override points. The frequency, content, and delivery are to be user adjustable.

2. All reports shall be capable of being delivered in multiple formats including text- and comma-separated value (CSV) files. The files can be printed, emailed, or saved to a folder, either on the server hard drive or on any network drive location.

M. Configuration/Setup

1. Provide means for operator to display and change system configuration. This shall include, but not be limited to system time, day of the week, date of daylight savings set forward/set back, printer termination, port addresses, modem port and speed, etc. Items shall be modified using understandable terminology with simple mouse/cursor key movements.

N. Field Engineering Tools

1. AWS shall include field engineering tools for programming all controllers supplied. All controllers shall be programmed using graphical tools that allow the user to connect function blocks on screen that provide sequencing of all control logic. Function blocks shall be represented by graphical displays that are easily identified and distinct from other types of blocks. Graphical programming that uses simple rectangles and squares is not acceptable.
2. User shall be able to select a graphical function block from menu and place on screen. Provide zoom in and zoom out capabilities. Function blocks shall be downloaded to controller without any reentry of data.
3. Programming tools shall include a real-time operation mode. Function blocks shall display real-time data and be animated to show status of data inputs and outputs when in real-time operation. Animation shall show change of status on logic devices and countdown of timer devices in graphical format.
4. Field engineering tools shall also include a database manager of applications that include logic files for controllers and associated graphics. Operator shall be able to select unit type, input/output configuration and other items that define unit to be controlled. Supply minimum of 250 applications as part of workstation software.
5. Field engineering tool shall include Device Manager for detection of devices connected anywhere on the BACnet network by scanning the entire network. This function shall display device instance, network identification, model number, and description of connected devices. It shall record and display software file loaded into each controller. A copy of each file shall be stored on the computer's hard drive. If needed, this file shall be downloaded to the appropriate controller using the mouse.
6. AWS shall automatically notify the user when a device that is not in the database is added to the network.
7. AWS shall include backup/restore function that will back up entire system to selected medium and then restore system from that medium. The system shall be capable of creating a backup for the purpose of instantiating a new client PC.
8. The system shall provide a means to scan, detect, interrogate, and edit third-party BACnet devices and BACnet objects within those devices.

O. Workstation Hardware

1. Provide operator's workstation(s) at location(s) noted on the plans.
2. AWS server
 - a. 10th generation Intel Core i7 processor, 3.4 GHZ, 8M cache.
 - b. 2 x 32" inches LED monitors of 4K resolution for schematics, alarms, reports,

trends, etc.

- c. 64-bit OS
- d. Windows 7, Windows 8, Windows 8.1, Windows Server 2016 or later
- e. 2 GHz (or better), one or more dual-core or quad-core processors
- f. 16 GB RAM or higher
- g. 1TB of hard drive space required for base installation without application data
- h. Network interface card (10/100/1000 Mbps)

P. Software

- 1. At the conclusion of the project, contractor shall leave with owner a CD-ROM that includes the complete software operation system and project graphics, setpoints, system parameters, etc. This backup shall allow the owner to completely restore the system in the case of a computer malfunction.

A. The BMS Specialist shall provide a site-wide software license that shall permit the Employer to add a PC to the BMS Management level Network without the requirement to obtain an additional software license.

B. BMS shall have an “off-the-shelf” software package (Middleware) that shall be able to import and export data in, at minimum, the following protocols over an Ethernet TCP/IP backbone:

- 1. BACnet.
- 2. LonTalk
- 3. Modbus
- 4. SNMP (Simple Network Management Protocol).
- 5. XML (eXtensible Markup Language) / SOAP (Simple Object Access Protocol).
- 6. OPC (Open Platform Communications)

C. It shall be output from the BMS in the appropriate protocol to the appropriate ELV Building System. Note that it shall be possible to output the same data in different protocols to different ELV Building Systems

D. Information shall be output from the BMS when:

- 1. Requested by the operator. This includes requests made by the operator for the immediate output of specific information and requests for information that are optional and have been pre- programmed by the operator for display.
- 2. Scheduled by the operator. An example of this might be the output of a report at an operator- defined time of day.
- 3. An alarm condition has been detected by the BMS. All alarm conditions detected by the BMS shall be annunciated as detailed elsewhere in this section.
- 4. Operator interfaces for the Operator Interface Workstation and the Portable Operator Terminal shall be the same.
- 5. Graphical user operator interface shall be "user friendly" using progressive disclosure

maps. The interface shall be such that there is no need for the operator to reference documentation other than "help" menus on the system in order for the operator to perform his normal duties after the training has been received, as detailed in these specifications.

E. Access to the BMS at the WS (WorkStation) and POT (Portable Operator Terminal) and any other I/O device shall be protected by a password based access system. An operator shall not be able to access information or perform any tasks at the BMS until a valid password has been entered. Access shall terminate when an operator signs off or after a predetermined time-out period, initially set at 10 minutes after the last operator access, whichever occurs first. The password shall not be echoed at any terminal when it is entered. Any operator functions, whether or not detailed within these specifications, shall be subject to the operator's password being of a sufficiently high level to enable the operator to perform the function.

F. If the system has fixed defined function levels then there shall be at least five (5) levels. Only the highest level shall be able to undertake changes to the passwords.

G. An operator with the highest level of password shall be able to make additions, deletions and changes to the passwords on-line using an interactive procedure including the changing of the time-out period. An operator with the highest level of password shall be able to obtain a report detailing the passwords assigned to each operator and all relevant details of the access privileges associated with each password.

H. An operator with the suitable level of access shall be able to access the network to perform the following BMS functions:

1. Observe values of BMS controlled points, BMS outputs, and BMS calculated points.
2. Issue commands to change the state of digital output or the value of analog output.
3. Observe and acknowledge BMS determined alarm conditions.
4. Generate trends and alarm reports.
5. Add, delete, and modify operating sequences.

I. Each monitored, controlled and calculated point in the system shall be accessed via the menu and the alphanumeric approaches as detailed in these specifications.

J. operator shall be able to place any output in the BMS manual control mode and when in that mode the operator shall be able to place any output to any required value subject to any hardwired interlocks and any software interlocks that are specified to apply in the BMS manual control mode.

K. operator shall be able to place any setpoint that is determined by software into the manual mode and to manually assign a value to the setpoint.

L. operator shall be able to initiate the restart of equipment following a fire alarm and following a power failure. Refer to the paragraphs titled "Equipment restart following a fire alarm" and "Equipment restart following a power failure".

M. Alarms shall be generated by BMS upon the occurrence of the following conditions

1. Failure of a PC, DDC controller, or any other BMS hardware components.
2. Failure of communications between nodes on any LAN or the BMS Management level Network.
3. A monitored status indicates a discrepancy between the actual and the required value.

4. A monitored value does not meet criteria established by the operator.
 5. The deviation of a variable from setpoint exceeds operator-established criteria.
 6. A digital input from equipment indicating alarm condition
 7. Software failure shall be diagnosed and annunciated by BMS.
- N. Alarms shall be automatically output at the alarm printer.
- O. Operator shall be able to designate one state of a digital input as an alarm state.
- P. Web based access to the BMS shall provide monitoring and limited control specific to the user. The information displayed on the web pages shall be refreshed at least once every 10 seconds.
- Q. BMS to be capable of all energy management functions specified herein to reduce energy consumption. System to provide energy supervisory management routine which is to improve energy efficiency of building HVAC systems. This routine to coordinate the execution of all energy management application features, acting as an executive software system for them, and as an interface for their control strategy with other application features in the system. Load control features shall be maintained as follow:
1. BMS to provide predefined control algorithms to accomplish peak demand control and power consumption reduction through Demand Limiting and Duty Cycle control application. Although the primary focus of these applications is towards electrical energy usage, the system to be capable of applying these predefined algorithms to other energy sources, such as steam, without software revisions.

2.4.2.3 Web Client

1. EMCS supplier shall provide an HTML5-based browser access to the AWS as part of standard installation. User must be able to access all displays of real-time data that are part of the AWS using a standard web browser. Web browser shall tie into the network through owner-supplied Ethernet network connection. The web client shall support a minimum of 200 users with a single license.
2. Browser shall be standard version of Microsoft Internet Explorer v10.0 or later, Firefox v19.0 or later, Chrome v24.0 or later, and Safari v7.1.1 or later. No special vendor-supplied software shall be needed on computers running browser. Data shall be displayed in real-time and update automatically without user interaction.
3. Web pages shall be automatically generated using HTML5 from the data display files that reside on the AWS. Any system that requires use of an HTML editor for generation of web pages shall not be considered.
4. Access through web client or thick client shall utilize the same hierarchical security scheme as the AWS. User shall be asked to log on once the client makes connection to the AWS. Once the user logs on, any and all changes that are made shall be tracked by the AWS. The user shall be able to change only those items he or she has authority to change. A user activity report shall show any and all activity of the users who have logged on to the system, regardless of whether those changes were made using a web client, thick client or through the AWS.
5. Shall provide User Session Management including the ability to view all connected user sessions to the web client, see how long they have been active/inactive for each unique session, and force log-out for any or all sessions.
6. Shall provide menu-style navigation access to primary features, i.e. alarm history, Data

Viewer, Search scheduled points and Zones, System Activity, User Session Management, and Top Display

7. Web client shall, at a minimum, support the following tablets:

a. Android platform:

1. Google Nexus
2. Samsung Galaxy Note

Apple platform

1. iPad
2. Apple iPad Mini

2.4.2.4 Field Engineering Tools

1. AWS shall include field engineering tools for programming all controllers supplied. All controllers shall be programmed using graphical tools that allow the user to connect function blocks on screen that provide sequencing of all control logic. Function blocks shall be represented by graphical displays that are easily identified and distinct from other types of blocks. Graphical programming that uses simple rectangles and squares is not acceptable.
2. User shall be able to select a graphical function block from menu and place on screen. Provide zoom in and zoom out capabilities. Function blocks shall be downloaded to controller without any reentry of data.
3. Programming tools shall include a real-time operation mode. Function blocks shall display real-time data and be animated to show status of data inputs and outputs when in real-time operation. Animation shall show change of status on logic devices and countdown of timer devices in graphical format.
4. Field engineering tools shall also include a database manager of applications that include logic files for controllers and associated graphics. Operator shall be able to select unit type, input/output configuration and other items that define unit to be controlled. Supply minimum of 250 applications as part of workstation software.
5. Field engineering tool shall include Device Manager for detection of devices connected anywhere on the BACnet network by scanning the entire network. This function shall display device instance, network identification, model number, and description of connected devices. It shall record and display software file loaded into each controller. A copy of each file shall be stored on the computer's hard drive. If needed, this file shall be downloaded to the appropriate controller using the mouse.
6. AWS shall automatically notify the user when a device that is not in the database is added to the network.
7. AWS shall include backup/restore function that will back up entire system to selected medium and then restore system from that medium. The system shall be capable of creating a backup for the purpose of instantiating a new client PC.
8. The system shall provide a means to scan, detect, interrogate, and edit third-party BACnet devices and BACnet objects within those devices.

2.4.3 Audit Trail

A transaction log of all user and system actions shall be provided with records. This file shall be

capable of being used in a circular mode i.e. overwriting the oldest data. It shall be possible to review individual user's actions per BMS terminal such as log on / off time, any points which have been manually over-ridden / placed to auto, created, edited, up/down loaded, file transfers, creation or amendment of log data files as a minimum. All background system activities such as controller file requests shall also be logged.

Audit trail shall be available in the ODBC option.

2.5 SCHEMATIC INTERFACE

- 2.5.1 A schematic event based interface shall be included and be an integral part of the standard operating software and included on all BMS terminals. Following the initial display of a particular schematic diagram, the status of plant and measured values will be obtained from the controller and displayed as appropriate. Subsequent refreshing of this data shall only occur either when the status of an item of plant changes (e.g.: on to off) or where the measured value changes by a pre-defined amount. The use of event based schematics ensures that, as the programmable controllers are updating the BMS terminal when they register an point value change being presented on the schematic.
- 2.5.2 It shall be possible to create schematic backgrounds or points utilising bitmaps with 24bit true colour resolution, these shall have a minimum resolution of 1024 x 768 pixels. Points shall be able to appear if a given plant condition exists.
- 2.5.3 It shall be possible to log on to the system via the schematic interface 2.5.4 It shall be possible to indicate current status or value of individual points in the system together with their location on the schematic diagram.
- 2.5.5 It shall be possible for the user to create new schematics or modify existing ones (subject to password authority) without reference to the BMS contractor or manufacturer.
- 2.5.6 Creation or modification of schematic diagrams shall be possible while the BMS terminal is online.
- 2.5.7 An integrated library of standard symbols representing fans, pumps, valves, etc. shall be provided for use when creating new schematics.
- 2.5.8 It shall be possible to allocate and display a schematic diagram at any and every level of the system structure, e.g. building, plant, zone and point.
- 2.5.9 It shall be possible to employ active areas of the schematic (hot spots) to permit a hierarchical or lateral linkage between diagrams without recourse to a subordinate or superior routine.
- 2.5.10 It shall be possible to manually override the state or value of an point or return it to its automatic state directly from the schematic diagram.
- 2.5.11 During normal operation it shall be possible to perform edit functions to the parameters to be modified, this shall include:
- Time schedules with graphical edit facility
 - Time extension of plant enabling up to 18 hours extension
 - Holiday schedules
 - Alarm high and low limits, transient time filters and hysteresis
 - Set Value and where applicable slope correction

- Control PID terms
- Optimiser occupancy, leaving and low limit set values

2.5.12 It shall be possible to automatically indicate the status of plant by a configured colour change of the symbol. In the case of analogue points it shall be possible to configure a colour change of the schematic as the measured value changes e.g.: pipe work to be shown changing in increments from dark blue to light red as the temperature increases.

2.5.13 It shall be possible to represent measured values as either symbols, which change according to the measured value, or text. Examples of a symbol representation are:

- A temperature might show as a thermometer with a graduated scale
- A water tank with water level rising and falling combined might be shown as with colour change to represent its temperature level.

Where text is used it shall utilize engineering values such as Temp 21.2°C.

2.5.14 A "zoom" facility shall be provided whereby a selected area of the schematic diagram may be enlarged to show greater detail.

2.5.15 Configurable, the schematic interface shall be automatically displayed following the power-up of the BMS Terminal.

2.5.16 Where the existing display is representative of only a part of the installed plant it shall be possible to enable a full composite display as a sub-display at the base of the screen for reference purposes.

2.6 TEXT INTERFACE

2.6.1 The BMS terminal shall provide a two-pane window with one pane showing the plant/point index and the other showing requested values.

2.6.2 The index pane shall be active to indicate the following by selection

- Points selected for logging
- Points currently in alarm
- Points selected for database alignment.

2.6.3 A toolbar shall be provided for regularly performed functions. A toolbar shall be selected for off/view to provide access to the following:

- Printing
- Saving information to disk
- Log in
- Log out
- System set-up
- Short form editing
- Logging set-up

- Historical log information
- Dynamic log information
- Transaction logging
- System diagnostics
- Schematic interface

2.6.4 A status bar shall be provided for regularly viewed information. The status bar shall be selected for off/view and be configurable per user to provide the following:

- Date, time, current username.
- Communications status.
- Mnemonic, controller number, controller Point number, Point type (e.g. AI, DI, DO, TIM, SPA, OPT etc.).
- Number of alarms in the file, total alarms, cleared, acknowledged, unacknowledged, delayed.

2.6.5 The layout of the system manger shall be configurable per user allowing a customized environment to meet individual user's needs. It shall be configurable for font type and size, window positions, status bar contents and both point and alarm review formats.

2.6.6 It shall be possible to carry out the following commands by use of dedicated function keys on the keyboard or by the mouse:

- Manual on - digital points
- Manual off - digital points
- Auto - analogue and digital points
- Temporary on - time schedules
- Temporary off - time schedules
- Set to value - analogue points
- Alarm review
- Point type review
- Point group review
- Plant review
- Hours run review
- Review cancel
- Schematic enable
- Printer on/off
- Log on

- Log off
 - Help
- 2.6.7 It shall be possible on a per user basis to create a number of points requiring manual control in a specific sequence with appropriate time delays into a single command routine. This routine shall be able to be initiated at any time by the operator and all events associated shall be recorded as part of the transaction logging function.
- 2.6.8 It shall be possible to address buildings, plant, zones, points etc., and using clear language descriptions. Each individual point may also be identified by a unique alphanumeric mnemonic. However, systems relying solely on mnemonic address entry shall not be accepted.
- 2.6.9 There shall be a minimum of four hierarchical levels of address available, typically:
- Level one shall identify the Building
 - Level two shall identify the Plant
 - Level three shall identify the Zone
 - Level four shall identify the Control Element (e.g. sensor)
- 2.6.10 Reviews shall be via drag and drop from within the pane allowing individual or a group of points to be displayed. It shall be possible to select an individual point type or selection of point types to be included within a review.
- 2.6.11 The review information display for each point type shall be configurable per user for font type and size and selection of information to be displayed, thus allowing the necessary level of information to suit the users requirement and shall include as a minimum the following:
- Date / time of review.
 - Mnemonic, controller number, controller point number, point type (e.g. AI, DI, DO, time, set-point etc.).
 - Short point title, long point title.
 - Primary value (e.g. Flow 1.5L/s), secondary value (e.g. totalised 2950 Litres).
 - Cross-referenced controlling point (reason for state).
 - Alarm and limits if applicable.
 - Totalised values and limits if applicable (e.g. hours run and maintenance levels).
- 2.6.12 In addition to the information above it shall be possible to provide clear language descriptions of each point extracted from the plant index. To avoid cluttering the screen only the last four most significant index sections shall be displayed.
- 2.6.13 It shall be possible to enter any of the subroutines such as the editor functions, alarm or logging function from the system manager without closing down the BMS terminal or the programs currently running.
- 2.6.14 It shall be possible to store separate display groups per user. These group reviews can be run on user request and shall provide a consistent review mechanism. It shall be possible to print or save to disk the contents of the review.

2.6.15 It shall be possible to add new points and reconfigure or modify existing points without recourse to the manufacturer and without taking any part of the system offline.

2.6.16 There shall be a facility to easily modify parameters from within the system manager via a graphical editing environment. This “short form” editing shall only present operational information for modification. As a minimum this shall include:

- Time schedules with graphical edit facility
- Time extension of plant
- Holiday schedules
- Alarm high and low limits, where multiple limits exist it shall be possible to edit all alarm transient time filters and hysteresis
- Set value and where applicable slope correction
- Control PID terms
- Optimizers occupancy, leaving and low limit set values

2.7 AUTOMATIC TIME CHANGE

It shall be possible to programme the system in advance to automatically modify both the terminal and controller time for time change or daylight saving mode.

It shall be possible on a daily basis to configure the system to synchronise the time from the main site server to a selection or all controllers on the site.

2.8 ALARMS

2.8.1 General

The BMS provides a vehicle for the visibility of plant failure, patterns, annunciation of alarms and the reporting function. In particular it shall provide the standard functions stated below:

- Alarms shall be generated under the following conditions:
- Digital input contact change from normal to alarm condition (either open or closed).
- Digital input condition either matches or mismatches another digital point type.
- Digital point has accumulated hours run tantalization limits.
- Analogue input has exceeded high or low limit alarm level.
- Analogue input tantalization has exceeded alarm level.
- Maximum demand alarm condition.

- Conditional based on a number of digital / analogue conditions being met.

2.8.2 All alarms shall have the ability to be filtered to eliminate “nuisance” alarms. Methods shall include:

- Transient inhibit timers to make allowance for fluttering contacts or slow internal responses
- inhibits timers checking that the condition has been active for a set time for both into alarm and alarm clear
- inhibits timers allowing for alarms which are cross referencing to other plant to stabilize

It shall be possible to enter a hysteresis value on analogue alarms.

It shall be possible to send alarms to a file at the BMS terminals without presenting them to the users until a predetermined time such as a shift change when all “delayed” alarms received will be presented to the users for appropriate action.

If alarm conditions are not cleared then these shall be retransmitted after a predetermined time configurable based on alarm priority. These shall be identified as a distinct alarm to represent their retransmission.

The annunciation of each alarm filter shall be configurable for both colour and text.

Alarm presentation shall be configurable per user, for font type and size and selection of information to be displayed for each point type thus allowing the necessary level of information to suit the user requirement. The alarm review format selection shall include as a minimum the following:

- Date / time of alarm
- Mnemonic, controller number, controller point number, point type (e.g. AI, DI, DO, time, set-point etc.).
- Short point title, long point title.
- Primary value (e.g. Flow 1.5 L/s), secondary value (e.g. totalised 2950 Litres).
- Alarm priority and limits if applicable.
- Totalised values and limits if applicable (e.g. hours run and maintenance levels).
- Name of the user who acknowledged the alarm, time of acknowledgement and BMS terminal number.
- Supporting text if entered at the time of acknowledgement.

The printout for alarms shall contain information as detailed above.

Alarm status shall be identified by a coloured icon representing unacknowledged alarms, acknowledged alarms, alarm clear and retransmitted alarms.

It shall be possible to extract information via filters from the alarm database for presentation in a management report for alarm activity. It shall be possible to filter for a given alarm type, start/end time and date, relative time, selection of either mnemonic, mask, controller number point number, etc. This shall present information in a dedicated window for analysis and printing. This window shall be live

with new alarm events that meet the filter requirements being displayed.

Alarms shall be able to initiate BMS terminal actions on specific alarm activity, such as high temperature alarm, Hours run alarm limit 1, etc. These BMS terminal actions shall include the ability to present a schematic associated with the alarm, a multimedia sound file being played, a Windows *.exe application being triggered and opening of documents.

Alarms shall automatically be printed on the BMS terminal printers allocated, or alternatively IT networks printer.

Where colour printers are utilized it shall be possible to print alarms in red and alarm clear in black.

It shall be possible to produce an extended text message to accompany the annunciation of any alarm. This shall provide further information about the alarm and any action required to be taken by the user or indicate that automatically programmed in the system. These messages shall contain multiple lines of text and automatically printed on a designated printer local to a BMS terminal. There shall be no practical limit to the length of messages created.

Acknowledgement of alarms shall be automatically printed and will indicate the time, date, and any message generated by the user.

It shall be possible to allocate alarm priorities into single or group acknowledgements allowing high priority alarms to be individually acknowledged by the user or several low priority alarms to be acknowledged in one action. It shall be possible to acknowledge a selection of the new alarms received.

2.8.2 Alarm Banner

Alarms shall be displayed via an alarm banner present across all system screens to provide instant visibility of system wide alarms and access to individual filters. This banner shall be configurable for always displayed or if hidden the banner shall be shown when an alarm event occurs.

The alarm status banner shall provide a display filters each detailing the current active alarm information and alarm Acknowledgement State per filter. It shall be possible by selecting the active alarm total to review that points are in alarm for the specific filter.

The alarm banner shall enable each filter to be customized by configurable alarm filter text and bell colour allowing definition such as Critical Alarms, General Alarms and Low level alarms.

The alarm banner shall show the alarm status per each filter with the total number of points currently in alarm as received at the BMS terminal, and acknowledgement state e.g. grey - no alarms, flashing grey / colour - unacknowledged alarms, colour stable - alarm condition present and acknowledged. It shall be possible by clicking on the totals active alarm button to display a summary of points currently in alarm for that filter.

2.9 DATA LOGGING

2.9.1 General

Data logging shall be available in the following forms:

- Controller/Server Based: Logging sets which are specified at the BMS terminal are loaded into multiple controllers for operation and whose data is automatically sent to selected servers.
- Real Time Displayed: Logging sets which are specified at the BMS terminal are loaded into

multiple controllers for operation and whose data is automatically displayed at selected BMS terminals.

- Controller: A mechanism in every programmable controller that automatically logs every point on a regular basis. This circular log is viewed on request from any remote or mobile user interface

2.10 CONTROL FUNCTIONS

2.10.1 General

The energy management software shall provide as a minimum the following functions:

2.10.2 Time Control Modes

There shall be no limit to the number of time schedules each of which can have any point or number of points assigned.

Each time programme shall operate over a seven-day week and allow for on/off periods per time programme.

There shall be a facility to cross reference to a core time schedule via a relative schedule allowing an offset of +/- time in minutes based on the core time schedule on/off periods.

A calendar schedule shall allow advance programming defining which time or relative schedules are to be utilized for the advance dates to provide a variable switching pattern over a calendar period.

There shall be no limit to the number of holiday schedules. Any time schedule or number of time schedules may be assigned to any holiday schedule.

It shall be possible to programme holiday's 365 days in advance.

2.10.3 Optimum start/stop control

There shall be no limit to the number of optimum start schedules. Any point can be assigned to any optimizer.

The system shall start the plant at the latest possible time to achieve the desired internal temperature by target time. Configurable, one of two algorithms shall be used:

- by use of a linear algorithm monitoring outdoor air and space temperatures
- by use of a logarithmic algorithm relating to inside temperature only Similarly the system shall stop the plant at the earliest possible time before the end of the occupancy without ambient conditions deteriorating below pre-set acceptable limits.

This shall operate in heating or cooling mode or a combination of both.

The optimizer programme shall be self-adjusting, the amount of adjustment being user defined between 0% and 100%.

The user shall have the option of defining the termination of plant run-up or boost on target temperature being achieved, target time being reached, or whichever occurs first.

The following reports shall be available for selection by the system user as required for each

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schedule:-

- Report at commencement of run up.
- Report at termination of run up.
- Report at commencement of run down.
- Report at leaving time
- Report at premature run down
- Report of low temperature stop/start
- Report at start of occupancy.

2.10.4 Anti-condensation/frost protection control

During plant off times the space and outdoor temperature shall be monitored continuously. When the temperature falls to a pre-set low limit, the plant shall start to prevent formation of condensation or water freezing.

It shall be possible to set the differential between plant on and plant off temperature under these conditions to minimize plant cycling.

2.10.5 Enthalpy control mode

For air-conditioned plant it shall be possible to monitor both fresh air enthalpy and extract air enthalpy.

It shall be possible for the comparison of fresh air and return air enthalpy to be used to set damper control mode for minimum energy usage.

2.10.6 Night Purge

When plant is operating in a cooling mode the system shall monitor both the external and internal temperatures (including humidity). Providing the external temperature is less than the desired and actual internal temperatures at a pre-set time, the plant shall operate in a full fresh air mode for fixed time duration. This is in order to “flush” the building with cooler fresh air, thus reducing chiller load at occupancy start and providing fresh air into the building.

2.10.7 Load Cycling

The system shall be capable of cycling loads to reduce electrical consumption. The programme shall have a user defined cycle period and maximum off time. A temperature detector shall be assigned to each plant on the load cycling programme. Should the space temperature exceed pre-set limits the load cycling shall be overridden, and the plant returned to the time control programme until the space temperature has returned to an acceptable level.

2.11 USER PROGRAMMABLE SOFTWARE

The system shall have a flexible software package to allow a user with minimal knowledge of software programming to construct unique programmes for plant control and management information.

The package shall provide, but not be limited to, the following functions:

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- Auto changeover of pumps, fans etc. on time programme, hours run or event (pump trip etc.).
- Damper enthalpy override
- Degree day calculations
- Boiler and fridge efficiency calculations
- Time delays and sequenced interlocks
- Cost savings calculations

Full arithmetic operators shall be available for use in the programmes as required e.g.: +, -, /, *, ().

Programmes shall permit the use of comparison statements such as: =, >, <, =< etc. Programmes shall permit logical operators to be used such as: NOT, AND, OR, XOR AND MASK.

Direct reference to any point shall be available to obtain its current value. Such references shall be using standard language such as, ON, OFF, OPEN, CLOSE, MANUAL, SETPOINT, EXPIRED TIME, PRIORITY, ALARM - e.g.: the programme shall support conditional statements (IF...THEN...ELSE) and branches (GOTO).

It shall be possible to refer to secondary values of points such as in optimisers to determine which mode they are currently operating e.g.: run-up, occupancy, frost, run-down etc.

It shall be possible to refer to time and calendar functions directly such that DATE, TIME, HOUR, MINUTE functions may be used.

It shall be possible to set timers for example so that a timed delay may be produced before an action is carried out.

2.12 PID CONTROL

The system shall be capable of performing proportional control, integral control, derivative control or any combination of the three utilizing direct digital control techniques.

The primary input to a control loop shall be either a measured value (°C, %RH, etc.) or a calculated value).

The secondary or reset input shall be either a measured value (°C, or %RH etc.) or a calculated value. The secondary reset input characteristic may be a 3-step segment function. It shall be possible to view this in a graphical format and modify characteristics by mouse control.

Additional sun or wind influence may be input.

The set-point of any control loop shall be capable of being adjusted by time, event or as a result of a calculation.

It shall be possible to reference a number of control loops to the same primary input point to achieve multi-stage control.

Where a step-controlled output is utilized, it shall be possible to rotate the order in which the steps are switched by time, or as the result of a calculation.

Valve and damper control shall be via 0-10 Vdc. Control loop set-up shall be via a graphical tool to

enable sampling test and trend display.

2.13 AUTO POWER UP

The system shall sequence the reinstatement of plant after a power failure. When a standby generator is in operation, the system shall limit the number of plants in operation according to the load on the generator.

It shall be possible to configure a fixed limit in addition to the floating limit determined by generator capacity.

It shall be possible to monitor any overload condition of the standby generator so that if an overload is detected reinstatement of the loads shall be prevented and loads switched off until the condition is cleared.

2.14 INTEGRATION AND INTERFACING WITH OTHER SERVICES

The system shall be capable of integration with other building services level by simple hardware (electrical) methods or by the use of a computer-based communications protocol.

Where a communications protocol is used, integration controllers shall provide this facility and shall be capable of integrating with systems such as fire, security, lifts, lights, etc. The integrity of the individual systems shall be maintained and the failure of one system will not adversely effect the operation of the others.

The integration controller shall provide a means of a consistent user interface for presentation of third-party information at each of the BMS terminals.

Cross-referencing points of third-party systems shall be required such that the points created may have similar attributes as the points in main system and used in the same way as other BMS points to create intelligent control strategies.

Thus it shall be possible for the integration controller points to be used by the system as a whole for control and monitoring purposes as well as for the annunciation of alarms, logging, display, maintenance etc.

The integration controller shall have a self-analysis feature with in-built diagnostics and shall transmit any malfunction messages to the designated BMS terminal.

As a minimum each integration controller shall provide a port for connection to the LAN.

Implementation of an integration controller shall be carried out in software and hardware. Hardware that is dedicated to this function is preferred and shall be capable of being connected directly to the main BMS LAN. Integration may equally be incorporated direct into a controller if required.

Alternatively the use of a PC which is running integration controller software shall be used.

This shall utilise standard developer's tools for mapping of parameters from systems providing OPC (OLE for process control) and DDE/NetDDE.

2.15 OPEN SYSTEMS

2.15.1 General

A fundamental requirement is that an open system architecture shall be provided.

The system offered shall be flexible and incorporate multiple proprietary systems / protocols such as

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Bacnet, Lontalk, LonWorks®, LonMark®, Modbus, EIB etc. with full interoperability.

Interoperability is defined that controllers from proprietary system “A” can be connected to the system architecture where there may be controllers from proprietary system “B”. Values from either system can be equally logged, alarmed, displayed on schematics, audit trailed for changes and interact to provide the most economical building control strategy. The open system shall be the middleware to achieve this integration.

The system shall be able to utilise different protocols and architectures to maximize the media infrastructure.

The system offered shall ensure that different vendors shall be able to equally tender competitively, add and amend or change the system for future new sections.

2.16 OPC

As an option, an interface shall be available which supports OPC (OLE for process control) for presenting values from other OPC compliant systems on the BMS schematic interface.

2.17 ODBC COMPLIANT DATABASE

As an option, an ODBC-compliant database shall be provided. It shall be possible to extract logging data records from this for open access for use by database packages such as Access, Oracle etc.

It shall be possible to use SQL to extract data from the ODBC-compliant database. Alarms, logged data and audit trail shall be available with this option

2.18 LONWORKS®

The BMS system shall have the ability to integrate with other systems that conform to LonWorks® networks and LonMark® equipment.

The integration shall be achieved by the use of an integration controller connected at either the backbone network or a controller network.

The connection to the LON devices shall utilize FTT-10A Network Transceivers complying with the requirements of EIA 790.3 Free Topology twisted pair channel specification (78Kbps) supporting star, home run, multi-drop and loop wiring technologies.

The integration shall provide 2 way communications between the connected systems and it shall be possible to access network variables for display at the BMS terminal.

2.19 EIB

The BMS system shall have the ability to integrate with other systems that conform to EIB. EIB is a system for Home & Building Automation based on a de-centralized, peer-to-peer, OSI-compliant network operating system. It uses a serial transmission protocol.

The integration shall be achieved by the use of an integration controller connected at either the backbone network or a controller network.

The connection shall be over either twisted pair (9,600 bps). The engineering shall be performed using ETS –EIB Tool Software.

The integration shall provide 2 way communications between the connected systems and it shall be possible to interface not less than 500 points per integration controller for display at the BMS terminal.

2.20 BACNET

The BMS system shall have the ability to integrate with other Building Control and Automation systems that conform to ASHRAE's BACnet communications standard "ANSI/ASHRAE 135-2016". The integration shall be achieved by the use of an integration controller connected at either the backbone network or a controller network.

The connection shall be over either Ethernet conforming to ISO standard IEEE 802.3

The integration shall provide 2 way communications between the connected systems and it shall be possible to transfer not less than 800 points for display at the BMS terminal.

A minimum of "ANSI/ASHRAE 135-2016" BACnet conformance class 3 shall be included.

2.21 ELECTRICAL AND MECHANICAL CONTROL DEVICES

Chilled Water Automatic control valves 2" and smaller shall be screwed type, of red brass material and valves 2½" and larger shall be cast iron type and flanged. Valves shall be ANSI-rated to withstand the pressures and temperatures encountered. Valves shall have stainless-steel stems and spring loaded Teflon packaging with replaceable discs. All valves shall have the manufacturers name/logo embossed on the valve.

2.21.1 Chilled Water Valves shall be sized for a pressure drop greater than or equal to the coil they serve.

2.21.2 Chilled Water valves shall be PICV type as specified in the sequence of operation. Stems shall be polished stainless-steel suitable temperature range of -10 to 120°C.

2.21.3 All automatically controlled devices, shall be provided with electric / electronic actuators sized to operate their appropriate loads with sufficient reserve power to provide smooth modulating action or two-position action and tight close-off.

2.21.4 Actuators shall be provided with suitable corrosion resistant linkages for valves or dampers. Except as specified herein, all actuators shall be sized for the load/close off encountered in strict accordance with manufacturer's recommendations. All actuators on outside air dampers, relief air dampers, and convertor steam valves shall be heavy duty type with oil immersed gear train. All actuators shall drive to their "normal" position anytime their associated AHU (etc.) is shut down.

2.21.5 Direct coupled modulating actuators shall be 24 volt type with 2 to 10 VDC control signal. Direct coupled on/off actuators shall be 24 VAC. The torque and the close off ratings shall match the application used.

2.21.6 Fire stats shall be 135 degrees manual reset type.

2.21.7 Duct Smoke Detectors shall contain an air sampling chamber with sampling tubes extending through the width of the air duct. Alarm status indicating lights shall be visible on the front of the detector.

2.21.8 DDC/FCU Controller

Controller shall consist of a one piece DDC controller complete with microprocessor, power supply, self-calibrating pressure transducer, and integral actuator. Provide DDC/FCU controller factory fit and calibrated for the minimum and maximum air flows scheduled on the drawings.

Controller housing shall be UL 94-5 5V or equivalent listed material for use in ceiling plenum applications without need of special metal enclosure. Controller shall work over an ambient temperature operating range of 0-55C (32-122F), 5- 95% non-condensing humidity. Power supply to unit shall be 20-30 VAC, 50 Hz, maximum power consumption of 5VA.

For reheat applications, provide factory fitted power relays rated for coils as required. Mount relay in electric control enclosure and wire to DDC/FCU controller.

Controller shall be UL 916 listed, CE mark compliant, and conform to IEC 801.4 transient and surge immunity standards. UL and CE marks shall appear on product labels.

Unit shall operate properly over inlet velocity pressure range of 0.00 to 2.00 in WC (0-500 Pa). End to end accuracy shall be +/- 5% of flow measured at 250 Pa inlet velocity pressure.

Cooling only model DDC/FCU controller shall be used in cooling only applications. Cooling and reheat model DDC/FCU controller shall be used in reheat applications and contain 3 Form A relays, pilot duty rated for control of floating proportional, time proportional or staged reheat valves and coils.

Integral damper actuator shall be 53 lb-in (6 N-m) torque, over the shaft mounting type. Stroke shall be fully adjustable from 0-90 degrees rotations with visible position indicator. Actuator shall be equipped with manual position override. The actuator stroke time shall be 3.6 minutes maximum for 90 degree rotation at 50 Hz operation.

Room temperature sensor shall be complete with digital LCD display. Temperature sensor shall permit local adjustment of temperature setpoint, minimum flow, and maximum flow.

Temperature sensor shall permit display of any of the four variables such as room temperature, room temperature set point, air flow and maximum flow setpoint.

The VAV room digital thermostat shall also display the following:

Room space temperature (°C or in °F freely configurable)

Room temperature setpoint (°C or in °F freely configurable)

Maximum flow

Minimum flow

The room temperature sensor shall be wired to the DDC/FCU controller using a two wire non polarity sensitive cable.

The room thermostat shall have plug-in facility to a portable operator terminal to get the following additional operation such as:

It shall be possible to modify the FCU controller parameters from the room thermostat key buttons without the use of any additional devices.

The controller should have the option of working along with a duct thermostat and any dry contacts.

2.22 ELECTRONIC DATA INPUTS AND OUTPUTS

2.22.1 General

Input/output sensors and devices shall be closely matched to the requirements of the remote panel for accurate, responsive, noise-free signal input/output. Control input response shall be high sensitivity and matched to the loop gain requirements for precise and responsive control. In no case shall computer inputs be derived from pneumatic sensors or thermocouple.

Temperature sensors shall be Resistance Temperature Detector (RTD) type of 100, 1000, or 3,000

ohm platinum, or 20,000 ohm.

Space temperature sensors shall be in attractive housing made from flame retardant ABS.

Duct temperature sensors shall be rigid stem or averaging type as specified in the sequence of operation. The terminal box shall be IP54 and of flame retardant plastic.

Water sensors shall be provided with a separable copper, monel or stainless-steel well. The sensor shall have an accuracy of class B as per DIN IEC 751.

Outside air, return air, discharge air, space and well sensors shall have + or - 1.0 degrees F accuracy between 32 degrees and 212 degrees.

Relative humidity sensors shall be capacitance type with 10% to 90% range. Duct mounted humidity sensors shall be provided with a sampling chamber. Wall mounted sensors shall be provided with decorative covers identical to temperature sensors.

Differential and Static Pressure Sensors and Switches:

Fan proof-of-flow switches shall be of the adjustable set point and differential pressure type. Switches shall be piped to fan discharge except where fans operate at less than one inch water column (WC), they shall be piped across the fan. For fractional horsepower and non-ducted fans, relays or auxiliary contacts may be used. Maximum pressure rating shall be at least 10 inches water column.

Pump proof-of-flow switches shall be of the adjustable differential pressure or flow type as specified in the sequence of operation or data point summary. Devices shall be 150 psi rated except chilled water flow switches shall be provided with totally sealed vapor tight switch enclosure on 300 psi body. Differential pressure switches shall have valved manifold for servicing.

Air flow and duct static pressure analog sensors shall be high accuracy suitable for the low pressures to be encountered.

Water flow analog sensors shall be IP66 magnetic type, cast steel up to 25 mm and Aluminum alloy for sizes 40mm and above. Operational accuracy shall be $\pm 5\%$. It shall be suitable for 240 VAC with a current output of 4-20mA. The electrodes shall be detachable for easy cleaning; it should not require the removal of the flow meter. The flow meter shall have a 3.5 segment 7 LED indicator.

Control relays and analog output transducers shall be compatible with PCP output signals. Relays shall be suitable for the loads encountered.

2.22.2 BMS NETWORK

BMS network shall be established using the Structured Cabling System. This shall be supplied by the specialist structured cabling system vendor, as part of the MEP contractor's scope of work. This shall be Level 4 type or higher as per latest standards for LON systems and Cat – 6, UTP for BACnet systems. Cables, F.O. Cables, Active Components, Termination, etc shall be provided for a high speed network connection.

BMS submittal without the design of BMS Structured Cabling System shall not be acceptable. Cable selections shall meet individual requirements as set out by the protocols.

2.23 SEQUENCE OF OPERATION

2.23.1 CHILLED WATER SYSTEM

The chilled water system comprises of District cooling (DC) heat exchangers with secondary CHW

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pumping system.

Tertiary circuits shall have heat exchangers and chilled water pumps as shown on drawings.

A. Plate Heat Exchangers

The plate heat exchangers and primary circuit in the ETS room shall be part of District Cooling provider scope. However, the heat exchangers at the mechanical floor levels shall be part of contractor's scope. The controls connections shall be as shown on BMS diagram/ CHW riser diagram.

The heat exchangers at the ETS room shall be controlled by the District Cooling Provider. BTU meter shall be installed at the main return chilled water pipe. Chilled water supply and return headers shall be provided with immersion temperature sensors and these temperatures shall be monitored by BMS. The heat exchangers at the mechanical floor levels shall be controlled by BMS. BTU meter shall be installed at the main return chilled water pipe for each circuit.

B. Chilled Water Secondary& Tertiary Pumps

The chilled water pumps shall be running in duty/standby mode as indicated on the drawings. Each of these pumps shall come complete with individual VFDs. Each duty pump shall be able to handle chilled water flow required for the 1 no. duty heat exchanger.

The BMS shall monitor all pressure differential at the index point based on the hydraulically longest circuit as indicated on the drawings. Based on the index pressure set point and the actual differential pressure at the index, the BMS shall provide the appropriate output to the secondary pump VFDs. The BMS shall also monitor the flow rate of the secondary system and use this value to maintain the minimum flow required in the system.

C. Proposed Peak Load and Off Peak strategy

1. Peak Load

The system shall attempt to maintain design conditions with the use of duty pump. The duty pump shall be enabled to run at the minimum speed as permitted by the VFD or 40% whichever is lower. If after an adjustable/preset time limit of 5 minutes, the system pressure remains below the set point, the duty pump speed shall be ramped up to maintain the differential pressure at the index point.

The speed of the duty pump shall be increased or decreased based on the differential pressure at the index and the secondary chilled water flow to ensure that the minimum required flow is maintained in the system.

2. Off Peak/Low Demand

During off peak and low demand conditions, the smaller pumps shall be used to meet the load requirements.

The system shall attempt to maintain design conditions with the use of first pump. If after an adjustable/preset time limit of 5 minutes, the system pressure remains below the set point with one pump running, the BMS shall start the second pump at low speed. The speed of the second pump shall continue to increase till the pump is operating at 50% capacity. The speed of the lead pump shall be simultaneously reduced to 50%.

The BMS shall then increase or decrease the speed of both pumps simultaneously to maintain the system pressure set point. The BMS shall continuously monitor the secondary system flow rate and

shall stop the large pump when the system flow rate or pump speed falls below 40% of the total flowrate of the combined pumps. The BMS shall then increase the speed of the lead pump to 100%. The speed of the lead pump shall be increased or decreased to maintain system pressure.

D. Run Time/Equal Run Hours (Applicable for Peak and Off Peak Demand)

The BMS shall continuously monitor the run time of each pump. The BMS shall rotate each pump to be the lead pump after every 200 hours of run time. The BMS shall select the next pump in the sequence as the lag pump.

If one of the pumps should fail then, the lag pump shall become the lead pump and the idle/standby pump shall become the lag pump.

E. Pumps Control and Other Monitoring points

The pumps starter panel shall be provided with a 'Hand-Off Auto' for each pump. BMS shall monitor the 'Auto' status for the pumps from the starter panel via volt free contact. The BMS shall control the pumps only when the 'Hand-Off Auto' switch is in 'Auto' position. The BMS shall enable the VFD for the pumps in 'Auto' position when the demand is established. The BMS shall control the VFD via 0-10v output from the respective DDC controller. In addition, the BMS shall also monitor the VFD speed feedback, VFD trip and pump run status. For all other detailed points refer to the data point schedule.

F. Delta T control:

In order to avoid low Delta T in chilled water system the following shall be considered by the contractor:

1- DPS (Differential pressure sensors) shall be located at the index circuit(s) with PID controller (proportional integral derivative) set point to be measured at site during T&C after chilled water balancing will keep all pumps run at 50HZ full load and open all PICVs of FCUs, AHUs & FAHUs in order to get proper set point.

2- TS (Temperature sensor) with PID controller shall be installed at the main return chilled water line coming from building and connected with chilled water VFD pumps to reduce the speed by 60% (30 Hz) once the temperature is <14 C Degree (in case of no demand).

3- Both DPS & TS with PID controller shall work together through BMS in normal condition but the DPS will override TS in partial demand and both shall be connected to control the pumps speed (via VFD) in order to achieve the design delta T and match the primary flow of the heat exchangers.

4- Chilled water pumps shall be installed before HEX (heat exchanger) suction from building and discharge to HEX in order to maintain supply Temp 5.5 C Degree.

5- All bypass lines to be canceled.

6- T&C for PICVs of FCUs, AHUs and FAHUs to be tested by the specialist supplier to make sure that actuator (0-10 V) is closed totally once achieve the set point temperature in order to avoid any bypass through the valves.

8- Chilled water VFD pump performance curve at 30 Hz in case of (partial demand / achieved demand). Head shall maintain the index circuit (farthest point) in order to achieve the design delta T and match the primary flow of the heat exchangers.

9- Using return chilled water lines to supply HEX serving domestic water tanks and pools cooling HEX with circulating pump.

2.23.2 PRESSURISATION UNIT

This system operates under its own dedicated controls; hence the operating sequence of this system is not covered under this document.

BMS monitors the pump run status, common fault status and panel healthy status through volt-free contact to DDC from the panel.

ALARMS:

Alarms will be generated on the BMS in case of following conditions:
Common fault

2.23.3 CHEMICAL DOSING UNIT

This system operates under its own dedicated controls; hence the operating sequence of this system is not covered under this document.

BMS monitors the pump run status, pump 'Auto' and pump trip status through volt-free contact to DDC from the panel.

ALARMS:

Alarms will be generated on the BMS in case of following conditions:

- Trip

2.23.4 PLATE HEAT EXCHANGERS (Where Applicable):

For each Heat Exchanger, there shall be a modulating PICV valve in supply line of the primary side of the heat exchanger.

HEAT EXCHANGER PRIMARY SIDE MODULATING VALVE CONTROL:

The temperature sensor shall be mounted in the supply line of the secondary side of the heat exchanger. The sensor relays the temperature signal to the DDC controller. In case the supply temperature is higher than the pre-defined set point the DDC will give a modulating signal to valve actuator in the primary side of the heat exchanger to open.

BMS shall also monitor the temperatures and pressures of the return line on the secondary side of the Heat exchanger unit as shown in the control schematic.

ALARMS:

Alarm will be generated on the BMS in case of high supply side temperature:

An alarm shall be initiated at the BMS for any of the following events:

- Programmable supply chilled water temperature high limit reached

- Programmable supply chilled water temperature low limit reached
- Programmable return chilled water temperature high limit reached
- Programmable return chilled water temperature low limit reached

2.23.5 FRESH AIR HANDLING UNITS (FAHUs)

2.23.5.1 PLANT CONFIGURATION

The Fresh Air Handling Unit shall comprise of an AHU supply air fan and extract fan provided with constant speed motor, CHW cooling coil with 2 way modulating PICV valve, Heat Recovery Wheels, 2 banks of filters (panel & bag filters) and 1 bank of filter on exhaust side, temperature & RH sensors.

2.23.5.2 AHU SUPPLY AND EXHAUST FAN MONITORING AND START/STOP

The motor control center serving the 100 % fresh air AHU and the extract fan shall have a "HAND-OFF-AUTO" switch. It shall be possible to start the FAHU and the extract fan directly when the switch positions are in manual. When this switch is in "AUTO" position, the DDC controller takes charge. For this purpose, the "AUTO" position of the switch is monitored.

The air flow switches AFS are mounted across the FAHU supply & extract fan to detect airflow. Each airflow switch is connected to a digital input of the DDC controller.

The FAHU fans and control status shall be controlled/ monitored on BMS as shown on the control schematics.

2.23.5.3 TEMPERATURE CONTROL AND HUMIDITY MONITORING

Temperature sensor shall be located in the supply air duct. The supply air temperature shall be maintained precisely at the desired set point via the 2 way modulating PICV valve located on the chilled water coil.

Temperature sensor shall be located in the return air duct and the return air temperature shall be monitored by BMS.

Temperature and RH sensors are also provided before and after the heat wheels for monitoring purpose.

2.23.5.4 DIRTY FILTER STATUS

For the FAHU, there shall be one differential pressure switch DPS, one across each filter bank. If the air pressure exceeds the setting on the differential pressure switch, a signal is relayed to the DDC, which shall give a dirty filter alarm.

2.23.5.5 OTHER TEMPERATURE / RH MONITORING POINTS

An outside air combined temperature and RH sensor is located in a shaded area that will monitor the ambient temperature and relative humidity.

A temperature sensor is located at the discharge of the enthalpy heat recovery wheels to monitor the coil entry temperature.

Pressure sensors shall be located on the supply air duct and return air duct to monitor the respective duct pressures.

2.23.5.6 FIRE MODE OPERATION:

In case of Fire alarm signal the system will be automatically shut down from the Fire Alarm system.

2.23.5.7 MOTORIZED DAMPER

The FAHU shall be provided with on/off motorized damper on the intake air duct which will be normally open in case of unit operation and closes in case of unit is off.

The BMS shall control the operation of the damper and monitor the status of the damper.

2.23.6 Recirculating Air Handling Units (Where Applicable)

This AHU Unit shall comprise of Supply fan, CHW cooling coil with two-way modulating PICV Control Valve & Filter banks (panel and bag filters as applicable).

Temperature Control

The BMS shall take over the operations of the AHUs when the Hand-Off-Auto switch is in the 'AUTO' position. The AHUs shall be switched on automatically as per the dictates of the time channels defined in the DDC controllers. Under normal conditions the AHUs shall run in the 'AUTO' mode. The AHUs shall run in the "hand" mode only for testing purposes.

The Temperature sensor shall be located in the return air duct. The Temperatures shall be maintained precisely at the desired set point by Modulating the 2 way PICV control valve located on the chilled water coil.

Dirty Filter Status

A Differential pressure switch is located across the filter. In case the filter gets clogged beyond the value set in the differential pressure switch, a contact is made which the DDC controller recognizes. The DDC controller raises an alarm after a pre-determined time delay.

The 'overload trip' condition of the motor is monitored at the DDC controller via a volt-free contact from the Motor Control Centre.

AHU Fans Monitoring and Start/Stop

The motor control centre serving the AHU shall have the "HAND-OFF-AUTO" switches. The airflow switch AFS is mounted across the Supply fan to detect airflow. The airflow switch is connected to a digital input of the DDC controller.

When the Fire Alarm Panel relays a Fire Alarm to the DDC Panel the DDC Controller will shut down the AHU.

For other points, which are to be monitored by the BMS, please refer to the Data Points Schedule and control schematics.

2.23.7 NOT USED.**2.23.8 BTU & WATER FLOW METERS**

- a. The BTU energy meter shall be comprise of an Ultrasonic flow meter, two temperature sensors, energy calculator, and serial communication ports. Each Meter shall be equipped with M-bus module that transmits data through M-bus network back to the central M-bus master, on which all meters data is retrieved and interfaced with BMS for reading of all data.
- b. The water flow digital meter shall be equipped with an internal data logger which can read out via IRDA or RS232 interface with Modbus
All the data shall be interfaced with BMS.

2.23.9 FAN COIL UNITS THERMOSTAT (FOR APARTMENTS):

The Fan Coil Unit shall have standalone control.

The Room Control Unit type shall have a built-in temperature sensor, set point potentiometer and fan 3-speed auto/manual selection with digital display. The room unit relays the actual room temperature and set point to the Controller, which accordingly modulates the 2 way chilled water PICV valves to achieve the desired room temperature. In the meantime, the controller varies the speed with temperature (when selection is in auto mode).

2.23.10 FAN COIL UNITS IN PUBLIC AREAS

FCU controllers for public areas and service / utility rooms shall be connected to the BMS workstation for monitoring and control of FCU's via Return air sensor behind the FCU.

2.23.11 DX SPLIT UNITS (Where Applicable)

The BMS shall monitor the DX split units for the unit panel healthy monitoring.

2.23.12 SMOKE EXHAUST / MAKE UP AIR FANS (CORRIDORS)

The smoke fans and make up air fans shall be directly controlled by the fire alarm panel. The fans shall be directly interlocked with the fire alarm panel. There shall be a motorized damper on the exhaust duct connecting the smoke exhaust fans and on the makeup air duct to each floor.

The fans shall operate directly on receipt of fire alarm signal. However, the fans run status is monitored on BMS through a digital input signal to DDC from differential pressure air flow switch mounted across each fan.

The smoke/ makeup air fans shall have a dedicated duct risers and branch ducts connected to each floor. The branch ducts shall have motorized on/off dampers. The dampers shall open at the fire floor based on a fire alarm signal. Motorized dampers on all other floors without fire shall remain closed.

ALARMS:

Alarms will be generated on the BMS in case of following conditions:

- Fan trip
- Fire Alarm

2.23.13 STAIRCASE/ LIFT PRESSURIZATION FANS:

AUTO MODE – FAN MONITORING & START/STOP:

Each staircase/Lift pressurization fan is directly controlled by Fire Alarm System. However, override facility is available to enable or disable the fan from BMS. In the event of smoke or fire detected by Fire Alarm System, first the fresh air motorized damper will be opened and the fan will be started by the Fire Alarm System. Fire Alarm signal will be reported to DDC.

In case the staircase/Lift shaft pressurization fan is to be controlled by dedicated BMS controllers for the VFD feedback to the fan for the ramp down, the following sequence shall be followed. The fan to be primarily operated by the FA signal sent to the MCC panels.

Pressure relief damper shall be provided at the supply duct to staircase to relief the excess pressure in the staircase beyond 50 Pa (limit to be adjustable). Refer section 07 for further details on the staircase pressurization fans to be used and their configuration.

The motor control center serving the fan has "HAND-OFF-AUTO" switches. It is possible to start the fan directly when the switch position is in "HAND". When this switch is in "AUTO" position, the DDC controller takes charge. For this purpose, the "AUTO" position of the switch is monitored.

ALARMS:

Alarms will be generated on the BMS in case of following conditions:

Staircase/Lift differential pressure exceeds pre-defined high limit of 65 Pa (adjustable)

2.23.14 GENERAL EXTRACT/ MAKEUP AIR FANS / JET FANS**AUTO MODE – FAN MONITORING & START/STOP**

The motor control center serving the fans has a "HAND-OFF-AUTO" switch. It is possible to start the fans directly when the switch position is in "HAND". When this switch is in "AUTO" position, the DDC controller takes charge. For this purpose, the "AUTO" position of the switch is monitored. The fans will be started automatically when the "HAND-OFF-AUTO" switch is in "AUTO" position and the time channel is activated.

A differential pressure air flow switch is connected across the fan, which proves the air flow status. The air flow switch is connected to digital input of the DDC controller. When the fan is commanded to switch-on and no air flow is detected across the fan, after a delay of 60 secs (adjustable) air flow fail alarm will be generated by the BMS and the fan will be switched-off.

ALARMS:

Alarms will be generated on the BMS in case of following conditions:

- No air flow detected across the fans in the event of fans being commanded to switch-on (Fan will be commanded to shutdown)
- Fan is not running in Auto Mode

2.23.15 WATER PUMPS**AUTO MODE – PUMP MONITORING & START/STOP:**

The motor control center serving the pump has a "HAND-OFF-AUTO" switch. It is possible to start the pump directly when the switch position is in "HAND". When this switch is in "AUTO" position, the DDC controller takes charge. For this purpose, the "AUTO" position of the switch is monitored. The pump will be started automatically when the "HAND-OFF-AUTO" switch is in "AUTO" position and the

time channel is activated.

BMS monitors the HOA and the run status through volt-free contact to DDC from the panel and the supply header pressure via water pressure sensor.

ALARMS:

Alarms will be generated on the BMS in case of following conditions:

- VFD (where applicable) / Pump Trip
- Supply header high pressure

2.23.16 FIRE PUMPS & FUEL TANKS

This system operates under its own dedicated controls; hence the operating sequence of this system is not covered under this document.

BMS monitors the pump HOA and the run status through volt-free signals to DDC from each pump panel and the supply header pressure through 0 – 10V signal from the pressure sensor.

ALARMS:

Alarms will be generated on the BMS for each Fire Pumps in case of following conditions:

- Pump Trip Alarm
- Fuel tank high level alarm
- Fuel tank low level alarm

2.23.17 WATER TANKS (Potable Water, Fire Water, Irrigation water)

BMS monitors the Tank Level and the motorized valve status through an analogue 0-10 VDC signal to DDC from the water / Fuel differential pressure mounted in the supply pipe.

The feed water pipe to the water tank shall have motorized on/off valve (Solenoid Valve) to be controlled by the level switch in the tank. The valve shall open at low level signal and closed at high level signal to the DDC.

ALARMS:

Alarms will be generated on the BMS in case of following conditions:

- Tank High Level Alarm
- Tank Low Level Alarm

IRRIGATION TANK:

For the Irrigation tank it shall have 4 level indicators linked to BMS: Low level (L/L), Mid-Level (M/L) , High Level (H/L), and Overflow Alarm Level. The irrigation tank shall have direct feed line from the condensate water of the FAHUs and also secondary feed line from the potable water with solenoid valve. The sequence of operation shall be as follows:

- If the water level inside the irrigation tank reaches the low level, the irrigation pumps shall be switched off for Dry run protection.
- If the condensate water level does not reach the mid-tank level then the solenoid valve on the potable water line shall open.
- If the water level reaches the high level indicator the solenoid valve shall close.
- If the water level reaches the overflow level then the BMS shall generate an alarm.

2.23.18 SUMP/SURFACE WATER PUMPS

This system operates under its own dedicated controls; hence the operating sequence of this system is not covered under this document.

BMS monitors the pumps for run status and trip alarms through volt-free signals to DDC from each pump panel as shown in the BMS control schematic

ALARMS:

Alarms will be generated on the BMS in case of following conditions:

- High Level Alarm for the sump pit.

2.23.19 GREASE SEPARATOR

This system operates under its own dedicated controls; hence the operating sequence of this system is not covered under this document.

BMS monitors the pumps for run status and trip alarms through volt-free signals to DDC from each pump panel as shown in the BMS control schematic

ALARMS:

Alarms will be generated on the BMS in case of following conditions:

- High Level Alarm for the grease separator tank.

2.23.20 LIFT MONITORING SYSTEM

This system operates under its own dedicated controls; the lift control panel will transmit VFC signals to the DDC controller to monitor the following status

- Run Status
- Failure Alarm
- Lift Cabin Alarm

The BMS shall transmit the signals to the lift control panels to shut down lifts as required by the building operator manually from the BMS workstation or automatically based on timing.

2.23.21 FIRE ALARM SYSTEM & EMERGENCY LIGHTING CONTROL

This system operates under its own dedicated controls; hence the operating sequence of this system is not covered under this document.

ALARMS:

Volt free contacts / BACNet integration will be available in each main control panel for status monitoring / alarm in the following conditions:

- Power On/Off Alarm
- Common Fault Alarm

These shall be connected to the DDC Controllers

2.23.22 ENERGY MONITORING UNITS / LV PANELS

EMU's shall be provided for all the LV panels as shown in the single line diagrams. These shall be communicating via industry standard open protocol and integrated directly with the BMS. EMU shall provide voltage, currents for each phase, power factor, active power, reactive power, instantaneous kW, harmonic analysis etc (at least 13 parameters)

ALARMS:

Alarms will be generated on the BMS for unit in case of following conditions:

- Common Fault Alarm

2.23.23 DIESEL GENERATOR & MAIN FUEL TANK

This system operates under its own dedicated controls; hence the operating sequence of this system is not covered under this document.

ALARMS:

The generator control panel will provide volt free contacts to BMS for status monitoring. These shall be connected to DDC Controllers.

- Generator Run Status.
- Generator Common Fault alarm indication.
- Battery Healthy / Battery charge fault
- Generator " NOT IN AUTO MODE" alarm
- Main fuel tank high level alarm
- Main Fuel tank low level alarm

2.23.24 UPS

This system operates under its own dedicated controls; hence the operating sequence of this system is not covered under this document.

ALARMS:

UPS shall be communicating via industry standard open protocol and integrated directly with the BMS for status monitoring and alarm in the following conditions:

- Battery Healthy / Battery charge fault

- Common Fault Alarm

2.23.25 CENTRAL MONITORING SYSTEM

This system operates under its own dedicated controls; hence the operating sequence of this system is not covered under this document.

ALARMS:

The above system shall be integrated to the BMS via Software integration through Industry Standard Open Protocols in each Panel. Contractor should ensure that the integration requirements mentioned are also incorporated in the relevant specification of Individual Systems

- System Common Fault Alarm
- Status Monitoring

2.23.26 LIGHTING CONTROL SYSTEM

Lighting control System shall be software integrated (EiB / CBus) with BMS for remote programming and control. External lights and sign boards shall also be controlled remotely. The timing functions, automatic operation functions etc shall also be achieved through BMS. All external elevation lights, landscape lights, etc. shall be connected to lighting control system and integrated to BMS. Dedicated lighting control systems are provided in the building. For operation of the systems refer the lighting control schematic and the relay schedule.

The lighting control system shall be fully integrated with the BMS.

2.23.27 CCTV & ACCESS CONTROL SYSTEM

Volt free contacts will be available in the CCTV system rack for status monitoring / alarm in the following conditions:

- Power On/Off Alarm
- Common Fault Alarm

Volt free contacts will be available in the Access Control system for status monitoring. These shall be connected to the DDC Controllers

2.23.28 POOL EQUIPMENT

This system operates under its own dedicated controls; hence the operating sequence of this system is not covered under this document.

BMS monitors the run status and trip alarms through volt-free signals to DDC from each pump panel of pool circulation and filtration pumps, pool heat pump unit.

PART 3 - EXECUTION

3.1 GENERAL

- 3.1.1 All wiring and tubing shall be properly supported and run in a neat and workmanlike manner. All wiring and tubing exposed and in equipment rooms shall run parallel to or at right angles to the building structure. All piping and wiring within enclosures shall be neatly bundled and anchored to prevent obstruction to devices and terminals.

- 3.1.2 The contractor shall be responsible for all electrical installation required for a fully functional system and not shown on the electrical plans or required by the electrical specifications. All wiring shall be in accordance to all local and national codes. All line voltage wiring, all wiring exposed, and all wiring in equipment rooms shall be installed in conduit in accordance to the electrical specifications. All electronic wiring shall be #18 AWG minimum THHN and shielded if required. All wiring in the central control room shall be concealed in an approved manner.
- 3.1.3 The contractor shall enter all computer programs and data files into the related computers including all control programs, initial approved parameters and settings, English descriptors, and color graphics complete with dynamic dispersed data. In addition, the following features, to be user implemented, shall have samples installed by the vendor for training and validation:
- a) Trend log
 - b) Alarm message (action taking message)
 - c) Run time maintenance message
 - d) Trouble action message
 - e) Dynamic Trend Plot (6 points)
- 3.1.4 The contractor shall maintain CD copies of all data file and application software for reload use in the event of a system crash or memory failure. One copy shall be delivered to the Employer during training session, and one copy shall be archived within a local software vault provided by the BMCS manufacturer within 15 miles of the OS.
- 3.1.5 The contractor shall completely check out, calibrate and test all connected hardware and software to ensure that the system performs in accordance with the approved specifications and sequences of operation submitted.
- 3.1.6 Witnessed validation demonstration shall consist of:
- Running each specified report.
- Display and demonstrate each data entry to show site specific customizing capability. Demonstrate parameter changes.
- Step through penetration tree, display all graphics, and demonstrate dynamic update and direct access to graphics.
- Execute digital and analog commands in graphic mode.
- Demonstrate DDC loop precision and stability via trend logs of inputs and outputs (6 loops minimum).
- Demonstrate scan, update, and alarm responsiveness.
- Demonstrate on-line user guide and help function and mail facility.
- Demonstrate digital system configuration graphics with interactive upline and downline load and demonstrate specified diagnostics.
- Demonstrate multitasking by showing dynamic curve plot and graphic construction operating simultaneously via split screen.

Demonstrate class programming with point options of beep duration, beep rate, alarm archiving and

color banding.

3.2 EXAMINATION

- 3.2.1 Prior to starting work, carefully inspect installed work of other trades and verify that such work is complete to the point where work of this section may properly commence.
- 3.2.2 Notify the consultant's representative in writing of conditions detrimental to the proper and timely completion of the work.
- 3.2.3 Do not begin work until all unsatisfactory conditions are resolved.

3.3 INSTALLATION

- 3.3.1 Install in accordance with manufacturer's instructions.
- 3.3.2 Provide all miscellaneous devices, hardware, software, interconnections, installation and programming required to ensure a complete operating system in accordance with the sequences of operation and point schedules.
- 3.3.3 All wiring and tubing shall be properly supported and run in a neat and workmanlike manner. All wiring and tubing exposed and in equipment rooms shall run parallel to or at right angles to the building structure. All piping and wiring within enclosures shall be neatly bundled and anchored to prevent obstruction to devices and terminals.
- 3.3.4 The contractor shall be responsible for all electrical installation required for a fully functional system and not shown on the electrical plans or required by the electrical specifications. All wiring shall be in accordance to all local and national codes. All line voltage wiring, all wiring exposed, and all wiring in equipment rooms shall be installed in conduit in accordance to the electrical specifications. All electronic wiring shall be #18 AWG minimum THHN and shielded if required. All wiring in the central control room shall be concealed in an approved manner.
- 3.3.5 The contractor shall enter all computer programs and data files into the related computers including all control programs, initial approved parameters and settings, English descriptors, and color graphics complete with dynamic dispersed data. In addition, the following features, to be user implemented, shall have samples installed by the vendor for training and validation:
 - a) Trend log
 - b) Alarm message (action taking message)
 - c) Run time maintenance message
 - d) Trouble action message
 - e) Dynamic Trend Plot (6 points)
- 3.3.6 The contractor shall maintain diskette copies of all data file and application software for reload use in the event of a system crash or memory failure. One copy shall be delivered to the Consultant during training session, and one copy shall be archived within a local software vault provided by the BMCS manufacturer within Dubai.
- 3.3.7 The contractor shall completely check out, calibrate and test all connected hardware and software to ensure that the system performs in accordance with the approved specifications and sequences of operation submitted.
- 3.3.8 Witnessed validation demonstration shall consist of:
 - a) Running each specified report.

- b) Display and demonstrate each data entry to show site specific customizing capability. Demonstrate parameter changes.
- c) Step through penetration tree, display all graphics, and demonstrate dynamic update and direct access to graphics.
- d) Execute digital and analogue commands in graphic mode.
- e) Demonstrate DDC loop precision and stability via trend logs of inputs and outputs (6 loops minimum).
- f) Demonstrate scan, update, and alarm responsiveness.
- g) Demonstrate on-line user guide and help function and mail facility.
- h) Demonstrate digital system configuration graphics with interactive upline and downline load and demonstrate specified diagnostics.
- i) Demonstrate multitasking by showing dynamic curve plot and graphic construction operating simultaneously via split screen.
- j) Demonstrate class programming with point options of beep duration, beep rate, alarm archiving and color banding.

3.4 LOCATION AND INSTALLATION OF COMPONENTS

- 3.4.1 Locate and install components for easy accessibility, in general, mount 60 inches above floor with minimum 3' – 0" clear access space in front of units. Obtain approval on locations from Consultant prior to installation.
- 3.4.2 All instruments, switches, transmitters, etc. shall be suitably wired and mounted to protect them from vibration, moisture and high or low temperatures.
- 3.4.3 Identify all equipment and panels. Provide permanently mounted tags for all panels.
- 3.4.4 Provide stainless steel or brass thermo wells suitable for respective application and for installation under other sections sized to suit pipe diameter without restricting flow.

3.5 INTERLOCKING AND CONTROL WIRING

- 3.5.1 Provide all interlock and control wiring. All wiring shall be installed neatly and professionally, in accordance with Specification Division 16 and all national, state and local electrical codes.
- 3.5.2 Provide wiring as required by functions as specified and as recommended by equipment manufacturers, to serve specified control functions. Provide shielded low capacitance wire for all communications trunks.
- 3.5.3 Control wiring shall not be installed in power circuit raceways. Magnetic starters and disconnect switches shall not be used as junction boxes. Provide auxiliary junction boxes as required. Coordinate location and arrangement of all control equipment with the owner's representative prior to rough in.
- 3.5.4 Provide auxiliary pilot duty relays on motor starters as required for control function.
- 3.5.5 All control wiring in the mechanical, electrical telephone rooms shall be in raceways. All other wiring to be installed neatly and inconspicuously per local code requirements. If local code

allows, control wiring above accessible ceiling spaces may be run with plenum rated cable (without conduit).

END OF SECTION

APPENDIX - A

**SCHEDULE OF APPROVED MANUFACTURES FOR
MECHANICAL WORKS**

SCHEDULE OF MANUFACTURERS

All the materials supplied under this contract shall be from one of the manufacturers listed below or equal approved. All proposed materials shall fully comply with specifications and contract documents requirements

S. No.	MATERIALS	MANUFACTURERS
HVAC & Heating Water System		
1	Plate Heat Exchangers (For Chilled water system & Heating water system)	Alpha Laval Sondex Swep Tranter Danfoss
2	Chilled water/Heating water pumps	ITT Bell & Gosset / Vogel / Xylem Grundfos Ritz KSB Armstrong Flowserve Ebara SPCO
3	Air Handling Units / Fan Coil Units (CHW)	Euroclima VTS Clima Mekar Trane Carrier York Trosten SKM Petra Fin Power
4	DX Split Units & Condensing Units	LG Daikin Mitsubishi O General Rheem Media
5	Chilled water/ Heating water Dynamic Pressurization Units. Dirt Separator / Vacuum Degasser	ITT Bell & Gosset Grundfos Armstrong Holden & Brook Pillinger IMI Hydronics Reflex Watts Flamco
6	Chemical Treatment (For Chilled water &	Metito

S. No.	MATERIALS	MANUFACTURERS
	Heating water systems)	Veolia Culligan Water Bird Emirates Water Technology
7	Electric Water Heater	Ariston Atlantic Lamborghini Rheem Ferroli Zanussi
8	Extract Fans	Ventaxia System Air / Sodeca /Metrico Penn, ACME Greenheck Dynair Nuaire S&P Elektrovent Loren Cook
9	Air Filters	AAAF Cambridge Envitec FCI
10	Ecology Units / Electrostatic filters	Purified Air / Bioclimatic Environ Air Trion Rydair May Air
11	Air Curtains	Berner National GTG DCM Wing Panasonic S&P
12	Building Management System & Controls	Siemens Honeywell/Alerton Johnson Controls Sauter Kieback & Peter
13	Air Outlets (Grilles & Diffusers)/ Fire Dampers / Volume Control Dampers/ Motorized Dampers/ NRD	Air Master Technalco Beta Flowtech Trox Ruskin Action Air Safid

S. No.	MATERIALS	MANUFACTURERS
		KBE Aldes Halton CVS Aerovac Excelair E.H. Price
14	Sound Attenuator (Silencer)	Kinetics Seagul Safid Acoustic Air Technology Energy Intl Leminar/Galloway Acoustics KBE EH Price
15	GI Ductwork	Nippon Cape Kawasaki Iscore Jindal Nishin AGIS
16	Phenolic Pre-insulated Ductwork	Kingspan Easy duct PAL Pearl Industry P3
17	Duct & Pipe Supports	Hilti Grinnel Flammco Mupro Fischer Weicco Diamond VMC
18	Valves (For Chilled water & Heating water)	Tour & Anderson Hattersley Oventrop Danfoss Nibco Crane Tyne Flow ABO valves/ITAP IMI Hydronics Conex-Banninger Albion-UK Maplef
19	Pressure Independent Control Valve-PICV	Danfoss Flowcontrol

S. No.	MATERIALS	MANUFACTURERS
		Belimo Frese Oventrop Tour Anderson Pettinaroli Conex-Banninger Maplef
20	Strainers	Mueller Steam Armstrong Sarco
21	Gauges / Thermometers	H.O. Trerice Weiss Dwyer Weksler Hunter Winters
22	Air Vent	Armstrong Crane Fischer
23	Black Steel Pipes and Fittings (For Chilled water & Heating water systems)	<u>Pipes:</u> Nippon / Sumitomo Trouvay Cauvin Interpipe Hyundai Steel Zenith / Surya PPF Any Equivalent Make
		<u>Fittings:</u> Crane Nefit Benken PPF BSI Shield
24	Grooved Fittings & Coupling	Victualic National Grinnel Smith Cooper
25	Vibration Isolators / Acoustic Floating Floors/Seismic Supports	Mason Kinetics Unison Weicco Christie & Grey VMC
26	Fibreglass Insulation	Kimmco Knauf Afico

S. No.	MATERIALS	MANUFACTURERS
27	Elastomeric Rubber insulation	K Flex Armaflex Kai Flex Aerofoam Gulf-O-Flex ApcoFlex
28	Sound Liner	Manville Manson Knauf Kimmco Afico
29	Pipe Flexible Connector	Mason Kinetics Tozen Weicco
30	Flexible Ducts	Safid Thermoflex ATCO Supaflex Aeroduct Flotek Aerovac
31	Flexible Duct Connector	Durodyne Clim/Climatech Cain Aeroduct Aerovac
32	Access Doors	Zest Safid Ruskin Aerovac
33	Adhesive and Coatings / Vapour Barrier (Low VOC & Low emitting)	Foster Miracle Childers Fevicol
34	Testing & Commissioning Agency	CML i2i Prime technology Sutton TAB Alpine ABL
35	Water Flow / BTU Meter	ABB-Kent Kamstrup Schlumberger Hersey

S. No.	MATERIALS	MANUFACTURERS
		Aquametro Sensus Siemens
	<u>Plumbing, Drainage</u>	
1	Polypropylene Random Pipes and Fittings, Multi-layer PPR	Banninger Polymuthan Poloplast Aquatherm Wefatherm BQ Rohrsystem ATP Novatherm
2	PEX Pipes and Fittings	Wirsbo Fripex Aquatherm Valsir/ Vassen/ Wavin Unidelta Vialli
3	Copper Pipe and Fittings	Crane Enfield Wednesbury Yorkshire Nibco MM Kembla Outokumpu Maksal Met tube
4	uPVC / HDPE Pipe and Fittings	Hepworth Terrain Marley Polyfab Durapipe George Fischer/ Uponor Geberit Hurner
5	Sound Proof Drainage Pipes	Argo- night sky Skolan-DB Poloplast-Polokal
6	Valves, Strainers & Accessories	Tour & Anderson Hattersley Oventrop Holmes NibcoTyne Flow Pegler Crane ITAP/CSA

S. No.	MATERIALS	MANUFACTURERS
		Danfoss Conex-Banninger Albion-UK Maple
7	Pressure Reducing Valve	Honeywell Baily Watts Zurn Wilkins Resideo
8	Gauges / Thermometers	H.O. Trerice Weiss Dwyer Weksler Hunter Winters
9	Water Hammer Arrestor (Where Applicable)	Sioux Chief Zurn PPP
10	Hangers & Supports	Hilti Grinnel Flamco Mupro IsoFix Fischer Unison Weicco Diamond VMC
11	Water Pumpsets	Bell & Gosset/ Vogel Xylem Ritz Grundfos KSB Armstrong Lowara Desmi LEO DAB Wilo
12	GRP Water Tanks	Sekisui Mitsubishi FTC Pipelink Sejin Sunnik
13	Water Filtration System / Chemical Treatment	Metito Culligan

S. No.	MATERIALS	MANUFACTURERS
		Veolia Corodex Group Emirates Water Technology
14	Sump/Sewage Pumps	ABS KSB Flygt Grundfos
15	Floor Drain / Roof Drain	Jay R Smith ACO Frost Zurn Wade Dallmer Geberit Kessel Aqualine
16	Manhole / Gulley Trap Covers / Catch Basin	RBA RSI Ducast MBBM Saint Gobain/Amana Carnation KAJ
17	Grease Separator	Kessel ACO
18	Water Flow / BTU Meter	ABB-Kent Kamstrup Schlumberger Hersey Aquametro Sensus Siemens
<u>Fire Fighting</u>		
1.	Fire Pumps (UL/FM)	Reddy Buffalo Armstrong Peerless Patterson SPP Armstrong American Marsh NAFFCO

S. No.	MATERIALS	MANUFACTURERS
2	Diesel Engine	Caterpillar Cummins Clarks Fire driver
3	Fire Pumps Controller	Firetrol Cutler Hammer Tornatech Joslyn / Naffco
4	Galvanized Steel Pipes & Fittings	Zenith Surya Shield Econosto Universal Tubes Tata Gujarat Steel
5	Grooved Fittings (UL/FM)	Grinnell Shield Central Victaulic
6	Hangers and Accessories	Flammco Hilti Mupro PHD Shield Fischer Unison
7	Fire Fighting Equipment	VIKING MARSHAL ANSUL ANGUS NAFFCO BRISTOL SEFECO AL ARABIA FIREX
8	Sprinkler System	Viking Angus Victaulic Central Active Shields

S. No.	MATERIALS	MANUFACTURERS
		Tyco Lifeco Reliable
9	Clean Agent Fire Suppression System (FM200)	ANSUL MINIMEX SIEMENS GINGE KERR FIKE NAFFCO TYCO KIDDE FENEWAL SAFETY HITECH
10	Foam/ Water System	Viking Angus Marriog LPG Chemguard NAFFCO TYCO
11	Fire Stop Materials	3M Hilti Astroflame Abesco Nullifier
12	LPG System	One of the DCD Approved LPG contractors

Note : All Fire related Materials e.g. Fire Alarm, Fire Fighting, Fire Stop Materials, Central Emergency Lighting, Fire Dampers etc., must have Local Civil Defence approval.

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PART - B

ELECTRICAL

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ELECTRICAL SECTION: 01

GENERAL REQUIREMENTS

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1.0 SCOPE OF WORKS

The Scope of works includes the supply, delivery to site, installation, testing and commissioning of the various complete systems outlined briefly below, and as described elsewhere in the specification documents.

The contractor shall be responsible to visit the site to familiarize himself with the scope of work, preparation of shop drawings and obtaining approval from the various authorities prior to execution of work. The contractor shall also be responsible for obtaining all materials and workmanship approval during execution and on completion of works including the connection of permanent power supply and KWH meter. All costs and charges required by the various authorities shall be included in the scope of work except the power connection charges which shall be reimbursed by the Client to the Contractor immediately upon production of actual payment receipts by the Contractor.

LV switchboards and Distribution boards, LV Panels, MDB, SMDB's, DB's, Capacitor Banks, MCC's, Control Panels, etc.

Generator(s) and UPS system with synchronizing panels

Bus bar System

Cable Management System.

Cables and Wires.

All general and special purpose electrical accessories.

Internal and external lighting and Lighting Control System

Aircraft Warning Lighting System

Earthing and Lightning Protection system.

Audio Intercom System

Emergency / EXIT lighting system

Integrated Fire Alarm and Voice evacuation system.

Security Systems

AV / ICT Systems

Complete Testing and Commissioning of each and every system

All coordination with other sub trade to supply and install power supply to the various equipment including all conduit, ducts for the installation of the control wires by the equipment supplier, ensuring a complete functioning system.

Any cost associated with removing existing services of electrical, control system and telephone system to be included by the contractor.

- 1.1 The work shall comprise the whole of the labour and, unless otherwise indicated, all the materials necessary to form a complete installation and such tests, adjustments and commissioning as are prescribed in subsequent clauses and as may otherwise be required to give an effective working installation to the satisfaction of the Engineer.

2.0 GENERAL

- 2.1 This specification is for the Mechanical and Electrical (MEP) works. However, the Instructions to Tenderers, Form of Tender, Conditions of contract, Note on Pricing, Preliminaries Section and any Supplementary Conditions of Contract applicable to Main Contract shall be applicable to this contract also.
- 2.2 The term Contractor used in the MEP documents and drawings is to be considered to mean the Contractor for MEP works.
- 2.3 Contractor shall be responsible for proper performance of all MEP systems in accordance with relevant standards and codes and design parameters listed hereinafter.
- 2.4 Provide all items, articles, materials, operations, sundries, labour, supervision, guarantees, allowances for overhead and profit, etc., to achieve a fully functionable and acceptable system.
- 2.5 The precedence of documents and drawings shall be as determined in the Main Contract. However, consider the specifications as an integral part of the work together with the drawings. Consider any item or subject omitted from one but mentioned or reasonably implied on the other as properly and sufficiently indicated and provide the same under the work of this division.
- 2.6 The Contractor is responsible for developing his own take off of materials and is to make this available as requested to the Consultant.
- 2.7 Electrical equipment installation in this Seismic Risk Zone shall be protected from earthquakes in accordance with the requirements of the Dubai Municipality and all applicable local codes.
- 2.8 Contractor shall be responsible for the design of seismic restraint systems for the entire project. Seismic protection system including labor, materials and design shall be included in the Contract Sum.
- 2.9 All material and equipment shall be new. Old manufactured / Stored material shall not be accepted Latest Edition of Dubai Building Code to be included as a major part for compliance. To be added in the specs

3.0 EXAMINATION OF SITE AND DRAWINGS.

- 3.1 Visit the site of the proposed works and obtain all information as to existing conditions and limitations and all proposed works on adjacent sites and in adjacent areas which might affect the works on this site, whether by Private Individuals or by Government Authorities or others.
- 3.2 Examine the documents including the Specifications and Drawings of all other Divisions before bidding and again before commencing any portion of the works.
- 3.3 Neither the Owner nor the Consultant will be responsible for any claim for extra work or expense resulting from the failure of the Contractor to be fully aware of Site Conditions.

4.0 PERMITS, FEES AND INSPECTIONS.

- 4.1 Arrange for inspection of all work by the Local Authorities as and where applicable. This is to occur on an on-going basis throughout construction to avoid delays to the project. On completion of the work, present to the Consultant, for the Owner, final unconditional approval certificates of the Inspecting Authorities.
- 4.2 Pay fees / charges as levied by the Local Authorities for inspections, approvals, temporary services, connections, etc. at no additional cost to contract unless noted otherwise.

- 4.3 Before commencing any work on site, submit the shop drawings (approved by the Consultant) to the Authorities, as and where applicable, for checking and approval. Follow the progress of such drawings to permit the timely approval of them by the Authorities. Comply with any changes requested by the Authorities but notify the Consultant immediately of any such change and obtain his approval.

5.0 CONTRACT DRAWINGS

- 5.1 The drawings for services works are design drawings, diagrammatic, and intended to convey the scope of work and indicate general arrangement and approximate locations of apparatus, fixtures, pipe and duct runs, etc. The drawings do not intend to indicate architectural or structural details.

These cannot be used as shop drawings. Contractor must develop own detailed shop drawings for work at site.

- 5.2 Do not scale drawings. Obtain accurate dimensions to structure and architectural items from drawings of those trades. Confirm by site measurement. Verify location and elevation of all services (Water, Electrical, Telephone, Sanitary, Storm Drainage, Gas, etc.) before proceeding with the work.
- 5.3 Make at no extra cost, any changes or additions to materials, and / or equipment necessary to accommodate structural conditions (pipes or ducts around beams, columns, etc.)
- 5.4 Alter, at no additional cost, the location of materials and/ or equipment as directed, provided that the changes are made before installation and do not necessitate additional material.
- 5.5 Install all ceiling mounted components (Diffusers, grilles, detectors, light fixtures, emergency lights, fire detectors, loudspeakers, camera points, etc.) in accordance with the reflected ceiling drawings which are to be prepared by the Contractor and coordinated with all trades. These must be submitted for approval and be approved before any work commences on site.
- 5.6 Leave space clear and install all work to accommodate future materials and/or equipment as indicated and/or supplied by other divisions of work of the contract. Install all pipe runs, conduit runs, cable trays, etc., to maintain maximum headroom and clearances, and to conserve space in shafts and ceiling spaces and under floors, and to provide adequate space for service and maintenance.
- 5.7 Confirm on the site the exact location of outlets and fixtures.

6.0 SHOP DRAWINGS, BUILDER'S WORK DRAWINGS AND MATERIAL SUBMITTALS

- 6.1 Submit shop drawings and samples for materials and equipment as listed in this and in each subsequent section. All shop drawings shall be produced on the latest version of AUTOCAD/REVIT. Shop drawings shall include, but not limited to, the following: sketch, schematic, detailed design, coordination, site installation, wiring and logic control diagrams, manufacturers, builder's work.

Provide transparencies wherever possible and provide printed copies in a number as specified elsewhere in the contract documents. The Consultant may retain the sample of each item at his discretion until the completion of the contract.

- 6.2 Present a schedule of shop drawings and material submittal after award of the contract and not later than what is specified in the contract documents, indicating the anticipated date when the drawings will be submitted for review. Assume full responsibility for timely submission of all drawings. Allow a minimum of two weeks for the Consultant's review.
- 6.3 Drawings production and presentation is a Contractual matter and any delay on the Contractor's behalf in making these submissions will be considered a Contractual delay and may be subject to Contractual penalties and other remedies as determined by the Consultant.

- 6.4 The Consultant will only consider shop drawings bearing the stamp of the Contractor and all Subcontractors involved. Check for all pertinent information such as physical dimensions, make, performance, electrical characteristics and indicate the intended use and location before submitting these drawings. Use reference symbols or enumeration to correspond to the design drawings.
- 6.5 Assume responsibility for accuracy of equipment dimensions related to space available, accessibility for maintenance and service, compliance with inspection authorities' codes. Ensure that shop drawings indicate the shipping and working weights of all equipment.
- 6.6 The submission of samples will be subject to the same procedure as that of shop drawings. One set of such samples shall be required to be brought to site and kept there after approval till substantial completion.
- 6.7 Prepare drawings in conjunction with all trades concerned, showing sleeves and openings for all passages through structure and all insert sizes and locations.
- 6.8 Prepare composite construction drawings, fully dimensioned, of piping and equipment in tunnels, shafts, mechanical equipment rooms and areas, and all other critical locations to avoid a conflict of trades.
- 6.9 Base equipment drawings upon shop drawings and include but do not necessarily limit to, all details pertaining to access, cleanouts, tapping's, sleeves, electrical connections, drains, location and elevation of pipes, ducts, conduits, etc., obtained from consultation with, and agreement of, all trades involved.
- 6.10 Prepare drawings of equipment bases, pump pits, anchors, inertia slabs, floor and roof curbs, wall openings, trenches, pertaining to mechanical work.
- 6.11 The Consultant is not responsible for any delays caused by the inadequacy of the Contractor's drawings or his failure to obtain initial or subsequent approval. Any time taken by the Contractor to obtain approval after the originally scheduled date will be considered as a delay to the contract caused by the Contractor.
- 6.12 When drawings are marked "resubmit specified item", the Contractor is to resubmit the item as originally specified or as may be determined to be equivalent by the Consultant. The Consultant is the sole arbiter of whether any item is satisfactory or equivalent.
- 6.13 When drawings are marked "rejected" a complete resubmission of the particular drawing is necessary, subject to the same conditions as outlined above.
- 6.14 The Consultant's review shall not relieve the Contractor from responsibility for deviations from the Contract documents, unless he has, in writing, called the Consultant's attention to such deviations at the time of submission of drawings. The Consultant's review shall be construed to apply to, and only to, general arrangements and shall not relieve the Contractor from the entire responsibility. Any approval by the Consultant shall be on the understanding that any item submitted shall be ordered with options and modifications to fully meet the specification. Any fabrication, erection, setting out or other work done in advance of receipt of stamped drawings shall be done entirely at the Contractor's risk and cost.
- 6.15 Furnish prints of the reviewed details to all other parties who may require them for proper coordination of their work and furnish all information necessary for the work as a whole.
- 6.16 Obtain Manufacturers' installation directions to aid in the proper execution of the work. Submit two copies of such directions to the Consultant prior to installation, for use in inspecting the work.
- 6.17 Bind one complete set of checked As-built drawings into each operating and maintenance manual.

- 6.18 In case of any updated site requirements that does not match with the approved Consultant's drawings, Contractor to produce a revised drawing after consultant/client approval and secure an approval from local authorities before installing at site.

7.0 RECORD (AS-BUILT) DRAWINGS

- 7.1 As the job progresses, mark on one set of prints to accurately indicate the status of installed work. Have the prints available for inspection at the site at all times. 30 days before commissioning, finalize the As-built drawings and submit 2 sets as draft to the consultant for checking and approval. Upon approval, submit one set of transparencies and three sets of prints to the consultant for onward transmission to the client.
- 7.2 Show on the record drawings the installed inverts of all services entering and leaving the building and the property. Dimension underground services at key points of every run-in relation to structure and building. Record all elevations for underground services in relation to floor level of the building and give reference datum to Municipal benchmarks.
- 7.3 All record drawings shall be produced on the latest version of REVIT available.

8.0 OPERATION AND MAINTENANCE MANUALS

- 8.1 Upon successful commissioning, submit one draft of Operation and Maintenance Manuals for review and approval of Consultant. All required documents of O&M Manual should be in English.

Separate binders shall be used as follows:

1. Electrical
 2. Fire alarm system.
 3. EM/ EXIT system.
 4. Low current Systems
 5. Security Systems
 6. ICT systems
 7. Any Services other than above.
- 8.2 The minimum information required is as follows:
1. Catalogues highlighting the Make, Model and other necessary details for all Material and Equipment installed.
 2. List of Local Agents / Suppliers for all Materials and Equipment with Telephone, Fax and Email address.
 3. Detailed description of systems operation.
 4. Procedures for Preventive, regular and the breakdown maintenance, with Manufacturer's Operation and Maintenance Catalogue for all Systems / Equipments.
 5. Commissioning data for all Systems / Equipment.
 6. List of recommended spares.

7. Country of Origin Certificate

8. Factory test data

8.3 Upon approval of the draft by the Consultant, submit 4 set of manuals to consultants for onward submission to the Owner.

9.0 TEMPORARY SERVICE

9.1 Do not use any of the permanent service facilities during construction, unless specific written approval is obtained from the Consultant and the Owner or where specifically allowed elsewhere in the Contract Documents.

9.2 In the event that mains power is not available or is not of sufficient capacity for commissioning and testing, it is contractor's responsibility to supply, install and operate diesel generator(s) of the correct capacity to carry out the T &C without any additional cost to Client.

Any generator must have full safety features and must be maintained regularly to ensure site power at all times.

10.0 COOPERATION

10.1 Confer with all trades installing equipment which may affect the work of this division and arrange equipment in proper relation with that equipment installed under all Divisions of the Contract.

10.2 Furnish all items to be built in by others, in time, complete with all pertinent information, commensurate with the progress of the work.

10.3 Store materials neatly and out of the way and clean up all refuse caused by the work daily.

11.0 EXISTING WORK AND EQUIPMENT

Before this Contractor undertakes work in any area, he must prepare a list of deficiencies in that area which affect his works, or which could possibly be construed as being caused by himself if not noted. In the event that such deficiency list is not prepared, then he shall be deemed responsible for such deficiencies and the Contractor shall shoulder all related cost implications any list shall be brought to the attention of the Consultant forthwith.

12.0 SUPERVISION

12.1 The Subcontractor will maintain at site, as necessary for the performance of the Contract, qualified personnel and supporting staff, with proven experience in erecting, testing, and adjusting projects of comparable nature and complexity.

12.2 Before commencing work the Contractor will submit details of the proposed Engineers and Supervisors, including copies of their Certificates. If in the Consultant's opinion the proposed Engineers or Supervisors are not adequately qualified or are otherwise unacceptable, the onus is on the Contractor to submit alternates until such approval is given.

12.3 Where the Contractor's staff becomes unavailable during the Contract deficient in performance, the Contractor is to remedy the situation by immediate and appropriate replacement, to Consultant's approval.

12.4 Approval of the Contractor's Engineers or Staff shall in no way prevent the withdrawal of that approval at any time during the Contract should the Consultant so desire. In the event of such disapproval, Contractor will be required to rectify the position as stated above within 14 days.

- 12.5 In the event of any negligent or severely detrimental behaviour the Consultant has the right to order the removal from site of any Engineer, Supervisor, or worker on a "forthwith" basis.

13.0 CLEANING

- 13.1 Each day as the work proceeds and on completion, clean up and remove from the premises all rubbish, surplus material, equipment, machinery, tools, scaffolds, and other items used in the performance of the work. Clean out dirt and debris and leave the buildings broom clean with no stains and in a condition acceptable to the Consultant.
- 13.2 Where electrical items form part of the visible finish in the rooms, protect from over-painting, etc. and give all items a final cleaning before handing over of the project.

14.0 ACCESSIBILITY, ACCESS PANELS AND DOORS

- 14.1 Any item of equipment requiring maintenance shall be located so as to be accessible for maintenance or repair without removing adjacent structures, equipment, pipings, ducts or other materials.
- 14.2 Install all concealed equipment requiring adjustment or maintenance in locations easily accessible through access panels or doors. Install systems and components to result in a minimum number of access panels. Indicate access panels on as-built drawings.
- 14.3 Provide the respective Division of work with panels, doors or frames, complete with all pertinent information for installation.
- Ensure that access doors are installed in a manner to match the building grids where applicable.
- 14.4 Prepare detail drawings showing location and type of all access doors in coordination with other trades before proceeding with installation and hand these to the Contractor to obtain approval.
- 14.5 Size all access doors to provide adequate access and commensurate with the type of structure and Architectural finish. Should it be necessary for persons to enter, provide a minimum opening of 600 x 450 mm.
- 14.6 Ensure proper fire rating of access doors in fire separations
- 14.7 Lay-in type ceiling tiles, if properly marked may serve as access panels.
- 14.8 The access panels shall be constructed and shall have finishing to match surrounding architectural construction and finishes and shall be to engineer's approval or as specified elsewhere in the contract documents.

15.0 CONTRACTORS RESPONSIBILITY

- 15.1 The appointed Contractor shall accept full functional responsibility for all electrical services to meet the requirements of the Client / Engineer and generally in accordance with this specification.
- 15.2 The Contractors tender submission shall include for all the necessary material, plant, equipment and systems to meet the Engineer's approval whether specifically mentioned herein or not.
- 15.3 Design Parameters

The electrical system equipment selection and installation shall comply with the particular requirements listed hereunder:

- 15.4 Site Conditions

All electrical equipment and components shall be selected and installed such as to operate satisfactorily and safely under the following climatic conditions.

Maximum shade temperature in summer	50 deg. C
Direct sun temperature in summer	84 deg. C
Minimum ambient temperature in winter	2.8 deg. C
Relative humidity max	100
Rusty atmosphere	

15.5 System of Supply

Three phase, neutral and separate protective conductor (TN-S).

The system of supply shall be at 400V/230V (Secondary), 3Ph/1PH, 50HZ at the main bus bar of the LV Switch Board with a maximum short circuit capacity of 50 KA for 1 second.

The electrical system shall have following system variations at the sub-station:

Voltage	+/- 10%
Frequency	+/- 4%

16.0 HIGH VOLTAGE NETWORK

- 16.1 The contractor shall be responsible for all negotiations with DEWA to ensure that supplies are available to meet the construction programme. It shall be the Contractor's responsibility to execute all civil construction work connected with the HV Distribution Network as required by the local authorities.. All costs associated with acquiring permanent incoming mains supplies including all civil works shall be included in the contract except for the power connection charges which shall be paid by the client.

17.0 AUTHORITIES APPROVAL

- 17.1 The contractor shall be responsible for liaising with DEWA, TAAWUN, DU / Etisalat, DM / TECOM and Civil Defence Authority (CDA), Dubai Civil Aviation Authority and any other authorities after obtaining the Consultant's approval

18.0 REGULATIONS

The installation shall be carried out in accordance with the current edition and supplements of the following Regulations and Standards:

- DEWA guidelines for building substation – 2017
- DEWA regulations for Electrical installation -2017
- DEWA circular for uninterrupted power requirement
- Requirements for Electrical installations, IET Wiring regulations and local authority standards.
- Latest UAE Fire Code – Local Civil Defence Regulations.
- International Electrical Code – IEC 2006
- National Fire Protection Association Standard for Emergency and Standby Power System NFPA110.
- National Fire Protection Association Life Safety Code – NFPA 101.
- British Standards
- Chartered Institute of Building Services Engineers Code for Interior Lighting (CIBSE).
- Illuminating Engineering Society (IES)
- Security Industry Regulatory Agency
- Dubai and Civil Aviation Regulations

- Dubai Municipality Green Building Regulations and Specifications (DMGBR)
- National Electrical Manufacturer's Association (NEMA)
- Underwriter Laboratories (UL) Guidelines
- Security Industry Regulatory Agency (SIRA) Laws and Regulations
- Etisalat Design Guide - 2013
- Telecom Regulatory Authority
- Latest Edition of Dubai Building Code
-

19.0 DEFINITIONS

The definitions given in the IET Regulations, UAE Fire code and DEWA regulations for Electrical Installations apply.

20.0 PHASE IDENTIFICATION

Phase identification shall be as per Local Authority Standard amendment no. 2, 2004 Appendix-7.

Function	Marking	New conductor colour
3 phase circuit		
Phase – 1	- R	- Red
Phase – 2	- Y	- Yellow
Phase – 3	- B	- Blue
Single Phase Circuit	- L	- R/Y/B
Neutral	- N	- Black
Protective conductor		- Green and Yellow

21.0 VOLTAGE DROP

The maximum voltage drop from the point of supply (Transformer - LV switchboard) and the extremity of any circuit shall not exceed 4% of the nominal supply voltage.

22.0 TELECOMMUNICATIONS INTERFERENCE

The whole of the electrical installation work shall be so designed that there is an absolute minimum of interference with telecommunications, and the reception of broadcasting to BS 55014 or any other relevant BS.

23.0 SEGREGATION OF SERVICES

The armour of all cables, cable trays, trunking, ducts etc., shall be prevented from coming into contact with non-electrical services by minimum spacing of 150mm. Where this is impracticable, they shall be bonded to the exposed metal of the other service through a protective conductor.

24.0 BUILDER'S WORK

To reduce the necessity for cutting away and the like, certain holes and chases will be provided in the structure by the Contractor to the Engineer's instructions.

The Contractor shall plan his work so that full advantage is taken of these provision's but it is the Contractor's sole responsibility to ensure that all holes and chases are in the required position and that any additional ducts, holes and chases necessary for the execution of his work in the in-situ concrete walls, floor slabs, columns and beams are executed in the early stage of construction of the building.

25.0 FOUNDATION BOLTS AND SUPPORTS

25.1 The Contractor shall arrange for the supply of, in advance of the delivery of the equipment, all necessary foundation bolts, nuts, plates, sleeves and anchorage's as and when directed.

25.2 All cable trays, cable trunkings, light fittings and any other electrical components shall be installed and supported using a standard product recommended by the manufacturer, where such support is not a standard product range. The contractor shall only use galvanized material to engineer's approval as a support.

26.0 DESIGN AND STANDARDISATION

26.1 The Works shall be executed to facilitate inspection, cleaning and repairs for use where continuity of operation is the first consideration. All equipment supplied shall ensure satisfactory operation under working conditions. All plant containing rotating parts shall be capable of operating at speeds up to the maximum duty specified without vibration or excessive noise.

26.2 Corresponding parts throughout the Contract Works shall be made to gauge and shall be interchangeable wherever possible. The Contractor may be required by the Engineer to prove interchangeability by actual interchanging of the various parts.

26.3 Suitable provision by means of eyebolts or other means are to be provided to facilitate handling of all items that are too heavy or bulky for lifting and carrying by two men (70 kg).

27.0 MATERIALS, SHOP DRAWINGS AND WORKMANSHIP

27.1 The contractor shall be responsible to obtain the Consultant's approval on the detailed working drawings prior to ordering any equipment and commencing the installation.

27.2 Separate drawings shall be provided for each electrical system.

27.3 The contractor shall be responsible for preparing a dimensional shop drawing showing the exact location and mounting height of the different components.

27.4 The contractor shall also be responsible for preparing enlarged scale details, sections and elevations wherever requested by the Engineer.

27.5 If requested by the Engineer, the contractor shall forward coordinated shop drawings showing the exact locations of all services.

27.6 In addition to the above, the Contractor shall forward schematic diagrams for all system. The schematic diagram shall include all details requested by the Engineer.

27.7 It is a requirement of the specification that the finished appearance of the plant in public areas is of a high architectural standard and all panels, covers, trim panels, finishes and the like shall be included to provide this required appearance to the satisfaction of the Engineer.

27.8 The whole of the equipment supplied shall be of durable finish and suitable for installation in a modern building.

- 27.9 The Contractor shall be responsible for ensuring that the components of each system are mutually compatible and integrated to form fully efficient systems complying with the Drawings and specifications.
- 27.10 All materials shall be, where applicable, in accordance with the IET, BS Regulations, UL Specifications, Factories Act and Insurance Company requirements, where such exists, unless otherwise specified or agreed by the Engineer in writing.
- 27.11 All articles and materials specified to conform to the Standards shall be clearly and indelibly marked and stamped with the Standard number specified and other details required by the regulations, except where marking is impracticable when the relevant advice / delivery notes shall include the Standard number with which they are to comply.
- 27.12 All materials and workmanship shall be to the satisfaction of the Engineer, particular attention shall be paid to a neat orderly well-arranged installation, carried out in a methodical competent manner.
- The contractor shall be responsible to replace all the unsatisfactory work to the engineer without any extra cost and to the standard required by the consultant. This also applies to any item which is found to be defective in service during the maintenance period or extended maintenance as appropriate.
- 27.13 No person shall be allowed to execute any type of work which is normally carried out by a skilled tradesman unless he is thoroughly experienced and proficient in the trade concerned. The Engineer shall have the option to require a tradesman to demonstrate his proficiency to the satisfaction of the Engineer.
- 27.14 The Contractor shall ensure to specify the latest models of the electronic equipment, considering the latest advance options in Communication.

28.0 SAMPLES

Whenever requested, the Contractor shall provide a sample properly labelled of all lighting fittings, switches, fittings and other like accessories described in this specification or as specified by the Engineer.

Such samples shall be submitted to the Engineer for his approval at his offices or elsewhere as directed, with all parts left loose, so that they may be taken apart for internal inspection by hand without the necessity of using spanners, screw drivers or wrenches.

29.0 STORAGE OF PLANT AND EQUIPMENT

- 29.1 All plant and equipment shall be stored off the ground under weather-proof cover until ready for incorporation in the works. All electrical apparatus shall be examined and cleaned before installation. All open conduit ends shall be fitted with plastic caps or suitable protective covering to prevent the ingress of foreign matter. All panel boards installed at site shall be maintained clean and shall not be left uncovered. All installed wires and cables shall be properly kept to ensure safety from any damage. All drums with cables shall be protected from direct sunlight.
- 29.2 Protect the building and structures from damage due to carrying out the work.
- 29.3 Protect all electrical works from damage. Keep all equipment dry and clean at all times.
- 29.4 The contractor shall be responsible for and make good any damages caused directly or indirectly to any walls, floors, woodwork, brickwork, finishes, services, etc.

30.0 PAINTING (ELECTRICAL PLANT AND EQUIPMENT)

- 30.1 Where it is the usual practice of the manufacturer of items such as electric motors, switchgear, control panels, and similar equipment, to apply a high standard of protective paint work in the shop before dispatch, this will be acceptable provided any damage to paint work on the plant and equipment which occurs is made good by the Contractor to the satisfaction of the Engineer. The interiors of control panels, switchboards and switchgear, finish paint (two coats work) shall comply with the appropriate standard for enamel finish and the exteriors of such panels shall be of an British Standard colour to give a minimum reflection value of 42%.
- 30.2 Instruments shall be finished dull black and control handles, push buttons and similar fittings shall be chromium plated or otherwise specially finished to the approval of the Engineer.
- 30.3 All items of equipment installed above suspended ceiling and exposed area shall be painted with a matt black finish and as required by Interior Designer.

31.0 LABELS

- 31.1 Identification labels of 'trifoliate' or other approved material engraved black on white both in Arabic and English language unless otherwise agreed, with not less than 5mm 'lino' style letters shall be fixed on or adjacent to all controls, switches and distribution gear by means of at least two brass screws. Socket outlets of voltage other than 230-volt AC shall be similarly identified or engraved.
- 31.2 The labels shall bear the identifications shown on the drawings, such as identification, designation, function and where necessary, phase and voltage.
- 31.3 Each distribution board shall be complete with a chart protected by transparent plastic and fixed securely to the inside of the lid indicating details of each circuit controlled by the board. These details shall contain points of utilization served, number and size of conductors and type of wiring.
- 31.4 A permanent warning labels durably marked "Earth Connection Do Not Disconnect" shall be permanently fixed in a visible position near each earthing and bonding connection.
- 31.5 A mimic diagram shall be provided on each main, submain panel board and on each motor control centre.

32.0 INSPECTION AND TESTS AT MANUFACTURER'S WORKS

- 32.1 The Engineer shall have at all reasonable times access to the Contractor's premises to inspect and examine the materials and workmanship of the plant and equipment during its manufacture there, and if part of the plant and equipment is being manufactured on other premises, the Contractor shall obtain for the engineer permission to inspect as if the plant and equipment was manufactured on the Contractor's own premises. Such inspection, examination or testing, if made, shall not relieve the Contractor from any obligation under the Contract.
- 32.2 All work, materials and the like rejected shall be corrected or replaced as necessary at the Contractor's expense to the satisfaction of the Engineer.
- 32.3 Where the plant and equipment are a composite unit of several individual places manufactured in different places, it shall be assembled and tested as one complete working unit. All equipment will be tested at the maker's works to the relevant standards where applicable.
- 32.4 The aforementioned works tests carried out before delivery to the Site shall not in any way relieve the Contractor of completing satisfactory site tests after erection as specified.
- 32.5 The contractor shall give the Engineer reasonable notice, at least seven clear days in writing of the date on and the place at which any plant or equipment will be ready for testing as provided in the contract and the Engineer shall thereupon at his discretion notify the contractor of his intention either to release such part of the plant and equipment upon receipt of the works tests certificates or of his intention to inspect

such part of the plant and equipment and shall then, on giving twenty-four hours notice in writing to the Contractor, attend at the place so named within seven days of the date by which the contractor has stated in his notice the said plant and equipment will be ready for testing. The Contractor shall forward to the Engineer six duly certified copies of the test readings.

- 32.6 Whether at the premises of the Contractor or of any of his sub-contractors, the Contractor except where otherwise specified shall provide, free of charge, such labour, materials, electricity, fuel, water, stores, apparatus and instruments as may be reasonably demanded, to carry out efficiently such tests of the plant and equipment, in accordance with the contract and shall give facilities to the Engineer to accomplish such testing.
- 32.7 Works tests shall also be carried out such that due consideration is given to the Site conditions under which the equipment is required to function. The test certifications shall give all details of such tests.
- 32.8 As and when any plant and equipment shall have passed the tests referred to in this clause the Engineer shall issue to the Contractor a notification to that effect.
- 32.8.1 The Contractor shall not pack for delivery or transport to site any part of the plant or equipment until he has obtained from the Engineer his written approval to the release of such part for delivery after any tests required by the Engineer in terms of this clause have been completed to his satisfaction.

33.0 TESTING AND COMMISSIONING AT SITE

- 33.1 Upon completion of the installation or part of the installation, the Contractor shall carry out and be responsible for the testing and commissioning all plant, equipment and integral systems, in stages if required, to ensure that it is in proper working order and capable of performing all of its functions in accordance with the specification and to the satisfaction of the Engineer. The Contractor shall be fully responsible for all equipment until each item of plant, equipment or system or part thereof has been tested, commissioned and accepted by the Engineer.
- 33.2 Any equipment damaged in commissioning shall be replaced by new plant by the Contractor at his own expense and the plant, equipment or system concerned shall be re-tested and commissioned. No instruction or action of the Engineer shall relieve the Contractor of this responsibility.
- 33.3 All testing and commissioning shall be carried out according to the requirements of the relevant Standards and regulations as may be stated or implied in this specification.
- 33.4 The Contractor shall give to the Engineer in writing at least ten days notice of the date after which he will be ready to make the specified tests on completion of installation. Unless otherwise agreed the tests shall take place within seven days after the said date on such day or days as the Engineer shall in writing notify the Contractor. The tests shall as far as possible be carried out under normal working conditions to the satisfaction of the Engineer and shall extend over such periods as he may direct.
- 33.5 If in the opinion of the Engineer the tests are being unduly delayed, he may by notice in writing call upon the Contractor to make such tests within ten days from the receipt of the said notice and the Contractor shall make the said tests within the said ten days and notify the Engineer of the days on which the said tests are to be made. If the Contractor fails to make such tests within the time aforesaid the Engineer may appoint a third party testing agency to make the tests. All tests so made by the Engineer shall be at the risk and expense of the Contractor.
- 33.6 The Contractor shall provide all skilled labour, supervision, apparatus and instruments required for commissioning and testing and within a reasonable time thereafter furnish to the Engineer six certificates of all tests performed and accepted, signed by the Engineer, the Contractor and an authorized person acting on behalf of local authorities having jurisdiction as prescribed in the appropriate Regulations and Specifications.
- 33.7 If any part of the plant or equipment fails to pass the specified tests, further tests shall, if required by the Engineer, be repeated. The Contractor shall, without delay, put in hand such modifications as are

necessary to meet the requirements as described in the Contract and any expense which the Employer may have incurred by reason of such further tests may be deducted from the Contract Price.

- 33.8 The Contractor shall include for submission of working drawings for the electrical installation to the local authorities having jurisdiction for approval and shall allow for the procurement of the local authorities test certificate upon completion of the building following inspection of the complete electrical installation by the local authorities having jurisdiction. Acceptance shall not in any way absolve the Contractor of his responsibility for the performance of the plant or equipment after erection as a complete working system in all respects.

34.0 SYSTEM ACCEPTANCE

- 34.1 The ultimate condition for system acceptance is that the Owner and Consultant have inspected the system and found it to be acceptable and indicated this in writing.

Issuance of the final payment certificate does not necessarily indicate system acceptance, neither does release of final payment holdback in whole or in part. The Consultant's acceptance may be contingent on any or all of the following if applicable:

- 34.1.1 Submit original copies of letters from manufacturers of all systems indicating that their technical representatives have inspected and tested the respective systems and are satisfied with the methods of installation, connections and operation.
- 34.1.2 Submit "as built" drawings and operation and maintenance manuals.
- 34.1.3 Train owner's maintenance staff.
- 34.1.4 Submit original of all guarantees and warranties of equipment and systems required as per contract.
- 34.1.5 Hand over all keys of panel boards and equipments, passwords for operation systems, control, etc.

35.0 CORRECTION AFTER COMPLETION / DEFECT LIABILITY.

- 35.1 Remedy all work in accordance with the General Conditions of Contract during the Maintenance period.
- 35.2 Attend immediately to any and all the defects occurring during the period defined above and repair in a manner to prevent recurrence. This contractor is responsible for all work required by other trades necessary to repair the works of this section, or necessary to repair damage caused by the failure of any part of this section.
- 35.3 Instruct all Suppliers and Manufacturers that guarantees on equipment will commence when the completed work is accepted and not from the date the equipment is put into operation. In the event that this condition is omitted by the supplier, or if subsequent cost to the Owner is involved. Contractor shall be liable for such costs.

36.0 GUARANTEES

- 36.1 The Contractor will guarantee all material and workmanship for at least 12 months after preliminary take over by the Owner.
- 36.2 All guarantees from equipment suppliers will be vested in the Owner, regardless of whether the Contractor who supplied the equipment is still associated with the project or not.
- 36.3 Guarantees will be full guarantees and will include all overhead, profit, incidental charges and sundries.
- 36.4 Where damage is caused to any other item by any failure of the item guaranteed, then the guarantee shall also include the costs incurred in rectifying that damage.

Recommendation for Warranty of Electrical Systems (All shall be started from the TOC)	
Light fitting	5 Years including full fixture body, LED, Drivers. etc
Generators	2 Years including the Synchronization Panel
Bus Bars	3 Years
Structured Cabling	25 Years By Manufacturer
BMS	2 Years
BGM	2 Years
Central Battery System	2 Years for System (Batteries 10 Years as specified) Light fixtures (4 Years)
Fire Alarm and Voice Evacuation System	3 Years (Batteries 10 Years to be included)
UPS	2 Years (Batteries 10 Years as specified)
CCTV and ACS	3 Years
Wiring Accessories	25 Years By Manufacturer
Audio Intercom System	3 Years

Recommended spares as per manufacturer / system specialist shall be provided as part of the contract

Sr. No.	Description	Spares and tools to be handed over to Employer after 1 year of mandatory maintenance	Availability of tools and spares with local agent or sole agency	Remarks
	<u>Electrical</u>			
1	LV switchgear	Yes	Yes	
2	Generator set	Yes	Yes	
3	Emergency lights	Yes	3% on each type to be handed over	
4	Automatic transfer switch (ATS)	Yes	Yes	
5	Lights (all types)		3% on each type to be handed over	

END OF SECTION

ELECTRICAL SECTION: 02

LOW VOLTAGE DISTRIBUTION

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1.0 GENERAL**1.1 REFERENCE**

Conform to General Requirements for Electrical Services.

1.2 DESCRIPTION OF WORK

- 1.2.1 Supply, installation, testing and commissioning of LV switchboards. Switchboard shall generically mean Low Voltage (LV) panels, Main distribution Boards (MDB's), Sub-main Distribution Boards (SMDB's), Motor Control centres (MCC's), Power distribution Units, (PDU's) Final Circuit Distribution boards (DB's), feeder pillars and all Factory Built Assemblies of switchgear (FBA's). And Capacitor Banks as specified here after and with ratings as indicated in the relevant electrical drawings and compliance with local codes and regulations. The proposed panel shall be Type tested assembly with the certification as per IEC 61439-1, in the name of the principal manufacturer and being assembled by certified authorized panel builders for the panels with OEM enclosures and in the name of panel builder for panels with local enclosures. LV Switchgear Assembly (Enclosure, ACB, MCCB etc.) shall be the product of a single manufacturer.

The principal manufacturer and the certified/authorized panel builder shall have switchboards that are BSEN/IEC 61439-1 certified and have certified quality assurance system to ISO 9001. Noncertified manufacturers would not be accepted.

Contractor to allow for software based Harmonic study simulation during the LV technical submission and harmonic measurements shall be performed during T & C and at the end of DLP. Based on the result recommendation harmonic active filters shall be added. This shall be included as part of Provisional Sum.

1.3 QUALITY CONTROL/ASSURANCE

Design, manufacture, testing and method of installation of all apparatus and materials furnished under requirements of these specifications shall conform to latest publications or standard rules of the following:

- A. Dubai Water and Electricity Authority (DEWA) regulations for electrical installation works and Department of Civil Defense requirements.
- B. Regulations for the Electrical Equipment of Buildings (IET), (BS) London.

The Quality Control/Assurance/Assurance/Quality Assurance duties shall be performed by the Contractor. The system of Quality Control/Assurance/Assurance verification shall be in accordance with ISO 9001 standards of quality assurance. The main switchboard manufacturer/panel builders shall be independently certified to BS EN ISO 9001 Quality Standard including in-house Engineering for product design activities. A copy of this certification shall be submitted to the Engineer for acceptance.

Manufacturer Qualifications: Company regularly engaged in the manufacturer of all materials specified in this section of types and sizes required, whose products have been in satisfactory use under similar service conditions for not less than 10 years.

Panel builder qualifications: The selected panel builder must have minimum of 5 years' experience / relationship with the principal manufacturer and offered enclosure system.

Contractor to use competent persons to install the equipment and provide them with documentary evidence of their competence, which the Engineer or Employer, or their nominees, shall be entitled to see at any time.

Acceptance by the Engineer of any switchboard shall not relieve the manufacturer from any of his performance guarantees or from any other obligations.

1.4 CODES AND STANDARDS:

The following codes and standards to the extent specified herein, form a part of this specification. When an edition date is not indicated for a Code or Standard the latest edition enforce at the time of proposal submitted shall apply

1. IEC 61439-1 Low-voltage switchgear and control gear assemblies – Part 1: General rules.
2. IEC 61439-2 Low-voltage switchgear and control gear assemblies – Part 2: Power switchgear and control gear assemblies
3. IEC61641 Ed-3 Enclosed low voltage switchgear and control gear assemblies Guide for testing under conditions of arcing due to internal fault
4. IEC 60044-1 Current transformers
5. IEC 60186 Voltage transformers
6. BS EN 60204 Safety of Machinery. Electrical Equipment of Machines
7. IEC60529 Degrees of protection provided by enclosures
8. IEC60947-2:2017+A1:2020 Low voltage switchgear & control gear– part 2 Circuit breakers
9. IEC60947-3 Switches, disconnectors, switch-disconnectors and fuse combination units.
10. IEC60068 Environmental testing
11. IEC61140 Protection against electric shock – Common aspects for installation and equipment – Basic safety publication
12. IEC60947-4-1 Contactors and motor starters
13. IEC 60947-1:2000 LV switchgear & control gear-General Rules.
14. IEC 337-2: Control devices.
15. IEC 51 Indicating measuring instruments.
16. BS 88 Cartridge fuses.
17. BS 142 Electrical protective relays.
18. BS 159 Busbar & connections.
19. BS 37 Electricity Meters-General
20. BS 6231 Electric Cables. Single core PVC insulated flexible cables of rated voltage 600/100 V for switchgear and control gear wiring
21. IEC 61000-41 Electromagnetic compatibility, testing and measurement techniques
22. IEC 61557-12 Combined performance measuring and monitoring devices for electrical parameters
23. IEC TR 61641:3.0 Enclosed low-voltage switchgear and control gear assemblies - Guide for testing under conditions of arcing due to internal fault
24. IEC 60664-1 Insulation Coordination for equipment within low voltage systems - Part 1: Principles, requirements and tests
25. IEC 62208 Empty enclosures for low voltage switchgear and control gear assemblies - General requirements
26. IEC 62053-21 Electricity metering equipment - Static meters for active energy (Classes 1 & 2)
27. IEC 60898-1 Electrical accessories – Circuit-breakers for over-current protection for household and similar installations – Part 1: Circuit-breakers for a.c. operation
28. BS EN 61008 / 61009 Residual current operated circuit-breakers without integral over-current protection for household and similar uses (RCCBs / RCBOs). General rules
29. IEC 60255-1 Measuring relays and protection equipment - Part 1: Common requirements
30. IEC 61811-1 Electromechanical telecom elementary relays of assessed quality - Part 1: Generic specification and blank detail specification
31. IEC 61921 Power capacitors - Low-voltage power factor correction banks
32. Power meter shall Comply with following standards:
33. Measurement accuracy comply as with IEC 61557-12, IEC 62053-22 IEC 62053-23
34. Electrical safety as per IEC 61010-1.

- 35. EMC Comply as per IEC 61326-1 (IEC 61000-3-2, IEC 61000-3-3, IEC 61000-4-2, IEC 61000-4-3, IEC 61000-4-4, IEC 61000-4-5, IEC 61000-4-6, IEC 61000-4-8, IEC 61000-4-11)
- 36. Enclosures shall have mechanical impact resistance of IK10 to IEC 62262

Where conflicts exist between the requirements of this Specification and other drawings, standards, codes and specifications, the most stringent shall be applied.

Total Type Test Certificate from ASTA/INTERTEK, KEMA/DEKRA or any other testing agency shall be submitted to prove the assembly performance and design

All materials shall be suitably selected and installed for use in Seismic Zone applied in the location of the project.

1.5 SUBMITTALS

1.5.1 MANUFACTURERS TECHNICAL DATA

The contractor shall submit the following drawings and documents related with each electrical system for Engineer's approval:

- A. Original catalogues for various proposed components.
- B. Schedule of Panels and details.
- C. Test Certificates.

Provide manufacturer's specifications and technical data including performance, construction and fabrication. Contractor shall submit product data for each manufactured component and include following information as a minimum:

- A. Service voltage, number of phases and frequency.
- B. Main bus-bars capacity (Ampere Rating) and short circuit rating.
- C. Over current protective device types with short circuit, frame and trip ratings, features and characteristics.
- D. Metering equipment types and functions.
- E. Time-current coordination curves for each circuit breaker.

During the time of the contract and before substantial completion of the electrical installation, submit three (3) copies of descriptive literature, technical data, maintenance recommendations (from the equipment manufacturer), catalogues etc.

1.5.2 DRAWINGS AND INFORMATION:

A drawing pouch / pocket shall be provided in each panel and laminated load schedules and approved schematic drawings (Framed) shall be providing.

The contractor / vendor shall furnish the following drawings and documents along this submittal for approval. Drawings shall have the following information:

- A. General Arrangement drawing showing front view, plan, foundation plan, floor cutouts and trenches for external cables, elevations, transport sections and weights.

- B. Sectional drawings of various types of feeders, panels showing general constructional features, mounting details of various devices, busbars, current transformers, cable boxes, terminal boxes for control cables, etc.
- C. Dimensioned drawings showing outline of the switch boards arrangement and position of important external features. Include required clearances and service space around equipment. Show tabulations of installed devices, equipment features, and ratings. These drawings shall show masses, crane lift necessary for un-tanking and size of lifting lugs or eyes.
- D. Schematic and control wiring diagrams for each type of feeder and protection including indicating devices, metering instruments, alarms, space heaters etc.
- E. Original catalogues for various proposed components.
- F. Drawings of nameplate and all termination arrangements.
- G. Drawing giving typical mechanical and electrical details of the voltage control apparatus.
- H. Details of cable terminations and fittings.
- I. Foundation plan, including foundation loading.
- J. Schematic and connection diagrams covering all equipment pertaining to the switch board.
- K. Type and special test certificates as applicable.
- L. Approved test certificate / test report.
- M. Schedule of Accessories and Fittings.
- N. Technical Manual giving installation, operation and maintenance instructions.
- O. Schedule of recommended spare parts.
- P. The switchgear manufacturer should provide a BIM model of the Main Low Voltage Switchboard and multi-starter Motor Control Centre at shop drawing stage.

1.5.3 CALCULATIONS:

Forward the following calculations and data:

- A. Calculations and charts proving that discrimination in the breaker characteristic is achieved. The calculations to be carried out using breaker manufacturer's software.
- B. Detailed short circuit calculation to select the short circuit rating of the various switchboards. The calculations to be carried out using breaker manufacturer's software.
- C. Capacitor Bank calculation to be included in the material submittal

1.5.4 INFORMATION AND CLOSEOUT:

- A. Qualification Data: For Installer and testing agency
- B. Field Quality-Control Reports: Provide test reports prepared in accordance with requirements of the Electricity supply. Authority having jurisdiction and to the Engineer's approval. Provide other test reports to establish the adequacy, quality, safety, complete status and suitable operation of each system.

The test reports to include the following:

1. Test procedures used.
2. Test results that comply with requirements.
Results of failed tests and corrective action taken to achieve test results that comply with requirements.
Busbar Sizes shall comply to the Type Test document.
- C. Comply with the Quality Requirements” and submit the following:
 1. Warranties: Submit written guarantee signed by the contractor or the manufacturer/panel builder for the period of 1 year or manufacturers standard warranty whichever is longer from the date of substantial completion. The guarantee shall cover the repair and replacement of defective materials and workmanship as directed by the engineer except where any specific warranty from a supplier or manufacturer extends for a longer time.
 2. Operation and Maintenance Data," include the following:
 - a) Routine maintenance requirements for switchboards and all installed components.
 - b) Manufacturer's written instructions for testing and adjusting over-current protective devices.
 - c) Time-current coordination curves for each type and rating of over-current protective device included in switchboards. Submit on translucent log-log graph paper; include selectable ranges for each type of over-current protective device. This should be done using breaker manufacturer's software.

2.0 PRODUCTS

2.1 MAIN LOW VOLTAGE SWITCHBOARD – FORM – 4 TYPE 2

- 2.1.1 The main LV Switchboard shall be of a free standing, cubicle type modular construction system conforming Type Tested Assembly, incorporating ACB's, MCCB's busbars, fuses, measuring equipment etc. all as required and complying with the latest editions of IEC (see clauses 1.4.1&2).

All circuit breakers and busbar ratings shall be as per the load schedules and drawings. All circuit breakers and busbars should be rated for 50°C. All auxiliary power supplies shall be provided as per the manufacturer's requirement. Main LV panels shall be from Original Equipment Manufacturer (OEM) Enclosure with Type Testing in the name of the OEM/Principal manufacturer.

The temperature rise test shall be done at 50deg C steady ambient temperature,

Contractor shall ensure that the LV panels are procured from manufacturers authorized local panel builder on authorization basis with high quality deliverables. Fault level calculations shall be provided to substantiate Proper selection of all devices. These fault calculations shall originate from the original manufacturer/panel builder selected by the contractor for supplying the LV Switchgear for this project

- 2.1.2 Design & Construction:

Valid test certificates issued by independent (third party) accredited laboratories, DEKRA (KEMA Quality), Intertek ASTA or other recognized body, to be provided for approval. No testing in the manufacturer own laboratory shall be accepted.

- 2.1.2.1 Switchboard construction shall be designed to FORM 4, Type - 2 standards for panels as shown on drawings in accordance with IEC (see clause 1.4.2). Busbars shall be in a separate chamber, fully segregated from the rest of the switchboard. Additional insulating material shall be provided behind the

- doors/plates of the busbar chamber. Individual functional units shall be located in independent compartments.
- 2.1.2.2 The switchboard shall be of a modular construction and be extendable on both sides. Welded construction switchboards will not be accepted. The degree of protection of enclosure shall be at least IP 42 or as per latest standard defined in IEC (see clause 1.4.7). Frame thickness and mechanical impact shall be as per the independent third party type test certificate.
- 2.1.2.3 Enclosures shall be made of sheet steel. The basic framework should be electro-galvanized steel coated with high-solid enamel, polyester electrostatic spray and oven baked, the thickness shall be as per type test compliance. The doors and panels shall be iron-phosphate steel, coated like the framework but also with primer. The color shall be as per manufacturer's standard finish. The base frame shall be of minimum 3mm thick Electro galvanized steel. Barriers shall be included between each compartment to contain an internal fault as per IEC for criteria 1-7 in BS / IEC standards of 65 kA for 300 milli-second
- 2.1.2.4 Compartments shall be easily accessible for maintenance purposes. Barriers shall be included between each compartment to contain an internal fault as per IEC/TR (see clause 1.4.23). for criteria 1-7 in BS / IEC standards to ensure safe maintenance on any outgoing circuit when the remainder of the board is live. Each compartment shall be provided with a separately fixed compartment thermostat controlled heater at the bottom of each compartment.
- 2.1.2.5 The maximum height of the enclosure shall be 2300 mm with plinth. All meters, lamps, operating handles shall be within a maximum height of 2000 mm. All dimensions shall be of uniform appearance. All live conductors shall be shielded in such a manner that they cannot be accidentally touched when the doors are open. All doors and plates shall be interchangeable and may be hinged left, right, top and bottom as standard. Doors shall be provided with an integral gasket and earth stud and a range of locks to meet all regulations.
- 2.1.2.6 All internal separations shall be from standard pre-fabricated plates and from manufacturer. The usage of Bakelite, fiber sheets or any other material to provide internal separation will not be acceptable. The internal separation between compartments and sections shall be at least IP2X.
- 2.1.2.7 Suitable cableways shall be provided for each functional unit section. There shall be front & rear access for cabling purposes. It shall be possible to safely terminate or work on the outgoing cables of any functional unit without having to switch off the main breaker or adjacent functional units. Switchboards shall be suitable for bottom or top cable entry as specified. All openings and entries shall be vermin proof. Switchboards shall be arranged such that safe access (front, rear, side) may be readily obtained. Adjustable gland plates, adequate channel frames and cable clamps for cable supports shall be provided.
- 2.1.2.2 MCCB's shall be fixed type for normal supply feeds. Essential, fire life and safety and retail feeds shall be plug-in or withdrawable type within LV MDB. The loads shall be identified by the contractor and the details of the same shall be included in the shop drawings that would be prepared by them.
- 2.1.2.3 Operating devices shall be incorporated only in the front of the switchgear with a hinged door. Only the handles of control and selector switches, push buttons, knobs and cut-outs for lamps and meters shall be arranged on the front doors of the respective compartments to permit operation without opening the door. Protective relays shall be mounted on the front door of the compartment. All cut-outs shall be provided with gaskets for the purpose of dust proofing.
- 2.1.2.4 Current transformers shall not be directly mounted on the buses. Current transformers shall be mounted on the fixed portion of the compartment using removable CT link for easy access and maintenance.
- 2.1.3 Busbars:
- 2.1.3.1 Busbars shall be of rectangular cross section HDHC tinned copper and suitably rated for continuous operation. The main busbar rating shall be the same throughout the entire length of the switchboard. The

busbars and primary connections shall comply with BS (see 1.4.18). The surface temperature of the busbar should not exceed 55°C over an ambient of 50°C as required by AHJ. Connections to incoming circuit breakers shall be of the same cross section as per the Type Tested configuration. Bus bars shall be made of 99.9 % electrolytic copper (type Cu ETP as defined by standards ISO1190-1), tinned (5mm thickness) to ensure correct thermal and electrical conductivity and shall be in accordance with BS EN 13601:2013.

- 2.1.3.2 Busbars shall be installed in a separate chamber placed at the TOP only and shall run along the whole length of the switchboard, which allows for easy extension in the future. Vertical droppers if any, shall run behind/adjacent to each functional unit section. Busbars shall be rigidly supported along their entire length. The busbar supports shall be of a high-grade polymer. The mechanical and dielectric strength of the busbars and supports shall be capable of withstanding the worst fault conditions and as per Type testing requirements. Split bus bar distributions shall not be used. Busbars shall be designed for mounting on insulated supports that are sufficient in number to withstand electrodynamic forces resulting from the flow of the peak asymmetric short-circuit current.
- 2.1.3.3 The main busbar system shall be of a parallel 2/3 bar per phase depending on the Type Tested arrangement, which eliminates the need for drilling or bending of busbars at joints. All switchboards shall be complete with a neutral and earth bar running along the full length of the panel. The cross section of the neutral bar shall be same as that of the phase conductor and that of the earth bar is at least half the size as that of the phase bar. The minimum clearance between phases, phase to neutral and neutral to earth busbar shall be as per type test certificate.
- 2.1.3.4 Busbars and connections shall be color coded for phase identification and shall conform to the phase sequence, counting from left to right, upper to lower and from near to remote when viewed from the operating side of the switchboard.
- 2.1.3.5 Under no circumstances will common earth / neutral bar be accepted. Earth bars, neutral bars and links shall be so located and mounted that access to them is not obstructed by the structure or wiring of the switchboard and so that all outgoing neutral and earth conductors can be readily and safely connected and disconnected without moving other cables or disconnecting the incoming supply to the switchboard.
- 2.1.3.6 The main busbar shall be certified to withstand a fault level of at least 50kA for 1 second. This certificate shall be from ASTA or an independent and recognized testing authority.
- 2.1.3 Panel wiring / Terminal Blocks / Labels:
- 2.1.4.1 All internal and control wiring shall be 600 / 1000V LSF insulated stranded copper cable to BS (see 1.4.20). All cables shall be adequately rated and not less than 2.5 sq.mm. Run on cables trays. Each cable shall be properly terminated at both ends with loops. When wiring is across door hinges, it shall be enclosed by a flexible tubing and be looped to permit the opening of doors and removal of components for inspection. Plastic ferrules with correct numbers shall be fitted on all wires to correspond with circuit number and diagram. The colour coding shall be as follows:
- Phase connections - red, yellow and blue
 - Neutral connections – black
 - Earth connections - green/yellow
 - A.C. voltages below 50V - brown; if the return connection is earthed, a green ferrule shall be fitted.
 - D.C. circuits - grey with red and black ferrules to denote polarity.
- 2.1.4.2 Extra low voltage wiring and terminals shall be separated from the wiring and terminals for higher voltages. Each screened cable shall have its screen wired to its own terminal on the outgoing terminal strip. Terminal blocks for power and control wiring shall be of Cage Clamp connection technique in which the only tool required is a flat – bladed screwdriver. Once the conductor is inserted, the stainless

steel cage clamp shall automatically provide the correct clamping force for a secure connection. Screws terminals shall not be used. Cables having the same number shall be terminated at adjacent terminals and connected by means of cable jumpers at the terminal block. Terminal blocks shall be grouped according to the working voltage, power circuit and control circuit. 10% of spare terminals and not less than two numbers shall be provided on each group of terminal block.

2.1.4.3 All Circuit breakers, Feeders' etc. shall be provided with labels of adequate dimensions. Labels shall be of traffolyte having a white background engraved letters. They shall be screws fixed on the panel.

2.1.4.4 Mimic diagram to be provided for all main LV Panels.

2.1.4.5 Anti-condensation heaters shall be provided within the assembly and shall be controlled by a panel-mounted hygrostat with a 50% to 100% setting range.

2.1.5 Metering and indication:

2.1.5.1 The Mains supply incomer should be provided with ammeter, a voltmeter and RYB phase indication lamps. kWh meter shall be provided and shall be of an approved type and make by the Utility. Ammeters and voltmeters shall be Digital type.. All VT secondary's shall be for 230V and all CT secondary's shall be 5A.

2.1.5.2 Indicating lamps shall be of the long-life LED. They shall be connected through individually fused circuits tapped from the outgoing side of the main switch.

2.1.5.3 Internal and auxiliary wiring shall be of the 600 / 1000V grade with a minimum cross section of 2.5-sq. mm.

2.1.5.4 All incoming and outgoing switching devices shall have provision for locking in the 'off position.'

2.1.6 Safety Shutters And Interlocks:

2.1.6.1 Where shutters are provided to maintain IP 20 (minimum), they shall automatically cover all bus bar and cable spouts or isolating contacts whenever removable or withdrawable units are removed. The safety shutters shall be automatically opened and closed by action of the removable or withdrawable unit movement. A positive drive to the shutters shall be provided in both directions. They shall be capable of being padlocked in the closed position.

2.1.6.2 Where multiple incoming supplies are provided then the incoming functional units and bus-section devices shall be provided with interlocks and controls including:

- a) Those noted in the particular specification schedules and drawings.
- b) A complete set of electrical and mechanical interlocks to prevent incorrect operation and to provide safety for personnel in addition to any external interlocks specified for operational reasons. This shall be taken to include:
 - i. such interlocks as required to prevent parallel operation of sources where such paralleling has not been agreed with the supply authority
 - ii. Check sync Controls that automatically prevent parallel operation of two supplies that are not synchronised.

2.1.6.3 On removal of withdrawable units, mechanical interlocks shall be provided to prevent the following operations and any other operations which could cause danger or damage:

- a) The unit being closed unless it is correctly located in the service, isolated or earthed positions or unless it is removed from the fixed part of the assembly.

- b) The unit being withdrawn from or inserted into the feeder spouts or isolating device when it is in the closed position.
- c) The unit being moved to any position other than that selected by means of the mechanical selector.
- d) The unit being withdrawn or inserted into the fixed part unless it is in the correct state for withdrawal or insertion.
- e) The unit being closed in the service position unless all auxiliary circuits between fixed and removable parts have been completed.
- f) Means shall be provided to prevent opening of cubicle doors or covers before disconnection of supply to exposed live parts within.

2.1.7 Power Monitoring Unit (PMU)

- A. Digital power monitoring unit (PMU) with MODBUS or equivalent communication capabilities shall be provided in each main LV Panel and as shown on drawings for integration with BMS. PMU's to be considered on all outgoing feeders, except the feeder to substation DB, within the main LV panel and on feeders with load more than 50kW for all other panels/MCC.
- B. Power Monitoring Units (PMU) shall be listed under UL508, CSA approved marking as per 61010-1 Class 1. All setup parameters required by the power monitoring units shall be stored in non-volatile memory and retained in the event of a control power interruption. PMU shall support applications in single retained in the event of a control power interruption. PMU shall support applications in single phase and three phase systems. PMU shall be capable of application without modifications at nominal frequencies of 45 to 65 hertz. The PMU shall have an LCD Display.
- C. The unit shall support measurements or calculations of the following data:
 - a. Real Time Readings Current
 - Current
 - a. Per Phase
 - b. Neutral (calculated)
 - c. 3-Phase Average
 - Voltage
 - a. Line to line per phase
 - b. Line to line, 3 Phase average
 - c. Line to neutral per phase
 - d. Line to neutral per phase average
 - Real Power
 - a. Per-phase
 - b. 3-phase total
 - Reactive Power
 - a. Per-phase
 - b. 3-phase total
 - Apparent Power
 - a. Per-phase
 - b. 3-phase total
 - Power Factor, 3-Phase Total
 - Frequency
 - Total Harmonic Distortion, Current and Voltage
 - Accumulated Energy
 - a. Real kWh
 - b. Reactive kVAr

- c. Apparent kVAh
 - Demand
 - a. Demand Current Calculations Per-Phase, Present and Peak
 - b. Demand Real Power Calculations 3-Phase Total, Present and Peak
 - c. Demand Apparent Power Calculations 3-Phase Total, Present and Peak
 - Power Analysis Values
 - a. THD Voltage, Line to Line and Line to Neutral
 - b. THD Current, per Phase and Neutral
 - c. Power Factor, 3-Phase
- D. All power demand calculation shall use block interval, with optional sub-intervals. The window length shall be set by the user from 1 to 60 minutes in 1 minute intervals. When the interval is less than 15 minutes, the sliding block demand calculation shall occur every 60 seconds.

E. Sampling

The current and voltage signals shall be digitally sampled at a rate high enough to provide true RMS accuracy to the 15th harmonic and shall be accessible on the PMU screen. Power monitoring unit shall provide continuous sampling at a minimum of up to 32 samples per cycle, simultaneously on all voltage and current channels in the meter.

F. Maximum and Minimum Values

The unit shall provide maximum and minimum values for the following:

- Line to Line Voltage
- Line to Neutral Voltage
- Current
- Power Factor
- Real Power Total
- Reactive Power Total
- Apparent Power Total
- THD Voltage Line to Line
- THD Voltage Line to Neutral
- THD Current
- Frequency

G. For each maximum and minimum value listed above, the unit shall record the following attributes.

- Maximum/Minimum Value
- Phase of recorded Maximum/Minimum (for multi-phase installations)
- Values shall be available by display and via network communications

H. Current Inputs

I. Accuracy

The unit shall comply with IEC (see 1.4.26), Class 1. The unit shall be accurate to 1% of reading for power and energy. Voltage and current shall be accurate to 0.5% of reading. Frequency metering shall be accurate to +/- 0.02 Hertz from 45 to 65 Hertz. These accuracies shall be maintained for both light and full loads. No annual calibration shall be required to maintain these accuracies

J. Firmware Upgrades

It shall be possible to field upgrade the firmware in the unit to provide added functionality or repair defects. Upgrades shall be accomplished via a communication connection. Disassembly or changing of integrated circuits shall not be required. It shall not be necessary to de-energize the circuit or the equipment to perform the upgrade

K. Control Power

The unit shall operate over a wide range of control power including 110 to 400VAC, +/- 10% or 125 to 250VDC, +/- 20%.

L. Communications

The unit shall communicate via RS485 Modbus with a 2-wire connection at speeds up to 19.2kBaud. The PMU devices shall be connected in a daisy chain fashion and brought on to Modbus TCP network via an IP enabled Protocol translator.

2.2 MOTOR CONTROL CENTRES (MCC) – FORM 2

2.2.1 General

MCC constructions shall be made of local enclosures confirming Type tested assembly from authorized panel builders. This specification applies to switchboard designed to feed fans and related equipment. The design, construction, busbar arrangement, circuit breaker details, etc., shall be in accordance with IEC (see 1.4.2).the relevant specifications stated under main distribution boards). All circuit breaker ratings busbar ratings and fault levels shall be as shown in the load schedules and drawings. MCC rating 800A and above shall be considered with Form 4 Type 2 compliance.

The temperature rise test shall be run at 50deg C steady ambient temperature.

Form 4 panels shall be used of panels rated 800A and above.

2.2.1.1 All chillers shall be provided with the following:

1. Suitable earth leakage protection with adjustable settings from 0.1 – 1A.
2. Ammeter with CT's and selector switch.
3. Undervoltage relay with built in timer and contactor.
4. Suitably rated control supply with ON/OFF switch.
5. Chiller ON / OFF indication lamps.
6. Hour meter for each chiller.

2.2.2 Starters

2.2.2.1 All motors shall be fed separately from a Motor Control Centre. Starters shall be in accordance with IEC/BS-EN (see clause 1.4.6&12) . They shall have the components as per the manufacturer's Type-2 or applicable coordination chart of the manufacturer. They shall be protected by circuit breakers and be generally complying with the below detail:

- a. 0.37kW and above - Star Delta or Soft Starter or VFD, as per SLD

Electric motor rating exceeding 0.37 kW shall be considered with control equipment incorporating means of protection against over load of the motor. Each starter panel shall be provided with a main incoming MCCB capable of carrying the full load current.

2.2.2.2 Each starter shall comprise as a minimum:

- a. On load triple pole MCCB with two normally open and one normally closed auxiliary switches. The MCCB to be mechanically interlocked with the starter component door and capable of breaking the motor starting current and of making onto a fault at the motor terminals. Breakers shall be motor protection type without overload release. .

Provision to be made for padlocking the MCCB in the 'OFF' position.

- b. Three Pole Contactor AC3 duty to be fitted with three normally open and three normally closed auxiliary contracts.

All the auxiliary contacts to be wired down to the starter terminal block.

- c. One triple pole thermal over current relay incorporating over current, phase sequence and single phase protection with adjustable trip settings.

External mechanical hand reset to be provided.

- d. One control circuit MCB or fuse and one copper link.
- e. Each starter shall be fitted with one set of local start/stop buttons and a local / remote / auto selector switch or hand / off / auto switch as required. Each starter control wiring terminal block shall include control terminals for connection of remote push buttons, auto circuits and local and remote emergency stop push buttons.
- f. Where more than one motor with a duty / standby or duty / backup functions used, the starter control wiring shall be arranged to automatically change duty upon the fault tripping of the selected motor. A duty / standby or sequence selector switch shall be provided.
- g. Each starter shall be fitted with built-in transformer operated lamps indicating:

Motor running (green)
Motor stopped (red)
Overload tripped (amber)

- h. Each starter panel shall be fitted with an ammeter, volt meter and hour meter and selector switch for motors 7.5kW and above.
- i. Each starter to be equipped with labels in English and Arabic giving drive designation, plant number and control operation.
- j. Each starter control wiring shall include control terminal and control relays as necessary to achieve any interlocking specified elsewhere.
- k. Adjustable time delay devices for Star Delta Starters.
- l. Reduced voltage electronic soft starters in addition to various motor starting and stopping options should include the following:
 - Stall protection and jam detection.
 - Excessive starts per hour.

- Underload protection.
- Undervoltage, overvoltage, voltage unbalance and phase reversal protection.
- Diagnostics features.

A TP electronic overload relay should be provided as detailed elsewhere in this specification. In addition, a bypass facility shall be provided.

2.2.2.3 Starter units for manual operation shall have push buttons for starting and stopping, together with auxiliary contacts for remote indicator lights where specified. Starter units arranged for automatic operation shall have Hand / Off / Auto controls and remote push buttons.

2.2.2.4 All Star Delta starter shall be provided with mechanical and electrical interlocks for star delta contactor assembly to prevent short circuiting.

2.2.3 Enclosure

Panels shall be constructed of a minimum 1.5mm to 2.0mm thick electro galvanized steel coated with high solid enamel polyester electrostatic spray with mechanical impact resistance of IK10 to IEC regulations and over backed and protected to IP-55 for wet areas and IP 42 for dry locations within the electrical and MCC rooms.

Front access shall be arranged for all panels with hinged lift off doors. Dust and damp proof gasket around the edge and fitted with lockable cam type chromium plated handle mechanically interlocked with the starting MCCB.

The enclosure for the motor starters are divided into two types as follows:

- a. Single starter panel.
- b. Multi – starter panels.

2.2.4 Single starter panel.

This specification shall be applied for starter panels used for starting a single equipment with a maximum of two motors.

All equipment shall be mounted on sheet steel mounting plate with screws into tapped holes and all live parts shall be shrouded to a minimum IP 2X.

The minimum short circuit rating (Ics) for the Motor Duty MCCBs shall be 30kA. However, the contractor shall forward a short circuit calculation and upgrade the same if found necessary.

2.2.5 Multi – Starter Panels.

This specification shall be applied for starter panels used for starting more than a single equipment.

The panel shall be constructed to IEC (see 1.4.2), Form-4, type-2 fixed starters modules with all internal partition in aluzinc of 1mm thick. It should be possible to achieve all three positions ‘Service’, ‘Test’ and ‘Isolated’ of the withdrawable starters modules with door closed to maintain the IP rating. The terminal for the external conductor shall be in separate enclosed protected space and shall be separated by insulated covering.

Each module shall be made of hinged front cover with side panels made of EGI/aluzinc sheet steel. The rear wall shall be made of insulating material.

Busbar shall be of hard drawn high conductivity tinned copper of the stated capacity.

The short circuit rating of the busbars, droppers and associated apparatus shall be 36kA for 1 second unless indicated otherwise on the drawings.

2.2.6 Variable Frequency Drives

Each VFD shall be designed and manufactured in accordance with the current edition of the following European and international standards and directives:

- EN 55011 - Limits and methods of measurement of radio disturbance characteristics (EMC Emission).
- EN 50178 - Electronic equipment for use in Power Installations.
- EN 61800-3 - Adjustable speed electrical power drive systems - Part 3: EMC product standard.
- EN 61000-3-2 - Limits for harmonic current emissions (equipment input current below 16 A per phase).
- EN 61000-3-4 - Limits for harmonic current emissions (equipment input current above 16 A per phase).
- ENV 50140 - EMC/Radiated radio frequency electrical field (simulation of impact from GSM telephones).
- EMC Directive 89/336/EEC.
- Low Voltage Directive 73/23/EEC.
- EN61000-6-2 - Generic Immunity, Industry.
- EN61000-4-2 / IEC 1000-4-2 – Electro Static Discharge (EMC Immunity).
- EN61000-4-3 / IEC 1000-4-3 - Radiated electromagnetic field (EMC Immunity).
- EN61000-4-4 / IEC 1000-4-4 - Burst (EMC Immunity).
- EN61000-4-5 / IEC 1000-4-5 - Surge (EMC Immunity).
- EN61000-4-6 / IEC 1000-4-6 - Cable born HF (EMC Immunity).
- UK Electricity Council Recommendation G5/3 Stage 2 (EMC Emission).

Design Criteria:

Variable frequency drives shall be designed for HVAC applications, shall be of the pulse width modulated voltage vector control type, and shall be able to operate any asynchronous AC motor with a variable torque load. Bypass function to be included in the VFD related MCC / CP (Built-in or external) to ensure operation incase of Failure of VFD.

Regulatory Requirements:

The variable frequency drives shall be CE marked in accordance with the EMC Directive 89/336/EEC and the Low Voltage Directive 73/23/EEC.

A Manufacturers Declaration shall be issued in accordance with the Machine Directive 89/392/EEC and amendments 91/368/EEC and 93/44/EEC.

Conducted Radio Frequency Interference (RFI) shall be in accordance with EN 55011 Group 1 Class B with built-in RFI filters.

Total Harmonic Voltage Distortion in the electrical power system shall not exceed 5% at the Point of Common Coupling (transformer secondary side).

The variable frequency drive immunity shall be in accordance with EN 61000-6-2.

The manufacturer shall be certified in accordance with ISO 14001: Environmental.

Quality Assurance:

Variable frequency drives shall be manufactured in accordance with ISO 9001: Quality Systems.

To ensure quality and minimise infantile failures at the site, every VFD shall be tested by the manufacturer. The VFD shall operate a dynamometer at full load, and the load and speed shall be cycled during the test.

All optional features shall be functionally tested at the factory for proper operation.

Submittals:

Submit installation and manufacturer's drawings and performance data including:

- 1) dimensional and layout drawings
- 2) power circuit diagrams
- 3) installation and maintenance manuals
- 4) warranty description
- 5) VFD's full load ampere rating
- 6) UL certification agency file numbers
- 7) catalogue information
- 8) VFD selection chart based on actual equipment loads
- 9) Protection settings schedule.

The specification lists the minimum VFD performance requirements for this project. Each supplier shall list any exceptions to the specification. If no departures from the specification are identified, the supplier shall be bound by the specification.

Where specified, necessary provision shall be made for interfacing with the SCADA system as required.

Motors for Equipment with VFD shall comply with the following:

- The variable frequency drive motors shall be tested in accordance with VFD and motor manufacturers specific recommendations.
- The motor shall be of a special high efficiency design specifically intended for variable load and variable speed applications to provide optimised efficiency performance at partial load conditions.

2.2.7 Control Panel Internal Wiring

The panels shall be fully wired internally at the manufacturer works to terminal strips.

The control panel internal wiring shall accord with BS 7671. All internal wiring of the control panel shall be utilizing 300 / 500V grade single core, circular LSF (HR) insulated and sheathed cable of appropriate current carrying capacity. Minimum size cable that shall be used is 32/0.20 (1 mm²).

The controlling circuits shall be wired using white sheathed cable for live feeds, with black sheathed neutral conductors.

All cabling within the panel shall be run within appropriately sized slotted side, plastic trunking, fitted with appropriate retaining clips and arranged to run continuously around inside the panel. Cables to items of control gear shall then leave trunking slot adjacent to their location, with any quantity of cables from trunking to control item being formed into a loom using plastic straps.

A 50% space factor shall be allowed in all trunking and loom strap to cater for both the cooling and spare capacity requirements.

2.2.8 Selector Switches

- 2.2.8.1 They shall be of the rotary type, rated to suit item controlled and fitted with pointer type switching knob. Terminations shall be screw grip to receive spade type crimped / soldered lugs.
- 2.2.8.2 Where starter / contactor is particularly detailed to be mounted remote from the main control panel then, in these instances, they shall be of the totally enclosed type, incorporating motor supply isolator, hand-off- auto-selector switch, latching emergency power 'off' button and incorporating auxiliary contacts for basic control panel indications.
- 2.2.8.3 Starters and contactors shall, however, be located within the control panel and be of the open type with readily replaceable operating contacts and coils. They shall incorporate auxiliary contacts as required to achieve controls operation and indications specified.
- 2.2.8.4 Starter for motors up to 5kW, rated output, shall be of the direct on-line start type with any motor above such limit arranged for star delta start (unless specifically requested otherwise).
- 2.2.8.5 Unless particularly detailed, all motors shall be assumed as operating from a 400 volt 50 HZ 3-phase supply.
- 2.2.8.6 All starters shall incorporate under volt release, single phasing and appropriately set thermal overload protection, together with appropriate auxiliary contact block for required signaling, etc.
- 2.2.8.7 Where motors are duplicated for standby purposes, then they shall be arranged to automatically change duty upon the fault tripping of the selected motor.

2.2.8 Compensated Equipment

- 2.2.8.1 Any motorized valve or damper controlled via a compensated system, shall be arranged such that a manual selector switch is provided to override compensated selection and enable manual opening / closing or on/off control, as appropriate. Any compensator unit mounted within the control cabinet shall remain in the manufacturer's purpose made enclosure and be installed strictly in accordance with the manufacturer's recommendations.

2.2.9 Time Switch Control

- 2.2.9.1 Each time switch shall be of the quartz movement type with minimum 24-hour battery reserve or operated by a 230-volt synchronous motor and of the type incorporating separate motor supply and control switch terminals and be with a minimum of a 24-hour self-winding clockwork spring reserve.
- 2.2.9.2 Unless more sophisticated operation is specified elsewhere, then each time clock shall be capable of two 'on' and two 'off' operations every twenty-four-hour period. Also, any control time switch incorporated within the panel, shall be complete with an associated bypass selection facility to manually override 'on/off'.

2.2.10 Control Relays

- 2.2.10.1 Shall be of the encapsulated plug-in type and incorporate visual indication that the operating coil is energized.
- 2.2.10.2 Bases for relays shall be mounted horizontally, to ensure relay is vertical above the base when the control cabinet is in its intended location, with screw type terminals, suitable for receipt of spade type crimped/soldered lugs.

2.2.11 Circuit Protection

- 2.2.11.1 Each item of controlled plant shall be separately protected. Circuit breakers shall be arranged in a composite bank.
- 2.2.11.2 All terminals and bus bars (which shall be solid copper) etc., shall be correctly shrouded to prevent exposed live contact.
- 2.2.11.3 Circuit breakers space within MCC's provided shall incorporate space allowance for 25% increase in capacity, by later modification, with space for at least three breakers of each size used in the initial panel circuits.
- 2.2.11.4 Isolators shall be rated to make or break the electrical supply to the particular item whilst such item is running on load.
- 2.2.11.5 All Life safety, lifts essential, IT, security to be provided with ELS + Alarm.
- 2.2.11.6 No Diversity shall be used in Bus Bar dropper sizing.

2.3 SUB MAIN DISTRIBUTION BOARD – FORM - 2:

- 2.3.1 SMDB constructions shall be made of local enclosures confirming Type tested Pan assembly from authorized panel builders. The panel board shall be Wall mounting type constructed to BS EN (see 1.4.1&2), Form 2 except 800A and above which shall be Form – 4 requirements similar and from the same make/type as the MDBs. It shall be constructed with electro-galvanized steel coated with high solid enamel polyester electrostatic spray and oven baked and the thickness shall be as per type test compliance. The color shall be as per manufacturer's standard finish. Doors locks shall be fitted. The degree of protection of the board shall be IP 41/42.
- 2.3.2 Busbars shall be of hard drawn, high conductivity tinned copper and rated for continuous operation. The phase bar, neutral bar and earth bar shall be identified by approved colour codes. The neutral bar shall be of the same cross section as the phase bus bar. A sub feed neutral assembly with a removable link for isolation shall be available as an integral part of the Distribution board. Earthing bar assembly shall also be available to facilitate the connection of earth continuity conductors. Detachable panels of insulating material shall be provided at all locations to avoid accidental contact with live metal on the opening of any cell. All outgoing and incoming MCCBs and isolators shall be fully shrouded and independently segregated to DEWA requirements. All equipment mounted on the door shall be fully shrouded with insulating material.
- 2.3.3 All Circuit breakers, Metering and Busbar ratings shall be as per the load schedules and drawings. The main MCCB and the feeder breakers shall be of appropriate capacity.
- 2.3.4 Panel wiring, Terminal blocks and labels shall be as specified for the Main Distribution Boards.
- 2.3.5 Tests to be conducted on the SMDB before dispatch and the Drawings to be furnished is the same as specified for Main Distribution Boards.
- 2.3.6 Wall mounted SMDB shall be fixed on a U – channel support, and not direct to the wall. Outgoing way shall be limited to 16-way maximum as the installation is wall mounted.
- 2.3.7 Submain panel boards should be provided with minimum 2 no. pad locks with locks and key. All handles for Breakers (Panel isolators) shall have the facility to open in ON Position in addition to padlocking facility
- 2.3.8 SMDB rating 800A and above shall be designed with Form 4 Type 2 rating.
- 2.3.9 Enclosure Requirements: Extendable accessories for Top/ bottom shall be available to accommodate metering and cable extension for larger cable termination if required. Panel board shall be confirming to IK10 for the mechanical impact.

- 2.3.10 Coordination Study: Submit coordination study along with setting of protective devices for overload, short-circuit and earth-fault currents as coordinated with upstream and downstream systems based on specific coordination curves of protective devices used and specific calculated prospective short-circuit currents at various points.

2.4 FINAL DISTRIBUTION BOARDS

2.4.1 Description

- 2.4.1.1 DB constructions shall be made of local enclosures with components from the principal manufacturer same as MDBs from certified/authorized panel builders. The specification hereunder shall be applied to all panel boards referenced on the drawings / load schedules as DB's. The distribution board shall be flush / surface mounted row type busbar and shall be provided with standard extension box to allow cable glanding if required. The distribution board shall be surface mounted when installed within the electrical room/service rooms.

Distribution boards shall be provided with an isolator incomer with common ELCB's for group of lighting circuits and common ELCB's for group of power circuits for each section of MCB's or RCBO's for each power circuit as shown on schedules.

- 2.4.1.2 As a standard item, all MCB boards shall be provided with permanent circuit identification charts to the approval of the engineer.
- 2.4.1.3 MCB's shall comply with IEC (see 1.4.27) and RCBO's and ELCB's shall comply with BSEN (see 1.4.28).
- 2.4.1.4 MCB's shall be considered as Type B, C, and D as per manufacturer recommendation.
- 2.4.1.5 DB's inside electrical rooms shall be surface mounted type fixed onto a U channel support.
- 2.4.1.6 ELCB's in each DB shall be provided as per requirements of DEWA without any variation.

2.4.2 Enclosures

- 2.4.2.1 The Final Distribution Boards shall be surface/flush mounted type. The enclosures within Electrical rooms must be surface mounting type. Powder coated rustproof sheet steel units, complete with circuit breakers providing thermal overload and magnetic short-circuit and earth fault protection as shown on the drawings. The enclosure protection shall be IP 41/42, while the complete assembly shall comply with IEC (see 1.4.1). The arrangement of the enclosure shall be such that the MCB and ELCBs cannot be operated without opening the hinged door but to obtain access to the terminals of the MCB and ELCB it shall be necessary to remove a second cover. Adequate clearance shall be maintained between phases and non-current carrying metals and the neutral block. The terminals of the neutral blocks and earth blocks of the boards shall be adequately sized to receive the main and outgoing circuit feeders. Incoming terminals shall be fully shrouded. Terminals shall be so located that in the final connected position there shall be no crowding of wire in close proximity of live metal. DB cover / door shall have earthing link for grounding with the DB body
- 2.4.2.2 Surface wall mounted DB's shall be fixed on U channel support and not direct to the wall.
- 2.4.2.3 Busbars construction and assembly shall be as per local authority regulation

- 2.4.2.4 The MCBs shall be mounted on either 3 phase or single phase or split busbars as appropriate. All spaces which are not provided with breakers shall be covered with blank plates.
- 2.4.2.5 A final sub-circuit neutral assembly and an earthing bar assembly shall be available to facilitate the connection of neutral conductors and protective conductors respectively.
- 2.4.2.6 Extendable accessories for Top/ bottom shall be available to accommodate additional components and cable extension for larger cable termination.
- 2.4.2.7 Distribution boards shall be provided with an internal metal shrouding such a way that circuit breaker toggle operator shall be accessible after opening the door. Internal components shall be accessible after opening the metallic shroud.

2.4.3 Breakers

- 2.4.3.1 Miniature circuit breakers, RCBO's and earth leakage circuit breaker and shall be provided as indicated on the load schedules.
- 2.4.3.2 The short circuit rating for the MCB's and RCBO's shall be minimum of 6KA for DB's
- 2.4.3.3 All circuit breakers shall be factory calibrated to ambient temperature of 50 deg. C.

2.5 AIR CIRCUIT BREAKERS

- 2.5.1 Circuit breakers shall be designed and constructed to have a short circuit breaking capacity of not less than 50 kA and shall comply with the latest editions of IEC and BS (see 1.4.8&23). They shall be of the fully withdrawable type and motor operated. ACB On / Off pushbuttons with On / Off indication lamps shall be provided. ACB shall communicate with BMS using Modbus protocol and would be communicating all the trip settings and other details from the ACB

3 pole / 4 pole ACB's shall be provided as shown on drawings, while 4 pole shall be provided for busbar trunking outgoing.

- 2.5.2 Suitable automatic safety shutters shall be provided to cover live parts when the ACB is in the withdrawn position. There shall be 3 distinct positions i.e.

"Connected"

"Test"

"Disconnected-Isolated"

There shall also be a "Withdrawn" position.

The withdrawable chassis shall move on suitable guides and on stainless steel rollers or balls to facilitate easy withdrawal. All withdrawable chassis housing Air circuit breakers shall be of the fully draw out type. It shall be possible to draw out the withdrawable chassis without having to unscrew or unbolt any connections to the equipment mounted on the withdrawable chassis. The power and control draw out type connections shall be of the slab-in or sliding type. All draw out contacts including for auxiliary and control wiring shall be of self-aligning type.

- 2.5.3 ACB's shall have adequately designed arc chutes and arc shields for fast and effective arc chopping. The protective release shall be of the solid state electronic type having protection against overloads, short circuits and ground fault with adjustable time and current settings. The following protections shall be included:

- (a) Overload protection (LT)

- (b) Short-circuit protection (ST)
- (c) Instantaneous short-circuit protection (I)
- (d) Earth Fault Protection (G)

The ACB shall have sufficient auxiliary contacts with minimum 2 NO and 2 NC contacts. ACB shall have a provision to communicate with BMS using Modbus.

- 2.5.4 The ACB's with the rating 800 Amp and above should be provided withdrawable type.
- 2.5.5 The ACB's shall be of utilization category B. ACB's shall have rated service short-circuit breaking capacity (Ics) equal to the ultimate breaking capacity (Icu) equal to the short circuit withstand capacity (Icw) for 1 sec.
- 2.5.6 All ACB shall be provided with a 4-line display to set breaker parameters.
- 2.5.7 The ACB shall be integrated with Building Management System. On, Off & Trip status of ACB shall be visible from single line diagram replicated in the BMS.
- 2.5.8 The circuit breaker shall be able to perform a minimum of 20000 mechanical operations up to 2500 A and at least 12000 from 4000 to 6300 A.
- 2.5.9 The electrical life of ACB shall be up to 25,000 cycles for rating 800A to 1600A and 17,500 cycles for rating 2000A to 3200A to ensure long life of the breakers.

2.6 MOULDED CASE CIRCUIT BREAKER

- 2.6.1 Moulded case circuit breakers shall conform to the latest editions of BS 4752 and IEC (see1.4.2). The short circuit breaking capacities (Ics) of the outgoing MCCB's installed in the main LV switchboards shall be at least 50kA and minimum 25kA in outgoings of SMDB's (shall be upgraded if required by the conclusions of the discrimination study).
- 2.6.2 MCCB's shall be of the heavy-duty quick make break type. All MCCB's shall be ambient temperature compensated to give the ratings indicated in the drawings at 50°C for which a Type Test certificate as per IEC (see1.4.1) confirming that MCCB's have been temperature compensated should be furnished. Each breaker shall have on each phase a thermal over current release and also an instantaneous magnetic trip both of the adjustable type. Four pole breakers shall have a common trip both with a single operating handle and designed so that any overload in one pole automatically causes all poles to open. Breakers shall be trip free and each shall have a trip indication independent of ON or OFF positions. To reset from the tripped position, the mechanism shall first pass through the off position. Tripped indication shall be clearly shown by the breaker handle taking a position between ON and OFF.
- 2.6.3 The MCCB's shall be of utilization category A. MCCB's shall have rated service short-circuit breaking capacity (Ics) equal to the ultimate short-circuit breaking capacity (Icu) for all ratings.
- 2.6.5 The MCCB shall be integrated with Building Management System. On, Off & Trip status of MCCB shall be visible from single line diagram replicated in the BMS.

2.7 DISCONNECT SWITCHES

- 3.7.1 Disconnect switches and isolator shall comply with IEC (see1.4.9)
- 3.7.2 Disconnect switches shall be provided at each motor controller and appliance location. Isolators shall be rated to connect and disconnect 5 to 7 times the rated current of the item being switched. Isolators shall have a fully visual braking characteristic and shall be provided with pad lockable handle

interlocked with the door. Isolators exposed to weather shall be housed within self-extinguished polyester loaded fiberglass panels to IP 65 / IP 66 based on site requirements and subject to approval.

2.8 EARTH LEAKAGE CIRCUIT BREAKERS (ELCB'S)

2.8.1 Current operated Earth leakage Circuit breakers shall comply with BS EN (see1.4.28).

2.8.2 The ELCB's shall have a residual current operated electromechanical release and shall incorporate a filtering device to eliminate unwanted tripping due to transient voltage. A test button is to prove operation of the ELCB.

The ELCB's shall have 100mA sensitivity for lighting circuits and 30mA for power circuits.

2.9 CONTACTORS / OVERLOAD RELAYS / CONTROL RELAYS / TIMERS

2.9.1 Contactors:

Contactors shall be in conformance to IEC (see1.4.12)

The magnets and contact movements of the contactor shall be horizontal. The contacts shall be double break, butt type and each moving contact shall be individually spring loaded. The fixed and moving contacts shall be of silver -cadmium oxide. The contactor coils shall be continuously rated and shall wound on moulded formers. Contactors shall be factory calibrated, ambient temperature compensated and sealed. It shall be possible to attach the contactors to 35mm DIN Mounting Rail (EN 50 022). Power contactors shall be provided with a minimum of 2 NO and 2 NC Auxiliary contacts.

The contactors shall provide greater performance in less panel space. There shall be flexibility in the selection of front mounting or side mounting of auxiliary contacts. The contactor shall design features for IP 2LX finger protection against accidental contact with live components, positive drive contacts and protection against manual operation, Auxiliary and relay contacts shall be compatible with electronic circuits to 17V, 5mA.

2.9.2 Overload Relays

Overload relays shall confirm to IEC (see1.4.12)

The motor overload protection should have a protection choice of trip class 10 or 20. As a standard, the overload relay shall offer a manually operated trip function, visible trip indicator and isolated N.O. alarm contacts are to be used to protect motor circuits against overloads, phase sequence, phase failure (single phasing) and under voltage conditions. The relay range selected shall be such that the full load current of the equipment falls well within the relay adjustment range. The energy consumption of the relay shall be not above 150mW.

2.9.3 Control Relays

Control relays shall conform to IEC (see1.4.30)

Control relays shall be of plug in type and shall have built in LED's. They shall have miniature square bases with blade terminals. The contacts shall be of silver cadmium rated for 5A continuous current. The control relays shall be suitable for wiring with spade type crimped lugs.

2.9.4 Timers

Timers shall conform to IEC (see1.4.29)

Timer relays with adjustable range for the Star delta starter shall ensure precise reproducibility of the timing function. The relay shall be equipped with Single pole double throw 1 Pole change over contact two electrically separated time delayed contact one normally open and one normally close.

2.10 INDICATION LAMPS / PUSH BUTTONS / SELECTOR SWITCHES.**2.10.1 Indicating lamps:**

The Indicating lamps shall be LED type.

2.10.2 Push Buttons:

Push button shall be flush type mushroom type, lockable with colors in accordance with BS4099 or relevant IEC Regulations. Push Buttons contact blocks shall have double break silver plated contacts in NO / NC configuration rated not less than 5A resistive at 230V, 50 Hz. Push Buttons shall feature a diaphragm seal to protect the button from the entrance of oil, water and other contaminants. O-ring seals, which rely on lubricants, shall not be used.

Start and stop push buttons shall be of shrouded type.

Emergency stop push buttons shall be of maintained contact, pull to reset and lockable at EM-stop to prevent reset. Engraved legend plates shall be fitted with all push buttons.

2.10.3 Selector switches:

Selector switches shall be positive action rotary switches, which shall incorporate double break contacts. Selector contacts shall be double break and of hard silver alloy. They shall have a minimum continuous current rating of 10A and be of a high breaking capacity. The switch positions shall be marked on all engraved legend plates, which shall be made from black, white, black traffolyte, and the minimum character size shall be 3 mm. Operating knobs of the rotary type with lever or key operated actuators. Hands -Off -Auto selector switch shall have three position with center position as Off.

2.11 VOLT METER, AMMETER & POWERFACTOR METER

2.11.1 Volt meters, ammeter and PF meters used throughout the project shall be of the digital type and shall be fitted into the front of the panels.

2.11.2 The meters shall have a dielectric strength of 2KV at 50Hz for 1 minute with 7 segments light emitting diode displays and the figure height shall be 20mm.

2.11.3 Meters shall have 4 digits display and shall have an accuracy of 1% or better.

2.12 AUTOMATIC TRANSFER FUNCTION USING MOTORISED BREAKERS INTEGRAL TO THE MAIN/SUB MAIN DISTRIBUTION BOARDS

The Automatic Transfer switch shall be of mechanically and electrically type interlocked to exclude any possibility of coupling together the normal and emergency sources. ATS shall be integrated with the LV MDB or other panel boards as per single line diagram. To ensure continuity of service both air circuit breakers shall have two stable positions, Closed and Open. It shall be possible to manually operate each circuit breaker (or unprotected circuit breaker) in the event of the absence of control voltage.

For maintenance purpose, the transfer switches shall have a neutral position with both circuit breakers in the Open position. It shall also be possible to lock each circuit breaker in the Open position.

Circuit breaker (or unprotected circuit breaker) of the transfer switch shall be equipped with auxiliary contacts and alarm contacts (Tripped status indication).

2.12.1 CONSTRUCTION

Operations of the air breakers (or unprotected circuit breaker) shall be by individual momentarily energized motor or solenoid mounted each breaker.

Automatic transfer switch shall include 2 circuit breakers with Electrical & Mechanical Interlock.

ATS switch shall be equipped with microprocessors to carry out the following functions:

- a. Switching from one source to another depending on the presence of voltage on the transformer generator.
- b. Control of engine generating set.
- c. Load shedding and reconnection of non-priority circuits, via auto ACB's & MCCB's in the outgoing section of LV panels, MDB's etc.
- d. Switching to the generator source in the event of failure of normal source.
- e. Circuit breaker status indications shall be displayed in front of the controller (On / Off fault).
- f. Test push button shall be provided on the front of the controller to check the transfer from normal to emergency and return to normal source.
- g. Controller shall be equipped with adjustment of time delays in front.

2.12.2 Operation

- a. Three phase control shall be provided for three phase power sources. Three phase controls shall include a selector switch to enable temporary operation on single phase power sources.
- b. The voltage of each phase of the normal source shall be monitored, with pickup adjustable from 85 to 100% and dropout adjustable from 75 to 98% of pickup setting, both in increments of 1%, and shall be fully field adjustable without the use of any tools, meters or power supplies. Repetitive accuracy of settings shall be $\pm 2\%$ or better over an operating temperature range of -20°C to 70°C . Factory set to pick up at 90% and drop out at 85%.
- c. The control module shall include four time delays that are fully field-adjustable in increments of at least 13 steps over the entire range as follows:
 - * Time delay to override momentary normal source outages to delay all transfer switch and engine starting signals. Adjustable from 0 to 6 seconds. Factory set a 1 second, unless indicated otherwise on the plans.
 - * Transfer to emergency time delay. Adjustable from 0 to 5 minutes. Factory set at 0 minutes unless indicated otherwise on the plans.
 - * Retransfer to normal time delay. Time delay is automatically bypassed if emergency source fails and normal source is acceptable. Adjustable from 0 to 30 minutes. factory set at 30 minutes unless otherwise indicated on the plans. Provide a toggle switch to manually bypass time delay on retransfer.
 - * Unloaded running time delay for emergency engine generator cooldown. Adjustable from 0 to 60 minutes. Factory set at 5 minutes, unless indicated otherwise on the plans.
- d. A set of DPDT gold-flashed contacts rated 10 amps. 32VDC shall be provided for a low-voltage engine start signal when the normal source fails. The start signal shall prevent dry cranking of the generator by requiring the generator to reach proper output, and to run for the duration of the cooldown setting regardless of whether the normal source restores before the load is transferred. Also provide a commit/no commit to transfer selector switch to select whether the load should be transferred to the emergency generator if the normal source restores before the generator is ready to accept the load.

- e. A momentary-type test switch shall be provided to simulate a normal source failure. Also, terminals for a remote contact which opens to signal the ATS to transfer to emergency and terminals for remote contacts which open to inhibit transfer to emergency and/or retransfer to normal shall be provided.
- f. A visual position indicator shall be provided to indicate bypass-isolation switch position. Pilot lights shall indicate availability of power sources and automatic transfer switch position. A prominent and detailed instruction plate shall be furnished.
- g. One set of auxiliary contact shall be provided rated 10 amps, 480V AC, consisting of one contact closed when the ATS is connected to normal and one contact closed when the ATS is connected to emergency. Output terminals to signal the actual availability of the normal and emergency sources as determined by the voltage sensing pickup and dropout settings for each source, shall be provided.
- h. Each switch shall be furnished with an operator's manual providing installation and operating instructions.

2.13 POWER FACTOR CORRECTION EQUIPEMNTS:

2.13.1 Power Factor Correction Capacitors

- A. PFC constructions shall be made of local enclosures confirming IEC (see1.4.1) and IEC (see1.4.31) Type tested assembly from manufacturers or certified/authorized panel builders using Capacitors, Reactors from the principal manufacturer same as that of the other switchgear components.
- B. The power factor correction capacitor shall be suitable detuned (14%) with a series reactor for each step to prevent any resonance or resulting amplification of harmonics present in the network and shall comply with IEC 60831 standard and shall be double insulated dry type with metallized polypropylene dielectric to IP 54. The capacitor shall be able to withstand up to 130% of the rated current caused by harmonics and shall incorporate a discharge resistance. Internal fuse combined with the secondary solid foil electrode or an overpressure disconnect switch shall be fitted on each capacitor as an additional safety measure.
- C. Capacitance variation due to ambient temperature shall be taken into consideration and the capacitors shall be de-rated to give the required capacitance at 50 deg. C.
- D. When capacitors are not directly connected to the motors, these shall be protected by fuses and wired through cables with capacity of 1.8 times the capacitor rating.
- E. A six terminal capacitor, switched at the same time as the motor shall be used for star delta starting of the motor.
- F. Where multi steps capacitor bank is used the power factor shall be maintained at a set value indicated by the reactive power regulator which shall control the opening and closing of the capacitor switching contactors. The number of capacitors to be connected through the regulator in the event of power failure or when the power is restored shall be in accordance with the system reactive power. To minimize the hunting, step sizes shall be 100KVAR, 50KVAR, 25KVAR & 12.5KVAR.
- G. The high inrush current due to capacitor switching shall be taken into consideration by using fast acting contactor designed for capacitor switching and which shall be able to handle 100 times the rated current. In addition to that, the cable connecting the contactor to the bus-bar shall be installed in a way that self-inductance is produced e.g. by coiling the cable into necessary number of loops along the run.

- H. The capacitor banks shall be separated from the contactors and controls by housing the same in a separate compartment. A ventilation fan shall be provided in order to ensure that the temperature rise will not exceed 35-degree C.
- I. The Capacitor Bank Panel shall be provided with reactive power controller commanded by microprocessor with working programs 1:1:1....:1 and shall be supplied with LED indicators showing the capacitors that are connected and also shall ensure uniform ageing of magnetic contactors and capacitors by using a circular connection sequence that takes into account the time that each capacitor has been switched on.
- J. The Capacitor Bank Panel shall be equipped with voltmeter, ammeter with selector switches, digital cos ϕ (Power Factor) display module and shall have the feature of automatic disconnection of all the capacitors in the case of a failure in the electric power supply. The reactive power controller shall be able to measure the reactive power needs of an installation and to give the necessary command for connecting and disconnecting the capacitors in order to maintain the desired power factor.
- K. The panel shall be equipped with a relay and a blinking lamp with trouble acknowledged and reset button incase the automatic power improvement plant does not achieve the desired power factor due to failure of the capacitor banks.
- L. The 3 phase series reactors to be connected in series with each capacitor stage for harmonic current suppression and to prevent resonance should be of iron cored type with copper/ Aluminum windings. The current rating of the reactors should be increased suitably to take into account the effect of harmonics in the network which could be generated by variable speed drives, elevators, UPS systems, fax machines, computers, fluorescent discharge lamps etc. The offered reactors should undergo tests wherein the reactors should be subjected to harmonic currents generated by harmonic generating test equipment. The magnitude of these harmonic currents should correspond to the harmonic current magnitude in the network.

The capacitors and the reactors combination should be tuned below the lowest harmonic present in the network. With the capacitors and suitable series reactors connected it should be proved through submission of relevant calculations that the Total Demand Distortion (Current) and the Total Harmonic Voltage Distortion is within the limits as specified in IEEE Std 519-2014, standards on IEEE recommended practices and requirements for Harmonic Control in Electric Power Systems.

- M. Three phase HRC fuses shall be provided for each capacitor step and mounted on fuse bases. They shall be of modern industrial design, current limiting type and with a breaking capacity of over 100kA. They should have a current rating of between 1.7 to 2 times the rated 3-phase capacitor units current. They shall be tested as per IEC 60269 with a test voltage 1.1x Un. They shall be plug in type with suitable handle to ensure safe operation to personnel. The fuses shall not be exposed under normal operating conditions.

Additionally, a Thermal Overload Relay shall be used for giving more accurate overload protection for the steps. Else in the place of fuses + overload relay, supplier could offer a MCCB to give combined overload and short circuit protection to comply with IEC (see1.4.31).

- N. Capacitor to be dry, encapsulated sealed type IEC 61921
- O. Protection for Capacitors as per DEWA Regulation to be included. Such as link between cover and main breaker

3.0 EXECUTION

3.1 INSTALLATION

3.1.1 Fixings

- 3.1.1.1 No item of equipment secured within the control panel shall incur penetration of the main cabinet for riveting, bolting, etc.
All fixings shall be achieved with adequately sized brass bolts and nuts (or threaded inserts).

Any fixing bolt visible from the front face of the panel shall be mushroom headed and bright chromium plated with similar sized visible bolt heads generally used, with the possible exception of main panel plates retaining bolts which may be larger.

3.1.2 Identification and Referencing

- 3.1.2.1 Each isolator switch, indicator lamp and panel component shall have its function clearly defined.
- 3.1.2.2 Adjacent to each external connection terminal block shall be a label defining the item to be connected and each terminal individually referenced to the main wiring diagram. Each circuit breaker and starter shall be labelled to define item protected and rating or overload setting to be used.
- 3.1.2.3 For ease of tracing cabling, at each termination position the cable shall be identified with a numbered Helerman identification ring co-related to the wiring diagram provided with the control panel.
- 3.1.2.4 All labels shall be screw fixed engraved black traffolite having white infilling and clearly legible letter, graded in size for emphasis and clear identification for control grouping by larger lettering with smaller lettering for selected operations and indications. 400 volt warning labels shall be provided as appropriate.

3.1.3 Control Panel Components

- 3.1.3.1 For simple/minimal function control panels, indicator lamps shall be of the low voltage 24-volt transformer (trani lamp) type and accord with the signaling colors laid down in BS-4099. Minimum lens diameter shall be 12.5mm and all lenses on a panel shall be of a matching size and appearance and removable from the front for lamp replacement. For general purpose medium to large scale control systems, the indicator lights shall be of the light emitting diode systems, the indicator (LED) type and train lamps are not to be used. Circuitry shall be incorporated to achieve (with a suitable test push button) verification of all lamps operability.
- 3.1.3.2 Terminations shall be screw grip to receive spade type crimped/soldered lugs.

3.2 SWITCHBOARD OPERATOR'S MAT

- 3.2.1 All electrical switch cupboards and plant room switchboard shall be provided with a 1M wide carbon free rubber mat or mats, having a ribbed upper surface and being of such length to suit the full operating extent of each switchboard.
- 3.2.2 The insulating mats shall each be certified to have sustained, without failure, a one-minute voltage test at 2kV.

3.3 COORDINATION

It is Contractor's responsibility to provide a coordination calculation to establish actual tripping time between breakers and for proper coordination level between downstream and upstream breakers. Contractor must upgrade breaker types if required to achieve the above without any variation.

Short circuit level calculation also must be submitted for approval.

3.4 HARMONIC ACTIVE FILTERS

GCE 596: (G+4P+28+RF) Residential Development on Plot No. 6117242 in Bukadra, Dubai - U.A.E.

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Consider the provision breaker in the LV panel and space allocation for future Harmonic filters. Contractor to allow for Harmonic study during the LV technical submission and harmonic measurements shall be performed during T & C and at the end of DLP. Based on the result recommendation harmonic active filters shall be added.

3.5 TESTING AND COMMISSIONING

3.5.1 Type Test

The main distribution board and the components as applicable shall be tested in accordance with the IEC standards to verify the specified fault level withstand capacity from a reputed and approved type testing laboratory and certified by a competent authority.

Testing shall be scheduled at times approved by the Client and the Engineer

Notify the Engineer 4 weeks in advance of final factory tests of the switchboards. The Employer and the Engineer shall be entitled to witness all or any of the factory tests. Bear all costs associated with the factory testing.

The main switchboard manufacturer/panel builder shall be independently certified to BS EN ISO 9001 / 14001 / 18001 Quality Standard. A copy of this certification shall be submitted to the Engineer for acceptance.

Switchgear and Panel boards shall be as per Tested Assembly Code IEC (see1.4.2), by quality certification mark UL® or KEMA-KEUR or ASTA Diamond certifying quality in manufacturing or ASEFA, or by another testing agency approved by Local Electricity Authority having jurisdiction.

Contractor to submit the following:

A. Detailed method statements for each test.

B. Copies of all certified test results after the tests.

Temperature rise test should have been carried out with at least one joint in main Horizontal busbar as per IEC (see1.4.1) standards. Temperature rise test should have been carried out in a chamber with ambient temperature of 50°C.

Type test certificate to be in the name of the original product manufacturer & Panel assembler. The following 12 test as specified in IEC (see1.4.2) standards to comply with requirements of TTA equipment shall be carried out at recognized test laboratories and certificates from approved test witnessing authority shall be provided for each type test:

Construction:

Clause 10.2	Strength of materials and parts: Corrosion, insulation materials & lifting.
Clause 10.2.2	Verification of resistance to corrosion
Clause 10.2.3	Properties of insulating materials
Clause 10.2.5	Lifting
Clause 10.2.6	Mechanical impact
Clause 10.2.7	Marking
Clause 10.3	Degree of protection of assembly
Clause 10.4	Clearances and creepage distances
Clause 10.5	Protection against electric shock and integrity of protective circuits
Clause 10.6*	Incorporation of switching devices and components

Clause 10.7	Internal electrical circuits and connections
Clause 10.8	Terminals for external conductors

Performance:

Clause 10.9	Dielectric properties
Clause 10.10	Verification of temperature rise
Clause 10.10.2.3.7	Verification considering individual functional units and the main and distribution busbars separately as well as the complete assembly
Clause 10.11	Short-circuit withstand strength
Clause 10.12	Electromagnetic compatibility (EMC)
Clause 10.13	Mechanical operation

Type test certificates shall be submitted to the consultant engineer for verification.

3.5.2 The panels shall undergo the following tests in the factory before dispatch:

Primary Injection Test for all Circuit Breakers

Secondary Test for all protective relays and meters, CT polarity tests.

Millivolt Drop Test

HV Test 2.5kV for 1 min.

Insulation test

Function test

Voltage test

Short circuit test

Mechanical test

These tests shall be carried out in presence of the Consultant.

3.5.3 MDB / SMDB / MOTOR CONTROL CENTRE

Testing and commissioning shall be in accordance with specification section 2 and including the following:

The main low voltage panels shall be tested in the Assembler's works. The Assembler will be required to carry out any one or all of the tests stated in this specification under witness of the Engineer and other representatives. The manufacturer's representative also must be present for such tests.

Acceptance by the Engineer of any switchboard shall not relieve the manufacturer from any of his performance guarantees or from any other obligations.

Test certificates for each switchboard shall be submitted prior to delivery of the switchboard.

The Engineer or other representatives reserve the right to perform checks during the manufacturing process at any time. It shall be at the discretion of the Engineer to witness tests on 100% or any percentage quantity of each switchboard for routine tests, apart from type tests, wherever called for.

Testing of the main low voltage panels shall include the factory and site tests.

The selected switchgear and control gear brands shall be equal to those mentioned in the type test reports for the equipment.

The contractor shall arrange for the witnessing of the following tests on the fully equipped switchboard including primary bus bars and connections at the factory, in accordance with

IEC 61439-1:

- A. High voltage power frequency tests on main and auxiliary circuits
 - B. Dielectric/insulation resistance tests
 - C. Ground fault tests
 - D. Electrical operation of circuit breakers control circuits at the appropriate voltage limits
 - E. Mechanical operations tests and tests to certify correct functioning of interlocks.
 - F. Primary injection tests
 - G. Secondary injection tests
 - H. Millivolt drop tests
- Any other tests required by the Consultant

The manufacturer who designed the equipment shall provide type test certificates showing evidence that the switchboard can withstand an internal arc of 50kA for 1 sec. The switchboard shall be designed to minimize the risks of occurrence of an internal arc, and whenever such an arc occurs it should prevent its adverse impact on operators and material/equipment surrounding the switchboard. Also to guarantee the surrounding of the switchboard and the buildings, and to facilitate fast repair, the switchboard shall be designed to limit risks related to the propagation of an "Internal arc" fault to its immediate area.

Type tests should have been performed successfully on the below locations as minimum:

- A. Horizontal busbar compartment
- B. Vertical bus bar compartment of Fixed column
- C. Vertical bus bar compartment of Withdrawable column
- D. On the cable connection compartment of outgoing Functional unit in the Fixed column.
- E. On the cable connection compartment of outgoing Functional unit in the Withdrawable column.

Assembly shall fulfill all the 7 criteria during type test as per IEC/ TR (see1.4.23, version3) as detailed below:

Criteria sequence	Assessment as per IEC/TR 61641
1	Correctly secured doors and covers, do not open and remain effectively in place and provide a minimum level of protection in accordance with the requirements of IP1X of IEC (see1.4.7). The objective is to minimize the risk of severe injury by impact from doors, covers etc. and ensure a minimum level of protection of persons against accidental contact with hazardous live parts.
2	No parts of the assembly are ejected which have a mass of more than 60grams except those which are dislodged and fall between the assembly and indicators. The objective is to minimize the risk of severe injury to persons by impact of ejected parts.
3	Arcing does not cause holes to develop in the external parts of the enclosure below 2m, at the sides declared to be accessible as a result of burning. The objective is to minimize the risk of severe injury to persons by direct burning from the arc.
4	The indicators do not ignite
5	The protective circuit for accessible part of the enclosure is still effective in accordance with IEC (see1.4.2)

6	The assembly is capable of confining the arc to the defined area where it was initiated and there is no propagation of the arc to other areas within the assembly. Effect of hot gases and shooting to adjacent units other than the unit under test are acceptable, as long as only cleaning is necessary.
7	After clearing of the fault or after isolation or disassembly of the affected functional units in the defined area, emergency operation of the remaining assembly is possible. This is verified by a dielectric test according to clause 10.9.2 of IEC (see 1.4.2): 2011, but with a test voltage of 1.5 times the rated operational voltage for 1 min. With the exception of the tested zone as declared by the manufacturer, all other units should remain fully operable mechanically and electrically and are essentially in the same condition before the test.

The following tests shall be carried out after the equipment has been completely erected and connected up on site:

- A. Power frequency voltage tests on the completed switchgear
- B. Insulation resistance tests on all main and secondary circuits
- C. Secondary injection tests
- D. Calibration checks on ammeters, voltmeters and any other instruments.
Testing at normal voltage to prove that closing and tripping from local and remote control points, tripping from relays and protective gear, interlocks, alarm and indications, etc are satisfactory.
- E. Any other test required to prove compliance with the specification
- F. At the end of tests, the protective relays and breaker tripping times shall be set in accordance with the Consultant approved discrimination coordination.

3.5.3.1 Detailed Check List

- A. Effectiveness of mechanical actuating elements
- B. Door interlocks
- C. Interconnection conductor sizes, laying and clamping/screwing
- D. Visual inspection of degree of protection, creepage and clearance distances
- E. Labels for feeders and circuit indications
- F. Supply of operational instructions, wiring diagrams, technical data of equipment used
- G. Inspection and operational testing of the factory-built assembly
- H. Electrical controls and sequence of operation
- I. Physical verification of drawings, including front and internal layouts
- J. Single line diagrams & Control schemes
- K. Check for cracks on components and housings
- L. Correctness of sizes and terminations. Torque tables shall be followed.

Bolt size	Tightening torque	Tightening Force
	Nm	kN
M6	8	3
M8	20	6
M10	40	13
M12	70	22
M14	110	26
M16	170	35

- M. Cable entry/tray provisions, top and bottom
- N. Ventilation, anti-condensation heaters, and fans

- O. Sealing/pad locking for meter cabinet
- P. Phase barriers for incoming and outgoing breakers
- Q. Size of bus bars - neutral/earth
- R. Gland plates
- S. Colour codes for cables
- T. Ferrules, labels etc.
- U. Termination of ECCs with copper lugs
- V. Wiring connections of CT meter - load side of incomer
- W. Mounting height of Meter, max 2M min 0.8M

3.5.3.2 Dielectric tests (excluding components, lamps etc.)

All control fuses and electronic components shall be isolated Test voltage of 2.5kV for 1 minute shall be applied between

- Phase to phase
- Phase to neutral
- Phase to earth
- Neutral to earth

3.5.3.3 Protective measures check/Electrical continuity of protective circuits

Capacity and continuity between:

- A. All metal parts and main earth bar
- B. Door and main earth bar
- C. Continuity test according to control circuit diagram
- D. Checking of indicator lamps (with transformers), voltmeter etc.
- E. Outgoing circuits check with hand lamp

3.5.3.4 Physical Check

Check general arrangements of the components, internal wiring and complete cleanliness.

Check the panel against approved shop drawing to ensure that right components (ACB, MCCB) of correct ratings are installed.

Check the tray riser, cable drops cable overlapping and dressing of cable to MDB.

Check all the bolts / nuts and bus bar connection to ensure that all are in good and tight condition.

Check earth bonding of components and other related connection.

Check all meters and selector switches to ensure the correct selection & rating.

Check the phase barriers are properly fixed for separation of circuits.

Check all cables and panels are properly labelled and identified.

3.5.4 Insulation Resistance Check

Check and measure insulation resistance between phases, neutral and earth with 500 volts megger (the values to be verified with the permissible limits).

Check the terminal insulation, cable gland termination and shrouding.

3.5.5 Functional Test (Panel to be temporarily energized)

Check the control circuit of each system for its correct operation.

Switch ON and OFF of all MCCBs, contactors, relays to verify for the desired operation.

Check draw out mechanism of main incomer ACB.

Check operation of the indication lamps & meters to ensure proper functioning.

Check functions of motor operated ACB/MCCBs

Check for any abnormal rise in temperature on bus bars, jumper cables, terminals and various devices by using an infrared scanner.

3.5.6 Earth Leakage Test

- A. Check earth leakage circuit breakers by means of RCCB tester, selecting 50%, 100%, 150% of the rated sensitivity currents and the respective trip time to be recorded.
- B. Check the rating of core balance CTS and EL relays for earth leakage protection.
- C. Tabulate the readings taken.

3.5.7 Earth Fault Loop Impedance Test

Measure the earth loop impedance across phase and protective conductor with earth loop impedance tester.

Tabulate the reading taken and check with permissible values.

3.5.8 Polarity Test

Check the polarity of incomer supply by polarity tester before energizing panels.

Check all fuses/circuit breakers and single pole control devices are connected to phase conductors only.

3.5.9 Test on Capacitor Panels

- A. Check the operation and indications of contactors in manual mode
- B. Check the operation and indications of contactors in auto mode by setting various power factor in the regulators.

3.5.11 Final DB Light & Power Points**3.5.11.1 Physical Check**

- A. Check the electrical components such as conduits, wires, trunking, light fixtures, socket outlets are connected to the final circuit of the distribution board.
- B. Check the installation of distribution boards and internal wiring.
- C. Check the identification and labeling of each final circuits.
- D. Check that all metallic components are earthed/bonded.
- E. Check the terminal insulation, cable gland termination and shrouding.

3.5.11.2 Polarity Check

- A. Check the polarity of power / light points to ensure the light switches (single pole) are installed in the line conductor.
- B. Check for screw type lamp holders. The outer or screwed conductor is connected to the neutral conductor.

3.5.11.3 Continuity Check

- A. Check the continuity of the protective conductor from the farthest point to main earth terminal of the DB for each final circuit by means of a DC ohm meter verify the values are within permissible limit).
- B. Check the continuity between the open ends of each of the three rings of circuit (i.e. phase, neutral and earth) for all ring final circuits conductors with a multi meter and the value to be within the permissible limit.

3.5.11.4 Insulation Resistance Test - Check and measure the insulation resistance between phase, neutral and earth through a 500 Volts, Megger.

3.5.12 RCCB and Polarity Test

- A. Switch on and off all MCB's and contactor relay to verify for the desired operation.
- B. Check earth leakage circuit breakers by means of RCCB tester.
- C. Check the circuit performance by switching on the MCB in the DB.
- D. Check the light and power points by operating all the switches connected in the circuits.
- E. Check all the sockets outlets of ring/radial circuits with 3 pin socket tester (13A)/test lamp.
- F. Tabulate the readings taken.

3.5.13 Earth Fault Loop Impedance Test

- A. Measure the earth loop impedance across phase and protective conductor with earth loop impedance tester.
- B. Tabulate the readings taken and compare with permissible values.

END OF SECTION

ELECTRICAL SECTION: 03

GENERATOR

1.0 GENERAL**1.1 DESCRIPTION OF WORK**

- A. The generator set shall be provided complete with all ancillary plant and apparatus, associated control and alarm panels, switchgear, interconnecting wiring, cabling, room acoustic treatment and insulation (walls and ceilings) and all other equipment required for the operation of the set. The complete system shall be endorsed and sourced from a single authorized manufacturer.
- B. The rating of the sets shall be adequate to meet the estimated load as shown on the drawing while taking into consideration the ambient temperature conditions of 50°C and 100% humidity not less than 8 hours.
- C. The generators shall also be capable of handling the starting current of the connected equipment with the voltage dip within limits acceptable to the equipment manufacturers and as per standard regulations.
- D. While taking into consideration the ambient temperature conditions referred to in this specification:

The generator rating indicated on the drawings is the minimum standby power rating of the diesel generator set and contractor shall apply derating factors to compensate for the ambient temperature and humidity indicated in the specification.
- E. The Schematic Single Line Diagram and genset sizing calculation shows the equipment required to be supported by the generator.
- F. The Contractor shall provide all details of these equipment to the Generator Manufacturers to ensure that they consider the characteristics of these loads (such as power factor, harmonic content, etc.) when selecting generator components. The Contractor shall verify the manufacturer selection and offer to the Engineer for approval. A complete selection strategy shall be prepared detailing step loads and timed delays to ensure minimal droop etc.
- G. Contractor shall be responsible for providing the temporary resistive load banks for the Testing of the completed installations prior to handover. Inductive Load Tests shall be conducted at the factory prior to dispatch.

1.2 QUALITY ASSURANCE

Design, manufacture, testing and method of installation of all apparatus and materials furnished under requirement of these specifications shall conform to latest publications or standard rules of the following:

- A. Dubai Water and Electricity Authority (DEWA) regulations for electrical installation works
- B. Dubai Civil Defense Regulations

The following are relevant IEEE, ANSI, NEMA, BS, IEC Specification recommendations except were modified by this specification: -

NEMA-ICS1	Industrial Controls and Systems
NEMA-MG1	Motors and Generators
NEMA-PB2	Dead Front Distribution
NEMA-RI2	General: Purpose and Communication Battery Chargers
NFPA-30	Flammable and Combustible Liquid Code
NFPA-37	Stationary Combustion Engine and Gas Turbine
NFPA-70	National Electrical Code
NFPA-110	Standard for Emergency and Standby Power Systems
NFPA-101	Safety to Life from Fire in Buildings and Structures
ANSI C39.1	Electrical Analog Indicating Instrument
DEMA	Standard Practice for Stationary Diesel and Gas Engines
IEEE C37.13-2015	Low Voltage AC Power Circuit Breakers used in Enclosures
IEEE-5G	Semi-Conductor Rectifier Components
IEEE C37.90.1-2012	Guide for Surge Withstand Capability Tests for Relays
IEC 60034, BS 5000-99	Rotating electrical machines
IEC/EN 60051-2, BS 89-2	Indicating Electrical Instruments
BS 5514	Performance of Reciprocating Compression Ignition Engines
BS 4999	General Requirement for Rotating Electrical Machines (Various Parts)
BS 5000	Specification for Rotating Electrical Machines of Particular Types or for Particular Applications
IEC 60095, BS 3911	Lead Acid Starter Batteries
BS 5514/77 for Class A1 or with ISO 3046/N class A governing accuracy for engine	
ISO 8528-1	Reciprocating internal combustion engine driven alternating current generating sets — Part 1: Application, ratings and performance

Local Representation

It should be noted that all equipment manufacturers shall have authorized representation in the Emirates of **Dubai** supported with maintenance Engineers and adequate stock of spare parts. The Specialist shall be one who is normally an agent representing of the approved makes of Diesel generator sets. Diesel Generator installation shall be supervised, checked and tested by a

qualified representative of the manufacturer and handover the works in perfect running order to the satisfaction of the Engineer.

Manufacturer's Quality System: Registered to ISO 9001 Quality Standards, including in-house Engineering for product design activities.

Generator set shall be assembled as one unit with the Engine and alternator coupled with the engine and alternator coupled at the OEM factory with all equipment mounted on a skid type rigid steel bed frame.

It shall be confirmed by the contractor that the rated full load test shall be carried out at the assembled generator set at the OEM factory and the test result shall be furnished to the Engineer for review prior to the dispatch.

The generator set shall be shipped and delivered to the site as a fully assembled unit.

The generator set should be able to pick up full load in single step.

All materials and equipment shall comply with NFPA specifications as regards quality of materials, performance and proving tests.

The emergency power supply system and its components shall be such as may be properly maintained and serviced without the necessity of carrying expensive spare part stocks, or being subjected to interrupted service due to the lack of spare parts.

The generator set shall be designed to allow easy replacement of major items subject to wear.

An engine of the same model, bore, stroke and configuration, rated at the same or higher bhp and rpm shall have performed satisfactorily (i.e. without failures involving a major component, such as crankshaft, valve, cam shafts, piston, injector and the like), warranted for at least 2000 hours of satisfactory service at a minimum average load of 90% over a four-year period, either at the manufacturer's factory or a stationary power plant.

Compliance with this specification shall not relieve the Contractor of responsibility to supply equipment suited to meet the specified service conditions and applicable regulations.

Where conflicts exist between this specification and other drawings, standards, codes and specifications, the most stringent shall be applied.

All equipment supplied under this scope shall be new and shall have, in addition to normal manufacturers' warranty, an assurance of continued availability of spare parts for minimum period of 10 years.

1.3 SUBMITTALS

1.3.1 Drawings, Operating and Maintenance Instructions

A. The following list of drawings to be submitted by the Contractor for approval:

1. Drawings corresponding to all drawings submitted by the Contractor with his Tender.
2. Foundation block of diesel generating set.
3. Foundation details for diesel generator house floor, including cable and pipe trenches.
4. Detailed drawings of all components of equipment, including fuel tanks, fresh air intake, exhaust, air discharge, engine exhaust pipe and silencers, acoustics, vibration control, etc.
5. Detailed layout of the control cubicles and Panel Integrated Auto Transfer Switch.

6. Details and schematic of any integral part of each unit.
7. Complete schematic diagrams of: -
 - Protective gear (both mechanical and electrical).
 - Diesel set control, interlocks, protection, instruments, indication and alarm circuits.
 - AVR's, Battery and charger
 - Circuit breaker control and indication circuits.
 - Earthing
 - Entire emergency transfer system showing all wiring between the engine starting panel, engine generator set and the automatic transfer switches
- B. Complete wiring diagrams to include control systems for auto synchronization of generator sets where specified.
- C. Detail of protective relays, accessories and current transformers including particulars of wiring and drilling dimensions.
- D. Details and arrangements of all auxiliary plant and kiosks, certified and weights (indicate any shipping splits and weights) and manufacturer's installation instructions inside container. Jacket water heater connection diagram shall be included.
- E. Engine mechanical data at varying loads up to full load, including heat rejection, exhaust gas flows, combustion air and ventilation air flows, noise data, fuel consumption, etc.
- F. General electrical data including temperature and insulation data, cooling requirements, excitation ratings, voltage regulation, efficiencies, waveform distortion, and telephone influence factor.
- G. Sound data for mechanical and engine exhaust in octave band frequencies between 31.5 Hz to 8,000 Hz and sound data for manufacturer engine exhaust silencer options, complying with all applicable regulations.
- H. Emissions data, complying with all applicable regulations.
- I. Instrument scales, include manufacturer, model number and serial number of test instruments and date of last calibration and calibration standards used.
- J. Materials lists and spare parts.
- K. Method Statements for installation and quality plan, documentation of Field Quality Control "Prior to Installation" requirements.
- L. Layout and schedule of main and multi-core cables. (Documentation of Field Quality Control "After Installation" requirements.
- M. Supplementary product literature such as, but not limited to, generator resistances, reactance's, and time constraints; current decrement and thermal damage curve; motor starting capability.
- N. List of type tests as per the BS ISO standards to be included. Name and location of certified testing agency.
- O. Independent Third Party Certified test data, factory and field test reports.

- P. Genset Sizing Calculation, Ventilation Calculation.
- Q. Calculations:
1. Submit electrical load calculations based on the approved mechanical and electrical equipment to be connected to the LV Emergency Power Generation System. Two scenarios (one during blackout and the other during fire) shall be submitted.
 2. Submit battery sizing calculations showing compliance with specifications at ambient conditions.
 3. Submit the generator(s) calculations and sizing using the software of a reputable manufacturer showing the loads by steps. The steps to be set according to priorities defined in NFPA 110, as shown on the contract documents and as agreed with the Engineer. The sizing shall be as per the actual connected loads with suitable diversity applied based on the above two.
- R. Written guarantee signed by the contractor or the manufacturer for the period of 1 year or manufacturers standard warranty whichever is longer from the date of substantial completion. The guarantee shall cover the repair and replacement of defective materials and workmanship as directed by the engineer except where any specific warranty from a supplier or manufacturer extends for a longer time. Manufacturers or authorized agents default product warranty shall be passed on to the client.

1.4 CHARACTERISTIC OF THE GENERATOR SETS - GENERAL

Ratings

- | | | | |
|-------|--------------------------------|---|--|
| 1.4.1 | Output | - | 400V / 230, 50Hz ,1500rpm ,0.8 p.f., standby-rated as shown on Drawings. 3 phase, capacity as mentioned on schematic Drawings. |
| | Performance Class | | G2 as per ISO8528 |
| | Ambient temperature capability | - | 50 degree centigrade.
(Manufacturer's deration curves shall be provided)
Manufacturer to provide individual temperature capability charts for – a) Engine; b) Alternator; c) Radiator to confirm to 50°C suitability |
| | Maximum speed | - | 1500rpm |
| | Maximum Starting Time | - | 10 seconds |
| | Total Harmonic Distortion | - | Less than 5% at full load, AC waveform. |
| | AVR | - | ± 1.5% (from no-load to 100% load and from 0.8 lag to unity power factor) |

A detailed derating chart from the manufacturer shall be provided. Derating calculation to be substantiated based on engine/alternator/radiator temperature capability charts published by manufacturer. For packagers who are not original manufacturer, published data from manufacturer to be provided.

Sets shall be standby rated in accordance with ISO 8528 taking into consideration the site climatic conditions (altitude, temperature, etc.). The rating of generator sets as shown on the drawings and schedules is minimum and subject to submittal and approval of final load calculations considering

the approved electrical and mechanical equipment fed from the emergency supply to justify the rating of the generator. 10% spare capacity shall be allowed for future expansion.

The offered genset shall be suitable for installation in closed purpose built generator room.

2.0 PRODUCTS

2.1 DIESEL ENGINE AND ANCILLARIES

2.1.1 Diesel engine

The diesel engine shall be 4-stroke, either normally aspirated or supercharged and after-cooled, 4-stroke, water cooled, equipped with the following:

- Heavy duty or dual element air-cleaner, with service indicators
- Crank case breather
- Lubricating - oil engine driven pump
- Lubricating - oil simplex filter, with replaceable elements
- Lubricating - oil cooler
- Engine - driven jacket water pump
- Fuel filter, with service indicators
- Flexible fuel lines
- Fuel priming pump
- Dry exhaust manifold, fitting and flange
- Flywheel and flywheel housing
- Governor, Electronic
- Manual and automatic shut-off
- Oil pressure gauge
- Cooling water temperature gauge
- Digital Tachometer
- Electric starting - 24V DC.

2.1.2 Air filters

Air filters shall be suitable for use in the environmental conditions which are likely to arise occasionally, with the atmosphere highly polluted by blown sand. They shall be fitted with an 'Air filter condition' indicator of the vacuum type which shall be fitted to the air inlet manifold.

2.1.3 Cooling

Generator set package shall be complete with skid mounted radiator with engine driven fan suitable for 50°C ambient. Radiator shall be fitted with stone guard. Generators shall be provided with local bearing temperature indicators and two thermocouples per bearing to monitor bearing temperature.

Thermostatically controlled bypass valves shall be fitted to the engine cooling water system to maintain cooling water at optimum temperature.

The cooling system shall be suitable for the addition of corrosion and scale inhibitors. Sampling points shall be provided.

2.1.4 Governor

The governor shall be Electronic type. It shall control with speed within the limits laid down in the respective Standards. The engine shall run steadily at any load within its rating at its rated speed, and the changes in speed due to change in load shall comply with BS and shall be suitable for isochronous and droop control, and for local/remote speed adjustment.

2.1.5 Engine shut - off.

The engine shall be fitted with a mechanically operated device which will shut off the fuel supply to the engine when any of the specified alarm conditions occur. The device shall be fitted with one normally open and one normally closed contact which will trip and lock-out the generator circuit breaker in the event of the automatic device operating. Micro switches shall not be used in any direct current circuit nor in any inductive circuit.

2.1.6 Starting System

The electric starting equipment shall be so designed that:

- (a) At engine speeds in excess of the minimum firing speed, it shall be impossible to complete the starting circuit and
- (b) not more than three consecutive attempts to start may be made in any period of 1 minute, each starting period and the period between such consecutive attempts being automatically timed controlled by the equipment. At the failure of the third attempt the starter circuit shall be locked out pending manual re-setting. Programmable crank cycling feature shall be provided.
- (c) Sizing shall be considered based on the single load pick up upon power failure. Loading shall be based on the power failure and fire scenarios stipulated in genset sizing.
- (d) Engine starting shall be manual using push-button or automatic through control system at control panel.
- (e) Cranking motor and battery shall be rated for cranking the engine when cold and at lowest temperature recorded. Starting pinion shall automatically disengage when engine fires

2.1.7 Engine Lubrication

The engine lubricating system shall conform to BS ISO 8528-2:2005 and operate under a predetermined and constant pressure. The primary lubricating oil pump shall be an engine driven, positive displacement pump. Provide a sump pump, relief valves and by-pass valves as required for proper operation.

One full flow strainer and duplex filter shall be provided in the lube oil line on the discharge side of the lube oil pumps. The strainer shall be of the metal edge or screen type with a maximum spacing of 0.007 inch (80 mesh). The filter shall be of the multiple cleaning and replaceable element type and the filter case shall be compatible with the filter element. A pressure relief valve shall be provided for instances of clogging. An alarm condition shall be annunciated when the relief valve is in the open position. The strainer and filter shall be mounted in accessible locations without disconnecting the piping or other engine equipment when changing the strainer and filter element. Indicating pressure gauges shall be provided to measure the pressure drop across the strainer and filter.

- a) Provide automatic engine lubrication using an integral gear driven pump.

- b) Provide coarse strainer on suction side and full flow filter on delivery side.
- c) Ensure engine lubricating oil dipstick is accessible and marked to indicate maximum and minimum levels.
- d) Provide engine sump with accessible drain point or drain pipe, fitted with plug.
- e) Provide electric motor driven lubricating priming pump with automatic intermittent time control to keep engine bearings lubricated.
- f) Ensure lubricating oil system enables the engine to run continuously for period indicated at any load without replenishment.
- g) Ensure system can be replenished while engine is running.

2.2 GENERATOR

The generator shall be of the brushless, revolving field, solid - state exciter type, single bearing construction, directly coupled to and sharing a common high bed plate with diesel engine and the radiator.

The generator shall be complete with all necessary cooling fans, excitation and voltage regulating equipment. The excitation system shall not include a commutator.

The generator shall be star connected.

The generator shall be capable of maintaining its continuous maximum rated output when operating within +/- 1% of rated voltage and at rated power factor.

The critical speed shall exceed twice the over speed test value.

All windings shall be tropicalized and suitably impregnated to withstand the site ambient conditions.

The generator shall meet the following requirements:

- IEC 34/1, NFPA standard requirements
- Local government requirements for Engine Exhaust Emissions
- G2 Requirements of BS ISO 8528
- Class H insulation with tropicalization and anti-abrasion
- Three - phase sensing.
- Drip - proof enclosure, IP23
- Pilot shaft alignment
- 150 % over speed capability.
- Less than 5 % wave form deviation
- Standard paralleling capability
- Volts per Hertz, generator mounted, digital automatic voltage regulator, three phase sensing, assuring less than +/- 1% voltage regulation
- Adjustable voltage droop for parallel operation
- Adjustable voltage gain to compensate for engine speed droop and line loss.
- Less than 50 Telephone Influence Factor (TIF)
- Maximum recovery time to return to frequency stability and voltage bandwidth shall not exceed 6-8 seconds after any load step up to the specified load step limits
- Less than 3% Total Harmonics Factor (THF)
- Radio interference suppressed to the limits specified in BS 800. The generator shall also be equipped with manual voltage control

2.2.1 Low Voltage Alternator

Low voltage, three phase, four wire, star connected/wired with a neutral or ground connection. The generator shall produce electrical energy at the Voltage and frequency noted in the drawings or schedules.

The alternator shall be a four (4) pole, synchronous brushless, dual bearing type coupled directly to the engine flywheel by means of a flexible disc coupling.

The alternator voltage regulator shall be solid state type, three phase true RMS sensing and shall incorporate an under-frequency protective circuit to limit generator excitation at lower than normal operating speeds. The voltage regulator shall be equipped with a voltage adjusting rheostat capable of plus or minus five (5) percent rated voltage adjustment. The voltage regulator shall maintain the voltage within the limits specified.

The alternator insulation system shall be suitable for the expected temperatures but with a minimum Class H insulation with a combination of epoxy coating and varnish. The alternator shall be sized and properly de-rated according to IEC 34 to yield a maximum temperature rise of 130°C by resistance above an ambient temperature of 50°C at rated altitude.

Excitation shall be provided by a direct connected brushless permanent magnetic rotating exciter/AREP. The armature shall be of the three phase design and connected to a full wave three phase rotating bridge. Diodes used shall be of the silicon type mounted on proper heat sinks with surge protector to prevent voltage spikes during parallel operation.

The core and poles of the exciter shall be made of high permeability low loss steel stamping tightly clamped and secured to minimize magnetic noise. Exciter field power shall be provided by a separate permanent magnet generator/AREP directly connected to the brushless exciter. The PMG (Permanent Magnet Generator)/AREP (Auxiliary Windings, Regulations, Excitation Principle) shall provide sufficient power to the excitations system to produce 300% short circuit from the main operator armature during a three phase fault with sufficient duration for protective devices to operate. The characteristics of the exciter shall comply fully with the operational requirements of the alternator in matching the required loading conditions.

The sub transient reactance shall be 15% or less.

The alternator shall be sized/ rated to tolerate the Harmonic currents produced by the electrical systems to be connected to it. The contractor shall note that of all classes of non-linear load, variable frequency drives, which are used to control the speed of induction motors, induce the most distortion in generator output voltage. Larger alternators are required to prevent alternator overheating due to the harmonic currents induced by the variable frequency drive, and to limit system voltage distortion by lowering alternator reactance.2.2.1.

2.2.2 GENERATOR ROOM VENTILATION & COOLING SYSTEM.

The generator room shall be equipped with air intake and outlet louvre panels, of sufficient free area to allow passage of combustion/cooling air within the parameters required for the selected generator set without significant pressure drop.

- a) The Air intakes and discharges to/from the generator rooms shall be located such that Inlet Air is drawn across the Alternator and Engine and then discharged, through the radiator (where it's in the room) to atmosphere.
- b) Air inlet and discharges shall be positioned such that discharge air does not recirculate in to the inlet.

Both intake and outlet louvre panels shall be of the sand trap type and fitted with filters on the inside in accordance with this specification.

Engine cooling shall be by a water jacket, with water circulating pump and heavy duty tropical radiator and an engine driven pusher fan. The radiator and fan shall be sized to maintain safe operation at 50°C. maximum ambient temperature. The radiator shall be equipped for a duct adapter flange.

The cooling fluid shall incorporate sufficient corrosion inhibitor to protect the cooling system from internal corrosion. Thermostatic control shall be incorporated.

Obstructions in path of cooling air flow (openings, sand trap louvres, grilles, mesh, ducts, bends etc.) shall not reduce airflow below that needed at full rated output. Fan and radiator characteristics shall be selected accordingly. Provide additional booster fans if required along with necessary control gear for automatic operation.

The fan shall be rated for the specified site conditions and provide sufficient pressure to maintain the required airflow through the radiator, ducting and sand trap louvres. Provide ductwork with flexible connecting section between radiator duct flange and exhaust louver at facade of generator room. Exhaust louver panels shall be demountable to enable access to generator where applicable. Ventilation louvres shall be provided as detailed on the drawings.

Engine shall be fully equipped to start and pick up initial load specified at specified minimum ambient temperature. Provide coolant heater.

Cold starting aids, such as engine jacket water heater, lubricating oil heater, intake-air heater, battery and all devices, accessories, connections, thermostatic switches and off-duty disconnects with pilot lights and necessary protection, shall be supplemented as recommended by the manufacturer and approved by the engineer, to suit the local ambient conditions where the Generator Set is installed.

2.2.3 Voltage Regulation

Continuously active, quick acting automatic excitation control equipment responsive to the generator line to line voltage and loading conditions shall be provided to maintain the alternator voltage within the prescribed limits. Equipment will be based wholly on static state devices. Variable Regulation capability regulators shall be preferred. Diode monitor and loss of excitation protection shall be provided.

The equipment shall include all selector switches, indicating instruments, rectifiers, resistors and any other auxiliary components necessary for the satisfactory operation of the plant.

A voltage setting device capable of adjusting the required voltage in steps of 1 volt between the limits of 90% and 110% of normal voltage at any load within the rating of the machine shall be mounted on the control panel.

When the generator maximum continuous load at rated power factor is rejected, the regulator equipment shall be capable of limiting the generator momentary over voltage to a value not exceeding 125% rated voltage and shall restore the voltage to within 1% of the nominal present value within 3 seconds.

Over-voltage / under voltage and over excitation protection shall be provided.

2.2.4 Earthing

The generator terminal boxes, base and base frame shall be provided with M12 external earth studs.

All metallic non-current carrying parts that are not inherently bonded together shall be bonded to the frame earth.

An Earth bar shall be provided along the full length of each equipment enclosure. All metallic non-current carrying parts shall be bonded together and connected for the earth bar. External M10 earthing terminals shall be provided to facilitate connection of the enclosures to the earthing system.

Panel mounted equipment with exposed metal parts such as indicating lights; pushbuttons and relays shall be separately earthed.

Provision shall be made for connecting the earth continuity conductors and cable armoring of all outgoing circuits.

Doors and hinged parts shall be bonded to the enclosure by means of a flexible copper connection having a minimum cross-sectional area of 6 mm², arranged so that it cannot be trapped as the door is opened or closed.

Connections from the main earth bar to components shall consist of green/yellow PVC sheathed stranded copper conductors. Termination lugs shall be of the compression type.

2.3 SET MOUNTINGS

The diesel engine, the generator and exciter, and the radiator shall be mounted a common welded steel bedplate. Where flange mounted generators are not employed, the bedplates shall be stress relieved before machining. Comply with requirements of the "Hydronic Piping Specialties" for vibration isolation and flexible connector's materials for steel piping and the "Metal Ducts" for vibration isolation and flexible connector materials for exhaust shroud and ductwork.

Lifting points shall be provided on the bedplate.

Suitable anti-vibration (Restrained Spring Isolators: freestanding, open-spring isolators) mounting of neoprene or linear spring type shall be supplied for the diesel generator unit, and all set mounted ancillary apparatus. Vibration isolation devices shall not be used to accommodate misalignments or to make bends.

Flexible connections shall be provided to all exhaust, air, fuel and water piping to avoid fracture due to vibration and to minimize conduction of noise. Raceway connections to the generator shall be in liquid tight flexible metal conduits.

Any necessary foundation bolts shall be supplied under this Contract.

The generator and any associated enclosures shall include anti-vibration mountings to reduce the transfer of vibration from the generators to the building structure. Mounts shall be designed to withstand seismic forces.

2.4 SAFETY GUARDS

All moving parts shall be adequately guarded in accordance with BS 1649 and to the satisfaction of the Engineer, in order to prevent danger to personnel.

2.5 GENERATOR SET CONTROL EQUIPMENT

The control equipment is designed to start and stop the set automatically from a remote signal. The signal will be issued by the Automatic mains failure panel as described elsewhere.

The generator set control equipment shall be set mounted and shall be based on microprocessor based system to provide maximum system flexibility. The generator set mounted control panel should have continuous operating temperature range of -20°C to +70°C. The control panel shall be of the electronic modular type utilizing environmentally sealed, solid state, micro-processor based modules for engine control and AC metering. Metering accuracy shall be 0.5% true RMS. Shall confirm to below standards:

- UL recognized
- CSA 22.2 No. 100,14,94
- CE certification
- 98/37/EC Machinery directive
- BS EN 6024-1
- 89/336/EEC EMC Directive
- BS EN 50081-1 Emission standard
- BS EN 50082-2 Immunity standard
- 72/23/EEC LVD Standard
- IEC 529 / 60034-5/61131-3
- MIL STND 461

Generating set, instruments, protection and controls are to be mounted preferably in one control cubicle.

Make of the generator set mounted controller shall be same as the manufacture of the original equipment manufacturer.

The control function shall be provided with an STOP / RUN / AUTO soft touch button with tactile feedback.

2.5.1 The following digital measuring instruments with liquid crystal display shall be fitted:

- an ammeter - simultaneous display 3 phase
- a voltmeter - simultaneous display of 3 phase L-L and L-N
- a 3 position ammeter/voltmeter selector switch
- a frequency meter, in Hz and rpm
- Service-hour running counter
- Power factor meter
- Coolant Temperature
- KW (total & per phase)
- KVA, KVAR, KW-Hr, kVAR-Hr
- Oil Pressure
- System Diagnostic
- Fuel Gauge
- Battery Voltage
- Battery charger, on/off switch, pilot lights
- Crank attempt and successful crank counter
- Anti- condensation heater
- Enhance Engine monitoring

2.5.2 Automatic shut-down and lock-out circuits shall be fitted for:

- Low oil pressure
- Over voltage / Under voltage (27/59)
- Over / Under frequency (81 O/U)
- Phase Sequence
- Low coolant level
- Over Current (50/51)
- Reverse Power- Active/Reactive (32/32RV)

2.5.3 Below controls should be provided

- Run / Auto / Stop control
- Speed and voltage adjust
- Local and remote emergency stop
- Remote start/stop
- Cycle crank
- Engine Control Switch
- Cool-down time setting controls
- BMS integration capability on MODBUS-RTU protocol

2.5.4 Engine Protection

- Control switch not in Auto (Alarm)
- High Coolant temp (Alarm and shutdown)
- Low coolant temp (Alarm)
- Low coolant level (Alarm)
- High Engine oil temp (Alarm and shutdown)
- Low, high and weak battery voltage (Alarm)
- Over Speed (Alarm)
- Over crank

2.5.5 16 light – 32 event local annunciator with LED type and sealed microprocessor control shall be provided as per the generator set mounted control panel. Following indications shall be possible:

- Low oil pressure / filter blocked / temperature high alarm and shutdown
- High water temperature alarm and shutdown
- Over speed shutdown
- Ready to start / Stopped / over crank
- Low/High / Weak Battery voltage
- Generator supply available / on load
- Electrical trip / Generator over voltage alarm and shutdown
- Driver coolant low level alarm and over-speed alarm and shutdown
- Generator fail to start / Not available to start alarm
- Emergency stop
- Fault shutdown/Alarm.
- 2 spare external faults.

- 2.5.6 Manual reset provision shall be provided on the control panel. It shall be possible to restore operation only after manual reset.

Simultaneously, an audible alarm shall be given, and the cause of the stoppage shall be displayed and remain displayed on the control panel and the building management system until the alarm device is manually re-set.

When the control selector switch is in the RUN (MANUAL) position, the audible alarm shall be operative.

The control panel shall have a self-diagnostic feature to assess the condition of the various senders and sensing components.

A proprietary annunciator system shall be utilized for visual and audible display of alarms and shutdowns.

Provide an volume sounder at panel with silencing (override) switch to silence alarm. Override switch shall have flashing pilot lamp labeled "Override" to indicate that alarm is silenced. Actuation of the Volume Override shall not cause the indicator lamps to change status until the cause of the fault/warning has been removed.

Provide primary power for the local alarm system from a local LV supply. Provide a separate battery backed secondary supply to maintain alarm functions in the event of a mains failure to the panel as per manufacturer's standard. Panels to include Battery chargers suitable for re-charging the batteries on restoration of the mains supply.

- 2.5.7 An adjustable run on "off load" timer, shall be fitted to provide cool down running.

All individual circuits such as voltmeter, shutdown circuits etc. shall be adequately protected by fuses.

All control wiring shall be carried out in Thermoplastic insulated cable termination and relative terminals shall be clearly marked. Terminal boards having pressure type terminal lugs or equivalent shall be used so that no terminal clamp is necessary. Terminal boards shall have pairs of terminals for Incoming and Outgoing wires and not more than two wires shall be connected to any one terminal.

Insulating barriers shall be provided between adjacent connectors. Labels for wiring designation marks shall be provided on the fixed portion of the terminal boards as well as wires. No live metal shall be exposed at the back of the terminal boards.

Engine wiring shall be protected by non-metallic conduit.

The control equipment shall also include a breaker as shown on drawings complete with short-circuit magnetic and thermal overload trips, separately housed in a sheet steel enclosure adjacent to the control panel. A shunt trip & Auxiliary contacts shall also be provided.

Separate busbar cubicle must be provided to enable termination of multi-core cables.

- 2.5.8 Main Line Circuit Breaker

Main line circuit breaker which shall be 4-pole, operate both manually as an isolation switch and automatically during overload, Fault and short circuit conditions. The trip unit for each pole shall have elements providing adjustable inverse time delay features during overload conditions and adjustable instantaneous magnetic tripping for short circuit protection. Circuit breaker shall be suitably rated. The circuit breaker shall be provided with an auxiliary contact to provide a warning indication should the

breaker be in the “OFF” or “tripped” position. The warning indication shall be displayed on the control panel and shall have remote indication to the BMS

- B. The circuit breakers shall be mounted in an IP54 (dust tight and suitable for humid conditions) generator mounted cabinet. The cabinet shall be formed frame design to support the weight of the installed devices. Cabinet shall be welded construction, no bolt together structures will be allowed. Box material shall be minimum 1.6mm construction. Cabinet will be provided with a bolt-on front access door. Cabinet shall be finish painted to match the generator. Circuit breaker shall be equipped with load side, full rated bus-bars to accommodate 2-hole, long barrel copper compression lugs for the cables specified. A neutral disconnect link shall be provided for the neutral connection. Bus shall accept fully rated feeders for neutral and ground/earth connections. Neutral and ground/earth connections shall be isolated from the box frame.

Circuit breaker shall be equipped with a minimum two auxiliary contacts (NO/NC) rated at 240V 50A and a 24 VDC-shunt trip.

A light shall be provided for local annunciation. The mainline circuit breaker shall be cabled from the line side to the alternator. The circuit breaker shall be provided with identification labelling.

2.5.9 Over-current and Fault Protection

Generator Circuit Breaker: Moulded case Circuit Breaker or Air Circuit Breaker, thermal magnetic-trip type; suitably rated; complying with required BS EN .

Generator Protector: Microprocessor-based unit shall continuously monitor current level in each phase of generator output, integrate generator heating effect over time, and predict when thermal damage of alternator will occur. When signaled by generator protector or other generator-set protective devices, a shunt-trip device on the moulded case circuit breaker/Air circuit breaker shall open the device to disconnect the generator from the load circuits. Protector shall perform the following functions:

1. Initiates a generator overload alarm when generator has operated at an overload for 60 seconds. Indication for this alarm is integrated with other generator-set malfunction alarms.
2. Under single or multiphase fault conditions (including faults to earth), regulates generator output to 300 percent of rated full-load current for 10 seconds.
3. As overcurrent heating effect on the generator approaches the thermal damage point of the unit, protector switches the excitation system off, opens the generator disconnect device, and shuts down the generator set.
4. Senses clearing of a fault by other overcurrent devices and controls recovery of rated voltage to avoid overshoot.

Ground-Fault Indication: Comply with NFPA 70, "Emergency System" signals for ground-fault. Integrate ground-fault alarm indication with other generator-set alarm indications.

2.6 FUEL SYSTEM

The engine to be supplied shall operate on fuel of the light distillate type. The genset shall be provided with a day tank or base tank suitable for 8 hours continuous operation at full load. A level switch also provided for the day tank and wired to genset control panel for low fuel level warning.

Day Tank: Freestanding, elevated (gravity) factory-fabricated fuel tank assembly, with double wall construction to protect from fuel leaks or ruptures, with integral, float-controlled transfer pump and the following features:

- a) Containment: Integral rupture basin or proper bonding (bund walls) for tanks with a capacity of 150 percent of nominal capacity of day tank. Leak Detector shall be located in rupture basin and connect to provide audible and visual alarm in the event of day-tank leak.
- b) Tank Capacity: As recommended by engine manufacturer for an uninterrupted period of up to 8 hours' operation at 100 percent of rated power output of engine-generator system without being refilled.
- c) Pump Capacity: Exceeds maximum flow of fuel drawn by engine-mounted fuel supply pump, including fuel returned from engine.
- d) Low-Level Alarm Sensor: Liquid-level device operates alarm contacts at 25 percent of normal fuel level.
- e) High-Level Alarm Sensor: Liquid-level device operates alarm and redundant fuel shutoff contacts at midpoint between overflow level and 100 percent of normal fuel level.
- f) Diesel tank level indication (sight glass) shall be provided in the assembly.
- g) Piping Connections: Factory -installed fuel supply and return lines from tank to engine; local fuel fill, vent line, overflow line; and tank drain line with shutoff valve.
- h) Redundant High-Level Fuel Shutoff: Actuated by high-level alarm sensor in day tank to operate a separate motor control device that disconnects day-tank pump motor. Sensor shall signal solenoid valve, located in fuel suction line between fuel storage tank and day tank, to close. Both actions shall remain in shutoff state until manually reset. Shutoff action shall initiate an alarm signal to control panel but shall not shut down engine-generator set.

Fuel fill system shall be considered only for areas where the generator room is at an inaccessible location for filling fuel directly from a fuel truck. Fuel fill connections to the Daily service tank shall be equipped with in line filters having replaceable elements which may be easily removed from their housing for replacing without breaking any fuel line connections or disturbing the fuel pumps or any other parts of the engine. Separate fuel tank room shall be provided for higher rating generators as per NFPA and DCD requirement.

Fuel lines between injection pumps and valves shall be of heavy grade seamless tubing and to eliminate irregularity of fuel injections shall be of the same length for all cylinders on a single engine.

Provide fusible link type safety shutoff valves which are to fail close. Shut-off valves to be provided on:

- 1. Automatic fuel fill systems for the daily services tank
- 2. Remote fuel fill systems for the daily services tank.
- 3. Fuel supply from daily service tank to engine

Each of these shall be connected via an interface to the fire Alarm systems such that an alarm is raised in the event of their operation.

The fuel system shall be equipped with a fuel filter having replaceable elements which may be easily removed from their housing for replacing without breaking any fuel line connections or disturbing the fuel pumps or any other parts of the engine. All fuel filters shall be conveniently located in an accessible housing, ahead of the injection pumps, so that the fuel will have been thoroughly filtered before it reaches the pumps. No screens or filters requiring cleaning or replacement shall be used in the injection pump or injection assemblies. Filters shall be duplex type and have a valve arrangement so filters can be changed during operation.

Provide crankcase fumes reclamation system. System shall collect engine crankcase emissions, filter out airborne lube oil, and reintroduce the emissions back into the engine combustion air system

The tank shall be supplied and installed, fully piped up, to provide the service requirements of the genset and as shown on drawings. The tank shall be complete with filler cap, contents gauge, overflow pipe, vent pipe and sludge valve. Contractor shall fill tanks with diesel for the generator for initial operation testing and commissioning. Fuel tank to be filled with diesel prior to handing over.

Fuel handling system shall facilitate ease of diesel tank refilling from a filling point located outside room, considering the vehicle access to the area near the filling point. System shall be in line with authority requirements. Please refer Plumbing drawings for exact location.

2.7 SOUND INSULATION OF THE ROOM AND VIBRATION CRITERIA

Contractor shall appoint Specialist Acoustic treatment Contractor for designing and installation of air inlet attenuator, outlet attenuator, louvers, critical grade silencers, and accosting linings for generator room and associated doors to provide a satisfactory generator installation to reduce the noise specified Noise level as per BS 8233:1999 (sound insulation & noise reducing)

The contractor shall provide sound level data of the proposed equipment for the consideration of the Engineer. The noise level outside the genset room shall not exceed 65 dBA at 1 meter . Attenuators to be provided for radiator discharge and air intake. The air intake requirements shall be coordinated at site.

The vertical noise propagation to the floor above should not exceed 55dB.

The generator base mounted springs should be able absorb 95% of the Engine vibrations. Any louvers provided in generator room shall be acoustic type.
IBC certified isolator shall be preferred

2.8 EXHAUST

All exhaust gases shall be suitably piped outside the genset room. The engine shall be efficiently, silenced and be complete with a silencer and all fixings to attach the silencer to the generator room ceiling. In addition to erecting the silencer the Contractor shall also be responsible for providing and erecting the fittings necessary to carry the tail pipe through the generator room wall. All fixings shall be anti-vibration type where required and galvanized. Exhaust pipes of the proper diameter, properly insulated with 50mm thick rock wool insulation and clad with 22G aluminum sheet, shall be supplied and installed. They shall run from the exhaust silencers to the outside. The exhaust pipes shall terminate with an adequate bend or cap to prevent ingress of rain.
The total exhaust noise attenuation shall be 25dBA.

2.9 BATTERY STARTING

The equipment shall include an adequately rated low maintenance lead acid starter battery together with a suitably rated charger. The battery charger will be supplied from the mains (standby supply) and an engine driven dynamo shall be used for boost charging. The battery shall be protected against accidental, mechanical or electrical damage and shall be housed in a purpose built rack.

The charger shall be mounted remote from the engine and shall include a battery charge fail alarm.

Without power supply to the charger, the battery shall be capable of providing no fewer than four starting cycles within a period of 5 minutes. Upon restoring power supply to the charger immediately thereafter, the charger shall be capable of restoring the battery, within 30 minutes, to a condition in which one more starting cycle can be completed.

The following instrumentation and protection shall be provided:

- * Charger failure
- * Battery voltmeter
- * Charge / discharge ammeter.
- * On/off switch.

2.10 CABLES

All cables shall be supplied in accordance with local Electricity Authority and civil defense regulations. A schedule of loads and cable sizes shall be submitted to the Engineer for approval prior to commencement of any work.

2.11 SELECTION

The capacity of Generator is shown for guidance Contractor shall submit Generator selection calculation from the proposed Vendor for approval and upgrade the size if required to meet the starting sequence and requirement of the type of loads connected, without any variation.

When the smoke fans are in operation during a fire situation all lifts shall be shut off automatically except firemen elevator. One no. passenger lift shall be kept in operation along with firemen elevator during normal testing sequence of smoke fans, generators etc. Final selection of sequence and operation shall be to the Engineers approval at site.

Coordinate with lift supplier to provide a control logic and to ensure 50% of the lifts in the respective bank or 50% of the lifts in one group will operate during power failure condition.

3.0 EXECUTION

3.1 INSTALLATION

Install equipment in accordance with the manufacturer's specification approved shop drawings and the contract document.

3.2 TESTING AND COMMISSIONING

3.2.1 Inspection and Testing

A. Static load test

During this test the generator set shall run for at least three hours under the following conditions:

- 30 minutes at 25% rated output.
- 30 minutes at 50% rated output.
- Two hours at 100% rated output.

During the load test all engine and generator parameters shall be recorded. These shall include, but not be limited to:

- Generator power output
- Generator phase currents
- Generator voltages and variations
- Generator power factor
- Generator frequency
- Engine oil pressure
- Cooling water temperatures
- Noise level

- Engine speed and variations
- Harmonic content
- Battery and battery charger
- Exhaust emission
- Back pressure

SAT test shall include 8 Hours continuous running with Full load.

B. The frequency and voltage regulation of the generator shall be verified. Voltage and frequency variations shall be recorded on oscillograms. The requirement for special dynamic load tests to simulate motor starting will be indicated in the requisition.

C. Functional test

Functional tests shall be performed on the complete generator set including the generator and engine control panels. Coordinate tests with tests for transfer switches and run them concurrently.

The functional tests to be performed shall be subject to approval by “the Engineer” and shall include, but not be limited to the following tests:

- Engine start up system, including measurement of the starting time. A minimum of two start and stop tests shall be performed.
- Automatic start up system.
- For emergency generator sets verification of the time between initiation of a start command and acceptance of the nominal load for a cold start.
- All generator alarm and trip functions, including the verification of protection relay operation.
- Operation of the battery charger.
- The status /alarm/trip/shutdown indication of the control panels.
- Local manual start/stop/emergency stop functions.
- Remote start/stop/emergency stop functions (input signals).
- The remote signaling contacts for Principal’s use
- Operation of the AVR
- Operation of synchronizing equipment (as applicable)
- Calibration of measuring and protective devices.

D. Insulation resistance tests

These tests shall be performed on the generator main stator.

E. Hydrostatic testing

All pressure containing parts, including auxiliaries, shall be factory tested hydrostatically in accordance with the relevant design codes.

F. Final Inspection

The following inspection and checks shall be performed:

- Verification against the approved drawings
- Overall dimensions of the skid and the location of all equipment
- Location of all piping and electrical interfaces
- Enclosure degree of protection of all equipment concerned
- Accessibility of components
- Lifting arrangement, including spreader bar, if applicable.
- Noninterference of the lifting arrangement
- Availability of eye bolts for lifting heavy components
- Terminal and wiring marking
- Available space for cable termination, the size and number of terminals, the cable supporting devices.
- Earthing and bonding
- Marking of components according to the relevant drawings
- All rating plates of main components.
- Construction materials of auxiliaries, such as base plate, piping, fittings, coolers, fuel day tank (for diesel engines) and control panels.
- Hook up of local instruments.
- Packagers welding procedures and welder qualifications
- Welding by non-destructive testing, where specified.
- Weld inspection reports
- Quality assurance documentation.

3.2.2 Work test

- A. These tests shall be conducted at manufacturer's authorized supplier/distributor premises in order to determine whether the material and apparatus comply with this specification.

3.2.3 Type Tests

- A. Tests shall be carried out on generator set using reactive load bank. For the purpose of these tests the rated power shall be as specified. All apparatus, instruments, fuel and lubricating oil for these tests shall be provided by the supplier.
- B. Governor tests in accordance with an appropriate national standard and to check the drop.
- C. Cold and hot starting tests.
- D. Tests on trips and alarms incorporated in the engine.
- E. When type tests have previously been carried out, the Contractors may submit accredited copies of the results instead of carrying out the tests again.

3.2.4 Routine Tests

The engine shall be run in the works to demonstrate satisfactory performance and the correct operation of all control ancillary apparatus.

The engine shall be run at full load for a sufficient period of time to enable steady conditions to be obtained, following which a run at 100% load shall be carried out with inductive load bank at 0.8 P. F at manufacturer's original assembly and report to be issued to engineer for verification prior to dispatch (FAT).

3.2.6 Electrical

- A. Load tests shall be carried out on the diesel generator unit complete to prove the efficiency at the loads stated in Schedule.
- B. A full load rejection test shall be carried out on the set to prove the capability of the excitation system in limiting the transient voltage swing.
- C. Routine tests shall be carried out in accordance with IEC recommendation or agreed National Standards.
- D. The Engineer reserves the right to call for such additional tests on any equipment as may be necessary to prove compliance with the specification.

3.2.7 Site Tests

Such tests as are required by the "the Engineer" to prove compliance with Specification independently of any tests which may already have been carried out at the manufacturer's works including circuit continuity, phasing, insulation resistance earth resistance tests and 100% load test carried out with resistive load bank at unity P.F with site-built installation.

After installation, charge exhaust, coolant, and fuel systems and test for leaks. Repair leaks and retest until no leaks exist.

After electrical circuitry has been energized, start units to confirm proper motor rotation and unit operation for generator and associated equipment.

3.2.8 Testing shall include static load tests, functional tests and visual inspection as specified.

Report results of tests and inspections in writing. Record adjustable relay settings and measured insulation resistances, time delays, and other values and observations. Attach a label or tag to each tested component indicating satisfactory completion of tests.

The Contractor shall supply all fuel for the testing. Upon acceptance by the Architect/Engineer, the day tank shall be filled to capacity after each testing.

Load banks shall be used for full load testing of the generator at manufacturer authorized supplier/distributor premise.

END OF SECTION

ELECTRICAL SECTION: 04

BUS BAR SYSTEM

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1.0 GENERAL**1.1 REFERENCE**

Conform to General Requirements for Electrical Services.

1.2 DESCRIPTION OF WORK

- 1.2.1 Supply, installation, testing and commissioning of complete bus bar system as specified here after and with ratings as indicated in the relevant electrical drawings and compliance with local codes and regulations. The proposed trunking shall be tested by a certified body (KEMA-DEKRA Or ASTA) to IEC61439-6 as a type tested assembly.

1.3 QUALITY CONTROL/ASSURANCE

Design, manufacture, testing and method of installation of all apparatus and materials furnished under requirements of these specifications shall conform to latest publications or standard rules of the following:

- A. Dubai Water and Electricity Authority (DEWA) regulations for electrical installation works and Department of Civil Defense requirements.
- B. Regulations for the Electrical Equipment of Buildings (IET), (BS) London.
- C. The manufacturer should have a proven track record of minimum 10 years in UAE in high rise buildings and malls.
- D. The manufacturer should have a Third Party Quality Control certification like KEMA KEUR or ASTA DIAMOND besides ISO 9001 and ISO 14001
- E. Manufacturer should have environmental certificates like GREENGUARD for his premise and all proposed products

The Quality Control/Assurance/Assurance/Quality Assurance duties shall be performed by the Contractor. The system of Quality Control/Assurance/Assurance verification shall be in accordance with ISO 9000 standards of quality assurance.

1.4 CODES AND STANDARDS:

The following codes and standards to the extent specified herein, form a part of this specification. When an edition date is not indicated for a Code or Standard the latest edition enforces at the time of proposal submitted shall apply

- 1. ASTA / KEMA certified and type tested assembly to IEC61439-6.
- 2. IEC 61439-1 Low-voltage switchgear and control gear assemblies – Part 1: General rules.
- 3. IEC 61439-2 Low-voltage switchgear and control gear assemblies – Part 2: Power switchgear and control gear assemblies
- 4. IEC 60439-1 Low voltage switchgear & control gear assemblies – Part 1 Power switchgear and control gear assemblies
- 5. IEC 61439-6 Busbar Trunking Systems: Fault-loop resistances and reactance's
- 6. IEC60529 Degrees of protection provided by enclosures

7. IEC60068 Environmental testing
8. IEC61140 Protection against electric shock – Common aspects for installation and equipment – Basic safety publication
9. IEC 51 Indicating measuring instruments.
10. BS 159 & IEC 695-2-1 Bus bar & connections.
11. IEC 60331-1- Circuit Integrity for 3 Hours under Fire conditions
12. BS EN 10346 - Continuously hot-dip coated steel flat products for cold forming. Technical delivery conditions.
13. Quality certification mark UL® or KEMA-KEUR or ASTA Diamond certifying quality in manufacturing.

1.5 SUBMITTALS

1.5.1 MANUFACTURERS TECHNICAL DATA

The contractor shall submit the following drawings and documents related with each electrical system for Engineer's approval:

- A. Original catalogues for various proposed components.
- B. Schedule of Bus bar and accessories.
- C. Test Certificates.

During the time of the contract and before substantial completion of the electrical installation, submit three (3) copies of descriptive literature, technical data, maintenance recommendations (from the equipment manufacturer), catalogues etc.

1.5.2 DRAWINGS AND INFORMATION:

A drawing pouch / pocket shall be provided in each panel and laminated load schedules and approved schematic drawings (Framed) shall be providing.

The contractor / vendor shall furnish the following drawings and documents along this submittal for approval.

- A. 3D Dimensioned drawing showing bus bar positioning and important external features.
- B. Original catalogues for various proposed components.
- C. Drawings of nameplate and all termination arrangements.
- D. Drawing giving typical mechanical and electrical details of the voltage control apparatus.
- E. Details of cable terminations and fittings.
- F. Installation details.
- G. Schematic and connection diagrams covering all equipment pertaining to the switch board.
- H. Approved test certificate / test report.
- I. Schedule of Accessories and Fittings.
- J. Technical Manual giving installation, operation and maintenance instructions.
- K. Schedule of recommended spare parts.

2.0 PRODUCTS

2.1. BUS BAR SYSTEM

- 2.1.1 Busbar trunking shall be installed so that it is inaccessible to unauthorized persons and totally enclosed in an earthed trunking duct. The proposed trunking shall be tested in a ASTA / KEMA / LOVAG certified or equivalent authorized laboratory as a type tested assembly under IEC61439-6 standards for Low voltage busway. A product safety mark (e.g. KEMA-KEUR, ASTA DIAMOND) should be on the product offering a visible assurance of full product safety testing, factory inspection and ongoing surveillance under independent authority to ensure the ongoing safety of product.
- 2.1.2 The conductors shall be manufactured from HD HC 99.9% Pure (100% conductive) silver plated copper and shall comply with BS 6071 or equivalent standard. Current rating shall be as shown on drawings as a minimum and upgrading the ratings due to derating shall be by contractor at no extra cost. Conductor size shall be based on current carrying capacity, voltage drop and temperature rise limits with full type test certification and local code regulations. Documents substantiating the same should be provided for review and approval. Use of bi-metallic conductors is explicitly prohibited. Only those busbar cross sections that the manufacturer has successfully type tested their product to IEC (see clause 1.4.1&2) standards can be proposed on the project.
- 2.1.3 The bus bar trunking system shall be of sandwich type construction without any air gap and designed to operate on a 400/230V 3ph 5bar and 50Hz supply system. The complete system shall comply with the requirements of IEC (see clause 1.4.5). Complete test certificates as per latest IEC requirements shall be submitted as part of the Technical Submittal for review. Busbar trunking system shall be minimum 600/1000V rated.

The earth conductors shall be properly bonded with the housing with at least 50% the size of the phase conductor and bonded with the joints as well as adjoining busbar piece.

- 2.1.4 The continuous-rated current shall be according to the designed load of each system and type tested for its rated current at ambient temperature at 50°C. The maximum temperature rise on the surface of the conductor should be ≤ 55 Deg K over an ambient temperature of 50deg C. The minimum short circuit withstanding capacity for one second shall be 50kA for main Busbar. Contractor to select the busbar cross section to satisfy the voltage drop of maximum 2.5% @ R 85 requirement on every run prior to submission for approval.
- 2.1.5 The conductors shall be of high conductivity silver plated copper bars of 99.9% purity the manufacturer shall product a mill certificate proving conductivity for each project. They shall be colour coded and hybrid (epoxy + mylar) insulated with Class B insulation over the entire length. Insulation voltage shall not be less than 1000V. Insulation material shall be self-extinguishing and non-fire propagating type for non-flammability. Manufacturer's data of the epoxy should be furnished to ensure compliance to this clause. These characteristics shall be certified by an independent testing authority member of LOVAG, ASTA/KEMA/DEKRA or equivalent. Neutral bars shall have the same cross section as phase conductors. An earth bar of half the phase bar size shall be provided as 5th bar. The busbar shall be totally enclosed to provide protection against mechanical damage and dust accumulation.
- 2.1.6 External fire barriers shall be provided for busbar trunking, and shall also be positioned when it passes through fire insulating ceilings and walls. These fire barriers shall be effective in the space between the bus duct and the protective sleeve of the ceiling.
- Fire barrier, when it is installed shall prevent fire, smoke and hot gases passing into the next storey and according to Civil Defense regulation.
- 2.1.7 The continuous busbar system shall be established as specified on the drawings. The feeder busbar including all joints shall be water proof to minimum IP-55 for vertical and IP 65 for the horizontal section. An independent testing authority member of LOVAG, KEMA / DEKRA/ ASTA or equivalent body shall

certify the degree of protection. In addition, the busbar trunking system should be tested for water spray test and be sprinkler-proof for 1 hour under live conditions. Busbar housing shall have provision of drain holes to avoid accumulation of water (in case of water leakage or sprinkler activation) within housing. Any length of the assembled bus ducts shall be possible to remove without affecting the other lengths, connected at the edges, to facilitate future additions and maintenance work. All the bolted and plug-in joints and terminations shall be checked for proper contact pressure; no loose connection shall be allowed in the system.

- 2.1.8 Vertical bus bars shall be IP 55 with IP 55 plug in feeders. All Horizontal bus bars shall be IP 65. All plug-in MCCBs shall be current limiting type. The breakers shall be proposed from the specified switchgear manufacturers. Busway vendors have to supply MCCBs that they have type tested product with and furnish details of relevant IEC type tests indicating Tap off units with MCCBs have been tested in conjunction with the busway.
- 2.1.9 The bus bar system shall be supported throughout the entire length as per manufacturer's requirements to withstand its own weight and abnormal forces created on short circuit conditions. The effect of Thermal expansions during short circuit conditions should be managed by Expansion Units in vertical installation. Bus bar expansion fittings shall be provided at all building expansions joints and for long lengths as recommended by the manufacturer. Spring type hangers (without penetration in to busway housing) shall be used for fixing of bus bars when they run vertically. Rigid type supports for vertical installations are explicitly not allowed. It should be possible to remove any piece of busbar without the need of any special hoist to support the busbar. Purpose made fittings, supports and accessories should be used throughout the installation.
- 2.1.10 The tap off boxes used at bus plug in units shall be of the current limiting MCCB with the type & ampere sizes and ratings shown on the plans. The tap off lock shall have a safety type so interlocked with the bus way that they cannot be added or removed unless the switching mechanism is off. The plug-in opening shall have automatic shutter design and be finger-proof to ensure the equipment operator safety. Plug in openings shall be additionally covered with padlock able hinged door to ensure unauthorized access.
- 2.1.11 Plugs shall be equipped with red/green indicator flag clearly visible to indicate the 'ON or OFF' position of the switching mechanism. It shall be possible to insert plug-in unit (no-load) while busbar trunking is under live condition (hot plug design). Plug in unit up to 400A shall use plug-in jaw connections. Bolt-on connections or Plug-in unit connected at joint is not permitted. Plug in jaw and breakers shall be connected using solid copper link. Cable connection are not permitted.
- 2.1.12 The contractor shall be responsible for routing the trunking to co-ordinate with other services. The contractor prior to release of the trunking for fabrication shall make final measurements. Bus ducts should be covered with plastic envelope until building is clean and the bus bars are ready to be tested and energized. Post removal of these envelope all surfaces and joints should be cleaned (without the use of pressurized water or any solvent) and vacuumed to ensure all debris and any existing condensation are removed,
- 2.1.13 Proper arrangement shall be provided to ensure that fire in one level is not propagated through the busbar to the next level. Fire barriers shall be provided to effectively achieve the same.
- 2.1.14 Busbar system shall be certified to comply with type tests as specified by LOVAG, IEC (see 1.4.5)..
- 2.1.15 The enclosure shall be boltless rivetted or bolted design or approved equal in compliance to the standards and should have passed an impact rating to IK10 as per IEC (see 1.4.5) standards with evidence furnished in the submittal.

- 2.1.16 The proposed manufacturer has to carry out voltage drop calculation based on the voltage drop data. The maximum voltage drop allowed from source to final outlet is 4%. The vendor and contractor to upgrade the busbar size if they cannot meet the voltage drop conditions without any additional cost to client.
- 2.1.17 The Cross – area section of busbar should be adequate and must be substantiated with necessary calculations based on DEWA regulations. Conductor size shall be based on voltage drop, temperature rise limits with complete type test certification and local code regulations. Documents substantiating the same to be provided for review and approval.
- 2.1.18 Complete site installation including alignments, joints, meggering etc. must be certified by the manufacturer representative and signed off by Contractor and Vendor prior to offering the installation for commissioning.
- 2.1.19 All busbar up to 2500A shall be single busbar per phase construction.
- 2.1.20 The busbar shall be of true sandwich configuration. No air gaps between conductors and between conductors and enclosure shall be allowed.

2.2 TAP OFF

Tap Off boxes for rating up to 400A shall be Plug -in type suitable for MCCB. Higher tap off rating shall be considered in critical cases with dedicated support and tap off rating shall be limited maximum 800A.

The tap off box enclosure shall be constructed from galvanized sheet steel with paint finish, and it should have hinged front cover. It shall incorporate an external operating handle with ON, OFF, TRIP indication for MCCB, enabling it to be operated with the cover closed. The MCCB shall incorporate padlocking facilities in OFF position for maintenance purpose. The plug-in units shall be suitable for mounting on front of the riser as required. The unit shall be supplied with all connections and accessories to enable connection to the busbars and the outgoings cable to be correctly and safely achieved. Dedicated intermediate support to be considered for all tap off points.

The units shall have clamps with interlocking facility to prevent the unit being removed from the busbar when MCCB is in the ON position. Additional interlock to be provided to prevent opening of the door of plug in when the MCCB is in the ON position.

The MCCB shall be connected to the Plug-In unit with copper busbars. MCCBs to be Switched 3 Pole with solid Neutral & ground bars within the Tap-off secured firm.

The tap off to be with earth terminal connection to the earth bar by plug in contact.

The earthing contact shall make first when tap off is plugged in.

2.3 HOUSING

The external casing shall be extruded aluminum and shall be treated in accordance to BS EN (see 1.4.12) regulation with thickness not less than 3.0mm and to be shaped in such a way as to give excellent electromechanical characteristic and heat dissipation and to guarantee total safety against contact with live parts. Continuous Thermal indication & monitoring facility shall be considered. Metal Housing shall be accepted with standard manufacturer finish. Housing must be act as Integral earth

(75% Earth capacity) by having one continuous piece (zero break point) without bolting on housing. Use of Aluminum is preferred over steel since steel housing in part or full is heavier and has a lower heat dissipation properties and hysteresis losses.

The construction shall be of sandwich configuration to prevent the circulation of air inside the casing, all plastic material to be self extinguishing and non fire propagating.

Housing to be tested to IK10 mechanical impact test rating under IEC (see 1.4.5).

2.4 EARTH BAR

The earth bar shall be inside the busbar enclosure.

The earth bar should be in pure copper and at least 50% of the phase busbar and should run along both sides or single side of the Busbar system and bonded to the busbar enclosure.

2.5 ACCESSORIES

The manufacturer should have a standard range of accessories including horizontal and vertical elbows, T elements and double Elbows, connecting elements to switchgear and transformer.

2.6 JOINT CONNECTIONS

Each busway section shall be furnished complete with a factory installed joint. The joint pack shall be a single bolt, non-rotating design which shall be removable from the side without disturbing adjacent sections of the busway. All bridge joints shall be furnished with torque-indicating, double-headed shear off bolts. The bridge joint shall utilize a captive nut retainer on the opposite side of the torque-indicating bolt.

The joint design shall ensure proper installation without the use of a torque wrench and provide visual indication that the joint is properly secured. Each bridge joint shall allow for a, minimum to maximum, 25mm (12.5 at both sides) adjustment in section length at each joint connection. De-energization of the busway shall not be required for safe testing of joint tightness.

Joints shall be accomplished by means of an insulated bolt passing through the conductors. Joint bolt shall provide an easily detected visual indication that the bolt has been tightened. Inspection plates shall be provided to permit periodic joint examination without disturbing joint pressure.

Joints should have a breather mechanism factory installed to allow of equilibrium of air inside and outside the busway to minimize failures due to condensation that lead to degradation of the insulation over time. Contractor to ensure the joints are protected from direct exposure to any wet services and / or encapsulation within a floor or wall opening and / or in fire-sealant material in electrical openings.

3.0 EXECUTION

3.1 INSTALLATION

- A. Support bus assemblies independent of supports for other elements such as equipment enclosures at connections to panelboards and switchboards, pipes, conduits, ceilings, and ducts.

1. Design each fastener and support to carry load indicated by seismic requirements (only if mandated by authority) and to comply with seismic-restraint details.
 2. Support bus assembly to prevent twisting from eccentric loading.
 3. Support bus assembly with not less than 12mm steel rods. Install side bracing to prevent swaying or movement of bus assembly. Modify supports after completion to eliminate strains and stresses on bus bars and housings.
 4. Fasten supports securely to building structure.
- B. Install expansion fittings at locations where bus assemblies cross building expansion joints. Install at other locations so distance between expansion fittings does not exceed manufacturer's recommended distance between fittings.
- C. Construct rated fire-stop assemblies where bus assemblies penetrate fire-rated elements such as walls, floors, and ceilings. Seal around penetrations according to Division 07 Section "Penetration Fire stopping."
- D. Install weather seal fittings and flanges where bus assemblies penetrate exterior elements such as walls or roofs. Seal around openings to make weathertight. See Division 07 Section "Joint Sealants" for materials and application.
- E. Coordinate bus-assembly terminations to equipment enclosures to ensure proper phasing, connection, and closure.
- F. Tighten bus-assembly joints with torque wrench or similar tool recommended by bus-assembly manufacturer. Tighten joints again after bus assemblies have been energized as recommended by manufacturers.
- G. Install bus-assembly, plug-in units. Support connecting conduit independent of plug-in unit.

3.2 CONNECTIONS

- A. Ground equipment according to relevant Earthing Section.
- B. Connect wiring according section Wires and cables

3.3 FIELD QUALITY CONTROL

- A. Retain one of three paragraphs below that identify who shall perform tests and inspections. If retaining second option in first paragraph, or if retaining second or third paragraph, retain requirement for field quality-control test reports in "Submittals" Article.
- B. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust components, assemblies, and equipment installations, including connections. Report results in writing. Perform tests and inspections and prepare test reports.
- C. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing. Full length of the busbar and all Joints should be tested with thermographic test and reports to be witnessed by the consultant.

3.4 TESTS AND INSPECTIONS

- A. Perform each visual and mechanical inspection and electrical test. Certify compliance with test parameters.
- B. Remove and replace units that do not pass tests and inspections and retest as specified above.
- C. Infrared Scanning (Thermo graph): Immediate after Substantial Completion on power energization, perform an infrared scan of bus assembly length including entire joints and plug-in units. Tests shall be conducted on manufacturer's recommended load and not less than 70% load.
- D. Thermal cycling test as per IEC 61439-1/6 clause 10.2.102. Tested single tap off rating must be exactly half of the busbar rating. Thermal cycling test carried at IP55 minimum for tapoff box enclosure.
- E. Use the latest infrared-scanning device designed to measure temperature or detect significant deviations from normal values. Provide documentation of device calibration.
- F. Perform 1 follow-up infrared scans of bus assembly at end of DLP, i.e., at 11 months after Substantial Completion.
- G. Prepare a certified report identifying bus assembly checked and describing results of scanning. Include notation of deficiencies detected, remedial action taken, and scanning observations after remedial action. Manufacturer's endorsement on the report to be submitted.
- H. Physical Check

Check the Main busbar installation and ensure the following:

Proper installation.

Supports are adequate.

Bus bar identification and labeling.

No physical damage.

Torque Test to be conducted as per manufacturer's recommended values.

The following are the types test to be carried out on the busbar system. (Recommended to be included in the specs)

1. Verification of temperature rise limits
2. Verification of the dielectric properties
3. Verification of the short circuit withstand strength
4. Verification of the effectiveness of the protective circuit
5. Verification of Clearance & creepage distances
6. Verification of mechanical operation

7. Verification of the degree of protection
8. EMC Tests (as applicable)
9. Verification of the resistance of insulating materials to abnormal heat and fire
10. Verification of the electrical characteristics
11. Verification of structural strength
12. Verification of crushing resistance
13. Verification of resistance to flame propagation
14. Verification of the fire resistance in building penetration
15. Verification of the endurance of trunking systems with trolley type tap-off facilities (if applicable)

- I. Each piece to be tested prior to installation for Insulation Resistance Test:

Insulation resistance test as per manufacturer recommendations, and test requirements.

Measure insulation resistance between cores of all phases with a 500/1000 V Insulation Tester (as per engineer approval) and tabulate readings, the readings to be verified with permissible limits (min. 10 Mega ohms)

Measure insulation resistance between cores and ECC and earth terminal with a 500/1000 V Megger (as per engineer approval) and tabulate readings. The readings to be verified with permissible limit.

- J. Phase Rotation Test- Phase rotation at all points should be same otherwise rectify.
- K. Seismic restraint Details to be provided by a qualified professional Engineer including the design Calculation requirements for selecting seismic restraints.
- L. Manufacturer Seismic Qualification Certification:

Submit certification that enclosed bus assemblies, plug-in devices, accessories, and components will withstand seismic forces for the Seismic zone applied in the location of the project.

3.5 ADJUSTING

- A. Set field-adjustable, circuit-breaker trip ranges as per discrimination study and as indicated in the authority approved layout.

3.6 CLEANING

- A. Vacuum dirt and debris; do not use compressed air to assist in cleaning.

3.7 PROTECTION

- A. Provide final protection to ensure that moisture does not enter bus assembly.

END OF SECTION

ELECTRICAL SECTION: 05

CABLES & WIRES

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1.0 GENERAL**1.1 REFERENCE**

Conform to General Requirements for Electrical Services

1.2 DESCRIPTION OF WORK

1.2.1 Supply, Installation, testing, and commissioning of complete cables and wires system as specified hereafter and as indicated in the related electrical system drawings.

1.2.2 Unless otherwise specified all wires and cables enclosed / concealed in conduits or trunking or the like, shall be single core PVC insulated 450 / 750V grade and the installation shall be carried out on the looping-in system with continuous mains throughout each circuit. All exposed cables in common public escape corridor including (ceiling void) shall be LSOH and other areas shall be PVC insulation. All connections to fittings and the like shall be of a solid nature and small connectors (other than porcelain enclosed heavy brass sleeves) will not be allowed. Porcelain connectors enclosing heavy brass sleeves shall be provided for heat resisting trails for enclosed fittings and for the drops for spot lights only.

1.2.3 All connections to flexible cords or cables shall be such that any strain on the cord or cable is not transmitted to the connections.

1.2.4 All wires/cables connections to switches, socket, circuit breaker etc. shall be carried out using suitable swivel lugs or cable ends.

1.2.5 Joints in the wiring cables will not be allowed in any part of the system

1.2.6 The number of cables run in conduits shall be in accordance with the IEC – wiring regulations and DEWA Regulations. With the exception of 3 phase power points or sub main supply circuits to 3 phase boards the circuits grouped in any one conduit shall be connected to one phase of the system only. This does not apply to wiring run in metal trunking.

1.2.7 All switch wires for lighting circuits shall be on the 'live' side.

1.2.8 When drawing the wires and cables into the conduits or trunking, the Contractor must see that proper appliances are used so that the wires and cables suffer no damage whatever in the process. All kinks must be rigidly avoided and care must be taken to ensure that the insulation, taping and brading of the cables receive no damage or abrasion.

1.2.9 During the progress of work or at completion, the engineer may require certain wires and cables to be drawn out in order to test that the conditions mentioned have been complied with. They may require that certain fittings and/or accessories be dismantled for examination of workmanship and quality.

1.2.10 This Section includes the following:

1. Building wires and cables rated 1000 V and less.
2. Terminations rated 1000 V and less.
3. Sleeves and sleeve seals for cables.

1.2.11 Definitions

- A. LV: Low Voltage: 1000V and less
- B. XLPE: Cross linked Polyethylene
- C. LSZH: Low Smoke Zero Halogen

- D. SWA: Steel Wired Armoured
- E. Fire rated cable: The ability of electrical cables or optical cables to maintain a reliable form of power supply or signal from the source to the life safety installation(s) when exposed to FLAME/SHOCK/WATER SPRAY for a period of 120 minutes.
- F. CWZ: This refers to BS (see clauses 1.4.3&21).. Cables comply with the following tests:
 - 1. C – Resistance to fire at 950°C for 3 hours
 - 2. W – Resistance to fire and water at 650°C for 30 minutes
 - 3. Z – Resistance to fire and mechanical shock at 950°C for 15 mins
- G. HOFR: Heat Oil and Flame retardant

1.3 QUALITY CONTROL/ASSURANCE

Design, manufacture, testing and method of installation of all apparatus and materials furnished under requirements of these specifications shall conform to latest publications or standard rules of the following:

- A. Dubai Electricity and Water Authority (DEWA) regulations for electrical installation works and Department of Civil Defense requirements.
- B. Regulations for the Electrical Equipment of Buildings (IEE), (BS) London.
- C. British Standards.

The Quality Control/Assurance/Assurance/Quality Assurance duties shall be performed by the Contractor. The system of Quality Control/Assurance/Assurance verification shall be in accordance with ISO 9001 standards of quality assurance. Cables shall also be certified by ESMA (Emirates Standards and Meteorology Authority).

All Cables shall be certified by BASEC for appropriate BS test methods with independent authorized 3rd party certification.

Manufacturing company regularly engaged in the manufacturer of all materials specified in this section of types and sizes required, whose products have been in satisfactory use under similar service conditions for not less than 10 years.

Installers Qualifications: Use competent persons to install the equipment and provide them with documentary evidence of their competence, which the Engineer or Employer, or their nominees, shall be entitled to see at any time.

1.4 CODES AND STANDARDS

The following codes and standards to the extent specified herein, form a part of this specification. When an edition date is not indicated for a Code or Standard the latest edition enforces at the time of proposal submitted shall apply.

All cables supplied and installed shall be selected from the following types with stranded copper conductors and shall comply with the appropriate British Standard referred to below as applicable.

- 1. DEWA and UAE Fire Code Regulations.

2. BS EN 50525-1:2011, BS EN 50525-2-42:2011, BS EN 50525-2-31 Electric cables.
3. BS 6387 - Test method for resistance to fire for cables required to maintain circuit integrity under fire conditions.
4. BS EN 50525-2-11:2011, BS EN 50525-2-12:2011, BS EN 50525-21:2011, BS EN 50525-2-51, BS EN 50525-2-71:2011 Electric cables.
5. BS 6724: 2016 - Electric cables. Thermosetting insulated, armoured cables for voltages of 600/1000 V and 1900/3300 V, for fixed installations having low emission of smoke and corrosive gases when affected by fire.
6. BS 7211 - Electric cables. Thermosetting insulated, and thermoplastic sheathed cables for voltages up to and including 450/750V for electric power and lighting, and having low emission of smoke and corrosive gases when affected by fire.
7. BS EN 50525-3-41 – Low Smoke Zero Halogen Wires 450/750V.
8. BS 7629-1 Electric cables. Specification for 300/500 V fire resistant screened cables having low emission of smoke and corrosive gases when affected by fire. Multicore and multi pair cables.
9. BS 7655 - Specification for insulating and sheathing materials for cables.
10. BS 8434-2+A2: 2009 - Methods of test for assessment of the fire integrity of electric cables. Test for unprotected small cables for use in emergency circuits. BS EN 50200 with a 930° flame and with water spray.
11. BS 8491 - Method for assessment of fire integrity of large diameter power cables for use as components for smoke and heat control systems and certain other active fire safety systems.
12. BS 8519 - Selection and installation for fire resistant power and control cable systems for life safety and firefighting cables. Code of practice.
13. BS EN 10143 - Continuously hot-dip metal coated steel sheet and strip. Tolerances on dimensions and shape.
14. BS EN 61034-1+A1: 2014 - Measurement of smoke density of cables burning under defined conditions. Test apparatus.
15. NFPA 72 - National Fire Alarm and Signalling Code.
16. BS 7671 – Wires / Cables - Testing and Commissioning
17. IEC 60502-1 - Power cables with extruded insulation and their accessories for rated voltages.
18. BS EN 60228 - Conductors of insulated cables.
19. All LV cables, wires and accessories furnished under these Specifications shall conform to the applicable standards of DEWA, DCD, IET, IEC, NEC, BASEC and BS.
20. Where conflicts exist between the requirements of this Specification and other drawings, standards, codes and specifications, the most stringent shall be applied.
21. Fire Rated Cables related standards:
 - BS 6387: 2013 (CWZ) – sizes 1.5mm² up to 95mm²
 - EN 60754-1: 2014 – All sizes
 - EN 61034-2: 2005+A1: 2013 – All sizes

- IEC 60331-21: 1999 – All sizes
 - IEC 60332-3-24: 2009 – sizes 10mm² up to 1000mm²
 - BS 8434-2: 2003+A2:2009 (120 mins) sizes 1.5mm² up to 95mm²
 - EN 50200: 2015 (Class PH 120) – sizes 1.5mm² up to 95mm²
 - BS 8592:2016 – sizes 1.5mm² up to 70mm² (Rated voltage 450/750V)
 - BS 7846:2015 - all sizes, 600/1000V
 - BS 8592:2016 - all sizes, 450/750V
22. BS 6121-1 - Mechanical cable glands. Armour glands. Requirements and test methods.
23. BS 6121-5 - Mechanical cable glands. Code of practice for selection, installation and inspection of cable glands and armour glands.
24. BS EN 62444 - Cable glands for electrical installations.
25. IEC 60332 - Tests on electric and optical fibre cables under fire conditions.
26. IEC 61238-1 - Compression and mechanical connectors for power cables
27. Dubai Building Code (G.4.7) Cables and conductors shall be followed

1.5 SUBMITTALS

The contractor shall submit the following drawings and documents related with each electrical system for Engineer's approval:

- A. Original catalogues for various proposed components.
- B. Schedule of Cable types and sizes
- C. Test Certificates.

2.0 PRODUCTS

2.1 GENERAL

In general, Type and minimum size of cable as indicated in electrical drawings, SLD, electrical load schedules and other design elements.

Cable specification and manufacture must be consistent throughout each wired system. All cable utilized must be continuously run from source to termination, without any through jointing included.

- 2.1.1 Cable must be adequately stored and protected from damage pending installation and also during installation until permanent protection is affected.
- 2.1.2 Cable reels must be supported on purpose formed support frames and under no circumstance shall cable be drawn from reels laid on the ground. This shall be strictly enforced as the Engineer's condemnation of any cable that is considered to have been abused will involve total replacement at cost to the Contractor.
- 2.1.3 The cross-sectional area of every cable shall be suitable for carrying the maximum sustained load current under normal conditions and shall be selected in accordance with related Regulations. The cross-sectional area of the neutral conductor for 3-phase circuits shall be equal to the cross-sectional area of the phase conductors. The cables shall be selected such that the drop in voltage from the origin of the installation to any point in the installation does not exceed 4% of the nominal voltage when the conductors are carrying the full load current, but disregarding starting conditions. Voltage drop calculation shall be submitted for approval and the cross-sectional area of the cable shall be increased accordingly if required to meet the above mentioned requirement without any extra cost.

- 2.1.4 The cables connected in parallel shall be of the same type, cross sectional area, length and disposition and be arranged so as to carry substantially equal load currents.
- 2.1.5 Where cables are to be connected to bus bars, breakers etc. The insulation and/or sheath shall be removed for a distance of 150mm from the connection and replaced by suitable heat-resisting insulation.
- 2.1.6 The wire armour of single core cables in the same circuit shall be bounded together at both ends.
- 2.1.7 All cables supplied and installed shall be selected from the following types with stranded copper conductors and shall comply with the appropriate British Standard referred to below as applicable. Cabling shall be as indicated on the cable schedules of this specification and as indicated on the drawings and as described in the particular specification. Where not indicated comply with the following cable types which are detailed in the following articles:

A. WIRES:

1. Final Circuits (General lighting and Power) PVC : BS EN 50525-2-31 (Wires)
2. Final Circuits on escape routes / common areas LSZH : BS EN 50525-3-41 (LSZH wires)

B. FLEXIBLE WIRES:

1. Final Connections to luminaires from ceiling rose: PVC / PVC white plastic (105 degree compliant)
BSEN- 50525-1:2011, BS EN 50525-2-11:2011, BS EN 50525-2-21:2011, BSEN 50525-2-51:2011 Electric cables

C. CABLES:

1. LV Distribution in all general areas: XLPE/SWA/PVC – BS 5467: 2008.
2. LV Distribution in all public areas: XLPE/SWA/LSZH – BS 5467: 2008
3. External landscape lighting: XLPE/SWA/PVC – BS 6724+A3: 2008
4. Sub main supplies for Life safety equipment: XLPE/SWA/ LSZH – Fire resistant
BS7846, BS EN 61034-1+A1,
BS8519, BS 8491.
5. Aircraft/Marine Navigation Lighting: XLPE/PVC
6. Mechanical services supplies from MCCs: XLPE/LSZH/SWA.- BS 6724+A3
7. BMS and control applications: PVC if enclosed/concealed, LSZH if exposed

D. RUBBER CABLES:

Final connection to mechanical equipment: HOFR – BS EN 50525-1:2011, BS EN 50525-2-41:2011, BS EN 50525-2-42:2011

HO7RNF – BS EN 50525-2-21:2011

HO7RN8F – BS EN 50525-2-21:2011 (Water Resistant)

- E. FIRE RESISTANT CABLES:** All fire rated cables shall meet the DCD and UAE fire and life safety code standard for fire alarm cabling in accordance with BS6387 Category CWZ, EN 50200 PH120, BS 8434-2, 120 mins.

1. Emergency Lighting: 2 Hour fire rated to DCD (white)
2. Fire Alarm final circuits: 2 Hour fire rated to DCD (RED)

3. Fire Alarm Network: 2 hour Fire rating (red) multicore
4. Voice Alarm: 2 hour Fire rating (red) multicore
5. Fire alarm/ Lift/MCC interface: 2 hour Fire rating (red) multicore

2.2 WIRES

2.2.1 PVC- Insulated Single Core Cables

- a. PVC cables (non-armoured, non-sheathed) for electric power and lighting.
- b. The cable shall be of 450/750 volts grade and manufactured to BS EN (see clause 1.4.7).

2.2.2 LSZH- Insulated Single Core Cables

- a. Thermosetting insulated, cables (non-armoured, non-sheathed) for electric power and lighting on escape routes and common areas (including ceiling void) with low emission of smoke and corrosive gases when affected by fire.
- b. The cable shall be of 450/750 volts grade and manufactured to BS EN (see clause 1.4.7).

2.3 PVC/PVC – MULTICORE FLEXIBLE CABLES

- a. All flexible cables used for pendant lighting fittings or for final connections to fixed or portable equipment shall be of 300/500 volts grade.
- b. Multi core Flexible cables shall be of annealed copper conductor insulated with heat resisting PVC and of circular construction to BS EN (see clause 1.4.4)
 - a. Light Duty – final connection to apartment Indoor lighting.
 - b. Heavy duty - 105 degrees C – final connection to, washing machines, dish washers, cookers and water heaters etc.

2.4 CABLES

2.4.1 XLPE/SWA/LSZH Multicore Cables

- a. Crosslinked polyethylene cables for 600/1000 V armoured electric cables having thermosetting insulation and low emission of smoke and corrosive gases when affected by fire.
- b. The cable shall be of 600/1000 volts grade and manufactured to BS (see clause 1.4.5)

2.4.2 XLPE/LSZH/SWA/LSZH - Insulated Armoured Multicore Fire Resistant Cables

- a. Electric cables. Specification for 600-1000V fire resistant screened cables having low emission of smoke and corrosive gases when affected by fire. And where fire resistance is required to comply with BS (see clause 1.4.3) CWZ.
- b. The cable shall be of 600/1000 volts grade and manufactured to BS (see clause 1.4.21) and BS EN (see clause 1.4.14).
- c. Fire Rated Single Core Power Cable - Al Armoured

1Cx630mm² - Generally to IEC (see clauses 1.4.17&21) specification.

0.6/1kV, Class 2, Copper Conductor, XLPE Insulated, LSZH Bedded, Aluminium Wire Armoured, BLACK LSZH Outer Sheathed, Generally to IEC (see clauses 1.4.17&21) specification.

Approved by:

1. LPCB
2. Dubai Civil Defense (DCD)

Notes:

1. Stranded Conductors only
2. Al Armoured cables shall meet the requirements of IEC (see clauses 1.4.21) when tested at a temperature of 950°C for a duration of 180 mins + 15 mins cool down, at a voltage of 600V.

- c. Fire Rated Single Core Wires – (BS 8592)

0.6/1kV Cu (Class 2)/XL-LSZH Fire resistant Cables to meet the requirement of BS (see clauses 1.4.3&21) CWZ. Insulation Color as per request.

2.4.3 HOFR – Heat Resisting Multicore Flexible Cables

- a. HOFR flexible cables shall be used for final connection to mechanical equipment in Boiler rooms, oil storage plant rooms or where equipment requires withdrawal from mechanical plant (e.g. final connections to electric heaters in Air Handling Units). The cable shall be of 450/750 volts grade BS EN (see clause 1.4.4).
- b. Flexible cables shall be of annealed copper conductor insulated with heat resisting insulation and of circular construction to (see clauses 1.4.2&4) Flexible cables shall comply with relevant standards and codes of practice mentioned in the specification.

2.5 FIRE RESISTANT CABLES (FIRE ALARM + CENTRAL EMERGENCY SYSTEM)

- 2.5.1 Fire resistant cable shall have stranded (if installed in conduits within slab) installed plain annealed copper conductors to BS EN (see clause 1.4.21). With extruded silicone rubber insulation to BS (see clause 1.4.8) and Aluminium / LSF laminate and LSF composite sheath with tinned copper circuit protective conductor. The cable shall be manufactured and tested to BS (see clause 1.4.8). The conductors shall be solid if installed externally on any surface, and clipped directly.
- 2.5.2 The cables shall be rated at 300 / 500V and shall be suitable for operation at sheath temperature of 70 deg. C.
- 2.5.3 The cable shall comply with the flame retardant requirements of BS (see clauses 1.4.3&21), category CWZ latest BS 5839 and IEC (see clause 1.4.25). The cable shall be 120 minutes' fire rated cable as per DCD / AHJ with certifications.
- 2.5.4 The Fire-resistant cables shall be used for the wiring of Fire Alarm System and Emergency Lighting System and to be laid in conduits through slab or wall.
All cables must have DCD certifications and valid listing.

2.6 TERMINATION AND SPLICE CONNECTORS (LV POWER)

- 2.6.1 Connector terminations and splices shall be of class A to IEC (see clause 1.4.26).
- 2.6.2 Connectors shall be compression, barrel crimped type.

- 2.6.3 Connectors for copper conductors shall be made of electrolytic copper.
- 2.6.4 Connectors for aluminium conductors shall be made of aluminium alloy of purity not less than 99.5%.
- 2.6.5 All termination and splice connectors specified above shall be tin plated with minimum thickness of 5 microns for copper material and 7 microns for aluminium material. Aluminium connectors shall incorporate neutral grease to avoid aluminium oxidation.
- 2.6.6 Termination connectors for aluminium-copper termination shall consist of barrel made of aluminium alloy of purity not less than 99.5% and palm made of forged copper connected using a friction stir welding.
- 2.6.7 Splice connectors for aluminium-copper termination shall consist of two barrels, one is made of electrolytic copper and the other is made of aluminium alloy of purity not less than 99.5% connected using a friction stir welding.
- 2.6.8 Splice connectors shall have a central stop to divide the connector into two equal parts to facilitate the correct cable positioning.

2.7 CABLE GLANDS

- 2.7.1 Cable glands shall comply with BS (see clauses 1.4.22&23) and BS EN (see clause 1.4.24). Glands for armoured or screened cables shall have suitable clamps. The gland shall be of such a design that during assembly no torque will be induced in the cable sheath.
- 2.7.2 Cable glands used with unarmored cables shall be made of brass, type A2 to BS (see clause 1.4.25), with mechanical retention class B to BS EN (see clause 1.4.24), impact resistance category 8 to BS EN (see clause 1.4.24) and an environmental seal to the cable outer sheath. Glands shall be deluge protected, with IP66 degree of protection to IEC 60529.
- 2.7.3 Cable glands used with single steel wire armored cables (Internal applications) shall be made of brass, type BW to BS (see clause 1.4.23), with mechanical retention class B to BS EN 50262, impact resistance category 8 to BS EN (see clause 1.4.24), electrical continuity via armor wire termination category B to BS EN (see clause 1.4.24), earth tag to BS EN (see clause 1.4.24) and an environmental seal to the cable outer sheath. Glands shall have IP66 degree of protection to IEC 60529.
- 2.7.4 Cable glands used with single steel wire armored cables (outdoor and Life Safety Equipment's) where sprinkler / water mist firefighting systems are used shall be made of brass, type CW/ CWZ to BS (see clause 1.4.23), with mechanical retention class B to BS EN (see clause 1.4.24), impact resistance category 8 to BS EN (see clause 1.4.24), electrical continuity via armor wire termination category B to BS EN (see clause 1.4.24), earth tag to BS EN (see clause 1.4.24) and an environmental seal to the cable inner and outer sheaths. Each gland shall be locked onto equipment with suitable brass backnuts. Also, every gland shall have a polychloroprene shroud fitted over the gland for additional protection.
- 2.7.5 Cable glands used with LSOH or fire-resistant cables shall be of the applicable type as stated above and provided with low smoke fume shroud.

2.8 CABLE SEALING COMPOUNDS

- 2.8.1 Shall be tropical grade to BS 1858:2002 or BS 6910 and approved by cable manufacturer, of an oil-resisting compound where the difference in level between cable ends exceeds 6 m.
- 2.8.2 Each cable shall be kept sealed except when termination and splicing work is being performed.
- 2.8.3 The ends of all cables shall be sealed with heat shrinkable caps. Cap sizes shall be as recommended by the cap manufacturer for the cable outside diameter and insulation.

- 2.8.4 Caps shall contain sufficient adhesive that shrinkage of the cap during application results in formation of a positive watertight seal capable of withstanding complete immersion or total exposure without permitting the entrance of moisture.
- 2.8.5 Before and after pulling, the cable where the same are to be installed in conduits, the leading end seal of each length of cable shall be examined and repaired if necessary. All cut cable ends shall be promptly sealed after cutting except those to be spliced or terminated immediately.

2.9 CABLE CLEATS

- 2.9.1 Shall be approved claw type cast aluminium, gunmetal, plastic or brass of approved type, two bolts fixing for cables greater than 50 mm diameter.
- 2.9.2 Where more than two power cables are run together, provide a cable tray. Secure cables with cleats, saddles or ties as appropriate for location and use spacers where more than one layer is required.
- 2.9.3 All single core phase cables shall be installed in die-cast, non-ferrous, trefoil cleats and neutral cables in similar single way cleats of the same manufacture. Flat formation may be used in specified cases when approved cleats shall be used
- 2.9.4 For cables of core area of and less than 25mm² multicore cleats on a common bolt or spindle shall be acceptable. Multiple cable shall be secured with PVC cleats at a spacing consistent with the size of cable used
- 2.9.5 Structural steelwork shall not be drilled for cable fixing. Individual cables other than MICC cable run on steelwork shall be supported in cleats fixed to back-straps clamped or welded to the structural steelwork. These cleats shall be spaced so as to avoid sagging of cables and in no case at more than 1m between centers
- 2.9.6 Fixings, supports, cleats etc. used for Fire rated cables shall have the same fire rating as specified for the cable

3.0 EXECUTION

3.1 CABLE INSTALLATION

- 3.1.1 Generally, the installation of cables shall conform to IET regulations and DEWA requirements. The final sub circuit connection and wiring shall be carried out in accordance with DEWA regulations and unless otherwise indicated, the following details shall apply throughout the works:

- a) Lighting Circuits up to 1000 Watts:

10amp MCB protection Wired with 3 x 2.5 mm² (PVC) cables.

- b) 13amp socket outlets:

20amp MCB protection on radial main basis wired with 3 x 4 mm² PVC cable shall be provided for up to four socket outlets serving one room only of less than 50 m² floor area.

32amp MCB protection on ring main basis wired with 4 x 4mm² and 2 x 4mm² PVC cables shall be provided for maximum 8 nos. 13A sockets.

- c) Unit water heaters up to 3 KW rating:

20 amp MCB protection Wired with 2 x 4mm² and 1 x 4mm² PVC cables.

- d) Fan coil units each:

20 amp MCB protection wired with 2 x 4mm² and 1 x 4mm² PVC cables.

e) Hand dryers:

20 amp MCB protection Wired with 2 x 4mm² and 1 x 4mm² (2kW rating) PVC cables.

The above details for connections shall apply throughout the Works, except where the related Regulations would be contravened in respect of permissible voltage drop. In such cases the contractor shall utilize a larger conductor size as appropriate.

- 3.1.2 All cables shall be protected against any risk of mechanical damage to which they may be liable in normal conditions, of service. Where cables are installed under floors or in ceiling voids, they shall be run so that they are not liable to be damaged by contact with the floor or the ceiling or their fixings. Where the cables pass through holes in metal work, care shall be taken to prevent abrasion of the cables on any sharp edges.
- 3.1.3 All cables rising through floors and trench covers shall be protected, for example, by a length of lead or steel or PVC pipe which should project at least 75mm above the finished surface level. The open end of the pipe shall be sealed with a suitable compound. Lead and PVC pipes shall be used where they will be free from mechanical damage or in situations where steel pipe would be subject to corrosion.
- 3.1.4 The cables installed throughout the system shall be enclosed in conduit, duct, ducting or trunking as specified.
- 3.1.5 All cables buried in the ground shall incorporate an armour and all cables installed in positions which may be exposed to direct sunlight shall be supported by heavy duty cable tray and shall be suitably covered by a standard ventilated stand and non perforated cover to protect from ultra-violet light.
- 3.1.6 Cable ties shall be utilized to fix cables into cable trays, Cable cleats shall be utilized to fix the cables in vertical cable trays/ladders (Nylon cleat for normal cables and claw cleat for fire rated cables), cable separator and cable retainer shall be utilized to fix wires/cables into cable trunking.
- 3.1.7 All conduits, ducting and trunking shall be properly supported and of a type suitable for any type of mechanical damage to which they may be liable in normal conditions of service or adequately protected against such damage. The internal radius of every bend of a non-flexible cable shall be as recommended by manufacturer and the related standards.
- 3.1.8 Cables under roads, and where deemed necessary under pipe tracks etc., shall be protected by concrete encased PVC ducts. Bitumen bonded fibre ducts shall not be used under any circumstances. Conduit buried in trenches or in the ground is not permitted.
- 3.1.9 Where existing cable sleeves are to be utilized, the contractor shall ensure that the lower most ducts are employed, thereby leaving any remaining ducts accessible for future use.
- 3.1.10 Cable ducts shall be sealed at both ends using materials that are not soluble in hydrocarbons, and that are resistant to any likely corrosive and insect attack in the area concerned.

3.2 FIRE BARRIERS

- 3.2.1 Where conduit, ducting and/or trunking pass through fire-resistant structural elements such as walls and floors designated as fire barriers, the openings made shall be sealed according to the appropriate degree of fire resistance. In addition to this external fire barrier, an internal fire-resistant barrier shall also be provided to prevent the spread of fires.

3.3 CABLES INSTALLED UNDERGROUND

- 3.3.1 Direct buried cables shall be bedded in 75mm of clean sand or riddled soil, covered with similar material, then protected by concrete slabbing or interlocking tiles. The trench shall be backfilled over a plastic identification tape laid for as detailed under cable and cable trench marking.

- 3.3.2 Prefabricated concrete troughs or preformed trenches in which cables are laid shall be filled with clean sand to provide the top layer of cables with a minimum of 75mm cover, or be completely filled with sand if this should be required by circumstances.
- 3.3.3 Prefabricated cable troughs, partly or wholly filled with sand shall be closed using prefabricated concrete covers. Such covers shall be provided, by the contractor, to meet any particular load bearing requirements but shall not be less than adequate for use as a footpath.
- 3.3.4 Preformed trenches shall be sand filled and closed by floating over with concrete of adequate strength and thickness.
- 3.3.5 The use of chequer plating and other materials for closing cable troughs, trenches etc., shall be confined to those conditions where such special treatment is fully justified technically and economically and shall be agreed by the Engineer.
- 3.3.6 Power cables operating at voltages of 400 volts and above shall not be routed below 'jacked-up' typed unfixed flooring systems in any internal area unless unavoidable for the purpose of supplying equipment installed on the flooring system. In such cases the power cables must be totally enclosed within a steel trunking, conduit, etc., over their entire route length below the flooring system irrespective of the type of cable used.
- 3.3.7 Cable trench and prefabricated trough depths should not normally have less than 500mm. Direct buried cables should not normally have less than 500mm of cover from cable to finished grade level. The above shall be varied according to the requirements of actual site conditions and cable routes to provide adequate safeguard against, for example, erosion, soft ground conditions and mechanical damage etc. The maximum thickness of any one group of cables shall not exceed 75mm unless otherwise agreed with the Engineer.
- 3.3.8 Cable Sheath of External Buried Cables
- All cables directly buried in external areas should have MDPE (medium density polyethylene) outer sheath where ever specified. Other characteristics of external cables shall be similar to XLPE insulated cables.
- 3.3.9 Cable in Duct
- All external cables shall be laid in 150mm dia UPVC underground ducts of 3.2mm wall thickness.

3.4 CABLE AND CABLE TRENCH MARKING

- 3.4.1 All cables shall be fitted with indestructible identification bands at each end and then over their entire length at 30 metre intervals, at all points where they enter and leave ducts, and at changes in direction, etc.
- 3.4.2 Concrete or slabs covering cable trenches shall be coloured for identification purposes. Similarly, where cables run under floors, markers shall be employed to clearly define the extent of the cable way over the entire route.
- 3.4.3 The route of underground electric cables shall be marked by permanent markers posts. They shall be reinforced concrete units fitted with lead inserts bearing cable sizes and references. The marker post shall extend above ground level or shall be flush with the final ground level according to the site and Engineer's requirements. Two or more markers shall be visible from any point on the route and markers shall be placed at any divergence from the straight.
- 3.4.4 The location of all underground power cables shall be clearly identifiable throughout their route length by means of a polyethylene board laid in the ground at 300mm above each cable. The board shall be coloured red and continuously marked, with black indelible lettering of not less than 12mm high, to read the following message in both Arabic and English:

"CAUTION BURIED ELECTRIC CABLE 300 mm BELOW"**"HV CABLE", "ELECTRIC CABLE", "ELV CABLE", "EARTH PIT" or "CABLE JOINT"**

- 3.4.5 A sample of the polyethylene board shall be submitted to the Engineer for approval prior to any cable laying works being carried out.
- 3.4.6 The cable trench as shown on the drawings between service block and main building cable termination room shall be decided taking into consideration of laying all cables within the trench in 3 or 4 layers on cable ladders with sufficient access space. The crossing of other services through the trench must also be considered while deciding the exact width and depth.

3.5 JOINTING AND TERMINATIONS

- 3.5.1 All cable conductors shall be fitted with a correctly sized cable socket or thimble and a means of identification. The cable sockets may be of the sweated or crimped compression types. If for the former the solder should have a melting point of not less than 185 deg. C and if for the latter, they must be the appropriate tools as specified by the manufacturers of the joint connectors. The cable terminations shall be made following the positive identification of the conductors in accordance with the specified phase rotation sequence.
- 3.5.2 Joints in cable are not allowed and shall not be carried out without the written approval of the Engineer. Low tension cable joints shall incorporate compression type ferrules with polyethylene tape insulation housed in a plastic joint box and sealed with cold pouring resin fitting.

3.6 SLEEVE AND SLEEVE-SEAL INSTALLATION FOR ELECTRICAL PENETRATIONS

- 3.6.1 Install sleeves and sleeve seals at penetrations of exterior floor and wall assemblies.
- 3.6.2 Sleeves consisting of plastic conduit shall be installed for all cables passing through brick, concrete and similar structures to permit future withdrawal without disturbing the structure. All sleeves shall be tightly packed with fire-resisting material of approved type.
- 3.6.3 All exposed or concealed cables rising from the ground or floor as well as horizontally run cables 2 meters or lower from finished floor level shall be protected. The protection shall be by means of galvanized steel tube or metal channel of minimum thickness 2mm

3.7 TESTING & COMMISSIONING**3.7.1 Physical Check**

Check the cable installation and ensure the following:

- A. Proper laying and dressing cable cleating /tying cable.
- B. Tray supports are adequate.
- C. Cable identification and labeling.
- D. Glanding and gland earthing termination is correct.
- E. No physical damage.

3.7.2 Insulation Resistance Test

- A. Conduct insulation resistance test as per cables manufacturer recommendations, and test requirements.
- B. Measure insulation resistance between cores of all LV cable with a 500 V Megger and tabulate readings, the readings to be verified with permissible limits (min. 10 Mega ohms).
- C. Measure insulation resistance between cores and ECC and earth terminal with a 500 V Insulation Tester and tabulate readings. The readings to be verified with permissible limit.

- 3.7.3 Phase Rotation Test- Phase rotation at all points should be same other wise rectify.
- 3.7.4 Performance Test (to be energized temporarily)
- A. Switch on all isolators and run all equipment, for 2 hours and verify complete performance.
 - B. Check for any unusual temperature rise in cables, terminals and switches and protective devices.
- 3.7.5 Provide DC test equipment and apply, after isolation, in the presence of the Engineer, the following DC test voltages on all cables between cores, cores and sheath and cores and armor:
- a) PVC SWA LSZH and XLPE SWA LSZH 1000V grade cable.
 - 1) Between cores: 3500V.
 - 2) Between core and armour: 7000V.
- 3.7.6 Demonstrate correct phasing of cores in all cables throughout the works and test the insulation of all cables, both between the cores and between the cores and earth, during installation with a 500V insulation tester.

3.8 TEMPORARY WIRING

- 3.8.1 No part of the new installations shall be connected temporarily to the Supply Authority's mains without the approval of the Engineer and the Authority.
- 3.8.2 All temporary lighting and electric wiring which the Contractor may require, must be provided by the Contractor who will be responsible for all charges for electricity consumed as stated in the Contract conditions.
- 3.8.3 Temporary wiring and connections executed by the contractor to the instructions of the Engineer shall be arranged in accordance with the IEE regulations. Where the ambient air temperature is in excess of 50 deg. C. heat resisting silicon rubber cables shall be used for temporary connection. Where such cables may be subject to mechanical damage they shall be protected by conduit.

END OF SECTION

ELECTRICAL SECTION: 06
CABLE MANAGEMENT SYSTEM

1.0 GENERAL**1.1 SUMMARY**

- A. This Section includes:
1. Conduits, fittings, boxes, enclosures, and cabinets for electrical wiring.
 2. cable trays, cable ladders, cable baskets and accessories
 3. Cable trunkings
 4. Man holes and underground ducts
- B. A containment system shall be provided for the distribution, protection and support of a combination of cables, electrical equipment and communication system installations as indicated on the drawings.
- C. Applications: Use containment as indicated on the drawings or where not shown comply with the design narrative report of this specification and the following general applications:
1. Galvanized Conduits: Use when services are surface mounted i.e. plant rooms and risers
 2. Heavy Duty PVC Conduits: Use when conduits are embedded / concealed in walls or floor screeds.
 3. Heavy Duty LSZH Conduits: Use when conduits are exposed / surface mounted within false ceiling applications
 4. Galvanized Cable trunking: Use for final LV circuits, single cables, Data, Comms, security and BMS cables.
 5. PVC Steel Flexible Conduit: Use for final connections to motors and other equipment subject to vibration or adjustment and to thermostats, motorized valves and similar items mounted in pipelines or ducts.
 6. Cable baskets: Use for ICT only for Cat 6 cabling.
 7. Cable tray : medium Duty for ICT Fibre cabling, Fire alarm and EM systems.
 8. Cable tray and cable ladder: Use for power cables with an allowance of 15% for future spare capacity.
 9. Perimeter trunking: Use for power and communication services above desk top level or at skirting level for office/workshop type areas without floor distribution systems.
 8. Underfloor trunking: Use for power and communication systems to reception and office type areas where a flexible/adaptable layout is required.
 9. Size conduits and trunking according to the number of cables being drawn in with an allowance of 10% for future spare capacity. Provide calculations to support the sizing of conduits and trunking systems in accordance with DEWA regulations.

1.2 SUBMITTALS

- A. Product Data: Provide product data for conduits, trunking, cable trays, fittings, floor boxes, hinged-cover enclosures, and cabinets. Include data indicating dimensions and finishes for each type of conduit, trunking, cable tray and cable ladder indicated.
- B. Shop Drawings: For the following conduit and trunking components. Include plans, elevations, sections, details, and attachments to other work
 - 1. Custom enclosures and cabinets.
 - 2. For manholes and boxes for underground wiring, including the following:
 - a. Duct entry provisions, including locations and duct sizes.
 - b. Frame and cover design.
 - c. earthing details.
 - d. Dimensioned locations of cable rack inserts, and pulling-in and lifting irons.
 - e. Joint details.
- C. Samples for Initial Selection: For cable trays, cable baskets, trunking and conduits with factory-applied texture and colour finishes.
- D. Samples for Verification: For each type of exposed finish required for cable trays, cable baskets, trunking and conduits, prepared on Samples of size indicated below.
- F. Coordination Drawings: Conduit and cable tray/basket/ladder routing plans, drawn to scale, on which the following items are shown and coordinated with each other, based on input from installers of the items involved:
 - 1. Structural members in the paths of conduit groups with common supports.
 - 2. HVAC and plumbing items and architectural features in the paths of conduit groups with common supports.
 - 3. Cable tray vertical and horizontal offsets and transitions.
 - 4. Clearances for access above and to side of cable trays.
 - 5. Vertical elevation of cable trays above the floor or bottom of ceiling structure.
- G. Source quality-control - provide test reports.

1.3 QUALITY ASSURANCE

- A. Source Limitations: Obtain cable tray components through one source from a single manufacturer.
- B. Electrical Components, Devices, and Accessories: Listed and labelled as required by DEWA, by a third party testing agency such as BSI, Kitemark, etc. acceptable to authorities having local jurisdiction, and marked for intended use.

C. Comply with the following standards :

- a. BS 4607-1:1984+A2: Non-metallic conduits and fittings for electrical installations. Specification for fittings and components of insulating material.
- b. BS 4662:2006+A1:2009: Boxes for Flush Mounting of Electrical Accessories. Requirements, Test Methods and Dimensions.
- c. BS 4678-2: Cable trunking. Steel underfloor (duct) trunking.
- d. BS 4678-4: Cable trunking. Specification for cable trunking made of insulating material.
- e. BS 5733:2010+A1:2014 - General Requirements for Electrical Accessories.
- f. BS 6701: Telecommunications equipment and telecommunications cabling. Specification for installation, operation and maintenance.
- g. BS 6946: Specification for metal channel cable support systems for electrical installations.
- h. BS 7371-1: Coatings on metal fasteners. Specification for general requirements and selection guidelines.
- i. BS 4174: Specification for Self-tapping Screws and Metallic Drive Screws.
- j. BS 476-7: Fire tests on building materials and structures.
- k. DEWA Wiring Regulations.
- l. BS EN 50085-1: Cable trunking systems and cable ducting systems for electrical installations. General requirements.
- m. BS EN 50085-2-2: Cable trunking systems and cable ducting systems for electrical installations. Particular requirements for cable trunking systems and cable ducting systems intended for mounting underfloor, flush floor, or on floor.
- n. BS EN 50174-2: Information technology. Cabling installation. Installation planning and practices inside buildings.
- o. BS EN 50174-1 Information technology. Cabling installation. Installation specification and quality assurance.
- p. BS EN 60529 - Specification for degrees of protection provided by enclosures (IP code).
- q. BS EN 60670-1 + A1: Boxes and enclosures for electrical accessories for household and similar fixed electrical installations. General requirements.
- r. BS EN 60670-22 - Boxes and enclosures for electrical accessories for household and similar fixed electrical installations. Particular requirements for connecting boxes and enclosures.
- s. BS EN 60670-23: Boxes and enclosures for electrical accessories for household and similar fixed electrical installations. Particular requirements for floor boxes and enclosures.
- t. BS EN 61386-21: +A11: Conduit systems for cable management. Particular requirements. Rigid conduit systems.
- u. BS EN 61386-23+A11: Conduit systems for cable management. Particular requirements. Flexible conduit systems.
- v. EN 45545-2:2013+A1:2015: LSZH Conduits and Accessories. Determination of burning behavior by oxygen Index, ambient temperature test, smoke generation. Determination of optical density.

- w. BS EN 61534-1:2011 + A1:2014 and BS EN 61534-22:2004: Power track systems. Particular requirements for power track systems intended for on floor or underfloor installation.
- x. BS EN 61537: Cable management. Cable tray systems and cable ladder systems.
- y. BS EN 61914: Cable cleats for electrical installations.
- z. BS EN 10346:2015 - Continuously Hot-Dip Coated Steel Flat Products for Cold Forming - Technical Delivery Conditions.
- aa. BS EN 10162:2003 & 10210: - Cold rolled steel sections. Technical delivery conditions. Dimensional and cross-sectional tolerances. And Hot finished steel structural hollow sections Part 2: Tolerances, dimensions and sectional properties.
- bb. BS EN ISO 1461: Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods.
- cc. BS EN ISO 2081: Metallic and other inorganic coatings. Electroplated coatings of zinc with supplementary treatments on iron or steel.
- dd. BS EN ISO 12683: Mechanically deposited coatings of zinc. Specification and test methods.
- ee. BS EN ISO 12944-2: Paints and varnishes. Corrosion protection of steel structures by protective paint systems. Classification of environments.
- ff. BS EN ISO 14713-1: Zinc coatings. Guidelines and recommendations for the protection against corrosion of iron and steel in structures. General principles of design and corrosion resistance.
- gg. BS EN ISO 17668: Zinc diffusion coatings on ferrous products. Sherardizing. Specification.
- hh. BS EN 124 Gully tops and manhole tops for vehicular and pedestrian areas.
- ii. IEC 60695-2-1: Fire Hazard Testing - Test Method.
- jj. BS 3505: UPVC Ducts
- kk. Department of Civil Defense
 - ll. Telecommunication Company's Standards and Requirements

2.0 PRODUCTS

2.1 G.I CONDUIT SYSTEMS AND ACCESSORIES

Provide heavy gauge hot dipped galvanized, threadable, after fabrication rigid steel Class 4 manufactured in accordance with BS EN. Use conduit of each type from one manufacturer. The system must have electrical continuity. Use accessories from same manufacturer. Mix and match not acceptable. All surface mounted conduits and conduit accessories in non-false ceiling areas shall be G.I conduits.

2.2 PVC CONDUIT SYSTEMS AND ACCESSORIES

Provide very heavy gauge high impact PVC type, minimum 1.7 mm wall thickness manufactured in accordance with BS EN. Use accessories from same manufacturer. Mix and match not acceptable. All concealed conduits and conduit accessories embedded in concrete, plaster and screed shall be PVC conduits. Any containment shall not be installed in the slab without the approval of structural engineer considering the type of the slab proposed. Coordinate closely with the structural engineer

2.3 LSF CONDUIT SYSTEMS AND ACCESSORIES

Provide very heavy gauge high impact LSF type, minimum 1.7 mm wall thickness manufactured in accordance with BS EN. The conduits should be tested by third party as per EN +A1 fire testing method. Use accessories from same manufacturer. Mix and match not acceptable. **All surface mounted conduits and conduit accessories in false ceiling areas shall be LSOH conduits.**

2.3.1 FLEXIBLE PVC / LSF CONDUITS SYSTEM AND ACCESSORIES

Low smoke and fume material. Use brass male adaptors to connect flexible conduit to motors and any other equipment having a threaded entry. Use male adaptors, solid couplers, flanged couplers with washer and externally screwed brass bushes to connect flexible conduit to trunking and equipment not having a threaded entry.

2.3.2 FLEXIBLE GI CONDUITS SYSTEM AND ACCESSORIES

Use plastic adaptors and bushes. Use male adaptors to connect flexible conduit to motors and other equipment having a threaded entry. Use female adaptors and externally screwed bushes to connect flexible conduit to trunking and equipment not having a threaded entry.

2.4 CONDUIT SUPPORTS AND FIXINGS

- A. Provide proprietary suspension systems comprising channel sections with return lips and compatible fixing accessories made of material to BS EN and/or slotted angles to withdrawn no replacement.
- B. Ensure support components for Class 4 conduit have the same finishing method as the conduit carried out after manufacture. Ensure components in direct contact with conduit match profile of conduit.
- C. Ensure all steel components such as studding, bolts and steel screws, bolts, nuts and washers are either cadmium plated and passivated or zinc electroplated to BS after manufacture. Do not use metal fixing components likely to deteriorate and/or cause damage through electrolytic action.

2.5 UNDERGROUND DUCT BANKS

- A. Ducts shall be Type heavy duty, Class D, uPVC complete with associated couplings, spacers and all required accessories. Duct shall be suitable to withstand a minimum pressure of 6 bar.

B. Concrete, reinforced concrete, materials methods of construction and methods of test shall comply with the requirements of this Division, as well as relevant civil specification section

Concrete encased UPVC ducts shall be installed at locations wherever indicated on the drawings including cable entry ducts to each building.

Size and type of ducts shall be as indicated in the related electrical system drawings and as per BS, BS EN.

2.6 CABLE TRAY AND FITTINGS

- A. The cable tray system shall be complete with all necessary trays, risers, elbows, supports, end plates, drop outs and fittings required for the installation.
- B. Cable trays shall be heavy duty perforated sheets to BS EN ISO for ships and marine technology, ship-shifting winches. Hot dip galvanized after manufacture to BS EN. All heavy duty trays supporting power cables shall be of folded edge with return flange. Thickness of tray shall not be less than:

Width of tray / accessory (mm)	Minimum Thickness (mm)
Up to and including 150	1.0
Up to and including 300	1.2
Above 300 up to and including 600	1.5
Above 600 up to and including 900	2.0

- C. Loading of cable tray and of cable ladder shall be in accordance with NEMA VE1. Contractor shall submit loading calculation for company review and approval.
- D. All supports and fixings shall be hot dip galvanized after manufacture to BS EN ISO
- E. All fixation hardware (bolts, nuts, washers shall be stainless steel).
- F. Cable ladder shall be produced of hot dip galvanized mild steel.
- G. Minimum size of cable tray shall be 100mm wide.
- H. The cable tray/ladder shall be sized such that a minimum spacing of twice the cable diameter is kept between two adjacent cables with the spare capacity of 15%.

2.6.1 Cable support system - perforated tray (Type - Flanged or return flanged.)

- a. Use factory made fittings throughout of same material, type, pattern, finish and thickness as cable tray.
- b. Use reducers, inside angles and outside angles as manufacturer's standard.
- c. Use flat bends, equal tees, unequal tees and crosses with corners gusseted.
- d. Join lengths of tray and fittings using manufacturer's standard shouldered ends, fish plates, or couplers, with galvanized or zinc plated slotted domed head 'roofing' bolts, nuts, washers and shake proof washers.

2.7 CABLE LADDER AND FITTINGS

- A. The cable ladder system shall be complete with all necessary trays, risers, elbows, supports, and end plates, drop outs and fittings required for the installation.
- B. Cable trays shall be heavy duty perforated sheets to BS EN ISO, Hot dip galvanized after manufacture to BS EN ISO.
- C. Loading of cable ladder shall be in accordance with NEMA VE1. Contractor shall submit loading calculation for company review and approval.
- D. All supports and fixings shall be hot dip galvanized after manufacture to BS EN ISO.
- E. All fixation hardware (bolts, nuts, washers shall be stainless steel).

- F. Cable ladder shall be produced of hot dip galvanized mild steel.
- G. The cable tray/ladder shall be sized such that a minimum spacing of twice the cable diameter is kept between two adjustment cables with the spare capacity of 15%.

2.8 CABLE BASKETS AND ACCESSORIES

- 2.8.1 Cable baskets shall be constructed from 5mm steel wires. The spacing between longitudinal wires shall be 50mm and between transverse wires 100mm. The steel wires shall be welded at all joints. Transverse wires shall be “T” welded to top edge longitudinal wires in such a way so as to minimise fraying of the cables when being laid.

The steel wire baskets shall be Electro-Zinc galvanized and corrosion resistant.

Wire baskets shall be minimum 50mm high and, 150mm / 200mm / 300mm / 400mm / wide as may be specified elsewhere or as shown in layout drawings. Wire baskets shall be used for telecommunication services in particular to Cat 6 cablings, while fibre cabling to be laid in cable trays only.

Use Proprietary system of wire basket accessories with compatible jointing and fixing accessories. Use factory made fittings throughout of same material finish as basket, for risers, bends, reducers, tees, crosses and drop outs.

2.9 TRUNKING SYSTEMS

2.9.1 GI cable Trunking

- A. Material - Steel trunking to BS EN
- B. Supply partitions and covers same material as trunking.
 - 1. Gauge of metal BS EN.
 - 2. Trunking type - Standard cable trunking with compartments.
 - 3. Use trunking manufactured with inward return edge flanges and fitted with flange couplers which ensure that when the cover is removed a minimum of 80% of the nominal trunking or compartment width is available for access.
 - 4. Protection: Constructed from Pre-galvanized sheet steel to BS EN specification DX51D+Z275. This is a bending and profiling steel grade with a 275gsm zinc coating for superior corrosion resistance. The thickness of the trunking body, compartment separators, flanges, junction boxes, vertical access boxes, etc, shall not be less than 1mm for 50mm to 75mm and 1.2mm for 100mm.

Width of Trunking (mm)	Minimum Thickness (mm)
Up to and including 75	1.0
Upto100	1.2
For 150	1.5

- 5. Finish - Manufacturer's standard, all surfaces.
- 6. Colour - Manufacturer's standard or as approved
- 7. Use purpose made brackets to fix to structural steel or suspension rods.

8. Provide external fixing lugs where specified protection for the installation is IP44 or greater.
9. Use bends, tees and angles of similar gauge, type and finish as trunking body and supplied by same manufacturer.
10. Ensure partitions are electrically continuous with the body of the trunking or provide a connector for a circuit protective conductor.
11. Ensure gap between partitions and lids maintains segregation of circuits.
12. Provide individual mounting plates for each accessory mounted on trunking covers.
13. Fittings shall be of the same material as the trunking.
14. Use purpose made jointing pieces fixed with screws into captive nuts. Ensure screws do not protrude through the nuts.
15. Ensure rigidity of trunking is maintained across joint.
16. Ensure external dimensions of trunking are maintained and not reduced by more than 4% across joints between trunking lengths and/or fittings.
17. Use purpose made fittings of the same manufacture where trunking connects to switchgear and distribution boards.
18. Provide flanges for connection of flush floor trunking to vertical trunking to maintain the cross sectional area of compartments with 50mm minimum radius.
19. Maintain electrical continuity at each joint by a copper link, (tinned copper for galvanized trunking), fixed on outside of trunking, secured by screws, nuts and shake proof washers. Screws must not project through the nut. Make provision for continuity to be achieved without need to remove paint from ferrous metal where trunking has a painted finish.
20. Do not use self-tapping screws. Use cheese or round head screws except where provision is made for the use of counter-sunk heads.
21. Screws, nuts and washers material - Use steel zinc coated (coating to BS).
22. Provide horizontal trunking with removable cable retainers or bridges to retain cables in situ.
23. Provide vertical trunking with insulated sleeved pin racks to support cables at 3000mm maximum spacing.

C. Cable trunking Support and Accessories:

1. Provide proprietary suspension systems comprising channel sections with return lips and compatible fixing accessories made of material to BS EN and/or slotted angles to withdrawn no replacement.
2. Ensure support components for Class 4 conduit have the same finishing method as the conduit carried out after manufacture. Ensure components in direct contact with conduit match profile of conduit.
3. Ensure all steel components such as studding, bolts and steel screws, bolts, nuts and washers are either cadmium plated and passivated or zinc electroplated to BS after manufacture. Do not use metal fixing components likely to deteriorate and/or cause damage through electrolytic action.

2.9.2 Skirting (Dado) Trunking System

- A. The Dado Trunking shall be manufactured in accordance with the requirements of BS /BS EN
- B. The UPVC material used shall be tested by approved laboratory in accordance with the requirements of the British Standards (BS EN). All extrusion material tested and comply in accordance with the requirements of BS and has achieved a Class 1Y classification. Moulding material tested conforms with IEC at a severity of 650 C.
- C. Dado Trunking shall comprise of 3 compartments, the side compartments are one for power and one for voice and data and central compartment is for fixing of all sockets.
- D. All the trunking shall be laid at the Dado/skirting level surface mounted or semi recessed in walls and according to Engineers approval.
- E. Trunking size shall be 220 mm x 50mm or equivalent subject to Engineers approval for all areas except at intersections with columns.
- F. Suitable accessories shall be used wherever required, like couplers, internal, external corners, cable dividers, cable retainers, flat tees, flat angles, stop ends etc. to give a neat and strong system.
- G. Suitable back boxes shall be used as 1 Gang, 2Gang etc. according to the requirements to fix the wiring accessories flush with the trunking system.
- H. The trunking color shall be to Engineer selection and the finish of the Wiring Accessories shall be with color to Engineer selection.
- I. The socket outlets for power, outlets shall be fixed on the middle compartment, through PVC back boxes of 1 Gang or 2 Gang according to the requirement.
- J. Samples of accessories shall be provided for approval by Engineer prior to ordering.

2.10 BOXES, ENCLOSURES, AND CABINET

- A. Sheet Metal Outlet and Device Boxes
- B. Cast-Metal Outlet and Device Boxes
- C. Non-metallic Outlet and Device Boxes
- D. Metal Floor Boxes: Cast or sheet metal, fully adjustable, rectangular.
- E. Non-metallic Floor Boxes: Nonadjustable, round.
- F. Small Sheet Metal Pull and Junction Boxes
- G. Cast-Metal Access, Pull, and Junction Boxes: BS EN cast iron with gasket cover.
- H. Hinged-Cover Enclosures: IP55 where located externally, with continuous-hinge cover with flush latch, unless otherwise indicated.
 - 1. Metal Enclosures: Steel, finished inside and out with manufacturer's standard enamel.
 - 2. Nonmetallic Enclosures: Plastic.

3.0 EXECUTION

- A. Ensure entire system is electrically and mechanically continuous, and in accordance with DEWA requirements.
- B. Fire Barriers: Comply with the requirements of DEWA and Dubai Civil Defense wherever the conduit or trunking passes through the perimeter of a fire compartment (wall, floor or ceiling).
- C. Appearance: Arrange conduit, trunking and ducting to present neat appearance, parallel with other service runs and lines of building construction, except where in screed or in-situ concrete. Ensure plumb vertical runs.
- D. Cable Installation: Install cable in conduit, trunking or equipment enclosures only when completely erected throughout its length.
- E. Do not use framework of partitions or similar unless indicated.
- F. Quality:
 1. Cut conduit clean and square with axis. Remove any burrs prior to erection.
 2. Site form 90°bends in conduit wherever practical or use circular or adaptable boxes.
 3. Construct bends and sets cold with a bending machine. Do not apply heat when forming sets or bends.
 4. Use bending tools complying with British Standards appropriate to conduit material.
 5. Ensure no indentation or reduction in cross sectional area occurs during installation.
 6. Use correct tools to assemble conduit. Ensure no tool marks or damage to components occurs.
- G. Spacing: Install conduit, trunking and equipment clear of other services. Measure distance from external surface of any thermal insulation. Notify instances where minimum clearance cannot be achieved and bond items concerned. Minimum general spacing between conduits, trunking, equipment and:
 1. Insulated steam services 300mm
 2. Other services excluding steam 150mm
- I. Access: Arrange trunking to allow access to wiring. Locate covers on top or sides of trunking if practicable. Arrange access so covers are on a continuous face and cables can be laid in throughout the length of the trunking. Notify client where either condition cannot be achieved.

- A. Use heavy duty conduits and comply with manufacturer's instructions for bending, setting and jointing of conduit. Use plastic conduit only where indicated. Do not install conduit when ambient working temperature is or will be below -5°C or above 60°C.
- B. Use solvents recommended by manufacturer of conduit when using solvent welded joints and ensure spigots enter full depth of sockets. Hold joints rigid and in position until weld sets. Remove excess solvent before surface damage occurs.

- C. Use slip joints as necessary, but not exceeding 6 metres on straight lengths to allow for expansion and contraction over temperature variation as indicated.
- D. Use semi-mastic adhesive where expansion joints are formed.
- E. Where fitments do not have shaped or smooth conduit entries connect with male bushes and external couplings.
- F. Ensure special care is taken to prevent mechanical damage or warping to conduit where mechanical loads are imposed on conduit system, e.g. light fittings.
- G. Keep conduits at least 300 mm away from parallel runs of flues and steam pipes and at least 150mm from hot water pipes. Install horizontal conduit runs above water and steam piping.
- H. Complete conduit installation before starting conductor installation.
- I. Support conduits as specified by the manufacturer's installation guidance or to meet the requirements of the DEWA Regulations.
- J. Arrange stub-ups so curved portions of bends are not visible above the finished slab.
- K. Install no more than the equivalent of three 90-degree bends in any conduit run except for communications conduits, for which fewer bends are allowed.
- L. Conceal conduit within finished walls, ceilings, and floors, unless otherwise indicated.
- M. Conduits Embedded in Slabs:
 - 1. Run conduit larger parallel or at right angles to main reinforcement. Where at right angles to reinforcement, place conduit close to slab support.
 - 2. Arrange conduits to cross building expansion joints at right angles with expansion fittings.
 - 3. Change from PVC to galvanized steel conduit before rising above the floor.
- I. Threaded Conduit Joints, Exposed to Wet, Damp, Corrosive, or Outdoor Conditions: Apply listed compound to threads of conduit and fittings before making up joints. Follow compound manufacturer's written instructions.
- J. Conduit Terminations at Locations Subject to Moisture or Vibration: Use insulating bushings to protect conductors.
- K. Install pull wires in empty conduits. Use polypropylene or monofilament plastic line with not less than 90-kg tensile strength. Leave at least 300 mm of slack at each end of pull wire.
- L. Conduits for Optical Fibre and Communications Cable: Install conduits, metallic and heavy duty non-metallic, rigid and flexible, as follows:
 - 1. Install conduits in maximum lengths of 20 m.
 - 2. Install with a maximum of two 90-degree bends or equivalent for each length of conduit. Separate lengths with pull or junction boxes or terminations at distribution frames or cabinets where necessary to comply with these requirements.

- M. Install conduit sealing fittings at suitable, approved, and accessible locations and fill them with listed sealing compound. For concealed conduits, install each fitting in a flush steel box with a blank cover plate having a finish similar to that of adjacent plates or surfaces. Install conduit sealing fittings at the following points:
 - 1. Where conduits pass from warm to cold locations, such as boundaries of refrigerated spaces.
 - 2. Where otherwise required by local regulations
- N. Expansion-Joint Fittings for rigid PVC: Install in each run of aboveground conduit that is located where environmental temperature change may exceed 17 deg C, and that has straight-run length that exceeds 7.5 m.
 - 1. Install expansion-joint fittings for each of the following locations, and provide type and quantity of fittings that accommodate temperature change listed for location:
 - a. Outdoor Locations Not Exposed to Direct Sunlight: 70 deg C temperature change.
 - b. Outdoor Locations Exposed to Direct Sunlight: 86 deg C temperature change.
 - c. Indoor Spaces: Connected with the Outdoors without Physical Separation: 70 deg C temperature change.
 - 2. Install fitting(s) that provide expansion and contraction for at least 0.06 mm per meter of length of straight run per deg C of temperature change.
 - 3. Install each expansion-joint fitting with position, mounting, and piston setting selected according to manufacturer's written instructions for conditions at specific location at the time of installation.
- O. Flexible Conduit Connections: Use maximum of 1800 mm of flexible conduit for recessed and semi-recessed lighting fixtures, equipment subject to vibration, noise transmission, or movement; and for transformers and motors.
- P. Use suitably IP rated flexible GI conduit in damp or wet locations not subject to severe physical damage. Use swivel adaptors to ensure safety of cables and conduits during operation. Minimum IP rating for areas susceptible to water and moisture should be IP65.
- Q. Use suitably IP rated Steel flexible conduit with appropriate sheathing in damp or wet locations not subject to severe physical damage. . PVC coated GI flrxible conduits with UV protection properties can be used in areas exposed to sunlight.
- R. Recessed Boxes in Masonry Walls: Saw-cut opening for box in centre of cell of masonry block, and install box flush with surface of wall.
- S. Set metal floor boxes level and flush with finished floor surface.
- T. Set non-metallic floor boxes level. Trim after installation to fit flush with finished floor surface.

3.2.1 Cleaning before wiring

- A. Clean inside of conduits and trunking with swabs immediately before wiring.

- B. Inspect all components and remove any foreign matter, fit temporary plugs to open ends of conduit and trunking to prevent ingress of water and solid material.

3.2.1 Wiring

- A. Comply with the requirements of DEWA Wiring Regulations and IET Guidance Notes.
- B. Segregate circuits as indicated.
- C. Ensure draw wires are left within empty conduits for use of specialist installers. Use draw wires comprising nylon tapes with fitted eyelets.
- D. For concealed conduit ensure system is installed to enable re-wiring to be carried out from boxes for fittings or accessories only. Draw-in boxes will only be permitted with prior permission in writing.
- E. Do not use tallow or any other substances to facilitate drawing-in of cables.

3.2.2 Builders Work

- A. Ensure conduit is not concealed until work has been inspected and approved.
- B. Obtain permission before horizontally chasing walls.
- C. Ensure that conduit and fittings buried in concrete or behind plaster are protected against corrosion or electrolytic action prior to rendering.
- D. Ensure conduit concealed in wall chases is covered by plaster and/or rendering to minimum depth of 12mm.

3.2.3 Conduit Boxes

- A. Ensure that, wherever conduit boxes are cast in, the face of the box is flush with the face of the concrete or plaster. Fit circular conduit boxes with extension rings to ensure a flush face with plaster or concrete or where terminal blocks are to be accommodated.
- B. Ensure fixing holes are countersunk where material thickness allows or use round head screws to prevent damage to cables and remove burrs before cables are drawn in.
- C. Use a minimum of two screw fixing for standard circular conduit boxes and four screws for large conduit boxes and adaptable boxes up to 150mm × 100mm.
- D. Use back outlet boxes where surface conduits pass through walls, to outside accessories or lighting points.
- E. Secure switch boxes and socket boxes using countersunk steel screws where provision is made for them or if not use round head screws. Use plug inserts and finally grout in position prior to plastering or screed pouring.

3.2.4 Fixing Conduit

- A. Support conduit in accordance with the requirements of DEWA Wiring Regulations and IET Guidance Notes.

- B. Ensure conduit is not under mechanical stress. Fix conduit boxes independently of conduit. Make allowance for any additional mechanical loading supported by conduit boxes.
- C. Where protection is specified as IP44 or greater ensure fixings of conduit boxes are suitable to maintain degree of protection.
- D. Use saddles while fixing the conduits above false ceiling areas, surface mounted and crampets / saddles while fixing in floor screed and buried in plaster.

3.2.5 Underground Conduit Installation

Where buried below ground, use heavy duty PVC conduits. Do not use any buried boxes unless prior permission in writing has been obtained. Wrap duct with PVC self-adhesive tape, half lapped. Extend taping 150mm beyond point where conduit leaves ground. Install circular through conduit boxes at the end of the tape. Fill conduit boxes after cable installation with inert, permanently plastic compound with high insulation value, and wrap in PVC self-adhesive tape. Install backfill as specified in the civil engineer's specification.

3.3 STEEL TRUNKING INSTALLATION

- A. Install steel trunking in accordance with the manufacturer's requirements and those of the DEWA Wiring Regulations.
- B. Use trunking to avoid multiple parallel conduit runs.
- C. Cut trunking clean and square with axis, prepare ends and remove burrs and sharp edges. Ensure inside of trunking is free from anything liable to damage cables either during installation or after covers are fitted.
- D. When trunking is held in a vice, ensure surfaces remain undamaged and components are not warped.
- E. Avoid tool marking or damage to trunking system components.
- F. Do not site fabricate trunking tees, bends, flanges and other accessories. Use only factory made accessories.
- G. Form circular holes over 6mm diameter in trunking body using correctly sized punch sets. Use twist drill for holes up to 6mm maximum diameter.
- H. Use only factory formed openings for accessories.
- I. Line unprotected apertures in trunking with PVC or nylon edging strip.
- J. Fit ends of runs with removable blanking plates.
- K. Ensure connections are not made to covers unless indicated or approval obtained.
- L. Provide fixed section of cover projecting 25mm either side of fabric where trunking passes through wall, floors or ceiling.
- M. Fit cable retaining straps at 500mm intervals except where cover is on top.

3.4 CABLE TRAY INSTALLATION

- A. Support from building fabric with minimum clearance behind of 20mm. Install fixings at regular intervals to prevent visible sagging when loaded, with maximum spacing 1.2m and 230mm from fittings.
- B. Keep cutting of cable tray to a minimum. Cut along a line of unperforated metal. Make good finish with zinc rich paint, primer and top coat, or two pack epoxy paste, as appropriate to tray material and finish.
- C. Fit holes cut in tray for passage of cables with grommets, bushes or other lining.
- D. Install all bolts, fixings and hangers with threaded portion away from cables.

3.4.1 Cable cleats, ties, saddles and clips installation

- A. For cables on horizontal tray use ties for each circuit. Use tie manufacturer's special tensioning tool where available. Crop off tie ends.
- B. For cables on vertical tray use cleats bolted to tray for paper, plastic or elastomeric insulated cables and saddles or clips for mineral insulated cables. Use cleats sized to grip cables firmly without undue pressure or strain on cable, but preventing slipping.
- C. For cables on vertical or horizontal rack use proprietary fixings to rack for paper, plastic or elastomeric insulated cables and saddles or clips for mineral insulated cables. On continuous flat surfaces of wood, plaster, brick etc.
- D. Use polypropylene surface fixing clips with prefixed hardened steel pin for PVC insulated and sheathed cables and sheathed or bright mineral insulated cables. Use round or flat or flat twin pattern as appropriate, manufactured specifically for cable being fixed.
- E. Use one hole 'P' clips or two way saddles of bright copper for unsheathed mineral insulated cable. Use PVC covered for sheathed mineral insulated cables.
- F. Space cleats, ties, saddles and clips - as per the requirements of the DEWA Wiring Regulations a

3.4.2 Cable Installation

- A. Install cables only when containment installation has been completed and inspected.
- B. Fasten cables on horizontal runs with cable clamps or cable ties as recommended by the manufacturer. Tighten clamps only enough to secure the cable, without indenting the cable jacket. Install cable ties with a tool that includes an automatic pressure-limiting device.
- C. On vertical runs, fasten cables to tray every 450mm. Install intermediate supports when cable weight exceeds the load-carrying capacity of the tray rungs.
- D. Install covers after installation of cable is completed.

3.4.3 Connections

- A. Earth cable trays according to manufacturer's written instructions and the earthing specification.

3.4.4 Field Quality Control

- A. After installing cable trays and after electrical circuitry has been energized, survey for compliance with requirements. Perform the following field quality-control survey:
 - 1. Visually inspect cable insulation for damage. Correct sharp corners, protuberances in cable tray, vibration, and thermal expansion and contraction conditions, which may cause or have caused damage.
 - 2. Verify that the number, size, and voltage of cables in cable tray do not exceed that permitted by DEWA. Verify that communication or data-processing circuits are separated from power circuits in accordance with DEWA regulations.
 - 3. Verify that there is no intrusion of such items as pipe, hangers, or other equipment that could damage cables.
 - 4. Remove deposits of dust, industrial process materials, trash of any description, and any blockage of tray ventilation.
 - 5. Visually inspect each cable tray joint and each ground connection for mechanical continuity. Check bolted connections between sections for corrosion. Clean and re-torque in suspect areas.
 - 6. Check for missing or damaged bolts, bolt heads, or nuts. When found, replace with specified hardware.
 - 7. Perform visual and mechanical checks for adequacy of cable tray grounding; verify that all takeoff conduits are bonded to cable tray.
- B. Report results in writing.

3.4.5 Protection

- A. Protect installed cable trays.
 - 1. Repair damage to galvanized finishes with zinc-rich paint recommended by cable tray manufacturer.
 - 2. Repair damage to PVC or paint finishes with matching touchup coating recommended by cable tray manufacturer.
 - 3. Install temporary protection for cables in open trays to protect exposed cables from falling objects or debris during construction. Temporary protection for cables and cable tray can be constructed of wood or metal materials until the risk of damage is over.

3.5 INSTALLATION OF UNDERGROUND BOXES

- A. Install boxes level and plumb and with orientation and depth coordinated with connecting conduits to minimize bends and deflections required for proper entrances.
- B. Unless otherwise indicated, support units on a level bed of crushed stone or gravel, graded from 12.5-mm sieve to 4.75-mm sieve and compacted to same density as adjacent undisturbed earth.
- C. Elevation: In paved areas, set so cover surface will be flush with finished grade. Set covers of other enclosures 25 mm above finished grade.
- D. Install boxes with bottom below the frost line.

- E. Install removable hardware, including pulling eyes, cable stanchions, cable arms, and insulators, as required for installation and support of cables and conductors and as indicated. Select arm lengths to be long enough to provide spare space for future cables, but short enough to preserve adequate working clearances in the enclosure.
- F. Field-cut openings for conduits according to enclosure manufacturer's written instructions. Cut wall of enclosure with a tool designed for material to be cut. Size holes for terminating fittings to be used, and seal around penetrations after fittings are installed.

3.6 SLEEVE INSTALLATION FOR ELECTRICAL PENETRATIONS

- A. Coordinate sleeve selection and application and provide fire stopping in accordance with the architectural specification
- B. Concrete Slabs and Walls: Install sleeves for penetrations unless core-drilled holes or formed openings are used. Install sleeves during erection of slabs and walls.
- C. Use pipe sleeves unless penetration arrangement requires rectangular sleeved opening.
- D. Rectangular Sleeve Minimum Metal Thickness:
 - 1. For sleeve cross-section rectangle perimeter less than 1270 mm and no side greater than 400 mm, thickness shall be 1.3 mm.
 - 2. For sleeve cross-section rectangle perimeter equal to, or greater than, 1270 mm and 1 or more sides equal to, or greater than, 400 mm, thickness shall be 3.5 mm.
- E. Fire-Rated Assemblies: Install sleeves for penetrations of fire-rated floor and wall assemblies unless openings compatible with firestop system used are fabricated during construction of floor or wall.
- F. Cut sleeves to length for mounting flush with both surfaces of walls.
- G. Extend sleeves installed in floors 50 mm above finished floor level.
- H. Size pipe sleeves to provide 6.4-mm annular clear space between sleeve and containment unless sleeve seal is to be installed or unless seismic criteria require different clearance.
- I. Seal space outside of sleeves with grout for penetrations of concrete and masonry and with approved joint compound for gypsum board assemblies.
- J. Interior Penetrations of Non-Fire-Rated Walls and Floors: Seal annular space between sleeve and containment, using joint sealant appropriate for size, depth, and location of joint. Refer to architectural specification for materials and installation.
- K. Fire-Rated-Assembly Penetrations: Maintain indicated fire rating of walls, partitions, ceilings, and floors at conduit penetrations. Install sleeves and seal with firestop materials. Refer to architectural specification for materials and installation.
- L. Roof-Penetration Sleeves: Seal penetration of individual conduits with flexible, boot-type flashing units applied in coordination with roofing work.

- M. Aboveground, Exterior-Wall Penetrations: Seal penetrations using sleeves and mechanical sleeve seals. Select sleeve size to allow for 25-mm annular clear space between pipe and sleeve for installing mechanical sleeve seals.
- N. Underground, Exterior-Wall Penetrations: Install cast-iron "wall pipes" for sleeves. Size sleeves to allow for 25-mm annular clear space between conduit and sleeve for installing mechanical sleeve seals.

3.7 SLEEVE-SEAL INSTALLATION

- A. Install to seal underground, exterior wall penetrations.
- B. Use type and number of sealing elements recommended by manufacturer for containment material and size. Position containment in centre of sleeve. Assemble mechanical sleeve seals and install in annular space between containment and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make watertight seal.

3.8 FIRE STOPPING

- A. Apply fire stopping to electrical penetrations of fire-rated floor and wall assemblies to restore original fire-resistance rating of assembly. Fire stopping materials and installation requirements are specified in the architectural specification.

3.9 PROTECTION

- A. Provide final protection and maintain conditions that ensure coatings, finishes, and cabinets are without damage or deterioration at time of Practical Completion.
 - 1. Repair damage to galvanized finishes with zinc-rich paint recommended by manufacturer.
 - 2. Repair damage to PVC or paint finishes with matching touchup coating recommended by manufacturer.

END OF SECTION

ELECTRICAL SECTION: 07

EARTHING SYSTEM

1.0 GENERAL**1.1 SUMMARY**

- A. This Section includes complete installations to earth every source of energy and to provide protective earthing and equipotential bonding in line with local electricity authority's regulations and requirements, including:
1. Transformer neutral earthing.
 2. Main earthing terminals or bars.
 3. Exposed conductive parts of electrical equipment.
 4. Extraneous conductive parts.
 5. Standby generator neutral earthing
 6. UPS Earthing
 7. Telecommunications System Earthing
- B. Scope of work shall include but not limited to the following:
1. Main earthing system of the building consists of Earth Continuity Conductors (ECC), earth bars, earth pits, connections to structural reinforcement and all related accessories including bonding and terminations.
 2. Earthed equipotential bonding of all exposed and extraneous conductive parts
 3. Earthing system for the telephone, GSM and data communication system including ECC conductors, earth bars, earth pits, all related accessories including bonding and terminations.
 4. Clean earth arrangement for the power distribution and power outlets feeding computer terminals and related equipment
 5. Substation earthing system including transformer neutral earthing
 6. Earthing system for the external electrical services.
 7. Earth termination for the Lightning protection system conductors.
- C. The complete installation covered by this specification shall be efficiently bonded back to the main switchboards through the metal sheathing of cables, the steel conduit, the trunking system and earth wires/bars as per DEWA regulations.
- D. Provide equipotential bonding connections to all exposed metalwork and other conducting materials to ensure that no differences in potential exist. Provide a safe path to earth for any electrical current created by an electrical fault within or external to the premises.
- E. The general electrical services contractor shall be responsible for provision of earthing conductors from earth bars to equipment/studs on ductwork, pipework and all mechanical and electrical systems including, electrical continuity at joints of ductwork, pipework etc.
- F. Generators: Where generators are specified at 400 Volts the generator neutral, generator frame, exposed conductive parts and extraneous conductive parts shall be bonded to the generator main earth terminal. This earth terminal shall be bonded to an independent earth electrode and to the main generator room dedicated earth bar which will also be earthed to an independent earth electrode.

- G. In general, all sub main cables shall be provided with a separate CPC clipped to the sub main cable, the CPC size shall be sized to provide the appropriate disconnection times and must comply with local DEWA requirements.
- H. 70mm² to 150mm² cables rising to local earth bar riser shall be installed in each electrical rooms and shall be connected to a 6mm x 50mm x length (as shown on the plan). Earth bar with enough number of ways and mounted on high impact uPVC channel with disconnecting link provided in each electrical room and shall be connected to the individual switchboard earth bar through a cable sized in accordance to IET Regulations and to the equipotential bonding conductor connecting metal conduits, trunking systems, metal frames, enclosures, cable sheaths, A/C duct work, pipe riser etc.
- I. A 'clean' earth 150mm² copper bar riser shall as well be installed and shall be connected to 'clean' earth bar similar to the earth bar specified above and installed in each electrical room. This bar shall provide clean earth to all the clean earth power outlets.
- J. All switchboards, metal conduit and trunking systems, metal frames, enclosures, lighting fittings and cables sheaths shall be bonded together and connected through 16mm² PVC cable to the earth bar of the appropriate switchboard. Similarly, all earth pins and metallic plates of socket outlets, switches, accessories and enclosures shall be bonded to earth with earth continuity conductors. Each individual earth path shall be electrically continuous throughout its length from the farthest point of the associated part of the system back to the main earth.
- K. Where separate pieces of apparatus connected to different phase are, of necessity erected or positioned less than two meters apart, an equipotential bonding conductor of single core 70mm² PVC sheathed shall be installed between such pieces of apparatus.
- L. The star point of the transformers shall be solidly earthed through separate earth well and to DEWA approval.
- M. All earthing cables shall be PVC insulated with green/yellow color and in accordance with the relevant requirements called for in the cables section of this specification.
- N. All wires and cables shall be protected against mechanical stress and corrosion. All joints between wires and earthing metal work shall be mechanically sound and bimetallic connector shall be used wherever necessary.
- O. If earth pits cannot be installed in a particular project earthing Grid (Mesh) system shall be provided. Ensure separate grid is provided for MV, LV, Generator, and ELV/UPS/IT/Security services. Two dedicated earth leads and earth bars shall be provided from the respective Grids to the service. The mesh conductor depth shall be at 900mm from FFL. Tinned copper conductor shall be provided with suitable corrosion resistance bonding seal. Area of mesh and conductor size shall be as per the fault discharge calculation in accordance with local authority regulations.
- P. All necessary earthing fault level calculations shall be provided. Adequate clearance shall be provided for the earthing between different services.
- Q. The contractor shall supply and install Main equipotential bonds and supplementary bonds in accordance with DEWA regulations. Main equipotential bonds shall be wired from substation main earth bars and installed to the following:
1. Steel structure
 2. Main Cable trays/Cable ladders
 3. Sprinkler Pipe work
 4. Lifts and Lift motor rooms

5. LV Switchgear
6. Metallic cable sheaths of all cables except Telecom Cabling
7. Lightning protection systems

1.2 SUBMITTALS

A. Product Data: The contractor shall provide submittals for all specified and proposed products for the earthing system. Provide manufacturer's specifications and technical data including performance, construction and fabrication. Contractor shall submit the following:

1. Operation and Maintenance (O+M) Manuals
 - a) Include component list with manufacturer's reference numbers, descriptions of materials and procedures for repairing.
 - b) Instructions for periodic testing and inspection of grounding features at test pit based on IEC.
 - i. Periodic tests shall determine if ground-resistance or impedance values remain within specified maximums, and instructions shall recommend corrective action if values do not match the defined values.
 - ii. Include recommended testing intervals.
 - iii. Recommend spares to make sure the system works without interruption or down time that could impair the operation of the services.
2. Original catalogues for various proposed components
3. Schedule of Accessories and Fittings
4. Test Certificates
5. Warranties: Submit written guarantee signed by the contractor or the manufacturer for the period of 1 year or manufacturers standard warranty whichever is longer from the date of substantial completion. The guarantee shall cover the repair and replacement of defective materials and workmanship as directed by the engineer except where any specific warranty from a supplier or manufacturer extends for a longer time.

B. Submittals:

The contractor shall submit dimensioned shop drawings for approval and as-built drawings identifying the location of installed earthing system components including the following:

1. Test pits/points
2. Earth Cable routes, cross-section, depth of laying and covering
3. Earth rods, pits and details of installation and connections
4. Earth rings, Cross sectional area of all earthing, protective and bonding conductors
5. Earthing arrangements and connections for separately derived systems
6. Earthing for sensitive electronic equipment
7. Earth Bars
8. Indicate dimensions, description of materials, general construction, specific modifications, component connections, anchorage methods, hardware, and installation procedures.

Supplementary Product Literature: Include a statement from the manufacturer for the design life of the system.

Calculation as specified in this section to ensure proper earthing system as per specified standard.

Outline technical specifications reflecting proposed materials and systems.

Compliance Clause: The contractor must submit a complete specification compliance statement for the proposed cables.

Summary of deviations from the Specification.

Manufacturer's Certificate: Certify that products meet or exceed specified requirements.

List of tests included, certified test data, method statements for installation and quality plan.

- C. Qualification Data: The contractor shall submit details of their proposed qualified testing agency including proposed testing agency's field supervisor for approval.
- D. The contractor shall submit field quality-control documentation prior to installation and reports upon completion of the earthing installations.

1.3 QUALITY ASSURANCE

- A. Ensure the construction, materials and finishes of all equipment are suitable for the location, climatic and operating conditions to which the installation shall be exposed.
- B. Acceptable manufacturers:
 - 1. Manufacturer Qualifications: Company regularly engaged in the manufacture of all materials specified in this section of types and sizes required, whose products have been in satisfactory use under similar service conditions for not less than 10 years.
 - 2. Manufacturer's Quality System: Registered to ISO 9001 Quality Standards, including in-house Engineering for product design activities.
 - 3. Manufacturers must be included in the Specification, drawings or schedules.
- C. Installers Qualifications: Use competent persons to install the equipment and provide them with documentary evidence of their competence, which the Engineer or Employer, or their nominees, shall be entitled to see at any time.
- D. Ensure that the whole installation complies with the relevant standards, including the following:
 - 1. DEWA –Regulations for Electrical Installations
 - 2. BS 7211 - Fixed Wiring Cables
 - 3. BS 7430+A1 : 2015 - Code of practice for protective earthing of electrical installations
 - 4. BS 1377 - Methods of test for soils for civil engineering purposes
 - 5. BS EN 12163 - Copper and copper alloys. Rod for general purposes
 - 6. BS EN 12164 - Copper and copper alloys. Rod for free machining purposes
 - 7. BS EN 12167 - Copper and copper alloys. Profiles and bars for general purposes
 - 8. BS 951 - Earthing clamps for earthing and bonding specification
 - 9. BS 970 - Earthing material, solid copper rods and accessories
 - 8. BS EN 50310 - Telecommunications bonding networks for buildings and other structures.
 - 9. BS 4444 - Guide to electrical earth monitoring and protective conductor proving

10. UL 467 - Safety Grounding and Bonding Equipment
 11. BS EN 13601 - Copper and copper alloys. Copper rod, bar and wire for general electrical purposes
 12. BS EN 62561 - Lightning Protection System Components
 13. BS EN 62305-3 - Protection against lightning. Physical damage to structures and life Hazard
 14. International Electrical Testing Association, NETA Standards
 15. National Fire Protection Association, NFPA Standards
 16. International Electro-Technical Commission, IEC Standards
- E. Provide testing and inspections as indicated under Field Quality Control and in accordance with the requirements of this document and those noted in the summary above. Preconstruction Testing/ Reports:
1. Submit reports of independent tests demonstrating that the products and systems comply with the specified performance requirements.
 2. Where test results for a material or product are not available, undertake testing to show compliance with the Specification at an independent testing laboratory acceptable to the Engineer.
 3. The provision of testing data or the carrying-out of tests does not relieve the Contractor of his responsibilities regarding the performance requirements, durability or service life requirements.
- F. Use materials and installation methods in accordance with DEWA regulations.

2.0 PRODUCTS

2.1 SYSTEM DESCRIPTION

- A. Component parts of earthing system are to include the following:
1. Conductors
 2. Connectors, Clamps and Lugs
 3. Protective Cable Terminations
 4. System Earthing / Earthing electrodes (rods, couplings, etc.)
 5. Earth Electrode Clamps
 6. Earth Electrode Inspection Facilities (earth pits, earth rod seals)
 7. Equipotential Bonds
 8. Main earthing terminals or earth bars.
 9. Earth Tapes
 10. Electrically independent earth electrodes for special systems.
 11. Earth seal component for earthing in the basement raft.
 12. Accessories and termination fittings and other materials.
- B. The incoming supply and its earthing arrangement shall be as shown on the drawings and schedules and shall be agreed with the Electrical Supply authority by the Contractor.
- C. The design and installation to ensure to meet the following:

1. Protection against indirect contact shall be by earthed equipotential bonding and automatic disconnection of supply.
 2. Earthing and Bonding conductors shall be sized to carry the fault current and to pass sufficient current to enable the protective devices to operate in the required times.
 3. Time Grading shall be provided between the Earth Fault protection devices.
 4. The non-current carrying parts of the installation (Metallic wireways, enclosures, cable armour etc.) shall be bonded together and to earth such that in the event of fault to earth the overcurrent and or earth leakage protective devices trip to clear the cause of the fault.
 5. The complete installation covered by this specification shall be efficiently bonded back to the main switchboards through the metal sheathing of cables, the steel conduit, the trunking system, and earth wires/ bars.
 6. The Electrical Systems Cable ways / containment (Conduit, trunking, Tray, Ladder etc.) shall be bonded to earth but not used as the primary Earth conductor for any circuit. The primary earth Conductors for all circuits shall be cables: -
 - a) Final circuits in conduit or trunking: separate wired earth conductor of the same size as the phase conductor. Earth to be contained in the same trunking/conduit.
 - b) Multicore un-armoured cables: wired earth conductor of the same size as the phase conductor. Earth to be contained in the same cable or cleated alongside.
 - c) Multicore armoured cables: Separately wired cable in accordance with the requirements in 7 below.
 7. The nominal cross sectional area of all earth continuity conductors shall be as specified by the local power supply authorities unless, but in no case shall the size be less than that required to accord with the IEE regulations and to Engineer approval.
 8. All switchboards, metal conduit and trunking systems, metal frames, enclosures, lighting fittings and cables sheaths shall be bonded together and connected through 16mm² PVC cable to the earth bar of the appropriate switchboard. Similarly, all earth pins and metallic plates of socket outlets, switches, accessories and enclosures shall be bonded to earth with earth continuity conductors. Each individual earth path shall be electrically continuous throughout its length from the farthest point of the associated part of the system back to the main earth.
 9. Where separate pieces of apparatus connected to different phases are, of necessity erected or positioned less than two meters apart, an equipotential bonding conductor shall be connected between their earthing terminals.
- D. Earthing Cables: Use copper conductors with insulation, rated for wet location. Unless otherwise indicated grounding conductors shall have a continuous green outer insulating jacket for size 16mm² and smaller and with green tape banding for size 25mm² and larger, marked at each access point, for example at junction boxes and enclosures.

2.2 CONDUCTORS

A. Insulated Conductors (Circuit Protective Conductors):

Material - Insulated cable, single core to BS - 6491b or integral conductor of multi-core cable, green/yellow colour insulation. Copper wire, for both insulated and bare earthing conductors in direct contact with earth, concrete, masonry, crushed stone and similar material.

Provide protective conductors sized in accordance with DEWA Regulations.

B. Bare Copper Earthing Conductors for bonding switchgear and electrical equipment together:

1. Use - Earthing conductor and shall conform to the following:
 - a) Solid Conductors: ASTM B33
 - b) Assembly of stranded conductors: ASTM B8
 - c) Tinned conductors: ASTM B33
2. Minimum dimension in accordance with BS and local regulations
3. Form - Strip.
4. Material - Copper, annealed.
5. Coverings – as required by installation
6. Accessories - Conductor clips, metallic.

C. Earthing electrode conductors: Standard copper cable.

D. Main Earth Bar: The contractor shall provide proprietary predrilled rectangular bars of hard drawn, tinned high conductivity copper bar, as indicated on drawings and schematics with holes sized as per earthing schematic and spaced 50mm apart. Provide stand-off insulators for earth bar mounting to give 30mm clearance at rear. Comply with BS 7430+A1 including testing.

Where earth bar requirements are not indicated elsewhere comply with the following:

1. Provide earth bar at incoming electrical service position adjacent to main LV switch panel.
2. Bond earth terminals and metallic structure of switch and control gear and plant.
3. Mount earth bar on insulated supports located at 300mm centers for 25mm bar and 450mm centers for 50mm bar, giving 30mm clearance at rear of bar.
4. Provide 25mm bar earth matrix in each plant room for bonding of plant room services. Plant room earth bars to be connected to main switch room earth bar by 25mm² cable.
5. Connect each earth terminal to all other earth terminals by a ring conductor sized in accordance with BS 7430 and local regulations

E. Miscellaneous Conductors:

1. Braided Bonding Jumpers: Copper tape, braided bare copper wire terminated with copper granules.
2. Bonding Straps: Soft copper, 1mm thick and 50mm wide except as indicated.

2.3 CONNECTORS

A. Connectors generally shall:

1. Comply with IEC / BS EN / UL where applicable

2. Clamped earthing connections shall not cause damage to the equipment, materials, structure or fabric to which they are to connect/bond.
 3. The type of clamp to be used for each purpose shall be agreed with the Engineers. Samples and full manufactures data shall be submitted for perusal and comment.
 4. Pressure Connectors: High conductivity plated units
 5. Bolted Clamps: Heavy Duty type
 6. Exothermic -welded connection: Selected as per manufacturers instruction for specific types, sizes and combinations of conductors and connected items.
- B. Shall be Cast copper, copper alloy, or bronze alloy suitable for use with aluminium and copper:
1. Double bolt type with formed shoe and U cable clamp for connection to pipe or conduit.
 2. Single bolt type with cable shoe and U cable clamp for connections to flat bar or metal.
 3. Double bolt, parallel conductor split clamp type for cable to cable connections.
- C. Equipment grounding conductors sized, 6mm² and smaller may be connected with twist-on connectors.
- D. Bonding connections to boxes and enclosures shall be attached with machine screws or approved head screws only. Sheet metal screws and push-on connectors shall not be acceptable.
- E. All connectors shall be proprietary type and supplied by manufacturers who specifically supply connectors for earthing systems. All connectors shall be fit for purpose and installed in accordance with DEWA regulations.
- F. Earthing Clamps: Use clamps complying with BS, for bonding pipes and lead sheathed cables.
- G. Welded Connectors: Use Exothermic-welding kits of types recommended by proprietary manufacturer for materials being joined and installation conditions. All uses to be agreed with the Engineers.
- H. Bus-bar Connectors: Use Mechanical type, cast silicon bronze, solderless compression or exothermic-type wire terminals, and long-barrel, two-bolt connection to earth bus bar.
- I. Copper Tape Joints
1. Provide waterproof protection at joints subject to moisture.
 2. Joint copper tapes by brazing, using zinc-free brazing metal with melting point at least 600°C or thermic welding.
- J. Stranded Conductor Joints
1. Provide waterproof protection at joints subject to moisture.
 2. Joint copper stranded conductors with compression joints to BS EN 61284.
 3. Provide lugs or tags to enable connection of bonding conductors to equipment earth terminals.
- K. Dissimilar metals: Where dissimilar metals are used for an earth system, that purpose made jointing materials is used such that corrosion and deterioration of the electrical connection are not caused. Ensure bonding connections to other metal parts of building are electrolytic

compatible with those metal parts. Use the guidance given in BS 7430+A1 Table 8 when bonding dissimilar materials.

- H. Clean Earth Distribution: Comply with BS EN 50310: “Application of equipotential bonding and earthing in buildings with information technology equipment” and TIA standard 568/569. Where clean earth distribution is required provide a separate earthing system from the clean earth busbar either integral to or adjacent to clean earth distribution boards. Mount all bus bars with insulators and separate from other earthing systems.

2.4 PROTECTIVE CABLE TERMINATIONS

- A. For bolted connections use crimp type lugs compressed by automatic tool to achieve correct pressure and crimp depth.
- B. Make connections between tape and equipment using high tensile grade brass bolts with brass nuts, washers and locking devices. Use phosphor bronze bolts, nuts and washers where connections are liable to corrosion.

2.5 SYSTEM EARTHING/EARTHING ELECTRODES

- A. Earth/Earth Rods: Solid copper rod. Type and number of earth rods shall be selected based on site earth conditions. The contractor shall employ a specialist to advice on the type of earth rod required for the site conditions.
- B. Minimum earth rod size as BS 7430+A1 Table 4.
 - 1. Form rod with female thread at each end to allow for extension if required.
 - 2. Dimensions:
 - a. Rod Diameter – Refer to table 4 of BS 7430+A1.
 - b. Rod Length – 2400mm minimum, extend as necessary in multiples of 1200mm long standard rod length using appropriate coupling mechanism to achieve the resistance to earth value of:
 - i. Functional earthing $<1.0\Omega$
 - ii. Equipotential bonding $<0.1\Omega$ and
 - iii. Lightning protection earth termination $<10.0\Omega$
 - iv. For special earthing requirement it should be as per the guideline from the specialist consultant. (Refer clause 2.1 (A) 10 above.
 - 3. Where the resistance to earth value is high due to the poor conductive characteristics of the soil mass, use the Conductive Earth Mix (CEM) to enhance the performance of the earth electrode and to provide longevity of the earth electrode. Proper calculation to justify the same needs to be submitted for engineer’s approval from the specialist manufacturer or supplier.
 - 4. Complete with detachable Driving head for forcing the rod into the ground.
 - 5. Connections to the top of the rod shall be via GUV Cable clamps contained in suitable Earth inspection Pits.
 - 6. Earth rod earth bars shall be provided in the earth pits where required to terminate the number of cables required by the particular specification. These shall be:
 - a. Minimum 25mm x 5mm and be sufficiently long to have separate cable studs for each of the cable/connections required to the earth bar.
 - b. Connected to the top of the rod via a 70mm² cable or 25 x 3mm copper tape.

- C. Earth electrode couplings:
 - 1. Use high strength driving cap in contact with driven rod and couplings of compatible material fully enclosing the rod threads.
 - 2. Interconnect electrodes using bare copper tape 25mm × 6mm.
- D. Provide concrete cover, permanently labelled, for electrodes installed through cable draw-pit bases.
- E. Connect main earth conductor to first electrode using heavy duty purpose made silicon aluminium bronze body conductor clamp and high tensile phosphor bronze bolt.
 - 1. Material, minimum size as BS 7430+A1 Table 4 - Copper.
 - 2. Accessories: Rod to tape clamp. Sized to suit earth rod and connector.

2.6 EARTH ELECTRODE CLAMPS

- A. Connect tape to electrode head using heavy duty purpose made silicon aluminium bronze body connector clamps or leaded gunmetal body connector clamps, and high tensile phosphor bronze bolts to BS EN 12163.

2.7 EARTH ELECTRODE INSPECTION FACILITIES

- A. Provide enclosure for each connection between earth conductor and associated earth electrode system. Install so that top is flush with finished earth or floor level. Ensure enclosure provides adequate access for testing purposes.
- B. 20mm diameter copper earth rod shall be driven to a minimum depth of 6M and an earth bar shall be provided to the top of the earth pit for branch connections. Earth pit shall be protected by concrete housing and MS or concrete cover at grade level. Bentonite shall be provided in the pit around the earth rod.
- C. Double flange Double sealed earth pits polycarbonate heavy duty shall be used for basement application, while standard heavy duty concrete pits shall be used for ground floor applications.
- D. Earth Pits:
 - 1. General:
 - a. Unless required otherwise by the electrical supply authority or noted otherwise in the specification, schedules or drawings heavy duty polymer earth pits with 5000 kg static weight withstanding polymer lid shall be used.
 - b. Earth pits shall be suitable for the pedestrian and vehicular traffic likely for the installation:
 - i. Pedestrian areas – 3500kg
 - ii. Light traffic areas – 5000kg
 - iii. Heavy traffic areas – 8000kg
 - c. Earth pits shall:
 - i. Be suitable for the installation of an integral Earth bar to facilitate the connection of multiple cables.
 - ii. Have ducted cable entry where they are installed cast into concrete.
 - iii. Have a brass plate, engraved "Electrical Earth below".
 - iv. Be of nominal dimensions of 300mm x 300mm x 212mmD.

2. Heavy Duty Polymer Earth pits shall be:
 - a. Compliant to BS EN and tested to IEC/BS EN.
 - b. Complete with heavy duty polymer lid that has a recessed lifting hook at the head of the earth rod to protect the earth rod and provide access.
3. Polymer Earth Pits shall be:
 - a. Compliant to BS EN and tested to IEC/BS EN.
 - b. UV stable and chemical resistant.
 - c. Rated for a load of 5000Kg.
 - d. Fitted with a lockable lid.
 - e. Suitable for use with Earth Rod Seals (puddle flanges) to prevent water seepage into buildings when used internally. (Polymer Earth pits shall only be used internally where permitted by the Particular specification, schedules and drawings and where expressly agreed with the electrical supply authority).

E. Earth Rod Seals:

1. Suitable for use with polymer earth pits and shall be used to prevent water seepage into buildings or into the pit when the pits are located within buildings in the basement penetrating the raft to the soil mass beneath it.
2. Specifically designed to work in conjunction with the Earth rod and earth pit that they will be used in conjunction with.
3. Manufactured by the same company as the Earth rods and earth pits.
4. Be pressure tested to 5.5 bar / 80psi and include:
 - a. Sealing Kit: Comprising compression rings and neoprene 'o' rings to seal to the rod into the flange and tube.
 - b. Main Flange and Housing: comprising a single piece high density polyethylene moulding accommodating the sealing kit and the protective tube for the earth rod.
 - c. Bottom seal to securely seal the water proofing membrane and avoid water leakage through the earth termination system.
 - d. Protective tube: made from non-degradable ABS and cut to length to suit the slab to be penetrated.

2.8 EQUIPOTENTIAL BONDS

A. Main (Building Elements)

1. Provide main equipotential bonds in accordance with BS 7430+A1 and DEWA wiring regulations.
2. Material – LSF Insulated single core cable to BS 7211 - 6491b.
3. Main equipotential bonds must be continuous (no joints allowed)

B. Supplementary (Extraneous metalwork)

1. Provide supplementary equipotential bonds to BS 7430 and DEWA regulations.
2. Material – LSF Insulated single core cable to BS 6004.
3. Main equipotential bonds must be continuous (no joints allowed)

2.9 EARTH BARS

A. General Electrical Earth Bars shall:

1. Be Manufactured from Hard Drawn High Conductivity copper to BS EN.
2. Be Mounted uPVC channel with insulating spacers.
3. Have a cross sectional area to safely carry the fault currents matching the higher cross section of the protective earth conductor connected to the earth bar. However, the minimum dimensions should not be less than:
 - a. 700 mm length
 - b. 50 mm width
 - c. 6 mm thick
 - d. length to accommodate all of the required earthing cables and to leave 30% capacity for the connection of future earthing cables
4. Have factory made holes for the connection of the earthing and bonding cables:
 - a. Main Electrical Room Earth Terminal Bar: Two rows of holes on 41mm centres for 21mm bolt, to receive cables from two directions
 - b. Electrical Closet Earth Bar: Two rows of holes at 41mm centres for 16mm bolts
5. Have suitable detachable links (to isolate sections) where required by the particular specification, schedules and drawings for testing of elements of the earthing systems. Such links shall in any case be provided for separation of the Main Earth Electrode(s) for testing.
6. Each connection to each Earth bar shall be labelled to identify its purpose. These labels shall be Engraved and tie wrapped to the Cables in a manner approved by the Engineers.

B. Clean (Data and communication System) Earth Bars:

1. Data systems and communications systems earth bars shall be provided where required on the drawings to provide a low impedance, low noise ('clean') earth reference for IT systems and similar electronic equipment.
2. Data systems and communications systems earth bars shall principally comply with the requirements for general electrical earth bars noted above. They shall have Holes for 21mm bolts for the connection of the earthing cables.
3. Where required by the particular specification these shall be incorporated into polycarbonate enclosures. The Enclosures shall:
 - a. Have a transparent screw retained lid with an Engineer approved identification label.
 - b. Be sized to accommodate the earth bar specified and the cables provided as part of the contract plus a further 30% at some time in the future.
 - c. Have sufficient working space to terminate the cables provided as part of the contract and a further 30% at some time in the future.

2.10 EARTH TAPES

A. Earth Tapes in general:

1. Shall be provided where shown on drawings and shall:
 - a. Be of the material and size indicated.
 - b. Include sheaths or be bare tape as shown on drawings.
 - c. All earth Tape and Tape fixings shall be from the same from the manufacturer.

- d. Joints and connections between tapes and tapes and cables shall be carried using proprietary methods specially designed to join the material being used.
- 2. Copper Earth Tapes shall be to BS EN and UL.
- 3. Sheathing:
 - a. Shall be available in colours to BS
 - b. Shall be PVC or LSF or LSZH as required by the specification schedules and drawings
 - c. Be UV stabilized where used in locations subjected to sunlight
- 4. Earth Tape Fixings:
 - a. Be specifically designed for use with the tape to be fixed
 - b. Where used in conjunction with sheathed tape they shall be of the same finish and colour as the tape
 - c. Normal fixings to masonry shall comprise a spacer bar saddle into which the tape is laid before the saddle cover is fixed in position to secure the tape
 - d. Fixing saddles for Bare Copper tape shall be manufactured from Copper Alloys.

3.0 EXECUTION

3.1 INSTALLATION OF EARTHING SYSTEM

- A. Carry out installation of earthing system in accordance with BS 7430+A1 and DEWA wiring regulations.
- B. Carry out soil tests in accordance with BS 1377 to establish soil electrical resistivity.
- C. Existing Installations: Check earth continuity conductors and loop impedance values of existing installation. Report defects and elements not in accordance with local regulations before connecting new or modified installations to existing supply.

3.2 INSTALLATION

- A. Earthing Conductors: Route along shortest and straightest paths possible unless otherwise indicated or required by Code. Avoid obstructing access or placing conductors where they may be subjected to strain, impact, or damage.
- B. Earth Bonding Common with Lightning Protection System: Comply with local wiring regulations when interconnecting with lightning protection system. Bond electrical power system earth directly to lightning protection system earthing conductor at closest point to electrical service earthing electrode. Use bonding conductor sized same as system earthing electrode conductor, and install in conduit, unless indicated on detailed lightning protection drawings.
- C. Earth Rods: Drive rods until tops are 50 mm below finished floor or final grade unless otherwise indicated.
 - 1. Interconnect earth rods with earthing electrode conductor below grade and as otherwise indicated. Make connections without exposing steel or damaging coating if any.
 - 2. For earthing electrode system, install earthing electrode system in accordance with BS 7430+A1.

- D. Test Pits: Earth rod driven through drilled hole in bottom of hand hole. Hand holes are specified in Division 26 Section "Underground Ducts and Raceways for Electrical Systems," and shall be at least 300 mm deep, with cover.
1. Test Pits: Install at least one test pit for each service unless otherwise indicated. Install at the earth rod electrically closest to service entrance. Set top of test pit flush with finished grade or floor.
- E. Bonding Straps and Jumpers: Install in locations accessible for inspection and maintenance except where routed through short lengths of conduit.
1. Bonding to Structure: Bond straps directly to basic structure, taking care not to penetrate any adjacent parts.
 2. Bonding to Equipment Mounted on Vibration Isolation Hangers and Supports: Install bonding so vibration is not transmitted to rigidly mounted equipment.
 3. Use exothermic-welded connectors for outdoor locations; if a disconnect-type connection is required, use a bolted clamp.
- F. Earthing and Bonding for Piping:
1. Metal Water Service Pipe: Install insulated copper earthing conductors, in conduit, from building's main service equipment, or earthing bus, to main metal water service entrances to building. Connect earthing conductors to main metal water service pipes; use a bolted clamp connector or bolt a lug-type connector to a pipe flange by using one of the lug bolts of the flange. Where a dielectric main water fitting is installed, connect earthing conductor on street side of fitting. Bond metal earthing conductor conduit or sleeve to conductor at each end.
 2. Water Meter Piping: Use braided-type bonding jumpers to electrically bypass water meters. Connect to pipe with a bolted connector.
 3. Gas Piping: Bond each portion of gas piping system downstream from equipment shutoff valve.
- G. Bonding Interior Metal Ducts: Bond metal air ducts to equipment earthing conductors of associated fans, blowers, electric heaters, and air cleaners. Install tinned bonding jumper to bond across flexible duct connections to achieve continuity.
- H. Earthing for Steel Building Structure: Install a driven earth rod at base of each corner column and at intermediate exterior columns at distances not more than 18m apart, Refer to drawings for details of earthing steel structure.
- I. Where earth rings are required Install a earthing conductor, electrically connected to each building structure earth rod and to each steel column indicated extending around the perimeter of the building.
1. Install tinned-copper conductor not less than 70mm for earth ring and for connections to building steel.
 2. Bury earth ring not less than 600 mm from building's foundation.
- J. Where Concrete-Encased Earthing Electrode are specified: Fabricate according to BS 7430+A1; use a minimum of 6 m of bare copper conductor not smaller than 25MM.

1. If concrete foundation is less than 6 m long, coil excess conductor within base of foundation.
2. Bond earthing conductor to reinforcing steel in at least four locations and to anchor bolts. Extend earthing conductor below grade and connect to building's earthing grid or to earthing electrode external to concrete.

3.3 LABELING & WARNING NOTICES

- A. Protective Conductors: Provide a permanent engraved laminated label durably marked in letters 75mm minimum height "SAFETY ELECTRICAL CONNECTION - DO NOT REMOVE" in Arabic and English in visible position, at each bonding conductor connection to extraneous conductive parts.
- B. Main Earth Conductor - Warning Tapes: Provide green / yellow PVC tapes with engraved laminated labeled "EARTHING CONDUCTOR" in Arabic and English over complete external lengths of main earth conductors at 300mm depth below finished earth.
- C. Earth Bar Label: Label earth bar "SAFETY ELECTRICAL CONNECTION - DO NOT REMOVE" in Arabic and English with wall mounted engraved laminated plastic tablet engraved in 10mm high red letters on white earth.
- D. Comply with requirements in Division 26 Section "Identification for Electrical Systems" Article for instruction signs. The label or its text shall be green.
- E. Install labels at the telecommunications bonding conductor and earthing bus bar and at the earthing electrode conductor where exposed].

Label Text: "If this connector or cable is loose or if it must be removed for any reason, notify the facility manager" in Arabic and English.

3.4 TEST LINKS

- A. Provide two test links, in connections between main earth conductors and earth bus bar. Fabricate each from two additional sections of earth busbar. Mount one section on stand-off insulators matching earth busbar; use remaining section as removable test link. Secure 12mm high tensile brass studs to fixed sections of bus bar and drill corresponding clearance holes in test links and provide brass washers, nuts and locking devices to secure frame/neutral earthing and test links.

3.5 FIELD QUALITY CONTROL

- A. Tests and Inspections:
 1. After installing earthing system but before permanent electrical circuits have been energized, test for compliance with requirements.
 2. The inspection and tests shall be carried out in the same sequence as set out in the IEE Wiring Regulations, Local Authorities and the Engineer requirements and in such time as to allow any remedial work to be completed prior to Practical Completion.
 3. Tests shall be carried out by an independent electrical testing organization.
 4. All the portal frames shall be ring earthed 50mm size and all the joint to be 35mm size and to be connected to the main earth pit - to be checked.
 5. Tests shall be carried out strictly in accordance with a programme to be agreed with the Engineer prior to commencement of such tests. Facilities shall be provided for witnessing of such tests.

6. All instruments necessary for inspection and testing shall be supplied and shall be properly calibrated and operated by personnel skilled in their use.
 7. Allowance shall be made for disconnection or similar operations to satisfy the requirements for testing, etc., and the reinstatement of the installation.
 8. Measure earth resistance not less than 2 full days after the last trace of precipitation and without the soil being moistened by any means other than natural drainage or seepage and without any chemical treatment or other artificial means of reducing natural earth resistance.
 9. Inspect physical and mechanical condition. Verify tightness of accessible, bolted, electrical connections with a calibrated torque wrench according to manufacturer's written instructions.
 10. Test completed earthing system at each location where a maximum earth-resistance level is specified, at service disconnect enclosure earthing terminal at earth test pits and at individual earth rods. Make tests at earth rods before any conductors are connected.
 - a. Measure earth resistance without soil being moistened by any means other than natural drainage or seepage and without chemical treatment or other artificial means of reducing natural earth resistance.
 - b. Perform tests by fall-of-potential method according to BS 7430+A1.
 11. Where resistance to earth exceeds specified values, notify the engineer and include recommendations to reduce earth resistance and to accomplish recommended work.
 12. Restore surface features, at areas disturbed by work of this section. Restore areas disturbed by trenching, storing of dirt, cable laying, and other activities to their original condition.
 13. Prepare dimensioned Drawings locating each test pit, earth rod and earth-rod assembly, and other earthing electrodes. Identify each by letter in alphabetical order, and key to the record of tests and observations. Include the number of rods driven and their depth at each location, and include observations of weather and other phenomena that may affect test results. Describe measures taken to improve test results.
- B. Earthing system will be considered defective if it does not pass tests and inspections.
- C. Prepare test and inspection reports.
- D. Report measured earth resistances that exceed the values indicated as per local wiring regulations.
- E. Excessive Earth Resistance: If resistance to earth exceeds specified values, notify Architect promptly and include recommendations to reduce earth resistance.

END OF SECTION

ELECTRICAL SECTION: 08
LIGHTNING PROTECTION SYSTEM

1.0 GENERAL**1.1 Summary**

- A. Faraday cage type lightning protection systems to BS EN shall be installed to cover the complete building. The specialist vendor shall undertake the lightning protection system design and risk assessment for the building. The level of protection (the roof copper mesh size & the down conductors spacing) shall be decided according to Risk Analysis Calculations.
- B. The lightning protection system shall encompass all structural elements within the building to form a Faraday cage, all structural elements such as piles, slabs, columns, walls and parapets shall be bonded together to ensure electrical continuity in the event of a lightning strike. The air termination network shall comprise surface mounted tapes and strike pads/finials as agreed with the architect.
- C. 25x3mm copper tape air termination network shall be fixed to the building roof using purpose made fixings clamps. No part of the roof within the air termination network shall be more than 5m from a conductor.
- D. Items of roof-mounted plant shall be bonded to the air termination network via copper tapes. All exposed metalwork on the roof or the structure shall also be securely bonded to the air termination network. A bimetal clamp shall be used to bind different metal.
- E. The intention is to utilize the steel rebars within structural columns as down conductors, MEP contractor should provide 25x3mm copper tapes at building periphery (behind cladding tiles) without any additional cost to client.
- F. Each down conductor should have a test point at 1.5m above the ground level or in the inspection pit. Test points shall be installed for future testing and maintenance and to be equipped with lightning counter wherever applicable.
- G. When Rebars are used as down conductors, the following conditions shall be fulfilled:
 - 1. The total resistance from roof till ground shall not exceed 0.2 Ohm when disconnected from the earthing system
 - 2. The overlap between each 2 rebars shall be 20 time the Rebar Dia.
 - 3. Use a rebar coupler to avoid the lap distance or crowding. The clamps should be tested as per BS EN
- H. When Rebars are not used as down conductors, the following shall be considered:
 - 1. Concrete Inspection Earth Pit shall be installed in the ground floor for earth electrodes termination
 - 2. Double flange triple earth seal shall be used if the Earth Pit termination is in the basement.
- I. If the building height is more than 60m Ring conductors shall be installed around the building as follows:
 - 1. Ring conductors shall be installed each 20m starting from 60m to 120m.

2. After 120m, Ring conductors shall be installed each 10m.
 3. For the last 5% of the building height, Ring conductors shall be installed at every slab level.
 4. Note, sizing of rolling sphere shall be according to class of LPS. Class I, 20m; Class II, 30m; Class III, 45m and Class IV, 60m.
- J. Ring conductors shall be connected to each down conductor by suitable clamps.
- K. The level of protection [the roof copper mesh size & the down conductors spacing] shall be decided according to Risk Analysis Calculations which shall be submitted by the specialist vendor for Engineer's approval.
- L. The connection between the down conductor and the earth electrode shall be made below ground level in a small purpose-made pit with removable concrete cover to facilitate future testing and maintenance. Each electrode shall consist of 1200x20mm solid copper rods complete with stud, spike, driving cap and a conductor strip clamp. Sufficient number of rods of suitable length shall be installed to provide an overall resistance to earth not exceeding 10 ohms at any point. With the test clamp disconnected, the resistance of each individual earth should be no more than ten times the number of down conductors in the complete system.
- M. The earth electrode for the lightning protection system shall be connected below ground level to the earthing system earth pit through single core copper uPVC cable or bare solid copper tape as shown on the drawings.
- N. All metal work on or around the building must be bonded to the lightning protection network to avoid side flashing.
- O. Where it is necessary to form tape to tape joints, they shall be of Exothermic weld. Soldering alone will not be accepted. Clamped or bolted joints will only be accepted at test points and for connection of down conductors to earth rods.
- P. All materials used throughout the installation shall be either copper or copper based components which are corrosion resistant and compatible with the application.
- Q. Electronic lightning protection (surge protection device) of suitable rating shall be provided in all data and communication cables interconnecting the building or entering into the building.
- R. The Contractor shall submit fully detailed shop drawing for the air termination network, down conductors, earth termination network and bonding and shall be responsible to provide all the necessary accessories to integrate the system with the architectural finish of the building.
- S. The system shall comprise the following elements:
1. Air terminal strike plates, safety studs
 2. Air termination network
 3. Façade metallic frames
 4. Roof steelwork

5. Down conductors
 6. Earth pits
 7. Bi-metallic test clamps
 8. Earth bridge or test points (steel rebar to copper conductor connection)
 9. Electronic surge suppression devices for power and telecommunication systems.
- T. The system and equipment shall be designed, supplied, installed, commissioned and tested in accordance with IEC, BS EN, BS +A1 and DEWA regulations.
- U. For the avoidance of any doubt, radioactive-type or early streamer emitter-type air terminals are not permitted.
- 1.2 Submittals
- A. Product Data: For each type of product indicated. Include manufacturer's specifications and technical data including performance, construction and fabrication and the design life of the system.
- B. Shop Drawings: For air terminals and mounting accessories.
1. The Contractor shall submit workshop drawings (layout of the lightning protection system, along with details of the components to be used in the installation) and documents related to lightning protection system for the acceptance of the Engineer.
 2. The Contractor shall submit fully detailed and dimensioned Shop Drawings for the air termination network, down conductors, earth termination network and bonding and shall be responsible to provide all the necessary accessories to integrate the system with the architectural finish of the building. Include indications for use of concealment requirements will be met, and calculations required by IEC / BS EN for bonding of grounded and isolated metal bodies.
- C. Qualification Data: Provide qualifications for proposed Installer and manufacturer. Include certification by back ground information confirming projects undertaken in region.
- D. Certification, signed by Contractor, that roof adhesive is approved by manufacturer of roofing material.
- E. Provide field quality-control reports.
- F. Operation and Maintenance (O+M) Manuals: Include component list with manufacturer's reference numbers, descriptions of materials and procedures for repairing and cleaning of finishes and cleaning frequency.
- G. Comply with recommendations in BS EN, for maintenance of the lightning protection system until handover.
- H. Warranties: Submit written guarantee signed by the contractor or the manufacturer for the period of 1 year or manufacturers standard warranty whichever is longer from the date of substantial completion. The guarantee shall cover the repair and replacement of defective materials and workmanship as

directed by the engineer except where any specific warranty from a supplier or manufacturer extends for a longer time.

- I. Other Informational Submittals: Plans showing dimensioned as-built locations of earthing features, including the following:

1. Soil resistivity test points layout plan
2. Earth rods
3. Earth loop conductor

1.3 Quality assurance

- A. Acceptable manufacturers: Firms regularly engaged in manufacture of equipment required for this installation, whose products have been in satisfactory use in similar service for minimum 10 years.
- B. Manufacturer's Quality System: Registered to BS EN ISO 9001 Quality Standards, including in-house Engineering for product design activities.
- C. Installers Qualifications: Use competent persons to install the equipment and provide them with documentary evidence of their competence, which the Engineer or Employer, or their nominees, shall be entitled to see at any time.

- D. Applicable Codes and Standards:

1. BSI Group, (BS) British Standards:

- a. IEC/BS EN 62305 - Protection against lightning, Part 1 - General principles
- b. IEC/BS EN 62305 - Protection against lightning, Part 2 - Risk management
- c. IEC/BS EN 62305 - Protection against lightning, Part 3 - Physical damage to structures and life hazard
- d. IEC/BS EN 62305 - Protection against lightning, Part 4 - Electrical and electronic systems within structures
- e. IEC/BS EN 62561 - Lightning Protection Components (LPC) Part 1 - Requirements for connection components
- f. IEC/BS EN 62561 - Lightning Protection Components (LPC) Part 2 - Requirements for conductors and earth electrodes
- g. BS 7671 - Requirements for electrical installations, IET Wiring Regulations
- h. BS 7430:2011+A1:2015 - Code of practice for protective earthing of electrical installations
- i. BS EN 12163 - Copper and copper alloys. Rod for general purposes
- j. BS EN 12164 - Copper and copper alloys. Rod for free machining purposes
- k. BS EN 12167 - Copper and copper alloys. Profiles and bars for general purposes
- l. BS EN 10083,84,85,87,95,10250 - For earthing material, solid copper rods and accessories
- m. BS 951 - Earthing clamps for earthing and bonding specification
- n. IEC /BS EN 60099 - Specification for surge diverters for alternating current power circuits

2. International Electro Technical Commission, IEC Standards

3. Institute of Electrical and Electronics Engineers, IEEE Standards: ANSI/IEEE C62.41 'Recommended practice on surge voltages in low voltage AC power circuits
4. Lightning Protection Institute, LPI Standards
5. Comply with requirements of DEWA regulations

E. Preconstruction Testing/ Reports

1. Submit reports of independent tests demonstrating that the products and systems are in accordance with the specified performance requirements.
2. Where test results for a material or product are not available, undertake testing to show compliance with the Specification at an independent testing laboratory acceptable to the Engineer.
3. The provision of testing data or the carrying-out of tests does not relieve the Contractor of his responsibilities regarding the performance requirements, durability or service life requirements

2.0 PRODUCTS

2.1 Component Parts

A. The principal components of lightning protection system shall be as follows:

1. Air terminations
2. Air termination network
3. Down conductors
4. Joints and bonds
5. Façade
6. Roof steelwork
7. Bi metallic test clamps
8. Electronic Surge Suppression Devices for power and telecommunication systems
9. Test joints / links and earth bridges
10. Earth terminations
11. Earth electrodes
12. Earthing pit
13. Pile foundation earth termination

2.2 Air Terminations

- A. Air terminations shall consist of horizontal conductors in the form of networks as per British standard: IEC /BS EN. These may comprise natural building components if available, refer to drawings for details.
- B. A Strike plates or safety studs shall be used where it is not possible to install to install conductor on the roof and or side of the building.

2.3 Down Conductors

- A. IEC /BS EN permits the use of down conductors of various types including the use of strips, rod, reinforcing bars and structural steel stanchions. Any good conductor which may form part of the building structure can be included, appropriately joined to the air and earth terminations and in general the higher the number of down conductors used, the lower the risk of side-flashing and other undesirable phenomena.
- B. The down conductor system should, where practicable, be directly routed from the air termination to the earth termination network and be symmetrically placed around the outside walls of the structure starting from the corners.
- C. Pattern and number of down conductors, joints, bonds and test joints shall be as per the recommendations of IEC /BS EN, refer to drawings for details.

2.4 Earth Termination Network

- A. Certain piles identified on the electrical and structural engineers drawings shall be used as earth electrodes and shall be connected to the relevant down conductor, these generally comprise reinforcing in the building structure.
- B. The whole of the earth termination network should have a combined resistance to earth not exceeding 10 (ten) ohms without taking account of any bonding to other services. Reference shall be made to IEC /BS EN and design drawings for further details.

2.5 Earth Electrodes

- A. Provide earth electrodes in accordance BS EN and DEWA regulations, refer to drawings for details.
- B. The earth electrodes should be driven into the ground beneath, or as close as practicable to, the structure and down conductor.
- C. Earthing electrodes shall be solid copper 20mm minimum diameter, 2.4m (metre) length.
- D. The calculation for the electrical characteristics of the soil mass should be performed by the specialists who use the Four point (probe) Wenner method to ascertain the soil resistivity at different location of the project site at probe distance varying from 1meter to 10meter at 1meter increment (total 10 readings per location or coordinate).

2.6 Connectors/Fittings/Fasteners and Hardware

- A. Provide all connectors, fittings, fasteners, hardware, clamps, guards, lugs, etc., as required to connect, interconnect and install all parts of the system (maximum cable fasteners spacing – 1 meter).
- B. Fabricated from copper metal and approved for use intended.
 - 1. Provide connections rated and approved for application between dissimilar metals.

2.7 Test Links

Two-bolt split-coupling, copper alloy, made to join two 2 ends of down conductor specified. Plate indicating position and number of electrodes is to be fitted above each test link.

2.8 Transient Overvoltage Protection

- A. All power, telephone, data and other ELV communication lines which enter or leave the building shall be protected from massive transient over voltage caused by lightning.
- B. Armour of the inter building voice and data back bone cables shall be earthed.
- C. Telephone/Data and other ELV System Protection:
 - 1. Transient over voltage protectors shall be provided for the protection of telephone, data, and other ELV system lines both entering and leaving the buildings. It shall provide effective protection by limiting the transient over voltage to less than twice the normal working voltage of the signal line, and shall provide effective protection between all conductors, line to line and line to earth.
 - 2. It shall be installed at both ends of the line to be protected. It shall be connected in series with the line to be protected and with a connection to the earth.
 - 3. Transient over voltage protectors should not affect in any way the performance and characteristics of the line to be protected and shall not introduce excessive resistance into the line.
 - 4. Average life time of the various protectors shall be more than 10 years.
 - 5. Armour of the inter building voice and data back bone cables shall be earthed

2.9 Power Distribution Protection

Power distribution feeding electronic equipment including those of servers, computers, UPS, PABX, etc shall be provided with electronic transient over voltage protectors to restrict the let through voltage to a low and harmless level, below the susceptibility of the various equipment. It shall be installed in parallel with the power distribution board to be protected. It shall have visual status monitoring indicators indicating the following three level of protection and shall have volt free contacts for remote indication:

- A. Full protection present
- B. Warning- Reduced protection
- C. No protection

2.10 Telephone/Data and other ELV System Protection

Transient over voltage protectors shall be provided for the protection of telephone, data, and other ELV system lines both entering and leaving the buildings. It shall provide effective protection by limiting the transient over voltage to less than twice the normal working voltage of the signal line, and shall provide effective protection between all conductors, line to line and line to earth.

It shall be installed at both ends of the line to be protected. It shall be connected in series with the line to be protected and with a connection to the earth.

Transient over voltage protectors should not affect in any way the performance and characteristics of the line to be protected and shall not introduce excessive resistance into the line.

Incoming telephone lines and PABX shall be protected using plug in type transient over voltage protectors matching with the type of terminal blocks of the MDF. It shall provide full protection between line to line and line to earth. It shall limit transient over voltages to below the damage threshold of PBX circuitry. It shall be designed to provide repeated protection and shall comply with CCITT IX.

Average life time of the various protectors shall be more than 10 years.

3.0 EXECUTION

3.1 Installation

- A. Install lightning protection system as indicated, in accordance with manufacturer's written instructions, and in accordance with DEWA regulations requirements to ensure that installation complies with requirements, the international standards and serves the intended function.
- B. Coordinate as necessary to interface installation of the lightning protection system with the building architecture, structure, roof equipment and earthing.
- C. All installation details shall be agreed with the Engineer and will be subject to his approval prior to commencement of work. The Specialist Lightning Protection Subcontractor shall obtain an inspection certificate from the manufacturer or its authorized representative towards the satisfactory installation of the system and that proper material in accordance to the BS EN standard has been used.
- D. Install conductors with more than one direct path from air termination to earth connections avoiding sharp bends and narrow loops. Conceal all conductor drops to earth within building structure as indicated. Provide flush cable support boxes with gaskets and bolted cast covers, finish as directed by Architect.
- E. Properly guard and protect all work installed in accessible locations.
- F. Install all material in a manner to prevent electrolytic action under the presence of moisture.
- G. Make all roof, wall or other penetration in a manner to prevent the ingress of water or moisture.

- H. Install lightning protection system so that completed work is unobtrusive and does not detract from the building's appearance.
- I. Earthing and Air Termination Conductors
1. Install conductors to earth pits (Test Pits only) within MDPE pipe or UPVC conduit, cover extent of cable route with cable warning tape and cable markers at ground level. Cable markers shall be provided every 20 metres and at change of direction.
 2. Cables on flat roof areas shall be installed surface run, fasteners shall be fixed in accordance with BS EN and manufacturer's recommendations. Color of tape sheath and fasteners shall be agreed with the Architect. Should there be metallic structures forming the roof of the required thickness (conforming to BS EN) and characteristics, they can be connected to form the desired air termination network.
 3. All metallic parts protruding outside the extended air termination network (e.g. balcony hand rails, PV array, BMU rail, BMU hoist/crane, cladding façade, etc.) shall be connected to the air termination using suitable clamps (material complying to BS EN 62561) The equipment's on the roof (i.e. extract fans, air conditioning units, etc) shall be protected using vertical air terminals of suitable height so that it provides the zone of protection. This air terminal shall be installed at a suitable separation distance so that the partial lightning current doesn't flow through the equipment. The cable from the equipment shall be protected using an over-voltage protector. The equipment shall not be bonded to the air termination network. A suitable separation distance calculation shall be provided by the Specialist Lightning Protection Installer once further details of services on the roof have been established - locations, equipment heights, etc. - following final selection by the Contractor.
 4. If air-terminals cannot be installed at the desired separation distance, then the air terminal shall be bonded to the equipment using a braid bond. In this case however, the cable from the equipment shall be protected using a current surge protector of 25KA current carrying capacity.
- J. Down Conductors:
1. The use of the building structure as alternative to the use of down conductors shall be approved by Engineer. Where appropriate routing and installation of down conductors between structural steel columns and the external building envelope shall be coordinated on Site between all relevant Specialist Subcontractors.
 2. Where column reinforcing bars instead of copper dedicated bars are used a minimum of 2 Nos. re-bars have to be used as down conductors in each column. The overlap dimension shall be at least 20 times the diameter of the rebar and shall be done using arc/ electric welding. If the welding is not permitted or not possible then this dimension of the overlap shall be achieved using suitable clamps at every 100 mm interval to ensure electrical continuity.
 3. The electrical continuity of the reinforcing bars shall be determined by electrical testing between the uppermost part and the ground level. The overall electrical resistance should not be greater than 0.2 Ω , measured using test equipment suitable for this purpose. If this value is not achieved, or it is not practical to conduct such testing, the reinforcing bar shall not be used as a dedicated down conductor. If this value cannot be achieved, in practical scenario then a dedicated copper conductor of 25mm x 3mm dimension shall be used as dedicated down conductors alongside the

rebar. However, this has to be determined before start of the construction of the column and accepted and approved by the engineer.

4. Connections to the steel reinforcement should be via a specially designed clamp pre welded to a test clamp two-hole earth point. Each down conductor(s) should have a test disconnection point.
5. All offset re-bar columns are bonded to each other by using 150mm² cable and proprietary joints wherever necessary and as per the details. Special large size "U" bolt clamps are also to be used wherever necessary.
6. Each down conductor bottom shall be connected to the slab re-bars using 150mm² cables and ensuring 120° above bending angle.
Where there is down conductor column terminated or is getting shifted due to the shape of the building proper routing measures as defined in the earlier points and as approved by the engineer should be carried out in the transfer to next nearest peripheral column of the building. No open end termination of the down conductor allowed.

K. Coordinated Surge Protection Devices:

1. The Lightning Protection System Specialist shall provide 25KA current SPDs at each incoming main switchboard, PV panel array combiner box or inverter panel. Suitable over-voltage SPDs with an extra low let - through voltage shall also be provided in the local power distribution boards feeding all IT and ELV systems, in order to protect these against transients generated downstream of the current SPD as per the requirements of BS EN. Surge protectors for a given Location Category shall be rated for a High Exposure Level.
2. The Lightning Protection System Specialist shall also provide transient over-voltage protectors on all data communication/ signal/ telephone lines in order to protect equipment connected to the line, against transient over-voltages. Where data lines link equipment in separate buildings, transient over-voltage protectors shall be installed at both ends of the line in order to protect both pieces of equipment.
3. The contractor shall coordinate with the specialist switchgear manufacturer to provide suitable SPDs for all applicable areas within the project as recommended by the specialist in the risk analysis report.

L. Earthing (ground) Rod:

1. Drive to full depth in soil undisturbed by building excavation; do not drive into excavation backfill area. The contractor shall be responsible for carrying out soil resistivity tests prior to the installation of any earth/ground rods.
2. Provide earthing grid as indicated on the drawings.
3. Install earth rods at least 75cm below finished grade complete with access. All ground rods to be installed within concrete lightning protection earth pit complete with stainless steel identification label. The final location of earth rods and pits shall be agreed with the Architects.

3.2 Field Quality Control

- A. Upon completion of lightning protection system, test the system to demonstrate to the Client compliance with IEC /BS EN, BS and DEWA regulations. Facilities shall be provided for witnessing of such tests.
- B. Check continuity of entire system as the installation progresses and upon completion of the installation. All instruments necessary for inspection and testing shall be supplied and shall be properly calibrated and operated by personnel skilled in their use.
- C. Inspect all connections prior to concealing same and ensure compliance with all applicable sections of this specification
- D. Perform an earth resistance test immediately after driving earthing (ground) rods and furnish test results to Client.
- E. Measure earth resistance not less than 2 full days after the last trace of precipitation, and without the soil being moistened by any means other than natural drainage or seepage and without any chemical treatment or other artificial means of reducing natural earth resistance.
- F. Where resistance to earth exceeds specified values, notify the Engineer and include recommendations to reduce earth resistance and to accomplish recommended work.
- G. All the over-voltage surge protectors shall be tested in accordance with BS EN and IEEE.
- H. Restore surface features, at areas disturbed by work of this section. Restore areas disturbed by trenching, storing of dirt, cable laying, and other activities to their original condition
- I. Install equipment under direct supervision of the equipment manufacturer's representative.
- J. Fully test the complete lightning protection system in the presence of the Client to demonstrate continuity of all conductors and submit a written test report confirming results.

END OF SECTION

ELECTRICAL SECTION: 09

LIGHTING

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1.0 GENERAL**1.1 SUMMARY**

- A. The work specified in this section shall be upheld with all the conditions of the Contract and all general requirements of the Project with regard to the fabrication, supply, delivery and installation of luminaire's designated in this section and in accordance with the Sub-contract Documents.
- B. Provide complete, a luminaire and lamp/light source for each symbol shown on the drawings unless noted otherwise. The luminaires shall comply in appearance, performance, quality and construction with the performance specifications, attached product specification sheets, luminaire diagrams and testing requirements.
- C. Provide all labour and materials required for complete installation and commissioning, including receipt, handling, installation, focusing and final cleaning.
- D. All luminaires specified in this section shall be designed for working from an electricity supply of 230 Volt single phase AC 50 Hertz, according to project requirements and in accordance with all applicable Project Country standards, unless otherwise specified.
- E. All hangers, cables, supports, channels, frames and brackets of all kinds for safely erecting this equipment in place, shall be furnished from the standard manufacturer's product range and shall be erected in place under this Section.
- F. Contractor shall protect all opening edge in gypsum false ceiling by a protection means to engineer's approval.
- G. Each lighting fixture shall have a manufacturer's label affixed to it and shall comply with the requirements of all authorities having jurisdiction.
- H. Fixtures shall be wired with approved fixture wire. Each fixture shall be wired to a single point with adequate slack for proper connections.
- I. All fixtures shall be protected from damage during the installation and any broken fixtures, globes, sockets, stems, and the like, shall be replaced with new parts, without additional cost of the Employer.
- J. The right to select other fixtures of the same quality, without additional cost to the Employer is reserved by the Engineer regarding the shape of the lighting luminaire.
- K. All lighting fixture shop drawings, cuts, etc., shall be complied and submitted in bound brochures. All loose pages, sheets, clippings, etc., will not be accepted. Catalogue cuts shall contain sufficient data to evaluate all properties of the fixtures submitted. If requested a sample of proposal fixture shall be submitted.
- L. Lux level calculation must be submitted by the manufacturer / supplier complying with Dubai building code and applicable standards as a part from the submittals based on the proposed / selected light fixtures with full consideration of mounting height and Finishing of the relevant area/location.
- M. It's contractor responsibility to full coordinate with the supplier prior selection / submission of the material to ensure applicability of installation and combatilby with all associated parties

1.2 CODES & STANDARDS

- A. BS 7671 Requirements for electrical installations.
- B. BS 7430:2011+A1:2015 Code of Practice for Earthing of Electrical Installations
- C. BS 6004 Electric Cables. PVC Insulated and Sheathed Cables Voltages up to and Including

- 300/500 V, for Electric Power and Lighting
- D. BS EN 50525 Electric Cables. Low Voltage Energy Cables of Rated Voltages up to and Including 450/750 V (U0/U)
 - E. BS EN 62717 LED modules for general lighting. Performance requirements
 - F. BS EN 62031 LED modules for general lighting. Safety specifications
 - G. BS EN 62504 General lighting. Light emitting diode (LED) products and related equipment
 - H. BS EN 13032-1 Light and lighting. Measurement and presentation of photometric data of lamps and luminaires
 - I. BS EN 61347-1 Lamp Control Gear, General and Safety Requirements
 - J. BS EN 61347-2-7 Lamp Control Gear, Requirements for Battery-Supplied Electronic Control Gear for Emergency Lighting
 - K. BS EN 62384 Electronic Control Gear for LED Modules, Performance Requirements
 - L. BS EN 55015:2006 + A2:2009 Limits and methods of measurement of radio disturbance characteristics of electrical lighting and similar equipment
 - M. BS EN 61000-3-2:2006+A2:2009 Electromagnetic compatibility (EMC). Limits. Limits for harmonic current emissions (equipment input current ≤ 16 A per phase)
 - N. BS EN 61547:2009 Equipment for general lighting purposes. EMC immunity requirements
 - O. BS EN 60598-1:2015 Luminaires. General requirements and tests
 - P. BS EN 60598-2-22:2014. Luminaires. Particular requirements. Luminaires for emergency lighting
 - Q. BS EN 60598-2-2:2012. Luminaires. Particular requirements. Recessed luminaires
 - R. BS EN 61167:2018+A1:2018 Metal Halide Lamps. Performance Specification
 - S. BS EN 62612:2013. Self-ballasted LED lamps for general lighting services with supply voltages ≤ 50 V. Performance requirements.
 - T. IEC 60529 Degrees of Protection provided by Enclosures (IP code)
 - U. IEC 62560 Self-ballasted LED-lamps for general lighting services by voltage > 50 V – Safety specifications
 - V. IEC 62612:2013 Self-ballasted LED lamps for general lighting services with supply voltages > 50 V - Performance requirements
 - W. IEC 62722-2-1:2014 Luminaire performance - Part 2-1: Particular requirements for LED luminaires
 - X. IEC 62386 The International Standard for DALI Technology
 - Y. IEEE C62.41 Guide on the Surge Environment in Low Voltage (1000V and Less) AC Power Circuits
 - Z. IESNA LM-79-08 Approved Method: Electrical and Photometric Measurements of Solid State Lighting Products
 - AA. IESNA LM-80-08 Approved Method: Measuring Lumen Maintenance of LED Light Sources
 - BB. IES RP-1610 Nomenclature and Definitions for Illuminating Engineering
 - CC. IES TM-21-11 Projecting Log Term Lumen Maintenance of LED Light Source
 - DD. CIBSE Chartered Institute of Building Services Engineers – Lighting Codes. The general lighting installation shall be designed in accordance with CIBSE lighting design guidelines particularly, LG5, LG7, and EN12464-1:2011. All lighting testing & commissioning shall fully comply with CIBSE commissioning Code, lighting code L-2003.

EE. IES Handbook of Illumination Engineering Society.

1.3 SUBMITTALS

A. Manufacturer Data:

1. Name of Manufacturer for each luminaire/lighting product or system.
2. Catalog cut sheets for all standard products. For non-standard, custom or modified products a complete datasheet must be provided.
3. Manufacturer product information including cut sheets and ordering code for all ballasts, lamps, drivers, transformers, test switch / integrated emergency lighting components.
4. If requested, qualification and experience of Manufacturer on similar projects or with similar luminaires/lighting product or system.

B. Shop Drawings and Equipment Tests:

1. Detailed shop drawings for any custom luminaire type or modified elements of non-standard products.
2. Photometric tests (where applicable).

C. Operation and Maintenance Instructions:

1. Lens and/or reflector cleaning procedure.
2. Procedure for replacement of any driver/ballast/transformer and any lamp/light source.
3. Ballast/driver/transformer specification.
4. Specification of any replaceable lamp/light source.

D. Samples and Prototypes:

- E. Working, wired sample in specified finish and colour, complying in all respects in appearance, performance, quality and construction with the performance specifications, attached product specification sheets, luminaire diagrams and testing requirements.

1.4 QUALITY ASSURANCE

- A. Compliance: Provide fixtures that comply with the following, and with any additional local regulations. All products related to lighting and lighting controls shall be sourced from authorized original manufacturers only. The specialist manufacturers shall be engaged in the lighting product manufacturing for a minimum period of 10 years.

1. Electrical

- a. Electrical supply on site is 230VAC (+10%/–6%), 50Hz, single phase and/or 400VAC (+6% and - 10%) three phase in accordance with local authority. All equipment should be compatible with this supply.
- b. Luminaires should exhibit no perceivable change in light output with a variation of plus/minus 10 percent supply voltage.
- c. Total Harmonic Distortion due to any lighting equipment shall be less than 20% percent and meet ANSI C82.11 maximum allowable THD requirements at full output.
- d. Final selection "Look&Feel" shall be coordinated and approved by the Engineer, ID & Client.

2. Physical**a. Materials and Finishes**

- 1). Where galvanically incompatible metals are used in the construction of a luminaire, design must eliminate galvanic corrosion, through separation and/or insulation.
- 2). Paints and powder coat finishes should be appropriate for the luminaire's proposed location. Equipment should employ finishes which are stable to UV light and other environmental factors.
- 3). Any gaskets should be made of materials which do not out-gas chemicals which can degrade the LED dies or phosphors.

b. Ingress Protection

- 1). IP Ratings stated in this specification refer to IEC standard.
- 2). Where IP68 is specified (continuous immersion), test conditions are assumed to be 3m depth in fresh water, for an indefinite duration after being switched on for 30 minutes, then switched off, unless otherwise specified.

c. Fixing

- 1). Removable front-plates, reflectors, trim rings, etc should be physically retained to the luminaire body by means of a wire or chain safety bond.

d. Adjustment and Lockability

- 1). Where luminaires feature pan and/or tilt adjustability, adjustment in both axes should be lockable by means of a knurled knob, allen screw or similar. Such locking mechanisms must not allow for any sag through gravity over time.

e. Assembly

- 1). Luminaires shall be completely factory-assembled including wiring and accessories up to the Point of Contact. All Point of Contact connections shall be installed in an accessible junction box of appropriate IP rating.

3. Maintenance**a. Cleaning**

- 1). It must be possible for all external surfaces of luminaires to be cleaned safely and easily without the use of chemicals or abrasives, and without disconnecting power.

b. Replacement Parts and Repair Services

- 1). Manufacturers should be capable of providing spare parts and/or repair services for specified equipment for a period of not less than ten years from purchase. Replacement LED modules and circuit boards should produce the same output in lumens, colour temperature and CRI, as the original units, regardless of improved conversion efficiency due to technological development - electrical load may be lower than the original units.
- 2). Any replacement of LED module/circuit board or driver electronics should maintain the same dimming curve as the original luminaire.
- 3). It should be made clear at time of order whether equipment is serviceable by end-user or requires Return-to-Base for repair at manufacturer's facility, so that this information can be incorporated into contractor's Operation and Maintenance

manuals at project handover.

4) 3% spares shall be included part from the contract for each type

4. Light Sources

a. LEDs/LED Modules

1). Colour Consistency

- a). The importance of colour consistency between luminaires and over time is critical to the success of the lighting scheme. The light produced by LEDs described as the same correlated colour temperature should produce light that is indiscernible in appearance from other LEDs of the same CCT. It is the responsibility of the Party producing the final specification to compare samples of the specified luminaires to ensure this consistency across the project.
- b). The benchmark for acceptable colour consistency between luminaires of the same type shall be 3 MacAdam ellipse SDCM (Standard Deviation Colour Matching) for all lighting fixtures. The consistency requirements for each specified type is indicated in the product specification sheets included in this document.
- c). For lights which are to be retained into legacy, colour of LED lightsources should not exhibit greater change than four SDCM steps after five years.
- d). LM79 / LM80 test report to be submitted for all the lighting products.

2). Colour Rendering

- a). Unless specifically stated, a Colour Rendering Index of less than 70 will not be acceptable for any lightsource.
- b). Luminous flux, colour (chromaticity, CCT, CRI) and intensity distribution shall be measured and stated in compliance with IESNA.

3). Lifetime

- a). The lifetime of LEDs shall be stated in terms of L70B10, unless explicitly stated otherwise within this specification and shall not be lesser than 50000hrs. Measurement shall be in accordance with IESNA, with extrapolated data in accordance with IES TM. Unless otherwise stated, data should relate to an ambient temperature of 25° C for interior luminaires and 50° C for exterior luminaires.

4). Thermal Management

- a). Luminaire manufacturers shall ensure that sufficient thermal management is provided within the luminaire to maintain the junction temperature of LEDs at or below the LED manufacturer's recommendations. Heat rejection should be by means of heatsink, convective heatpipes and similar passive means. Active cooling by means of fan or oscillating diaphragm will not be acceptable unless explicitly identified in the specification.

5) Dimming Compatibility

- a). LED lamps which are dimmable by "mains dimming" shall be compatible with phase-cut leading-edge Triac dimmers, trailing edge IGBT (transistor) dimmers, and sinewave dimmers. Lamp manufacturers should provide sample lamps to the nominated control system supplier so as to verify dimming compatibility.

6) Form Factors

- a). LED replacement lamps should accurately follow the form factor of the traditional lamp which they are designed to replace, allowing for direct replacement in luminaires designed for conventional lamps without modification.

5. Control Gear

a. Constant Current LED Drivers

- 1). Constant Current LED drivers shall deliver the designated current (at full output) $\pm 5\%$. 2). Drivers shall have an efficiency of at least 80% and power factor of at least 0.9.
- 2) Drivers which are dimmable via a control signal should have a parasitic load whilst in “off” state not exceeding 0.5w.
- 3) Drivers which are dimmable by “mains dimming” shall be compatible with phase-cut leading-edge Triac dimmers, trailing edge IGBT (transistor) dimmers, and sinewave dimmers. Luminaire manufacturers should provide sample luminaires containing final specification components to the control system supplier so as to verify dimming compatibility.
- 4) Control Protocols
 - a). LED drivers must be controllable by means of DALI, , as specified item-by-item in this document. Relevant standards describing each protocol should be followed.
- 5) Flicker
 - a). Drivers and LEDs shall deliver illumination that is free from objectionable flicker as measured by flicker index (IES RP). At all points within the dimming range from 100 percent to specified minimum, luminaires shall have flicker index less than 5% at all frequencies below 1000 Hz.
- 6) Electrical Protection
 - a). All drivers should be able to withstand up to a 1,000 volt surge without impairment of performance as defined by Category A.
 - b). Drivers should have in-built protection against damage due to a short circuit or open circuit on the output, and should recover automatically after fault condition is removed.
 - c). Drivers must incorporate protection against overload on outputs, such that no damage is sustained to the driver itself, and should recover automatically after fault condition is removed.
- 7) Thermal Protection
 - a). All drivers should incorporate over-temperature protection. In the case of ambient temperatures exceeding normal operational conditions, drivers should reduce output, recovering automatically once ambient temperature falls back within operational range.

b. Constant Voltage LED drivers/power supplies

- 1). Constant voltage power supplies shall deliver the designated voltage with ripple of not more than 10 % (P-P).
- 2). Power supplies shall have an efficiency of at least 80% and power factor of at least 0.9.
- 3). Drivers which are dimmable via a control signal should have a parasitic load whilst in “off” state not exceeding 0.5w.
- 4). Control Protocols
 - a). LED drivers must be controllable by means of DALI, DMX, as specified item-by-item in this document. Relevant standards describing each protocol should be followed.
 - b). DALI devices must carry the DALI logo, indicating compliance testing with IEC.
- 5). Flicker
 - a). Drivers and LEDs shall deliver illumination that is free from objectionable flicker as measured by flicker index (IES RP).). At all points within the dimming range from 100 percent to specified minimum, luminaires shall have flicker index less than 5% at all frequencies below 1000 Hz.
- 6). Electrical Protection
 - a). All drivers should be able to withstand up to a 1,000 volt surge without impairment of performance as defined by Category A.
 - b). Drivers should have in-built protection against damage due to a short circuit or open circuit on the output, and should recover automatically after fault condition is removed.
 - c). Drivers must incorporate protection against overload on outputs, such that no damage is sustained to the driver itself, and should recover automatically after fault condition is removed.
- 7). Thermal Protection
 - a). All drivers should incorporate over-temperature protection. In the case of ambient temperatures exceeding normal operational conditions, drivers should reduce output, recovering automatically once ambient temperature falls back within operational range.
- c. Ballasts and Transformers
 - 1). Transformers for low-voltage lamps shall be electronic and have a stable output voltage of 11.5V/23V, under varying loads, where the output voltage is nominally quoted at 12V/24V.
 - 2). Ballasts for lamps shall operate lamps at high frequency (>20kHz) and shall be
 - 3). Flicker
 - a). lamps and ballasts shall deliver illumination that is free from objectionable flicker as measured by flicker index (IES RP-16-10). At all points within the dimming range from 100 percent to specified minimum, luminaires shall have flicker index less than 5% at all frequencies below 1000 Hz.
 - 4). Electrical Protection

- a). All ballasts and transformers should be able to withstand up to a 1,000 volt surge without impairment of performance as defined by Category A.
 - b). Drivers should have in-built protection against damage due to a short circuit or open circuit on the output, and should recover automatically after fault condition is removed.
 - c). Drivers must incorporate protection against overload on outputs, such that no damage is sustained to the driver itself, and should recover automatically after fault condition is removed.
- 5). Thermal Protection
- a). All drivers should incorporate over-temperature protection, shutting down when required and recovering automatically once temperature falls back within operational range.

B. Samples and Mock-Ups

- 1. Should a sample (in addition to the submittal sample) of a standard, or prototype of a special luminaire or variant of a standard luminaire be ordered, its cost shall be incorporated into a tender price.
- 2. Any such sample prototype or variant shall be a sample of the luminaire type specified, and shall be submitted to the Lighting Consultant, together with full photometric and dimensional data if specifically requested, to enable both its accurate assessment and the detailing of any adjustments deemed necessary prior to the commencement of full production.
- 3. Prototypes are required of a series of luminaires where noted in the Luminaire Equipment Specification.
- 4. Should any further samples be required for use in mock-ups, their cost shall be incorporated into a tender price.
- 5. Sample board shall be submitted along with material submittal including all types. Complete approved/signed-off samples shall be kept at sample room till handing over for the project.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. All equipment shall be supplied with adequate means of protection to ensure its preservation during transport to site and any subsequent storage prior to installation. All vulnerable finished parts liable to scratching or other abrasion during handling and installation shall be further protected by a removable film applied prior to dispatch.
- B. Products shall be stored in manufacturer's unopened packaging, out of extreme heat or direct sunlight, until ready for installation.
- C. The luminaires shall be handled with care and as scratches or finger prints are unacceptable. At the time of acceptance by the Employer, the luminaire's must be completely clean.
- D. Luminaires shall not be switched ON until any and all protective peel is removed from reflectors and/or lenses.

1.6 FOCUSING AND ADJUSTMENT

- A. All aiming and adjusting shall be carried out after the entire installation is completed and where possible, units shall be focused during the normal working day. Where daylight interferes with seeing, aiming shall be accomplished during hours of darkness. Sub-contractor shall have sufficient manpower and equipment available during aiming hours to permit aiming to be done efficiently. Aiming shall be done under the direction of Lighting Consultant.

1.7 PROJECT CONDITIONS

- A. Coordinate locations of lighting fixtures with ceilings, ceiling mounted components, fire protection and mechanical components, and partitions.
- B. Maintain environmental conditions within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's recommended limits.

1.8 WARRANTIES AND SERVICE

- A. All luminaires shall be warrantied for a minimum of five years, or longer if requested by the client under separate communication. Warranty shall cover the premature failure or degradation of performance of all components, including LEDs and drive electronics.
- B. Any luminaires found to be defective upon energization post-installation shall be completely replaced as a whole assembly from the point of contact

1.9 LIGHT FIXTURE SCHEDULE FOR RESIDENTIAL BUILDING

Refer to drawings EL-203-LIGHT FIXTURE SCHEDULE for the same

Generally, all the proposed fixtures must comply with the following:

1. All fixtures must carry 5 years warranty for the entire product and shall have third party independent certification. Luminaire must have any of the approvals or marking such as CB, CE, ENEC, VDE, ETL, Energy Star, RoHS, etc.
2. Life span of the lamp shall be LM-80. 80% lumen maintenance at 50,000hr and the drive current within limit published as per LM 80 report of the used LED chip
3. LED chip used shall be binned to 3 step SDCM or better.
4. Wattage to be as per the lighting layout.
5. Control gears shall have protection against overheating and short circuits, mains voltage spikes, overloads.
6. Tests on ingress protection against dust and water according to declared IP protection degree to EN 60529 or IEC 60529 version applicable at the time of testing or equivalent for all outdoor luminaires
7. Evaluation of luminous efficacy & output and electrical parameter of each luminaire type to EN 13032-4 version applicable at the time of testing or equivalent for all outdoor/indoor luminaires according to declared data.
8. The led chips make and model number used inside each luminaire to be verified and confirmed in the third-party test report to ensure they are the same as declared in the technical submittals vide led chip manufacturer's LM-80 data submitted, and be reviewed in conjunction with the verified temperature data.
9. Third party certificate should be submitted for the fitting including the make, model number of the led driver/PSU used with the test sample as declared to be supplied by manufacturer in the technical submittals.

END OF SECTIONS

ELECTRICAL SECTION: 10
LIGHTING CONTROL SYSTEM

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1.0 GENERAL**1.1 REFERENCE**

Conform to General Requirements for Electrical Services.

1.2 DESCRIPTION OF WORK

- 1.2.1 Supply, installation, testing and commissioning of the Programmable Lighting Management System shall be supplied, installed and commissioned in accordance with the European Installation Bus technology (EIB)/ KNX or Equivalent technology.

The following specification details the minimum standards for performance of components, material and operation of lighting control and dimming system proposed for this project.

The development will be equipped with an Intelligent Lighting Control, Management providing flexible and efficient control and management of lighting. The system shall operate using any of the international lighting control protocol for communication between all items of hardware and shall be supplied, tested and commissioned by the system supplier or his appointed partner and shall be in conformity with the standards as stated under clause 1.4 of this section.

The contractor shall supply and install the lighting control system suitable to project requirement as indicated in the specification as well as in drawings.

The manufacturer shall have local support available through distributor/ system integrators and local office presence in the country to support their supplier in the local market. Systems which are single vendor based and supporting proprietary protocols shall not be acceptable. The system shall ensure that equivalent devices from different manufacturers are interoperable and compatible thus providing the end user flexibility of operation. System shall be capable of expanding to multiple functions

It shall be the contractor's responsibility to ensure that the equipment installed meets the specification requirements. Equipment not meeting this performance specification will not be accepted.

The Programmable Lighting Control System shall be designed & developed in accordance with the European Installation Bus technology (KNX open protocol system) or Equivalent technology shall conform to EN CENELEC standard to cover lighting control (on/off) based on applications such as time, daylight, manual switches, etc.

Lighting control system based on different internationally accepted open protocol shall be acceptable.

A two-wire bus cable shall link all sensors (push buttons, brightness sensors, motion sensors, timers, etc.) and actuators (on/off controllers, dimming controllers, etc.) to each other. The bus cable shall be a twisted pair, screened & shielded with solid conductors and shall be capable of handling information exchange and supplying power to the bus devices. Separate power supply to any of the sensors and actuators shall not be acceptable. The bus cable shall be laid in the building in the form of a linear, star or tree structure similar to the power mains. Systems requiring fixed wiring configurations shall not be acceptable. All the sensors & actuator devices shall be coupled to the bus via easy to install 2-wire bus connectors capable of accepting up to 4 bus connections. Systems requiring crimping, lugging or RJ45 bus connections using CAT 5 UTP or Cat 6 cable also acceptable.

The systems shall be compatible with the luminaires to be controlled and with the luminaire control gear control interfaces (DALI (main lobby, gym, entertainment and multipurpose room), dimming key pad as required to provide the functionality required by the Schedules and Drawings.

The system shall be suitable for interfacing with the fire alarm system so as to switch all controlled lights ON in the case of fire.

The system shall be completely de-centralized and programmable. The programming shall be implemented via a PC or notebook computer located anywhere in the system and having access level passwords. Programming via local system switches or with no password security shall not be acceptable. Any device in the system shall be accessed for programming from the PC location without having to manipulate the device locally.

Each device in the system shall be addressable via a software programmable physical address. Any device in the system shall communicate with any other device via software programmable group addresses (telegrams). Each individual device will respond to only those group addresses for which they are programmed to do so. There shall be an EEPROM storing the physical address, group addresses & other software parameters for every device, thus making it intelligent. No centralized processors or centralized memory storage devices shall be permitted.

It shall be possible to program any of the devices on-line at the working site without affecting any of the system devices or the system operation as well as off-line prior to dispatch of the material to site.

The entire system shall consist of bus lines each consisting of up to 64 devices. Two consecutive lines shall be connected to each other via line couplers that act as network filters and also provide communication between devices in different lines. In the event of failure of a device in one line, only the control functions controlled by that device shall be affected and all other devices shall continue to operate as normal.

The power supply module feeding power to the network shall consist of a built in back-up power to compensate short voltage interruptions of up to 200ms. This back-up power shall enable the system to put all actuators in a fail-safe position (either on or off or as it is) in the event of power failure.

Each of the devices shall have a built-in push button along with a LED. The LED shall lit in the event the push button is pressed and if there is power to the device, thus depicting that the device is communicating in the system.

All devices in the system should conform to CE standards and should be manufactured in accordance with the EMC guidelines and the low voltage guidelines.

System shall be interfaced with the BMS via OPC Server or by BACNET open protocol. The BacNet Interface module shall be from the same manufacturer which shall be compatible with IP network. The system shall be able to integrate with other systems such as Building Management System,

The Multi-protocol module is capable of bridging various protocols such as BacNet, Modbus, and IP with KNX or Equivalent technology system and shall be able to support web based application to support smart phones and tablets to view / control the KNX or Equivalent technology devices in the network. The module shall be from same manufacturer. Third party manufacturers are not acceptable.

Manual ON/OFF controlling of motion detectors shall be provided for the Open/Public area lighting. The maximum spacing between the motion detectors shall be no more than 12 meters. Upon field

detection by body motion, the lighting shall be turned-on automatically for duration of 10 to 15 minutes software adjustable delay-off timer from the central lighting control system.

The motion sensor shall be flush mounted into a ceiling and have a 360-degree field of view. Coverage shall be a minimum of 9 meters and maximum 40 meters diameter circular at the working plane with a ceiling height of 2.5 meters. The outdoor motion sensors shall be either wall mounted or ceiling mounted. The motion sensors shall be with software programmable time delay function and adjustable sensitivity on the unit.

The wall mounted button plate shall support 2-Button, 4-Button and 8 Button with LED indicator, which is used to indicate the operation status of each gang. The wall mounted button plate shall have the ability to be controlled remotely via infrared hand held remote control units.

The lighting control system shall be provision for its own light control PC with graphical user interface to remote monitoring, ON/OFF control and dimming. The system shall utilize web based graphical user interface system so as to ensure that the programming is not stored in the PC, in case of PC failure still there will not be any single point of failure in the system. The system shall be possible to monitor segment wise, floor wise or area wise from the smart tablet to control and monitor which will ease the operator not to fully depending up on the central PC.

The system shall have a facility to upgrade all software automatically on any point of time. The system/network design and components shall be flexible to allow expansion or reduction in size at design or post installation with no loss of system capability and no replacement of network devices due to size or capacity.

The lighting control modules and relays shall be designed in a metal enclosure and shall be able to house the system devices and the related control equipment depending on the number of circuits being controlled. The lighting system proposed should be having the direct connectivity with the same protocol & platform.

Manufacturer shall be in the market for at least 10 years and must provide the product warranty.

1.2.2 Lighting Control Concept

The programmable lighting management and control system shall consist suitable sets operation level.

It shall be possible to pre-program the time channels in advance for any upcoming events as per the requirement

Next level of operation is based on manual override switches. Other circuits are controlled based on, movement detection, local timers, etc. All the Relay module, dimmer modules, switches, sensors and other controlling elements shall be addressable and programmable. Dry output switches and central processor units are not accepted.

The LCP interface module shall be connected to BMS PC including the complete lighting control graphics programmed to the dedicated software for lighting control and monitoring.

It shall also be possible to override each of these control relays manually from the Lighting Control Panel or from the lighting control modules located in the electrical room.

These control modules will be located in Lighting Control Panel located (LCP) next to the distribution boards in the electrical room each of the floors or according to project requirement.

The lighting control system should operate at the extra low voltage of 24Vdc, 9.6kbit/s transmission. The bus cable is laid as a line, star or tree structure in the building.

The programming of the system is carried out with the Engineering Tools Software (ETS) parameterization software via an RS232 or USB interface. They can be used in any location within the bus system. A programming of the KNX or Equivalent technology devices by the Ethernet shall also be possible via IP gateway.

1.2.3 Area Control Concept

The Lighting Control system shall be programmed in such a way that unnecessary usage of power is minimized. Lighting control in different areas shall be as follows;

Ground Floor & Entrance Lighting: lights should be controlled via a local dimmer key pads. Daylight harvesting should also be implemented in areas near to the peripheral of the building towards glassed exteriors. Unnecessary lights should be switched OFF during non activity hours to conserve energy, wherever indicated by the consultant /Client.

Common Area / Public Area lighting: All lights in the lift lobbies, corridors, should be controlled using a movement/presence detector, Time schedule and central control. Lighting Circuits should be designed to achieve a minimum light level (say 30%) during non-activity period. Staircases shall be controlled with Timer push button. During an emergency situation the lights should automatically be switched ON until the situation is restored.

Staircase lights: All staircase lights shall be controlled from motion sensor and an override switch in case of an emergency.

Plant rooms and shafts: manual / local switching.

External lights are switched through a timer and a photocell.

The Apartments will be controlled locally via manual switches as shown on the layouts.

Shell and core Retail and F&B lighting controls to be designed and supplied under fit out contract.

The program concept of the lighting control should achieve switching percentage as below.

30% switching ON in Off-peak activity hours, 70% switching ON in mid activity hours, 100% switching ON in peak activity hours subject to control strategy approval for all FOH areas

At any part of the time the light switching should not be less than 30%. Contractor should change the wiring pattern of the lighting circuits to achieve the above control logics.

Covered Car park and BOH areas: All the covered car park should have lights controlled in such a way that 30% switching ON in Off-peak activity hours, 70% switching ON in mid activity hours, 100% switching ON in peak activity hours subject to control strategy approval and through motion sensors.

All ramp lights should be kept ON 50% always & 100% on peak hours and emergency scenarios. Driveway lights can be designed to have split circuits to achieve 30% on no activity, 70% on medium activity and 100% on activity period. Manual override switches should also be provided. During emergency situation all the lights should be switched ON until the situation is restored. The lighting to external facades shall be controlled via photocell and time clock control. All façade luminaires will automatically turn ON at dusk and switch OFF at midnight.

External Landscape: All lighting to external Landscape area, street and promenades will be controlled via photocell and time clock control. All luminaires will turn on at dusk. Feature luminaires will turn off at midnight whilst luminaires providing functional illumination will remain on until dawn: Pool/tree & planter up-lights - OFF, orientation & signage lights - ON.

The networked lighting control system shall also interface with the façade lighting system and landscape lighting system to guarantee proper interaction.

Temporary PC will be made available by the specialist during testing and commissioning of the system in case the BMS PC is not ready to use.

1.3 QUALITY CONTROL/ASSURANCE

Design, manufacture, testing and method of installation of all apparatus and materials furnished under requirements of these specifications shall conform to latest publications or standard rules of the following:

- A. Dubai Water and Electricity Authority (DEWA) regulations for electrical installation works and Department of Civil Defense requirements.
- B. Regulations for the Electrical Equipment of Buildings (IET), (BS) London.
- C. All devices shall be CE listed specifically for the required loads (i.e., incandescent, fluorescent and magnetic low voltage transformer, etc.). Manufacturer shall provide valid supporting document upon request.
- D. Energy Efficiency in Building Control System BS EN.
- E. Installation (TH35) and Safety clearance must comply with IEC and BSEN.

The Quality Control/Assurance/Assurance/Quality Assurance duties shall be performed by the Contractor. The system of Quality Control/Assurance/Assurance verification shall be in accordance with ISO 9000 standards of quality assurance.

1.4 CODES AND STANDARDS

The following codes and standards to the extent specified herein, form a part of this specification. When an edition date is not indicated for a Code or Standard the latest edition enforces at the time of proposal submitted shall apply

- 1. DM / DCCA Green building regulation
- 2. DEWA regulations.
- 3. BS EN 62717 LED modules for general lighting. Performance requirements

4. BS EN 62031 LED modules for general lighting. Safety specifications
5. ISO/IEC 14543-3 - IT - Home electronic system (HES) architecture
6. EN 13321 - Home or Building Electronic Systems (HBES) - KNXnet/IP
7. CENELEC EN 50090 and CEN EN 13321-1 - Home and building electronic system
8. BS EN ISO 9001 - Quality Management Systems
9. IEC 61000-4-2 - Electromagnetic Compatibility - Electrostatic Discharge
10. IEC 60664-1 - Insulation coordination for equipment: Principles, requirements and tests (Safety clearances)
11. BS EN 60669 - Switches for household and similar fixed electrical installations – electronic switches
12. IEC/BS EN 60715 - Dimensions LV panels: Standardised mounting on rails
13. IEC/BS EN 60730-2 - Automatic Electrical Controls: Particular requirements for timers and switches
14. IEEE C62.41 - Recommended Practice for Surge Voltages in Low-Voltage AC Power Circuits
15. ASHRAE 135- Building Automation and Control Network Protocol
16. SAC GB/T 20965 - Chinese Standard for HBES
17. BS EN 15232-1 - Energy performance of buildings. Impact of Building Automation, Controls and Building Management
18. British Standards

1.5 SUBMITTALS

1.5.1 MANUFACTURERS TECHNICAL DATA

The contractor shall submit the following drawings and documents related with each electrical system for Engineer's approval:

- A. Original catalogues for various proposed components.
- B. Schedule of Panels and details.
- C. Test Certificates.
- D. Warranties: Shall be 1 year from the date of provisional handover of all the works and shall include all parts, labour, and the service technician's travel and living expenses for the entire 1-year period

During the time of the contract and before substantial completion of the electrical installation, submit three (3) copies of descriptive literature, technical data, maintenance recommendations (from the equipment manufacturer), catalogues etc.

Manufacturer's Quality System: Registered to BS EN ISO 9001 Quality Standards, including in-house Engineering for product design activities.

Lighting Control System Components: Listed by CSA specifically for the required loads. Provide evidence of compliance upon request.

The provision of testing data or the carrying-out of tests does not relieve the Contractor of his responsibilities regarding the performance requirements, durability or service life requirements.

1.5.2 DRAWINGS AND INFORMATION:

The contractor / vendor shall furnish the following drawings and documents along this submittal for approval.

- A. General Arrangement drawing showing sensors, panels, wiring details. The following types of drawings to be submitted are:
 - 1. Shop Drawings: Detail assemblies of standard components, custom assembled for specific application on this Project. the drawing shall indicate the description of materials and finishes, general construction, specific modifications, component connections, anchorage methods, hardware and installation procedures, including specific requirements indicated. Submit Shop Drawings for each system or assembly. It shall also the dimensions, weights, arrangement of components, and clearance and access requirements.
 - 2. Block Diagram: Show interconnections between components specified in this Section and devices furnished with power distribution system components. Indicate data communication paths and identify networks, data buses, data gateways, concentrators, and other devices to be used. Describe characteristics of network and other data communication lines.
 - 3. Wiring Diagrams: Power, signal, and control wiring. Coordinate nomenclature and presentation with a block diagram.
 - 4. Coordination Drawings: Submit evidence that lighting controls are compatible with connected monitoring and control devices and systems specified in other sections. Show interconnecting signal and control wiring and interfacing devices that prove compatibility of inputs and outputs.
- B. Original catalogues for various proposed components.
- C. Method Statements for installation. Details of cable terminations and fittings.
- D. Approved test certificate / test report.
- E. Schedule of Accessories and Fittings.
- F. Technical Manual giving installation, operation and maintenance instructions.
- G. Schedule of recommended spare parts.

2.0 PRODUCTS

2.1 LIGHTING CONTROL MODULES

The relay unit shall allow control of 2/4/8/12 independent loads, equipped with a voltage free, magnetically latched relay per channel or equivalent solid state electronic relay according to the project requirement. No derating shall be acceptable for inductive lighting loads. The load rating shall be of 16A per channel or 10A per channel as per project requirement, each relay operation shall be indicated on channel LED indicator or via the override knob. Local ON/OFF toggle knob shall allow individual channels to be manually overridden at each unit. The relays shall have interlocking facility feature for specific operation, and this shall be possible via software from central PC. The individual relays shall be DIN TH35 rail mounted within the LCP enclosure according to EN.

Nominal voltage: AC 230 V, 50-60 Hz and each contact shall have Nominal current rating of 16 A, with $\cos\phi = 0.6$, with Capacitive load: AC 230 V, 16 A, max. 200 μF , and for 10A rated load, the nominal current shall be 10A in each channel with $\cos\phi = 1$ & 0.6 and for capacitive load of AC230V, it shall stand 10A with max of 105 μF . Peak inrush current (150 μs) 600A, Peak inrush current (230 μs) 480A, Peak inrush current (600 μs) 300A.

The module width shall have 4 modules width which is approx. 72 mm.

In the event of power failure or bus wiring failure or control module failure, each of the relays shall attain a pre-programmed fail-safe position ('Switch On', 'Switch Off' or 'Remain as it is').

The control modules shall be capable of been programmed with different applications to suit site requirements for e.g. staircase lighting function which switches 'Off' the relays after a preprogrammed time from the time it has switched 'On'. The application for which a relay has been programmed shall apply irrespective of the signal from which it is controlled.

All switching modules should have different levels of override, overrides for master control, separate overwrite object for BMS and separate overwrite for safety/ security function. By no ways a common object to be used for different override applications.

Each of the relays shall be capable of being programmed with its own 'On' and 'Off' delays that shall be applicable irrespective of the signal from which the relays are controlled.

Each of the control module shall have its own individual address and shall be capable of being programmed from the central PC for the purpose of changing parameters without the need to access the module locally.

2.2 MOVEMENT / OCCUPANCY SENSOR – DUAL FUNCTION(PIR+AMBIENT LIGHT SENSING)

The presence detection / Movement detection sensor shall directly work on KNX or Equivalent technology bus as an integrated system, which shall send commands in KNX or Equivalent technology bus to switch on or switch off the lights based on the occupancy. Sensors with dry contact output is not acceptable, each sensor shall have an EEPROM memory to store its address and programming information.

The sensors shall have targeted connection and disconnection of lights bands depending on the room brightness. Regulation in response to the movement possible. Using the device as presence and/or movement detectors. Detectors application with 2 power off stages. Detectors application with integrated monitoring function. Constant light switch with up to two independent channels. Constant light switch with max. 2 outputs for brightness-dependent switching of two light bands in the area. With integrated KNX bus coupler.

If sufficient day light and natural light brightness is available inside the room, the sensor shall have an intelligence to keep the lights in off condition though it detects the movement / presence in the room.

Ceiling mounting sensor in a size 60mm mounting box, with optimal installation at 2.4 to 2.6 m. Can also be mounted to wall using the surface mounting housing.

It shall be possible to program the sensor from ETS software.

a. **Indoor Short range Flat Motion sensor – Shall be dual function PIR and ambient light sensing For FOH / Common Circulation Areas**

Low profile KNX or Equivalent technology occupancy detector with integrated KNX bus connector.

When using the product database, several operation modes should available:

1. Full automatic mode
2. Semi-automatic mode
3. Slave mode
4. Occupancy-independent regulating mode

Up to three additional switching channels – selectively either daylightdepending or not – for controlling lights, or for transmitting occupancy telegrams. Set values and follow-up times can be changed for all channels using communication objects .Burn-in function with adjustable burn-in time from 1 to 100 hours, activatable via communication object or remote control, in order to profit from the entire operating time of the controlled lamps

With activatable motion LED, deactivatable via ETS parameter, communication object or remote control.Spring clips for quick and easy installation in suspended ceilings.

If enough day light and natural light brightness is available inside the room, the sensor shall have an intelligence to keep the lights in off condition though it detects the movement / presence in the room.

The presence detector can also be set and tested without the ETS, by using remote control during installation.

The sensor shall comply to Low voltage directive and EMC directive

The sensor shall have the following characteristics:

Sensors: 4 x passive infrared

Detection brightness: internal light sensor adjustable from 2 to 1000 Lux

The Sensor must support true detection technology with built-in temperature sensor. The temperature Sensor within the Sensor must be capble of understanding different temperature levels around its vicinity for better sesnivity and must have the capabilities to work independently like a Temperature sensor for HVAC application.

The sensors shall have targeted connection and disconnection of lights bands depending on the room brightness. Regulation in response to the movement possible. Using the device as presence and/or movement detectors. Detectors application with 2 power off stages. Detectors application with integrated monitoring function. Constant light switch with up to two independent channels. Constant light switch with max. 2 outputs for brightness-dependent switching of two light bands in the area. With integrated KNX bus coupler.

Technical Data

Input Voltage:24 V DC from KNX or Equivalent technology BUS

Current Cosumption: 8mA .

Detection range Angle : 360° max.

Detection Range: 9 m diameter when walking across (tangential), 8m Diameter when walking towards

3m Diameter while sitting.

Protection Grade: IP20 / Class II -25 °C to +55 °C Polycarbonate, UV- and shock resistant.

IR adapter for Smartphones, IR-PD-KNX 5 - 1200 Lux ,Mixed light measuring.

b. **Indoor Long range Motion sensor – Shall be dual function PIR and ambient light sensing FOH / Corridors, lift lobby, Indoor stations etc.**

KNX motion detector with integrated KNX bus connector

Switching mode, control mode, slave mode, occupancy-independent regulating mode

When using the product database, several operation modes are available:

1. Full automatic mode.
2. Semi-automatic mode
3. Slave mode
4. Occupancy-independent regulating mode

Set values and follow-up times can be changed for all channels using communication objects

Up to three additional switching channels – selectively either daylightdepending or not – for controlling lights, HVAC devices (with time delay) or for transmitting occupancy telegrams. If presence sensors are also used for HVAC application it must have its own built in Temperature, Humidity and Air Quality sensor for perfect lighting and HVAC controls.

Burn-in function with adjustable burn-in time from 1 to 100 hours, activatable via communication object or remote control, in order to profit from the entire operating time of the controlled lamps.

With activatable motion LED, deactivatable via ETS parameter, communication object or remote control. Additional functions can be set via optional remote control.

The presence detector shall include two detection sensors (passive infrared), a brightness sensor, IR receiver and LED to indicate a detected movement, in test mode indication of the activated programming mode.

The presence detector can also be set and tested without the ETS, by using remote control during installation.

One piece of remote control shall be provided for the project to do the testing, trouble shooting and for maintenance purpose.

The presence detection / Movement detection sensor shall directly work on KNX bus as an integrated system, which shall send commands in KNX bus to switch on or switch off the lights / do constant light regulation based on the occupancy. Sensors with dry contact output is not acceptable, each sensor shall have an EEPROM memory to store its address and programming information.

The sensor shall comply to Low voltage directive and EMC directive .

The sensors shall have targeted connection and disconnection of lights bands depending on the room brightness. Regulation in response to the movement possible. Using the device as presence and/or movement detectors. Detectors application with 2 power off stages. Detectors application with integrated monitoring function. Constant light switch with up to two independent channels. Constant light switch with max. 2 outputs for brightness-dependent switching of two light bands in the area. With integrated KNX bus coupler.

Technical Data

Input Voltage:24 V DC from KNX or Equivalent technology BUS

Current Consumption: 7mA .

Detection range Angle : 360° max.

Detection Range: 40 m diameter when walking across (tangential), 20m diameter when walking towards.

Detection brightness: internal light sensor adjustable from 2 to 1000 Lux.

Protection Grade: IP20 / Class II -25 °C to +55 °C Polycarbonate, UV- and shock resistant.

IR adapter for Smartphones, IR-PD-KNX 5 - 1200 Lux, Mixed light measuring.

c. **Out Door Motion Sensor – Shall be dual function PIR and microwave sensing / BOH – Carpark, Stairs, Driveways etc.**

KNX or Equivalent dual technology motion detector with integrated KNX or Equivalent technology bus connector

Switching mode, control mode, slave mode, occupancy-independent regulating mode

When using the product database, several operation modes are available:

1. Full automatic mode.
2. Semi-automatic mode
3. Slave mode
4. Occupancy-independent regulating mode

Set values and follow-up times can be changed for all channels using communication objects

Up to three additional switching channels – selectively either daylight depending or not – for controlling lights, HVAC devices (with time delay) or for transmitting occupancy telegrams.

Built-in function with adjustable burn-in time from 1 to 100 hours, activatable via communication object or remote control, in order to profit from the entire operating time of the controlled lamps.

With activatable motion LED, deactivatable via ETS parameter, communication object or remote control

Additional functions can be set via optional remote control. Wall, ceiling and corner mounting.

Factory settings 3 min and 20 lux.

Technical Data

Input Voltage: 24 V DC from KNX or Equivalent technology BUS

Current Consumption: 7mA .

Detection range Angle : 230° max.

Detection Range: 20 m when walking across (tangential)

Protection Grade: IP54 / Class II -25 °C to +55 °C Polycarbonate, UV- and shock resistant.

IR adapter for Smartphones, IR-PD-KNX 5 - 1200 Lux, Mixed light measuring.

The Dual technology sensor Microwave + PIR motion detector, designed for surface mount

Temperature-independent detection

2.3 INTEGRATOR MODULE/ MULTI PROTOCOL GATEWAY

The unit shall have following functionality to support the building lighting control system to integrate with BMS and Graphical user interface, the unit shall also work as embedded graphical user interface which enables to easily view the state of lighting control system in a building.

It should support: KNX, Modbus , TCP/IP, BacNet, RS232, USB (the connections should be rapid connection terminals, screw terminals are not acceptable).

The unit shall work as a BACnet server for 500 points for TCP/IP bi-directional communication, the unit must be certified for BTL (BACnet Technical Laboratory) as a B-ASC (BACnet – Application Specific Controller)

The unit shall be integrating the following functions:

Webserver / Internet gateway, Weekly and annual clock, KNX or Equivalent technology IP router, Logic module, Energy and technical dashboard.

For high level integration and customization such as webservices or different communication protocols, the unit shall be flexible enough to program through LUA language

2.4 INTELLIGENT TIMER MODULE

A four channel timer device shall control circuits on a particular network group (line) so as to have local standalone operation. It shall be possible to send signals from a timer channel in one network to circuits in another network if programmed to do so. This is to ensure that all timer channels are available for control of all circuits in the system. Each timer shall be capable of being programmed locally with at least 128 timer programs for any of the four channels combined together, without the need to connect a PC. Each of the four channels shall have a manual push-button to over-ride the channel to 'On' or 'Off' irrespective of the time programs. In addition, each channel shall have a 'Off-Auto-Manual' switch for local over-ride purpose. The timer shall have a built-in battery back-up to ensure that the timer is always operating even in the event of power failure.

2.5 LIGHTING CONTROL PANELS (LCP's)

The LCP shall house the system devices and the related control equipment depending on the no. of circuits being controlled. This is to ensure the power wiring between the DB's and the control modules inside the LCPs is kept to a minimum. The LCPs shall be surface mounted IP 41 (Elec. room), IP 55 (Wet room), similar to final DB metal enclosure together with built-in DIN-rails for easy installation of the control equipment.

The External Brightness Sensor shall have built-in temperature sensor for logic operation in order to avoid false trigger arising due to sudden artificial brightness change.

The wiring of the lighting circuits must be re-arranged without any variations to meet all the specified lighting control requirements.

2.6 LUX LEVEL SENSOR- EXTERNAL LIGHT

The appliance switches upto three lighting groups on and off depending on the daylight. The switching points for the individual channels are set at the potentiometers. Interference is screened out by a parameterizable switching delay. A led show the relevant switching state of the three channels. A light sensor must be connected to detect daylight.

The following functions can be set or selected with the aid of the parameterization software:

- Switching behavior, switching ON / OFF
- Delay time factor
- Once – only / cyclical transmission

Adjustment range	-	200.....20,000 lux
(Switchable by wire jumper)		20...2,000 lux
Power supply	-	via EIB or KNX bus
Inputs	-	1, for light sensor LF/A 1.1
Cable length		max. 100m

Operating and display elements

LED and push button	-	for entering the physical address
3 potentiometers	-	setting switching values
3 LEDS	-	switching state of the 3 channels
Operating temperature range	-	5 °C to + 45 °C
Wire jumper	-	2 screw terminals
I – bus®KNX	-	Pressure contacts on data bus.

Light sensor for light value switch, 3 – channel with mounting bracket for wall mounting

Protection	-	IP 54 acc. DIN 40 050
Operating temperature range	-	40 °C to + 70 °C
Connection lighting sensor	-	3 screw terminals connection cross section

CO2-, Humidity and Temperature Sensor - -25°C to +55°C with +/- 5% or +/- 1°C tolerance.
It shall be possible to set threshold in the devices and based on that predefined triggers shall be initiated.

Brightness Sensor	-	1 to 100,000 lux with +/- 20% or +/- 5 lux tolerance
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2.7 DIMMING SYSTEM

2.7.1 LED Dimmers:

The Unit shall allow controlling up to 2/4 Dimmable circuits for LED lights and various other load types Independent of the circuits which they are feed from. It shall be able to operate different operating mode such as RC, RL, and RL-LED.

The dimmer shall have manual override the circuits with the buttons provided even without programming.

The Device shall have the capability to dim LED (light-emitting diode)
The unit shall have LED indicators to identify various faults such as Overload, Short Circuit, mains Voltage failure, Bus voltage failure, etc.

The dimmer unit shall have the facility to program a minimum level setting and power-up delay time via the lighting control software.

The insulation voltage between the bus and mains shall be tested with minimum 4KV AC.

The unit shall be DIN rail mounted within the unit enclosure.

The unit shall be programmed to return to ON, OFF or No change in case of the bus voltage failure.

All LED dimmers must have an independent dimming curve that can be defined by the operator based on the dimming characteristic. This dimming curve must be defined during commissioning stage based on the behaviors of the LED lamp to avoid any type of load flickering.

The Specialized 2 channel or 4 channel LED dimmer must ensure flicker free dimming for any LED lamps at site.

2.7.2 KNX - DALI Module

If the light fittings are provided with DALI interface, appropriate switching/Dimming actuators with digital primary switching devices that have DALI Interface and connected to the KNX shall be used. There shall be eight channels, of which one shall have the capacity of a switching/dimming actuator. The DALI operating devices of each individual channel shall be directly wired so that no further additional commissioning procedure is necessary for the primary switching devices. The status (brightness & error messages of lamps and ballast) of the DALI primary switching devices shall be transmitted on the EIB/KNX BUS.

The KNX DALI gateway shall connect the cross-function KNX bus with the DALI bus designed exclusively for lighting control. Luminaries with DALI electronic ballasts shall be integrated into a KNX system using this unit. The gateway shall serve as the DALI master and power supply for the connected electronic ballasts. Maximum number of electronic ballasts according to IEC groups with a provision to connect DALI presence sensors and DALI powered switching and dimming interface or set to a defined value based upon 1 or 2 channel DALI gateways.

The device shall be installed on a DIN-rail TH 35 according to EN with the bus connection made via a bus connecting terminal.

Failure indication with LED on request, the DALI side programming shall be done through the ETS embedded tool.

It shall be possible to see the Operating hours per DALI group is very useful for preemptive maintenance. Also, the unit shall provide the indication once the lifecycle operating hour are crossed.

Indication of ECG- and lamp-failure per DALI-ballast by KNX object. Indication of 3 failures per group, also information object if threshold of failures per group is crossed.

The DALI Gateway shall support of Normal-, Permanent-, Night- and Panic-Mode, Selectable switch-On value. Optionally object to switch Off the power the power of groups which are off to save energy.

DALI Controllers should support additional built in features like Scenes, Effects/ Sequence without additional hardware. KNX DALI controllers must support Human centric lighting control with advance Tunable lights on DALI device type 8.

It shall be able to support 16 DALI Scenes within the gateway itself without using an external scene controller. Naming of scenes with up to 20 characters.

Weekly timer to control brightness and color of groups optionally with fading times with up to 300 commands, controllable objects, for repeating lighting applications. The weekly timer file can be importer - and ex-exported as xml-file. KNX object to synchronize the weekly timer daily.

3.0 EXECUTION

3.1 INSTALLATION

Maintain minimum working clearance at equipment according to manufacturer's written instructions.

Install interconnections between system components and EIB/ KNX or Equivalent technology cable terminations.

Install the phase bypass wiring between lighting DBs and switch actuators. All circuits should be properly terminated using cable glands and clearly marked with ferules for the circuit number, section details etc. Make connections to the load circuits in accordance with the manufacturer's wiring diagrams, unless noted otherwise on the Drawings.

Align all motions sensors to achieve maximum field of sensing.

On completion of installation, inspect system components. Remove paint splatters, other spots, dirt and debris. Repair scratches and mars to match original finish. Clean components internally using methods and materials recommended by the manufacturer.

All the Cables used for the lighting control System to be LSZH.

3.2 TESTING AND COMMISSIONING

After the installation is complete, the contractor will conduct operating and commissioning tests. The equipment will be demonstrated to operate in accordance with the requirements of the specification. The system installation, testing and commissioning will be as per the approvals and requirements.

A specialist from the manufacturer will attend and demonstrate the complete system. A company trained representative will personally supervise the complete installation and final testing of the system.

All tests will be carried out in the presence of the Client or persons authorized by the consultant / client. Upon the completion of the acceptance tests, the representatives will instruct operatives in the proper operation, maintenance programming, configuration, and testing of the system. The vendor will provide equipment and /or software which is necessary to allow field modification of the programming and configuration.

3.3 FINAL INSPECTION

At the final inspection, a manufacturer-trained representative shall demonstrate that the system functions properly in every respect.

Instruction:

Provide instruction as required for operating the system. "Hands-on" demonstrations of the operation of all system components and the entire system including program changes and functions shall be provided. The Contractor and/or the Systems Manufacturer's representatives shall provide a typewritten "Lighting control concept program" to the Owner if required

END OF SECTION

ELECTRICAL SECTION: 11
TELECOMMUNICATION SYSTEM

1.0 GENERAL**1.1 Introduction**

This document defines the services and supplies necessary for the installation of a standardized data cabling system to EIA / TIA Standards, CENELEC and ISO 11801 Class E with a fibre backbone.

The characteristics of the cabling system must allow for the widest range of protocols, and support all applications up to and including IEEE 802.x (Gigabit- Ethernet)

The Contractor shall be responsible for providing conduiting and cable tray and trunking system including Telephone Wiring, all to EITC approval.

The Contractor shall liaise with Local Telephone Dept. to ensure that cable entry ducts and internal conduit/trunking tray installations are approved prior to carrying out the installation work.

All external cable ducts, manholes and draw-pits shall be of a size and specification that is approved by EITC.

JRC-12 manholes to be constructed by Contractor based on EITC requirements.

Complete installation including wiring, backbone cable sizes etc. shall meet EITC requirements.

Risers shall be provided as shown on drawings including, cabinets, terminal, riser cables, etc.

Riser cable shall be provided as per requirement of EITC.

GSM containment shall be provided as required. The power supply requirement in GSM Room shall meet with Etisalat /Du requirements and to be provided as per Etisalat / Du GSM Department.

Contractor to submit the detailed shop drawing to EITC prior to carrying out any work at site.

1.2 Standards and Regulations

The installation must comply with the laws and standards in force at the time. In particular:

EITC requirements

Local regulations or codes of practice shall be followed at all times.

The contractor shall be certain that all installation work areas are secure and made safe in accordance with local Health and Safety regulations.

The Contractor should take the following standards into account:

ISO/IEC 11801	Information technology – Generic cabling for customer premises
ISO/IEC/TR3 8802-1	Information technology – Telecommunications and information exchange between systems – local and metropolitan area networks – Specific requirements – Part 1 Overview of Local Area Network Standards.
ISO/IEC/8802-3	Information technology – Telecommunications and information exchange between systems – local and metropolitan area networks – Specific requirements – Part 3 Carrier sense multiple access with collision detection access method and physical layer

	specifications
ISO/IEC 61935-1	Generic specification for the testing of generic cabling in accordance with ISO/IEC 11801 – Part 1: Installed cabling
IEC 60364-1	Electrical installation of buildings - Part 1: Scope, object and fundamental principles
IEC 60950	Safety of information technology equipment, including electrical business equipment
EN50173	Information technology - Generic cabling for customer premises
EN50174-1	Information technology – Cabling Installation. Specification and quality assurance.
EN50174-2	Information technology – Cabling Installation. Installation planning practices inside buildings
ANSI/TIA/EIA-568	Commercial Building Telecommunications Cabling Standard
ANSI/TIA/EIA569	Commercial Building Standard for Telecommunications Pathways and Spaces
TIA/EIA TSB-72	Centralized Optical Fibre Cabling Guidelines
TIA/EIA TSB-75	Additional Horizontal Cabling Practices for Open Offices

The Contractor will have to take account of all addendums and the new standards which could come into effect during the progress of the design and installation.

1.3 Responsibilities of the Contractor

The successful contractor will need to include all essential work in the estimate to ensure the completion of the project to specification and on time. On completion, the contractor will provide final cabling documentation which will include:

- Site plans to include the identifications and locations of nodes, pathways, cables, termination points, closures, patch panels, protection devices;
- As-built information to include nodal, route and cable connectivity, closures (in a schematic or report based format);
- Installed cabling acceptance test records;
- Evidence of conformance to the installation specification from cabling installer or main contractor;
- Handover certification;
- Earthing and equipotential bonding details.

The contractor will ensure that they have all relevant information before tendering. A lack of precision in the plans and specification of the installation cannot be used as an excuse not to complete the installation in full working order.

1.4 Scope of Work

Supply and install complete Active and Passive end-to-end structured cabling system with all accessories for transport of data, video and voice / IP telephony signals through the network from designated demarcation points to outlets located at various desk, workstation and other locations as indicated on the drawings and described herein.

Supply and installation of MDF / MDT, IDF's / IDT, CP's and ACP's as indicated on the drawings and described herein.

Wiring, terminations and patch panels between these designated demarcation points and telecommunication outlets as indicated on the drawings.

Supply and installation of all telecommunication outlets as indicated on the drawings and as specified herein.

The system shall utilize a network of fiber optic backbone cables, UTP Cat-6 A horizontal cables, tie, etc. as indicated on the drawings and described herein.

Supply and installation of active components as required to form a complete system in coordination with the clients IT department.

All cables and terminations shall be identified at all locations.

1.5 Power Supply and Earthing

The Contractor shall provide power supplies as shown in the drawing in Etisalat room.

The earth bar shall be a 60 cm long copper bar 50 mm. X 6 mm. mounted to the wall on insulated moulded pedestals. The earth bar shall be connected using 10 mm² LSOH insulated cable connected to appropriate earthing pit.

1.6 Conduits, Trays, Trunking and Pipes

The Contractor shall provide a complete system of containment, Pipes and conduits to facilitate the installation of Telephone cables and the Data system

All duct shall be provided with a draw wire. Cable duct shall be installed for main distribution runs between M.T.J.B. & J.R.C and shall be heavy-duty type of adequate size to accommodate the cables without stacking.

1.7 Bonding

All conduits, cable trunking and cable tray shall be electrically continuous and bonded to the electrical earth system, from a point adjacent to every junction box.

1.8 Installation:

The contractor shall allow for the design, supply, install and commissioning of a voice, data and CATV as here in specified

Voice & Data system containment shall be installed and tested by the MEP contractor. The installation of Voice & Data system shall be as per the Etisalat requirement and their regulations.

The contractor shall submit a set of documentation comprising of layouts showing cable route & position of all outlets, labelling scheme, layout of cabinets and test results. Contractor shall be consider separate out let boxes for Data as shown in the drawing
Contractor shall submit the workshop drawing for the complete system for the entire satisfaction of the Engineer / Etisalat.

2.0 VOICE & DATA SYSTEM

2.1 General:

The proposed solution must be the latest available technology for the structured cabling system.

The proposed solution of Class D (99) must be provided with a minimum of 15 years performance warranty for the full bandwidth.

Wherever the standards (EIA/TIA, Cenelec, and ISO) are not ratified, the latest "Draft" shall be referred.

All components pertaining to structured cabling system shall be manufactured by one OEM and mixing of components from different manufacturers shall not be accepted.

The offered system shall match or better the Class D (99) channel requirements based on the limits from:

- ISO/IEC 11801: 1995
- Amendment to 11801:1999
- EN 50173:1999
- Amendment to EN 50173:1999
- Draft ANSI/TIA/EIA-568-A-5: May 1999
- Draft ISO/IEC 11801 (rev. 2000) Cancun 1/99 and Berlin 6/99

2.2 Product Specifications:

Horizontal Cable:

The horizontal cable shall constitute of four (4) Unshielded Twisted Pair (UTP), AWG24, with an impedance of 100 Ohms. The outer sheath shall be LSOH and the cable shall be specified up to 100MHz and have the following values or better for Attenuation and NEXT:

Frequency (MHz)	Attenuation (dB)	NEXT (dB)
1.00	2.4	65.0
4.00	3.9	55.0
10.00	6.3	51.0
16.00	8.0	46.5
20.00	9.4	45.0
31.25	11.5	40.0
62.50	16.0	35.5
100.00	21.9	32.0

- The horizontal cabling shall be star-wired.
- The maximum horizontal distance shall be 90Mtr. between the outlet and the patch panel (excluding patch cords and line cords).

- The horizontal cabling shall be free of taps and bridges.
- One cable shall be strictly used for one application; cable sharing will not be acceptable.

Connecting Hardware for Horizontal Cable:

Connecting hardware shall be based on widely used' RJ45 interface. The hardware shall feature excellent transmission performance and mechanical strength. Integral wire management and cable strain relief shall be integrated into the hardware. It should be possible to use the same RJ45 interface module at the outlet end as well as the patch panel (IDF) end. Termination of the cables on the hardware shall be accomplished without the use of special tools. Wherever single RJ45 outlet is used, it should be possible to easily upgrade to dual RJ4S, without any major changes to the faceplate.

Patch Cords & Line Cords:

The patch cords and line cords shall constitute of two modular RJ45 plugs assembled on unshielded twisted pair stranded cable. The UTP stranded cable shall be constructed using AWG26 flexible cable. The sum length of all patch cords & line cords for one outlet shall not exceed 10Mtr. Under no circumstances shall locally assembled patch cords and line cords will be acceptable; all must be factory made.

Voice Backbone Cable:

The voice backbone cable within the building shall be constructed of 25Pair UTP, bare copper conductor AWG24. The cable shall be specified up to 100MHz, have characteristic impedance of 100 Ohm and bundled into 6 units (5 units of 4 pairs, and 1 unit of 5 pairs).

The cable shall support multimedia applications (voice, image and data communication). Apart from the traditional applications like ISDN, Ethernet and Token Ring, the cables shall support applications operating at higher frequency like Fast Ethernet. The Cable shall have the following electrical characteristics or better:

Frequency (MHz)	Attenuation (dB)	NEXT (dB)
1.00	2.1	62.0
4.00	4.3	53.0
10.00	6.6	47.0
16.00	8.2	44.5
20.00	9.2	42.0
31.25	11.8	40.0
62.50	17.1	35.5
100.00	22.0	32.0

The cable when installed shall have a minimum bending radius ≥ 8 times the outer diameter of the cable.

Between the buildings, if required, the backbone cable shall be normal telephone cable (Category3). Depending on where the cable is installed, it shall be either indoor or outdoor jelly filled type. These cables shall be to the specifications of Etisalat.

Connecting Hardware for Voice Backbone Cable:

The voice backbone cables shall be terminated on 110 IDC panels, which shall be housed in the 19" IDF cabinet. These panels shall accept UTP cables having AWG22-26 solid or stranded wire. Highly

visible, colour coded strips shall be used to identify individual pair of the backbone cable. Apart from the performance of Class D (99) specifications, these panels shall have features like gas-tight connection, integral strain relief mechanism.

Data Backbone Cable:

The backbone cable for data applications shall comprise of fiber optic cables. The cables shall have minimum 6 cores of Multimode 50 μ /125 μ graded index fiber strands. The fiber optic cables shall have the following characteristics:

Wavelength (nm)	Max. Attenuation (dB)	bandwidth (MHz x Km)
850	3.5	400
1300	1.0	800

If the backbone distance exceeds 550Mtr, single mode fiber cables shall be installed. These cables when installed, shall have the following characteristics:

Wavelength (nm)	Max. Attenuation (dB)	Bandwidth (MHz x Km)
1300	0.4	100
1550	0.2	100

The fiber optic cable shall be tight buffered, indoor type, LSOH when used within the building and outdoor type, rodent protected when used to connect between the buildings.

The fiber optic cable shall be installed with a bending radius $\geq$ 15 times the outer diameter of the cable.

Connecting Hardware for Data Backbone Cable:

The fiber optic (data backbone) cable shall be field terminated with ST connectors (with ceramic ferrule) on each fiber strand. These connectors shall be installed in fiber optic patch panel, mounted in 19" IDF cabinet. These shall feature sliding drawer type panels for better administration of the fiber cables. The panels shall have holes at the rear for inserting and strain-relieving cables with PG glands. Cable and fiber guides for buffer management within the panels shall be built-in the panel. The insertion loss on each ST connectors shall be $\leq$ 0.2dB.

Fiber Optic Patch Cords:

Fiber optic patch cords need to be provided to enable rapid connection and disconnection repeatedly on a non-destructive basis. Fiber optic patch cords shall be supplied with each IDF. The sum total quantity of IDFs shall be supplied at the data MDF (FO-CD). The patch cords shall be Multimode Graded Index duplex 50 μ /125 μ , with ST connectors on one end and the other end will have connectors to suit the network equipment. Provide for Single mode patch cords, wherever necessary. The patch cords shall have a mating cycle of 1000 and maximum tensile load on the cable of 100N. Under no circumstances shall locally assembled patch cords and line cords will be acceptable; all must be factory made.

42" IDF/MDF Cabinet:

The cabinet shall provide a controlled environment to house telecommunications equipment, connecting hardware and to terminate the entire horizontal cabling, voice backbone cabling and data backbone cabling. These cabinets shall be of high quality and include locking doors. The cabinet shall be either wall mounted (with access from back and front) or free standing (with wheels) depending on

the size of the cabinet. Provide sufficient space in the cabinet for telecommunications/network equipment and future expansion. At IDFs where data backbone is not installed, allow sufficient space in the cabinet for installing data backbone in the future. Wherever data backbone is installed, the IDF cabinet shall be supplied with Fan Tray and UK type powerstrip. These cabinets shall be supplied with sufficient quantities of cable organisers for management/dressing of patch cords.

2.3 Installation:

The contractor shall not engage any sub-contractor to carry out the structured cabling work. The installation shall comply with the requirements of ISO-11801 Class D (99) as appropriate. The structured cabling system (cables, RJ4S outlets, panels etc) shall be labelled in accordance with EIA/TIA-606.

2.4 Testing:

The contractor shall test and demonstrate on site, to the satisfaction of the engineer, that the wiring is in full compliance with specific requirements of a Class D (99) installation (100MHz). All the tests shall be carried out in accordance with TSB-67 standard. The contractor shall provide test results (both hard copy and soft copy) for every data and every voice outlet.

2.5 Documentation:

The contractor shall submit a set of documentation comprising of layouts showing cable route & position of all outlets, labelling scheme, layout of cabinets and test results.

2.6 Warranty:

The OEM through the specialist contractor shall provide 25 years component and system performance warranty

4.0 GSM SYSTEM:**4.1 General:**

Contractor shall consider to design, supply, install, test and commission the complete containments and wiring for the complete GSM system. The system shall be as per the Etisalat requirements. The GSM system shall be provided to enhance and improve the full geographical coverage for the mobile phones

In order to improve the coverage and the availability of channels of the mobile system in the building. The contractor should co ordinate with Etisalat to furnish uniform mobile signal in the whole building. The Contractor shall establish contact with and liaise, closely to establish a mutually agreeable programme of work and their detailed requirements.

4.2 Installation:

GSM system containment shall be installed and tested by the MEP contractor. The installation of GSM system shall be as per the Etisalat requirement and their regulations.

The contractor shall submit a set of documentation comprising of layouts showing cable route & position of all outlets, labelling scheme, layout of cabinets and test results.

END OF SECTION

ELECTRICAL SECTION: 12

WIRING ACCESSORIES

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1.0 GENERAL**1.1 REFERENCE**

Conform to General Requirements for Electrical Services

1.2 DESCRIPTION OF WORK

Supply, Installation, testing, and commissioning of complete wiring accessories system as specified hereafter and as indicated in the related electrical system drawings.

The approximate schematic location of devices is given on the drawings. The exact location shall be determined at the building as the work progresses. Refer to Architectural plans for any special details, elevations, and reflective ceiling plan. Verify door swings at job site.

Verify equipment locations and adjust device and outlet locations to avoid inaccessibility. In no case shall switches be located behind door swings. Any switch so located shall be changed. Relocate inaccessible outlets.

All device boxes shall be installed plumb and level. All similar wiring devices in a given project shall be similarly positioned with respect to surrounding architectural features, to ensure consistency of final appearance throughout project (i.e. uniform distance of back boxes from doorframes, uniform position on wall column, uniform location under window, etc). While Contract Drawings indicate approximate location of outlet boxes, this attention to consistency shall dictate final box placement.

The Engineer reserves the right to change the location of any outlet, before it has been installed.

1.3 QUALITY CONTROL/ASSURANCE

Design, manufacture, testing and method of installation of all apparatus and materials furnished under requirements of these specifications shall conform to latest publications or standard rules of the following:

- A. Dubai Water and Electricity Authority (DEWA) regulations for electrical installation works and Department of Civil Defense requirements.
- B. Regulations for the Electrical Equipment of Buildings (IET), (BS) London.
- C. All products should be designed and manufactured to allow the designer and installer to comply in all respects with the latest IET Wiring Regulations.
- D. The wiring regulations are now dual numbered and have been incorporated into the British Standard System under BS 7671: 1992 incorporating Amendment No.1, 1994.

The Quality Control/Assurance/Assurance/Quality Assurance duties shall be performed by the Contractor. The system of Quality Control/Assurance/Assurance verification shall be in accordance with ISO 9000 standards of quality assurance.

The manufacturing of accessories should be in accordance with ISO 9002: 1994. The Third Party Approval Authority ASTA should internally audit the quality procedures once a quarter annually.

1.4 CODES AND STANDARDS

The following codes and standards to the extent specified herein, form a part of this specification. When an edition date is not indicated for a Code or Standard the latest edition enforces at the time of proposal submitted shall apply.

Product	Standard
Plate Switches/ DP switches	BS EN 60 669-1:1999+A1:2008
13A Sockets (switched and unswitched)	BS 1363-2:2016+A1:2018
5A/15A sockets	BS 546
Universal sockets	BS 5733:2010+A1:2014
Shaver socket outlet	BS 4573+A5:2016
Shaver supply units (isolated)	BS EN / IEC 61558-1 & 2-5
Fused Connection Units	BS 1363-4:2016+A1:2018
20A DP Switches	BS EN 60 669-1:1999+A1:2008
Cooker control units	BS 4177
BT Telephone outlets	BS 6312 Pt2
Co-axial outlets	BS 3041, BS6312 Pt2,
Ceiling Accessories	BS EN 61184, BS 7895
Plug in Ceiling roses	BS 6972, BS 5733:2010+A1:2014
Light Distribution boxes	BS 5733:2010+A1:2014
Installation boxes	BS 5733:2010+A1:2014
Plug tops	BS 546
Ingress Protection (General IP rating)	BS EN 60529
Industrial Plugs and Sockets	BS EN 60309
Powered Smoke and Heat Control Ventilators (Fans)	BS EN 12101-3
Switches, Disconnectors, Switch Disconnectors, and Fuse-combination Units	BS EN IEC 60947-3
Quality Management Systems. Requirements	BS EN ISO 9001
	NFPA Standards

1.5 SUBMITTALS

The contractor shall submit the following drawings and documents related with each electrical system for Engineer's approval:

- A. Original catalogues for various proposed components.
- B. Schedule of Accessories and Fittings.
- C. Manufacturer's Certificates
- D. Test Certificates.
- E. Operation and Maintenance (O+M) Manuals
- F. Warranties

2.0 PRODUCTS

2.1 SWITCHES

2.1.1 LOCAL SWITCHES

The local switches shall be 10AX one-way, two-way, intermediate and override as indicated on the drawings. Where more than one switch is indicated at any position multiple gang units shall be used. Light Switches shall be of the plate switch type complying with BS EN as shown under clause 1.4.

Switches shall be of the quick start make, slow break type specially designed for AC circuits to BS Standards. The operation of the switch shall not depend wholly on the action of the spring. The switches shall generally be of the rocker operated type. Switches shall be modern and compact design and with suitable thickness from wall.

All switch boxes shall be supplied with adjustable steel grids and earthing terminals. The range selected must be available in 1,2,3,4 & 6 gang versions. 1 to 3 gang plate switch dimensions at 85mm x 85mm with 4 and 6 gang dimensional at 86mm x 146mm. Switches must also be available in single pole one way, two way and intermediate versions.

The contacts must be rated 10A minimum with a 3mm contact gap such that no derating is necessary when used to switch fluorescent lamps.

Generally, switch units shall be of the adjustable grid pattern and to be secured to the adjustable grid by means of screws. For flush mounting switches the switch-plate shall overlap all edges of the box by not less than 7mm. For surface mounting switches the switch plate shall finish flush with the edges of the switch boxes. Switches for water heaters and fan coil units shall be complete with neon indicator lights.

In Plant rooms the switch units shall be weatherproof type, surface or flush mounted as required.

Local switches shall be arranged in convenient positions for switching the various circuits and generally as indicated on the drawings.

The switches shall be of the same manufacture for a particular type of switch throughout the installation. All accessories in wet and damp areas shall be of the splash proof type to IP 56 protection standard.

All switch boxes should be galvanized steel.

To ensure easy and correct connection of the conductors during installation, the necessary terminal shall be easily identified, grouped in line, upward facing, and captive and backed out prior to the installation.

Accessories Plate finish:

- a) The outlets in external locations, pump room, wet areas and parking shall be weatherproof to IP-56. Weatherproof range should be suitable for semi recessed mounting and supplied with back boxes.
- b) FOH areas / public areas such as main Entrance Lobby, Entrance lobby, Corridors shall be coloured PVC but the colour and shape shall be as per ID selection.
- c) FOH areas/ apartment wiring accessories shall be coloured PVC but the colour and shape shall be as per ID selection.
- d) BOH area, above false ceiling shall be standard white plastic.
- e) Outlets in Electrical rooms, Mechanical floors, Tel. Rooms, etc. shall be metal clad.
- f) Wiring accessory for FLS shall be metal clad.
- g) Metal clad box shall be provided where surface mounted wiring devices installed above ceiling and at all areas

2.1.2 W/P SWITCHES AND ENCLOSURES

Switches must comply with specific BS as shown under clause 1.4. The switch shall have an international protection rating of IP56. A full range of switch modules must be available including 10A & 20A, SP and DP, push to make/break and intermediate, switches, as well as one way and two-way switch modules. These switch modules will snap into polycarbonate enclosures.

A range of enclosures must be available including one gang, one gang and neon, and two gang; with the capability of joining two or more enclosures together with purpose designed parts. The enclosure shall have a brass earth terminal and have 20mm conduit knockouts on all four surfaces.

2.2 POWER / SOCKET OUTLETS**2.2.1 SOCKETS SINGLE GANG SWITCHED**

The Sockets shall be single gang switched rated at 13A, 230 volts and comply with the BS as shown under the table of clause 1.4, and the general section of the specification covering wiring accessories together with the following:

The Sockets safety shutter mechanism should require all three pins of the plug to drive open the shutter. The action requires the earth pin to lead the drive immediately followed by simultaneous entry of the live and neutral pins.

The switch will be double pole with the contacts arranged such that the neutral pole makes before and breaks after the live pole. When the switch is in the open position the contact gap must be a minimum of 3 mm. The switch rocker must include an indelible red or blue flash, preferably a moulded insert, to provide indication that the rocker is in the 'ON' position.

Terminal screws are to be backed out and captive. Terminal markings to be printed in white on grey rear mouldings for clearer identification.

The socket should provide a suitable earth terminal facility in accordance with the local code.

2.2.2 SOCKETS TWO GANG SWITCHED

The Sockets shall be single gang switched rated at 13A, 230 volts and comply with the BS as shown under clause 1.4, and the general section of the specification covering wiring accessories together with the following:

The Sockets safety shutter mechanism should require all three pins of the plug to drive open the shutter. The action requires the earth pin to lead the drive immediately followed by simultaneous entry of the live and neutral pins. The rockers should be inboard.

The switch will be double pole with the contacts arranged such that the neutral pole makes before and breaks after the live pole. When the switch is in the open position the contact gap must be a minimum of 3mm. The switch rocker must include an indelible red or blue flash, preferably a moulded insert, to provide indication that the rocker is in the 'ON' position.

Terminal screws are to be backed out and captive. Terminal markings to be printed in white on grey rear mouldings for clearer identification.

The socket should provide a suitable earth terminal facility in accordance with the local code.

2.2.3 13A SWITCHED SOCKET OUTLETS WITH USB - SINGLE AND DUPLEX

Shall comply with BS standard with 3 rectangular pin (2 P+E) shuttered, rated 13A.

Single pole switched / unswitched (per model). USB charger input voltage is 110-230V.

Nominal output voltage at 5VDC, +5/-5%; Output current at 2.6A (2-gang) or 2.1A (1-gang); AC +/- 10%, 50/60 Hz.

2.2.4 5A SWITCHED SOCKET OUTLETS

5A Switched Socket outlets shall be in accordance with BS standard. They shall be switched and shuttered, mounted in single or multiple assemblies and shall be of the type and rating as specified/ indicated in the drawings and schedules.

These will be 3 round pin (2 P+E) shuttered with combined switch rated 5 amps at 230 volts.

Socket outlets shall be complete with boxes not less than 35mm deep B. and the box shall have a suitably mounted brass protective conductor terminal:

1. Socket outlets installed in flush installations shall have overlapping front plates.
2. In surface installations front plates shall be flush with the sides of the box and shall be metal clad unless otherwise specified.

The earth pins of the sockets shall be connected to the back-box earth terminal with a green and yellow PVC-insulated protective conductor. Cover screws are not acceptable for earth continuity.

Where socket outlets are installed on perimeter panelling, metal frames for panel mounting shall be provided.

Socket outlet shall be suitable for back and side wiring with 1 x 6mm² or 2 x 4mm² standard conductors.

2.2.5 COMMUNICATION OUTLET

For Telephone Outlet: Subject to compliance with requirements, products that may be incorporated are as listed in the Client ICT specification. Finish of the wiring accessory should match with the other outlets.

2.2.6 FLEX OUTLET PLATE UNSWITCHED AND UNFUSED WITH NEON

Single gang 20A Flex Outlet plate is to be unswitched unfused and comply with the BS under clause 1.4, the General Section of this specification covering wiring accessories together with the following requirements.

The flex outlet shall be supplied with three pairs of terminals each suitable for 3 x 4.0mm² conductors and a 1.5mm² flexible cord. A cord grip should also be fitted.

A screw less cord grip, which upon insertion of the cord automatically holds and clamps the cord.

A flex outlet, supplied blanked off with a removable plug, positioned in the lower edge of the front plate.

This shall be used for water heater, dishwasher and washing machines connection unit.

2.2.7 FUSED CONNECTION UNIT UNSWITCHED AND FUSED WITH NEON:

An unswitched 13A 230 volt fused Connection unit complying with the BS, as shown under clause 1.4, the general section of this specification covering wiring accessories and the following requirements.

A fuse holder incorporating a BS compliant 13A fuse link. The holder being screw clamped into the front plate and must be released by operating the clamping screw as a 'jack out' device. When in the fully released position the fuse holder must remain attached to the front plate and shutter the fixed fuse connection such that they cannot be probed. The fuse holder must also be capable of being padlocked in the released position.

A screw less cord grip which upon insertion of the cord automatically holds and clamps the cord.

A flex outlet, supplied blanked off with a removable plug, positioned in the lower edge of the front plate.

All FLS related spur outlets shall be metal clad type, e.g. MSFD, etc.

This shall be used for life safety and security connection unit.

2.2.8 SWITCHED AND FUSED CONNECTION UNIT STANDARD WITH NEON

The switched 13A 250 volt fused Connection Unit, complete with neon indication must comply to BS as shown under clause 1.4.

Fuse holder should incorporate a BS compliant 13A fuse link. The holder being screw clamped into the front plate and must be released by operating the clamping screw as a 'jack out' device. When in the fully released position the fuse holder must remain attached to the front plate and shutter the fixed fuse connection such that they cannot be probed. The fuse holder must also be capable of being padlocked in the released position.

A screw less cord grip upon insertion of the cord should automatically hold and clamp the cord. A flex outlet, supplied blanked off with a removable plug, positioned in the lower edge of the front plate.

The switch will be double pole with the contacts arranged such that the neutral pole makes before and breaks after the live pole. When the switch is in the open position the contact gap must be a minimum of 3mm. The switch rocker must include an indelible red flash, preferably a moulded insert, to provide indication that the switch rocker is in the 'ON' position.

A neon indicator is to illuminate when the switch is placed in the closed position. The neon indicator assembly and lens should form an integral part of the switch rocker.

This shall be used for FCU, cooker and Gas indicator connections.

2.2.9 20 AMP Double Pole (DP) SWITCH WITH NEON INDICATOR

A 20A DP Switch complete with neon indicator all supplied within a 86mm x 86mm front plate complying with the BS as indicated in the table of clause 1.4, the General section of this specification covering wiring accessories and the following requirements.

The switch will be double pole with the contacts arranged such that the neutral pole makes before and breaks after the live pole. When the switch is in the open position the contact gap must be a minimum of 3mm. The switch rocker must include an indelible red flash, preferably a moulded insert, to provide indication that the switch rocker is in the 'ON' position.

A screw less cord grip which upon insertion of the cord automatically holds and clamps the cord.

A flex outlet, supplied blanked off with a removable plug, positioned in the lower edge of the front plate.

A neon indicator is to illuminate when the switch is placed in the closed position. The neon indicator assembly and lens should form an integral part of the switch rocker. The switch must have engraved of the purpose of the connected equipment.

2.2.10 45 AMP Double Pole (DP) SWITCH WITH NEON INDICATOR

A 45A DP Switch complete with neon indicator complying with the BS as indicated in the table of clause 1.4, the General section of this specification covering wiring accessories and the following requirements.

The switch will be double pole with the contacts arranged such that the neutral pole makes before and breaks after the live pole. When the switch is in the open position the contact gap must be a minimum

of 3mm. The switch rocker must include an indelible red flash, preferably a moulded insert, to provide indication that the switch rocker is in the 'ON' position.

A screw less cord grip which upon insertion of the cord automatically holds and clamps the cord.

A flex outlet, supplied blanked off with a removable plug, positioned in the lower edge of the front plate.

A neon indicator is to illuminate when the switch is placed in the closed position. The neon indicator assembly and lens should form an integral part of the switch rocker. The switch must have engraved of the purpose of the connected equipment.

2.2.6 FLEX OUTLET PLATE UNSWITCHED AND UNFUSED WITH NEON

Single gang 45A Flex Outlet plate is to be unswitched unfused and comply with the BS under clause 1.4, the General Section of this specification covering wiring accessories together with the following requirements.

The flex outlet shall be supplied with three pairs of terminals each suitable for 3 x 10.0mm² conductors and a 6mm² flexible cord. A cord grip should also be fitted.

A screw less cord grip, which upon insertion of the cord automatically holds and clamps the cord.

A flex outlet, supplied blanked off with a removable plug, positioned in the lower edge of the front plate.

This shall be used for electrical cooker / hob unit.

2.3 WEATHER PROOF SWITCHED SOCKETS

All products shall be manufactured in accordance with BS EN ISO 9001. The socket outlets shall be rated at 13A and comply with BS. The socket will have an Ingress Protection rating of IP56, and have the capability of accepting a standard BS plug top by completely enclosing the plug top. This IP rating will be maintained whilst the socket is in use. The polycarbonate enclosure will incorporate a sealing device, to seal around the flexible cord. The socket outlet is to be switched with a separate switch module enclosure beside the socket, fitted with a suitable switch. The socket outlet shutter mechanism shall be operated by all three pins of the plug top and should be capable of accepting virtually any British standard 13A plug. See clause 1.4 for the specific code and standard reference numbers.

The enclosure shall come complete with a brass earth terminal. Plastic cover protection is not accepted

2.3.1 INDUSTRIAL TYPE / COMMANDO SWITCH DISCONNECTORS

20A / 32A / 45A / 63A, 2Pole / 4Pole. They should conform to BS EN of clause 1.4 table, and manufactured from polycarbonate for maximum impact resistance and to have an IP 66 rating. They should be supplied with blanking plugs and neutral terminals. Terminal capacity should be minimum 4 mm².

2.4 CEILING ACCESSORIES

2.4.1 PLUG IN CEILING ROSE

1. Supply and install ceiling switches of suitable ratings, on locations showed on the drawings.
2. Supply and install ceiling accessories like batten holders, ceiling roses etc. on locations showed on the drawings.

3. All lamp holders' batten or pendant shall be of Shock guard type i.e. disconnect automatically on bulb removal.
4. Wherever shown on the drawings, use 3 pins (normal lighting) and 4 pins (emergency lighting) plug in ceiling roses for lighting point connection. The plug in ceiling roses shall comply with BS (see clause 1.4).

2.5 GI BOXES

- 2.5.1 GI boxes to be provided with brass earth terminal to facilitate earth wire connection. The boxes to have sufficient number of 20mm and 25mm knockout. The boxes thickness shall be 1.1mm minimum and shall comply with BS+A1 (see clause 1.4). Boxes to have adjustable lug for proper installations of wiring accessories. Extension ring to be used along with GI boxes, in places where the box is deep inside the wall, marble or concrete. Boxes used for ICT applications must have deeper box with at least 47mm while others should be minimum 35mm. Provide the degree of ingress protection specified for the accessory.

2.6 ISOLATORS

- 2.6.1 All switch disconnecter should comply with BS EN standard, shall be Heavy Duty, Kilo watt rated to AC-23A utilization category, capable of full load operation.
- 2.6.2 Switches shall be non-fusible, single throw, Safety type, hand operated, DP, TP&N or 2P & 4 poles as shown on the drawings.
- 2.6.3 The Neutral contact shall be arranged to be switched or linked as specified and in the switched neutral units, the neutral contact shall make before and break later than the phase contacts.
- 2.6.4 The switch handle shall be front operated and mechanically interlocked so that the door is normally prevented from being opened when the switch contacts are in the closed position. Subject to project / Engineers requirement, Provision shall be provided on the handle to defeat this interlock by a skilled person using the necessary tool while the switch contacts are in closed position.
- 2.6.5 Provision for Padlocking should be possible in OFF position for normal equipment's. All FLS equipment's isolators must be able to padlock in ON and / or OFF position.
- 2.6.6 Switch disconnecter for (Variable Frequency Drive) VFD's Motors shall have an auxiliary contact with 1NO+1NC that can interface with the motor's VFD to switch Off the VFD drive before disconnecting the motor or also providing ON/OFF status for BMS system.
- 2.6.7 Enclosures:
Indoor Applications: Enclosures used shall be made of Die cast aluminium or sheet steel to IP56. For non-armoured cables, high quality Polycarbonate housing can be used depending on the location.
Outdoor Applications: All cable termination should be from Bottom only. Enclosures used shall be of Die Cast aluminium & higher rating shall be in GRP enclosure. Isolators should be derated for 50°C Ambient Temperature.
- 2.6.8 Isolators used for the Life Safety Equipment's shall be Fire rated & tested at 400°C for 120 mins and shall comply with the Thermal requirements of BS standard.

2.7 DOOR BELLS

2.7.1 PUSH BUTTONS

Pushbuttons shall be IP 20 for indoor use and IP 65, suitable for sustained outdoor use.

Pushbuttons shall have feedback LEDs that confirm signal transmission.

Wired or wireless pushbuttons may be provided; if wired, pushbuttons shall be extra low voltage, used with a 16V or 24V transformer purpose-made by the same manufacturer for use as part of a complete system. If wireless, battery life must be at least 12 months at 5 activations per day.

Wired or wireless pushbuttons may be provided; if wireless, pushbuttons shall have a rated range of no less than 125 meters, and shall have a low battery indicator.

Rated operating temperature shall be from -10°C to 60°C inclusive.

2.7.2 CHIMES or INTERFACE WITH INTERCOM UNIT

Wired or wireless chimes may be used; if wired, chimes shall be extra low voltage, used with a 16V or 24V transformer purpose-made by the same manufacturer for use as part of a complete system. If wireless, chimes shall have battery life of at least 12 months at 5 activations per day.

Chimes shall have at least three tones, selectable on the chime without tools and without moving from installed location.

Rated operating temperature shall be from 0°C to 40°C inclusive.

Door bell push button shall be interfaced with the intercom unit, with suitable tone selection.

2.8 FLOOR BOXES

Service floor boxes shall be constructed from high pressure Zinc Alloy die casting base frame and fixed to heavy gauge galvanized steel base plate adequate in strength and performance and to be protected against corrosion.

The design of the base frame shall cast the whole unit in one piece without any joints/fabrication to prevent corrosion, concrete slippage into the box during casting of concrete or screeding. The boxes shall be constructed with provisions for conduit access on all four sides. Unwanted entries shall be blanked off with detachable side blanks.

Cover for service floor boxes shall be made of high pressure Zinc Alloy die casting with recess to accept the same floor finish and provided with suitable hinges designed to enable the trap cover to open through 180 degrees and giving access at all times to the power, telephone and ELV outlets.

The boxes shall be adjustable in height independently of the ducting system to take account of difference in floor thickness.

Cables emerging from floor boxes shall be protected against damage by means of nylon cables exit grommets or equivalent and approved material and shall be reversible to close position when not in use. The lifting handle and cable grommets shall be provided in opposite sides of the trap cover to allow easy access to lifting handle and to avoid damage of the cable insulation.

Floor boxes shall be complete with waterproof gasket and circuit protective conductors between the covers and the boxes.

Single gang floor box shall have stainless steel or chrome finish.

Floor boxes shall be used only in amenity areas like meeting rooms, gym, main reception and other FOH circulation areas. No floor boxes shall be used inside the apartments.

All floor boxes will be subject to the Engineer approval.

3.0 EXECUTION**3.1 INSTALLATION**

When required to do so the selected manufacturer of the wiring accessories will provide installation maintenance and disposal instructions to enable the client to comply with the CDM Regulations.

Where possible all wiring devices shall be obtained from the same manufacturer to ensure uniformity and visual compatibility.

All wiring device front plates shall be manufactured from a white urea plastic material having properties, which include anti tracking and non-flame propagation.

Examine areas for compliance with requirements for installation and conditions affecting performance of the Work. Identify conditions detrimental to a proper and timely completion and notify the Engineer of the unsatisfactory conditions. Proceed with installation only after unsatisfactory conditions have been corrected.

Coordinate wiring device heights in vending, kitchen and utility areas with benches and counters. Verify that outlet boxes are installed at proper height. Verify that wall openings are neatly cut and shall be completely covered by wall plates. Verify that branch circuit wiring installation is completed, tested, and ready for connection to wiring devices. Socket Outlets shall be away from kitchen sink / taps of minimum 1.5m as per DBC and DEWA regulations.

It is a requirement of the contract to reduce the amount of dust collection upon the accessory and where possible effectively reduces the prominence of the accessory. To this end the front face of the accessory should chamfer away at the top to a minimum projection of 5mm before installation. The front plates of all accessories should have radius corners and a chamfered top edge to confer a slim line and unobtrusive appearance, whilst preventing unsightly dust collecting. All socket outlets should have unique 3 pin operated safety shutter mechanism. All socket outlets should be Double pole switched as standard with the contacts arranged such that the neutral pole makes before and breaks after the live pole.

All switch plates including SSOs are to be coordinated with tiling layouts by prior agreement with the Architect on site.

Restrictions for spacing between outlets and devices shall be followed as per Dubai Building Code and authority regulations

To ensure easy and correct connection of conductors during installation and maintenance, the accessory terminals shall: -

1. Be easily identified by White marking on a dark ground.
2. Be grouped together and where possible like terminals will be in line.
3. Terminal screws to be upward facing, captive and backed out prior to installation.
4. Terminals should be so shaped to provide a generous lead in to the conductor.
5. Should have minimum 3mm contact gap.

3.2 MOUNTING HEIGHTS

The mounting heights for the electrical equipment and accessories shall be coordinated with the furniture layout and shall be as per site requirements to Engineer's instruction and approval.

3.3 TESTING

Test all switches, socket outlets etc. for correct polarity and continuity of conductors in the presence of and to the entire satisfaction of the Engineer.

Carry out live phase to earth loop impedance tests at all switches and socket outlets with an approved earth loop impedance tester to the entire satisfaction of the Engineer. Ensure that all device plates have satisfactory earth continuity to the protective conductor system.

Test all socket outlets for instantaneous tripping of associated distribution board current operated earth leakage circuit breaker using testing equipment, approved by the Engineer.

3.4 SITE QUALITY CONTROL

Perform tests and inspections and prepare test reports.

Test Instrument for Socket outlets: Digital wiring analyzer with digital readout or illuminated LED indicators of measurement.

Inspect each wiring device for defects. Operate each wall switch with circuit energized and verify proper operation. Verify that each socket device is energized. Test each socket device for proper polarity.

Adjust devices and wall plates to be flush and level.

Clean exposed surfaces to remove splatters and restore finish.

END OF SECTION

ELECTRICAL SECTION - 13

UNINTERRUPTABLE POWER SUPPLY

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1.0 GENERAL**1.1 Summary**

- A. The UPS system shall provide an uninterruptible power supply to defined loads within the building in the event of mains failure. The UPS shall be N configuration complete with automatic and manual bypass, batteries and will be rated to support the load for a minimum period of 30 minutes at full load.

For Security systems the UPS must be 3 phase, N configuration, complete with batteries and 30 minutes back up in accordance with SIRA requirements. In addition, each IDF racks must be provided with single phase rack mounted 30 minutes' backup solution to be provided.

For Fire Command Centre, standalone, local UPS, single phase, with 15-minutes backup to be provided as part of the system and as required by local civil defence.

BMS and DDC panels must have its own UPS system as part of the system requirement.

B. Section Includes:

1. Three-phase, on-line, double-conversion, static-type, IGBT UPS units with the following features:
 - a. Surge suppression
 - b. Input harmonics reduction
 - c. Rectifier-charger
 - d. Inverter
 - e. Static bypass transfer switch
 - f. Battery and battery disconnect device
 - g. External maintenance bypass/isolation switch
 - h. Output isolation transformer
 - i. Remote UPS monitoring provisions
 - j. Battery monitoring
 - k. Remote battery monitoring.
 - l. The UPS shall be online type.
- C. Co-ordinate sensor-communication module package with data network and with monitoring equipment specified in "Building Monitoring and Management Systems" for successful transmission and remote readout of remote monitoring data specified in this Section.

1.2 Submittals

- A. Product Data: For each type of product indicated. Include data on features, components, ratings, and performance.
- B. Shop Drawings: Include plans, elevations, sections, details, and attachments to other work:
1. Detail equipment assemblies and indicate dimensions, weights, components, and location and identification of each field connection. Show access, workspace, and clearance requirements; details of control panels; and battery arrangement.

2. Wiring Diagrams: For power, signal, and control wiring.
 3. Certification and proof that all components shall be available for at least 10 years by Manufacturer
- C. Qualification Data: Provide details of proposed qualified power quality specialist and testing agency.
- D. Manufacturer Certificates: Provide certificates for each product, from manufacturer.
- E. Factory Test Reports: Comply with specified requirements.
- F. Field quality-control reports.
- G. Performance Test Reports: Indicate test results compared with specified performance requirements, and provide justification and resolution of differences if values do not agree.
- H. Operation and Maintenance Data: For UPS units to include in emergency, operation, and maintenance manuals.
- I. Warranties
- 1.3 Quality Assurance
- A. Power Quality Specialist Qualifications: A registered professional electrical engineer or manufacturer's authorised representative shall manage the UPS performance testing, testing shall be witnessed by the CA.
- B. Electrical Components, Devices, and Accessories: Listed and labelled as defined in BS EN IEC62040-2. UPS Systems shall be tested by the manufacturer of the system and by a qualified testing agency, and marked for intended location and application.
- C. Where UPS units are installed ensure that the whole installation complies with the relevant standards, including the following:
1. BS EN IEC 62040-1 - Uninterruptible power systems (UPS). General and safety requirements for UPS
 2. BS EN 62040-2 - Uninterruptible power systems (UPS). Electromagnetic compatibility (EMC) requirements
 3. BS EN 62040-3 - Uninterruptible power systems (UPS). Method of specifying the performance and test requirements
 4. BS EN IEC 62485-2 - Safety requirements for secondary batteries and battery installations. Stationary batteries
 5. IEC 60364 - Low Voltage Electrical installations of buildings
 6. IEC 61439-1 - Low-voltage switchgear and control-gear assemblies - Type-tested and partially type-tested assemblies
 7. BS EN 60529 - Degrees of protection provided by enclosures (IP Code)

8. BS EN 60896-21 - Stationary lead-acid batteries. Valve regulated types. Methods of test
 9. BS EN 60896-22 - Stationary lead-acid batteries. Valve regulated types. Requirements
 10. BS EN 62368-1 – Audio / Video, Information and Communication Technology Equipment Part 1 – Safety Requirements
- F. Directives:
1. Low Voltage Directive 2014/35/EU
 2. Electromagnetic Compatibility Directive 2004/108/EC.
- G. Pre-construction Testing / Reports
1. Submit reports of independent tests demonstrating that the products and systems are in accordance with the specified performance requirements.
 2. Where test results for a material or product are not available, undertake testing to show compliance with the Specification at an independent testing laboratory acceptable to the Engineer.
 3. The provision of testing data or the carrying-out of tests does not relieve the Contractor of his responsibilities regarding the performance requirements, durability or service life requirements.
- H. Source Quality Control
1. The UPS manufacturer shall have a minimum of 15 years of experience in the design, manufacture and testing of solid-state UPS. A list of installed UPS of the same type as the manufacturer proposes to furnish for this application shall be supplied with the proposal.
 2. The UPS manufacturer shall have ISO 9001 certification for engineering/ R&D, manufacturing facilities and an efficient field service organization.
- I. Testing and Inspection: Provide testing and inspections in accordance with this document and those noted in the summary above.
- J. Safety: The UPS shall carry the CE mark, indicating the equipment complies with European Community standards EN for Safety and IEC for Electromagnetic Compliance.
- K. Spare Parts
1. The contractor shall consider the required spare parts for the equipment specified and provide a schedule of spare parts with costs separate to the tender for the MEP works.

2.0 PRODUCTS

2.1 Operational Requirements

- A. Automatic operation includes the following:
1. Normal Conditions: Load is supplied with power flowing from the normal power input terminals, through the rectifier-charger and inverter, with the battery connected in parallel with the rectifier-charger output.

2. In the event of a power outage the UPS will serve the IT systems specified for a period of 15 minutes.

2.2 Performance Requirements

- A. The UPS shall be N on-line double conversion, Voltage and Frequency Independent (VFI) technology in accordance with IEC (see 1.3.C.3). The UPS shall be provided with the facility for paralleling for capacity or redundancy without limitation on the number of UPS units connected in parallel. The input to the UPS and the output characteristic shall be defined on the equipment schedules.
- B. The UPS shall perform as specified herein while supplying rated full-load current, composed of any combination of linear and nonlinear load, up to 100 percent nonlinear load with a load crest factor of 3.0, under the following conditions or combinations of the following conditions:
 1. Inverter is switched to battery source
 2. Steady-state ac input voltage deviates up to plus or minus 10 percent from nominal voltage
 3. Steady-state input frequency deviates up to plus or minus 5 percent from nominal frequency
 4. THD of input voltage is 4 percent or more with a minimum crest factor of 3.0, and the largest single harmonic component is a minimum of 4 percent of the fundamental value
 5. Load is 30 percent unbalanced continuously.
- C. Minimum Duration of Supply: If battery is sole energy source supplying rated full UPS load current at 80 percent power factor, duration of supply is 30 minutes.
- D. Input Voltage Tolerance: System steady-state and transient output performance remains within specified tolerances when steady-state ac input voltage varies plus 10, or minus 10 percent from nominal voltage.
- E. The overall efficiency (ac-dc-ac, on-line mode) shall not be less than the figures shown in the table below (+/- 1%):

UPS unit rating	Load %			
	100%	75%	50%	25%
≤ 60 kVA	96	96	95	91
> 60 kVA	96	96	95	92

Measurement with linear load (Cos Ø = 0.8 ind)

- F. Maximum Acoustical Noise: Noise level shall be from 45 to 60 Max depending on the location and occupied rooms.
- G. Maximum Energizing Inrush Current: Six times the full-load current.
- H. Maximum AC Output-Voltage Regulation for Loads up to 50 Percent Unbalanced: Plus or minus 2 percent over the full range of battery voltage.

- I. Limitation of harmonic distortion of input current to the UPS shall be as follows:
 - 1. Description: Either a tuned harmonic filter or an arrangement of rectifier-charger circuits shall limit THD to 5 percent, maximum, at rated full UPS load current, for power sources with X/R ratio between 2 and 30.
- J. Maximum Harmonic Content of Output-Voltage Waveform: 5 percent rms total and 3 percent rms for any single harmonic, for 100 percent rated nonlinear load current with a load crest factor of 3.0.
- K. The harmonic characteristics of the input circuits of UPS modules when operating under full load shall be within the limits stipulated by G5/4 published by the Energy Networks Association regardless of the load harmonic content. Full galvanic isolation of the load shall be provided by the UPS systems.
- L. Minimum Overload Capacity of UPS at Rated Voltage: 125 percent of rated full load for 10 minutes, and 150 percent for 60 seconds in all operating modes.
- M. Maximum Output-Voltage Transient Excursions from Rated Value: For the following instantaneous load changes, stated as percentages of rated full UPS load, voltage shall remain within stated percentages of rated value and recover to, and remain within, plus or minus 2 percent of that value within 100 ms:
 - 1. 50 Percent: Plus or minus 2 percent
 - 2. 100 Percent: Plus or minus 2 percent
 - 3. Loss of AC Input Power: Plus or minus 1 percent
 - 4. Restoration of AC Input Power: Plus or minus 1 percent.
- N. EMC Emissions: Comply with BS EN (see 1.3.C.2).

2.3 UPS systems

- A. Basis-of-Design Product: Subject to compliance with requirements, provide UPS units as indicated on the drawings, schedules and in this specification
- B. Electronic Equipment: Solid-state devices using hermetically sealed, semiconductor elements. Devices include rectifier-charger, inverter, static bypass transfer switch, and system controls.
- C. Enclosures: Comply with IP21 as a minimum, unless otherwise indicated.
- D. The UPS shall have built-in protection against under-voltage, overcurrent, and overvoltage conditions, including low-energy surges introduced on the primary a.c. source and the bypass source.
- E. Surge Suppression: Protect internal UPS components from surges that enter at each ac power input connection including main disconnect switch, static bypass transfer switch and maintenance bypass/isolation switch. Protect rectifier-charger, inverter, controls, and output components.

- F. The UPS system shall provide a level of reliability mean time between failures (MTBF) in excess of 160000 hours.

2.4 Rectifier-charger

- A. Capacity: Adequate to supply the inverter during rated full output load conditions and simultaneously recharge the battery from fully discharged condition to 80 percent of full charge within 10 times the rated discharge time for duration of supply under battery power at full load.
- B. Output Ripple: Limited by output filtration to less than 0.5 percent of rated current, peak to peak.
- C. Control Circuits: Immune to frequency variations within rated frequency ranges of normal and emergency power sources:
 - 1. Response Time: Field adjustable for maximum compatibility with local generator-set power source.
- D. Battery Float-Charging Conditions: Comply with battery manufacturer's written instructions for battery terminal voltage and charging current required for maximum battery life.

2.5 Inverter

- A. Description: Pulse-width modulated, with sinusoidal output. Include a bypass phase synchronization window adjustment to optimize compatibility with local engine-generator-set power source.
- B. The inverter shall convert D.C. power from the input converter output, or the battery, into precise regulated sinusoidal wave a.c. power for supporting the critical a.c. load. The inverter shall be a transformer less free running rectifier type to maximise efficiency and limit weight and cost.
- C. The output frequency of the inverter shall be controlled by an oscillator. The oscillator shall hold the inverter output frequency to $\pm 0.1\%$ for steady state and transient conditions. The inverter shall synchronise with the bypass supply assuming the bypass supply stays within the selected range. If the bypass source fails to remain within the selected range, the inverter shall revert to the internal oscillator.
- D. The inverter shall incorporate an overload shut down facility to provide protection to its components such that, prior to damage occurring, the output shall be disconnected from the load. Reset shall be possible without having to access the inverter module directly or replace items such as fusible links.

2.6 Static Bypass Transfer Switch

- A. Description: Solid-state switching device providing uninterrupted transfer. A contactor or electrically operated circuit breaker automatically provides electrical isolation for the switch.
- B. Switch Rating: Continuous duty at the rated full UPS load current, minimum.

2.7 Battery

- A. The battery system shall consist of gas recombination, valve regulated, lead acid cells, compliant with IEC (see 1.3.C.8 & 9). Flame retardant batteries shall be provided.
- B. For each UPS system the battery system shall consist of a minimum of 2 parallel strings of multiple cells. Each individual parallel string shall have its own dedicated means of electrical protection. The batteries shall be configured so that in the event of a battery malfunction the affected string is automatically isolated from the system thereby ensuring battery autonomy is retained.
- C. Factory assembled in an isolated compartment of UPS cabinet, complete with battery disconnect switch.
- D. Floor mounted racks shall be bolted into concrete structure.
- E. Battery racks shall be earthed directly back to Mains Earth Facility.
- F. Battery system / racks shall be provided with Battery MCCB's of adequate size and type, which shall be suitably located within the battery room such that it is accessible from the door opening at a height in accordance with DEWA regulations. The battery MCCB's shall be labelled as to their purpose.
- G. Batteries shall have a 10 year life and shall be provided such that the system shall be capable of providing its full autonomy at full load at that time.
- H. A fully discharged battery system shall be capable of being recharged to 80% of the UPS output capacity within a maximum period of 10 times the normal total discharge time period.
- I. Provide batteries as indicated on the drawings and schedules and as described in this specification

2.8 Controls and Indications

- A. Description: Group displays, indications, and basic system controls on a common control panel on front of UPS enclosure. Incorporate an LCD display and control keypad to the inverter/charger cubicle to permit access to the system monitoring functions which shall be provided as a detailed on the equipment schedules.
- B. Minimum displays, indicating devices, and controls include those in lists below. Provide sensors, transducers, terminals, relays, and wiring required to support listed items. Alarms include audible signals and visual displays.
- C. Indications: Plain-language messages on a digital LCD screen:
 - 1. Quantitative indications shall include the following:
 - a. Input voltage, each phase, line to line
 - b. Input current, each phase, line to line
 - c. Bypass input voltage, each phase, line to line
 - d. Bypass input frequency
 - e. System output voltage, each phase, line to line
 - f. System output current, each phase

- g. System output frequency
 - h. DC bus voltage
 - i. Battery current and direction (charge/discharge)
 - j. Elapsed time discharging battery.
- 2. Basic status condition indications shall include the following:
 - a. Normal operation
 - b. Load-on bypass
 - c. Load-on battery
 - d. Inverter off
 - e. Alarm condition.
- D. In addition to a visual fault signal (alarm), the UPS shall also record fault occurrences in a rolling event log. The event log shall record up to 64 occurrences as a minimum, with the oldest events discarded first. The user shall have access to the event log through the LCD display. Every alarm and/or event recorded in the event log will contain a time and date stamp.
- E. The volume of all audible alarms shall be at least 65 dBA at a distance of one meter. An audible alarm shall be used in conjunction with the LED/LCD display to indicate a change in UPS status.
- F. The audible alarms shall warn loss of mains or generator supply, low battery (whilst on battery), and all other alarm conditions. For all audible alarm conditions, the display shall identify the cause of error/alarm. All alarm tones shall be a continual tone until the condition rectifies itself or the alarm is silenced. Once silenced, the audible alarm shall not sound until a new alarm condition is present, but the LED indication shall continue to identify the alarm condition.
- G. As a minimum the following conditions shall raise an alarm:
 - 1. Bypass ac input overvoltage or under voltage
 - 2. Bypass ac input over frequency or under frequency
 - 3. Bypass ac input and inverter out of synchronization
 - 4. Bypass ac input wrong-phase rotation
 - 5. Bypass ac input single-phase condition
 - 6. Bypass ac input filter fuse blown
 - 7. Internal frequency standard in use
 - 8. Battery system alarm
 - 9. Control power failure
 - 10. Fan failure
 - 11. UPS overload

12. Battery-charging control faulty
13. Input overvoltage or under voltage
14. Input transformer over temperature
15. Input circuit breaker tripped
16. Input wrong-phase rotation
17. Input single-phase condition
18. Approaching end of battery operation
19. Battery under voltage shutdown
20. Maximum battery voltage
21. Inverter fuse blown
22. Inverter transformer over temperature
23. Inverter over temperature
24. Static bypass transfer switch over temperature
25. Inverter power supply fault
26. Inverter transistors out of saturation
27. Identification of faulty inverter section/leg
28. Inverter output overvoltage or under voltage.
29. UPS overload shutdown
30. Inverter current sensor fault
31. Inverter output contactor open
32. Inverter current limit
33. Mains Input Failure
34. Load on inverter
35. Battery low
36. Load on mains
37. Battery test failure
38. System over temperature
39. Bypass supply fail
40. Earth fault.

- H. The UPS shall incorporate voltage-free relay contacts suitable for direct communication the clients BMS system and an RS-232 communication port for serial communications and to enable communication via modem equipment using an industry standard open protocol. Any manufacturer's software required for monitoring shall be provided at no additional cost.
- I. The UPS unit(s) shall include a facility for installing an optional SNMP adapter card to the UPS to permit one or more network management systems (NMS) to monitor the UPS in TCP/IP network environments. 10/100 Mbit Ethernet support shall be included.
- J. Controls shall include the following:
 - 1. Inverter on-off
 - 2. UPS start
 - 3. Battery test
 - 4. Alarm silence/reset
 - 5. Output-voltage adjustment.
- K. Dry-form "C" contacts shall be available for remote indication of the following conditions:
 - 1. UPS on battery
 - 2. UPS on-line
 - 3. UPS load-on bypass
 - 4. UPS in alarm condition
 - 5. UPS off (maintenance bypass closed).
- L. Emergency Power Off Switch: Capable of local operation and operation by means of activation by external dry contacts.

2.9 Maintenance Bypass/Isolation Switch

- A. Description: Manually operated switch or arrangement of switching devices with mechanically actuated contact mechanism arranged to route the flow of power to the load around the rectifier-charger, inverter, and static bypass transfer switch:
 - 1. Switch shall be electrically and mechanically interlocked to prevent interrupting power to the load when switching to bypass mode
 - 2. Switch shall electrically isolate other UPS components to permit safe servicing.
- B. Key interlock requires unlocking maintenance bypass/isolation switch before switching from normal position with key that is released only when the UPS is bypassed by the static bypass transfer switch. Lock is designed specifically for mechanical and electrical component interlocking.

2.10 Basic Battery Monitoring

- A. Basis-of-Design Product: Subject to compliance with requirements, provide product indicated on Drawings and described in specification

- B. Battery earth-Fault Detector: Initiates alarm when resistance to earth of positive or negative bus of battery is less than 5000 ohms.
- C. Battery compartment smoke/high-temperature detector initiates an alarm when smoke or a temperature greater than 75°C occurs within the compartment.
- D. Annunciation of Alarms: At UPS control panel.

3.0 EXECUTION

3.1 Factory Testing

- A. Initial testing of the UPS Module shall be performed at the manufacturer's works.
- B. Factory test complete UPS system before shipment. Use actual batteries that are part of final installation simulated battery testing. Include the following:
 - 1. Test and demonstration of all functions, controls, indicators, sensors, and protective devices
 - 2. Full-load test
 - 3. Transient-load response test
 - 4. Overload test
 - 5. Power failure test.
- C. Report test results. Include the following data:
 - 1. Description of input source and output loads used. Describe actions required to simulate source load variation and various operating conditions and malfunctions.
 - 2. List of indications, parameter values, and system responses considered satisfactory for each test action. Include tabulation of actual observations during test
 - 3. List of instruments and equipment used in factory tests.

3.2 Examination

- A. Examine areas and conditions on site, with Installer present, for compliance with requirements for conditions affecting performance of the UPS.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.3 Installation

- A. Equipment Mounting: Install UPS on concrete base. Comply with requirements for concrete base specified in Structural engineer's specification:

1. Install dowel rods to connect concrete base to concrete floor. Unless otherwise indicated, install dowel rods on 450-mm centres around the full perimeter of concrete base
 2. For supported equipment, install epoxy-coated anchor bolts that extend through concrete base and anchor into structural concrete floor
 3. Place and secure anchorage devices. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded
 4. Install anchor bolts to elevations required for proper attachment to supported equipment.
- B. Maintain minimum clearances and workspace at equipment according to manufacturer's written instructions and local wiring regulations.
- C. Connections: Interconnect system components. Make connections to supply and load circuits according to manufacturer's wiring diagrams unless otherwise indicated.

3.4 Earthing

- A. Separately Derived Systems: If not part of a listed power supply for a data-processing room, comply with the local wiring regulations requirements for connecting to grounding electrodes and for bonding to metallic piping near isolation transformer.

3.5 Identification

- A. Identify components and wiring according to the specification. 1. Identify each battery cell individually.

3.6 Battery Equalization

- A. Equalize charging of battery cells according to manufacturer's written instructions. Record individual-cell voltages.

3.7 Field Quality Control

- A. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and commissioning all components, assemblies, and equipment installations, including connections.
- B. Tests and Inspections:
1. Comply with manufacturer's written instructions
 2. Inspect interiors of enclosures, including the following:
 - a. Integrity of mechanical and electrical connections
 - b. Component type and labelling verification
 - c. Ratings of installed components.
 3. Inspect batteries and chargers according to manufacturer's requirements
 4. Test manual and automatic operational features and system protective and alarm functions.
 5. Test communication of status and alarms to remote monitoring equipment

6. Load the system using a variable-load bank to simulate kilovolt amperes, kilowatts, and power factor of loads for unit's rating. Use instruments calibrated within the previous six months:
 - a. Simulate malfunctions to verify protective device operation
 - b. Test duration of supply on emergency, low-battery voltage shutdown, and transfers and restoration due to normal source failure
 - c. Test harmonic content of input and output current less than 25, 50, and 100 percent of rated loads
 - d. Test output voltage under specified transient-load conditions
 - e. Test efficiency at 50, 75, and 100 percent of rated loads.
 - f. 100% full load test for 1 hour on rectifier/inverter
 - g. Demonstrate auto-bypass function
 - h. Demonstrate the overload at 125%, 150%, 300%
 - i. Demonstrate the soft start
 - j. Carry out a full battery test
 - k. Test remote status and alarm panel functions
 - l. Test battery-monitoring system functions
 - m. Test mounting and anchorage devices.
 - C. Include for providing an on-site inductive load bank to carry out site tests. During the course of the tests the following parameters shall be measured and recorded:
 1. Input line voltage, line current, power failure frequency input terminals per phase
 2. Output line voltage, line current, power factor, frequency per phase
 3. Record the harmonics currents and voltage distortion for mains input of all three phases
 4. All of the above readings shall be taken and recorded every 10 minutes and tabulated into common table against time.
 - D. The UPS system will be considered defective if it does not pass tests and inspections.
 - E. Record of Tests and Inspections: Maintain and submit documentation of tests and inspections, including references to manufacturers' written instructions and other test and inspection criteria. Include results of tests, inspections, and retests.
 - F. Prepare test and inspection reports.
- 3.8 Demonstration
- A. Engage a factory-authorized service representative to train Client's maintenance personnel to adjust, operate, and maintain the UPS.

END OF SECTION

ELECTRICAL SECTION: 14

INTEGRATED FIRE ALARM AND VOICE EVACUATION SYSTEM

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1.0 GENERAL**1.1 REFERENCE**

Conform to General Requirements for Electrical Services

1.2 DESCRIPTION OF WORKS

The contractor shall supply and install an integrated fire alarm and voice evacuation system as described herein and as shown on the plans to be wired, connected and left in good operating condition. This section of the specifications includes the furnishing, installation, and connection of the microprocessor controlled, addressable reporting fire alarm equipment required to form a complete coordinated system ready for operation. It shall include, but not be limited to, alarm initiating devices, alarm notification appliances, fire alarm control panel, auxiliary control devices, annunciators, and wiring as shown on the drawings and specified herein.

The System consist of main fire alarm control panel, optical smoke/heat sensors, heat sensors, and optical smoke/heat sensor with integral sounder units, manual call points, electronic sounders, repeat panels, and interface units, all initiating devices has its own bi directional short circuit built-in isolators. The components and accessories deemed necessary for a safe, reliable and satisfactory system will confirm to the relevant and applicable requirements and relevant standards with BS EN 54-2 & 4 BS EN 50130-4 LPCB to the requirements of EN54 Parts 2 & 4 and EN-54-13, VDS, LPCB, UAE Life Safety Codes or UL Listed & UAE Life Safety Codes as per specification.

Panel mounted printer must be provided to print out all Alarm and trouble events. LCD repeater panel with Alarm Acknowledge, Alarm silence and System Reset buttons shall be provided as shown in the drawings.

Fire alarm system manufacturing under the appropriate category by LPCB, ETL or underwriter's laboratories (UL) or kite mark in the approved vendor list approved by local authorities will only be considered.

The complete installation shall confirm to the applicable sections of NFPA-72. NFPA-71BS EN and Electrical codes.

The fire alarm system shall allow for loading and editing instructions and operating sequences as necessary. The system shall be capable of storing, and downloading while the system is in operation. In addition, the system shall be capable of on-site programming to accommodate system expansion and facilitate changes in operation. All software operation shall be stored in a non-volatile programmable memory within the fire alarm control unit. Loss of primary and secondary power shall not erase the instructions stored in memory.

Full flexibility for selective input/output control functions based on ANDing, ORing, NOTing timing and special coded operations shall also be incorporated in the resident software programming of the system.

Resident software shall allow for full configuration of initiating circuits so that additional hardware shall not be necessary to accommodate changes in, for instance, sensing of normally open contact devices to sensing of normally closed contact devices or from sensing of normally open contact devices to sensing a combination of current limited and no current limited devices on the same circuit.

The system shall have the capability of recalling alarms and trouble conditions in chronological order for the purpose of creating an event history of 5000 events.

The control panel voice command center shall be equipped with master telephone for two way voice combination with remote phones. Fire fighter's phone jacks shall be provided at remote locations as shown in the drawings. 4 nos. pluggable phone shall be provided by the supplier. The phone shall be ruggedly constructed of a thermal ABS material like Red Cycloc, type T. A five foot coiled line cord and jack shall be attached to each phone.

Alarm indicating application shall operate selectively by building / floors as called for in the sequence of operation. Any subsequent zone alarm after silencing of initial condition shall reactivate all the alarm indicating application.

The activation of any system smoke detector shall initiate an alarm verification operation whereby the panel will reset the activated detector and wait for a second alarm activation. If within one minute after resetting, a second alarm is reported from the same or any other smoke detector, the system shall process the alarm as described in the sequence of operation. If no second alarm occurs within one minute the system shall resume normal operation. The Alarm verification shall operate only on smoke detector alarms. Other activated initiating devices shall be processed immediately. The Alarm verification operation shall be selectable by zone.

A manual evacuation switch shall be provided to operate the alarm indicating appliances without causing other control circuits to be activated. However, should a true alarm occur, all alarm functions service conditions including the time of each occurrence.

The system shall have provisions for disabling and enabling all circuits individually for maintenance or testing purposes.

The system batteries shall be supervised for disabling and enabling all circuits individually for maintenance or testing purposes.

The panels shall be capable of networking up to 200 nodes as nodes as and when required without modification of hardware except adding network cards.

An automatic alarm monitoring system for 24x7x365 monitoring by the local Civil Defence shall be provided as part of the fire alarm system.

The interface panel shall be provided for interfacing with the Civil Defence monitoring panel.

- The interface panel shall have the following capability
- Open protocol connectivity from the addressable fire alarm panel in the event of Fire alarm system being connected/monitored by BMS system.
-

Separate volt free contacts (VFC) shall be provided for the following points. The state of the VFC should be Normally Closed (NC) during normal condition and Normally Open (NO) during alarm condition.

- Common Fire alarm within the building
- Common Fire alarm System fault

Detectors shall be provided in floor voids as required as per standards. Wherever the ceiling void exceed 800 mm detector with remote indicator with test switch shall be provided without variation.

1.2.1 System Basic Performance

- Alarm, trouble and supervisory signals from all intelligent addressable reporting devices shall be encoded on to a Class A (NFPA Style 6, 7) Signaling Line Circuit (SLC).
- Initiation Device Circuits (IDCs) shall be wired Class A (NFPA Style D).
- Notification Appliance Circuits shall be wired Class A (NFPA Style Z).
- Built-in Horn Strobe Synchronization w/ selective silence.
- Digitized electronic signals shall employ check digits or multiple polling.
- A single ground or open on the system Signaling Line Circuit (SLC) shall not cause system malfunction, loss of operating power or the ability to report an alarm.
- Alarm signals arriving at the main FACP shall not be lost following a power failure (or outage) until the alarm signal is processed and recorded.

1.2.2 Basic System Functional Operation

When a fire alarm condition is detected and reported by one of the system initiating devices the following functions shall immediately occur:

- The System Alarm LED shall flash.
- A local piezo electric signal in the control panel shall sound.
- A 240 x 64 -character, backlit LCD display shall indicate all information associated with the Fire Alarm condition, including the type of alarm point and its location within the protected premises.
- Printing and history storage equipment shall log the information associated with each new Fire Alarm Control Panel condition, along with time and date of occurrence. Minimum of 5000 event and diagnostic logs and 500 Fire logs.
- All system output programs assigned via control-by-event equations to be activated by the particular point in alarm shall be executed, and the associated System Outputs (alarm Notification Appliances and/or Relays) shall be activated.

1.2.3 Sequence of Operation

Upon actuation of any manual station, or automatic detector or sprinkler flow switch, or air conditioning and ventilation duct return and exhaust air smoke detector. The system is to operate as follows:

On the Main Panel, the local panels and the annunciator (repeater) panels the green normal LED is to extinguish and the red alarm LED is to light. An audible single is to sound. Visually indicate on the LCD control panel label of the addressable device or circuit of alarm initiation. The first line is to display the user specified message indicating the floor and zone that initiated. The first line is to display the user specification message indicating the floor and zone that initiated the alarm. The second line of the LCD is to indicating real time, number of messages waiting, type of alarm, zone of alarm and time the alarm occurred. Red LED corresponding to the zone in alarm in the main panel shall also be lit.

The alarm indicators on the MFACP and repeater panel to continue to flash until the alarm is acknowledged. If a subsequent alarm is received after acknowledgment, the alarm is to sound again. The operator is to acknowledge the alarm by pressing a dedicated button and the buzzer is to silence provided that isn't an additional alarms the operator is to acknowledge all pending alarms before the buzzer is to silence. To silence audible to resound. To reset the system, the device is to be cleared first then the reset button is to be pressed.

Sound evacuation alarm on floor of incidence. Floor above and below shall sound an alert speech message for standby.

The alarm shall consist a "slow whoop" alarm tone, for ten second followed by thirty second of an automatic pre-recorded voice evacuation message in Arabic and English. The message is to ask the tenants to evacuate the building. The message and tone shall repeat continuously (unless manually silenced) until the alarm initiating device is restored to normal and system reset. The silencing of an alarm condition is not to prevent the resounding of alarm devices if a subsequent condition occurs. A time delay feature is to be provided to sound a general evacuation alarm automatically throughout the building if the initiating alarm condition is not responded to within a predetermined time. Visual indication at the panels, corresponding to activated voice alarm circuits is to illuminate.

Sound a 20 pulses per minute alarm tone in the remainder of the building for 15 seconds. The message is to draw attention to the possibility of a fire condition that is being checked out. The message and tone shall repeat continuously (unless manually silenced) until the alarm initiating device is restored to normal and system rest. The silencing of an alarm conditions is not to prevent the resounding of alarm devices if a subsequent condition occurs.

Automatically following functions to start as per cause and effect matrix,

1. Turn on the flashing strobes for the floor of incidence, the floor above and below.
2. Shut down all fans serving the areas affected by the alarm condition.
3. Start all stairwell pressurization fans.
4. Start smoke extraction fan system
5. Open Gate barriers
6. Start Makeup air fans
7. Close main Gas Cut off Valve
8. Switch on Egress lighting
9. Dial the civil defense authorities

Recall all passenger elevators to ground floor or as designated by the Local Fire Department. If the alarm originates from the ground floor, redirect the elevators to the other floors or as designated by the Local Fire Department. A common pre-recorded message is to be sounded in all seized elevators. Lifts are then to held for the exclusive use of the fire fighter by use of a special key.

Transmit signal to radio paging system to display zone of alarm.

Printer is to print text message on the event log indicating the device which initiated a trouble alarm, time and date associated with alarm. All follow up information regarding this alarm, such as acknowledge, reset and the like is to be printed afterwards.

All emergency stair and exit doors will release.

1.3 QUALITY CONTROL/ASSURANCE

Design, manufacture, testing and method of installation of all apparatus and materials furnished under requirements of these specifications shall conform to latest publications or standard rules of the codes and standards mentioned in clause 1.4:

The Quality Control/Assurance/Assurance/Quality Assurance duties shall be performed by the Contractor. The system of Quality Control/Assurance/Assurance verification shall be in accordance with ISO 9000 standards of quality assurance.

1.4 CODES AND STANDARDS

The following codes and standards to the extent specified herein, form a part of this specification. When an edition date is not indicated for a Code or Standard the latest edition enforces at the time of proposal submitted shall apply.

A. National Fire Protection Association (NFPA) - USA:

- No. 70 National Electrical Code (NEC)
- No. 72 Central Station Signaling Systems
- No. 72 Protective Signaling Systems
- No. 72 Automatic Fire Detectors
- No. 72 Notification Appliances for Protective Signaling Systems.
- No. 72 Testing Procedures for Signaling Systems.
- No. 101 Life Safety Code

B. Underwriters Laboratories Inc. (UL) - USA:

- No. 268 Smoke Detectors for Fire Protective Signaling Systems
- No. 864 Control Units for Fire Protective Signaling Systems
- No. 268ASmoke Detectors for Duct Applications.
- No. 521 Heat Detectors for Fire Protective Signaling Systems
- No. 464 Audible Signaling Appliances.
- No. 38 Manually Actuated Signaling Boxes.
- No. 346 Water flow Indicators for Fire Protective Signaling Systems.
- No. 1971 Visual Notification Appliances for the hearing impaired.

C. BS EN Standards - BS EN 54-2 & 4 BS EN 50130-4 LPCB to the requirements of EN54 Parts 2 & 4 and EN-54-13, VDS, LPCB. etc.

D. All requirements of the Authority Having Jurisdiction (AHJ) & latest UAE Fire Code.

E. The system shall have proper listing and/or approvals from Dubai Civil Defence Authority.

1.5 SUBMITTALS

The contractor shall submit the following drawings and documents related with fire alarm and voice evacuation system for Engineer's approval:

- A. Original catalogues for various proposed components.
- B. Schedule of Accessories and Fittings.
- C. Test Certificates.
- D. Shop drawings
- E. Manuals
- F. Grantee / Warrantees
- G. Maintenance
- H. Post Contract Expansions etc.

1.5.1. General

- 1. Two Original and Two copies of all submittals shall be submitted to the Architect/Engineer for review.
- 2. All references to manufacturer's model numbers and other pertinent information herein is intended to establish minimum standards of performance, function and quality.
- 3. For equipment other than that specified, the contractor shall supply proof that such substitute equipment equals or exceeds the features, functions, performance, and quality of the specified equipment.

1.5.2. Shop Drawings

- 1. Sufficient information, clearly presented, shall be included to determine compliance with drawings and specifications.
- 2. Include manufacturer's name(s), model numbers, ratings, power requirements, equipment layout, device arrangement, complete wiring point-to-point diagrams, and conduit layouts.
- 3. Show remote annunciator(s) layout, configurations, and terminations.

1.5.3. Manuals

- 1. Submit simultaneously with the shop drawings, complete operating and maintenance manual listing the manufacturer's name(s) including technical data sheets.
- 2. Wiring diagrams shall indicate internal wiring for each item of equipment and the interconnections between the items of equipment.
- 3. Provide a clear and concise description of operation that gives, in detail, the information required to properly operate the equipment and system.

1.5.4. Certifications

Together with the shop drawing submittal, submit a certification from the major equipment manufacturer indicating that the proposed supervisor of installation and the proposed performer of contract maintenance is an authorized representative of the major equipment manufacturer. Include names and addresses in the certification.

2.0 PRODUCTS

2.1 MAIN FIRE ALARM CONTROL PANEL

A. The FACP shall contain a microprocessor based Central Processing Unit (CPU). The CPU shall communicate with and control the following types of equipment used to make up the system: addressable detectors, addressable modules, printer, annunciators, and other system controlled devices.

B. System Capacity and General Operation

1. The control panel shall provide, or be capable of expansion to 200 nodes. The system should support peer to peer fault tolerant network.
2. The Fire Alarm Control Panel shall include a full featured operator interface control and annunciation panel that shall include a backlit, 240x64-character Liquid Crystal Display, individual, color coded system status LEDs, and an alphanumeric keypad for the Field Programming and control of the Fire Alarm System.
3. All programming or editing of the existing program in the system shall be achieved with hand held programmer or directly from the panel and without interrupting the alarm monitoring functions of the Fire Alarm Control Panel.

The FACP shall provide the following features: Maintenance Alert to warn of excessive detector dirt or dust. Detector sensitivity read/test information and System Status Reports to display or print. Smoke Detector Alarm Verification. Pre-signal, meeting NFPA 72 requirements. Rapid manual station reporting (under 3 seconds). Periodic Detector Test, conducted automatically by the control panel every two hours. March time, temporal (ANSI Cadence) and California Code coding options. Walk Test will check for two detectors set to same address.

4. The main CPU shall contain Form-C relay contacts rated at 2.0 amps/30VDC for the following: Alarm, Trouble, and Supervisory.
5. The CPU shall contain two Class A (NFPA Style Y or Z) programmable Notification Appliance Circuits.
6. Automatically detect and report open circuit, short circuit, and ground fault of wiring for initiating device, signaling line, and notification-appliance circuits. System shall detect ground fault by panel, loop circuit, and group of devices.
7. Integral battery back-up capacity in line with NFPA 72

C. Central Microprocessor

1. The Microprocessor shall communicate with, monitor, and control all external interfaces with the control panel. It shall include EPROM for system program storage; non-volatile memory for building-specific program storage; and a "watch dog" timer circuit to detect and report microprocessor failure.
2. The Microprocessor shall contain and execute all programming for specific action to be taken if an alarm condition is detected by the system. Such programming shall be held in non-volatile programmable memory and shall not be lost if both the system primary and secondary power failure occurs.

3. The Microprocessor Unit shall also provide a Real- Time Clock for time annotation of system displays, printer, and history file.
4. The Microprocessor Unit shall contain flash memory capabilities for easy upload/download for upgrades of software.
5. All clock, date and history files shall be maintained during power loss.

D. Display

1. The Display shall provide all the controls and indicators used by the system operator and may also be used to program all system operational parameters.
2. The Display shall include status information and custom alphanumeric labels for all Addressable Detectors, Addressable Modules and Software zones.
3. The Display shall provide a 240x64 -character backlit alphanumeric Liquid Crystal Display (LCD). It shall also provide 9 Light-Emitting-Diodes (LEDs), consisting of and not limited to the following: ac power, fire alarm, supervisory, system trouble, maintenance, alarm silenced, disabled, battery, and ground.
4. The Display shall provide a touch key-pad with control capability to command all system functions, entry of any alphabetic or numeric information, and field programming. Two different password levels shall be provided to prevent unauthorized system control or programming.
5. The Display shall include the following operator switches: acknowledge/step alarm silence, drill, and system reset (also serving as a lamp test switch).

E. Signaling Line Circuit Interface

1. The SLC Interface shall provide power to, and communicate with, all of the Addressable Detectors and Addressable Modules over a single pair of wires. This SLC Loop shall be capable of NFPA Style 4, Style 6, or Style 7 operation.
2. The SLC interface shall receive information from all Addressable Devices. This information shall be processed to determine whether normal, alarm, or trouble conditions exist for each detector. This information may also be used for automatic detector testing and for the automatic determination of detector maintenance requirements.
3. The Signaling Line Circuit shall be capable of distances of 1.2 Km CLASS A circuit.

F. Serial Interfaces

1. An EIA-232 interface between the Fire Alarm Control Panel and UL Listed Electronic Data Processing (EDP) peripherals shall be provided. The EIA-232 interface shall allow the use of printers, or for an interface to an off-line PC programmer.
2. An EIA-485 port shall be available for the serial connection of optional remote led-type annunciators. EIA-485 in terminal mode shall allow serial connection of optional LCD, English language remote system displays. LED (per zone or point) annunciators shall also be provided. The maximum distance to the furthest annunciator shall be 3,000 feet. The system shall support a maximum of 32, remote annunciators on a single twisted, shielded pair. The maximum distance to the furthest annunciator shall be 6,000 feet.

3. A PS2/PC keyboard connection shall be provided to support the connection of a PC keyboard for local programming of the fire alarm system.

G. Enclosures

1. The control panel shall be housed in a UL/BS-EN listed cabinet suitable for surface or semi-flush mounting. Cabinet and front shall be corrosion protected.
 2. The door shall provide a key lock and shall include a glass or other transparent opening for viewing of all indicators.
 3. An optional semi-flush trim ring shall be available for a neat cabinet dress.
- H. All interfaces and associated equipment are to be protected so that they will not be affected by voltage surges or line transients consistent with UL standard 864 or BS-EN Standards.
- I. Optional plug-in modules shall be provided for NFPA 72 auxiliary and remote station fire alarm systems as well as a Digital Alarm Communicator Transmitter for NFPA 72 Central Station systems.
- J. Optional modules shall provide eight Form-C relays rated at 2.0 amps (Relays shall track programmable software zones).

K. Power Supply

1. The Power Supply shall operate on 230V AC, 50 Hz.
2. It shall provide a minimum of 2.0 amps of usable Notification Appliance power.
3. It shall provide a battery charger for 24 hours of standby using dual-rate charging techniques (UPS) for fast battery recharge.
4. It shall provide a very low frequency sweep earth detect circuit, capable of detecting earth faults on sensitive addressable modules.
5. It shall be power-limited using fuse-less, quick-acting electronic circuitry meeting the latest UL/BS-EN requirements.

L. Operators Controls

1. Acknowledge Switch
 - a. Activation of the control panel Acknowledge switch in response to new Alarms and/or Troubles shall silence the local panel piezo electric signal and change the Alarm and Trouble LEDs from flashing mode to steady-ON mode. If multiple Alarm or Trouble conditions exist, depression of this switch shall advance the 240x64 -character LCD display to the next Alarm or Trouble condition.
 - b. Depression of the Acknowledge switch shall also silence all remote annunciator piezo sounders.
2. Signal Silence Switch: Activation of the Signal Silence Switch shall cause all programmed Notification Appliances and relays to return to the normal condition after an alarm condition. The selection of Notification circuits and relays that are silence able by this switch shall be fully field programmable within the confines of all applicable standards. The FACP software shall include silence inhibit, auto-silence timers, and an option to silence horns and keep strobes flashing.

3. System Reset Switch: Activation of the System Reset Switch shall cause all electronically latched initiating devices, appliances or software zones, as well as all associated output devices and circuits, to return to their normal condition. Holding the RESET switch shall perform a Lamp Test function.
4. Drill (Evacuate) Switch: Press and hold of the Drill switch shall activate all Silence Able Notification Appliance circuits. The Drill function shall latch until press of Signal Silence or Reset.

M. Field Programming

The system and its respective devices (i.e. smoke detectors and modules) shall be programmable, configurable and expandable in the field without the need for special tools or electronic equipment and shall not require field replacement of electronic integrated circuits.

All programming may be accomplished through the standard FACP built-in keypad. As well through using a PC keyboard. All field-defined programs shall be stored in non-volatile memory and shall not be lost if AC mains and/or battery is lost.

The programming function shall be enabled with a password that may be defined specifically for the system when it is installed. Two levels of password protection shall be provided in addition to a key-lock cabinet. One level is used for status level changes such as zone disable or manual on/off commands. A second (higher-level) is used for actual change of program information.

Program edit shall not interfere with normal operation and fire protection. If a fire condition is detected during programming operation, the system shall exit programming and perform fire protection functions as programmed.

A special program check function shall be provided to detect common operator errors.

For flexibility, an optional off-line programming function, with batch upload/download, shall also be available.

2.2 SMOKE CONTROL SYSTEM

Provide Fire Operators panel at the Main control panel for control and monitoring of AHU, LIFTS, PUMPS, flow switches, supervisory valves and all smoke extract function in co- ordination with mechanical smoke control and extract system. All functions should be programmed in the sequence of operation or Cause and Effect programming. There should be a manual and auto switch for the fire operator to control the smoke control equipment's through the control panel in the FCC room.

The Cause and Effect matrix chart should be submitted for Engineer's approval prior to implementation. The smoke control system and interfaces should be as per local authority requirements.

In Smoke Control System all MSFD's should be directly controlled by Fire Alarm panel through a dedicated relay individual addressable module with built in short circuit isolator installed in a fire rated enclosure as per UAE FLS code or through a high level software interface. A dedicated monitoring module can be provided for BMS for status indication only.

Fire Alarm System shall be integrated with 24x7, based on their requirements.

2.3 INTEGRATED VOICE ALARM & COMMUNICATION

A central digital audio control module shall be provided for the necessary alarm message / tone generation main and remote microphone connections, music inputs, and mixer / pre-amplifier circuits and shall seamlessly integrate with the Fire Alarm Control Panel. Continuous supervision shall be provided along with specific information as to the type of failure should a problem occur (e.g. Main microphone trouble, tone trouble, etc.) Audio outputs shall have individual gain control. For European products similar products complying BS5839 part 8 could be proposed.

A hand-held push-to-talk microphone shall be provided in the Voice Command Center, recessed within a protective panel-mounted enclosure. The microphone shall be a noise-canceling communication type with a frequency range of 200 Hz and shall be equipped with self-winding five foot coiled cable. An LED indicator shall be provided for the circuits ready for transmission. The microphone shall be supervised for disconnection.

An audio control switch- module shall be furnished to provide manual access to audio operations for authorized personnel. The module shall include an "ALL CIRCUITS" switch, "Aux Tone 1" which, "Aux Tone 2 switch, tone generator stop switch, "Audio Trouble Reset" switch. These switches and associated LED indicators shall be supervised for disarrangement or failure.

A. Audio power amplifiers:

Audio power amplifiers shall be furnished with a self-contained filtered 24VDC power supply, transformer, and amplifier monitor circuits. The amplifiers shall provide a 25/70/100 Volt RMS output with a frequency response of 120 to 18,000 Hz. Provide sufficient amplification to operate all system speakers simultaneously plus ten (10) percent spare capacity. The Amplifiers power shall be in the range from 50W to 100W as per requirements.

Each amplifier shall have built-in Back-up amplifier. In case of Primary amplifier failure, speaker circuits shall automatically get connected to back-up amplifier and system operation continues without interruption. In the event of a total loss of audio data communications, all amplifiers will default to the local "HORN" tone. If the local panel has an alarm condition, then all amplifiers will sound the Horn signal on their connected speaker circuits. Audio amplifiers shall automatically detect a short circuit condition on the connected speaker circuit wiring, and shall inhibit itself from driving into that short circuit condition

The speaker circuits shall be capable of supplying 25/70/100Volt RMS audio power from the system amplifiers. Supervision for open, short, or ground fault conditions shall be provided. Individual and distinct trouble indications shall be provided for each fault. Provide one circuit for each zone or area of distinct communication using individual Amplifier.

Digital tones for alarm (show whoop) and auxiliary requirements (wail, horn, chime, etc.) shall be provided.

A pre-recorded digitized voice message capability is to be provided for automatic transmission to building occupants during alarm conditions. The automatic message player shall not rely on a tape or other mechanical means of transmitting the evacuation message. A standard evacuation message shall be provided under this contract, however, the message player must be capable of transmitting a custom message of up to five (5) minutes long.

The system shall be configured to allow selective voice paging. Upon activation of any speaker manual control switch, two attention getting beeps shall sound over the speakers indicating an impending voice message will occur.

If any speaker manual control switch is activated, the control panel operator shall be able to make announcements via the push-to-talk paging microphone over the pre-selected speakers.

Facility for total building evacuation and paging shall be provided to allow for activation of all speakers. This shall be accomplished by the means of an "All Circuit" switch.

B. Power Requirements

The system shall be provided with sufficient battery capacity to operate the entire system upon loss of normal 230V AC power in a normal supervisory mode for a period of 24 hours with 30 minutes of alarm operation at the end of this period. The system shall automatically transfer to the standby batteries upon power failure. All battery charging and recharging operations shall be automatic. **Contractor shall submit standby and alarm power calculations in support of the selected battery size**

All external circuits requiring system operating power shall be 24V DC and shall be individually fused at the control panel.

C. Printer

Printer shall receive English language text from the fire alarm control panel in an industry standard ASCII format via an EIA RS-232-C connection. All printed information shall include time and date. Printer shall be quiet in operation, with automatic paper take. Printer shall be panel mounted. External printers are not acceptable.

2.4 ADDRESSABLE INITIATING DEVICES

A. General

The system control panel must be capable of communicating with the types of addressable devices specified below. Addressable Devices shall be located as shown on the drawings. The system shall identify when a smoke sensor becomes too dirty to operate properly. It shall also identify sensors which are almost dirty which need cleaning before they drift beyond their selected sensitivity. In short, a review of the front panel display or the printed status report quickly identify sensor that need cleaning. The intelligent detectors should have automatic environmental compensation and double stage dust protection to avoid false alarms.

Sensitivity of the sensor shall be programmable, photoelectric shall be variable with four sensitivity levels. It shall also be possible to program for timed, automatic sensitivity selection such as less sensitive during working hours and more sensitive when quite.

The panel shall provide the following features:

- Individual sensitivity selection for each sensor
- Peak value logging allowing accurate analysis for sensitivity selection
- Automatic, once per minute individual; sensor calibration checks to verify sensor integrity
- Display of sensitivity directly in per cent per foot
- Multi-stage alarm operation
- Ability to display and print detailed sensor information

B. Sensor Bases

Each sensors output shall be digitized and transmitted to the control panel every four seconds. The bases shall be available with connections for remote LED alarm indicator or connections for supervised remote relay.

C. Addressable Sensors:

All analogue sensors and bases shall be provided by the manufacturer of the control equipment. The sensor bases for interfacing between the loop wiring and the sensor head shall not contain any electronics. All bases shall have the necessary connections for sending repeat fire signals to a remote LED unit.

The sensors provided shall be lockable into position if required and removal of locked sensors shall be achievable only through the use of the appropriate removal tool. Sensor removal tools should be provided on completion of the contract as part of the spare parts profile. The removal of a sensor from its base shall not affect the continuity of the detection loop.

The following types of analogue sensors will be available as standard:

- Optical Smoke sensor
- Optical sensor with integral sounder and/or strobe
- Heat sensor
- Multi-sensor.
- Specialist sensor housing designed to monitor air conditioning ducting –Duct Smoke Sensor.

All of the above shall be compatible with the aforementioned base providing inter-changeability between sensor heads without the requirement for switch settings. All sensors will also have an integral short circuit isolator, which in the event of a single cable fault will isolate the faulty section of cable and retain all devices on the loop operationally in less than 1 second. The sensitivity off all sensors will be adjustable from the control panel. This may be carried out manually to manage false activation issues or automatically using the system clock i.e. day/night settings for specific risks. It shall be possible to program sensors within a range of sensitivity levels from State 0 (high level) through to State 15 (disabled). Each sensor will possess an integral Red LED giving a flashing indication of a fire signal or a continuous indication for certain fault conditions. The integral LED can also be enabled/disabled to provide operational status i.e. short flashes that indicate the unit is powered and communicating with the control panel.

The addressable sensor shall plug in unit which is mounted to a twist lock base. Sensor shall have 360 degree Omni view LED indications to display device polling, Alarm and Trouble conditions.

Each detector shall contain an environmental compensation algorithm that identifies and sets ambient “Environmental Thresholds” approximately six times an hour. The microprocessor shall continually monitor the environmental impact of temperature, humidity, other contaminates as well as detector aging. The process shall employ digital compensation to adapt the detector to both 24 hour long-term and 4 hour short-term environmental changes. The detector shall monitor the environmental compensation value and alert the system operator when the detector approaches almost dirty, dirty and excessively dirty of the allowable environmental compensation value.

All devices shall be supervised or trouble conditions. The system control panel will be capable of displaying the type of trouble condition (open, short, device missing, failed).

Should a device fail it will not hinder the operation of other system devices.

D. Heat Sensors

The low-profile Intelligent Heat detector, with built-in isolator shall offers a wide range of capabilities to suit any commercial, industrial or institutional application. Distinguished by its clear alarm LEDs which illuminate red during alarm the detector shall provide a complete 360-degree view of device status. The detectors LEDs shall also be programmed for flashing during quiescent mode.

The heat sensor shall be self-restoring and provide a combination of rate and fixed temperature rate compensated sensing. It shall have low thermal mass to accurately and quickly measure the local temperature at the fire alarm panel.

It shall be possible to select the rate of rise temperature detection for either 15°F or 20°F per minute. Fixed temperature sensing and shall be programmable to operate at 135°F or 155°F.

E. Photoelectric/Optical Smoke Sensors

The low-profile Intelligent Photoelectric detector, with built-in isolator shall offers a wide range of capabilities to suit any commercial, industrial or institutional application. Distinguished by its clear LEDs which illuminate red during alarm the detector shall provide a complete 360-degree view of device status.

The head be designed to allow 360 deg. Smoke entry for optimum response to smoke from any director. A built-in screen shall keep insects from entering the smoke chamber. There shall be 4 sensitivity settings and drift compensation.

Sounders shall be provided in the sensor base or head as required and shown on drawings.

F. Multi Sensors

The low-profile Intelligent Multi Sensor Detector shall utilize a combination of photoelectric and thermal sensing technologies to monitor two types of fire signatures (smoke and heat), offering increased immunity to potential false alarms. The signals from the photoelectric sensing element and the temperature sensor shall be completely independent and represent the smoke level and the air temperature respectively in the vicinity of the detector.

Distinguished by clear LEDs which illuminate red during alarm the detector shall provide a complete 360-degree view of device status. The detector shall have built in isolator.

G. Sensor with Integral Sounders

Install as or if shown in the drawings. These combined units will in addition to the features listed above by the optical/heat sensor, incorporate a high output sounder. The output will be 85dBA or 75dBA at the bedhead.

These sensor sounder combined units will be installed on the same 2 core loops as the other devices. Additional 2 core power cable from the Power Supply Units in the FACPs / from remote external power supply units for the operation of these units will not be permitted, for better system integrity.

The integral sounders will be capable of being individually programmed from the main fire alarm control panels. Integral sounders operating on the initiation of its corresponding detector only, will not be acceptable.

Sensor-sounder combined units mounted in the false ceilings will be provided with semi-flush mounted kits.

Heat sensor sounder combined units will be used in the kitchen areas only. & it shall have inbuilt short circuit isolator.

H. Duct Mount Sensor with sampling tube

Duct smoke detectors are - flexible enough to fit configurations from square to rectangular and everything in between. These units sense smoke in the most challenging conditions, operating in air-ow speeds of 0.5m/s to 20m/sec, temperatures of 0°C to 70°C, and a humidity range of 0 to 95 percent (non-condensing). An improved cover design isolates the detector head from the low-- ow feature for simple maintenance. The unit incorporates cover tamper feature to indicate a trouble signal for a removed or improperly installed sensor cover. The duct detector housing provides a 20mm conduit knockout and ample space to facilitate easy wiring and mounting of relay module. The duct smoke detector can be customized to meet local codes and specifications without additional wiring.

NFPA 90A, 2012, A.6.4 supports this by stating: “Protection provided by the installation of smoke detectors and other related requirements is intended to prevent the distribution of smoke through the supply air duct system and, preferably, to exhaust a significant quantity of smoke to the outside. Neither function, however, will guarantee either early detection of fire or the detection of smoke concentrations prior to dangerous smoke conditions if smoke movement is other than through the supply air system.

Applications NFPA 90A2, “Standard for the Installation of Air Conditioning and Ventilating Systems”, specifies that “Smoke detectors listed for use in air distribution systems shall be located as follows:

(1) Downstream of the air filters and ahead of any branch connections in air supply systems having a capacity greater than 2,000 cfm (944L/sec).

(2) At each story prior to the connection to a common return and prior to any recirculation or fresh air inlet connection in air return systems having a capacity greater than 15,000 cfm (7080 L/sec), and serving more than one story.

Return system smoke detectors are not required when the entire space served by the air distribution system is protected by a system of area smoke detectors.” (NFPA 90A, 6.4.2.2) 6.4.2.3 “Fan units whose sole function is to remove air from inside the building to outside the building.” (NFPA 90A, 6.4.2.3)

Application Documents the International Mechanical Code (IMC) specifies that “If the space is air conditioned and the A/C unit is over 2,000 CFM, a duct smoke detector is required to be installed in the return air duct. IMC 2012, 606.2.1 *Exception: Smoke detectors are not required in the return air system where all portions of the building served by the air distribution system are protected by area smoke detectors connected to a fire alarm system in accordance with the International Fire Code.

The performance, application and installation of duct smoke detectors as per: • U.L. Standard 268A, Standard for Smoke Detectors for Duct Applications • NFPA Standard 90A, Installation of Air Conditioning and Ventilating Systems • NFPA 92A, Standard for Smoke-Control Systems Utilizing Barriers and Pressure • NFPA Standard 72, National Fire Alarm Code • NFPA Standard 101, Life Safety Code • ASHRAE Handbook and Product Directory, “Fire and Smoke Control” • International Mechanical Code.

I. Remote Indicator with test switch for void area Smoke detectors:

Remote indicator shall consist of red LED status indicator and a remote test key switch mounted on a single gang stainless steel plate. Turning the test switch to “TEST” will initiate an alarm and allow the resulting system responses to be verified. Power Requirements Alarm LED 15mA, Pilot LED 15mA.

J. Manual Call Points:

The manual call points shall be electrically compatible with all of the aforementioned sensor types. Each device shall contain its own microprocessor giving a 1 second response time from initiation as required within BS5839. The MCP shall be available as a semi-flush mounting unit fixing to a standard single gang recessed box or as surface mounting unit on a matching red plastic back box.

The MCP will have the ability to be tested functionally without the need to remove the front cover or breaking the glass with a special test key (supplied as standard). The key shall insert in the front fascia of the MCP ensuring easy access of the key at all times. The key will also be used to reset the MCP when fitted with a resettable plastic element. The option to retrofit a clip-on transparent plastic cover to prevent accidental or malicious activation should be available as standard and give the unit an IP55 ingress rating when fitted to the plastic back box. The MCP shall be double action with break glass and a lever to pull or Single action with retrievable glass and switch.

The device shall integral electronics for constantly monitoring the status of the device and communicating the same to the control panel. Address of the device shall be set by electronic addressing or by SAFE addressing.

All Manual call point is a critical life safety device. Hence device without have built-in bi directional Short circuit isolator are not acceptable. External isolator modules are accepted if it is provided on both sides of the MCPs in a loop.

2.5 NOTIFICATION APPLIANCES**A. Alarm Sounders**

Notification appliances shall include visible, audible or Audible / visible as shown in the drawing. Audible appliance shall be loudspeaker or electronic sounders and the visible appliance shall be strobes.

Install the Notification Appliances as shown in the drawings. These will comply with the requirements of BS EN 54-3 BS EN 50130-4. Alarm Sounders are all Electronic Sounders and Standard Evacuate & Alert tone with voice messaging Programmable tones selectable by control panel. Programmable tones are selectable by control panel Sounder frequency as defined in BS5839.

Synchronization of all sounders to be fully synchronized with all other analogue addressable loop powered sounder speech devices on the system.

The Alarm Sounders will be loop wired or separate power supply circuit to be provided and loop signaled and provided with built-in short circuit isolation and will be sited in the emergency stairwell staff/utility areas and plant rooms. The sounders will be configured via software to operate individually or in sectorised groups, totally independent of the way they have been connected to the loops. The sounders will have the synchronization feature to ensure that all the sounders give alert and evacuate tones that are totally in phase.

The Sounders will have Minimum sound pressure level 103dbA at 1 meter with frequencies of 970 Hz and 910 Hz. Variety of sounds will be available.

B. Sounder with Strobe/ Flasher Unit.

Strobes shall be suitable for wall or ceiling mounting as shown in the drawings. Xenon flash tubes shall be 24VDC powered from the panel. Visible output shall be 15/30/75/95/115 candela. The reflective design shall provide light output in key axis directions allowing vertical or horizontal mounding. The unit shall be of red finish with white "FIRE" lettering. Flash rate shall be 1 Hz as defined in BS5839.

Electrical Ratings Typical Operating Voltage Range 23 - 29.5 VDC, Special Application Supervisory Requirements 1-unit load (= 0.8 mA control panel current).

Temperature Range 32° to 120° F (0° to 55° C) Humidity Range 10% to 93%, non-condensing @ 104° F (40° C)

The Sounders will have Minimum sound pressure level 103dbA at 1 meter with frequencies of 970 Hz and 910 Hz. Variety of sounds will be available.

Connections Terminal blocks on mounting plate for 18 AWG to 12 AWG (0.82 mm² to 3.31 mm²); two wires per terminal for in/out wiring

C. Voice Alarm Speakers

Voice Alarm speakers shall be either wall or ceiling wall or ceiling mounting type to suite the site conditions. These shall provide high quality tone and voice reproduction. Round or square models shall be installed, to the approval of the Engineer. Generally, wall mounted units shall be of square type and round housing shall be constructed of high impact, flame retardant ABS thermoplastic or metal enclosures. All speaker shall have 25 V line or 70V or 100V transformer with tapings at 3W and 6W.

Frequency response of the speakers shall be **5000Hz to 18KHZ** for general signaling and sensitivity of the speakers shall be 85dBA 1W, 10ft. The speakers shall be high profile from reputed manufacturers and subject to local authority approvals and architects choice. There should not be any visible screw terminal or connection knock out entries on the sides. The Ceiling speakers color should match with the ID requirements and screw less and subject to Dubai Civil Defense Authority approval.

Speaker circuits shall be wired in **Class A** wiring topology in every floor and each Fire zone. All ceiling mounted speakers shall have 160° dispersion angle with fire dome, Ceramic terminals.

D. Voice Alarm Speakers with integral Strobe unit.

Audio -Visual units shall be provided on a common enclosure for the fire alarm audible and visual alarm devices. The housing shall be designed to accommodate speakers as well as strobe unit.

The Ceiling Mounted Speaker/Strobe is a selectable candela speaker/strobe and the Ceiling or Wall Mounted Speaker is designed to meet code requirements for audio, visual and voice communications. The Ceiling or Wall Mounted Speaker Series are quality speaker products that offer both dependable evacuation signaling and visual alarms, and a combination of both.

The Ceiling Mounted Speaker / Strobe has high output tamper proof candela selections are 15, 30, 75, 95 and 115.

The Ceiling or Wall Mounted Speaker Series can be mounted in a 4" square x 2-1/8" deep back box, an extension ring is should not be needed.

These shall provide high quality tone and voice reproduction. Round or square models shall be installed, to the approval of the Engineer. Generally, wall mounted units shall be of square type and round housing shall be constructed of high impact, flame retardant ABS thermoplastic or metal enclosures. All speaker shall have 25 V line or 70V or 100V transformer with tapings at 3W and 6W.

Frequency response of the speakers shall be **5000Hz to 18KHZ** for general signaling and sensitivity of the speakers shall be 85dBA 1W, 10ft. The speakers with strobe shall be high profile from reputed manufacturers and subject to local authority approvals and architects choice. There should not be any visible screw terminal or connection knock out entries on the sides. The Ceiling speakers with strobe color should match with the ID requirements and subject to Dubai Civil Defense Authority approval.

All field wiring connections shall be made via separate in-out terminal connections and the speaker or speaker strobe shall be UL, CSFM and BS&A/MEA listed and comply with all local, state and federal fire alarm codes/standards.

Speaker circuits shall be wired in **Class A** wiring topology in every floor and each Fire zone.
All ceiling mounted speakers shall have 160° dispersion angle with fire dome, Ceramic terminals.

2.6 INTERFACE MODULES

Fire detection interface units will be directly connected to the loop to provide both inputs and outputs for the control or annunciation of other life safety, security and building management systems. These units shall be either self-contained wall mountable units or DIN rail mounting units for fitting within 3rd party control equipment/panels. Each device will incorporate a short circuit isolator as standard to maintain system integrity in the event of an equipment failure or wiring fault. As standard six variants will be available:

- (i) 4 channel input and/or output interface
- (ii) 1 channel Low Voltage input/output interface
- (iii) 1 channel Low Voltage input only interface
- (iv) Mains Voltage 230V 13A switched output interface
- (v) Mains powered 4 channel interface with monitored integral power supply unit, battery standby and 250mA output circuits
- (vi) Key switch operated single channel interface.

Control modules shall be provided as shown in the drawings to supervise and operate notification appliances, speakers and telephone circuits. Output capacity shall be up to 24V DC. All control modules shall be powered from the panel via 2 core cables. Local powering of control modules shall not be acceptable.

The control module shall have built in short circuit isolator.

Supervised individually Addressable Modules shall be installed in the addressable network for monitoring the status of lifts, etc.

Monitoring modules shall have built in short circuit isolator.

These devices shall provide specific addressability to a single initiating device or multiple devices by monitoring normally open dry contacts shall cause an alarm. An open in the initiating circuit wiring shall causes a trouble to be reported at the panel. Monitoring modules shall have LED indications.

2.7 FIRE FIGHTER TELEPHONE AND FIRE TELEPHONE JACK

Remote phone jack station shall be recessed to the wall on a single gang box for use with portable phones. Face plate shall be engraved "Fire Emergency Phone". The Fire Telephone unit shall have a call LED indicator at the panel. 5 sets of fire telephone also should be provided for use.

2.8 OPEN AREA SMOKE IMAGING DETECTION/(OSID)

Open-area Smoke Imaging Detection (OSID) is a new innovation in projected beam smoke detection technology. By using advanced dual wavelength projected beams and optical imaging technology for early warning smoke detection, OSID provides a low-cost, reliable and easy-to-install solution that overcomes typical beam detection issues such as false alarm incidents and alignment.

The OSID system measures the level of smoke entering beams of light projected over an area of protection. A single OSID Imager can detect up to seven Emitters to provide a wide coverage area. Two innovations in smoke detection technology have been developed for the revolutionary OSID smoke detector:

Dual Wavelength Particle Detection.

The beam projected from each Emitter contains a unique sequence of ultraviolet (UV) and infrared (IR) pulses that are synchronized with the Imager and enable the rejection of any unwanted light sources. By using two wavelengths of light to detect particles, the system is able to distinguish between particle sizes. The shorter UV wavelength interacts strongly with both small and large particles while the longer IR wavelength is affected only by larger particles. Dual wavelength path loss measurements therefore enable the detector to provide repeatable smoke obscuration measurements, while rejecting the presence of transient dust particles or solid intruding objects.

Optical Imaging with a CMOS Imaging Chip.

An optical imaging array in the OSID Imager provides the detector with a wide viewing angle to locate and track multiple Emitters. Consequently, the system can tolerate a much less precise installation and can compensate for the drift caused by natural shifts in building structures. Optical filtering, high-speed image acquisition and intelligent software algorithms also enable the OSID system to provide new levels of stability and sensitivity with greater immunity to high level lighting variability.

Mode of Operation.

Status information (Fire Alarm, Trouble and Power) is communicated through the Imager via Status LEDs, dedicated Trouble and Alarm relays, and the Remote Indicator interface. Specific Trouble (Fault) conditions are identified through coded flashes of the Trouble LED. An internal heating option is also provided on the Imager to prevent condensation on the optical surface, and a reset input enables an external signal to reset the device.

Features

- Maximum detection range of 150 m (492 ft) for the OSI-10
- Dual wavelength LED-based smoke detection
- Optical imaging with CMOS imaging chip
- On-board event log for fault and alarm diagnostics
- High tolerance to false alarm due to building movement
- High tolerance to transient dust, steam and intrusion of solid objects
- Easy alignment with large adjustment and viewing angles

- No need for precise alignment
- Tolerant of alignment drift
- Automatic commissioning in under ten minutes
- Simple DIP switch configuration
- Simple and easy maintenance requirements
- Three selectable alarm thresholds
- Status LEDs for Fire, Trouble and Power
- Conventional alarm interface for straightforward fire system integration Listings/Approvals
- UL, ULC, FM, AFNOR, CE - EMC and CPR, VDS, ActivFire, BOSEC.

Technical Specifications.

Supply Voltage 20 to 30 VDC (24 VDC nominal).

Imager Current Consumption Nominal (at 24 VDC): 8mA (1 Emitter) 10mA (7 Emitters) Peak (at 24 VDC) during training mode: 31mA.

Emitter Current Consumption Wired Version (at 24 VDC): 350µA Std Power, 800µA High Power Battery Version (1.9 - 3.2 VDC): OSE-RBA Built-in Battery: -SP Version: 5-year battery life -HP Version: 3-year battery life Battery life time only valid for use at room temperature.

Field Wiring Cable Gauge 0.2 - 4mm² (26-12 AWG).

Alarm Threshold Levels: Low - Highest sensitivity / earliest alarm: 20% (0.97 dB) Medium - Medium sensitivity: 35% (1.87 dB) high - Lowest sensitivity / maximum resistance to nuisance smoke conditions: 50% (3.01 dB).

Adjustment Angle $\pm 60^\circ$ (horizontal) $\pm 15^\circ$ (vertical).

Maximum Misalignment Angle $\pm 2^\circ$.

Dimensions (WHD) Emitter / Imager: 208 mm x 136 mm x 96 mm (8.19 in. x 5.35 in. x 3.78 in.)

Operating Conditions* Temperature: -10 °C to 55 °C (14 °F to 131 °F)* Humidity: 10 to 95% RH (non-condensing).

Storage Conditions (non-operational) Humidity: Dry (> 95%)

Temperature range: 0 to 85 degree. Must not exposed to sunlight,

Protection Rating: IP44 for Electronics and IP 66 optical enclosure.

Historical Event Logs: 10,000 Events.

Dedicated LED indicators for Fire and Trouble events.

2.9 DISABLED REST ROOM ALARM SYSTEM

All disabled areas / facilities shall have audio / visual alarm system, using flasher sounder devices.
Provide disabled toilet alarm.

2.10 GRAPHIC COMMAND CENTER

A color graphic command center shall be installed in the FCC room. When a fire alarm status changes, color graphics shall display an indication of the type of alarm or other activity and its location. The operator then uses the mouse control to access a more detailed view of the alarmed zone or device. With proper access code, the operator shall be able to acknowledge alarm conditions, silence audible alarms and extinguish visible notification appliances, and perform system reset directly from the color graphic screens. The bit map images of floor plans should be loaded in each graphics page. The Graphic pages should clearly indicate the device location and type of device with address. The GUI should act as a SCADA- Supervisory Control and Data Acquisition system.

The Historical data logs of Fire alarm. Trouble alarm and panel healthy status data's should be retrieved through the GUI.

The graphic command center shall use compatible PC. The PC shall have an I5 processor operating at 4GHZ minimum, 8GB RAM, 500GB HDD and a 32" LCD monitor with graphics hardware acceleration. The PC and Peripherals shall be from manufacturer such as Dell, Compaq etc. Locally assembled equipment will not be accepted.

2.11 BMS INTEGRATION

Fire alarm system shall be integrated with BMS for status monitoring, fire & fault, trouble, supervisory through Open protocol Software integration.

2.12 WIRING

Wiring shall be in accordance with local, state and national codes (e.g., NEC Article 760) and as recommended by the manufacturer of the fire alarm system. Number and size of conductors shall be as recommended by the fire alarm system manufacturer, but not less than 18 AWG (1.02 mm) for Initiating Device Circuits and Signaling Line Circuits, and 14 AWG (1.63 mm) for Notification Appliance Circuits.

All wire and cable shall be listed and/or approved by a recognized testing agency for use with a protective signaling system.

Wire and cable not installed in conduit shall have a fire resistance rating suitable for the installation as indicated in NFPA 70 (e.g., FPLR).

Wiring used for the multiplex communication loop shall be twisted and shielded and installed in conduit unless specifically accepted by the fire alarm equipment manufacturer. The system shall permit use of IDC and NAC wiring in the same conduit with the communication loop.

All field wiring shall be completely supervised.

2.12.1 TERMINAL BOXES, JUNCTION BOXES AND CABINETS:

All boxes and cabinets shall be UL listed for their use and purpose.

The Fire Alarm Control Panel shall be connected to a separate dedicated branch circuit, maximum 20 amperes. This circuit shall be labeled at the Main Power Distribution Panel as FIRE ALARM. Fire Alarm Control Panel Primary Power wiring shall be 12 AWG. The Control Panel Cabinet shall be grounded securely to either a cold water pipe or grounding rod.

2.12.2 Batteries:

- A. Batteries shall be 12 volt, Gell-Cell type (two required).

- B. Batteries (two required) shall have sufficient capacity to power the fire alarm system for not less than twenty-four hours plus 30 minutes of alarm upon a normal AC power failure.
- C. The batteries are to be completely maintenance free. No liquids are required. Fluid level checks refilling, spills and leakage shall not be required.

3.0 EXECUTION

3.1 INSTALLATION

- A. Installation shall be in accordance with the NEC, NFPA 72, local and state codes, as shown on the drawings, and as recommended by the major equipment manufacturer.
- B. All conduit, junction boxes, conduit supports and hangers shall be concealed in finished areas and may be exposed in unfinished areas. Smoke detectors shall not be installed prior to the system programming and test period. If construction is ongoing during this period, measures shall be taken to protect smoke detectors from contamination and physical damage.
- C. All fire detection and alarm system devices, control panels and remote annunciators shall be flush mounted when located in finished areas and may be surface mounted when located in unfinished areas.

3.2 PRE COMMISSIONING

Provide the service of a competent, factory-trained engineer or technician authorized by the manufacturer of the fire alarm equipment to technically supervise and participate during all of the adjustments and tests for the system.

- a. Before energizing the cables and wires, check for correct connections and test for short circuits, ground faults, continuity, and insulation.
- b. Close each sprinkler system flow valve and verify proper supervisory alarm at the FACP.
- c. Verify activation of all flow switches.
- d. Open initiating device circuits and verify that the trouble signal actuates.
- e. Open and short signaling line circuits and verify that the trouble signal actuates.
- f. Open and short Notification Appliance Circuits and verify that trouble signal actuates.
- g. Ground all circuits and verify response of trouble signals.
- h. Check presence and audibility of tone at all alarm notification devices.
- i. Check installation, supervision, and operation of all addressable smoke detectors using the Walk Test.
- j. Each of the alarm conditions that the system is required to detect should be introduced on the system. Verify the proper receipt and the proper processing of the signal at the FACP and the correct activation of the control points.
- k. When the system is equipped with optional features, the manufacturer's manual should be consulted to determine the proper testing procedures. This is intended to address such items as verifying the controls performance by individually addressed or grouped devices, sensitivity monitoring, verification functionality and similar.

3.3 FINAL INSPECTION

At the final inspection, a manufacturer-trained representative shall demonstrate that the system functions properly in every respect.

3.4 INSTRUCTION

Provide instruction as required for operating the system. "Hands-on" demonstrations of the operation of all system components and the entire system including program changes and functions shall be provided. The Contractor and/or the Systems Manufacturer's representatives shall provide a typewritten "Sequence of Operation" to the Owner if required.

3.5 TESTING AND COMMISSIONING

After the installation is complete, the contractor will conduct operating and commissioning tests. The equipment will be demonstrated to operate in accordance with the requirements of the specification. The system installation, testing and commissioning will be as per Local approvals and requirements.

The fire alarm system will be completely programmed in accordance with Fire Department requirement and a specialist from the manufacturer will attend and demonstrate the complete system. A company trained representative will personally supervise the complete installation and final testing of the system.

All tests will be carried out in the presence of the Client or persons authorized by the consultant / client. Upon the completion of the acceptance tests, the representatives will instruct operatives in the proper operation, maintenance programming, configuration, and testing of the system. The vendor will provide equipment and /or software which is necessary to allow field modification of the programming and configuration.

3.5.1 Visual Checks

- A. Check installation of fire alarm system cables for proper laying, termination, clipping, labeling, and identification.
- B. Check cable connection to the F.A.C.P. battery, repeater panel.

3.5.2 Cable Continuity and resistance test

- A. Check cable continuity and resistance using a calibrated Multimeter.
- B. Tabulate readings.

3.5.3 Functional Checks**Fire Alarm:**

- A. Put the system into operation as per manufacturer's recommendation.
- B. Simulate fire conditions by generating smoke/heat to activate the detectors.
- C. Check display of detector number check operation of LED on the detector
Check display on network.
Check the printer operation.

Fault Alarm:

- A. Simulate fault situations like power failure, battery on, open circuit.
- B. Check visual display in the panel.

3.5.4 Interlock Check

- A. Simulate fire conditions in each zone.
- B. Check that the FAHU/AHU in that zone cuts out and all other interfacing equipments.
- C. Check the operation of fire roller shutters pressurization fans & extract fans.

3.5.5 Monitoring Features

- A. Activate the sprinklers/fire hose reel
- B. Check panel monitoring facility for fire pumps/sprinkler pump/zone control valves.

END OF SECTION

ELECTRICAL SECTION: 15

CENTRAL BATTERY EM / EXIT SYSTEM

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1.0 GENERAL**1.1 Description of Works:**

The Emergency lighting system and all its components shall be designed, supplied, installed, test, commissioned and maintained during the warrantee period of a system of fully addressable central battery emergency lighting systems complete with all components and central monitoring facility, meeting the requirements of UAE fire code, local civil defense requirements & the respective EN, BS standards and NFPA regulations applicable on the project. The product shall bear CE certification for electromagnetic compatibility. The manufacturer shall have ISO Certification.

Emergency lighting shall fulfill the following functions:

Illuminate the escape routes.

Indicate the escape route direction clearly

Provide EXIT signs on escape doors and routes to such doors.

Ensure fire alarm call points, firefighting equipment and other lifesaving equipment on the premises are illuminated.

Provide additional illumination to meet the requirements of the specifications, drawings and schedules.

Permit operations concerned with safety measures & to shut down all hazardous process.

Cross Zoning, Lux level and System as a whole as required and approved by Civil Defense.

Each emergency supply panels shall be used for supply, automatic testing & monitoring of maximum 20 Nos. Escape route, EXIT & Safety on each circuit. Panels should be distributed on the building based on Security Zone or Typical Areas etc., to minimize wiring. The product shall have local civil defense authority approvals. The Exit lights, downlights and EL shall be connected in same circuit using latest smart technology. The Central Battery Panel shall be according to project requirements.

Input Voltage: 230V – 400V AC

Output Mains: 230V AC

Output Emergency: 240V DC/AC

All products used in the emergency lighting system shall be manufactured by an ISO9001 certified company, complying with the relevant VDE/DIN/BS/EN standards & shall bear the CE certification for Electromagnetic compatibility. The product shall be field proven in the Middle East.

Battery backup required for Duration of Three Hours under emergency operation, with 100% light output on connected fittings.

The system shall be interfaced to the BMS for reporting the alarm and status functions of all system components.

1.2 Quality Control & Assurance

Design, manufacture, testing and method of installation of all apparatus and materials furnished under requirements of these specifications shall conform to latest publications or standard rules of the codes.

The Quality Control/Assurance/Assurance/Quality Assurance duties shall be performed by the Contractor. The system of Quality Control/Assurance/Assurance verification shall be in accordance with BS EN ISO 9001 standards of quality assurance, including in-house Engineering for product design activities.

All products used in the emergency lighting system shall be manufactured by an ISO 9001 certified company, complying with the relevant VDE/ DIN/ BS/ EN standards and shall bear the CE certification for Electro-Magnetic compatibility. The product shall be field proven for hot and humid climatic conditions.

Test reports/certificates by recognized test laboratories (such as DEKRA/KEMA, KITEMARK) shall be submitted to prove compliance with the specifications Comply with BS EN.

The provision of testing data or the carrying-out of tests does not relieve the Contractor of his responsibilities regarding the performance requirements, durability or service life requirements.

1.3 Code and Standards

The following codes and standards to the extent specified herein, form a part of this specification. When an edition date is not indicated for a Code or Standard the latest edition enforces at the time of proposal submitted shall apply.

The Emergency lighting system and all its components shall be designed & installed to meet the requirements of UAE FLS code, local civil defense requirements & the respective EN, BS standards and NFPA regulations applicable on the project and as listed below:

- A. BS 5266 - Emergency lighting
- B. BS 5499 - Graphical symbols and signs, Safety signs, including fire safety signs
- C. BS EN 13032 - Light and lighting. Measurement and presentation of photometric data of lamps and luminaires
- D. BS EN 50171 - Central power supply systems
- E. BS EN 50172 - Emergency Escape Lighting systems (VDE 0108-100 - German Version)
- F. BS EN 60529 - Degree of protection provided by enclosures
- G. BS EN 60598: Part 2-22 - Luminaires. Luminaires for emergency lighting
- H. BS EN 60896: Part 11 - Stationary lead acid batteries. General requirements and method of test (IEC 896-11)
- I. IEEE 485-2020 - Recommended Practice for Sizing Lead-Acid Batteries for Stationary Application
- J. BS EN 60929 - AC-supplied electronic ballasts for tubular fluorescent lamps. Performance requirements
- K. IEC / BS EN 61347 Lamp control gear
- L. IEC 60331-21 - Tests for Electric Cables under Fire Conditions - Circuit Integrity
- M. EN 61000 (3-2), (4-2) and (4-4) and EN 55015, 50081 - EMC compatibility
- N. DIN 4102-1 - Fire Testing to Building Material (German Standard)
- O. The Industry Committee for Emergency Lighting (UK), ICEL Standards: a) ICEL 1003 Emergency lighting design guide

The product shall bear CE certification for electromagnetic compatibility. The manufacturer shall have DCD Certification.

All requirements of the Authority Having Jurisdiction (AHJ) & latest UAE FLS Code.

The system shall have proper listing and/or approvals from Dubai Civil Defence Authority.

1.4 Submittals

The contractor shall submit the following drawings and documents related with Central Battery System for Engineer's approval:

- A. Original catalogues for various proposed components with
- B. The Civil Defense approved certificates for the products and the proposed model numbers of emergency and Exit luminaires and Central Battery control panels.
- C. Schedule of Accessories and Fittings.
- D. Battery size and calculation along with charger sizing calculations
- E. Test Certificates.
- F. Shop drawings, Method Statements for installation, Quality Plan
- G. Manuals (O+M)
- H. Guarantee / Warrantee.
- I. Maintenance
- J. Post Contract Expansions etc.

1.4.1. General

- 1. Two Original and Two copies of all submittals shall be submitted to the Architect/Engineer for review.
- 2. All references to manufacturer's model numbers and other pertinent information herein is intended to establish minimum standards of performance, function and quality.
- 3. For equipment other than that specified, the contractor shall supply proof that such substitute equipment equals or exceeds the features, functions, performance, and quality of the specified equipment.

1.4.2. Shop Drawings

- 1. Sufficient information, clearly presented, shall be included to determine compliance with drawings and specifications.
- 2. Include manufacturer's name(s), model numbers, ratings, power requirements, equipment layout, luminaires arrangement, complete wiring point-to-point diagrams, and conduit layouts.
- 3. Show luminaires layout, configurations, and terminations.

1.4.3. Manuals

- 1. Submit simultaneously with the shop drawings, complete operating and maintenance manual listing the manufacturer's name(s) including technical data sheets.
- 2. Wiring diagrams shall indicate internal wiring for each item of equipment and the interconnections between the items of equipment.
- 3. Provide a clear and concise description of operation that gives, in detail, the information required to properly operate the equipment and system.

1.4.4. Certifications

Together with the shop drawing submittal, submit a certification from the major equipment manufacturer indicating that the proposed supervisor of installation and the proposed performer of contract maintenance is an authorized representative of the major equipment manufacturer. Include names and addresses in the certification.

2.0 PRODUCTS

2.1 Central Battery Control

The CBS central power supply system must be state-of-the-art, reliable and easy-to-operate central battery system constructed in accordance with the requirements of BS EN standards. The system is capable of monitoring circuits, luminaires and configuration in mixed layout.

The CBS unit is equipped with a controller to supervise the operation of the entire system and archive any information on the occurred events and system condition. It has an intelligent charger to supervise the entire battery charge process and automatically stop the process in the event when battery is damaged.

Depending on the type of facility, it is possible to connect substations to diversify the central battery system functionalities, thus reducing the installation costs by shortening circuits with installed luminaires.

Damage to the central unit does not result in complete system failure, as substations take control of final circuits and luminaires.

The SD card used in the system allows saving the periodic test results, event log and system configuration. The above-mentioned information is stored in non-volatile memory of the controller.

The battery housing shall be sheet steel, powder coated housing IP20- IP40 with removable screwed front, rear & top doors to provide complete access to the batteries. The electronics section shall have IP20-IP40 sheet steel enclosure with transparent polycarbonate window for all central systems IP20-40 sheet steel enclosure with full metal door for all substations. Design and fabricate internal supports for assemblies, subassemblies, components, supports and fastenings for batteries to withstand static forces in accordance with the Authority Having Jurisdiction (AHJ).

The electronic compartment shall have large cabling compartments with cable entries from top via four or more undrilled removable metal flange plates. All incoming cables & looping circuits shall be connected on protected & fused terminals as per EN standards. All outgoing luminaire circuits shall be connected direct to relevant components via plug-in type terminals 2.5 sq. mm. The electronic components shall have 19" slide in technology or as per manufacturers design for installation and maintenance.

- a. Modular design for quick assembly
- b. Freely-programmable operating modes for each circuit (circuit monitoring)
- c. Freely-programmable operating mode for each luminaire, irrespective of the circuit settings
- d. Monitoring of each luminaire and circuits
- e. Controlling addressable luminaires
- f. Monitoring up to 20 luminaires per circuit
- g. Freely programmable work mode
- h. Independent control of each circuit
- i. Independent control of each luminaire
- j. Identification of each luminaire
- k. Fault, module status and circuit status indicators
- l. Supplying luminaires with ballasts and luminaires with LED light sources.
- m. Adjustable AC/DC changeover time.

The technology should use maintained, non-maintained and switchable, within a single circuit. Programming as well as control of luminaires takes place through power supply cables, therefore no additional communication cabling is required.

Access to subassemblies shall be front only. Doors shall be removable. All components shall have a modular design and a quick disconnect means to facilitate maintenance and parts replacement.

Provisions for cable access shall be at the sides, top and bottom.

The enclosure shall be protected against corrosion by suitable treatment, such as zinc Electroplating, power coating, epoxy paint or an equivalent. The enclosure(s) interior finish color shall be off-white.

2.1.1 Sub Station Panels

The substations shall be installed in areas with a maximum room temperature of 30 deg. C. The substations shall be located based on the requirements of the lighting loads and optimizing the field wiring. Substation panel shall be connected and networked through the data bus from the central panel located. It shall be possible to connect up to 100 substation panel in the network and any panel failure should not affect the function of the other panels in the network. Quantity, type and size of output breakers called for on the Construction Drawings.

2.2 Control Module

The control module is used to supervise and manage the operation of other modules included in the central battery. The keyboard and LCD display in front panel allow the user to configure and operate the whole system. System configuration may also be carried out using USB interface from PC with service application. Front panel LEDs allow the operating state of the central battery to be evaluated quickly. The module automatically supervises the following: mode of operation – mains/battery power supply, battery charging, system voltage, condition of insulation, protection against deep discharge. Detection of a failure or error is signaled at once and recorded in the event log. The occurrence of short-circuit or break in communication cables results in automatic switch of all circuits into mains power supply. In addition, the module allows automatic searching for and adding all the luminaires connected to the system. The controller allows upgrading firmware of any internal system modules as well as address modules.

The control module should have the following features:

Monitor & control all test cycles & functions

Conduct automatic / manual Function and Battery Duration test

Display & program every status of the system & the connected luminaries

Conduct an Earth Leakage test on each circuit.

Luminaries' failure will be displayed with text location

User programmable buttons for easy operation.

It also displays battery voltage level, withdrawn battery current, interrupted battery charge circuit, luminaire failure with address with text location. The control module has 3 LEDs for status indication of main supply ON/OFF, luminaire or unit failure and battery supply. Each battery panel shall be fitted with mode to facilitate communication to the BMS system.

Conformity with standards BS EN and VDE.

TFT-Touch-Display
Intuitive, single button operating
Password protected programming
Integrated network connection interface.
USB-interface to

- Load/save the system configuration.
- Load/save the circuit configuration (operation mode, monitoring depth, destination texts.)
- connect a standard printer or key-board.

Administration of up to 200 (128 minimum) number of circuits (max. 4000 luminaires - 20 addresses per circuit).

Automatic function monitoring of system including but not limited to single luminaire- and circuit failures without additional data-line, battery, etc.

Integrated logbook for central compilation and > 2-year storage of all system related data
Circuit-/luminaire function modes free programmable.

Connections for phase monitor, blocking switch, potential free contacts
Various languages available.

2.3 Changeover Modules

The system shall be built up in a modular format with changeover modules. The outgoing circuits shall be fed through programmable changeover modules and DIN rail mounted MCBs. Emergency & Exit lights will be connected to each circuit, thereby allowing connection of luminaires of all switching modes, namely Maintained, Non-maintained and Switched Maintained in the same circuit. All outgoing circuits shall be no load and short circuit protected with independent fuses & MCBs for mains and battery supply and will have LED status indication for operation & failure. Each changeover module communicates with the control module through the internal data bus. All outgoing circuit shall be protected with independent fuses. The unit shall not have a floating DC circuit and shall not require separate changeover relays for both L/DC (+) and N/DC (-).

Each panel shall have a spare fuse carrier with at least three spare fuses for each size used in the panel. It shall be possible to turn on individually or a group of luminaire from the Central Battery system based on the input given to the system.

The Changeover Modules/ Inverter modules shall be accessible from the front of the panel by opening a hinged door. (Fixed output type modules fixed to the back plate of the panels are not acceptable as they prevent future expansion/ addition of circuits/modules at site)

Hot swap facility to enable replacement of Changeover module/ inverter module shall be available to avoid the need for de-energising the panel during faulty module replacements.

Automatic changeover to Emergency operation shall be available in the event of any failure of the control module in the panel. This is for fail-safe mode.

2.4 Battery

Central Battery panel will have maintenance free sealed lead acid batteries with 10 years' service life at 25 °C and capable of supporting the connected load for 3 hours. The room shall be without any special battery room ventilation requirement. Where the space required for the batteries does not permit the installation of same in standard enclosure, the batteries shall be installed in a fire rated room, on purpose built support stand / enclosure.

Battery catalogues and calculation details should be submitted as part of Submittal. Battery calculation for SLA batteries for storage application should comply IEEE standards.

Batteries shall be low gas generation, low self-discharge and have permanently sealed pressure valve.

Battery sets shall be sized with 100% light output from each emergency and exit lights for 3 hours. Battery sets shall be provided with 15% spare where separate Spare requirement is not added in the load/DB schedules. In addition, additional 25% shall be considered as ageing factor for 10 years period.

Batteries shall be housed in a common cabinet with the CBS system. Cabinets shall have provisions for draw-out removal of battery assembly to allow for inspection and testing. Locally procured battery cabinets are not permitted.

Batteries shall also be provided with Battery Management System which is internally connected with own controller and directly interface with central battery panel. The own controller shall be accessed through the LAN network and monitoring /control with individual battery block status, such as voltage, current, temperature and log book entries. This also ensures that when individual block temperature exceeds safe limits, the charging function can be automatically switched off to prevent battery damage/explosion.

The battery assembly shall be provided with a battery disconnect switch/circuit breaker.

Batteries shall be arranged in strings such that individual strings can be removed/ replaced without disconnecting the other strings from the systems or rendering the whole emergency lighting system inoperative:

The number of batteries in each string shall be such that their nominal voltage sums to the required output voltage.

All Batteries shall be provided in a dedicated cabinet not part of the main or sub panel. Only central battery system is only acceptable. Distributed batteries within the sub panels are not acceptable.

The number of battery strings shall be agreed with the Engineers and should be as per the connected load and the rated hours of operation. The system is expected to include:

Battery Monitoring shall be provided such that individual batteries in each string are monitored to display the Voltage and temperature of each block. The display shall appear on the own controller and shall be logged to monitor effective battery maintenance.

The control module shall register:

Warnings whenever the battery conditions are not optimal

Alarms when the batteries need replacing

Shut Down Charger of the Central Battery panel where temperature of an individual battery block is noted to be excessively high.

Additionally, the control unit shall be through software interface with the BMS such that each alarm condition is displayed at the BMS console.

2.4.1 Charger

The charger module ensures battery charging based on UI characteristics with temperature compensation according to BS-EN. The charging algorithm executed by the charger is supervised by the control module. The charger is equipped with internal active PFC module, which provides power factor close to one ($\lambda \approx 1$). The charger is used for charging batteries with rated voltage in the range of 120V - 216V. The maximum power of charger is 980W. If there is a need to charge a battery with higher capacity, the BST-980 charge boosters are used.

Charger module shall be boost charge type, microprocessor controlled module with temperature controlled voltage charging to IU characteristics. It shall have charging & supervising method via battery monitoring for failure indication of battery circuit as well as battery charger. The charger shall be designed for a 80% recharge of a fully discharged battery set within 12 hours of commencement of recharge

For all available 216V lead-acid accumulators the charger should be 230V, 7.5A - 10.0A. The Charger should be modern microprocessor controlled design with high efficiency and low weight. Battery charge and discharge current information is permanently available at the controller. The Constant charger and battery circuit monitoring with fault display at the controller and the following cases the indication should be displayed in controller 1. Blown battery 2. Fuse missing or defect battery block - faulty charger, 3. Open battery circuit. 4. Battery voltage to high / low 5. Temperature controlled charging to ensure longer battery life.

The battery chargers must have Earth Leakage Monitoring function, with pushbuttons (or menu options in the Control Module) for manual simulation of the battery circuit in accordance to DIN VDE with fault indicated on control module in Text. It must also have Floating contacts for charging malfunction, boost charging and isolation fault; alternate activation of charging modules at trickle charge to optimize the service life of the charger(s); constant charge & battery monitoring facilities at intervals of 5minutes or less; LED indications for charger operation and charger failure and must indicate remaining battery capacity in the form of graphic display on the control module.

Shall have fuse protection for Mains and Battery Supply on the front face of the Charger module. Potential free contacts shall be provided in the panel to communicate Charger failure including Insulation failure. Shall have adequate ventilation facility to ensure safe operation. When more than one charger is utilized, the chargers shall operate independent of each other such that any failure of one charger does not affect the operation of the other charger.

2.4.2 Automatic Cyclic Function and Logging Facility

An automatic monitoring and testing facility shall be installed with permanent cyclic charger and battery monitor at intervals of five minutes or less with daily/weekly function test of the complete load supplied from the battery, charger, wiring, luminaires, registration and logging of tests and results. The panel should also have the facility to conduct automatic battery duration test once in a year.

2.5 Sub Circuit Monitoring

Phase monitoring operation of emergency lights.

Under emergency operation, the connected emergency light shall be switched on with 100% light output. monitoring modules DIN rail mounted inside the relevant lighting DB's. The Bus module shall also enable display The monitoring of relevant lighting DB's shall be done using intelligent Bus driven phase of exact text location failure of the monitored lighting DB at the Control module in the panel.

Dedicated Emergency lights shall be in Non-Maintained mode of operation while Exit lights shall be in Maintained mode of operation.

The monitoring of Switched Maintained, Non-Maintained and Maintained lights shall be carried out in such a way that the battery backup is utilized only under total mains failure in the building. Relevant outgoing circuits from the central battery panel/Substations/ELDB's shall be energized with AC Mains input from the central battery panel, upon power failures detected in respective local lighting DB's.

Final non maintained / switched individual lighting circuits shall be monitored for operation & when failure is detected in supply voltage. The circuit shall be automatically be transferred from normal supply to emergency supply with 100% light output on the luminaires. It shall be capable to monitor 8 sensing inputs including phase sensing.

2.5.1 Light Switch and Sensing Module

For mains operation common switching of general lighting & safety lighting by means of existing light switches a compact fully encapsulated module shall be provided. In built timer facility should be provided for controlling stairwell lights via push button. The module shall be suitable for third party luminaire.

2.5.2 Local Controllers

Local controllers should be connected to every circuit to monitor local normal lighting DBs and switch on the non-maintained luminaires during DB supply failure. There shall be no separate data cabling required between the local controllers, luminaires and panels. All necessary data transfer shall take place through the standard power supply cabling of the luminaires. Continuous monitoring of the lighting sub distribution panels shall be made by Local Controller units, each having 3-phase monitoring input. Each circuit shall have its own Local Controller so that a possible failure of a single controller does not affect the other circuits. The non-maintained luminaires at sub distribution panel area shall be lit during a local supply failure. However, the emergency supply panel shall not switch to battery use as long as its own mains supply is available. The Local Controllers should comply to BS EN.

2.5.3 Switch Controllers

Switch controller modules shall be provided to allow emergency luminaires to be used for general lighting purpose with ON & OFF switch operations via switches, lighting control system, etc. where necessary switch controllers shall be fully compatible with the Local Controller.

External switch module supplied from the central battery system (CBS). It is used for switching single luminaires or a group of luminaires connected to outputs. If a phase loss sensor and a 216V DC emergency lighting switch is triggered, the module always activates both outputs. In other circumstances, the activation of those outputs depends on the manually set up operating modes. In the case of a 230V AC power supply.

2.5.4 An Audible Alarm

An audible alarm sounds should a fault condition be found during a test. The alarm will sound for 3 seconds every 30 minutes until the fault condition is rectified and the unit reset.

2.6 LUMINAIRE DATA

2.6.1 General

Unless otherwise stated all EXIT luminaires shall be operated via a Central Emergency Battery System (ECBS).

All Exit luminaires shall have pictogram legends as per DIN 4844/CEN TC 169, EN or markings as per local civil defense requirements with viewing distance of 22 - 32 meters. All Exit luminaire and electronic ballast shall be built and comply according to EN standards..

Safety luminaires shall be operated by Central Battery System. This shall be achieved by using high frequency electronic ballasts operating at full load or by the use of compact fluorescent lamp down lights. The ballast in compact fluorescent lamps shall be part of normal lighting designated as emergency lighting.

All Exit and safety luminaires shall have a monitoring component with easy accessible addressing module. All luminaires with high frequency ballasts shall have a built in or separate monitoring facility. Outside of every installed exit and safety luminaire a label (Red / White) with the circuit and luminaire reference number shall be installed.

All Emergency luminaries shall be non-maintained type meet following requirements:

Supply voltage 230V AC / DC

The luminaire, electronic ballast and EMC or EMI protection shall comply to the requirements of EN standards.

2.6.2 Retail Area Emergency and Evacuation Lighting

The Central monitored self-contained evacuation lighting system shall be installed in small and medium-sized retail areas in the building. It determines the condition of fittings by performing function and autonomy tests, which are configured by the user through a clear display and alphanumeric keyboard.

The system should be centrally monitored by the main central battery monitoring system through a dedicated monitoring module for panel status.

All luminaires in retail areas should be addressable with built in battery or self-contained system. The battery should be Nickel cadmium batteries with five years' life time.

2.6.3 EM luminaires - FOH / ID / False Ceiling areas

3W / 5W / 8W / 10W LED luminaires, or equivalent to satisfy the DCD lux level requirement, fully recessed compact Circular or Square Emergency luminaires with a diameter of 40mm to 110mm. Housing material shall be sheet metal / polycarbonate with white trim with electronic ballast and addressable module with built in switch sensing. The protection rating should be IP20. Wet areas shall be IP 55 fixtures. FOH luminaires must be similar finish of adjacent fixture and must be approved by ID consultant and Architect.

2.6.4. EM luminaires - False Ceiling Wet areas

3W / 5W / 8W / 10W LED luminaires, or equivalent to satisfy the DCD lux level requirement, fully / semi recessed Circular / rectangular / square Emergency luminaires. Housing material shall be sheet metal or polycarbonate with white trim with electronic ballast and communication module. The protection rating should be IP55. Wet areas shall be IP 55 fixtures.

2.6.5 EM luminaires - Electrical Rooms, Plant Rooms, Staircase and BOH areas etc.

5W/6W/8W LED or equivalent to satisfy the DCD lux level requirement. The luminaires should have built in addressable low loss electronic ballast which can communicate with central control module. Luminaire housing shall be sheet metal or polycarbonate. Luminaire shall be suitable for wall / ceiling mounting, with rectangular shape and rated to IP 54.

2.6.6 EM luminaires for Substation.

5W/6W/8W LED Non Maintained self-contained emergency luminaire or equivalent to satisfy the DCD lux level requirement. The luminaires should have built in Ni-cd battery backup for 3hrs duration as specified by UAE FLS code. The battery life time should not be less than 4 years. Luminaire housing shall be sheet metal or polycarbonate. Luminaire shall be suitable for wall / ceiling mounting, with rectangular shape and rated to IP 54.

2.6.7 Exit luminaires –Lobbies and Corridors

2W/3W/4W Maintained LED, or equivalent to satisfy the DCD lux level requirement, decorative type, glass finish, ceiling recessed mounted luminaire extremely flat housing with Rectangular slim design. Exit luminaire shall have 'Running Man' pictogram for edge-lit type fittings and for ceiling mounted fittings at the Lobby areas. Wall mounted Exit luminaires shall be compact decorative internally illuminated or edge lit type as per site and ID requirements. Legends should be manufacturer laser printed with uniformity in the print without patches as required by the local authorities. The protection rate should be IP20 for FOH areas and luminaire housing shall be sheet metal or poly carbonate housing.

2.6.8 Exit luminaires Car Parking / Driveways

2W/3W/4W Maintained LED or equivalent to satisfy the DCD lux level requirement. The luminaires should have built in addressable low loss electronic ballast which can communicate with central control module. Luminaire housing shall be sheet metal or poly carbonate housing. Luminaire shall be suitable for wall / ceiling mounting/pendulum mounting, with rectangular shape and rated to IP 55.

2.6.9 Exit luminaires for Substation.

2W/3W/4W Maintained LED self-contained emergency luminaire or equivalent to satisfy the DCD's lux level requirement. The viewing distance should be as per the specified international standards and codes BS and BE EN, the UAE FLS codes. The pictogram in each luminaire must be factory printed type & any pictogram stickers are not acceptable.

The luminaires should have built in Ni-Cd battery backup for 3hrs duration as specified by UAE FLS code. The battery life time should not be less than 4 years. Luminaire housing shall be sheet metal or polycarbonate. Luminaire shall be suitable for wall / ceiling mounting, with rectangular shape and rated to IP 54.

2.7 Central Monitoring PC

A central monitoring PC shall be provided unless specifically noted otherwise in schedules and drawings.

The PC shall be provided with a monitor, latest Intel generation processor, keyboard, mouse and a printer. The unit shall operate with latest visualization software to give detailed addresses & information of all luminaires and circuits connected to system. The program shall be user friendly with visually clear fault indication screens on the monitor.

The Central Computer should have the following features:

1. Monitor & control all test cycles & functions
2. Indicate every status of the system & the connected luminaries
3. Display the status of each circuit
4. Luminaire status & failure information will be displayed with text location
5. Up to 15 min' Delay on Emergency lighting on mains return.
6. Timer facility for Stairwell Lighting
7. Logbook function: log the events of the last 4 years on the system in a separate window to facilitate easy access.
8. Program each individual luminaires from the PC to maintained; non-maintained and switched maintained mode.
9. Each luminaires shall be programmed with its own name (20 characters)
10. Each luminaires shall have additional text information of another 100 characters

Entire data from each individual panel, including end luminaire status with text location, shall be available at the central monitoring PC.

The central monitoring Software shall permit complete system status display when connected to the central monitoring network. It shall be possible to conduct Function & Duration tests, access log books as well as connect to more than one hundred Central Battery panels in the network. The software used for central monitoring shall be non-proprietary so that any future upgrades or maintenance or change in operating system of the central monitoring PC will not require extensive re-installation or purchase of this software from the emergency lighting manufacturer.

It shall be possible to connect any Building Automation or Management System in the same Central monitoring network so that the information available on the dedicated central monitoring PC of the Emergency lighting system is available also at the Building Automation PC. The protocol to enable this shall be Modbus over IP/LON works and shall be open protocol. The BMS PC as well as the Central Monitoring PC shall be working in parallel so that failure of one shall still enable the same data available at the other.

2.8 Bus Technology

Bi-directional data bus integrated in the control module shall be used for data communication between the Central Battery System and connected substations and monitoring devices or the visualization software.

Each of the Central Battery panels shall be capable of connecting directly to the local network. It shall be possible to connect up to 150 panels in the same network if required and a suitable network is provided locally. All Central Battery panels shall have the facility to transfer complete data to the Central monitoring PC.

The System Data information should be seamlessly provided to the BMS from the PC such that both PC and the BMS can provide the same fault reporting. Since both BMS & Central Battery systems will be communicating to the Central Battery or Substation panels it is essential that conflicting messages are avoided in the above configuration.

2.9 Equipment Cabinets

Emergency lighting equipment, excluding Luminaires and Central monitoring PC's which are detailed elsewhere in this section, shall be contained in cabinets. Standard cabinets may be used where the equipment is to be located in 2-hour Fire rated rooms containing only lighting and power supplies to the equipment. Fire Rated Cabinets shall be used in all other circumstances. All equipment cabinets doors shall be fitted with standard half cylinder locks. The lock/key suiting shall be agreed with the Engineers but is expected to include different keys such as the central battery and control cabinets.

Floor standing cubicles shall incorporate a plinth and shall be designed to enable movement by pallet truck. Cooling Fans shall be provided as required to ensure correct operation at the ambient temperatures. Fan loads to be added to the duty required to maintain the emergency lighting for the specified period.

Unless noted otherwise in the particular specification schedules and drawings, the audible noise < 55dB at 1m distance at operating temperature 0 to 50°C; relative humidity 0 to 95% non-condensing; operating height above sea level is 1000m.

The standard cabinets shall be a sheet steel, powder coated enclosures and meeting the requirements to IP21 in accordance with IEC/ BS EN. They shall be complete with hinged lockable door(s). Battery sections shall have Front, rear & top doors to provide complete access to the batteries. Electronic Sections shall have front lockable doors. The electronics shall be a Safety Class I sheet steel enclosure with transparent polycarbonate window(s) in hinged doors and lock.

All modules (Control, charging, Switch-over etc) mounted in the cabinets must be plug-in type and must be fully encapsulated for extra protection. NON-encapsulated modules will not be acceptable.

The electronic compartment shall have large cabling compartments with cable entries from top via undrilled removable metal flange plates. The flange plates must be removable so that drilling can be done externally to avoid metal parts falling into the cabinet and its electronic parts. All incoming and outgoing cabling shall be connected to a terminal strip placed at the top of the enclosure. The final end circuits are factory wired for connection to triple deck installation terminals with spring connection, including a disconnecting Neutral facility. These terminals must be suitable for the cable sizes/ types indicated in the particular drawings specifications and schedules and for a minimum of 2.5 mm² cable diameter.

The enclosure shall be of robust sheet steel outer construction, lined with thermally insulating material, have a separate insulated wiring chamber through which the connecting cabling is installed and have a door and back box frames that overlap to increase fire resistance and secure locking mechanisms that effectively wedge the door onto its frame.

2.10 High Frequency Control Gear

All safety and exit luminaires with their corresponding electronic ballasts and EMC protection shall be built according to BS EN. The electronic ballast with monitoring facility and easy accessible address module shall have an automatic cut-out fuse for failures within the lamp circuit.

All electronic ballasts for the safety luminaries supplied by others and acceptable to civil defense, but connected to Central battery system must comply with above clauses.

The self-testing modules shall be compatible with all known high frequency ballasts, utilizing 4 pole change over relays, and incorporating ballast disconnection to ensure that ballasts are powered down during tests.

This will enable them to be reset as if they had been switched off. Self-test luminaries should be left powered off until the installation is finally commissioned.

2.11 Integration

2.11.1 BMS Integration

All central battery panels shall transfer complete data to the building management system using local network protocol integration for panel healthy status, fault status, luminaire failure status, battery failure status etc.

2.11.2 Fire Detection System Integration

All central battery panels should be integrated with Fire Detection panel through programmable relay contacts. The interface should be such a way that all emergency lights near to critical Fire Alarm initiating devices and Fire Fighting should be switched on independent of mains failure in an emergency evacuation scenario.

3.0 EXECUTION

Factory Testing

Factory tests shall be performed on the complete Central Battery system, including battery, before shipment.

Include the following tests:

Functional test and demonstration of functions, controls, indicators, sensors and protective devices.

- Full load test.

- Transient load response test.

- Overload test.

- Power failure test.

Provide fourteen (14) days written notice of factory testing to allow the Client representative to witness testing.

Factory test results to include the following data:

- Description of input source and output loads to be used. Describe actions required to simulate source load variation and various operating conditions and malfunctions.

- List of indications, parameter values and system responses considered satisfactory for each test action.

- Include tabulation of actual observations during testing.

- List of instruments and equipment required for performing a duplication of factor.

3.1 Central Battery System Self-Testing

The self-testing shall routinely test all the fundamental aspects of the emergency lighting luminaire, reporting status through the LED. The unit shall comply with the latest International Standard for Emergency Lighting Testing systems, and exceed its testing requirements.

A continuous green LED indication specified for Auto Test emergency modules shall confirm the following:

- Main present and connected.
- Battery present and in circuit.
- Satisfactory operation of the battery charger.
- A flashing red LED indicates a faulty lamp.

The self-testing shall go into self-test mode automatically at a suitable time, the test schedule commences within a period of 37 days of commissioning. This is programmed into the modules during manufacturing to ensure that no more than 15% of luminaires in an installation are on test in any one day. After this period, tests will be carried out on a 30 days' time frame as required, at a period of low risk, normally when the switched supply is off. If a period of low risk does not arise within 7 days of the required test, then the test will be carried out immediately. A longer duration test will be carried out every 6th and 12th test.

Three durations of test are provided. Monthly the unit is activated for 5 minutes and the output to the lamp tested at the start and finish, 6 monthly the test will be conducted for one hour, and annually the test will be conducted for the full duration of 3 hours.

3.2 User Reset Facility

To reset the LED, after the fault condition has been rectified, the user simply switches the switched or unswitched supply ON / OFF twice within 10 seconds.

3.3 Test Report Status

After each monthly test, the memory status is incremented by one. The current status is displayed by the extinguishing of the LED each time the mains is turned on. After a delay of 30 seconds the LED extinguishes (in groups of three to facilitate counting), to indicate the number of tests carried out, and reset after each annual test.

In this way, user can be assured that essential monthly, 6 monthly and annual test have been carried out, and complete any supporting records that they may require. For example, 7 extinguishes at switch on followed by a steady LED indicates 6 previous monthly tests have been executed, and the unit is fully operational.

3.4 Installation

Maintain minimum working clearance at equipment according to manufacturer's written instructions.

Install interconnections between system components and luminaires. Make connections to supply and load circuits in accordance with the manufacturer's wiring diagrams, unless noted otherwise on the Drawings.

Identify each battery cell individually.

On completion of installation, inspect system components. Remove paint splatters, other spots, dirt and debris. Repair scratches and mars to match original finish. Clean components internally using methods and materials recommended by the manufacturer.

3.5 Wiring

Wiring, both power and control wiring between the central battery, substation panels and luminaires shall be carried out in 120-minutes fire resistant cable conforming to IEC test and as per Civil Defense authority requirements.

3 core fire resistant cables with earth wire shall be provided to feed substations from central battery panel if the batteries are provided in main panel. Cable size selection details must be provided based on the electrical load of each substation.

Wiring pattern shall be followed to ensure 50% redundancy on circuit failure as required by Local Authorities.

Maintained and Non-Maintained lights shall be connected in same circuit as shown on drawings.

Methods of cable support and fixings (for both Power and Control cables) shall be noncombustible and such that circuit integrity will not be reduced below that afforded by the cable used, and shall withstand a similar temperature, duration and water application to that of the cable, while maintaining adequate support. Fixings and clipping distances shall be in accordance with the manufacturer's recommendations.

3.6 Field Testing and Commissioning

Site Support, Inspection and Testing: The manufacturer shall include in his bid the provision of assistance by the Technical Representative. The assistance shall include, but not be limited to, supervision of unloading, setting, support, installation, and inspection of the Central Battery system. After inspection the Technical Representative shall test the equipment in accordance with an approved site test procedure. Tests shall be made in the presence of an Electrical / AHJ's designated representative. The application or interruption of power shall be programmed and directed in accordance with the approved EVAP, inclusive of the Equipment Energization Plan and necessary permits, work tasks and safety compliance steps.

The Project Manager reserves the right to require that the Subcontractor perform and repeat tests that are deemed necessary to complete or check the tests or the certified records of the Subcontractor at any time during the course of the work. The Subcontractor shall correct unsatisfactory portion of his work that is revealed by the tests or that may be due to progressive deterioration during this period, unless the item in question was a direct specification.

Factory-authorized service representative shall pretest system functions, operations and protective devices and perform tests listed below. Use instruments that have been calibrated according to Dubai Central Laboratory / NIST standards within six (6) months previous to field testing.

Simulate malfunctions to verify protective device operation.

Test duration of supply on emergency, low battery voltage shutdown and transfers and restoration due to normal source failure.

Test output voltage under specified transient load conditions.

Test remote status and alarm panel functions.

Correct deficiencies and retest until specified requirements are met.

Perform an equalization charge on the battery cells according to manufacturer's written instructions. Record individual cell voltages.

3.7 Identification

Provide an identification nameplate for each main panels.

Provide a typewritten directory card indicating load served by type and location for each branch circuit.

Mount directory in frame on outside of equipment door.

Current ratings of overcurrent protection devices shall be displayed on the device.

3.8 Acceptance and Energization

Final acceptance shall depend upon the satisfactory test results as performed in accordance with the manufacturer's instructions. After tests have been reviewed and approved by the Consultant/Client, energization may proceed in accordance with the approved Equipment Energization Plan and the Energization Validation and Authorization Package (EVAP).

END OF SECTION

ELECTRICAL SECTION: 16

AUDIO INTERCOM SYSTEM

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1.0 GENERAL**1.1 REFERENCE**

Conform to General Requirements for Electrical Services

1.2 DESCRIPTION OF WORK

Supply, Installation, testing, and commissioning of complete audio intercom system as specified hereafter and as indicated in the related electrical system drawings.

The contractor shall supply, install, test and commission an audio entry phone system as specified hereunder and as shown in the drawing. The system shall enable a visitor to call the apartment and shall have two-way communication possible. It shall be possible to release the entrance door electronic lock from the apartment. Contractor shall consider Electronic latch for the main entrance, necessary conduits, power supply units, Audio decoders.

The system shall consist of an intercom panel, digital dialing keypad, and microphone and loudspeaker unit with detachable handset with coiled cord. Necessary door control of building entrance doors shall be possible from the “release” button in each indoor unit.

The Audio Entry System shall be suitable for voice communications of the entrances to the building. The communication shall be full duplex between the handset inside the apartment with the Panel at the entrance of the building.

Calls to an indoor station inside an apartment shall be possible by dialling the corresponding flat number on the tower Intercom Panel. The system shall block out calls to other apartments while a call is in progress.

Unlocking of the Tower entrance door wherein the Intercom System is provided, shall be possible by pressing the door lock button on the Monitor installed in each apartment. This shall be possible only when the communication channel is active between the Indoor unit & the Entrance Panel. The door locks shall also be opened by entering a secret code on the keypad of the entrance Panel.

Guard units shall be provided in the security room / reception of the building. It shall be possible to program the Guard unit to ensure that all calls made from the entrance Panel are first intercepted by Guard/Concierge before transferring them to the respective Monitors inside the apartments. When the system is in standby mode, by pressing the door lock button on the Monitor, shall establish an audio communication with the Guard unit.

The system shall be capable to communicate via the NFC system configurator application ran on an Android device with NFC installed. The system shall be configured through mobile application and if the call is un answered through the indoor call station, the system has the possibility transfer the call to mobile devices.

1.3 QUALITY CONTROL/ASSURANCE

Design, manufacture, testing and method of installation of all apparatus and materials furnished under requirements of these specifications shall conform to latest publications or standard rules of the following:

- A. Dubai Water and Electricity Authority (DEWA) regulations for electrical installation works and Department of Civil Defense requirements.
- B. Regulations for the Electrical Equipment of Buildings (IET) & (BS).
- C. All products should be designed and manufactured to allow the designer and installer to comply in all respects with the latest Edition of the IET Wiring Regulations.

The Quality Control/Assurance/Assurance/Quality Assurance duties shall be performed by the Contractor. The system of Quality Control/Assurance/Assurance verification shall be in accordance with ISO 9000 standards of quality assurance.

1.4 CODES AND STANDARDS

- A. Dubai Water and Electricity Authority (DEWA) regulations for electrical installation works and Department of Civil Defense requirements.
- B. Regulations for the Electrical Equipment of Buildings (IET), (BS) London.

1.5 SUBMITTALS

The contractor shall submit the following drawings and documents related with each electrical system for Engineer's approval:

- A. Original catalogues for various proposed components.
- B. Schedule of Accessories and Fittings.
- C. Test Certificates.

2.0 PRODUCTS

2.1 OUTDOOR PANEL

The digital audio intercom outdoor panel shall be wall-mounted type with a keypad, microphone, and loudspeaker unit.

2.2 INDOOR UNIT

2.3 AUDIO TERMINAL

The Audio Terminal Installed in homes to enable communication with the street-level unit and allow the door to be opened.

The indoor unit is a compact and slim door entry that works in hands-free communication with the outdoor panel. Configurable in Push-to-talk mode if needed for very noisy surroundings.

The integrated digital audio technology (2 non-polarized wires) ensures a high quality, hands-free communication without interferences.

Equipment manufactured in ABS plastic with ultraviolet protection and elegant matt finish for easy cleaning. Surface installation.

Features

The following features should be added in the Audio terminal.

Intercommunication with:

- Outdoors panels.
- Call reception and communication with Reception and Security Terminal.
- Auto ON. You can pick up and listen to what is happening outside without having been called.

- Melody Selection.
- Recall.
- The last device in conversation can be recalled.

2.4 CONCIERGE PANEL

- The Guard unit shall form the communication link between the indoor unit inside the apartment and the Intercom Panel at the entrances. The unit shall have the provision for mounting on the desktop by using the desktop support included with the kit.
- All the events shall be displayed in the Guard unit using a 4inch display.
- The Guard unit shall be provided with the following terminals connections:
 - Call Queue Control Keys (Arrows keys)
 - Call button (Bell): To call Monitors
 - Dialling Keypad: shall be located beneath the phone handset of the Guard unit. The keypad shall include alpha numeric buttons in order to enter the house phone number, panel number and other data as required.
 - Guard unit → Apartment handset communication channel selection button
 - Main Panel → Apartment handset communication channel selection button
 - Secondary Panel → Apartment handset communication channel selection button
 - Guard unit → Main Panel communication channel selection button
 - Guard unit → Secondary Panel communication channel selection button
 - Door Lock button: When the communication is established with the main entrance panel, pressing the door lock button shall activate the electric latch connected to this panel.
- It shall be possible to program the Guard unit to ensure that all calls made from the entrance Panels are first intercepted by Guard/concierge before transferring them to the respective Monitors in the dwellings.
- The unit shall be programmable in three modes Day/Night/Automatic depending on the requirement.
- Each Guard unit shall have the facility of queuing up to 20 unattended calls.

All functions on the Guard unit shall be battery backed up to prevent loss of data in case of accidental power loss.

2.5 DOOR RELEASE BUTTON

When a call is received from the outdoor panel, it is possible to open the door at any time by pressing lock release button, even when the phone is ringing. Door release “Automatic opening” mode (DOORMATIC) With Doormatic activated, when a call is received from the outdoor panel, lock release will be activated after 5 seconds. You will have 30 seconds to answer the call. The call melody will continue to sound for 30 seconds, if continuous mode is selected, or if the call has not already been cancelled. The door release function is disabled by default. It must have been enabled (by the installer), for the user to be able to activate it.

2.6 POWER SUPPLY

Electronically stabilized digital power supply unit shall be provided to power the system. The input and output voltages shall be 230V / 50 Hz. and 12V DC respectively or as recommended by the manufacturer.

2.7 AUDIO DISTRIBUTOR

Audio distributors shall be used to distribute the audio signal to each apartment. Standard cable recommended by manufacturer shall be used for audio signal distribution.

3.0 EXECUTION**3.1 WIRING & INSTALLATION**

CAT 6 UTP cables or insulated PVC protected and twisted pair cable of suitable size or equivalent control cable recommended by the manufacturer shall be used for the wiring of the system.

The power supply unit and the decoders shall be installed in the electrical room. 20% spare cable capacity shall be provided in the riser.

3.2 TEST

Provide the service of a competent, factory-trained engineer or technician authorized by the manufacturer of the entire equipment to technically supervise and participate during all of the adjustments and tests for the system. Before energizing the cables and wires, check for correct connections and test for short circuits, ground faults, continuity, and insulation.

3.3 FINAL INSPECTION

At the final inspection, a manufacturer-trained representative shall demonstrate that the system functions properly in every respect.

3.4 TESTING AND COMMISSIONING

After the installation is complete, the contractor will conduct operating and commissioning tests. The equipment will be demonstrated to operate in accordance with the requirements of the specification. The system installation, testing and commissioning will be as per approvals and requirements.

A company trained representative will personally supervise the complete installation and final testing of the system.

All tests will be carried out in the presence of the Client or persons authorized by the consultant / client. Upon the completion of the acceptance tests, the representatives will instruct operatives in the proper operation, maintenance programming, configuration, and testing of the system. The vendor will provide equipment and /or software which is necessary to allow field modification of the programming and configuration.

END OF SECTION

ELECTRICAL SECTION: 17

FIRESTOPPING FOR ELECTRICAL SYSTEMS

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1.0 GENERAL**1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions and Division-1 Specification Section, apply to work specified in this section.

1.2 DEFINITIONS

- A. Firestopping: Material or combination of materials used to retain integrity of fire-rated construction by maintaining an effective barrier against the spread of flame, smoke, and hot gases through penetrations in fire rated wall and floor assemblies.

1.3 GENERAL DESCRIPTION OF THE WORK OF THIS SECTION

- A. Only tested firestop systems shall be used in specific locations as follows: Penetrations for the passage of cables, conduit, and other electrical equipment through fire-rated vertical barriers (walls and partitions), horizontal barriers (floor/ceiling assemblies), and vertical service shaft walls and partitions.

1.4 RELATED WORK OF OTHER SECTIONS

- A. Coordinate work of this section with work of other sections as required to properly execute the work and as necessary to maintain satisfactory progress of the work of other sections,

1.5 CODES AND STANDARDS

- A. Test Requirements: ASTM E814, "Standard Method of Fire Tests of Through Penetration Fire Stops"
- B. Test Requirements: UL 1479, "Fire Tests of Through-Penetration Firestops"
- C. Underwriters Laboratories (UL) of Northbrook, UL publishes tested systems in their "FIRE RESISTANCE DIRECTORY SET" that is updated annually.
 - 1. UL Fire Resistance Directory:
 - a. Firestop Devices
 - b. Fire Resistance Ratings
 - c. Through-Penetration Firestop Systems
 - d. Fill, Voids, or Cavity Material
 - e. Forming Materials
- D. International Fire stop Council Guidelines for Evaluating Firestop System Engineering Judgments
- E. Inspection Requirements: ASTM E2174, "Standard Practice for On-site Inspection of Installed Firestop Systems."
- F. ASTM E84, "Standard Test Method for Surface Burning Characteristics of Building Materials."

- G. All major building codes: ICBO, SBCCI, BOCA, and IBC.
(Note to specifier: Retain or delete building codes listed above as applicable)
- H. NFPA 101 - Life Safety Code
- I. NFPA 70 - National Electric Code
- J. UAE Fire and Life Safety Code of Practice

1.6 QUALITY ASSURANCE

- A. A manufacturer's direct representative (not distributor or agent) to be on-site during initial installation of firestop systems to train appropriate contractor personnel in proper selection and installation procedures. This will be done per manufacturer's written recommendations published in their literature and drawing details.
- B. Firestop System installation must meet requirements of ASTM or UL (see items 1.5.B & E) tested assemblies that provide a fire rating equal to that of construction being penetrated.
- C. Proposed firestop materials and methods shall conform to applicable governing codes having local jurisdiction.
- D. Firestop Systems do not reestablish the structural integrity of load bearing partitions/assemblies, or support live loads and traffic. Installer shall consult the structural engineer prior to penetrating any load bearing assembly.
- E. For those firestop applications that exist for which no UL tested system is available through a manufacturer, a manufacturer's engineering judgment derived from similar UL system designs or other tests will be submitted to local authorities having jurisdiction for their review and approval prior to installation. Engineering judgment drawings must follow requirements set forth by the International Firestop Council.

1.7 SUBMITTALS

- A. Submit Product Data: Manufacturer's specifications and technical data for each material including the composition and limitations, documentation of UL firestop systems to be used and manufacturer's installation instructions to comply with Section 1300.
- B. Manufacturer's engineering judgment identification number and drawing details when no UL system is available for an application. Engineering judgment must include both project name and contractor's name who will install firestop system as described in drawing.
- C. Submit material safety data sheets provided with product delivered to job-site.
- D. Sound Insulation Test Reports: Provide Test Certificates from qualified independent laboratories showing clearly a minimum of 54 dB sound reduction for Firestop Systems is achieved.
- E. Ageing Test Report showing Firestop Systems which have been tested and rated to withstand aging in normal service life between 25 and 30 years in accordance with DafStb Guidelines for

the repair and protection of concrete building components part 4, Section 2.4.5.6 with subsequent fire testing to ensure long term functionality.

- F. Submit Material safety data sheet of the firestopping material and volatile organic compound VOC have to be less than 150 g/l

1.8 INSTALLER QUALIFICATIONS

- A. Engage an experienced Installer who is certified, licensed, or otherwise qualified by the firestopping manufacturer as having been provided the necessary training to install manufacturer's products per specified requirements. A manufacturer's willingness to sell its firestopping products to the Contractor or to an Installer engaged by the Contractor does not in itself confer qualification on the buyer.

1.9 DELIVERY, STORAGE, AND HANDLING

- A. Deliver materials undamaged in manufacturer's clearly labeled, unopened containers, identified with brand, type, and UL label where applicable.
- B. Coordinate delivery of materials with scheduled installation date to allow minimum storage time at job-site.
- C. Store materials under cover and protect from weather and damage in compliance with manufacturer's requirements, including temperature limitations.
- D. Comply with recommended procedures, precautions or remedies described in material safety data sheets as applicable.
- E. Do not use damaged or expired materials.

1.10 PROJECT CONDITIONS

- A. Do not use materials that contain flammable solvents.
- B. Scheduling
 - 1. Schedule installation of CAST IN PLACE firestop devices after completion of floor formwork, metal form deck, or composite deck but before placement of concrete.
 - 2. Schedule installation of other firestopping materials after completion of penetrating item installation but prior to covering or concealing of openings.
- C. Verify existing conditions and substrates before starting work. Correct unsatisfactory conditions before proceeding.
- D. Weather conditions: Do not proceed with installation of firestop materials when temperatures exceed the manufacturer's recommended limitations for installation printed on product label and product data sheet.
- E. During installation, provide masking and drop cloths to prevent firestopping materials from contaminating any adjacent surfaces.

2.0 PRODUCTS**2.1 FIRESTOPPING, GENERAL**

- A. Provide firestopping composed of components that are compatible with each other, the substrates forming openings, and the items, if any, penetrating the firestopping under conditions of service and application, as demonstrated by the firestopping manufacturer based on testing and field experience.
- B. Provide components for each firestopping system that are needed to install fill material. Use only components specified by the firestopping manufacturer and approved by the qualified testing agency for the designated fire-resistance-rated systems.
- C. Any requirement for protection & safety such as chequered plates/supports for riser opening shall be provided by the contractor.

2.2 MATERIALS

- A. Use only firestop products that have been UL or ASTM (see items 1.5.B & E) tested for specific fire-rated construction conditions conforming to construction assembly type, penetrating item type, and annular space requirements, and fire-rating involved for each separate instance.
- B. Cast-in place firestop devices are installed prior to concrete placement for use with non-combustible and combustible plastic pipe (closed and open systems), conduit, or electrical cable bundles penetrating concrete floors, the following products are acceptable:
 - 1. Cast-In Place Firestop Device
 - 2. Cast-In Place Firestop Device for use with noncombustible penetrants
- C. Sealants, foams or caulking materials for use with non-combustible items including rigid steel conduit and electrical metallic tubing (EMT), the following products are acceptable:
 - 1. Intumescent Firestop Sealant
 - 2. Fire Foam
 - 3. Flexible Firestop Sealant
 - 4. Elastomeric Firestop Sealant
- D. Intumescent sealants, caulking materials for use with combustible items (penetrants consumed by high heat and flame) including PVC jacketed, flexible cable or cable bundles, and plastic pipe, the following products are acceptable:
 - 1. Intumescent Firestop Sealant
- E. Foams, intumescent sealants, or caulking materials for use with flexible cable or cable bundles, the following products are acceptable:
 - 1. Intumescent Firestop Sealant
 - 2. Fire Foam
 - 3. Flexible Firestop Sealant

- 4. Elastomeric Firestop Sealant
- F. Non curing, re-penetrable intumescent putty or foam materials for use with flexible cable or cable bundles, the following products are acceptable:
 - 1. Firestop Putty Stick
 - 2. Firestop Plug
- G. Wall opening protective materials for use with U.L. listed metallic and specified nonmetallic outlet boxes, the following products are acceptable:
 - 1. Firestop Putty Pad
- H. Materials used for large openings and complex penetrations made to accommodate cable trays and bundles, multiple steel and copper pipes, electrical busways in raceways, the following products are acceptable:
 - 1. Firestop Mortar
 - 2. FIRE BLOCK
 - 3. Fire Foam
 - 4. Firestop Board
- I. Non curing, re-penetrable materials used for large openings and complex penetrations made to accommodate cable trays and bundles, multiple steel and copper pipes, electrical busways in raceways, the following products are acceptable:
 - 1. FIRE BLOCK
 - 2. Firestop Board
- J. Provide a firestop system with a "F" Rating as determined by UL or ASTM (see items 1.5.B & E) which is equal to the time rating of construction being penetrated.

3.0 EXECUTION

3.1 PREPARATION

- A. Verification of Conditions: Examine areas and conditions under which work is to be performed and identify conditions detrimental to proper or timely completion.
 - 1. Verify penetrations are properly sized and in suitable condition for application of materials.
 - 2. Surfaces to which firestop materials will be applied shall be free of dirt, grease, oil, rust, laitance, release agents, water repellents, and any other substances that may affect proper adhesion.
 - 3. Provide masking and temporary covering to prevent soiling of adjacent surfaces by firestopping materials.
 - 4. Comply with manufacturer's recommendations for temperature and humidity conditions before, during and after installation of fire stopping.
 - 5. Do not proceed until unsatisfactory conditions have been corrected.

3.2 COORDINATION

Coordinate location and proper selection of cast-in-place Firestop Devices with trade responsible for the work. Ensure device is installed before placement of concrete.

3.3 INSTALLATION

- A. Regulatory Requirements: Install firestop materials in accordance with UL Fire Resistance Directory.
- B. Manufacturer's Instructions: Comply with manufacturer's instructions for installation of through-penetration materials.
 - 1. Seal all holes or voids made by penetrations through fire zone areas to ensure an air and water resistant seal.
 - 2. Where cable raceways and busbars pass through other floor slabs or firewalls, seal into the concrete or pass through a sleeve. Sleeves shall be high density plastic, with flange for attaching to form work or structure. Pack the sleeve with fireproof packing and seal approved by the Engineer's safety department.
 - 3. Sleeves for Future Cable raceways: Install capped sleeves for future cable raceways through firestop-sealed cable tray penetrations of fire and smoke barriers.
 - 4. Protect materials from damage on surfaces subjected to traffic.

3.4 FIELD QUALITY CONTROL

- A. Examine sealed penetration areas to ensure proper installation before concealing or enclosing areas.
- B. Keep areas of work accessible until inspection by applicable code authorities.
- C. Inspection of through-penetration firestopping shall be performed in accordance with ASTM, "Standard Practice for On-Site Inspection of Installed Fire Stops" or other recognized standard (see item 1.5.E).
- D. Perform under this section patching and repairing of firestopping caused by cutting or penetrating of existing firestop systems already installed by other trades.

3.5 ADJUSTING AND CLEANING

- A. Remove equipment, materials and debris, leaving area in undamaged, clean condition.
- B. Clean all surfaces adjacent to sealed holes and joints to be free of excess firestop materials and soiling as work progresses.

END OF SECTION

ELECTRICAL SECTION: 18

IDENTIFICATIONS FOR ELECTRICAL SYSTEM

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1.0 GENERAL**1.1 SUMMARY****A. Section Includes:**

1. Identification for conduits and trunking.
2. Identification of power and control cables.
3. Identification for conductors.
4. Underground-line warning tape.
5. Warning labels and signs.
6. Instruction signs.
7. Nameplates, Equipment identification labels, including arc-flash warning labels
8. Miscellaneous identification products.

B. Comply with BS (Requirements for Electrical Installations) Labels and Notices and DEWA wiring Regulations.

1. Identification of protective devices.
2. Warning notices voltages in excess of 250 volts.
3. Warning notices voltages in excess of 10000 volts.
4. Warning notices and labels for all substation doors
5. Warning notices and labels for all MV equipment
6. Warning notices and labels for all Gas suppression systems and associated control equipment
7. Warning notices and labels for all LV equipment
8. Warning notices and labels for all LV sub distribution rooms
9. Warning notices and labels for all Life Safety equipment
10. Periodic inspection and test notices.
11. Residual current device notices.
12. Earth electrode safety electrical connection label.
13. Bonding conductor connector points to extraneous conductive parts label.
14. Earth free local equipotential bonding areas warning notice.
15. Electrical separation areas warning notice.
16. Outdoor equipment socket outlet notice.
17. Equipment labels and isolator labels must indicate the circuit reference and where they are fed from.

1.2 SUBMITTALS

- A. Product Data: For each electrical identification product indicated.
- B. Samples: For each type of label and sign to illustrate size, colors, lettering style, mounting provisions, and graphic features of identification products.
- C. Identification Schedule: An index of nomenclature of electrical equipment and system components used in identification signs and labels.

1.3 QUALITY ASSURANCE

- A. Ensure that the design, construction, materials and finishes of all equipment are suitable for the location, climatic and operating conditions to which the installation shall be exposed.
- B. Ensure that the whole installation complies with the relevant codes and standards, including the following:
 - 1. DEWA wiring regulations
 - 2. UAE Fire and Life safety code of practice
 - 3. BS 5499-4:2013 - Safety signs. Code of practice for escape route signing
 - 4. BS 5499-10:2014 - Guidance for the selection and use of safety signs and fire safety notices
 - 5. BS ISO 3864-4:2011 - Graphical symbols - safety colours and safety signs. Colorimetric and photometric properties of safety sign materials
 - 6. BS 1710:2014 - Specification for identification of pipelines and services
 - 7. BS 3858:1992 - Specification for binding and identification sleeves for use on electric cables and wires
 - 8. BS 5472, EN 50005 - Specification for low voltage switchgear and control gear for industrial use. Terminal marking and distinctive number. General rules
 - 9. BS ISO 3864-1:2011 - Graphical symbols. Safety colours and safety signs. Design principles for safety signs and safety markings
 - 10. BS EN ISO 7010:2020 Graphical symbols. Safety colours and safety signs. Registered safety signs
 - 11. BS 6272, EN 50042 - Specification for low voltage switchgear and control gear for industrial use. Terminal marking. Terminals for external associated electronic circuit components and contacts
 - 12. BS 7671:2018+A1:2020 - Requirements for electrical installations. IET Wiring Regulations.
 - 13. BS EN 60034-8:2007+A1:2014 - Rotating electrical machines. Terminal markings and direction of rotation
 - 14. BS EN 60073 - Basic and safety principles for man-machine interface, marking and identification. Coding principles for indicators and actuators
 - 15. BS EN 61439 all parts - Specification for low-voltage switchgear and control gear assemblies

16. BS EN 60445 - Basic and safety principles for man-machine interface, marking and identification. Identification of equipment terminals, conductor terminations and conductors (including general rules for an alphanumeric system)
17. BS ISO 17398 - Safety colours and safety signs. Classification, performance and durability of safety signs.
18. BS 4800 - Schedule of paint colours for building purposes
19. 29 CFR & 1910.145 - Specifications for accident prevention signs and tags

1.4 COORDINATION

- A. Coordinate identification names, abbreviations, colours, and other features with requirements in other Sections requiring identification applications, drawings, shop drawings, manufacturer's wiring diagrams, and the Operation and Maintenance Manual; and with those required by codes, standards, and the specifications for accident prevention signs and tags. Use consistent designations throughout the project.
- B. Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be applied.
- C. Coordinate installation of identifying devices with location of access panels and doors.
- D. Install identifying devices before installing acoustical ceilings and similar concealment.

2.0 PRODUCTS

2.1 MATERIALS

- A. Use new materials and of the highest quality available for tags or labels, nameplates, tapes, signs and markers with a predicted life equal to or greater than the design life of the electrical cables, plant, equipment or installation to which it refers and to BS ISO (see 1.3.B.17).
- B. External: Engraved thermosetting plastic laminate.
- C. Internal: Engraved thermosetting plastic laminate.
- D. Colour:

Background	-	White.
Lettering	-	Black (non-essential services)
		Red - (Essential Services)

2.2 ARRANGEMENT

- A. Obtain approval prior to manufacture, with regard to style, colour, lettering, size and position of all labels and notices.

2.3 LETTERING AND SIZE OF LABELS AND NOTICES

- A. Ensure that all lettering and symbols comply with the requirements laid out in the DEWA Wiring Regulations and BS ISO (see 1.3.B.9). Use the same BS ISO, or 5mm minimum high letters for height of lettering where not otherwise indicated. Ensure labels and notices of adequate size for the lettering required, and allow a minimum margin around all lettering of one-line space vertically and two letter spacing horizontally.

- B. Font - Helvetica Medium

2.4 LANGUAGE

- A. The Arabic and English languages shall be used for all labelling and charts.

2.5 CONDUCTOR ARRANGEMENT

- A. Arrange circuit polarity so that phases read in phase rotation order followed by the neutral, if any, from top to bottom in horizontal conductor layouts and left to right in vertical conductor layouts. Ensure flat horizontal arrays have leading phase to the left and neutral to the right from left to right when viewed from supply point. Arrange phase or live pole of two wire apparatus at top or left hand and neutral and earth both at bottom or right hand side. In all cases, ensure conductor arrangements defined are when viewed from front face of all equipment and terminating facilities. Apply identification markers in accordance with DEWA Wiring Regulations to all conductor termination points.

2.6 SAFETY SIGNS

- A. Label all electrical plant and equipment using safety sign W012 of BS EN ISO (see 1.3.B.10) where voltages above ELV exist.



- B. Provide supplementary or text signs complying with BS EN ISO (see 1.3.B.10)
- C. Label all electrical plant and equipment with the labels specified in the appropriate British Standards for that plant or equipment.
- D. Identify each substation and main switch room with safety sign W012 and with supplementary signs to the same BS EN ISO, notices and signs required by the same BS EN ISO for any fire extinguishing system and notice giving details of:
1. Name of the substation or switch room
 2. The presence of medium and low voltages
 3. Administrative instructions for access
 4. Location and method of contacting controlling authority
 5. Actions to be taken in an emergency

2.7 PLANT AND EQUIPMENT LABELS

- A. Fit labels on all items of plant, equipment, switches, etc., include the following information:

1. service controlled,
2. circuit reference,
3. voltage,
4. type of supply and phase etc.,
5. circuit protection type and rating.

B. Maintenance Notices

1. Fix notices giving warning of, and instructions for, any special maintenance procedures to plant and equipment.

C. Colour Corrected Light Fittings

1. Fix a warning or identification disc to light fittings containing colour corrected sources to ensure that maintenance staff install the correct lamps.

D. Motors and Starters Labels

1. Fit identification labels to all motors, starters and starter panels.
2. Ensure positive identification of respective motors and starters.
3. Provide motors with non-corrodible labels attached adjacent to each bearing giving details of the lubricant to be used.
4. Mark direction of normal rotation on motor casing.
5. Provide labels to identify motor equipment fitted with surge suppressors and thermistors stating that insulation test voltages must not be applied to thermistors and thermistor control units.
6. Ensure labelling is compatible with schematic and wiring diagrams, and complies with BS EN (see 1.3.B.13).

E. Engraved Accessory Plates: Engrave the following using 5mm high letters with engraving:

1. Switch plates,
2. Spur units,
3. Call systems,
4. Fire alarms, etc.

F. Switchgear: Fit labels on switchgear as required by BS (see 1.3.B.12) which indicate:

1. duty of unit,
2. voltage, phase and current rating,
3. protective device rating size of conductor involved,
4. all other necessary details.

5. Use an agreed serial coding system,
6. provide at the switch a key to the coding system.

2.8 DISTRIBUTION BOARDS

- A. On each distribution board identify every outgoing way with a renewable circuit chart in a transparent plastic envelope permanently fitted inside distribution board cover. Clearly indicate in typed script, circuit identification number, cable size, fuse or circuit breaker rating and a description of item supplied and area supplied by circuit.

2.9 SCHEMATIC DIAGRAMS

- A. Provide a purpose made schematic diagram permanently fixed showing the connections of the equipment and plant adjacent to each MDB, MCCP and SMDB. Heat sealed and block mounted.

2.10 SPECIAL PURPOSE EARTHING

- A. Fit labels to special purpose earthing conductors and connection points, describing their purposes and any instructions necessary for their operation and maintenance:
 1. IT equipment "Clean Earths"
 2. Telecommunications functional earths
 3. Lightning Protection earth
 4. Generator earth
 5. Transformer earth

2.11 INDICATOR LAMPS AND PUSH BUTTONS FOR POWER SYSTEMS

- A. Use indicator lamp and push button colours in accordance with BS EN (see 1.3.B.14).
 1. Indicator lamp - Red, danger or alarm; yellow, caution; green, safety.
 2. Push buttons - Red, emergency action; red, stop or off; yellow, intervention; green, start or on.

2.12 CONDUIT AND TRUNKING COLOUR CODING

- A. In areas of mechanical plant or voids accommodating mechanical services, or where otherwise indicated, identify electrical conduits and ducts in accordance with BS(see 1.3.B.6). Apply colour orange to BS (see 1.3.B.18) by painting on service as a band over 150mm or applying an adhesive tape type wrap around services over a length of 150mm.
- B. Place identification colours at bulkheads, wall penetrations and any other place where identification is necessary.

2.13 CABLE IDENTIFICATION

- A. Provide all cables, other than final sub-circuit wiring enclosed in conduits or trunking, with labels fixed at each end of cable either side of wall and floor penetrations at approximately 12m

intervals at convenient inspection points by means of non-releasable plastic straps, minimum width 4mm. Labels are made of adhesive bands coloured orange and shall be not less than 100 mm long.

- B. Ensure labels show the reference number of cable.
- C. Except where cables are buried or enclosed in conduit, trunking or ducting; they shall be permanently identified by discs. Discs shall be of laminated plastic material with black characters on white. Character shall be not less than 3 mm high. The inscription shall indicate the nominal voltage, the designation of the load, the number and cross sectional area of cores and the rated voltage of the cable.
- D. Cable identification discs shall be attached to the cables with ties. Discs shall be located within 500 mm of terminations and joints, at least once within each separate compartment through which the cable passes, and at intervals not exceeding 24 m. Locations of discs shall coincide with the colour bands

2.14 TERMINAL MARKING AND CONDUCTOR IDENTIFICATION

- A. Provide for switchgear and control gear elements whose terminals are marked in accordance with BS standards (see 1.3.B.8 & 11).
 - 1. Use a unique reference to identify each element in the switchgear or control gear.
 - 2. Mark on or adjacent to each element its reference.
 - 3. Identify each terminal for connection to external wiring or cabling using a reference system complying with BS EN (see 1.3.B.16) based on the element reference and the appropriate element terminal reference.
- B. Adjacent to terminals:
 - 1. Use lettered or numbered ferrules or sleeves to BS (see 1.3.B.7)
 - 2. Mark each auxiliary conductor or control cable core with the identity of the terminal to which it is connected
 - 3. Include: the reference of plant or equipment to which it is connected the identity of the terminal at the remote end.
 - 4. Ensure that main circuit conductors are identified in accordance with Local Wiring Regulations.
 - 5. Ensure that all identification of terminals and conductors is recorded and included on record drawings and in operation and maintenance documentation.

2.15 UNDERGROUND CABLE IDENTIFICATION

- A. For buried LV and MV cables use warning tapes according to the standard practice of DEWA.
- B. Identify external underground cable routes by means of approved markers along their length at distances not exceeding 50m and where a change of direction occurs on such routes. Provide cables markers with a brass plate or impress concrete to clearly indicate the reference of group of cables or reference number of cable and operating voltage of cable. Provide key to any reference system used at switchgear. Mark and protect direct buried cables with plastic tape yellow printed black letters of not less than 30mm high, "DANGER ELECTRIC CABLES" or similar in Arabic

and English. In addition, the word “JOINT” shall be added to above words, where applicable. Cable warning tapes shall be of polythene, not less than 150 mm wide and at least 0.25 mm thick.

- C. For buried low current/communication cables or duct banks, use warning tapes in accordance with the requirements and standards of local authorities having jurisdiction and applicable international standards

2.16 CABLE CONDUCTOR COLOUR CODING

- A. Identify cable conductors in accordance with local wiring Regulations.
- B. Colour code is Red, Yellow, Blue Black, and Green/Yellow as per DEWA regulations
- C. Ensure that all mechanical and electrical equipment installed also complies with these requirements to ensure no ambiguity between the equipment and the wiring installations. It will not be permitted to identify equipment or wiring colours using identification tape or markers.

2.17 CABLE JOINTING AND TERMINATION

- A. Connect all cables in the installation so that the correct sequence of phase rotation is maintained throughout. Where straight through joints are approved on low voltage cables, whether power cables or control or auxiliary cables, joint conductors strictly in accordance with their colour or numeric coding.

2.18 CABLE SHEATH IDENTIFICATION

- A. Internal: Identify coloured cables sheaths for various services as follows:

1. Fire alarm - red
2. Telecommunications - grey
3. Data - as system suppliers requirements
4. Control - black
5. LV – black
6. MV red
7. Emergency Lighting – white fire rated

Code cables for various services using alpha numeric symbols as follows. Code letters preceding cable reference:

1. Fire alarm - FA
2. Telecommunications - T
3. Data - D
4. Control - C
5. Low voltage - LV

6. Extra low voltage - ELV
7. LV Essential circuits E
8. Medium voltage - HV

B. External: Identify cable sheaths for various services in accordance with DEWA, as follows:

1. MV - Red
2. LV - Black
3. Telecommunications and data - Grey

2.19 ADDITIONAL SAFETY SIGNS

A. For main switch and electrical plant room access doors complete with supplementary signs as shown:

1. With supplementary sign "Authorized persons only"

B. For use with permit to work systems, complete with supplementary signs as shown:

1. Printed on rigid plastic, with hanging loop, with supplementary wording "Do not operate. Work in progress" or similar
2. For use at each emergency stop. With supplementary sign "Emergency stop push-button" or similar.

2.20 BUS BAR IDENTIFICATION TAGS

A. Phase identification of bus bars on tag holder shall be as follows:

1. Tag holder type of identification shall be used to install at line terminations, at the ends of a bus run, bus tap-off positions, circuit breakers, disconnect switches and transformer positions.
2. The phase identification of bus bars shall be identified by designations (R, Y, B), and these designations (letters) shall be indented with 150 mm (6 in.) high and filled with black baked enamel, and these designations shall be inscribed on tag holder.
3. The tag holder shall be fabricated from anodized aluminium plate, vertical mountable, of size 174 x 124 x 2 mm.
4. The above mentioned tag holder shall be fabricated with 4 fixing holes of 10 mm diameter, 2 holes at each end of the aluminum plate.

B. Phase identification of bus bars by colors shall be as follows:

1. In indoor substations, each bus bar shall be identified by red, yellow, and blue (R, Y, B) to indicate the phase to which it is connected.
2. For identification, each bus bar shall be painted with (R, Y, B) at intervals of not more than 600 mm on all line termination, at the ends of a bus run and at bus tap-off positions.

2.21 EQUIPMENT WARNING / DANGER SIGNS

A. For external use, pressure sensitive danger signs shall be used. Dimensions shall be as approved by the Engineer. The signs shall be heavy duty vinyl with a self-adhesive backing which can be applied

to curved or irregular surfaces. Danger signs shall be weather-resistant and shall not discolour or deteriorate with age.

- B. Danger signs shall be inscribed with the equipment voltage level along with an internationally recognized danger sign.
- C. Warning/Danger signs made of red plastic (vinyl) with white letters at least 25 mm high reading “DANGER High Voltage” shall be fixed to the entrance doors of all 11 kV switchgear and transformer rooms.
- D. Warning/Danger signs made of red plastic (vinyl) with white letters at least 15 mm high reading “DANGER 400V” or “DANGER 230V” as appropriate, shall be fixed to the lids, covers or doors of any equipment which contains terminals or conductors connected to more than one phase of a low voltage supply.

3.0 EXECUTION

3.1 APPLICATION

- A. Apply identification labels and notices in accordance with Local DEWA Wiring Regulations and BS (see 1.3.B.12) (IET Wiring Regulations), Clause 514 to all electrical cables plant and equipment including components of mechanical systems.
 - 1. Labels and Notices
 - 2. Identification of protective devices.
 - 3. Diagrams, charts or tables to comply with Clause 514-09 of BS (see 1.3.B.12) wiring regulations.
 - 4. Warning notices, voltages in excess of 230 volts.
 - 5. Periodic inspection and test notices.
 - 6. Residual current device notices.
 - 7. Earth electrode safety electrical connection label.
 - 8. Bonding conductor connector point to extraneous conductive parts label.
 - 9. Earth free local equipotential bonding areas warning notice.
 - 10. Electrical separation areas warning notice.
 - 11. Outdoor equipment socket outlet notice.

3.2 INSTALLATION

- A. Verify identity of each item before installing identification products. Before beginning installation in any area, examine all parts of the adjoining work into which applicable work is to be placed. Should any conditions be found which will prevent the proper execution of the work, installation shall not proceed in that area until such conditions are corrected by the contractor.
- B. Location: Install identification materials and devices at locations for most convenient viewing without interference with operation and maintenance of equipment.

- C. Apply identification devices to surfaces that require finish after completing finish work.
- D. Fix labels and notices using materials compatible with label or notice and surface to which it is fixed by screws into tapped hole or bolted complete with washer nut and locking device.
- E. Self-Adhesive Identification Products: Clean surfaces before application, using materials and methods recommended by manufacturer of identification device.
- F. Attach signs and plastic labels that are not self-adhesive type with mechanical fasteners appropriate to the location and substrate.
- G. System Identification Colour-Coding Bands for trunkings and Cables: Each colour-coding band shall completely encircle cable or conduit. Locate bands at changes in direction, at penetrations of walls and floors, at 15m maximum intervals in straight runs, and at 7.6m maximum intervals in congested areas.
- H. Aluminium wrap-around Marker Labels and Metal Tags: Secure tight to surface of conductor or cable at a location with high visibility and accessibility.
- I. Cable Ties: For attaching tags. Use general-purpose type, except as listed below:
 - 1. Outdoors: UV-stabilized nylon.
 - 2. In Spaces Handling Environmental Air: Plenum rated.
- J. Underground-Line Warning Tape: During backfilling of trenches install continuous underground-line warning tape directly above line at 150mm to 200mm below finished grade. Use multiple tapes where width of multiple lines installed in a common trench or concrete envelope exceeds 400mm overall.
- K. Painted Identification: Comply with requirements in Division 09 painting Sections for surface preparation and paint application.
- L. Preparation: Ensure surface is suitable to receive a label or notice.
- M. Handling and Storage: Deliver component parts to site, completely identified in accordance with shop and certified drawings prepared for this work. Store in accordance with manufacturer's instructions, above ground, properly protected from the weather and construction activities.
- N. Fabrication: All labels shall be made by a specialist using special machinery designed to manufacture permanent labels.
- O. Provide Field Quality Control.
- P. This standard specification is intended to be used as a helpful indication of work within this section. It is not necessarily comprehensive or complete and it is the Contractor's responsibility to ascertain and complete the full works intended and as shown within the drawings, specification and Contract or thereby implied.

3.3 IDENTIFICATION SCHEDULE

- A. Concealed trunkings, Duct Banks, More Than 1000 V, within Buildings: Tape and stencil 100-mm wide black stripes on 250-mm centres over orange background that extends full length of raceway

or duct and is 300 mm wide. Stencil legend "DANGER CONCEALED HIGH VOLTAGE WIRING" or similar in Arabic and English with 75-mm high black letters on 500-mm centres. Stop stripes at legends. Apply to the following finished surfaces:

1. Floor surface directly above conduits running beneath and within 300 mm of a floor that is in contact with earth or is framed above unexcavated space.
 2. Wall surfaces directly external to raceways concealed within wall.
 3. Accessible surfaces of concrete envelope around raceways in vertical shafts, exposed in the building, or concealed above suspended ceilings.
- B. Accessible Raceways, Armoured and Metal-Clad Cables, More Than 600 V: Self-adhesive vinyl or Snap-around labels. Install labels at 10-m maximum intervals.
- C. Accessible Raceways and Metal-Clad Cables, 1000 V or Less, for Service, Feeder, and Branch Circuits More Than 30 A, and 230V to ground: Identify with self-adhesive vinyl label or self-adhesive vinyl tape applied in bands. Install labels at 10-m maximum intervals.
- D. Accessible Raceways and Cables within Buildings: Identify the covers of each junction and pull box of the following systems with self-adhesive vinyl labels with the wiring system legend and system voltage. System legends shall be as follows:
1. Emergency Power.
 2. Power.
 3. UPS.
- E. Power-Circuit Conductor Identification, 600 V or Less: For conductors in vaults, pull and junction boxes, manholes, and hand holes, use colour-coding conductor tape to identify the phase.
1. Colour-Coding for Phase and Voltage Level Identification, 600 V or Less: Use colours listed below for ungrounded service, feeder and branch-circuit conductors.
 - a. Colour shall be factory applied or field applied for sizes larger than No. 8 AWG, if authorities having jurisdiction permit.
 - b. Colours for 400/230-V Circuits:
 - 1) Phase A: Red.
 - 2) Phase B: Yellow.
 - 3) Phase C: Blue.
 - c. Field-Applied, Colour-Coding Conductor Tape: Apply in half-lapped turns for a minimum distance of 150 mm from terminal points and in boxes where splices or taps are made. Apply last two turns of tape with no tension to prevent possible unwinding. Locate bands to avoid obscuring factory cable markings.
- F. Install instructional sign including the colour-coded for grounded and ungrounded conductors using adhesive-film-type labels.
- G. Conductors to Be Extended in the Future: Attach write-on tags to conductors and list source.
- H. Auxiliary Electrical Systems Conductor Identification: Identify field-installed alarm, control, and signal connections.

1. Identify conductors, cables, and terminals in enclosures and at junctions, terminals, and pull points. Identify by system and circuit designation.
 2. Use system of marker tape designations that is uniform and consistent with system used by manufacturer for factory-installed connections.
 3. Coordinate identification with Project Drawings, manufacturer's wiring diagrams, and the Operation and Maintenance Manual.
- I. Locations of Underground Lines: Identify with underground-line warning tape for power, lighting, communication, and control wiring and optical fibre cable.
1. Limit use of underground-line warning tape to direct-buried cables.
 2. Install underground-line warning tape for both direct-buried cables and cables in raceway.
- J. Workspace Indication: Install floor marking tape to show working clearances in the direction of access to live parts. Workspace shall be as required by local wiring Regulations unless otherwise indicated. Do not install at flush-mounted panel boards and similar equipment in finished spaces.
- K. Operating Instruction Signs: Install instruction signs to facilitate proper operation and maintenance of electrical systems and items to which they connect. Install instruction signs with approved legend where instructions are needed for system or equipment operation.
- L. Emergency Operating Instruction Signs: Install instruction signs with white legend on a red background with minimum 10-mm high letters for emergency instructions at equipment used for power transfer, load shedding.

END OF SECTION

ELECTRICAL SECTION: 19

AIRCRAFT WARNING LIGHTING SYSTEM

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1.0 GENERAL**1.1 SCOPE OF WORKS**

- A. The Contractor shall supply, install, test, connect and commission a high-quality, high-performance state-of-the-art Aircraft Warning Lighting System, comprising, but not limited to, appropriate lighting fixtures, distribution boards/control units,; conduits; sub-circuit wiring; accessories etc., all in compliance with these Specifications, details in Drawings, Load Schedules, Schedules of Luminaires, and in strict compliance with the current ICAO rules, regulations and recommendations stipulated in Volume II of ICAO Annexure 14. Obstruction lights shall be fed via the generator distribution board.
- B. Provide and install all interconnecting wiring and power supplies to the main control panel, luminaires and photocell. Provide and install dedicated on line Uninterruptible Power Supply Unit and associated batteries and all associated power and controls cabling associated with the systems, as detailed on the layout and schematic drawings and within the specification. Note that UPS shall be provided for the building top light only, all other obstruction lights shall be fed via the generator distribution board.
- C. All aviation obstruction lighting installations shall be detailed, installed and tested in accordance in accordance with the current issue of the Dubai Civil Aviation Authority (DCAA) and the ICAO (International Civil Aviation Organization) and any other relevant authorities.
- D. The Contractor is responsible for obtaining approvals and NOC from the DCAA and DEWA for all equipment, wiring and controls. Include for all submissions and meetings with approval authorities and specialist supplier.
- E. Provide aircraft obstruction LED lighting on the external façade of the tower construction to highlight the presence of the building.
- F. Provide and install all interconnecting wiring and power supplies to the main control panel, luminaires and photocell. Provide and install dedicated on line Uninterruptible Power Supply Unit and associated batteries and all associated power and controls cabling associated with the systems, as detailed on the layout and schematic drawings and within the specification.
- G. The system must be fully synchronized such that flashing luminaires operate simultaneously.
- H. Provide and install the obstruction lighting system in accordance with the manufacturer's instructions.
- I. Provide and install low maintenance Stainless steel supports on the external building façade to support all luminaires. All supports shall be manufactured by the luminaire manufacturer.
- J. The Contractor shall co-ordinate all façade fixings, wiring pathways, apertures, detailing and making good prior to installation. The Contractor shall submit all shop drawing detailing this work to the Engineer for review.

Allow for fine-tuning of exact location of the luminaires to suit the facade manufacturer's fixing requirements and confirm acceptance of final positions with relevant authorities.

- K. Demonstrate the system to the official representatives of the DCAA during daylight and night time hours.
- L. System handing over to authority shall be included.
- M. Omissions and Clarifications: Where any fitting type or lamping designated has been omitted or cannot be determined by the Contractor, request a clarification from the Owner's Representative and provide the fitting type as directed. Where an inconsistency in the specifications and/or the drawings is determined by the Contractor, request a clarification from the Owner's Representative and provide as directed. There will be no increase in cost permitted to correct any inconsistency within the documents so long as the overall intent of the work scope is not changed.

1.2 PERFORMANCE REQUIREMENTS

- 1. Integrated System Design: The Building Management System (BMS) shall monitor the Aviation warning light control system and initiate alarms for the following:
 - a. Beacon lamp failure, intensity error,
 - b. Obstruction lamp failure, intensity error
 - c. Flasher failure (force beacon on steady in the event of failure)
 - d. Power failure
- 2. The contractor shall be responsible for the design of the integration between the Aviation warning light system and BMS system including all necessary cabling and containments.

1.3 CODES and STANDARDS

- 1. DCAA - Dubai Civil Aviation Authority
- 2. ICAO (Annex 14) – International Civil Aviation Organization
- 3. FAA – Federal Aviation Administration:
 - a. FAA AC150/5345-43H - Specification for Obstruction Lighting Equipment
- 4. IEC Standards:
 - a. IEC 60364 Electrical Installations of Buildings
 - b. IEC 60529 Degrees of Protection provided by Enclosures (IP code)
 - c. IEC 60598-1 Luminaires Safety – General Requirements and Tests
 - d. IEC 60598-2 Luminaires – Part 2, Particular Requirements (Various Parts)

1.4 SUBMITTALS

- A. General Submittal Requirements:

Submittals shall be approved by the Dubai CAA prior to submitting them to the Client's representative. Shop Drawings shall be prepared by persons with the following qualifications:

Trained and certified by manufacturer in Aviation Hazard Warning System design.

Licensed or certified by Dubai Civil Aviation Authority.

- B. Product Data: For each type of product indicated shall be provided with ICAO and DCAA approved certificate and test reports performed by manufacturer and witnessed by a qualified testing agency. Include data on features, accessories, finishes, and the following:

1. Physical description of lighting fixture including dimensions
2. Light Output
3. Catalogue sheets, specifications and installation instructions
4. Summary of deviations from the Specification

All lighting equipment shall be currently available; working samples shall be required for submission and are to be accepted by the Engineer.

- C. Shop Drawings: Include plans with dimensions, elevations, sections, details such as description of materials and finishes, general construction, specific modifications, component connections, anchorage methods, hardware, and installation procedures, and attachments to other work.
- D. Qualification Data: The contractor shall only employ companies and persons licensed to supply and install Aviation hazard warning light systems in Dubai.
- E. Warranties: Submit written guarantee signed by the contractor or the manufacturer for the period of 5 year from the date of substantial completion. The guarantee shall cover the repair and replacement of defective materials and workmanship as directed by the engineer except where any specific warranty from a supplier or manufacturer extends for a longer time.
- F. Operation and Maintenance (O+M) Manuals: Submit under the provisions of Section 01 78 23. Include component list with manufacturer's reference numbers, descriptions of materials and procedures for repairing. Submit the complete manufacturer's installation instructions of all the products used.

1.5 QUALITY ASSURANCE

- A. Installer Qualifications: Personnel shall be trained and certified by manufacturer for installation of units required for this Project. Installation shall be by personnel certified to operate as a hazard warning system installer in the region.
- B. Product Quality: All equipment and materials supplied and installed shall be new, of the best quality and in accordance with the relevant standards currently in force in the project's country of origin. All fittings, lamping, accessories, and materials, shall be new, of good quality, and free from defects which in any manner may impair their character, appearance, strength, durability and function. All blemished, damaged, or unsatisfactory fittings shall not be installed and shall be replaced in a satisfactory manner as directed by the Owner's Representative. All fittings shall have FAA certifications and CE/ ENEC and/ or IEC and/ or BS markings or other testing agency as acceptable to local code authorities.
- C. Source Limitations for System and Components: Obtain system from single source from single manufacturer.

- D. Electrical Components, Devices, and Accessories: Shall be provided in accordance with DEWA Wiring Regulations, and marked for intended location and application.
- E. The cost of obtaining any necessary approvals shall be included in any Tender price, detailed as a separate item.
- F. Testing, Inspection and Reports:
 - 1. Submit reports of independent tests demonstrating that the products and systems comply with the specified performance requirements.
 - 2. Where test results for a material or product are not available, undertake testing to show compliance with the Specification at an independent testing laboratory acceptable to the Engineer.
 - 3. The provision of testing data or the carrying-out of tests does not relieve the Contractor of his responsibilities regarding the performance requirements, durability or service life requirements.

2.0 PRODUCTS

2.1 GENERAL

- 2.1.1 Type A, Medium Intensity White LED Flashing Light Beacon shall be installed as Aircraft Warning Lights at locations indicated in the drawings and as per Civil Aviation authority requirements.
- 2.1.2 The quantity, types and placements of Beacons indicated in the drawings are notional and these may be altered to suit site conditions and to the approval of local Civil Aviation Authority. All lighting fixtures shall comply with protection requirements of IP66.
- 2.1.3 The System Control supplying power to the Beacons shall be integrated type functioning as one unit supplied from an essential (genset back-up) power supply source that will provide full operational functionality.
- 2.1.4 The Contractor shall provide certificates for all components, devices and equipment proposed for installation.
- 2.1.5 Prior to submitting his offer, the Contractor shall coordinate with the manufacturer of proposed Aircraft Warning System to ensure all necessary components, accessories, cables etc., other than those described above, shall be included in the offer, as no claims for variations shall be allowed for lack of knowledge in this respect.
- 2.1.6 Prior to placing order for any Beacons and associated equipment, the Contractor shall submit, for Engineer's study, comments and final approval, relevant Specifications; catalogues/descriptive technical literature; complete installation, operation and maintenance manuals relating to the various components and accessories proposed for the system.
- 2.1.7 Prior to commencing any installation works, the Contractor shall submit comprehensive workshop drawings for Engineer's study and approval.
- 2.1.8 All wiring shall be high temperature rating to 90°C or higher, PVC insulated and installed in PVC embedded conduits or surface mounted GI conduit. All wiring pathways shall be installed concealed from view.

- 2.1.9 The Contractor shall ensure that all Beacons, related equipment, accessories and components supplied shall be manufactured to stringent standards based on the latest technology available in the market, complying with standards and DCAA regulations.
- 2.1.10 All equipment, component, accessories and materials shall be guaranteed by the Contractor/Manufacturer for a minimum period of 5 years after handing over. The guarantee shall cover repair and replacement of defective materials.
- 2.1.10 Particulars of Beacons, together with respective rating of lamps and its power consumption, shall be as described and specified in Schedule of Luminaires. Samples of proposed Beacon and other related equipment together with photometric data for the Beacon shall be submitted for Engineer's final approval prior to placing orders for the same.

2.2 BEACONS

- 2.2.1 The proposed Beacons shall be designed, manufactured and tested to comply with the relevant and respective requirements of FAA and ICAO.
- 2.2.2 The fixture shall be 115V/230V AC, 50 Hz Medium Intensity White Flashing Beacon with design based on state-of-the-art high flux LED optics technology.
- 2.2.3 The Beacon shall be designed for durability with a cast aluminium base stainless steel (steel grade 316/316L shall be used , hardware and shall be resistant to shock and vibration. The dome of the fixture shall be made of Fresnel lens, UV resistant acrylic and shall be sealed to ensure maximum life of the product.
- 2.2.4 The Beacon shall be designed to withstand the dynamic conditions experienced in the most severe environments including temperatures in excess of 55 degrees Celsius, direct sunlight, winds in excess of 150 MPH, surge/lightning, high humidity, salt fog etc.
- 2.2.5 The Beacon shall meet or exceed industry EMI/RFI Standards.

2.3 CONTROL UNIT

The Beacon Control Panel shall be complete with 230V 50Hz 4P 40A 40A/100 mA RCCB, 4P Contactor, 10A SP MCBs, Control Relay, Fuse, 3-position switch, RYB indication lamps, appropriate terminals etc. all housed inside an appropriate sized IP66 wall-mounted lockable enclosure with transparent door. The unit shall be designed to be protected against vibrations and harsh conditions as described above. The Control unit shall provide the following functions and shall have the following additional features:

- A. Solid state alarm modules and flashers with zero voltage switching for longer lamp life, encapsulated to protect against harsh conditions and vibration.
- B. Line voltage transient protection.
- C. Fused output to all lighting circuits.
- D. Failure detections for Beacon lamp failure, Power and flasher failure, amps ON.
- E. Individual isolated alarm contacts for remote alarm and local indications for Beacon lamp failure, Power failure, Lamps ON.

- F. PEC control override switch on enclosure door.
- G. The Controller shall be password protected against non-authorized access.

3.0 EXECUTION

3.1 EQUIPMENT INSTALLATION

- A. Comply with ICAO, FAA, FCC Rules & Regulations and the Dubai Civil Aviation Authority guidelines for installation of aviation warning equipment as mentioned under clause 1.3 above.
- B. Provide and install low maintenance stainless steel supports on the external building façade to support all Flash-heads.
- C. Coordinate with the main contractor for all fixings, wiring pathways, apertures, detailing and making good prior to installation. Submit shop drawing details of the same for Engineer's review.
- D. Allow for fine-tuning of exact location of the Flash-heads to suit the façade fixing requirements.

3.2 IDENTIFICATION

- A. Identify system components, wiring, cabling, and terminals as detailed in the "Identification section" of the electrical specification.
- B. Install framed instructions in a location visible from the control unit.

3.3 EARTHING

- A. Provide earth connection to the aviation warning control unit and associated circuits; comply with local authority (DEWA) Wiring Regulations.

3.4 FIELD QUALITY CONTROL

- A. Field tests shall be witnessed by a Client's representative and Dubai Civil Aviation Authority Field Service: Engage a Testing and commissioning company to manage the inspection, test, and commissioning of all components, assemblies, and equipment installations, including connections.
- B. Perform tests and inspections.
- C. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.
- D. Examination, Tests and Inspections:
 - 1. Verification of Conditions: Examine areas for compliance with requirements for installation and conditions affecting performance of the Work. Identify conditions detrimental to a proper and timely completion and notify the Engineer of the

unsatisfactory conditions. Proceed with installation only after unsatisfactory conditions have been corrected.

2. Visual Inspection: Conduct visual inspection prior to testing. Inspection shall be based on completed Record Drawings and system documentation and in accordance with ICAO, FAA, FCC Rules & Regulations and the Dubai Civil Aviation Authority guidelines, as mentioned in clause 1.3 above. Test appliances according to manufacturer's written instructions.
3. Aviation warning system shall be considered defective if it does not pass tests and inspections.
4. Prepare test and inspection reports.
5. Maintenance Test and Inspection: Perform tests and inspections listed for weekly, monthly, quarterly, and semi-annual periods. Use forms developed for initial tests and inspections.

3.5 DEMONSTRATION

- A. The Aviation Warning System Contractor shall provide a factory-authorized service representative to train the owner's maintenance personnel to adjust, operate, and maintain the system.
- B. The Aviation Warning System Contractor shall allow for one full day visits for training purposes

END OF SECTION

ELECTRICAL SECTION: 20

IP BASED CCTV MONITORING SYSTEM

1.0 GENERAL**1.1 REFERENCE**

Conform to General Requirements for Electrical Services.

1.2 DESCRIPTION OF WORKS

- 1.2.1 Contractor shall allow for the supply, installation, testing and commissioning of the complete CCTV system as specified and as shown on related drawings and as per SIRA requirements, SIRA preventative system manual 2020 and latest amendments as per February 2023 guidelines and regulations.

The system shall be designed for ultimate reliability and maximum operating efficiency using only the highest quality, fully field proven products.

The system components shall be of modular design to allow ease of installation, service, future expansion, upgrades and addition to the system.

The system components shall have operating environments to allow complete functionality at a temperature of 0 to 60 degree centigrade and a relative humidity of 90% (non-condensing).

All the cameras and recording devices shall be approved from SIRA. The VMS and SDK shall be approved from SIRA.

The system shall be capable of connecting to the authority or Video Guard connectivity as required by SIRA guidelines.

1.3 SCOPE OF WORK

Provide complete closed circuit surveillance system including but not limited to the components and equipment as specified in this document and as per SIRA/ Dubai Police Requirements.

The CCTV system shall provide visual images from the cameras located throughout the facility to the security control room .The cameras shall be connected to Hard Disk Digital Recorders for storing the video images.

The system shall consists of but not limited to the following:

- A. Fixed dome cameras, Outdoor bullet camera with enclosure IP66, ANPR cameras, cameras with video analytics on critical entry and exit points, building entry, elevator lobbies, amenities floor, critical utility areas, security control and FCC room etc
- B. All cameras shall be IR, Day/Night upto 40M, WDR with operating temperature upto 55Deg.
- C. 55" Bezel less industrial grade LED monitor
- D. Networked video servers with in built storage and fail – over servers with redundant storage.
- E. All other related accessories and equipment including camera mounts, various camera lenses, etc.
- F. Accessories for interfacing with the building fire alarm system.
- G. Video transmission / control cabling and linking cables
- H. Power supplies as required to the various system components.
- I. Latest release of multiuser version system software.

The control / guard room shall have a custom-built control desk to accommodate the colour LCD monitors and all related accessories.

At the time of installation and commissioning the contractor shall provide all additional components required to eliminate any noise and video signal attenuation problems found at site.

All specific CCTV camera locations shall be selected to provide a view of the maximum amount of space.

All CCTV system equipment supplied shall be compatible to include similar characteristics of bandwidth, impedance synchronization, polarity and sensitivity.

The system shall be an integrated CCTV control system based on an open protocol communications network. All cameras shall be IP based and work on POE.

The works under this specification shall include all shop drawings and submittals as required for the CCTV surveillance system. Further, the works shall include such items or details as testing, adjusting, identification tags on equipment, training and manuals for operation, training and maintenance of this system.

All the recording and monitoring, storage equipment and components shall be capable of operating from a power source of 240 volts AC. Single phase. 50 Hz. unless noted otherwise. All components and equipment shall be supplied with the appropriate power supply to operate from this voltage.

1.4 CODES AND STANDARDS

The following Codes and Standards to the extent specified herein, form a part of this specification. When an edition date is not indicated for a code or Standard the latest edition enforce at the time of proposal submitted shall apply. SIRA-Preventative system manual 2020 codes shall be applied.

Under writer laboratories listing or equally approved international listing laboratories.

1.5 SUBMITTALS

1.5.1 Contractor shall submit all items in accordance with the requirements of Division 1, Submittals and shall include but not be limited to the following:

- A. Model numbers of all components furnished on the job.
- B. Manufacturer's original catalog and technical data sheets for all components.
- C. Input power requirements for all system components.
- D. Complete engineered drawings indicating:
 - 1. Manufacturer model numbers and specifications.
 - 2. Dimensions, layouts, installation details.
 - 3. Point-to-point wiring diagrams for all system devices.
 - 4. Termination details for all system devices.
 - 5. Single-line system architecture drawings representing the entire system.
 - 6. Interfaces with all sub-systems.
- E. Four (4) sets of the Manufacturer's User's Manuals and Installation Manuals.
- F. Course outlines for each of the end user training programs. The course outlines shall include the course duration, location, pre-requisites, and a brief description of the subject matter.

- G. Sample for the proposed cameras.
- H. Certificate for the country of origin.

1.5.2 QUALITY ASSURANCE

- A. Manufacturer:
 - 1. Minimum of [10] years of experience in manufacture and design Video Surveillance Devices.
- B. Video Surveillance System:
 - 1. List certifying bodies (UL, etc.)
 - 2. Provide evidence of compliance upon request.
- C. Installer:
 - 1. Minimum of [5] years of experience installing Video Surveillance System.

2.0 PRODUCTS

2.1 SYSTEM DESCRIPTION

The entire system will be connected on a high speed Ethernet Network. All the Digital Disk Recorders should be Network compatible. Remote Monitoring Software should be able to integrate multiple digital recorders for operating the recorder by users remotely via LAN or internet connection.

The video and control signals shall be transmitted through the coaxial cabling system. 16 inputs and four outputs shall be available in each NVR for external devices, with user programmable input/output and camera selection for recording on such alarms.

Each Network Video Recorder shall be capable of supporting high capacity HDD storage of up to 12TB in each recorder or calculated as per SIRA requirement for 31 days storage. A storage calculation shall be submitted by the contractor at the time of technical submission as per the guidelines of SIRA for video recording storages.

The NVR shall be provided with suitable Hard disk capacity for live recording of all cameras simultaneously. Recording shall also be possible based on the motion. The multiplexer shall have own camera identification titling, electronic zoom, and provision for mechanical pan / tilt controls. Necessary correctional amplifiers and correctors shall be included in the system for a good quality distortion free live or recorded image on all monitors. The main control and monitoring shall include keypads with mouse, monitors for sequential or split screen monitoring. The display modes shall be selectable for monitors.

UPS Power supply for each camera. The UPS shall provide back up as specified by SIRA and shall be rack mounted type.

A remote Client shall be able to remotely call up any camera from the Switcher and display that requested video on the monitor screen. In the event that it is desirable to output a video signal to a Client's HD monitor screen, the requesting Client shall be able to command a composite video output from the Client's switcher, which will be displayed on an associated video monitor(s).

2.3 SYSTEM COMPONENTS (All cameras/recording devices and storage units shall comply SIRA guidelines 2020)

2.3.1 Indoor Fixed Camera (Dome)

The camera shall meet the following performance specifications. The camera resolution shall be minimum 4MP. The camera must comply the minimum specification as mentioned below.

Camera	Image Sensor	1/1.8" CMOS
Camera	Max. Resolution	2688 (H) × 1520 (V)
Camera	ROM	4 GB
Camera	RAM	2 GB
Camera	Scanning System	Progressive
Camera	Electronic Shutter Speed	Auto/Manual 1/3 s–1/100,000 s
Camera	Min. Illumination	0.001 Lux@F1.2 (Color, 30 IRE) 0.0002 Lux@F1.2 (B/W, 30 IRE) 0 Lux (Illuminator on)
Camera	S/N Ratio	> 56dB
Camera	Illumination Distance	40 m (131.23 ft.)
Camera	Illuminator On/Off Control	Auto; Manual; Zoom prio
Camera	Illuminator Number	3 (IR LED)

Camera	Pan/Tilt/Rotation Range	Pan: 0°–355° Tilt: 0°–80° Rotation: 0°–355°
Lens	Lens Type	Motorized vari-focal
Lens	Lens Mount	Module
Lens	Focal Length	2.7 mm–12 mm
Lens	Max. Aperture	F1.2
Lens	Field of View	H: 114°–48°; V: 59°–27°; D: 141°–55°
Lens	Iris Control	Auto
Lens	Close Focus Distance	1 m (3.28 ft.)
Lens	DORI Distance	D: 60.1–128.7 m (197.18–422.24 ft.) O: 24–51.5 m (78.74–168.96 ft.) R: 12–25.7 m (39.37–84.32 ft.) I: 6–12.9 m (19.69–42.32 ft.)
Smart Event	IVS	Abandoned object; missing object
Intelligence	IVS (Perimeter Protection)	Intrusion, tripwire, fast moving (the three functions support the classification and accurate detection of vehicle and human); loitering detection, people gathering, and parking detection
Intelligence	Face Detection	Face detection; track; snapshot; snapshot optimization; optimal face snapshot upload; face enhancement; face exposure; face attributes extraction including 6 attributes (gender, age, glasses, expressions, mask, and beard) and 8 expressions (angry, sad, disgusted, scared, surprised, calm, happy, confused); face snapshot set as face or one-inch photo; snapshot strategies (real-time snapshot, quality priority and optimization snapshot); face angle filter; optimization time setting.
Intelligence	Face Recognition	Face detection; track; snapshot; snapshot optimization; optimal face snapshot upload; face enhancement; face exposure; face attributes extraction including 6 attributes (gender, age, glasses, expressions, mask, and beard) and 8 expressions (angry, sad, disgusted, scared, surprised, calm, happy, confused); face snapshot set as face or one-inch photo; snapshot strategies

		(recognition priority and optimization snapshot); face angle filter; optimization time setting. Support 5 face databases; register one by one or in batches; support face recognition similarity settings and face comparison on the basis of 100,000 face databases.
Intelligence	People Counting	Tripwire people counting and people counting in area; generating and exporting report (day/month/year); queue management and exporting report (day/month); 4 rules can be set for tripwire, people counting in area and queue management.
Intelligence	Video Metadata	Motor vehicle, non-motor vehicle, face, human detection; track; snapshot; snapshot optimization; optimal face snapshot upload. Motor vehicle attributes: license plate, vehicle type, vehicle color, vehicle logo and other attributes: seatbelt, smoking, calling. Non-motor vehicle attributes: type, color, number of people, top type and color, helmet. Human body attributes: top/bottom type and color, bag, hat and umbrella. Face attributes: gender, age, expressions, glasses, face mask and beard.
Intelligence	Smart Search	Work together with Smart NVR to perform refine intelligent search, event extraction and merging to event videos
Video	Video Compression	H.265; H.264; H.264H; MJPEG (Only supported by the sub stream)
Video	Smart Codec	Smart H.265+; Smart H.264+
Video	Video Frame Rate	Main Stream: 2688 × 1520@(1-50/60 fps) sub stream: 704 × 576@(1-25 fps)/704 × 480@(1-30 fps) third stream: 1920 × 1080@(1-25/30 fps) fourth stream: 1920 × 1080@(1-25/30 fps) fifth stream: 704 × 576@(1-25 fps)/704 × 480@(1-30 fps) *The values above are the max. frame rates of each stream; for multiple streams, the values will be subjected to the total encoding capacity.
Video	Stream Capability	5 streams
Video	Resolution	4M (2688 × 1520); 3M (2304 × 1296); 1080p (1920 × 1080); 1.3M (1280 × 960); 720p (1280 × 720); D1 (704 × 576/704 × 480); VGA (640 × 480); CIF (352 × 288/352 × 240)
Video	Bit Rate Control	CBR/VBR
Video	Video Bit Rate	H.264: 32 kbps-10240 kbps H.265: 12 kbps-9984 kbps
Video	Day/Night	Auto(ICR)/Color/B/W
Video	BLC	Yes
Video	HLC	Yes
Video	WDR	140 dB
Video	Scene Self-adaptation (SSA)	Yes
Video	White Balance	Auto; natural; street lamp; outdoor; manual; regional custom
Video	Gain Control	Manual; Auto
Video	Noise Reduction	3D NR
Video	Motion Detection	OFF/ON (4 areas, rectangular)
Video	Region of Interest (RoI)	Yes (4 areas)
Video	Image Stabilization	Electronic Image Stabilization (EIS)
Video	Smart Illumination	Yes
Video	Defog	Yes
Video	Image Rotation	0°/90°/180°/270° (Support 90°/270° with 2688 × 1520 resolution)

		and lower)
Video	Mirror	Yes
Video	Privacy Masking	4 areas
Audio	Audio Compression	PCM; G.711a; G.711Mu; G.726; G.723
Alarm	Alarm Event	No SD card; SD card full; SD card error; SD card durability warning; network disconnection; IP conflict; illegal access; security exception; motion detection; video tampering; tripwire; intrusion; fast moving; abandoned object; missing object; loitering detection; people gathering; parking detection; scene changing; audio detection; defocus detection; external alarm; face detection; video metadata; people counting in area; people counting; queue management
Network	Network Port	RJ-45 (10/100/1000 Base-T)
Network	SDK and API	Yes
Network	Network Protocol	IPv4; IPv6; HTTP; TCP; UDP; ARP; RTP; RTSP; RTCP; RTMP; SMTP; FTP; SFTP; DHCP; DNS; DDNS; QoS; UPnP; NTP; Multicast; ICMP; IGMP; NFS; SAMBA; PPPoE; SNMP
Network	Interoperability	ONVIF (Profile S/Profile G/Profile T); CGI; Milestone; P2P
Network	User/Host	20 (Total bandwidth: 80M)
Network	Storage	FTP; SFTP; Micro SD card (support max. 256 GB); NAS
Network	Browser	IE: IE9 and later Chrome: Chrome 42 and later Firefox: Firefox 48.0.2 and later
Network	Management Software	Smart PSS; DSS; DMSS
Network	Mobile Client	iOS; Android
Network	Cyber Security	Video encryption; Firmware encryption; Configuration encryption; Digest; WSSE; Account lockout; Security logs; IP/MAC filtering; Generation and importing of X.509 certification; syslog; HTTPS; 802.1x; Trusted boot; Trusted execution; Trusted upgrade
Port	RS-485	1 (baud rate range: 1200 bps–115200 bps)
Port	Audio Input	1 channel (3-pin terminal with 2.54mm clearance, with audio cable)
Port	Audio Output	1 channel (3-pin terminal with 2.54mm clearance, with audio cable)
Port	Alarm Input	3 channels in: 5mA 3V–5V DC
Port	Alarm Output	2 channels out: 1,000mA 30V DC/500mA 50V AC
Power	Power Supply	12 VDC/24 VAC/PoE+ (802.3at)
Power	Power Consumption	Basic power consumption: 6 W (12 VDC); 5.52 W (24 VAC); 5.24 W (PoE) Max. power consumption (Max. stream + intelligent function + IR LED + focus): 14.5 W (12 VDC); 13 W (24 VAC); 14.8 W (PoE)
Environment	Operating Temperature	-40°C to +65°C (-40 °F to +149 °F)
Environment	Operating Humidity	≤95%
Environment	Storage Temperature	-40°C to +70°C (-40°F to +158°F)
Environment	Protection	IP67, IK10, Anti-Corrosion Protection: NEMA 4X (optional)
Structure	Casing Material	Metal + plastic
Structure	Product	129.1 mm × Φ157.9 mm (5.08" × Φ6.22")

GCE 596: (G+4P+28+RF) Residential Development on Plot No. 6117242 in Bukadra, Dubai – U.A.E.

PARKWAY BY PRESTIGE ONE

Electrical Section 20: IP Based CCTV Monitoring System Rev-01

E - 7

e	Dimensions	
Structure	Net Weight	1.5 kg (3.31 lb.)
Structure	Gross Weight	1.6 kg (3.53 lb.)

Indoor Identification Cameras

Camera	Image Sensor	1/8" CMOS
Camera	Max. Resolution	2688 (H) × 1520 (V)
Camera	ROM	4 GB
Camera	RAM	1 GB
Camera	Scanning System	Progressive
Camera	Electronic Shutter Speed	Auto/Manual 1/3 s–1/100,000 s
Camera	Min. Illumination	0.0006 lux@F1.2 (Color, 30 IRE) 0.0003 lux@F1.2 (B/W, 30 IRE) 0 lux (Illuminator on)
Camera	S/N Ratio	> 56 dB
Camera	Illumination Distance	40 m (131.23 ft) (IR)
Camera	Illuminator On/Off Control	Auto;Manual;Zoomprio
Camera	Illuminator Number	3 (IR LED)
Camera	Angle Adjustment	Pan: 0°–355° Tilt: 0°–80° Rotation: 0°–355°
Lens	Lens Type	Motorized vari-focal
Lens	Lens Mount	φ14
Lens	Focal Length	2.7 mm–12.0 mm
Lens	Max. Aperture	F1.2
Lens	Field of View	H: 114°–48°; V: 59°–27°; D: 143°–56°
Lens	Iris Control	Auto
Lens	Iris Control Type	P-Iris
Lens	Close Focus Distance	1.75 m (5.74 ft)
Lens	DORI Distance	W: D: 60.4 m; O: 24.2 m; R: 12.1 m; I: 6.0 m (D: 198.16 ft; O: 79.40 ft; R: 39.70 ft; I: 19.69 ft) T: D: 128.7 m; O: 51.5 m; R: 25.7 m; I: 12.9 m (D: 425.52 ft; O: 168.96 ft; R: 84.32 ft; I: 42.32 ft)
Intelligence	IVS (Perimeter Protection)	Intrusion, tripwire, fast moving (the three functions support the classification and accurate detection of vehicle and human); loitering detection, people gathering, and parking detection
Intelligence	Smart Object Detection	Smart abandoned object; smart missing object
Intelligence	SMD 3.0	Less false alarm, longer detection distance
Intelligence	AI SSA	Yes
Intelligence	Face Detection	Face detection; track; snapshot; snapshot optimization; optimal face snapshot upload; face enhancement; face exposure; face attributes extraction including 6 attributes and 8 expressions; face snapshot set as face, one-inch photo or custom; snapshot strategies (real-time snapshot, quality priority and optimization snapshot); face angle filter; optimization time setting.
Intelligence	People Counting	Tripwire people counting, generating and exporting report

nce		(day/week/month/year); people counting in area and queue management, generating and exporting report (day/week/month); 4 rules can be set for tripwire, people counting in area and queue management.
Intelligence	Heat Map	Yes
Intelligence	Video Metadata	Motor vehicle, non-motor vehicle, face, human body detection; track; snapshot; snapshot optimization; optimal face snapshot upload. Motor vehicle attributes: vehicle type, vehicle color, vehicle logo and other attributes : seatbelt, smoking, calling. Non-motor vehicle attributes: type, color, number of people, top type and color, hat. Human body attributes: gender, top/bottom type and color, bag, hat and umbrella. Face attributes: gender, age, expressions, glasses, face mask and beard.
Intelligence	Smart Search	Work together with Smart NVR to perform refine intelligent search, event extraction and merging to event videos
Intelligence	Smart Sound Detection	Yes
Video	Video Compression	H.265; H.264; H.264H; H.264B; MJPEG (Only supported by the sub stream)
Video	Smart Codec	Smart H.265+; Smart H.264+
Video	AI Coding	AI H.265; AI H.264
Video	Video Frame Rate	Main stream: 2688 × 1520@(1-25/30 fps) Sub stream: D1@(1-25/30 fps) Third stream: 1080p@(1-25/30 fps) Fourth stream: 720p@(1-25/30 fps) *The values above are the max. frame rates of each stream; for multiple streams, the values will be subjected to the total encoding capacity.
Video	Stream Capability	4 streams
Video	Resolution	4M (2688 × 1520); 3.6M (2560 × 1440); 3M (2048 × 1536); 3M (2304 × 1296); 1080p (1920 × 1080); 1.3M (1280 × 960); 720p (1280 × 720); D1 (704 × 576/704 × 480); CIF (352 × 288/352 × 240)
Video	Bit Rate Control	CBR/VBR
Video	Video Bit Rate	H.264: 32 kbps-16384 kbps; H.265: 12 kbps-11008 kbps
Video	Day/Night	Auto(ICR)/Color/B/W
Video	BLC	Yes
Video	HLC	Yes
Video	WDR	140 dB
Video	Scene Self-adaptation (SSA)	Yes
Video	White Balance	Auto; natural; street lamp; outdoor; manual; regional custom
Video	Gain Control	Auto
Video	Noise Reduction	3D NR
Video	Motion Detection	OFF/ON (4 areas, rectangular)
Video	Region of Interest (RoI)	Yes (4 areas)
Video	Image Stabilization	Electronic Image Stabilization (EIS)
Video	Defog	Yes
Video	Image Rotation	0°/90°/180°/270° (Support 90°/270° with 4M (2688 × 1520) resolution and lower)
Video	Mirror	Yes
Video	Privacy Masking	8 areas
Audio	Built-in MIC	Yes

Audio	Audio Compression	G.711a; G.711Mu; PCM; G.726; G.723
Alarm	Alarm Event	No SD card; SD card full; SD card error; network disconnection; IP conflict; illegal access; motion detection; video tampering; tripwire; intrusion; fast moving; abandoned object; missing object; loitering detection; people gathering; parking detection; scene changing; audio detection; voltage detection; external alarm; face detection; video metadata; SMD; people counting in area; stay detection; people counting; people number error detection; queue people number alarm; queue time alarm; security exception
Network	Network Port	RJ-45 (10/100 Enhancement Base-T)
Network	SDK and API	Yes
Network	Network Protocol	IPv4; IPv6; HTTP; TCP; UDP; ARP; RTP; RTSP; RTCP; RTMP; SMTP; FTP; SFTP; DHCP; DNS; DDNS; QoS; UPnP; NTP; Multicast; ICMP; IGMP; NFS; SAMBA; PPPoE; SNMP; P2P
Network	Interoperability	ONVIF (Profile S/Profile G/Profile T); CGI
Network	User/Host	20 (Total bandwidth: 80 M)
Network	Storage	FTP; SFTP; Micro SD card (support max. 512 GB); NAS
Network	Browser	IE: IE 9 and later; Chrome: Chrome 102 and later; Firefox: Firefox 88 and later
Network	Management Software	SmartPSS Lite; DSS; DMSS
Network	Mobile Client	iOS;Android
Network	Cyber Security	Video encryption;Firmware encryption;Configuration encryption;Digest;WSSE;Account lockout;Security logs;IP/MAC filtering;Generation and importing of X.509 certification;syslog;HTTPS;802.1x;Trusted boot;Trusted execution;Trusted upgrade
Port	RS-485	1 (baud rate range:1200 bps–115200 bps)
Port	Audio Input	1 channel (terminal)
Port	Audio Output	1 channel (terminal)
Port	Alarm Input	3 channels in: wet contact, 5mA 3V–5V DC
Port	Alarm Output	2 channels out: dry contact, 1,000mA 30V DC/500mA 50V AC
Port	Analog Output	1 channel (CVBS output: built-in 2 pin (testing))
Power	Power Supply	12 VDC/24 VAC/PoE (802.3af)/ePoE
Power	Power Consumption	Basic: 4.1 W (12 VDC); 5.2 W (24 VAC); 5 W (PoE) Max. (Focus+WDR+ Intelligence on+IR on): 12.9 W (12 VDC); 15.8 W (24 VAC); 15.9 W (PoE) Heating: 3 W (12 VDC); 3.8 W (24 VAC); 3.9 W (PoE)
Environment	Operating Temperature	-40°C to +60°C (-40 °F to +140 °F)
Environment	Operating Humidity	≤95%
Environment	Storage Temperature	-40°C to +60°C (-40°F to +140°F)
Environment	Storage Humidity	10%–95% (RH), non-condensing
Environment	Protection	IP67; IK10
Structure	Casing Material	Metal

Indoor Recognition cameras

Camera	Image Sensor	1/3" CMOS
Camera	Max. Resolution	2688 (H) × 1520 (V)
Camera	ROM	128 MB
Camera	RAM	512 MB
Camera	Scanning System	Progressive
Camera	Electronic Shutter Speed	Auto/Manual 1/3 s–1/100,000 s
Camera	Min. Illumination	0.005 lux@F1.5 (Color, 30 IRE) 0.0005 lux@F1.5 (B/W, 30 IRE) 0 lux (Illuminator on)
Camera	S/N Ratio	> 56 dB
Camera	Illumination Distance	40 m (131.2 ft) (IR)
Camera	Illuminator On/Off Control	Auto;Manual
Camera	Illuminator Number	3 (IR LED)
Camera	Angle Adjustment	Pan: 0°–355° Tilt: 0°–80° Rotation: 0°–355°
Lens	Lens Type	Motorized vari-focal
Lens	Lens Mount	φ14
Lens	Focal Length	2.7 mm–13.5 mm
Lens	Max. Aperture	F1.5
Lens	Field of View	H: 104°–29°; V: 54°–16°; D: 125°–34°
Lens	Iris Control	Fixed
Lens	Close Focus Distance	0.8 m (2.62 ft)
Lens	DORI Distance	W: D: 64.0 m; O: 25.6 m; R: 12.8 m; I: 6.4 m (D: 209.97 ft; O: 83.99 ft; R: 41.99 ft; I: 21.00 ft) T: D: 210.0 m; O: 84.0 m; R: 42.0 m; I: 21.0 m (D: 688.98 ft; O: 275.59 ft; R: 137.80 ft; I: 68.90 ft)
Intelligence	IVS (Perimeter Protection)	Intrusion, tripwire (the two functions support the classification and accurate detection of vehicle and human)
Intelligence	AI SSA	Yes
Intelligence	Quick Pick	With AI NVR, quickly pick up the human/vehicle targets that users are interested in from SMD events
Intelligence	Smart Search	Work together with Smart NVR to perform refine intelligent search, event extraction and merging to event videos
Video	Video Compression	H.265; H.264; H.264H; H.264B; MJPEG (Only supported by the sub stream)
Video	Smart Codec	Smart H.265+; Smart H.264+
Video	AI Coding	AI H.265; AI H.264
Video	Video Frame Rate	Main stream: 2688 × 1520@(1–25/30 fps) Sub stream: 704 × 576@(1–25 fps)/704 × 480@(1–30 fps) Third stream: 1920 × 1080@(1–25/30 fps) *The values above are the max. frame rates of each stream; for multiple streams, the values will be subjected to the total encoding capacity.
Video	Stream Capability	3 streams
Video	Resolution	4M (2688 × 1520); 3.6M (2560 × 1440); 3M (2048 × 1536); 3M (2304 × 1296); 1080p (1920 × 1080); 720p (1280 × 720); D1 (704

		× 576/704 × 480); VGA (640 × 480); CIF (352 × 288/352 × 240)
Video	Bit Rate Control	CBR/VBR
Video	Video Bit Rate	H.264: 3 kbps–8192 kbps H.265: 3 kbps–8192 kbps
Video	Day/Night	Auto(ICR)/Color/B/W
Video	BLC	Yes
Video	HLC	Yes
Video	WDR	120 dB
Video	Scene Self-adaptation (SSA)	Yes
Video	White Balance	Auto; natural; street lamp; outdoor; manual; regional custom
Video	Gain Control	Manual;Auto
Video	Noise Reduction	3D NR
Video	Motion Detection	OFF/ON (4 areas, rectangular)
Video	Region of Interest (RoI)	Yes (4 areas)
Video	Image Stabilization	Electronic Image Stabilization (EIS)
Video	Smart Illumination	Yes
Video	Defog	Yes
Video	Image Rotation	0°/90°/180°/270° (Support 90°/270° with 2688 × 1520 resolution and lower)
Video	Mirror	Yes
Video	Privacy Masking	4 areas
Audio	Built-in MIC	Yes
Audio	Audio Compression	G.711a; G.711Mu; PCM; G.726; G.723
Alarm	Alarm Event	No SD card; SD card full; SD card error; network disconnection; IP conflict; illegal access; motion detection; video tampering; tripwire; intrusion; scene changing; audio detection; voltage detection; defocus detection; external alarm;SMD; security exception
Network	Network Port	RJ-45 (10/100 Base-T)
Network	SDK and API	Yes
Network	Network Protocol	IPv4; IPv6; HTTP; TCP; UDP; ARP; RTP; RTSP; RTCP; RTMP; SMTP; FTP; SFTP; DHCP; DNS; DDNS; QoS; UPnP; NTP; Multicast; ICMP; IGMP; NFS; SAMBA; PPPoE; SNMP
Network	Interoperability	ONVIF (Profile S/Profile G/Profile T); CGI; P2P; Milestone; Genetec
Network	User/Host	20 (Total bandwidth: 64 M)
Network	Storage	FTP; SFTP; Micro SD card (support max. 256 GB); NAS
Network	Browser	IE: IE 11 Chrome Firefox
Network	Management Software	Smart PSS;DSS;DMSS
Network	Mobile Client	iOS;Android
Network	Cyber Security	Video encryption;Firmware encryption;Configuration encryption;Digest;WSSE;Account lockout;Security logs;IP/MAC filtering;Generation and importing of X.509 certification;syslog;HTTPS;802.1x;Trusted boot;Trusted execution;Trusted upgrade
Power	Power Supply	12 VDC/PoE

Power	Power Consumption	Basic: 2.8 W (12 VDC); 3.6 W (PoE) Max. (WDR+IR intensity+Intelligence on): 7.9 W (12 VDC); 9.2 W (PoE)
Environment	Operating Temperature	-30 °C to +60 °C (-22 °F to +140 °F)
Environment	Operating Humidity	≤ 95%
Environment	Protection	IP67; IK10
Structure	Casing Material	Metal
Structure	Product Dimensions	94 mm × Φ122 mm (3.7" × Φ4.8")
Structure	Net Weight	713 g (1.6 lb)
Structure	Gross Weight	986 g (2.2 lb)
Structure	Installation	Ceiling mount; junction mount

Lenses: High quality coated optics attached to the camera and designed specifically for CCTV camera applications. Units shall have neutral-density spot filter. All lens shall be auto-iris with motorized vary-focal lens.

2.3.2 Outdoor Fixed Camera – (IR cameras)

Camera and Lens Specification same as 2.3.1

The Outdoor cameras shall be provided with IP66 rated Housing which should be Weatherproof, Vandal proof and tamper proof. Using the Ceiling/Wall Mount Brackets, the housing should be capable of installing on the Ceiling or Wall based on the site requirement. The Housing should be provided with a sunshield for exposure to direct sunlight and anti-condensation heater/blower to ensure high level of camera picture quality.

The camera resolution shall be minimum 4MP. 4MP stands for 4 Million Pixels. 4MP is 2560 x 1440 or 3,686,400 total pixels

Camera	Image Sensor	1/1.8" CMOS
Camera	Max. Resolution	2688 (H) × 1520 (V)
Camera	ROM	128 MB
Camera	RAM	1 GB
Camera	Scanning System	Progressive
Camera	Electronic Shutter Speed	Auto/Manual 1/3 s–1/100,000 s
Camera	Min. Illumination	0.003 lux@F1.8 (Color, 30 IRE) 0.0003 lux@F1.8 (B/W, 30 IRE) 0 lux (Illuminator on)
Camera	S/N Ratio	> 56 dB
Camera	Illumination Distance	60 m (196.85 ft.) (IR)
Camera	Illuminator On/Off Control	Auto
Camera	Illuminator Number	4 (IR LED)

Camera	Pan/Tilt/Rotation Range	Pan: 0°–360° Tilt: 0°–90° Rotation: 0°–360°
Lens	Lens Type	Motorized vari-focal
Lens	Lens Mount	φ14
Lens	Focal Length	2.7 mm–12 mm
Lens	Max. Aperture	F1.8
Lens	Field of View	H: 114°–47°; V: 62°–26°; D: 136°–56°
Lens	Iris Control	Auto
Lens	Close Focus Distance	1.2 m (3.94 ft.)
Lens	DORI Distance	W: D: 57.5 m; O: 23.0 m; R: 11.5 m; I: 5.7 m (D: 188.65 ft.; O: 75.46 ft.; R: 37.73 ft.; I: 18.70 ft.) T: D: 130.7 m; O: 52.3 m; R: 26.1 m; I: 13.1 m (D: 428.81 ft.; O: 171.59 ft.; R: 85.63 ft.; I: 42.98 ft.)
Smart Event	IVS	Abandoned object; missing object
Smart Event	Heat Map	Yes
Intelligence	IVS (Perimeter Protection)	Intrusion, tripwire, fast moving (the three functions support the classification and accurate detection of vehicle and human); loitering detection, people gathering, and parking detection
Intelligence	SMD 3.0	Less false alarm, longer detection distance
Intelligence	Face Detection	Face detection; track; snapshot; snapshot optimization; optimal face snapshot upload; face enhancement; face exposure; face attributes extraction including 6 attributes (gender, age, glasses, expressions, mask, and beard) and 8 expressions (angry, sad, disgusted, scared, surprised, calm, happy, confused); face snapshot set as face or one-inch photo; snapshot strategies (real-time snapshot, quality priority and optimization snapshot); face angle filter; optimization time setting
Intelligence	People Counting	Tripwire people counting and people counting in area; generating and exporting report (day/month/year); queue management; 4 rules can be set for tripwire, people counting in area and queue management
Intelligence	Smart Search	Work together with Smart NVR to perform refine intelligent search, event extraction and merging to event videos
Video	Video Compression	H.265; H.264; H.264H; H.264B; MJPEG (Only supported by the sub stream)
Video	Smart Codec	Smart H.265+; Smart H.264+
Video	AI Coding	AI H.265; AI H.264
Video	Video Frame Rate	Main stream: 2688 × 1520@(1 fps–25/30 fps) Sub stream: D1@(1 fps–50/60 fps) Third stream: 1080p@(1 fps–50/60 fps) *The values above are the max. frame rates of each stream; for multiple streams, the values will be subjected to the total encoding capacity.
Video	Stream Capability	3 streams
Video	Resolution	2688 × 1520; 2304 × 1296; 1080p (1920 × 1080); 1.3M (1280 × 960); 720p (1280 × 720); D1 (704 × 576/704 × 480); VGA (640 × 480); CIF (352 × 288/352 × 240)
Video	Bit Rate Control	CBR/VBR
Video	Video Bit Rate	H.264: 32 kbps–8192 kbps H.265: 32 kbps–8192 kbps
Video	Day/Night	Auto(ICR)/Color/B/W
Video	BLC	Yes

Video	HLC	Yes
Video	WDR	140 dB
Video	Scene Self-adaptation (SSA)	Yes
Video	White Balance	Auto; natural; street lamp; outdoor; manual; regional custom
Video	Gain Control	Auto
Video	Noise Reduction	3D NR
Video	Motion Detection	OFF/ON (4 areas, rectangular)
Video	Region of Interest (RoI)	Yes (4 areas)
Video	Image Stabilization	Electronic Image Stabilization (EIS)
Video	Smart Illumination	Yes
Video	Defog	Yes
Video	Image Rotation	0°/90°/180°/270°
Video	Mirror	Yes
Video	Privacy Masking	8 areas
Audio	Audio Compression	PCM; G.711a; G.711Mu; G.726; G.723
Alarm	Alarm Event	No SD card; SD card full; SD card error; network disconnection; IP conflict; illegal access; motion detection; video tampering; defocus detection; tripwire; intrusion; fast moving; abandoned object; missing object; loitering detection; people gathering; parking detection; scene changing; input error; voltage detection; face detection; people counting in area; stay detection; people counting; people number error detection; queue management; intensity change
Network	Network Port	RJ-45 (10/100 Enhancement Base-T)
Network	SDK and API	Yes
Network	Network Protocol	IPv4; IPv6; HTTP; TCP; UDP; ARP; RTP; RTSP; RTCP; RTMP; SMTP; FTP; SFTP; DHCP; DNS; DDNS; QoS; UPnP; NTP; Multicast; ICMP; IGMP; NFS; SAMBA; PPPoE; SNMP; Bonjour
Network	Interoperability	ONVIF (Profile S/Profile G/Profile T); CGI; P2P; Milestone; Genetec
Network	User/Host	20 (Total bandwidth: 88 M)
Network	Storage	FTP; SFTP; Micro SD card (support max. 256 GB); NAS
Network	Browser	IE: IE8 and later; Chrome; Firefox; Safari: 12 and later
Network	Management Software	Smart PSS, DSS, DMSS
Network	Mobile Client	iOS; Android
Network	Cyber Security	Video encryption; Firmware encryption; Configuration encryption; Digest; WSSE; Account lockout; Security logs; IP/MAC filtering; Generation and importing of X.509 certification; syslog; HTTPS; 802.1x; Trusted boot; Trusted execution; Trusted upgrade
Port	Audio Input	1 channel (RCA port)
Port	Audio Output	1 channel (RCA port)
Port	Alarm Input	2 channels in: 5mA 3V–5V DC
Port	Alarm Output	1 channel out: 300mA 12V DC
Power	Power Supply	12 VDC; PoE

Power	Power Consumption	Basic: 3 W (12 VDC); 3.7 W (PoE) Max. (max. stream + IR LED + focus): 9.9 W (12 VDC); 10.9 W (PoE)
Environment	Operating Temperature	-30 °C to +60 °C (-22 °F to +140 °F)
Environment	Operating Humidity	≤ 95%
Environment	Storage Temperature	-40 °C to +60 °C (-40 °F to +140 °F)
Environment	Protection	IP67;IK10
Structure	Casing Material	Metal + plastic
Structure	Product Dimensions	273.2 mm × 95 mm × 95 mm (10.76" × 3.74" × 3.74") (L × W × H)
Structure	Net Weight	1110 g (2.45 lb.)
Structure	Gross Weight	1510 g (3.33 lb.)

Outdoor Identification camera.

Camera	Image Sensor	1/8" CMOS
Camera	Max. Resolution	2688 (H) × 1520 (V)
Camera	ROM	4 GB
Camera	RAM	1 GB
Camera	Scanning System	Progressive
Camera	Electronic Shutter Speed	Auto/Manual 1/3 s–1/100,000 s
Camera	Min. Illumination	0.0006 lux@F1.2 (Color, 30 IRE) 0.0003 lux@F1.2 (B/W, 30 IRE) 0 lux (Illuminator on)
Camera	S/N Ratio	> 56 dB
Camera	Illumination Distance	60 m (196.85 ft) (IR)
Camera	Illuminator On/Off Control	Auto;Manual;Zoomprio
Camera	Illuminator Number	4 (IR LED)
Camera	Angle Adjustment	Pan: 0-360° Tilt: 0-90° Rotation: 0-360°
Lens	Lens Type	Motorized vari-focal
Lens	Lens Mount	φ14
Lens	Focal Length	2.7 mm–12.0 mm
Lens	Max. Aperture	F1.2
Lens	Field of View	H: 114°–48°; V: 59°–27°; D: 143°–56°
Lens	Iris Control	Auto
Lens	Iris Control Type	P-Iris

Lens	Close Focus Distance	1.75 m (5.74 ft)
Lens	DORI Distance	W: D: 60.4 m; O: 24.2 m; R: 12.1 m; I: 6.0 m (D: 198.16 ft; O: 79.40 ft; R: 39.70 ft; I: 19.69 ft) T: D: 128.7 m; O: 51.5 m; R: 25.7 m; I: 12.9 m (D: 425.52 ft; O: 168.96 ft; R: 84.32 ft; I: 42.32 ft)
Intelligence	EPTZ	Yes
Intelligence	IVS (Perimeter Protection)	Intrusion, tripwire, fast moving (the three functions support the classification and accurate detection of vehicle and human); loitering detection, people gathering, and parking detection
Intelligence	Smart Object Detection	Smart abandoned object; smart missing object
Intelligence	SMD 3.0	Less false alarm, longer detection distance
Intelligence	AI SSA	Yes
Intelligence	Face Detection	Face detection; track; snapshot; snapshot optimization; optimal face snapshot upload; face enhancement; face exposure; face attributes extraction including 6 attributes and 8 expressions; face snapshot set as face, one-inch photo or custom; snapshot strategies (real-time snapshot, quality priority and optimization snapshot); face angle filter; optimization time setting.
Intelligence	People Counting	Tripwire people counting, generating and exporting report (day/week/month/year); people counting in area and queue management, generating and exporting report (day/week/month); 4 rules can be set for tripwire, people counting in area and queue management.
Intelligence	Heat Map	Yes
Intelligence	Video Metadata	Motor vehicle, non-motor vehicle, face, human body detection; track; snapshot; snapshot optimization; optimal face snapshot upload. Motor vehicle attributes: vehicle type, vehicle color, vehicle logo and other attributes : seatbelt, smoking, calling. Non-motor vehicle attributes: type, color, number of people, top type and color, hat. Human body attributes: gender, top/bottom type and color, bag, hat and umbrella. Face attributes: gender, age, expressions, glasses, face mask and beard.
Intelligence	Smart Search	Work together with Smart NVR to perform refine intelligent search, event extraction and merging to event videos
Intelligence	Smart Sound Detection	Yes
Video	Video Compression	H.265; H.264; H.264H; H.264B; MJPEG (Only supported by the sub stream)
Video	Smart Codec	Smart H.265+; Smart H.264+
Video	AI Coding	AI H.265; AI H.264
Video	Video Frame Rate	Main stream: 2688 × 1520@ (1-25/30 fps) Sub stream: D1@ (1-25/30 fps) Third stream: 1080p@ (1-25/30 fps) Fourth stream: 720p@ (1-25/30 fps) *The values above are the max. frame rates of each stream; for multiple streams, the values will be subjected to the total encoding capacity.
Video	Stream	4 streams

	Capability	
Video	Resolution	4M (2688 × 1520); 3.6M (2560 × 1440); 3M (2048 × 1536); 3M (2304 × 1296); 1080p (1920 × 1080); 1.3M (1280 × 960); 720p (1280 × 720); D1 (704 × 576/704 × 480); CIF (352 × 288/352 × 240)
Video	Bit Rate Control	CBR/VBR
Video	Video Bit Rate	H.264: 32 kbps–16384 kbps; H.265: 12 kbps–11008 kbps
Video	Day/Night	Auto(ICR)/Color/B/W
Video	BLC	Yes
Video	HLC	Yes
Video	WDR	140 dB
Video	Scene Self-adaptation (SSA)	Yes
Video	White Balance	Auto; natural; street lamp; outdoor; manual; regional custom
Video	Gain Control	Auto
Video	Noise Reduction	3D NR
Video	Motion Detection	OFF/ON (4 areas, rectangular)
Video	Region of Interest (RoI)	Yes (4 areas)
Video	Image Stabilization	Electronic Image Stabilization (EIS)
Video	Defog	Yes
Video	Image Rotation	0°/90°/180°/270° (Support 90°/270° with 4M (2688×1520) resolution and lower)
Video	Mirror	Yes
Video	Privacy Masking	8 areas
Audio	Built-in MIC	Yes
Audio	Audio Compression	G.711a; G.711Mu; PCM; G.726; G.723
Alarm	Alarm Event	No SD card; SD card full; SD card error; network disconnection; IP conflict; illegal access; motion detection; video tampering; tripwire; intrusion; fast moving; abandoned object; missing object; loitering detection; people gathering; parking detection; scene changing; audio detection; voltage detection; external alarm; face detection; video metadata; SMD; people counting in area; stay detection; people counting; people number error detection; queue people number alarm; queue time alarm; security exception
Network	Network Port	RJ-45 (10/100 Enhancement Base-T)
Network	SDK and API	Yes
Network	Network Protocol	IPv4; IPv6; HTTP; TCP; UDP; ARP; RTP; RTSP; RTCP; RTMP; SMTP; FTP; SFTP; DHCP; DNS; DDNS; QoS; UPnP; NTP; Multicast; ICMP; IGMP; NFS; SAMBA; PPPoE; SNMP; P2P
Network	Interoperability	ONVIF (Profile S/Profile G/Profile T); CGI

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Network	User/Host	20 (Total bandwidth: 80 M)
Network	Storage	FTP; SFTP; Micro SD card (support max. 512 GB); NAS
Network	Browser	IE: IE 9 and later Chrome: Chrome 102 and later Firefox: Firefox 88 and later
Network	Management Software	SmartPSS Lite; DSS; DMSS
Network	Mobile Client	iOS;Android
Network	Cyber Security	Video encryption;Firmware encryption;Configuration encryption;Digest;WSSE;Account lockout;Security logs;IP/MAC filtering;Generation and importing of X.509 certification;syslog;HTTPS;802.1x;Trusted boot;Trusted execution;Trusted upgrade
Port	RS-485	1 (baud rate range:1200 bps–115200 bps)
Port	Audio Input	1 channel (terminal)
Port	Audio Output	1 channel (terminal)
Port	Alarm Input	3 channels in: wet contact, 5mA 3V–5V DC
Port	Alarm Output	2 channels out: dry contact, 1,000mA 30V DC/500mA 50V AC
Port	Analog Output	1 channel (CVBS output: BNC)
Power	Power Supply	12 VDC/24 VAC/PoE (802.3af)/ePoE
Power	Power Consumption	Basic: 3.7 W (12 VDC); 4.7 W (24 VAC); 5 W (PoE) Max. (Focus+WDR+ Intelligence on+IR on): 12.9 W (12 VDC); 15.1 W (24 VAC); 16.1 W (PoE) Heating: 3 W (12 VDC); 3.6 W (24 VAC); 3.8 W (PoE)
Environment	Operating Temperature	-40°C to +60°C (-40 °F to +140 °F)
Environment	Operating Humidity	≤95%
Environment	Storage Temperature	-40°C to +60°C (-40°F to +140°F)
Environment	Storage Humidity	10%–95% (RH), non-condensing
Environment	Protection	IP67; IK10
Structure	Casing Material	Metal + plastic
Structure	Product Dimensions	345.6 mm × 133.8 mm × 135.1 mm (13.61" × 5.27" × 5.32") (L × W × H)
Structure	Net Weight	1.6 kg (3.53 lb)
Structure	Gross Weight	2.4 kg (5.29 lb)
Structure	Installation	Wall mount; pole mount; ceiling mount

Outdoor Recognition cameras

Camera	Image Sensor	1/3" CMOS
Camera	Max. Resolution	2688 (H) × 1520 (V)
Camera	ROM	128 MB
Camera	RAM	512 MB
Camera	Scanning System	Progressive
Camera	Electronic Shutter Speed	Auto/Manual 1/3 s–1/100,000 s
Camera	Min. Illumination	0.005 lux@F1.5 (Color, 30 IRE) 0.0005 lux@F1.5 (B/W, 30 IRE) 0 lux (Illuminator on)
Camera	S/N Ratio	> 56 dB
Camera	Illumination Distance	60 m (196.85 ft) (IR)
Camera	Illuminator On/Off Control	Auto;Manual
Camera	Illuminator Number	4 (IR LED)
Camera	Angle Adjustment	Pan: 0°–360° Tilt: 0°–90° Rotation: 0°–360°
Lens	Lens Type	Motorized vari-focal
Lens	Lens Mount	φ14
Lens	Focal Length	2.7 mm–13.5 mm
Lens	Max. Aperture	F1.5
Lens	Field of View	H: 104°–29°; V: 54°–16°; D: 125°–34°
Lens	Iris Control	Fixed
Lens	Close Focus Distance	0.8 m (2.62 ft)
Lens	DORI Distance	W: D: 64.0 m; O: 25.6 m; R: 12.8 m; I: 6.4 m (D: 209.97 ft; O: 83.99 ft; R: 41.99 ft; I: 21.00 ft) T : D: 210.0 m; O: 84.0 m; R: 42.0 m; I: 21.0 m (D: 688.98 ft; O: 275.59 ft; R: 137.80 ft; I: 68.90 ft)
Intelligence	IVS (Perimeter Protection)	Intrusion, tripwire (the two functions support the classification and accurate detection of vehicle and human)
Intelligence	AI SSA	Yes
Intelligence	Quick Pick	With AI NVR, quickly pick up the human/vehicle targets that users are interested in from SMD events
Intelligence	Smart Search	Work together with Smart NVR to perform refine intelligent search, event extraction and merging to event videos
Video	Video Compression	H.265; H.264; H.264H; H.264B; MJPEG (Only supported by the sub stream)
Video	Smart Codec	Smart H.265+; Smart H.264+
Video	AI Coding	AI H.265; AI H.264
Video	Video Frame Rate	Main stream: 2688 × 1520@(1–25/30 fps) Sub stream: 704 × 576@(1–25 fps)/704 × 480 @ (1–30 fps) Third stream: 1920 × 1080@(1–25/30 fps) *The values above are the max. frame rates of each stream; for multiple streams, the values will be subjected to the total encoding capacity.
Video	Stream Capability	3 streams
Video	Resolution	4M (2688 × 1520); 3.6M (2560 × 1440); 3M (2048 × 1536);

		3M (2304 × 1296); 1080p (1920 × 1080); 720p (1280 × 720); D1 (704 × 576/704 × 480); CIF (352 × 288/352 × 240)
Video	Bit Rate Control	CBR/VBR
Video	Video Bit Rate	H.264: 3 kbps–8192 kbps H.265: 3 kbps–8192 kbps
Video	Day/Night	Auto(ICR)/Color/B/W
Video	BLC	Yes
Video	HLC	Yes
Video	WDR	120 dB
Video	Scene Self-adaptation (SSA)	Yes
Video	White Balance	Auto; natural; street lamp; outdoor; manual; regional custom
Video	Gain Control	Manual;Auto
Video	Noise Reduction	3D NR
Video	Motion Detection	OFF/ON (4 areas, rectangular)
Video	Region of Interest (RoI)	Yes (4 areas)
Video	Image Stabilization	Electronic Image Stabilization (EIS)
Video	Smart Illumination	Yes
Video	Defog	Yes
Video	Image Rotation	0°/90°/180°/270° (Support 90°/270° with 2688 × 1520 resolution and lower)
Video	Mirror	Yes
Video	Privacy Masking	4 areas
Audio	Built-in MIC	Yes
Audio	Audio Compression	G.711a; G.711Mu; PCM; G.726; G.723
Alarm	Alarm Event	No SD card; SD card full; SD card error; network disconnection; IP conflict; illegal access; motion detection; video tampering; tripwire; intrusion; scene changing; audio detection; voltage detection; defocus detection; external alarm; SMD; security exception
Network	Network Port	RJ-45 (10/100 Base-T)
Network	SDK and API	Yes
Network	Network Protocol	IPv4; IPv6; HTTP; TCP; UDP; ARP; RTP; RTSP; RTCP; RTMP; SMTP; FTP; SFTP; DHCP; DNS; DDNS; QoS; UPnP; NTP; Multicast; ICMP; IGMP; NFS; SAMBA; PPPoE; SNMP
Network	Interoperability	ONVIF (Profile S/Profile G/Profile T); CGI; P2P; Milestone; Genetec
Network	User/Host	20 (Total bandwidth: 64 M)
Network	Storage	FTP; SFTP; Micro SD card (support max. 256 GB); NAS;
Network	Browser	IE: IE 11 Chrome Firefox

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Network	Management Software	Smart PSS;DSS;DMSS
Network	Mobile Client	iOS;Android
Network	Cyber Security	Video encryption;Firmware encryption;Configuration encryption;Digest;WSSE;Account lockout;Security logs;IP/MAC filtering;Generation and importing of X.509 certification;syslog;HTTPS;802.1x;Trusted boot;Trusted execution;Trusted upgrade
Power	Power Supply	12 VDC/PoE
Power	Power Consumption	Basic: 2.8 W (12 VDC); 3.6 W (PoE) Max. (WDR+IR intensity+Intelligence on): 7.9 W (12 VDC); 9.2 W (PoE)
Environment	Operating Temperature	-30 °C to +60 °C (-22 °F to +140 °F)
Environment	Operating Humidity	≤ 95%
Environment	Protection	IP67
Structure	Casing Material	Metal
Structure	Product Dimensions	244.1 mm × 79.0 mm × 75.9 mm (9.6" × 3.1" × 3.0") (L × W × H)
Structure	Net Weight	820 g (1.8lb)
Structure	Gross Weight	1090 g (2.40 lb)
Structure	Installation	Ceiling mount; junction mount; pole mount (vertical)

2.3.3 ANPR Cameras

Basic	Illuminator Number	6 white light illuminators, brightness adjustable
Basic	Image Sensor	1/1.8" CMOS
Basic	Image Resolution	2688 × 1520 (OSD black strip excluded)
Basic	Video Resolution	Main stream: 4M (2688 × 1520)/ 1080p (1920 × 1080)/ 720p (1280 × 720) Sub stream: 720p (1280 × 720)/ D1 (704 × 576/ 704 × 480)/ CIF (352 × 288/ 352 × 240)
Basic	Video Frame Rate	PAL: Main stream (2688 × 1520@25 fps, 1920 × 1080@25 fps, 1280 × 720@25 fps), sub stream: (1280 × 720@25 fps, 704 × 576@25fps, 352 × 288@25 fps) NTSC: Main stream (2688 × 1520@30 fps, 1920 × 1080, 1280 × 720@30 fps), sub stream (1280 × 720@30 fps, 704 × 408@30 fps, 352 × 240@30 fps)
Basic	Video Bit Rate	H.264B: 32 Kbps–32768 Kbps H.264M: 32 Kbps–32768 Kbps H.264H: 32 Kbps–32768 Kbps H.265: 32 Kbps–32768 Kbps MJPEG: 512 Kbps–65536 Kbps
Basic	Video Compression	H.265;H.264B;H.264M;H.264H;MJPEG
Basic	Image Encoding Format	JPEG

Basic	Noise Reduction	3D NR
Basic	White Balance	Full-automation;Auto Color Temperature Range;Customized Color Temperature
Basic	Electronic Shutter Speed	Auto/Manual 1/50s~1/10,000s
Basic	WDR	96 dB
Basic	Edge Enhancement	Yes
Basic	Exposure Mode	Full auto, customized auto, customized
Basic	Iris Control	Auto (W:1.33—T:1.34)
Function	Trigger Mode	Video detection; I/O coil
Function	Shutter Mode	Single shutter
Function	Storage	1 built-in TF card port Kingston 16 G, 32 G, 64 G, and 128 G; 16 G, 32 G, 64 G, and 128 G; SanDisk 16 G, 32 G, 64 G, and 128 G
Function	Image Tampering Prevention	Yes. Video/image can be checked through watermark and verification
Function	Field of View	H: 40.55° (W)–9.33° (T) V: 22.7° (W)–5.3° (T) D: 46.6° (W)–10.63° (T)
Function	Illumination Distance	30 m (98.43 ft)
Function	Capture Distance	8 m–30 m (26.25 ft to 98.42 ft)
Function	Alarm Event	Alarm for events of no storage card, storage card no space, storage card error, network disconnection, IP conflict, and illegal access
Function	Security	Authorized user name and password, MAC address binding, HTTPS encryption, IEEE 802.1x, and network access control
Function	OSD Overlay	Supports overlay of time, address, plate, vehicle (color, type, accuracy), vehicle posture, counterfeit, checkpoint direction, trigger source, and custom
Function	Automatic Network Replenishment (ANR)	Yes. Transfer through platform/FTP
Function	Auto Registration	Yes
Function	Max. Detection Speed	60 km/h (196850.39 ft/h)
Function	Detected Width	6 m–7 m (19.69 ft to 22.97 ft)
Function	Min. Illumination	0.002 lx (built-in illuminator on)
Intelligence	Vehicle Detection	Vehicle detection rate $\geq 99\%$
Intelligence	Vehicle Recognition	Supports recognizing vehicle type, vehicle color, features of driver and front-seat passenger, licensed/unlicensed vehicle. Vehicle recognition rate $\geq 95\%$
Intelligence	Video Metadata	Yes
Intelligence	Smart TrackFrame	Intelligent frame display. Supports displaying plate and vehicle path
Port	Lens Mount	Module
Port	Analog Output	1 channel
Port	Network Port	1 × 100/1000 M Ethernet port (RJ-45)

Port	RS-485	2
Port	RS-232	1
Port	Alarm Input	3, optocoupler input (on-off input)
Port	Alarm Output	3 ports. 2-channel optocoupler output, 1-channel relay output, which can connect to devices such as barriers
Port	Audio Input	1 channel (RCA port)
Port	Audio Output	1 channel (RCA port)
General	Power Supply	12 VDC, 24 VAC, PoE+
General	Power Consumption	< 20 W
General	Operating Temperature	-30°C to +65°C (-22°F to +149°F)
General	Operating Humidity	10%–90% (RH), non-condensing
General	Casing Material	Metal + plastic
General	Protection	IP67
General	Product Dimensions	515.2 mm × 124.7 mm × 183.2 mm (20.28" × 4.91" × 7.21") (L × W × H)
General	Net Weight	1.9 kg (4.19 lb)
General	Gross Weight	3.8 kg (8.38 lb)
General	Installation	Bent pole bracket (included)
General	Lens	Included
General	Lens Type	Vari-focal
General	Focal Length	10 mm–50 mm

2.3.4 Intelligent Video surveillance servers

System	Main Processor	Industry embedded micro-controller
System	Operating System	Embedded Linux OS
System	Operating Interface	Web (PCAPP), local GUI
Face Detection	Face Detection	1. AI by camera: Supports. 2. AI by device: Supports. Supports detection zone settings, the event can trigger record, snapshot storage (full image), local alarm output, IPC external alarm output, access control, audio prompt, buzzer, preset, email.
Face Recognition	Face Database Capacity	1. Up to 50 human face databases with maximally 500 thousand face images in total, which means that one faces database can contain up to 500 thousand face images. 2. Up to 5 passerby databases with maximally 500 thousand face images in total, which means that one passerby database can contain up to 500 thousand face images. The human face database and the passerby database share the total capacity and number of face images
Face Recognition	Face Recognition	1. Recognition mode: front FR, front FD+ Back FR, front FR+ back FR, back FD+ back FR. 2. AI by Camera: Supports. 3. AI by Device: Function configurations: Supports arming face database and passerby database. Supports enabling general/stranger mode. Supports deploying several human face databases and setting similarity separately. Once the similarity reaches or does not reach the (stranger) threshold, the client software can generate the alarm prompt and trigger record, snapshot storage (full image), local alarm output, IPC external alarm output, access & control, audio prompt, buzzer, log, preset, email.
Vehicle License Plate	License Plate Database Capacity	1. Max. 50 plate databases. One database contains up to 5 hundred thousand plate numbers. 2. Supports black list/white list.

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Comparison		
Vehicle License Plate Comparison	Vehicle License Plate Comparison	Enables plate recognition after adding the ITC camera. Function settings: Deploys several plate databases, once the device detects the plate is matching the information on the plate database (black list/white list), the client can output alarm prompt and then trigger record, snapshot (full image), local alarm output, IPC external alarm output, access and control, audio prompt, buzzer, log, preset, email.
IVS Analytics	IVS	1. AI by camera: Tripwire, intrusion, abandoned object, missing object, fast moving, crowd gathering, loitering, parking, and fence-crossing. 2. AI by device: The general model supports tripwire and intrusion. The advanced model supports tripwire, intrusion, crowd gathering, parking detection, loitering. 10 rules at most. Configuration: When the relevant alarms are triggered, the system responses with linked actions such as PCAPP client message notification, video recording, snapshot storage (background picture), alarm output from IPC or the device, access control, audible warning, buzzer, logs, PTZ preset, and email.
Audio and Video	Access Channel	256-channel
Audio and Video	Network Bandwidth	Incoming bandwidth: 512Mbps, recording bandwidth: 384Mbps, outgoing bandwidth: 128Mbps
Audio and Video	Resolution	24MP/16MP/12MP/8MP/6MP/5MP/4MP/3MP/1080P/960P/720P/D1/CIF/QCIF
Audio and Video	Decoding Capability	2-channel 24MP/3-channel 16MP/6-channel 12MP (20 fps)/7-channel 12MP (15 fps)/6-channel 8MP (30 fps)/12-channel 8MP (15 fps)/7-channel 6MP (30 fps)/9-channel 5MP (30 fps)/11-channel 5MP (25 fps)/12-channel 4MP (30 fps)/18-channel 4MP (20 fps)/19-channel 3MP (25 fps)/12-channel 1080p (60 fps)/24-channel 1080p (30 fps)/36-channel 720p (30 fps)
Audio and Video	Video Output	1-channel VGA output, 3-channel HDMI output, VGA 1/HDMI 1 outputs the same video source. Supports 4K output.
Audio and Video	Multi-screen Display	Customized display. Max. 36-channel local live view. Max. 16-channel PCAPP live view.
Audio and Video	Third-party Camera Access	Onvif, RTSP, Sony, Panasonic, Axis, Arecont, Pelco, Canon, Samsung
Network	Mobile Phone Access	DMSS
Network	Interoperability	ONVIF (profile S), CGI, SDK
Network	Browser	Chrome, PCAPP, IE9 or higher, Firefox
Network	Network Mode	NIC binding mode such as multiple-address mode, load balance, fault-tolerance, etc.
Recording Playback	Multi-channel Playback	Max. 16-channel playback
Recording Playback	Record Mode	Search video detection/manual record/IO alarm/intelligent event/all record file
Recording Playback	Storage Method	Support internal HDD and SAS disk array enclosure.
Recording Playback	Backup Method	HDD, peripheral USB storage device
Recording Playback	Playback Function	1. Play/pause/stop/fast forward/fast backward/reverse play/frame by frame play. 2. Full-screen, backup (clip/file), snapshot, digital zoom, audio on/off.
Storage	Disk Group	16
Storage	iSCSI	support
Storage	RAID	RAID0/1/5/6/10/50/60
Storage	Record Management	Supports record control (continuous, event-based, scheduled, scheduled & event-based).
Alarm	General Alarm	Motion detect, tampering, local alarm, IPC external alarm, fire alarm, temperature alarm, temperature difference alarm, hot spot alarm, and cold spot

		alarm
Alarm	Anomaly Alarm	IPC offline alarm, storage error, HDD full, SSD exception, IP conflict, MAC conflict, login lock, AI module temperature alarm, AI module offline, fan malfunction, no HDD, network security exception, and power exception.
Alarm	Intelligent Alarm	Human face detection, human face recognition, video metadata (person/vehicle/Non-motor vehicle), metadata attributes alarm by device, IVS, plate recognition, people number alarm, stay number alarm, queue alarm.
Alarm	Alarm Linkage	Record, snapshot (full image), local external alarm output, uploading alarms, IPC external alarm output, access & control, audio prompt, buzzer, log, preset, email smart tracking.
Port	Audio Input	1-channel(reserved)
Port	Audio Output	2-channel, RCA port, speaker output, audio linkage output.
Port	Alarm Input	16-channel
Port	Alarm Output	8-channel
Port	HDD Interface	16 slots, SATA3.0/SAS. Max.16T/HDD
Port	eSATA	1 port
Port	SAS	2 SAS3.0 ports
Port	RS-232	1 port, for debug or COM data transmission
Port	RS-485	1 port, control peripheral PTZ and so on. Supports various protocols.
Port	USB	4 ports. 2 USB2.0 port at the front panel and 2 USB3.0 port at the rear panel.
Port	HDMI	3 ports. Support 4K output.
Port	VGA	1 port
Port	Network Port	4 RJ-45 10/100/1000Mbps self-adaptive Ethernet ports
Port	Power	1 port
General	Power Supply	AC100-127V/200-240V 50/60Hz
General	Power Consumption	100W (without HDD, idling) 292W (all HDDs connected)
General	Net Weight	16Kg
General	Gross Weight	17Kg
General	Product Dimensions	Chassis: 444.8mm×522.4mm×133.2m(W*L*H). Max. size: 485mm(with hanger)×522.4mm×133.2mm (with feet mat)(W*L*H)
General	Packaging Dimensions	With Packaging: 689mm×779mm×464mm (W*L*H).
General	Operating Temperature	0°C to 45°C
General	Operating Humidity	10%–90%
General	Operating Altitude	5000m
General	Installation	Desktop/rack
Intelligent Task Analytics	Intelligent Task Analytics	1. Video source: Local video records, import DAV videos from external devices. 2. Execution time: Set the start and end time of the task for each day. 3. Result search: Support searching by attributes, by image and by task. 4. Performance: 1-channel analytics at one time.
Thermal	Thermal	1. Alarms by Camera: Fire alarm, temperature alarm, temperature difference alarm, hot spot alarm, cold spot alarm. 2. Search: Supports searching for thermal alarms. 3. Linkage actions: Video recording, snapshot storage, alarm output from IPC or the device, audible warning, buzzer, logs, PTZ preset, and email.
Fisheye DE warp	Fisheye DE warp	1. Supports fisheye dewarp by the device for up to 3 fisheye cameras. 2. Fisheye mounting mode: Ceiling mount, wall mount, and ground mount. 3. Dewarp mode: 7 modes for ceiling mount: 360°1P+1, 2P,1+3,1+4,1P+6,1+8; 5 modes for wall mount:360°, 1P, 1P+3, 1P+4, 1P+8; 7 modes for ground mount: 360°1P+1, 2P,1+3,1+4,1P+6,1+8. For a 180° fisheye camera, the mounting mode is wall mount by default and cannot be changed. It supports 5 modes:

		180°, 1P, 1P+3, 1P+4, 1P+8.
People Counting	People Counting	1. Alarms: People counting, in area No., queue alarm. 2. Reports: People counting, in area No., and queue report by day or month. Supports searching for reports by area, type, and stay time. Reports can be exported.
Bullet-PTZ Smart Track	Bullet-PTZ Smart Track	Working with cameras to support manual tracking, automatic tracking, positioning by click, positioning by frame selection.
Cluster	Cluster	Supports N+M cluster.
AI	AI by Camera	Supports AI by camera function: Human face detection, human face recognition, video metadata (person/vehicle/Non-motor vehicle), vehicle recognition, perimeter protection
AI	AI by Recorder	Supports AI by IVSS function: Human face detection, human face recognition, entries frequency alarm, video metadata (person/vehicle/non-motor vehicle), metadata attributes alarm (AI by Recorder), IVS (Tripwire, Intrusion, Fast moving, Crowd Gathering, Loitering Detection, Parking), Plate recognition.
General	Disk Bay	16
General	Form Factor	3U

2.3.5 The Embedded storage system

System	Main Processor	64-bit high-performance multi-core processor
System	Operating System	Embedded Linux OS
System	Operating Interface	Web (PCAPP)
System	Controller	Dual controllers
System	Cache	16 GB by default (extendable to 64 GB)/controller
AI Applications	AI by Camera	Face detection, face recognition, IVS (abandoned object, missing object, loitering, crowd gathering, parking, fence-crossing, fast moving, tripwire and intrusion), video metadata (human, vehicle, non-motor vehicle), vehicle recognition, people counting (fisheye or people counting camera required), face & body detection, crowd distribution, smart thermal (call detection, visible channel- smoking detection, thermal channel-smoking detection)
Camera Compatibility	Thermal	Works with camera for fire alarm, temperature alarm, temperature difference alarm, hot spot, cold spot, smoking and calling
Camera Compatibility	People Counting	Works with camera for tripwire people counting, regional people counting and queue people counting
Camera Compatibility	Bullet-PTZ Smart Track	Works with camera for manual tracking, auto tracking, click positioning and select positioning
Camera Compatibility	Fisheye Dewarp	Supports dewarping for fisheye cameras
Audio and Video	Third-party Camera Access	Onvif, Onvif S, RTSP, Sony, Panasonic, Axis, Arecont, Pelco, Canon, Samsung, Hikvision
Compression Standard	Video Compression	Smart H.265+; H.265; Smart H.264+; H.264; MJPEG
Compression Standard	Audio Compression	G.711A; G.711U; PCM; G726
Network	Network Protocol	HTTP; HTTPS; TCP/IP; IPv4; RTSP; UDP; SMTP; NTP; DHCP; DNS; DDNS; P2P; iSCSI; FTP ; SMB ; NFS
Network	Mobile Phone Access	DMSS

Network	Interoperability	ONVIF (Profile S); CGI; SDK
Network	Browser	Chrome; PCAPP; IE9 or higher; Firefox
Network	Network Mode	NIC binding mode such as multiple-address mode, load balance, fault-tolerance, link aggregation, etc.
Recording Playback	Multi-channel Playback	Web playback, max. 16-channel playback
Recording Playback	Record Mode	Auto record; manual record; search video detection; IO alarm; thermal imaging; intelligent event; all record file
Recording Playback	Storage Method	Internal HDD and network disk
Recording Playback	Backup Method	HDD, peripheral USB storage device
Recording Playback	Playback Function	1. Play, pause, stop, fast forward, fast backward, reverse play, frame by frame 2. Full-screen, backup (clip; file), snapshot, digital zoom, audio on; off
Storage	Video Direct Storage (Private Protocol)	Incoming: 512-channel (bandwidth: 1024 Mbps) Recording: 512-channel (bandwidth: 1024 Mbps) Outgoing: 512-channel (bandwidth: 1024 Mbps) Playback: 32-channel (bandwidth: 64 Mbps)
Storage	Video Direct Storage (Onvif)	Incoming: 512-channel (bandwidth: 1024 Mbps) Recording: 512-channel (bandwidth: 1024 Mbps) Outgoing: 512-channel (bandwidth: 1024 Mbps) Playback: 32-channel (bandwidth: 64 Mbps)
Storage	Video Direct Storage (Auto Register)	Incoming: 512-channel (bandwidth: 1024 Mbps) Recording: 512-channel (bandwidth: 1024 Mbps) Outgoing: 512-channel (bandwidth: 1024 Mbps) Playback: 32-channel (bandwidth: 64 Mbps)
Storage	Video Direct Storage (International Protocol)	Incoming: 512-channel (bandwidth: 1024 Mbps) Recording: 512-channel (bandwidth: 1024 Mbps) Outgoing: 512-channel (bandwidth: 1024 Mbps) Playback: 32-channel (bandwidth: 64 Mbps)
Storage	Picture Direct Storage	Incoming: 250-channel (550 KB/picture) Recording: 250-channel (550 KB/picture) Outgoing: 250-channel (550 KB/picture)
Storage	Disk Group	Yes
Storage	iSCSI	Supports client and Server
Storage	Network Disk	Supports client
Storage	RAID	RAID 0/1/5/6/10/50/60; (with enterprise-grade HDD)
Storage	Record Management	Supports record control (continuous, event-based, scheduled, scheduled & event-based).
Storage	Storage Pool	Yes
IPSAN	IPSAN Performance	360-channel × 2 Mbps video stream writing and 24-channel × 2 Mbps video playback (I/O size of VMS is 64 KB)
Alarm	General Alarm	Motion detect; tampering; IPC external alarm
Alarm	Anomaly Alarm	IPC offline alarm; storage error; HDD full; video frame loss; SSD exception; IP conflict; MAC conflict; login lock; fan malfunction; no HDD; network security exception; power exception; disk health exception; RAID exception; storage pool exception; temperature alarm; share service
Alarm	Intelligent Alarm	Face detection, face recognition, IVS (abandoned object, missing object, loitering, crowd gathering, parking, fence-crossing, fast moving, tripwire and intrusion), video

		metadata (human, vehicle, non-motor vehicle), vehicle recognition, people counting (fisheye or people counting camera required), face & body detection, crowd distribution, smart thermal (call detection, visible channel- smoking detection, thermal detection-smoking detection)
Alarm	Alarm Linkage	Record; snapshot (full image); alarm uploading; remote device alarm output; camera audio; buzzer; log; preset; email, smart tracking and warning light. Disarming alarm linkage action by period or one-click sync disarm config with channels
Port	HDD Interface	48 slots, SATA; SAS Max.18 T/HDD hot swapping CMR
Port	eSATA	1 x eSATA/controller
Port	SAS	1 × SAS (up to 12 Gbps)/controller
Port	RS-232	1 × DB9 /controller
Port	USB	2 × USB3.0 and 2 × USB2.0/controller
Port	HDMI	1 x HDMI/controller, only for debug
Port	Network Port	1 x 1-GbE management port/controller; 4 × 1-GbE LAN ports/controller
Port	Power	1+1 redundant power
Port	Network Port Extension	4×1-GbE LAN ports/controller (optional) 2×10-GbE optical fiber ports/controller (optional)
General	Disk Bay	48
General	Form Factor	6U
General	Power Supply	100–127 VAC/200–240 VAC, 50/60 Hz, 8 A/4 A
General	Power Consumption	<220 W (without HDD, idling) <700 W (all HDDs connected)
General	Fan	Intelligent speed regulation fan
General	Net Weight	36 kg (79.37 lb.)
General	Gross Weight	75 kg (165.35 lb.)
General	Product Dimensions	Chassis: 446 mm × 261.4 mm × 736.5 mm (17.56" × 10.33" × 29.00") (W × L × H) Max. size: 482.6 mm (with hanger) × 261.4 mm × 736.5 mm (19.00" × 10.33" × 29.00") (W × L × H) Height: 6U
General	Packaging Dimensions	With packaging: 562 mm × 599 mm × 235 mm (22.13" × 23.58" × 9.25") (W × L × H) Protection box: 491 mm × 584 mm × 631 mm (19.33" × 22.99" × 24.84") (W × L × H)
General	Operating Temperature	0 °C to 45 °C (+32 °F to +113 °F)
General	Operating Humidity	10%–80% (RH), non-condensing
General	Operating Altitude	≤5000 m (16404.20 ft.)
General	Installation	Standard 19 inch rack

2.3.6 Video wall Display.

Panel Dimensions	55"
Viewing Size	1,209.6 mm × 680.4 mm (47.62" × 26.79")
MTBF	50,000 h
Backlight	Direct LED
Aspect Ratio	16:9

Color Temperature	10,000 K $\pm$ 1000 K
Brightness	500 cd/m ²
Static Contrast	1200:1
Resolution	1920×1080
Refresh Rate	60 Hz
Response Time	8 ms
Display Color	8Bit
Viewing Angle	H: 178° V: 178°
Pixel Density	40 dpi
LCD Technology	ADS
Standard Signal Input Port	1 × CVBS, 1 × VGA 1 × DVI-D, 1 × HDMI, 1 × RS-232, 2 × USB, 1 × Audio in, 1 × IR
Standard Signal Output Port	1 × CVBS, 1 × RS-232, 2 × HDMI, 1 × VGA, 1 × Audio out
Power Supply	100–240 VAC, 50 Hz/60 Hz
Power Consumption	195 W (Max.) 175 W (Typical) 0.5 W (Standby)
Control Mode	RS-232, IR Remote
Operating Temperature	0°C~+40°C (+32°F~+104°F)
Operating Humidity	10%–80% (RH)
Storage Temperature	+5 °C to +40 °C (+41 °F to +104 °F)
Storage Humidity	35%–75% (RH)
Product Dimensions	1213.7mm×684.5mm×121.8mm(47.8"×26.9"×4.8")
Bilateral Splicing	3.5 mm (0.14")
Packaging Dimensions	1360mm×945mm×280mm(53.5"×37.2"×11.0")
Net Weight	27kg(59.5lb)
Gross Weight	34.3kg(75.6lb)
Wall-mounting Holes Position	600 mm × 400 mm (23.62" × 15.75"), M6
Installation	Floor-standing; wall mount
Accessory	1 × AC power cable, 1 × HDMI cable, 1 × Network cable, 1 × Remote control, 2 × AAA Battery

Video wall controller features.

- 19" standard 4U ATCA industry box
- Easy and flexible configuration with pluggable card module design
- Supports input of analog/HD camera and various IPCs
- Achieves matrix switch, encoding and decoding with abundant interfaces
- 80ch@1080P or 320ch@D1 signal encoding (full configured)
- 80ch@4K or 320ch@1080P or 1500ch@D1 signal decoding, H.264&H.265 (full configured)
- Support max 60 screens splicing
- Supports network storage that leads to centralized storage management
- Supports online real-time preview
- Convenient controlling mode with web/Local GUI/Network Keyboard/iPad

Embedded LINUX OS: safe, reliable, stable, efficient, easy development and maintenance

Matrix Switch Control

- Analog, network, digital video signal input and switch output
- Signal non-compression direct switch output
- Keyboard control switch
- Modular input, output card design, switch matrix for different specifications of digital videos

A/V Encoding Input

- Adopts MPEG4 coding standard, H.264 video compression standard, dual stream technology, VBR, composite streaming, video streaming encoding, and A/V sync during composite streaming encoding
- A/V encoding card supports 32-ch BNC (CVBS signal), 8-CH HD-SDI, 4-CH DVI (supports DVI, VGA, HDMI signal), 8-ch HDCVI, 4-ch HDMI (supports DVI, HDMI signal)
- Max supports 80-ch HD video encoding capacity or 320-ch SD video encoding capacity
- Supports non-standard stream
- Supports MPEG4/H.264/MJPEG/H265 video standards

A/V Decoding Output

- HDMI output (supports DVI, VGA with convertor)
- 1/4/6/8/9/16/25/36 free splits
- Fluency function doubles original video 25 fps or 30 fps into 50 fps or 60 fps respectively, and thus enhances fluency to view high-speed moving objects
- A single card supports 8-ch 4K HD decoding capacity, 32-ch 1080P HD video decoding capacity, 72-ch 720P HD video decoding capacity and 128-ch D1 and lower SD video decoding capacity
- Supports 1200W, 800W, 500W and 300W HD video decoding
- Supports 8M&12M fisheye video output and adjustment
- Supports max 40-ch 4K output and 60-ch HD output
- Supports 30 preset scenes; User may customize each TV wall layout

Video Wall Splicing

- Random splicing among 60 screens
- Digital zoom in
- 1/4/6/8/9/16/25/36 free splits
- Supports virtual LED & point-to-point HD Base Map

Record and Storage

- Scheduled record and mobile detection record
- Pre-record and delay record
- Redundant record, supports to lock and unlock record file
- Lock and unlock record files
- HDD pack management
- IPSAN, ISCSI standard network protocol storage, support NVR, NAS, EVS and other centralized storages

Network Function

- 6 RJ45 ports, supporting 1000M network
- Supports TCP/IP protocol stack, including TCP, UDP, RTP, RTSP, PPPoE, DHCP, DNS, DDNS, NTP, SADP, SMTP, ISCSI etc
- Supports NAS, IP SAN network centralized storage, and support receiving system log remotely
- Supports management software to achieve remote switching between analog and digital videos on video wall and control with keyboard
- Supports remotely receiving and configuring parameter, remotely rebooting and remotely inputting/outputting parameters

Other Functions

- Complete set of operation, alarm, abnormality and log recording facilitates user and technical staff's maintenance
- Complete user authority management and storage management, while the authority can be subdivided into a channel and a single HDD, making the device more user-friendly
- Supports local and remote online upgrade, guarantee timely update to meet changing market demand
- Supports network storage to accommodate demand from medium to large monitoring systems
- Multiple users and clients login, convenient for users to preview and manage monitoring whenever and wherever possible

2.3.6 Cable

CCTV camera cable: CAT 6 Shielded twisted pair cable. STP cables to maintain a high degree of immunity to RF (Radio Frequency), electrical and magnetic disturbances as well as provide the lowest possible degree of radiated and conducted Radio Frequency emission.

2.3.7 Networking system and components for video surveillance.**Materials**

- A. All materials and products shall be appropriate for the intended use and shall meet all applicable UL, EN, BS, ISO and IEC Regulations.
- B. All system components shall be manufactured by ISO 9001:2000 certified companies and all packaging shall carry clear product identification (catalogue number, quality control number, etc.)
- C. All products shall be new, of the latest version at time of Installation, and brought to the job site in original manufacturer's packaging. Electrical components shall bear the Underwriter's Laboratories label. This listing requirement applies to the entire assembly. Only systems and equipment that meet or exceed the level of quality and capabilities stated within this document will be considered for acceptance. All active and passive components 20% spare capacity shall be provided.
- D. The contractor is to provide a written statement from all manufacturers that all installed hardware and software are the latest models at the time of handover.
- E. Materials shall be listed and approved for the particular application and permitted by the authority having jurisdiction for the application
- F. All components will be approved by the Consultant and afford the most aesthetic value possible while maintaining specified functionality. Hardware will adhere to the following guidelines:
 - 1. Fit and finish to the existing surrounding structure.
 - 2. Unobtrusiveness
 - 3. Functionality
- G. All workstation termination hardware shall match the existing wall surface color as closely as possible, to include mounting boxes, faceplates, and outlets.
- H. Used equipment or damaged material will be rejected.

Core switches for video surveillance servers

- I. Architecture
 - 1. The Core switches shall be based on Cisco Meraki series
 - 2. Guest Core Switches will be based on the Cisco Meraki platform
 - 3. Admin Core Switches will be based on the Cisco Meraki platform
- J. Core Switches shall be provided for the Hotel Network and shall provide connections to the access switches in a collapsed core tier structure
- K. Performance and Scalability
 - 1. The Core/Aggregation switches shall be based on Cisco Meraki series
 - 2. Switches shall provide the redundant 10G uplinks to the access switches
 - 3. Please refer to the Network Schematic for the distribution strategy

- a. Core switch shall be with appropriate switching capacity and shall be scalable to the future switching performance
- b. Core switch shall be able to support 10/100/1000Mbps, 10GE and future 40GE ports density
- c. The switch shall support both Ingress and Egress Multicast Replication modes.
- d. Shall support up to 256K IPv4 and 128K IPv6 entries in the forwarding table.

Access Switch**L. Architecture**

- 1. Access switches shall be based on Cisco Meraki series
- 2. Guest access Switches will be based on the Cisco Meraki platform
- 3. Admin access Switches will be based on the Cisco Meraki platform
- 4. Twenty Four (24) Ethernet RJ-45 PoE+ interfaces of 10/100/1000 speed with dedicated 2 x 10GE SFP+ ports for uplink.
- 5. Forty Eight (48) Ethernet RJ-45 PoE+ interfaces of 10/100/1000 speed, with dedicated 2 x 10GE SFP+ ports for uplink.
- 6. Twenty Four (24) Ethernet RJ-45 interfaces of 10/100/1000 speed, with dedicated 2 x 10GE SFP+ ports for uplink.
- 7. Forty Eight (48) Ethernet RJ-45 interfaces of 10/100/1000 speed, with dedicated 2 x 10GE SFP+ ports for uplink.

Server Farm Switches.

- M. Server farm switches shall be based on Cisco Meraki series
- N. Dynamic layer 3 routing
- O. 24 and 48 ports as required
- P. Multi gig and 10G uplinks

Firewall and Penetration Prevention system

Firewall appliances shall be provided by the ICT standard.

- Q. 2 x Firewalls should be cisco 5515x Firepower Series and ISP Router should be Cisco 2951.

Network Management System

- R. NMS will be based on Cisco Meraki platform.

UPS – Rack mounted (IDF)

- S. Rack mounted UPS required in each IT rack located in the IDF's
- T. 1 – hour UPS autonomy
- U. Battery type: Maintenance-free sealed Lead-Acid battery with suspended
- V. Communications & Management: USB, LED status indicators, Alarm when on battery
- W. Emergency Power Off (EPO)
- X. Network manageable
- Y. Hot-swappable batteries

- Z. Battery failure notification
- AA. Provides the ability to adapt the UPS for optimal performance in specific power environments or generator applications.
- BB. Operating Environment: 0 - 40 °C

2.3.8 Control Console / CCTV rack

A standard 19 inch rack for the server / backup and a custom built panoramic rack with control desks and monitors shall be provided as a part of the system to accommodate all control and monitoring equipment. Suitable rack mounting brackets shall be used for proper fixing of all rack-mounted items. The final design and locations for the control and monitoring stations shall be decided later as per project requirement and to the approval of the Engineer. The PC server rack shall be supplied from Compaq, Honeywell or equivalent makes.

The custom build control desk shall have enough number of 21' and 19" cubicles. Size and type shall be as per the requirements of providing monitors and control equipment for all the ELV system as specified and as shown on the related drawings.

3.0 EXECUTION

3.1 INSTALLATION

Location of the cameras shall be as per the detailed design and minimum requirements of SIRA/ Dubai Police. All Installation shall be carried out by SIRA licensed Engineer and Technician from SIRA licensed system installer.

However, the necessary containment and electrical power to support the Security system shall be the responsibility of the MEP Contractor.

3.2 TESTING AND COMMISSIONING

The Contractor shall be responsible for testing and commissioning of the installation in accordance with all applicable documents in the Contract set.

- 3.2.1 Testing shall be comprehensive and sufficient to demonstrate compliance with each requirement and as per system manufacturer recommendations.
- 3.2.2 a proposed test plan shall be submitted to the Engineer's approval.
- 3.2.3 Final tests shall be conducted in the presence of the engineer's / Client.
- 3.2.4 The complete installation shall be tested and commissioned by the Specialist Subcontractor, who is responsible to demonstrate the operation of the system to the satisfaction of the Client's personnel.
- 3.2.5 The Contractor shall prepare fully detailed Operation and Maintenance Manuals for the system equipment and submit to the Engineer for his approval.

3.3. TRAINING AND OPERATION MANUAL

3.3.1 Operator training shall consist of a two days course conducted on-site by a factory trained professional instructor. Training conducted by installers, technicians, or project managers is unacceptable.

3.3.2 Training materials shall consist of the following:

- A. Formal course outline and agenda.
- B. Operator training student guide for each student.
- C. Hands-on practice with on-line equipment.
- D. Written examinations.

The training course shall be a minimum of two continuous business days.

3.4 WARRANTY

3.4.1 All equipment furnished under this contract shall be warranted for a period of twelve (12) months from the date of final Final handing over of the building.

- A. Respond to service requests on-site, if required.
- B. Replace or repair defective components as required.

END OF SECTION

ELECTRICAL SECTION: 21

ACCESS CONTROL SYSTEM

1.0 GENERAL**1.1 REFERENCE**

Conform to General Requirements for Electrical Services of Division 16.

1.2 DESCRIPTION OF WORKS

- 1.2.1 Contractor shall allow for the supply, installation, testing and commissioning of the complete security access control system as specified and as shown on related drawings.

The system shall be designed for ultimate reliability and maximum operating efficiency using only the highest quality, fully field proven products.

The system components shall be of modular design to allow ease of installation, service, future expansion, upgrades and addition to the system.

The system components shall have operating environments to allow complete functionality at a temperature of 0 to 60 degree centigrade and a relative humidity of 90% (non-condensing).

Scope of works shall include but not limited to the following:

- A. Security management system server and software
- B. Main access control panel
- C. Multiple door controllers
- D. Alarm input panels
- E. Card readers and cards
- F. Door contacts
- G. Exit push buttons
- H. Electric door locks of the various doors
- I. Power supplies
- J. System printer
- K. All related accessories and equipment including network cabling and terminations.
- L. Smart Card Programmer
- M. Badge Maker System
- N. Integration with Fire Alarm system

At the time of installation and commissioning the contractor shall provide all additional components required to eliminate any noise and signal attenuation problems found at site. The work described in this section and the work described under the sections below shall be carried out by a single specialist security subcontractor as the Building Security System (BSS) Sub-Contractor. The BSS Subcontractor shall take end to end responsibility for the complete security system as defined in these specifications and the associated drawings. Equipment not mentioned herein nor shown on drawings but necessary to meet this requirement shall be provided without claim for additional payment.

Specifications and drawings are detailed only to the extent necessary to show design intent and signal flow. It is understood and agreed by the contractor that the work herein described shall be

complete in every detail so as to provide a complete and fully functional system even though every item necessarily involved is not specifically mentioned.

Complete all requirements of the Building security system subcontract prior to the scheduled substantial completion date for each portion of the work.

Provide to the General Contractor a schedule indicating the sequence of work, durations of individual tasks, delivery dates for all material, devices and equipment and detail any interface that must be coordinated with any other subcontractors.

Attend all project meetings as requested by the Owner and the General Contractor.

Provide written status reports at required intervals and in a format acceptable to the Owner.

An updated schedule of work shall be included in each status report.

Comply with the Project Construction Schedule. Provide additional staff and work overtime as required to comply with the Project Schedule and so as not to interfere with other on-site subcontractors in their effort to comply with the Project Schedule.

Provide a complete end to end Building security system (IP Based) solution including all equipment, mounting brackets, accessories and adapters of any kind necessary to accommodate the system installation, operation, testing, or maintenance.

Provide a computer-based Building security system having monitoring and control computers, operator workstations, annunciation devices, and other facilities. Provide field devices located throughout the development as indicated on the drawings.

Furnish all materials, labour and any engineering services to provide complete and professionally installed Building security systems in working order as described herein. Labour furnished shall be specialized and experienced in Building security system installation.

The Contractor is responsible for verifying the completeness of parts lists, the correctness of the type numbers and the overall suitability of the equipment to meet the main purpose of the specifications. The Contractor, even if not specifically mentioned herein or on drawings, shall supply any additional equipment needed in order to meet the requirements stated above, without claim for additional payment. The Contractor must obtain in writing explicit approval from the Engineer for any changes or substitutions to the specification.

Coordination with the Electrical Contractor, door supplier for mortised bolt lock installation in FOH areas and electric latches in back of house areas and it is required to ensure correct containment and conduit, back box locations, and power circuit locations are provided in the correct locations.

Security Control Room layout shall be provided in an efficient and ergonomic manner that conforms to the human factors requirements set out in ISO 11064. The console design, layout and configuration shall take into consideration to the orientation of equipment to suit its functionality, operation in relation to other equipment size, finish, and quality of equipment, heat dissipation and ventilation, cable access, ergonomic placement for visual and physical access and durability. Submit control room layout design for client approval. Submit drawings of all console panel layout, equipment cut outs, materials, finishes, and construction for approval prior to manufacture. All desk and console mounted security equipment shall be uniform in appearance, layout, construction and the like. No variation will be acceptable from different security system suppliers and manufacturers.

All security systems shall be on a dedicated security network LAN (Local Area Network) provided by the security contractor. This includes active (core and edge network switches etc.) and

passive (copper and fibre cables and connections etc.) All edge network devices (cameras, intercoms, readers etc.) will be PoE unless prior approval is given by the Security consultant.

System shall be interfaced with the fire alarm system of the building in order to operate the door locks automatically in case of fire as per Engineer's approved cause and effect logic and fire philosophy.

System shall be interfaced with the CCTV system for real time viewing of CCTV camera outputs to be used for ID verification.

The contact less smart cards of the system shall have an art work imprinted on it based on the design provided by the end user and approved by the Engineer.

System shall support full backward compatibility such that future expansion and upgradation is possible without replacing any of the existing components and making major changes to the existing system

1.3 Quality Control/Assurance

1.3.1 The Quality Control/Quality Control/Assurance duties shall be performed by the Contractor. The system of quality control verification shall be in accordance with ISO 9000 Standards of Quality Control/Assurance.

A. Design, manufacture, testing and method of installation of all apparatus and materials furnished under requirement of these specification shall conform to latest publications or standard rules of the following:

1. Dubai Electricity and Water Authority (DEWA) regulations for electrical installation works
2. Department of Civil Defence requirements.
3. DIC – Telecom Department (DIC-EITC-DU) standards and requirements
4. SIRA- Security Industry Regulatory Agency.
5. IEC 62262 and IEC 60068-2-75 for Vandal resistance IK10.
6. Premises cabling standard EIT/TIA568A.
7. IEEE 802.3 digital data network standard.

B. British Standards

1.4 CODES AND STANDARDS

The following Codes and Standards to the extent specified herein, form a part of this specification. When an edition date is not indicated for a code or Standard the latest edition enforce at the time of proposal submitted shall apply.

Under writer laboratories listing or equally approved international listing laboratories.

1.5 SUBMITTALS

1.5.1 Contractor shall submit all items in accordance with the requirements of Division 1, Submittals and shall include but not be limited to the following:

- A. Model numbers of all components furnished on the job.
- B. Manufacturer's original catalog and technical data sheets for all components.
- C. Input power requirements for all system components.
- D. Complete engineered drawings indicating:
 - 1. Manufacturer model numbers and specifications.
 - 2. Dimensions, layouts, installation details.
 - 3. Point-to-point wiring diagrams for all system devices.
 - 4. Termination details for all system devices.
 - 5. Single-line system architecture drawings representing the entire system.
 - 6. Interfaces with all sub-systems.
- E. There shall be engineer's acceptance form with a check box associated with each card reader and input point to indicate that each point was checked off as properly installed and reporting back to the control panel and the host computer prior to Engineer's acceptance of the system.
- F. Three (3) sets of the Manufacturer's User's Manuals and Installation Manuals.
- G. Course outlines for each of the end user training programs. The course outlines shall include the course duration, location, pre-requisites, and a brief description of the subject matter.
- H. Subcontractor shall include a full statement of compliance exactly as detailed below for ALL clauses, to acknowledge each and every clause, sections and part contained in this document by using one of the following terms only:
 - **Read and Understood** – where a contractual condition is agreed to, or the clause is for information only.
 - **Comply** – where the proposed equipment or services complies with the stated requirement. If the equipment or services have to be modified to comply, state what modifications are required.
 - **Non-Compliance** – Where the proposed equipment, services or contractual condition do not comply with the stated requirement. The contractor shall give full details of any non-compliance.
 - **Partially Compliant** – Where the contractor states the equipment, services or contractual condition are in part compliance. The contractor shall provide a full explanation for any part compliance. If the contractor has an alternative to the stated requirement, fully describe the alternative.
 - **N/A Not Applicable:** The specification paragraph is not complied by the proposed product.

2.0 PRODUCTS

2.1 SYSTEM DESCRIPTION

- 2.1.1 The Security Management System shall be capable of integrating multiple building functions including access control, alarm management, intrusion detection, video imaging and badging, and database partitioning.
- 2.1.2 ACP panels shall be mounted in the locations shown on the security drawings. Where space is not available for wall mounting the ACP controllers shall be provided with horizontal sliding rack mounts.
- 2.1.3 The system shall be modular in nature, and shall permit expansion of both capacity and functionality through the addition of control panels, card readers, and sensors.
- 2.1.4 The system shall incorporate the necessary hardware, software, and firmware to collect, transmit, and process alarm, tamper and trouble conditions, access requests, and advisories in accordance with the security procedures of the facility. The system shall control the flow of authorized personnel traffic through the secured areas of the facility.
- 2.1.5 The user interface at the host computer (server) and at the operator workstation terminal computers shall be a mouse driven graphical user interface (GUI) allowing the user to open and work on multiple windows simultaneously.
- 2.1.6 The System shall be capable of connecting two Buildings using the TCP/IP network and manage the entire facility through a Central Control room.
- 2.1.7 All security systems shall be on a dedicated security network LAN (Local Area Network) provided by the security contractor. This includes active (core and edge network switches etc.) and passive (copper and fibre cables and connections etc.) All edge network devices (cameras, intercoms, readers etc.) will be PoE unless prior approval is given by the Security consultant.
- 2.1.8 If the security LAN is to be connected to an external network, then it shall be protected by a Firewall to be approved by the security consultant and the employer's IT department. The supply, coordination and testing of the firewall will be in the security contractor's scope.
- 2.1.9 Structured cabling shall be independent of all other disciplines and shall be a dedicated security LAN
- 2.1.10 The ACS/ Access control system shall be based on contactless RFID technology. Electronics must support 13.56 MHz and 125KHz multi standard RFID platform that supports 14443A (Mifare), 14443B and 15693 ISO card standards.

- 2.1.11 The ACMS shall be capable of sending Short Message Service (SMS) text messages to users. Provide all hardware and software required to broadcast messages.

All Building security system head end equipment in the Security Equipment Room and Security Control Room shall be fed from emergency generator backed up circuits and provided with UPS power for a period not less than 4 Hrs. The UPS shall be provided by the BSS contractor and shall be of the same brand as the other UPS systems in this project.

2.2 OPERATIONAL REQUIREMENT

System Capabilities

- 2.2.1 General: The system shall be a true Client Server architecture. All system software and firmware required to provide the following system functions shall be existing and fully tested system application software.
- 2.2.2 Database management: The system shall create and maintain a master database of all cardholder records and all system activity for all connected points.
- 2.2.3 Audit trail: The system shall maintain an audit trail file of operator activity, and provide the ability to generate a report by operator, time and date, and type of activity (audit code). The system shall allow the operator to direct the audit trail report to screen, printer, or file. The audit trail feature shall record the following system events at a minimum:
- A. Site parameters modified.
 - B. System login or logout.
 - C. System restart.
 - D. Cardholder added, deleted, or changed.
 - E. Event added, deleted, changed, or executed.
 - F. Alarm message added, deleted, or changed.
 - G. Communications initiated or terminated.
 - H. Field device/points added, deleted, or changed.
 - I. Access privileges added, deleted, or changed.
- 2.2.4 Input point monitoring: Collect and process status information from all monitored points.
- 2.2.5 Alarm annunciation: Audibly and visually annunciate all alarm, tamper and trouble conditions, and advisories.
- 2.2.6 Input point supervision: The system shall electrically supervise all 2-state and 4-state input point circuits as specified or shown on the drawings.
- 2.2.7 Reports: the system shall have built-in report writer module that shall have access to the system database fields to allow users to create custom reports and shall have the following features as a minimum:
- A. Mouse driven graphical user interface with the ability to select database fields from a list of system database fields.
 - B. User definable reports that can be saved and re-run as required without redefining the report fields and format each time the report is run.
 - C. The system shall also ship with predefined reports.
- 2.2.8 On-line help system: The system shall provide online help, which shall be available at any time and from any screen
- 2.2.9 Operator Menu Access: The operator password shall control which menu items that the individual operator may access. It shall also be possible to restrict operators such that certain specified menu commands do not appear on the screen, or are grayed-out (disabled) for a given operator. All user passwords are fully encrypted, even while being stored and transmitted across the network.

- 2.2.10 Alarm Input Point Reporting Delay: The system shall allow the operator to apply an input point reporting delay period from 0-60 seconds for each input point terminal. The default setting for each input point reporting delay shall be 0 seconds.
- 2.2.11 Alarm Input Point Suppression: The system shall provide an alarm input point suppression facility such that the operator may define a time zone suppression period for each individual input point. Alarm conditions for suppressed input points shall not be recorded or archived by the system, however, trouble conditions will be recorded.
- 2.2.12 Alarm graphics: The alarm-graphics portion of the system shall provide dynamic color alarm graphic maps with the following functions:
- A. User definable graphic maps to depict input point conditions in the system.
 - B. The system shall support the importing of most bitmap file format graphics produced with any graphic drawing program such as .TIF, .BMP or .JPG file format. Vector file formats are not acceptable.
 - C. The system software shall be capable of storing a number of graphic maps. The quantity shall be limited by available hard disk storage space only.
 - D. The system shall provide a palette that includes categories of pre-defined alarm map icons for Input, Output, Map Layer, Reader Terminals, Panels, I/O Terminals.
- 2.2.13 Alarm handling: The alarm handling portion of the system shall provide the following functions:
- A. User definable alarm message/instructions name.
 - B. User definable alarm message/instructions description. The system shall provide the ability to assign an alarm message/instructions to each state of an input point ('Open', 'Short', 'Alarm/Active', and 'Secure.')
 - C. Alarm message "pick list": all alarm message names and associated descriptions shall appear in the form of a pick list from which the operator may select an appropriate alarm name and message from all alarm messages defined in the database by the operator.
 - D. Alarm input points: the system shall support up to 2,000 alarm-input points.
 - E. Alarm input point maintenance: the system shall provide the option for the operator to 'Add', 'Edit', or 'Delete' an alarm input point. The 'Delete' option shall require the user to confirm deletion of input point(s). All maintenance functions shall be logged to the audit trail and archived to hard disk of the host PC.
 - F. The system shall support both 2-state and 4-state alarm input point monitoring as called for in this specification or as shown on the drawings.
 - G. Alarm priority: the system shall provide an alarm priority queue from 0-9.
 - H. Alarm popup: alarm inputs that are designated, as "pop-up" by the operator shall take priority over any active "non-alarm" window. If the operator is viewing a non-alarm window when a popup alarm occurs, the alarm queue window shall be automatically placed on top all other windows to allow the operator to respond to the alarm condition.
 - I. Alarm instruction display: the system shall provide a window with up to ten lines of user defined instructions, which shall tell the operator how to respond to a selected alarm.

- J. Alarm condition history display: the system shall provide a window displaying the previous states of the alarm point with a time and date stamp of each condition.
- K. Alarm response entry: the system shall provide a window in which the operator may enter free form text describing how he/she responded to a given alarm or select from a user defined list of pre-defined response descriptions.
- L. The alarm instruction display, alarm condition history display, and the alarm response entry box shall all be a part of one summary window. Separate windows or applications to support any of these three functions is unacceptable.

2.2.14 Event processing:

- A. Host event triggers: the system shall provide the operator with a scrolling list of the event sequence triggers as a minimum that may be combined with the event sequence logical operators to program a custom sequence of events.
- B. Host event actions: the system shall be provide a scrolling list of the event sequence actions as a minimum, and allow the user to attach one or more actions to one or more of the event sequence triggers to program a custom sequence of events.
- C. Logical operators for trigger conditions: the system shall provide a scrolling list of the following logical operators for event trigger conditions
 - 1. = (Equal to)
 - 2. != (not equal to)
 - 3. > (Greater than)
 - 4. < (Less than)
 - 5. >= (Greater than or equal to)
 - 6. <= (Less than or equal to)
- D. Logical operators for triggers (Macro Logic): The system shall provide all or some of the following event trigger logical operators to allow the user to attach one or more of the logical operators with one or more of the event triggers and card actions listed above to program a custom sequence of events:
 - 1. NAND
 - 2. NOR
 - 3. AND
 - 4. OR

2.2.15 Time zones: The system shall provide the capability for the user to define time zones.

2.2.16 Communications:

- A. Communications between the server (Host) and the sub-controllers panels can optionally support redundant paths. Thus the loss of communications on one path automatically causes communications to be established via the other path without operator intervention.
- B. Should the sub-controller(s) lose communications with the Host, the sub-controllers shall continue to control access and monitor inputs for all connected points without any degradation in system performance. Local history of all transactions shall be buffered at the

sub-controller and automatically uploaded to the Host for alarm reporting and long-term historical storage once communications is re-established.

- C. The contractor shall be responsible for the design of a system that will compensate for all signal level losses in the trunk wiring. This shall include any power supplies for the field devices and any signal level converters or repeaters for the proper amplification of electrical signals.
- 2.2.17 User defined cardholder database fields: The system shall support up to 128 user defined data fields, which may be used to store information for each cardholder. Each field may be of a type: alphanumeric text, numeric, date, toggle (Yes/No). The system shall provide standard menu items, which shall allow the operator to define these cardholder database fields at any time. The system shall remain on-line while user defined cardholder database fields are added or edited. It shall be possible, using standard system menu commands to search and report on all user defined cardholder fields.
- 2.2.18 Video Imaging Integration: The system shall include an integrated video imaging system. It shall be possible to view a captured and stored cardholder image as part of a cardholder's data record on an OPERATOR WORKSTATION. The captured image will also be displayed in the image recall window with a statement indicating whether access was granted or denied.
- 2.2.19 Event and Transaction History: The system shall maintain a record of all alarm, card transaction, and system exceptions which take place, and provide a means for a user to access this information. It shall be possible to print information in the log in real-time or by a report.
- 2.2.20 Anti-Pass back Control: The system shall provide the capability to prevent more than one person from gaining access to a controlled area by recognizing when a cardholder who is granted access is passing back the card to another person to use the same card to gain access. If so programmed, an alarm may be generated if the anti-pass back rules are violated by the cardholder. It shall be possible to define on a reader by reader basis, which readers are subject to anti-pass back rules.
- 2.2.21 Anti-Tailgate Control: The system shall provide the capability to prevent more than one person accessing a controlled area as a result of a single card transaction.
- 2.2.22 Entry/Exit Control: The system shall support the capability to control a card's entry into or exit from an area based on the previous transaction status of the card. An alarm may be generated if the In-X-It conditions are violated by the cardholder.
- 2.2.23 Duress Processing: The system shall permit cardholders to indicate that they are requesting access to an area under some forced or duress situation. An alarm may be generated if a duress condition occurs, and the cardholder will be granted access.
- 2.2.24 Cardholder Definition: The system shall provide the capability for the user to define Cardholders with the following identification and operating parameters.
- A. Cardholder name (first, middle, last)
 - B. Cardholder address
 - C. Cardholder phone number and extension number
 - D. Validation period using start and void dates
 - E. User defined cardholder fields. The system shall provide the capability to use these fields in filtering reports
- 2.2.25 Digitized Photo Image Recall. When integrated video imaging is called for, a real time Image Recall window shall be provided on the system. The system shall recall and display cardholder pictures and following information.

- A. Picture
- B. First name
- C. Middle name
- D. Last name
- E. Card reader name
- F. Card number
- G. Card issue level
- H. Transaction type

2.2.26 Real-time system activity window. A real time system activity monitor window shall be available for display on any OPERATOR WORKSTATION screen whenever the system host is on-line.

The real time window shall have the following capability:

Be able to selectively display the following items at the operator's discretion:

- A. Input point alarms
- B. System Exception messages
- C. Access Grant
- D. Access Deny
- E. Access Trace
- F. Entry/Exit Central Mode of operation
- G. Audit Trail

Be able to toggle the display ON and OFF.

2.2.27 System Status Display. The system shall provide a dynamic system status summary display that graphically indicates the following status information, filtered by panel or terminal. All status display information shall be summarized in a single window:

- A. Terminal up/down
- B. Panel up/down
- C. State of input points (alarm, secure, short, open)
- D. Indication of whether each sub-controller, terminal, reader is disabled or not reporting.

2.2.28 Control points. The system shall provide the ability to define input points as control points to be used in input/output linking and event processing sequences of operation. Control points shall not enter the alarm queue and shall not require that an operator acknowledge them when they change state. The control point activity will however, be automatically logged to the history file.

2.2.29 Workstation Control. Workstations shall have the ability to be assigned a Name and have an IP address for network TCP/IP connections. The Workstation will have the following capabilities:

- A. Be identified as a Workstation only, or Workstation/Video Badging.
- B. Have an enable/disable toggle button to allow or disallow operator login at the workstation.

2.2.30 Real Time Printer. The system shall be capable of printing to a network accessible printer as well as printing from an LPT port. The system shall be capable of printing with the following parameters:

- A. Be able to specify printing of the following items, independent from each other:
 - 1. Input Point Alarms
 - 2. System Exception and Event Messages
 - 3. Access Trace
 - 4. Access Deny
 - 5. Access Grant
 - 6. Entry/Exit Central
 - 7. Audit Trail
- B. Have a toggle button to enable or disable Real Time Printing.

2.3 SOFTWARE REQUIREMENTS

2.3.1 The software shall be a minimum true 32 bit alarm and access control software and shall have an installed capacity to accommodate the following at a minimum:

- A. A central database on the host server able to support up to 1,000 Badges maximum.
- B. Unlimited number of access groups.
- C. Unlimited number of password groups each with an unlimited number of operator passwords.
- D. 2-state alarm input points and 4-state alarm input points (or any combination in-between) shall be in compliance with actual site requirements.
- E. Central on-line data storage of historical transactions of capacity defined by hard disc capacity expandable as system resources allow, with a local panel storage capability of up to 1,000 cardholders and events defined by storage capacity.
- F. Minimum Ten (10) levels of alarm priority.
- G. Up to One Hundred Twenty Eight (128) user-defined cardholder fields. The system shall be capable of reporting on any or all of the user-defined fields. Each user-defined field may be defined by the user as alphanumeric, numeric, date, or logical (yes/no).
- H. The Access control system shall be provided as a function within the Security Management System or a Unified Security Platform to assist in the control and monitoring of authorized access to the security access-controlled doors as documented on the tender drawings.
- I. ACS shall be of open architecture with a freely available SDK's and able to support multiple active access control hardware manufacturers.
- J. The ACS shall be able to integrate with multiple non-proprietary interface modules and controllers, access readers and other third-party applications. The ACS shall be an enterprise class IP access control software solution.
- K. The ACS shall have the flexibility to switch to a different access control software or door controller without the need of replacing the hardware or part of the hardware.
- L. The ACS shall be capable of providing lockdown and lockout functions in the event of a high-level threat from inside or outside the building respectively. This feature shall be programmed during the installation phase and its implementation shall be closely

- coordinated with the building operator. Only security operators with the highest level of access rights shall be able to implement this mode of operation. Coordinate exact requirements with building operator.
- M. Full system redundancy shall be built into the system architecture to provide reliable operations at both the equipment and the system workstation. This includes the distribution and structure of the system database and all relative functions and access requirements. Contractor to describe the redundancy level of the system in his commercial offer prior to award.
- N. Upon communication loss, Access Main Controller shall continue functioning in a stand- alone mode, and the onboard buffer shall be capable of locally storing up to 500,000 events until communication is restored.
- O. The ACS shall allow for automatic call up of personnel video image upon request, upon an ACS transaction and as defined by the integrated operations section. The Video Badging System images and database shall be resident on the ACMS server. Retrieval of video images and card data shall be possible from any ACMS workstation. An operator with the necessary password privileges shall be able to edit, add, or delete video images and card data at an ACS workstation.
- P. Guard tour package shall allow the use of access control readers as the checkpoints and shall not require additional hardware installation.
- Q. All schedules shall be defined by day, hours, and minutes. The access control system shall be a complete system, consisting of the following:
1. Access Control Software Application.
 2. Server to host application software.
 3. Workstations and badging workstations.
 4. Door Controllers
 5. Input devices to include card readers, keypads, biometric readers and other devices mentioned within this specification used for obtaining credential information.
 6. Access cards and tags
 7. Request to Exit (REX) Devices and door contact switches
 8. Emergency Break Glass Unit.
 9. The ACS shall provide access validation and prevent unauthorized access at designated facility portals
 10. The ACS shall be of true multi-user design and capable of simultaneous operations from multiple client interfaces.
 11. All ACMS wiring shall be contained within conduit (type as specified elsewhere within the Contract Documentation) and located at least 300mm from any mains power systems in order to avoid potential interference.

2.3.2 System Software

- A. The Host server operating system shall be Microsoft Windows® NT Server version 4.0. It shall have multi-tasking and multi-user capability and support workstations with Windows NT Workstation, Windows 98 or Windows 95 operating systems.
- B. The system database shall be SQL Server, Version 7.0 or equivalent.
- C. The SYSTEM software features shall be fully documented in the form of a complete user's manual including operation and installation sections, and a detailed description of the major system functions.

- 2.3.3 The system shall be capable of partitioning (segmenting) the database which must include at least, but not limited to the following items:
- A. Access Groups
 - B. Panels
 - C. Readers/Terminals
 - D. Workstations

2.4 HARDWARE REQUIREMENTS

- 2.4.1 The minimum System Server requirements shall be a standard name brand personal computer with proper capacity for the intended purpose. The Host computer shall ship factory configured with all software pre-loaded and tested. All computer hardware replacement components shall be available from multiple third party sources. It shall be the latest state of art technology available at the time of installation, however the minimum configuration for the host PC shall be:
- A. Intel Core I7 CPU with a clock speed of 4.9 GHz or greater
 - B. 16.0 Gb of RAM
 - C. SCSI tape drive
 - D. 3Xusb and HDMI ports.
 - E. DVD
 - F. 2 TB SCSI hard disk.
 - G. 3840 x 2160 resolution 4K resolution color, video card with 8 GB RAM
 - H. 28" color LED monitor
 - I. Standard 101- type keyboard and mouse
 - J. 2 network interface controllers (10Base-T Network Controller port) or equivalent.
- 2.4.2 One number printer shall be provided as part of the system. Printer shall be dot matrix, 180 characters per second, bi-directional printers.
- 2.4.3 Sub-controllers shall comply with the following functionality:
- A. The sub-controller shall be a fully stand-alone processor capable of making all access control decisions without the involvement of the Host computer.
 - B. The sub-controller shall support up to sixteen (16) card readers in addition to either 256 input points or 128 input points and 128 output points. It shall further support up to 12 facility codes per reader, 40 unique holidays, 8 access group and time zone pairs.
 - C. Memory Requirements:
 - 1. Minimum number of cards: 1,000 and expandable
 - 2. Minimum number of historical transactions: 5,000 expandable to 75,000 at full card capacity.
 - D. The controller shall require no firmware changes and shall use flash memory modules to provide non-volatile storage of both data and operational code.

- E. The controller shall support the direct connection of a standard dot matrix printer for local transaction and report printing. The printer shall connect to the controller via a built-in serial (RJ45) port.
 - F. Each controller shall be provided with built-in hardware to support hard-wired communications between the controller(s) and readers of up to 4000 feet.
 - G. An alarm summary relay shall be built-in to the controller motherboard. If so programmed, the alarm relay shall be activated whenever a connected alarm point transfers to the alarm state and whenever soft alarms become active.
 - H. A SPDT tamper switch shall be attached to the inner surface of the controller enclosure. The tamper switch shall change state whenever the enclosure door is opened to signal the system of the condition. The tamper switch input shall be user programmable to be suppressed, to be recognized as an input point to be processed by the alarm queue at the host computer, to printout at an optional printer connected directly to the controller, and to activate the alarm summary relay described above.
 - I. The standard AC line power supply version of the controller shall include a battery module to back-up the controller's applications programs and database for 30 days after the failure of the primary AC power service. The controller database, the time clock, the transaction history, and all operator entered parameters shall be backed-up by the battery.
 - J. The controller shall provide built-in LED to indicate whether the controller is properly communicating with the host computer.
- 2.4.4 Alarm monitoring and Output Control terminal boards. Intelligent alarm monitoring and output control terminal boards shall support the following functionality:
- A. Sixteen two-state/four-state supervised alarm input points.

2.5 CARDS AND CARD READERS

2.5.1 General

All readers shall be configured with the reader electronics mounted separately, on the "secure" side of the door such that only the reader head and pilot lights are mounted in the reader housing on the "entry" side of the door.

2.5.2 Smart Card Reader

The Card Reader technology shall be contactless smart card reader designed with the practical site installation requirements in mind, and shall be subjected to CONSULTANT/PROJECT MANAGER final approval.

The concealed fixings shall allow the reader to be mounted directly on to a flush or surface single gang back box. The reader shall also be capable of mounting directly to metal surfaces such as turnstiles etc. with no effect on the reader operation.

Hard anti-pass back (access card shall not be valid at entry card reader until the same access card has been presented at the exit card reader).

Soft anti-pass back (access card shall not be valid at same card reader until an operator definable time period has elapsed).

Shall support reader output to suit the specific application required, i.e. Wideband 26 to 40 bit.

The reader shall also operate in the standalone and RS485 formats as an encoder, writing data to the Mifare card either from its own programmed memory or from a PC or other such device.

All readers when sited externally to a building entrance shall be vandal proof and externally rated. The unit shall be flush mounted where possible into the external fabric of the building and securely fixed. All reader types must be submitted for approval before installation.

The system compatibility to programme the same card for access doors and gate barriers long range reader or separate card for each.

The reader shall have single piece design for easy installation.

IP65 standard finish for external readers.

A single LED that changes color from red to green is also acceptable.

The card reader shall have a built-in tamper detection feature.

The reader shall be available in card only and card plus PIN pad versions. Furnish and install in the style and quantities as shown on the Low Current System drawings, and architectural drawings.

The UHF RFID reader is an RFID long-range proximity card reader which can simultaneously read multiple passive UHF tags at ranges up to 12m for car park entry and exit system.

The reader is waterproof and is suitable for use in a wide range of RFID applications, such as transport management, vehicle management, car parking, production process control, and access control.

- Read range up to 12 meters (adjustable)
- Read Sensitivity: Dual polarization read mode
- Wiegand 26-Bit (Can change to 34-Bit via software)
- Two working mode: Trigger read & Always read
- Comes with the mounting plate and RS-232 cable
- Weatherproof (IP68)
- Maximum Cable Distance: 100m

2.5.3 Read range shall be up to 15cm.

2.5.4 Smart Card Technology

- A. The Contactless smartcard shall not need a direct contact to the reader.
- B. The card which shall have no contact plates for the data to be exchanged with the reader as specified in ISO 7816. It shall communicate using wireless technology with the reader.
- C. The phase signal and the energy to power the card, shall be transmitted via a high frequency signal from the reader to the card.
- D. Data shall then be transmitted bi-directionally by this signal providing a fast and very efficient data exchange.
- E. The Card Shall feature Quick transaction time, simple usage and robustness as standard.
- F. Subject to the style of reader used the card shall transmit data over a distance of up to 15cm.
- G. The Card shall support Contactless technology.

- H. The Card shall support memory sectors for multiple applications.
- I. The Card shall be protected by two software keys per sector for read and write.
- J. The Card Shall support High speed read/write communication
- K. The Card Shall be ISO credit card size
- L. Shall support High security protection of data held within the memory sectors.
- M. Shall feature Anti-collision (multitag operation)
- N. The Smart Card may also be capable of storing the complete two-dimensional structure of the fingerprint image.
- O. Design shall be suitable for semi-recessed and fully recessed installation requirements. Surface mounted units shall be available with:
 - a. Smartcard only
 - b. Smartcard + Proximity only
 - c. Smartcard + PIN keypad
 - d. smartcard, proximity + Pin
 - e. Outdoor readers shall be IP65 rated
- P. The cards shall be constructed of top quality, highly durable and resilient PVC plastic or a PVC/Polyester composite material for use with Smart Card readers. At least 5000 cards shall be supplied along with the system and badge printer shall be provided along with system with spare cartridge for badge printing and all necessary hardware's for full function of the system. The cards to be printed according to the client engineer/ consultant approved logo and apartment number or as requested by the client at the time of testing and commissioning without any additional cost.
- Q. Card shall have an artwork to the requirement and based on the design approved by CONSULTANT. Contractor shall carry out the necessary artworks on the total required quantity of cards based on the design provide by the end user.

2.5.4**DOOR ACCESSORIES**

1. Smart Locks- The lock shall be bolt lock cascaded inside the door frame. The contractor shall co-ordinate with door supplier for compatible type locks. External visible latch type locks are not acceptable in the FOH areas.
2. All Electric locking devices shall operate from 12/24 VDC and shall be failsafe where required by legislation. Provide power supplies for all security doors. Where possible field adjustable Fail Safe/Secure electric strike locks shall be provided.
3. The electric locking devices shall be positioned and mounted in accordance with the manufacturer's instructions, international standards and to the approval of the consultant.

The BOH can be provided with electromagnetic latch type lock.

The door lock shall have the following features:

- Power: 12-24V DC/AC
- Holding Force: 1200 Lb double leaf door / 600 lbs for single wooden door
- No residual magnetism
- Easy access wiring cavity

Be tested for operability for a minimum of 14,000,000 cycles.

Be able to lock and unlock electronically.

Have a lock case that is easy to mortise and have a snug and solid fit in door.

4. Magnetic Door Contact

Magnetic door contacts have a magnet which is to be mounted on the leaf of the door and reed switch mounted on the door frame. Door contacts shall be suitable for the type of the door.

5. Egress Device (Exit Button)

The Delayed Egress Device shall have the following features:

- Power: 12-24V DC/AC
- Finish shall match with door hardware
- Spring loaded momentary action
- “PUSH TO EXIT” in bold lettering on stainless steel faceplate.
- Mushroom button style.
- Local lock power interruption shall be for an adjustable time from 5-40 seconds.
- The push button shall be suitable engraved with "PRESS TO Exit: in black lettering of minimum 6mm height in both English and Arabic languages.
- Dual separate contact monitoring type (double pole) providing independent contact for alarms.
- Push Button activation shall be logged at the ACMS at all times.
- An End of Line resistor is to be installed at the device side to allow monitoring of line fault test.

6. Emergency Break Glass Device

Indoor Call Point – Green Color with NO/NC Contact. Directly connected to Access Control –Door locks for emergency opening and as per civil defense regulation.

1. Provide break glass at locations shown on the drawing. Break glass unit should be installed adjacent the door and when activated should break the direct power to the lock, giving free exit and an audible alarm should sound until reset.
2. Break Glass Unit shall meet the following minimum requirements:
 - a. Green in color with white lettering.
 - b. Double pole contacts and be monitored by the access control system and generate an alarm when activated.

- c. An End of Line resistor is to be installed at the device side to allow monitoring of line fault test.

The unit shall be protected by a clear plastic removable cover.

- d. Hinged cover engraved with "Emergency Used Only" in both English and Arabic languages.

- e. Visible warning flag confirms

activation

- f. Low-profile, heavy-duty termination.

- g. BGU shall require only finger pressure to activate.

- h. UL Listed

2.6 SECURITY INTERCOM SYSTEM

- A. Provide a SIS with Master Control Stations and remote stations at locations indicated on the Security System Drawings and as indicated in these documents. Refer to the Security System Drawings. Provide all software licenses as required. The SIS shall include parking Video & voice intercom stations.
- B. Provide all required installation and components including central exchange modules and all necessary boards, power supplies, master and sub-master control stations, receptacles, mounting boxes, terminal boards, cable, connectors, programming, accessories, signal conditioning equipment, amplifiers, connectors, wire and cable, installation work, mounting, etc. as required for a fully functional turn-key installation.
- C. Provide installation of the SIS only by manufacturer factory trained technicians. The BSS subcontractor shall have no less than two factory trained and certified staff personnel working on the installation of the SIS. Installation of the SIS by individuals not factory certified will not be acceptable.
- D. The SIS shall be controlled through a microprocessor-based controller capable of providing the required monitoring, switching and control of the specified equipment and future expansion requirements. Provide the required number of controller cabinets to incorporate all required inputs and outputs. The SIS provided shall incorporate modular central exchange cabinets for mounting of plug-in boards and modules. Central exchange cabinets shall be modular and shall allow expansion and networking. The SIS controller and all of its components shall be provided as standard 19-inch console mounted units along with racks, face plates, power supplies, mounting hardware and appurtenances. The SIS controller shall be mounted in the Security control Room. There will be a 10" TFT display shall be provided in the security control room for monitoring when a call initiated from outdoor security intercom station.
- E. The system shall be integrated with the ACS. Intercom stations shall be managed directly from the ACS at each operator workstation. The outdoor security intercom station shall be IP68 and mounted on a pedestal bracket as shown in the drawing. The guests/visitors can access the outdoor call station without getting out of the vehicle.

3.0 EXECUTION**3.1 INSTALLATION**

- 3.1.1 All consoles, terminals, and controllers shall be factory wired before shipment to the job site.
- 3.1.2 Controller cabinet doors shall open a minimum of 170 degrees to avoid blocking personnel movement. Each door shall be equipped with a cylinder lock, a tamper switch and a piano-type hinge with welded tamperproof pins.
- 3.1.3 Provisions shall be made for field wiring to enter the cabinet via standard knock-outs at the top, bottom and sides of controller cabinets.
- 3.1.4 Each wire shall be identified at both ends with the wire designation corresponding to the wire numbers shown on the wiring diagrams.
- 3.1.5 All exposed wiring within the cabinets, consoles, and terminals shall be formed neatly with wires grouped in bundles using non-metallic, flame-resistant wiring cleats or wire ties.
- 3.1.6 All ferrous metal work shall be painted, in accordance with the manufacturer's standards.
- 3.1.7 Input and output wire and cable requirements shall be application specific and shall use the proper shielded cable as required by the specific application / code and as per manufacturer requirement

3.2 TESTING AND COMMISSIONING

- 3.2.1 The Contractor shall be responsible for testing and commissioning of the installation in accordance with all applicable documents in the Contract set.
 - A. Testing shall be comprehensive and sufficient to demonstrate compliance with each requirement and as per system manufacturer recommendations.
 - B. A proposed test plan shall be submitted to the Engineer or Owner's representative for approval before commencement of final test.
 - C. Final tests shall be conducted in the presence of the Engineer or Owner's representative.

3.3 TRAINING AND INSTRUCTION

- 3.3.1 Operator training shall consist of a two days course conducted on-site by a factory trained professional instructor. Training conducted by installers, technicians, or project managers is unacceptable.
- 3.3.2 Training materials shall consist of the following:
 - A. Formal course outline and agenda
 - B. Operator training student guide for each student.
 - C. Hands-on practice with on-line equipment.
 - D. Written examinations.

The training course shall be a minimum of two contiguous business days.

Additional equipment Video Imaging training sessions shall be made available to the Owner if necessary, at additional cost.

3.4 WARRANTY

3.4.1 All equipment furnished under this contract shall be warranted for a period of twelve (24) months from the date of final Owner acceptance of the system.

- A. Respond to service requests on-site, if required.
- B. Replace or repair defective components as required.

END OF SECTION

ELECTRICAL SECTION: 22**AV-BGM SYSTEM**

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1 PART 1 – GENERAL**1.1. Scope**

Scope of Work in this section includes the following:

1. Supply, delivery, supervision, coordination, installation of equipment items specified herein and shown on The Contract Drawings as well as incorporation of Employer Furnished Equipment (E.F.E), testing, documentation, and instruction related to complete Audio, Video, Racks, Consoles, Cables, and Control wiring for the Audio-visual System.
2. Complete Audio-Visual Systems for Residential Towers shall include Public Spaces like gym, cinema, pool area, outdoor garden.
3. The SPL/STI level for each space shall be complied and contractor shall submit SPL/STI calculation using manufacturer approved software's.
4. Provide all equipment, materials, labour, and services, not specifically mentioned or shown, which may be necessary to complete or perfect all aspects of the installation. Ensure that they comply with requirements stated or reasonably inferred by the specifications.
5. Perform all work; provide all products, systems integration, engineering, design work, and testing required for the project in order to ensure fully operative systems and proper installation of equipment. Provide calculations and analysis to support design and engineering decisions as specified in submittals.
6. The Contractor shall ensure that all of their works are coordinated both in terms of design and on-site integration with the works of other contractors, including but not be limited to the contractors for other systems installations, any sub-contractors employed by the Contractor and other relevant parties including the Employer and / or the Consultant and / or its representatives.
7. The Contractor shall designate qualified person(s), other than the Project Manager, to be responsible for the on-site quality control and attendance of regular design with Employer implementation progress meetings. The Contractor shall furnish adequate information and drawings in a timely manner to the Employer and / or the Consultant in ensuring smooth AV systems implementations.
8. The Contractor shall provide AV system documentation including system schematics, design topology, wiring schemes, labelling conventions, rack layout plans, Equipment layout plans, and also other information relevant to utilizing this infrastructure.
9. Contractor shall ensure that all ducts and conduits till display screens are done as required and shall consider any type of drill work under his scope.
10. The Contractor shall be responsible in developing requirements into a fully functional and complete system to produce a detailed AV design that is fully coordinated with other building services, and architectural features.
11. The AV installations must be fully coordinated with the interior design and with other technologies such as voice/data and the fire alarm systems. AV control system must have an audio override by the fire alarm system.

12. The Contractor shall conduct on-site user training on the System to the Employer's operation team and relevant representatives under the supervision of the consultant.
13. The Contractor shall ensure that all of their works are coordinated both in terms of design and on-site integration with the works of other contractors, including but not be limited to the contractors for other systems installations, any sub-contractors employed by the Contractor and other relevant parties including the Client and / or the Consultant and / or its representatives.
14. Specialist shall be responsible to ensure all system component & requirements not shown in drawings are included to form a fully functional system without additional cost to the client.

1.2. CODES AND STANDARDS

1. Codes:

Work should be performed in accordance with applicable requirements of governing codes, rules and regulations including the following minimum standards, whether statutory or not:

- Local Building Code
- National Electric Code (NEC)
- National Fire Protection Association (NFPA)
- Federal Communications Commission (FCC)
- City, and other local codes and requirements

2. Standards:

Equipment and materials specified should conform to the current edition of the following standards where applicable:

- UL – Underwriters' Laboratories
- ASTM – American Society for Testing Materials
- NEMA – National Electrical Manufacturer's Association
- ANSI – American National Standards Institute
- ETL – Electrical Testing Laboratories
- SMPTE – Society of Motion Picture and Television Engineers
- EIA – Electronic Industries Association
- ISO – International Standards Organization
- SCTE – Society of Cable Television Engineers

2 PART 2 – SYSTEM & PRODUCT

2.1. Design Topology

The Residence project expects good quality BGM and Audio-Visual system.

2.1.1. BGM System Requirement

The following is a general outline of minimum requirements for the Background Music system:

- Digital music Server with minimum 04 output channels
- CD/MP3/USB Professional player with RS232
- Amplifiers, speakers, and volume controls with channel selector

- Equipment rack with complete accessories.
 - UPS rack mounted
 - Audio mixer on table inside BGM/AV Server room
 - AV Tie-in points with BGM DSP to play local music, follow drawings for locations.
1. As a design principle, ensure that guests are not able to hear more than one music source at any time, in public spaces.
 2. BGM system should have scheduling feature, so client can prearrange music changes according to time of day. As per program changes occur automatically and do not require client intervention or action.
 3. Provide central matrix audio distribution system. The matrix system shall also provide digital signal processing capabilities, volume controls and similar functions should be available over Ethernet.
 4. Contractor should provide Separate Network system for AV and BGM system. BGM music should be provided to all AV racks through Network switch.
 5. BGM, Ceiling speakers shall be 5.5“, 6” or bigger size, depending on the ceiling height (Refer Drawings)
 6. Speaker should have minimum tapping of 25W, 30W, 50W, 60W, 100W.
 7. Speaker frequency should be 50hz to 21khz.
 8. Speaker IXP lynxpro shall have an amplitude response of 50 Hz to 20 kHz. Pressure sensitivity, 93 dB SPL at 1 meter (3.3 feet) on axis with 1 watt of pink noise (ref. 20μPa). Minimum impedance, 6.5 ohms.
 9. Proposed System should have Dante Amplifier and can be expanded easily.
 10. BGM Rack should have integration with Fire evacuation system. During Trigger from fire evacuation BGM music should be muted.
 11. In Cinema provide ultra HD short throw projector and 75 “motorized screen to be provided. The Cinema are sound system shall be based on Dolby ATMOS 7.1 surround sound system. There will be an AV plate for local media connectivity and 7” touch control panel interfaced with AV rack for cinema selection and media selection and control. The complete cinema system shall be interfaced with dimming system for dimming the lights inside the cinema room in the event of projection.
 12. Wall mount Rack should be opened from behind for easy maintenance. Center section swing is reversible and can be both key locked and padlocked. Floor mount rack should have dual door perforated.
 13. IT/AV rack combined should have proper lock and keys and adequate distance between equipment for heat dissipation.
 14. Minimum 20% Spare channel capacity should be kept in each BGM & AV Amplifiers.
 15. Provide Facility panel/Audio Tie line to connect local source & provide Touch Control panel to select source and control volume. Location should be as per drawing and client requirements. Below area should be consider for facility panel:
 - Gym, Cinema, Outdoor Garden, pool area.
 16. Provide a secured system to prevent unauthorized access to the audio system electronics when an authorized operator is not present.

17. Contractor should coordinate with ID & client for reflected ceiling plans.
18. The Front Office personnel should have convenient access to the touch control panel to select source and volume
19. During emergency voice evacuation messaging; BGM system should be automatically muted
20. Sufficient Cabling (Fiber/AV/Cat6) should be considered between BGM/Fire Evacuation/Voice Evacuation panel (located elsewhere in project) and AV Room for Integration functionalities
21. Contractor should co-ordinate with interior designer and architect for BGM/AV niche, jack locations and speaker types.
22. Contractor should coordinate with the ID and architect for Finishes.

2.1.2. BGM & Video scope of work

Background music should be provided the following areas:

Level 5 Cinema , Outdoor Pool.

Level 12 Outdoor Garden.

Level 21 Gym, Yoga, Pool.

Speaker Type

SP1 – Ceiling mount 6inch .

SP2 – floor mounted mushroom type speakers.

END OF SECTION

APPENDIX - A

**SCHEDULE OF APPROVED MANUFACTURES FOR
ELECTRICAL WORKS**

SCHEDULE OF MANUFACTURERS

All the materials supplied under this contract shall be from one of the manufacturers listed below and subject to Engineer's approval. All proposed materials must comply fully with the contract documents requirements. All materials country of manufacturing shall be from USA / Europe unless specified.

ELECTRICAL SERVICES:

S. No.	MATERIALS	MANUFACTURERS
1	LV Panels, MDB's, SMDB's, DB's, ATS & CB's, Synchronizing Panel	ABB – (Germany / Sweden) Schneider Electric (UK) EATON- Moeller (UK) Siemens (Germany) AlFanar - KSA
2	MCCB / ACB's / MCB's	ABB – (Germany / Sweden) Schneider Electric (UK) EATON- Moeller (UK) Siemens (Germany) AlFanar - KSA Protective devices used shall be the same manufacturer as distribution equipment with selected manufacture used throughout.
3	Panel Assembler	Vergar De Porte – UAE Danway Switchgear - UAE Ever shine Switchgears- UAE Faisal Jassim Switchgears- UAE ALFanar Switchgears- UAE Or Equivalent approved
4	Enclosures for LV Panel Boards, MDB's & SMDB's & DB's	Same as component manufacturer, Assembled and tested only by the above respective Authorized Local Agent.
5	Digital Meters	CELSA- Germany SELEC – USA SOCOMECE-France
6	UPS	EATON (Powerware) - Europe Libert – France Legrand – France Schneider – France Riello – Italy BORRI - Italy
7	VFD / Variable frequency Drives	ABB -- Germany / Sweden Siemens – Germany Eaton Vican – Finland Omron- Japan Allen Bradley (USA) Eaton – UK / USA
8	Generators	Caterpillar (USA) SDMO (France)

S. No.	MATERIALS	MANUFACTURERS
		CUMMINS (Northern America / Europe) FG Wilson (UK / Larne) Perkins – UK MTU (France) Leroy Somer (France) Alternator
9	Floor Boxes	MK - UK Ackerman - UK Legrand – UK Cupra – Malaysia
10	Isolators (Metallic IP65) Disconnect Switches. Fire rated accessories and Isolators (Civil Defense Approval required)	ABB – (Germany) Eaton - (Northern America / Europe) SCAME -Italy Siemens – (Germany) Craig & Derricott – (UK)
11	Sub Main Cables & Wires	Ducab – (UAE) Riyadh Cables – (Saudi Arabia) Oman Cables – Oman National Cable - (UAE) AlFanar
12	Single Core Wires	Ducab – (UAE) Riyadh Cables – (KSA) Oman Cables – (Oman) National Cables – (UAE) Al Fanar
13	Fire Rated / Heat Resistant / Fire Alarm / Emergency Cables (CD Approved make)	Ducab – (UAE) Firecell – (UK) Pirelli Gold (Prysmian) – (UK) Firetuff TEKAB FireTek – (UAE)
14	Co-Axial Cables	Raydex Belden California
15	Flexible Cables	Ducab - DuFlex (UAE) Pirelli (UK) Tecab (UAE)
16	Cable Termination and Accessories (Glands, Lugs.....)	Ducab- Ducab Connect – (UAE) BICC – (UK) BIAX – (USA) Beldon – (USA) CMP- (UK) Simplex -UK
17	Wiring Accessories Subject to ID / Architect selection of wide rocker switches	Legrand – (France) – Synergy, Plexo & Clipper MK – UK (Avant & Aspect range) British General –Evolve range Schneider (Unica, Pieno, AvatarOn, Vivace)
18	PVC Conduits and Accessories	Decoduct – UAE

S. No.	MATERIALS	MANUFACTURERS
		LanRic – Malaysia EgaTube – UK Marshal Tuflex- UK
19	Cables Sleeves and Ducts (to BS 3505 class c)	Hepworth - UAE Gulf Eternit - UAE National Plastic - UAE Cosmoplast - UAE
20	GI Boxes	Eletra / Alfamar - (KSA) Barton (UK) Decoduct MK – UK LanRic Malaysia
21	GI / Cable Trays, Ladders, Trunking	PSI UniTech EMI LanRic – Malaysia Barton (UK) OBO-Betterman
22	Cable tray / trunking – FRP Type (External Use)	CoveTech – Sweden Intech- Malaysia Cupra – Malaysia LanRic - Malaysia
23	UPVC Skirting Trunking	Marshal Tuflex – UK MK – UK Ackerman- Germany
24	GI Conduits & Fittings	Barton-UK Burn Tube - UK LanRic – Malaysia Caparo - UK
25	Metallic Flexible Conduits and Accessories. (PVC Coated)	Barton - (UK) Ackerman - (UK) Legrand – (France) Cupra – Malaysia SMARTr (Singapore) LanRic Malaysia
26	Lightning Protection System	Furse / ABB (UK), Wallis (UK), DEHN (Germany), ERICO (USA). OBO-Betterman (Germany) Al Bahra - KSA
27	Earthing System	Furse / ABB (UK), Wallis (UK), DEHN (Germany), ERICO (USA). OBO-Betterman (Germany) Al Bahra - KSA

S. No.	MATERIALS	MANUFACTURERS
28	Cable support System	Fisher Hilti Erico –Caddy SMARTr (Singapore) LanRic Malaysia
29	CEILING ROSES	MK LEGRAND British General - UK
30	Light Fittings: (as per the schedule of light fittings) All USA / EU	As per the Architect Choice & the Technical Approval of the Electrical Engineer. Sylvania – UK, France (Concord) Thorn– UK /USA COOPER LIGHTING - UK DISANO LIGHTING – ITALY FAHERHULT LUCECO TRILUX
31	LED chips for Fixtures	Cree Lumiled Nichia Samsung Citizen
32	Electronic Ballasts / Control gears for light fittings	Meanwell - Taiwan Tridonic -Austria Vossloh Schwabe - Germany Philips - EU
33	Lamps	Osram – EU Philips - EU Sylvania – UK Samsung - Korea GE - EU
34	Bus Bar System	Gersan - Turkey PowerBar - UAE EAE – Turkey Eaton – UK Poligiano– UAE
35	Aircraft Warning Lights	Thorn Cooper Siemens Air Field light Honey well
36	Electrical Vehicle Charger	SCAME- Italy SCHNEIDER -Australia ABB- Germany Eaton – U.K EVGo – USA Blink – USA

S. No.	MATERIALS	MANUFACTURERS
	<u>Fire Alarm & Life Safety Services</u>	
1	Integrated Fire Alarm & Voice Evacuation System	Edwards Systems Tech (EST) - Canada Simplex – (USA) Gent-U.K Potter- USA Triga- Honeywell Cerebrus -Siemens
2	Central Battery System	Inotec- Germany EATON –CEAG, U.K Gessler – Germany RP Technic- Germany. Awex (Poland) Shield - EU
	<u>ELV System</u>	
1	Lighting Control System	BEG- Germany ABB – Switzerland Merten- Germany. F&G – Germany. Schneider-KNX system
2	Gate Barriers & Parking Management	ZKTeco- U.K Federal APD - USA BFT – Italy CAME - Italy PARK ASSIST – Europe Designa (Germany)
3	Disabled toilet alarm system- Standalone with interface to FAS	Sysscall. X-Tec Zeta Hoyles
	<u>Security / IT Services</u>	
1	CCTV System	Bosch – (Holland) CITA-U. K Axis- Sweden VANGUARD Uniview Technologies Panasonic Dahua Technologies. Hanwha
2	Access Control System	Bosch – (Holland) Dahua Technologies. SUPREMA- SOUTH KOREA Honeywell – (USA) Siemens Axis- Sweden

S. No.	MATERIALS	MANUFACTURERS
3	Public Address/BGM/AV system	BOSCH – Holland/Germany BOSE -USA JBL QSC - USA TOA- Japan
4	Audio Intercom system -	Comelit- U.K Fermax – Spain. Urmet- Italy. AI Phone- Japan. Dahua Technologies
5	Structured Cabling System (Passive components)	R&M – (USA) Furukawa Electric - Japan Belden - (UK) Schneider – (France) Nordex – (USA) CommPort Infinique -Canada 3M - USA Excel – UK Datwyler -Swiss Panduit- USA
6	IT Cabinets	Logitec Rittal Furukawa Electric Legrand Schneider Avayo

NOTE: - ALL FIRE RELATED MATERIAL, EXAMPLE FIRE ALARM, FIRE FIGHTING, FIRE STOP MATERIALS, CABLES, CENTRAL EMERGENCY LIGHTING ETC. MUST HAVE LOCAL CIVIL DEFENCE APPROVAL.

ALL PRODUCTS COUNTRY OF ORIGIN SHALL BE FROM USA / EUROPE UNLESS OTHERWISE INDICATED
